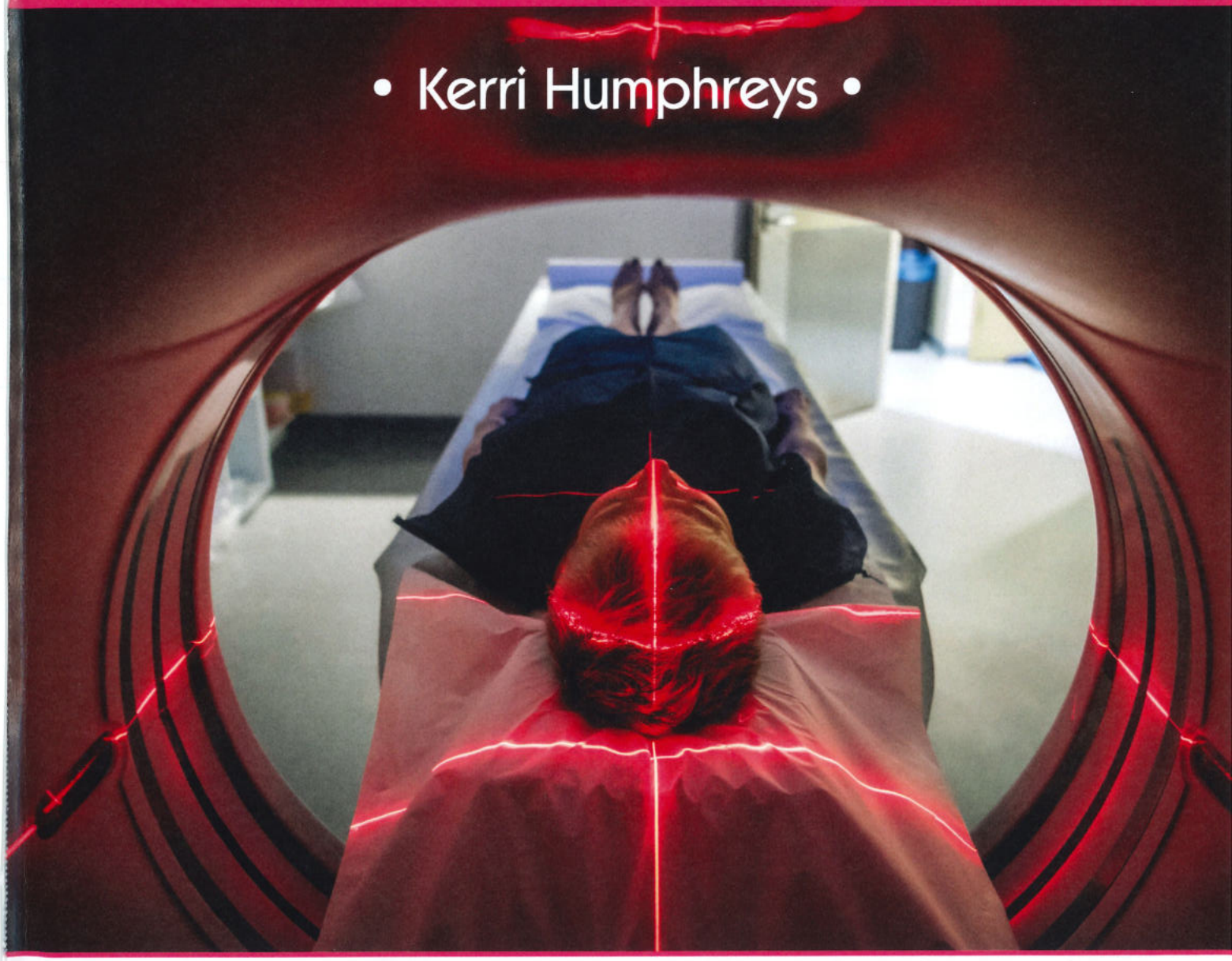
SURFING

WACE HUMAN BIOLOGY

Unit 3 Homeostasis and Disease

• Kerri Humphreys •



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Introduction

Each book in the *Surfing* series contains a summary, with occasional more detailed sections, of all the mandatory parts of the syllabus, along with questions and answers.

All types of questions – multiple choice, short response, structured response and free response – are provided. Questions are written in exam style so that you will become familiar with the concepts of the topic and answering questions in the required way.

Answers to all questions are included.

A topic test at the end of the book contains an extensive set of summary questions. These cover every aspect of the topic, and are useful for revision and exam practice.

Words To Watch

account, account for State reasons for, report on, give an account of, narrate a series of events or transactions.

analyse Interpret data to reach conclusions.

annotate Add brief notes to a diagram or graph.

apply Put to use in a particular situation.

assess Make a judgement about the value of something.

calculate Find a numerical answer.

clarify Make clear or plain.

classify Arrange into classes, groups or categories.

comment Give a judgement based on a given statement or result of a calculation.

compare Estimate, measure or note how things are similar or different.

construct Represent or develop in graphical form.

contrast Show how things are different or opposite.

create Originate or bring into existence.

deduce Reach a conclusion from given information.

define Give the precise meaning of a word, phrase or physical quantity.

demonstrate Show by example.

derive Manipulate a mathematical relationship(s) to give a new equation or relationship.

describe Give a detailed account.

design Produce a plan, simulation or model.

determine Find the only possible answer.

discuss Talk or write about a topic, taking into account different issues or ideas.

distinguish Give differences between two or more different items.

draw Represent by means of pencil lines.

estimate Find an approximate value for an unknown quantity.

evaluate Assess the implications and limitations.

examine Inquire into.

explain Make something clear or easy to understand.

extract Choose relevant and/or appropriate details.

extrapolate Infer from what is known.

hypothesise Suggest an explanation for a group of facts or phenomena.

identify Recognise and name.

interpret Draw meaning from.

investigate Plan, inquire into and draw conclusions about.

justify Support an argument or conclusion.

label Add labels to a diagram.

list Give a sequence of names or other brief answers.

measure Find a value for a quantity.

outline Give a brief account or summary.

plan Use strategies to develop a series of steps or processes.

predict Give an expected result.

propose Put forward a plan or suggestion for consideration or action.

recall Present remembered ideas, facts or experiences.

relate Tell or report about happenings, events or circumstances.

represent Use words, images or symbols to convey meaning.

select Choose in preference to another or others.

sequence Arrange in order.

show Give the steps in a calculation or derivation.

sketch Make a quick, rough drawing of something.

solve Work out the answer to a problem.

state Give a specific name, value or other brief answer.

suggest Put forward an idea for consideration.

summarise Give a brief statement of the main points.

synthesise Combine various elements to make a whole.

1 Assumed Knowledge

QUESTIONS

1. What is meant by the term 'metabolism'?
2. Distinguish between the internal and external environment of an organism.
3. Define receptor and effector.
4. List the five main senses found in humans.
5. Name the five sense organs found in humans.
6. Figure 1.1 shows most of the sense receptors found in the skin.

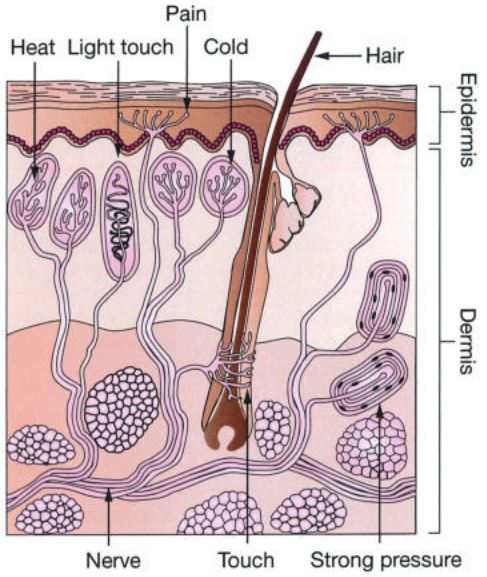


Figure 1.1 Senses in the skin.

Identify the different types of receptors found in the skin.

7. What is interstitial fluid?
8. Humans are endothermic. What does this mean?
9. Outline the function of the nervous system.
10. The nervous system is often divided into the central nervous system and the peripheral nervous system. Compare these two systems.
11. What is meant by ambient temperature?
12. Describe the function of the circulatory system.
13. The following diagram shows cellular components found in blood.

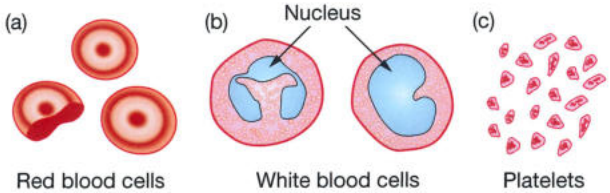


Figure 1.2 Blood cells.

Outline the function of each of the three types of cellular components shown in the diagram.

14. Define artery, vein and capillary.
15. Define diffusion.
16. In an experiment, what is a control?
17. Briefly describe the lymph system.
18. The following diagram shows a cross-section of the human brain.

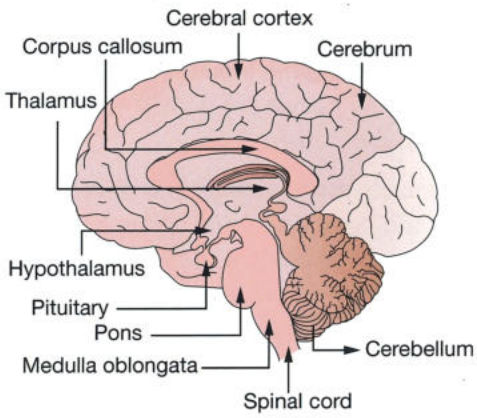


Figure 1.3 Cross-section of human brain.

Outline the function of the spinal cord and cerebrum.

19. The following diagram shows gas exchange in the lung.

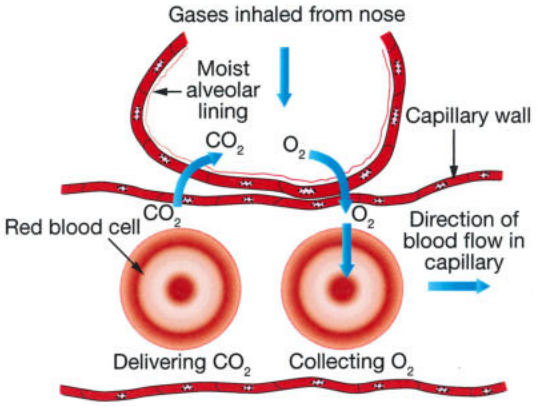


Figure 1.4 Gas exchange.

Describe the movement of gases in and out of the alveoli of the lungs.

20. The following diagram shows the distribution of body water.

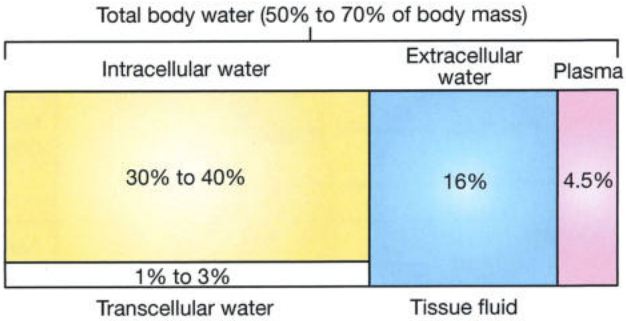


Figure 1.5 Distribution of body water.

Describe why water is an important molecule in organisms.

21. Define parasite.
22. Briefly describe bacteria.
23. Briefly describe viruses.
24. What is a pathogen?
25. Define a hormone.

2 The Endocrine System

The **endocrine system** is a body system composed of different endocrine glands that are ductless glands which secrete hormones directly into the bloodstream or body fluids. The glands are widely separated from each other.

Hormones are chemical signals that travel in the blood and body fluids and act on specific target cells to cause a specific response for internal communication and regulation and maintaining homeostasis. There are three main types of hormones – the peptide hormones, the steroid hormones and the amine hormones.

Many hormones are relatively slow acting with their effects being long lasting.

The **target cells** have **receptors** for that particular hormone which means that the hormone circulating in the bloodstream will only affect cells in particular locations. The receptors for water soluble hormones are on the cell membrane while the receptors for lipid soluble hormones are inside the cell. The binding of the hormone to the receptor initiates a chemical or metabolic reaction in the cell.

Hormone levels in the blood vary and are controlled by a feedback system.

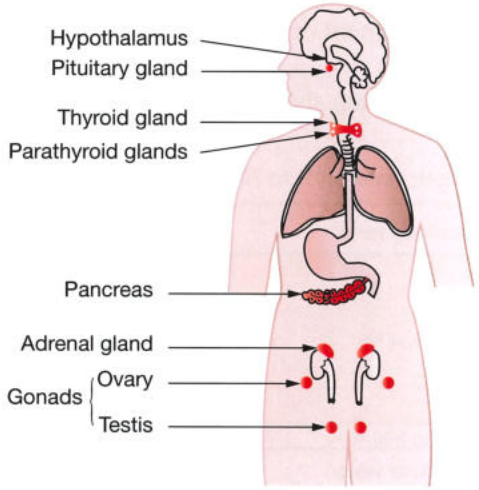


Figure 2.1 Endocrine system.

Homeostasis is the maintenance of a constant internal environment within narrow limits, e.g. homeostasis maintains constant conditions for blood pH, carbon dioxide concentration, blood glucose concentration, body temperature and water balance.

Table 2.1 Some human hormones.

Hormone	Synthesis site	Target cell/ organ	Effect
Prolactin	Anterior pituitary	Breast gland cells.	Milk production.
Growth hormone (GH)	Anterior pituitary	All body cells, e.g. liver	Growth, protein, carbohydrate and lipid metabolism.
Thyroid stimulating hormone (TSH)	Anterior pituitary	Thyroid.	Thyroid produces thyroxine.
Adrenocorticotrophic hormone (ACTH)	Anterior pituitary	Adrenal cortex.	Glucocorticoid production.
Follicle stimulating hormone (FSH)	Anterior pituitary	Male testes. Female follicles.	Male production of sperm. Maturation of primary follicle.
Luteinising hormone (LH)	Anterior pituitary	Male testes. Female Graafian follicle.	Male – production of testosterone. Female ovulation, formation of corpus luteum.
Antidiuretic hormone (ADH)	Hypothalamus	Kidney collecting ducts and distal tubule.	Water reabsorption when body water levels low.
Oxytocin	Hypothalamus	Ducts of breast. Uterus.	Milk ejection, labour contractions.
Testosterone	Testes	Testes, body cells.	Sperm production, male secondary sexual characteristics.
Oestrogen	Ovaries – follicles, corpus luteum	Uterus – endometrium. Body cells.	Development of endometrium, female secondary sexual characteristics.
Progesterone	Ovaries – corpus luteum	Uterus – endometrium.	Maintaining endometrium.
Thyroxine	Thyroid	Body cells.	Speeds up chemical reactions.
Calcitonin	Thyroid	Bone cells. Kidney.	Reduces loss of calcium from bones, inhibits reabsorption of calcium and phosphate.
Parathyroid hormone	Parathyroid	Bones. Kidneys. Intestines.	More reabsorption of calcium into blood, e.g. from urine and more absorption from intestines.
Adrenaline and noradrenaline	Adrenal medulla	Liver. Heart.	Prepares body for fight or flight.
Aldosterone	Adrenal cortex	Kidneys – nephron.	More reabsorption of Na^+ and water and more K^+ secretion.
Cortisol	Adrenal cortex	Liver.	Stimulates gluconeogenesis.
Insulin	Beta cells in pancreas	Body cells, e.g. liver muscle.	Lowers blood glucose levels with synthesis of glycogen.
Glucagon	Alpha cells in pancreas	Liver.	Raises blood glucose levels and glycogen breakdown.

Pituitary gland

The pituitary gland is often called the 'master gland' and has two lobes that secrete different hormones which can stimulate other endocrine glands to secrete hormones, e.g. anterior pituitary produces thyroid stimulating hormone that stimulates the thyroid gland to produce thyroxine.

QUESTIONS

1. What is the endocrine system?
2. Define a hormone.
3. Both the nervous system and the endocrine system are involved in coordination. Construct a table to compare these two systems comparing the speed of transmission, the mode of transmission, response time, response duration and specificity.
4. The diagram shows the some endocrine glands.

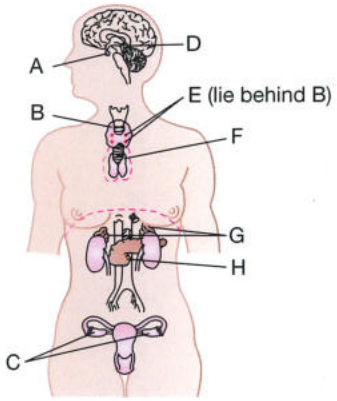


Figure 2.2 Some endocrine glands.

Identify the labelled glands.

5. The diagram shows the control and release of thyroid hormones.

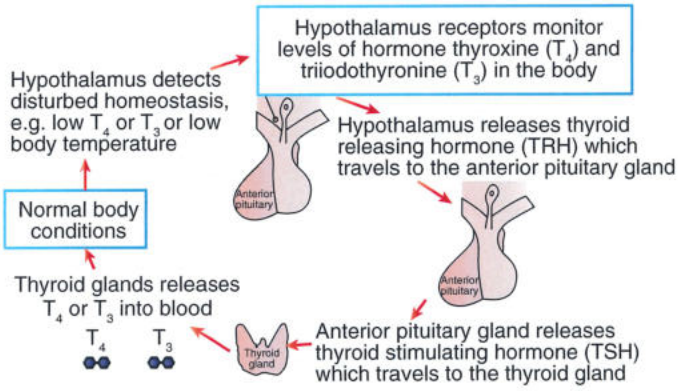


Figure 2.3 Control and release of thyroid hormones.

(a) Use the diagram to show how hormones control body functioning.
(b) Use the diagram to show how hormones travel in the body.

6. Explain why insulin and glucagon are antagonistic hormones.
7. Why is the pituitary gland often called the 'master' gland?
8. Which hormones are known as the 'fight or flight' hormones?
9. The diagram shows the anterior pituitary gland and some of its secretions.

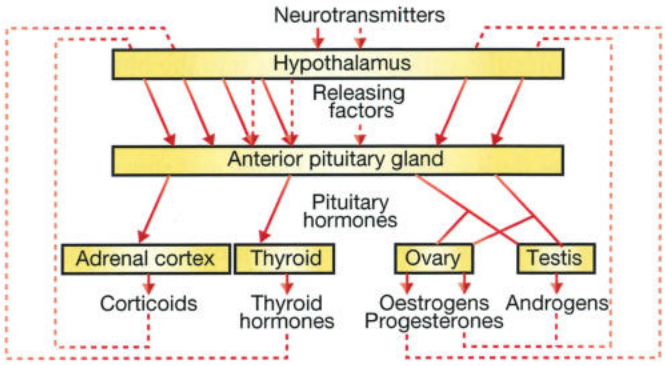


Figure 2.4 Anterior pituitary gland secretions.

(a) Outline the steps involved in the secretion of thyroxine from the thyroid gland.
(b) How is negative feedback involved in hormone secretion and shown in the diagram?
(c) The hypothalamus receives information from nerves throughout the body and is important in integrating the endocrine and nervous systems. Outline one way the hypothalamus interacts with the pituitary gland.

10. What are the three types of hormones?
11. Outline how the structure of a hormone affects its functioning.
12. A tropic hormone has an endocrine gland or cells as a target while non-tropic hormones do not regulate endocrine glands or cells. Classify each of the following hormones as tropic or non-tropic and explain your reasoning.
    (a) Thyroid stimulating hormone (TSH).
    (b) Prolactin.
    (c) Adrenocorticotropic hormone (ACTH).
13. What is a function of thyroxine?
    (A) Control biological rhythms.
    (B) Raise blood glucose level.
    (C) Stimulate ovaries and testes.
    (D) Speed up chemical pathways.
14. Adrenaline and noradrenaline are amine hormones synthesised from the amino acid tyrosine. They are produced in response to stress. What is a likely action of these hormones?
    (A) Increase heart rate and volume stroke.
    (B) Decrease heart rate and volume stroke.
    (C) Constriction of bronchioles in the lungs.
    (D) Increase blood flow to digestive organs.
15. Where is oxytocin synthesised?
    (A) Posterior pituitary gland.
    (B) Anterior pituitary gland.
    (C) Hypothalamus.
    (D) Ovaries.

3 Types of Hormones

The endocrine system consists of **endocrine glands** which are ductless glands and specialised cells that release chemical messengers, e.g. protein, amine or steroid molecules called **hormones** into the bloodstream. There are three main types of hormones – proteins and peptides (small polypeptides with up to 30 amino acids), and amines derived from amino acids and steroids.

Hormones act on specific target cells to change their functioning. Some hormones affect most cells of the body, e.g. growth hormone and thyroxine while other hormones are very specific, e.g. follicle stimulating hormone (FSH) stimulates the development of the follicles that contain eggs in females and the production and maturation of sperm in the testes of males. The target cells have specific receptor molecules on their surfaces. If a cell lacks an appropriate receptor site it will not respond to the hormone, e.g. the hormone prolactin acts on the milk producing cells of the breast.

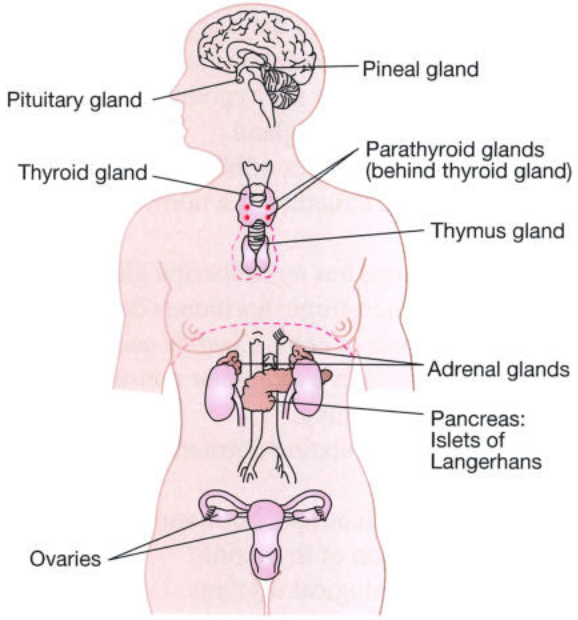


Figure 3.1 Main endocrine glands.

Steroid hormones

The **steroid hormones** are concerned with long term responses of the body, e.g. oestrogen, testosterone, aldosterone. The steroid hormones are lipids formed from cholesterol and travel around the body bound to carrier proteins. The steroid hormones are lipid soluble and can easily pass through cell membranes. The receptors for these hormones are usually in the cytoplasm or nucleus. Many steroid hormones directly regulate gene expression of target genes, e.g. oestrogen and cortisol.

Amine and peptide hormones

The **protein/peptide groups** include hormones secreted by the anterior and posterior pituitary gland, the pancreas (insulin and glucagon) and the parathyroid gland (parathyroid hormone PTH). The **amine hormones** are derivatives of the amino acid tyrosine and include hormones secreted by the thyroid gland, e.g. thyroxine and triiodothyronine and the adrenal medulla (adrenaline and noradrenaline). These hormones usually bring about a response within a few minutes and their actions are of relatively short duration. They are often made in advance and held in storage in the endocrine cells, available for immediate release. Some of these hormones are made in a longer form called a prehormone which is then broken down into the mature form. Most amine and protein/peptide hormones are water soluble and bind with receptors on the cell membrane. The water soluble hormones dissolve in the plasma to travel in the blood and can diffuse out of the capillaries into the interstitial fluid to reach their target cell.

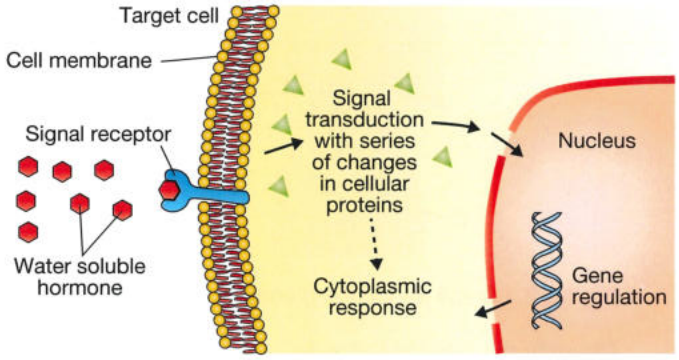


Figure 3.2 Signal transduction for a water soluble hormone.

Endocrine signalling

The hormones secreted by endocrine cells into the extracellular fluids move into the bloodstream and travel around the body. When they reach their specific target cells they bind to receptors to produce the desired response.

The water soluble hormones trigger signal receptors in the cell membrane which can activate an enzyme, change the uptake or secretion of specific molecules or rearrange the cytoskeleton in a signal transduction pathway. **Signal transduction** is the transmission of a molecular signal from the exterior of the cell into its interior using cell-surface receptors.

There are intracellular receptors for the lipid soluble hormones and signal transduction occurs entirely within the target cell.

Control of hormone secretion

The secretion of hormones from endocrine glands is highly controlled to regulate the concentrations of hormones in the blood and the rate of metabolic reactions. Stimulation can occur due to a hormone from another endocrine gland or by an impulse from the nervous system.

For most hormones there is a **negative feedback system** that stops further release of a hormone to prevent overactivity of the target cells.

Hormone secretion is also affected by other factors such as daily rhythms, seasonal changes, stages of development and ageing and sleep patterns, e.g. growth hormone has higher secretion rates during early sleep time than in later sleep time.

Hormones are produced in very small quantities but their effects can last for long periods of time.

Hypothalamus

The hypothalamus in vertebrates plays an important role in the functioning of the endocrine system. It is part of the ventral section of the forebrain and is involved in maintaining homeostasis especially in coordinating the endocrine and nervous systems.

The hypothalamus secretes the hormones of the posterior pituitary gland and the releasing factors that regulate the anterior pituitary gland.

QUESTIONS

1. What is the endocrine system?
2. What is a hormone?
3. Name some endocrine glands in vertebrates.
4. Give an example of hormones that can affect several types of cells.
5. Identify the three main types of hormones.
6. Describe steroid hormones.
7. Outline how steroid hormones travel in the body.
8. Where are the receptors for steroid hormones?
9. How do many steroid hormones, e.g. oestrogens and cortisol act?
10. Name some amine hormones.
11. Identify some protein/peptide hormones.
12. Compare the response time for steroid hormones and amine and peptide/protein hormones.
13. Compare how steroid hormones and peptide hormones travel in the body.
14. What is a prehormone?
15. Define signal transduction.
16. Outline how hormone secretion is controlled.
17. What is the hypothalamus?
18. Outline the function of the hypothalamus.
19. Study the diagram which shows the secretion of a peptide hormone.

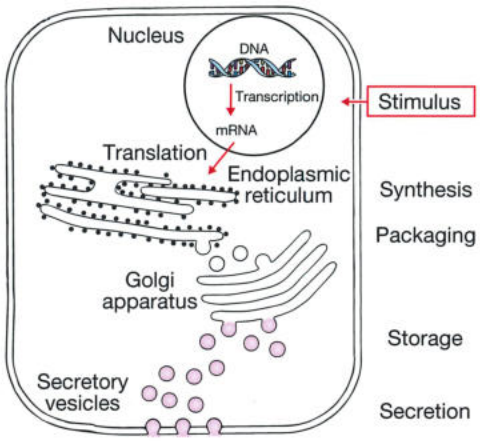


Figure 3.3 Hormone secretion.

From this diagram draw a flow chart to show the steps in the secretion of a peptide hormone.

20. The diagram shows the series of events that occur when a water soluble hormone contacts a target cell.

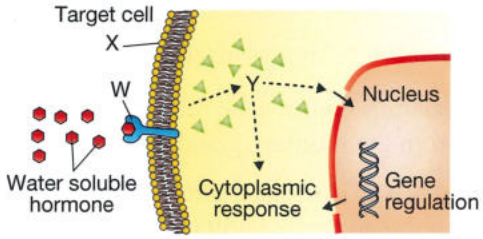


Figure 3.4 Hormone contacts target cell.

Identify W, X and Y.

	W	X	Y
(A)	Receptor	Membrane	Signal transcription
(B)	ATP	Receptor	Signal translation
(C)	Receptor	Membrane	Signal transduction
(D)	Membrane	Receptor	Signal translocation

21. The diagram shows a synapse between two neurons.

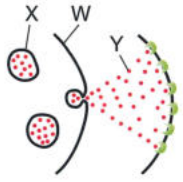


Figure 3.5 Hormone contacts target cell.

What is represented by W, X and Y?

	W	X	Y
(A)	Presynaptic terminal	Vesicle	Neurotransmitter
(B)	Presynaptic terminal	Lysosome	Acetylcholine
(C)	Postsynaptic terminal	Mitochondrion	Neurotransmitter
(D)	Postsynaptic terminal	Golgi Body	Acetylcholine

4 The Pituitary Gland

The pituitary gland is a small gland that lies below the hypothalamus and is often called the 'master gland' as many pituitary hormones regulate the activity of other endocrine glands. It is connected to the hypothalamus by a stalk of nervous tissue. It is the size of a large pea and weighs about 0.5 gram.

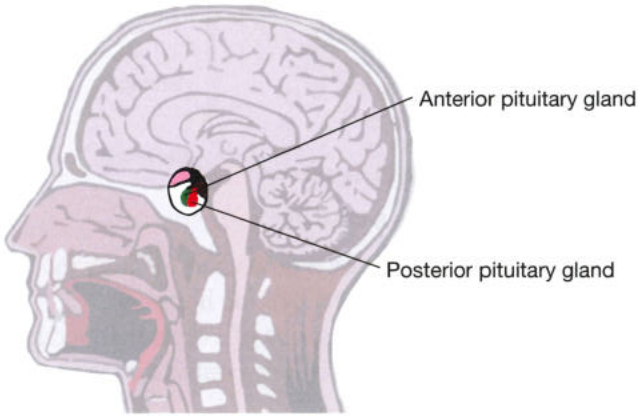


Figure 4.1 Location of the pituitary gland.

The hypothalamus controls the release of many of the hormones from the pituitary gland.

The pituitary gland is divided into two lobes – the anterior pituitary and the posterior pituitary. In the embryo the two parts of the pituitary gland arise from different sources.

Anterior pituitary

The anterior pituitary produces a number of hormones that regulate a range of body activities and other endocrine glands.

- **Growth hormone** (GH) acts on body tissues especially during childhood and adolescence increasing the rate at which amino acids are taken up by cells and built into proteins.
- **Prolactin** works with other hormones to initiate and maintain milk secretion in females and promotes development of female breasts.
- **Luteinising hormone** (LH) works with FSH in females to bring about ovulation and form the corpus luteum after ovulation. In males LH stimulates interstitial cells in the testes to secrete male sex hormones.
- **Thyroid stimulating hormone** (TSH) stimulates the production and release of hormones from the thyroid glands.
- **Adrenocorticotrophic hormone** (ACTH) controls the production and release of some of the hormones from the cortex of the adrenal glands, e.g. cortisol, androgens.

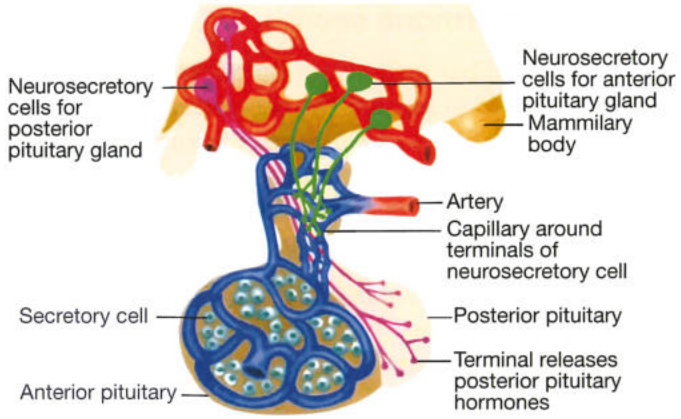


Figure 4.2 Pituitary gland.

Releasing hormones and inhibiting hormones produced by neurosecretory cells of the hypothalamus regulate the release of hormones from the anterior pituitary. These neurohormones enter capillaries that pass through special portal veins that connect two sets of capillaries and from the second set of capillaries the neurohormones pass into the anterior lobe.

Table 4.1 Neurohormones from the hypothalamus regulate the anterior pituitary gland.

From hypothalamus	Regulation of anterior pituitary
CRH	Causes anterior pituitary gland to secrete ACTH and beta-endorphins
TRH	Stimulates anterior pituitary to release TSH and prolactin
GnRH	Stimulates anterior pituitary to release FSH and LH
GHRH	Stimulates anterior pituitary to secrete GH
PRH	Stimulates anterior pituitary to produce prolactin
Dopamine	Inhibits the release of prolactin
Somatostatin	Inhibits the release of several hormones, e.g. GH, TSH.

Posterior pituitary

The posterior pituitary does not produce hormones but stores oxytocin and antidiuretic hormone (ADH) produced by the hypothalamus.

- **ADH** causes the kidneys to reabsorb more water to maintain body water balance and prevent dehydration.
- **Oxytocin** stimulates the contraction of muscles of the uterus, e.g. during childbirth and also stimulates contraction of cells in mammary glands in the release of milk while suckling.

Pituitary disorders

Pituitary disorders can involve **hyposecretion** (reduced amounts of hormone produced) or **hypersecretion** (increased amount of hormone produced). Pituitary gland tumours occur in many adults. They are not a brain tumour and are usually not cancerous (are benign tumours). Abnormal secretion of a hormone will interfere with normal body functioning. Secretory tumours cause hypersecretion of a particular hormone(s). If the tumour mass becomes large it can press against other areas, e.g. brain causing headaches, vision problems and other effects.

QUESTIONS

1. Where is the pituitary gland?
2. Why is the pituitary gland often called the 'master gland'?
3. What controls the release of many of the hormones of the pituitary gland?
4. Construct a table to summarise the hormones produced by the pituitary gland.
5. The diagram shows the hormones from the anterior pituitary.

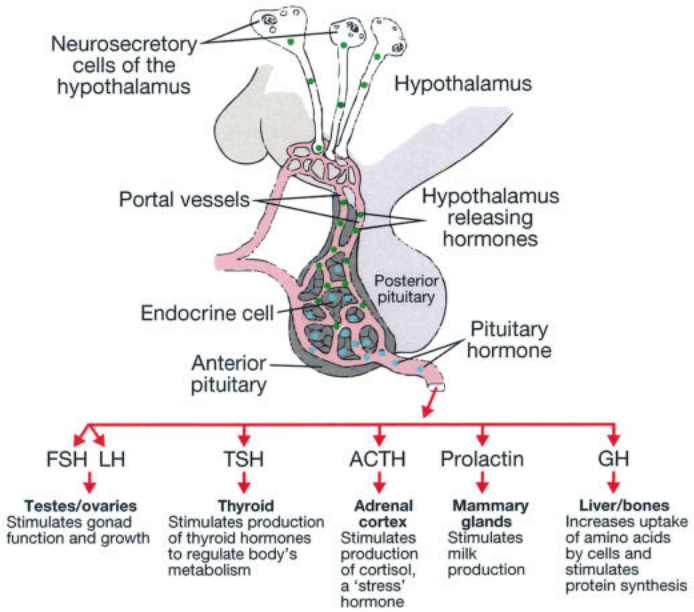


Figure 4.3 Anterior pituitary gland.

Draw a flow chart to show the sequence of events from stimulation of the hypothalamus to:

(a) Increase in uptake of amino acids by cells.
(b) Stimulation of milk production.
(c) Production of cortisol from adrenal cortex.

6. Distinguish between hyposecretion and hypersecretion.
7. Outline what happens when there is a secretory tumour of the pituitary gland.
8. Where is antidiuretic hormone (ADH) produced?
   (A) Anterior pituitary gland.
   (B) Posterior pituitary gland.
   (C) Adrenal gland.
   (D) Hypothalamus.
9. The diagram shows the glands of the endocrine system for males and females.

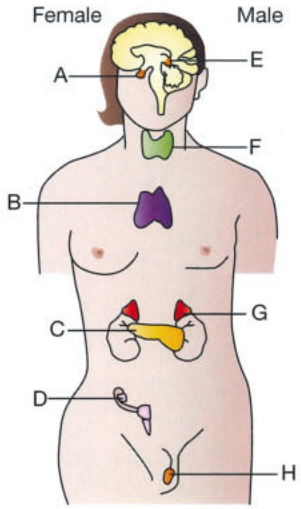


Figure 4.4 Endocrine system for males and females.

Which part is the pituitary gland?

(A) Part A
(B) Part E
(C) Part F
(D) Part B

10. The diagram shows the hormones released by the posterior pituitary gland.

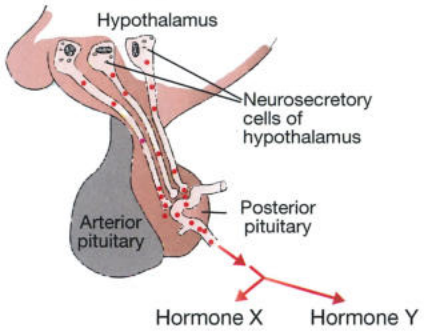


Figure 4.5 Posterior pituitary gland.

Identify hormones X and Y

(A) ADH and growth hormone.
(B) Oxytocin and ADH.
(C) Growth hormone and TSH.
(D) TSH and oxytocin.

11. What controls nearly all secretion by the pituitary gland?
    (A) Only hormonal signals from the hypothalamus.
    (B) Only nervous signals from the hypothalamus.
    (C) Either hormonal or nervous signals from the hypothalamus.
    (D) Signals from the sensory organs.

5 The Thyroid Gland

The thyroid gland lies below the larynx on each side of and anterior to the trachea. It consists of two lobes connected by an 'isthmus' and is one of the largest endocrine glands.

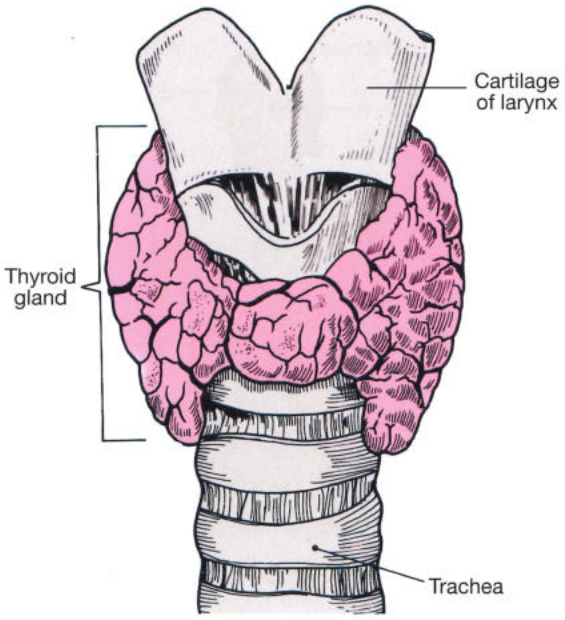


Figure 5.1 Thyroid gland.

Thyroid hormones

The thyroid gland secretes two main hormones:

- **Thyroxine** (T_4) which has four iodine atoms and **triiodothyronine** (T_3) which has three iodine atoms. T_4 and T_3 affect the metabolic rate of the body and bring about the release of energy when complex molecules are synthesised from simple molecules. They are also necessary for cellular differentiation. These hormones are synthesised from the amino acid tyrosine and from iodine and unlike many other hormones the thyroid hormones are stored in follicles in the gland. Iodine is needed in the diet to make the thyroid hormones and is the main site of iodine usage by the body. To prevent iodine deficiency common table salt is iodised.
- The thyroid gland also secretes **calcitonin** which is important in calcium metabolism as it inhibits the release of calcium ions from bone. Calcitonin works antagonistically to parathyroid hormone as parathyroid hormone stimulates the release of calcium from bones and calcium reabsorption from the kidney tubules.

Regulation of thyroid secretion

Thyroid secretion is mainly controlled by the hypothalamus and the pituitary gland by thyroid stimulating hormone (TSH). There is a negative feedback system between the anterior pituitary gland and the thyroid gland.

If there is a **low concentration** of thyroid hormones in the blood the anterior pituitary is stimulated to secrete more TSH.

If there is a **high concentration** level of thyroid hormones in the blood then the anterior pituitary is inhibited and will secrete less thyroid stimulating hormone (TSH). High thyroid hormone levels also affect the hypothalamus to inhibit the secretion of the releasing hormone THRH. Stressful cold weather can also stimulate the secretion of TSRH to raise body temperature by increased metabolism.

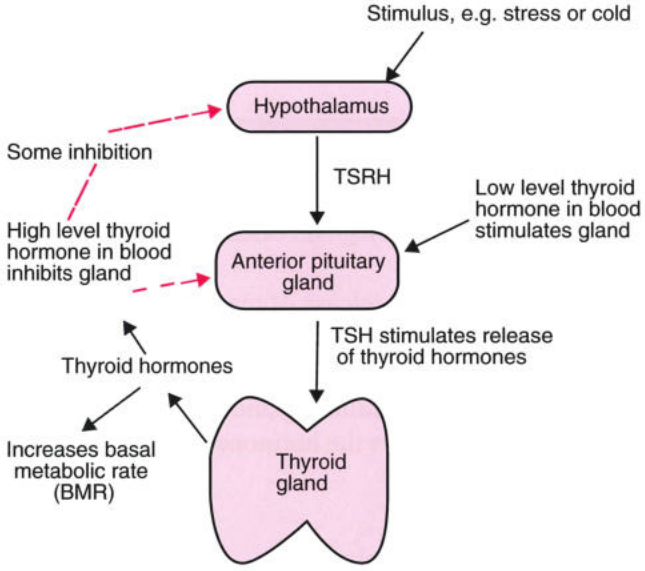


Figure 5.2 Regulation of the thyroid gland.

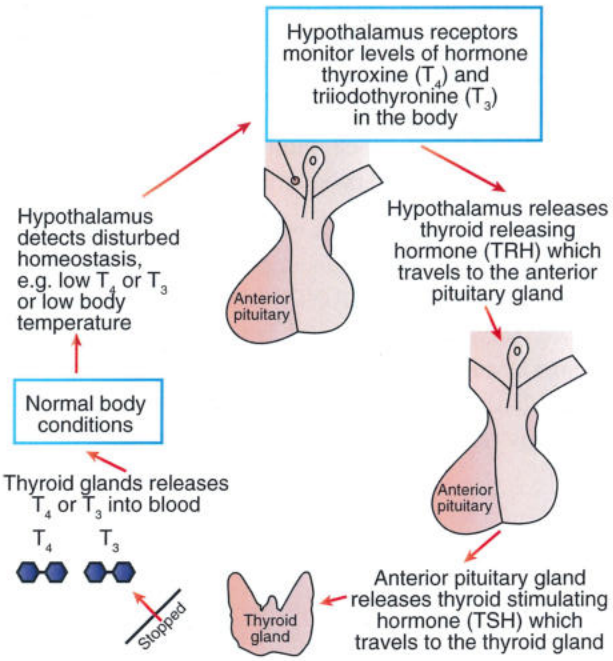
About 93% of metabolically active hormones secreted by the thyroid gland is thyroxine (T_4) and 7% triiodothyronine (T_3). But nearly all the thyroxine is converted to triiodiothyronine in the tissues. Both are important hormones with T_3 about four times more potent than T_4. The thyroid hormones activate gene expression and the transcription of a large number of genes, e.g. to cause the synthesis of protein enzymes, structural proteins and transport proteins.

Disorders of the thyroid gland

Hyperthyroidism is the overproduction of thyroxine and causes nervousness, insomnia and excessive excitability; heat intolerance and excessive sweating; increased heart rate and blood pressure; and weight loss. Most effects of hyperthyroidism are observable with the thyroid gland increasing in size to two or three times its normal size. Many patients have extreme fatigue with an inability to sleep.

Hypothyroidism is the underproduction of thyroxine. If it occurs during infancy it affects the development of brain cells and can lead to permanent mental deficiency and dwarfism. In adults it causes dry skin, intolerance to cold and lack of energy. If there is insufficient iodine in the diet, thyroxine cannot be produced causing the pituitary to make large amount of TSH which stimulate the thyroid gland which grows in size to form a goitre.

(a)



Lack of iodine means the thyroid gland cannot make T_4 or T_3. This means the body cannot return to normal conditions and more TSH is produced to stimulate the thyroid. The thyroid grows in size to form a goitre.

(b)

Figure 5.3 (a) Lack of thyroid hormones forms a goitre. (b) Goitre.

Cretinism is caused by extreme hypothyroidism during foetal development, infancy or childhood. The low levels of thyroid hormones cause failure of body growth and mental retardation and can be due to lack of iodine in the diet or a genetic defect of the gland. As skeletal growth is more affected by low thyroid hormones than soft tissue growth a child with cretinism will be stocky, short and obese.

QUESTIONS

1. Describe the location of the thyroid gland.
2. Name the hormones produced by the thyroid gland.
3. Describe thyroxine.
4. Outline the main functions of T_4 and T_3.
5. Draw a flow chart to show what happens if there is a low concentration of thyroid hormones in the blood.
6. (a) What is hyperthyroidism?
   (b) Describe some symptoms of hyperthyroidism.
7. (a) What is hypothyroidism?
   (b) Describe some symptoms of hypothyroidism.
8. What directly controls the secretion of thyroxine?
9. Explain why common salt is iodised.
10. Describe cretinism.
11. Outline how thyroid hormones affect gene expression.
12. Which element deficiency causes hypothyroidism?
    (A) Iodine deficiency. (B) Fluorine deficiency.
    (C) Potassium deficiency. (D) Bromine deficiency.
13. The diagram shows the glands of the endocrine system for males and females.

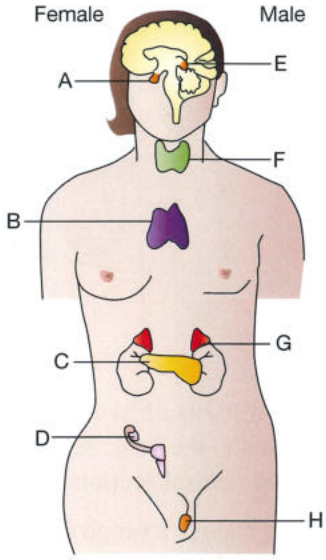


Figure 5.4 Endocrine system for males and females.

Which part is the thyroid gland?
(A) Part A (B) Part E (C) Part F (D) Part B

14. The graph shows the effect of a large dose of thyroxine on the basal metabolic rate.

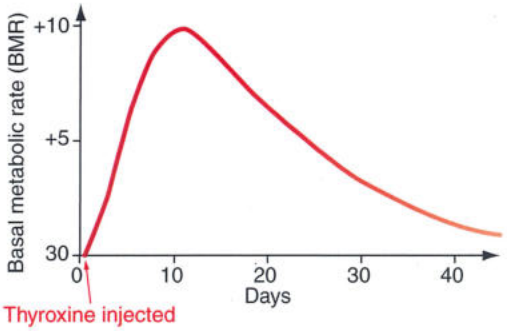


Figure 5.5 Effect of thyroxine on BMR.

From the graph, when is the BMR most affected by the injection of a large dose of thyroxine?
(A) Day 0 (B) Day 10
(C) Day 20 (D) Day 40

15. Which of the following would indicate hyperthyroidism?
    (A) Low level of free thyroxine in the blood.
    (B) High level of adrenaline in the blood.
    (C) Decrease of 60% of basal metabolic rate.
    (D) Increase of 60% of basal metabolic rate.
16. Where is the thyroid gland?
    (A) Immediately below the larynx.
    (B) Immediately above the kidneys.
    (C) At the base of the brain.
    (D) Next to the stomach.

6 The Adrenal Glands

The adrenal glands are small yellow masses of endocrine tissue on either side of the spinal column above each kidney. Each adult adrenal gland weighs around 4 grams.

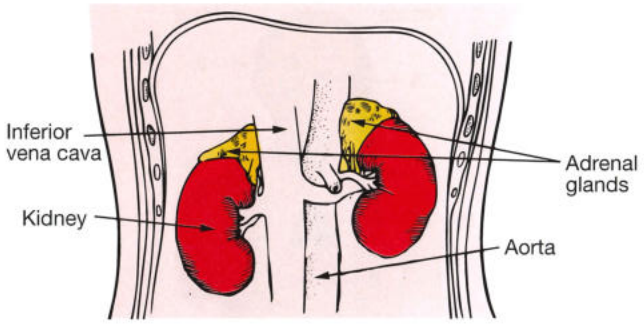


Figure 6.1 Adrenal glands.

Each adrenal gland consists of two distinct parts – an **adrenal cortex** which is the outer section and the **adrenal medulla** which is the central section.

The adrenal glands produce a range of hormones. The two parts of the adrenal gland are joined but develop from different tissue types in the embryo and function as distinct glands.

Adrenal cortex

The adrenal cortex is composed of three distinct layers, and it responds to endocrine signals and produces corticosteroid and androgen hormones. Stressful signals cause the hypothalamus to secrete a releasing hormone that stimulates the anterior pituitary to release adrenocorticotropic hormone (ACTH). ACTH causes the adrenal cortex to secrete the corticosteroids. In humans there are two main corticosteroids – the **glucocorticoids** which affect glucose metabolism and **mineralocorticoids** which affect mineral metabolism, maintaining salt and water balance. The adrenocortical hormones are steroids derived from cholesterol. More than 30 steroids have been isolated from the adrenal cortex.

The two most important hormones are **cortisol** which is the main glucocorticoid and **aldosterone** which is the main mineralocorticoid. Cortisol and aldosterone are regulated by independent mechanisms.

Cortisol has multiple metabolic functions and is involved in the metabolism of proteins, carbohydrates and fats. It also has anti-inflammatory effects. It responsible for about 95% of the glucocorticoid activity of the adrenal cortex. Glucocorticoids are used in medical treatments in allergic reactions, infections, arthritis and some types of cancer.

Aldosterone increases the reabsorption of sodium in the kidney nephrons and increases the secretion of potassium and hydrogen ions from the kidney.

Addison's disease is caused by insufficient production of adrenocortical hormones, e.g. lack of aldosterone. This causes a loss of sodium ions, chloride ions and water in urine and a drop in plasma volume. Cardiac output decreases and the person can die of shock within two weeks from the time when mineralocorticoid production stopped.

Cushing's syndrome occurs when there is hypersecretion of the adrenal cortex leading to excess cortisol in the blood. It is often caused by excess ACTH secretion. It causes fat from the lower body to mobilise and deposit in the upper abdomen and thorax causing a buffalo torso appearance. The excess steroids also cause increased facial hair and fat deposits on the face giving the face a full-moon appearance.

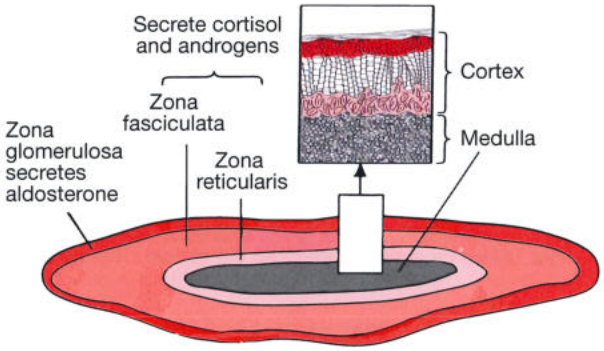


Figure 6.2 Cross-section of adrenal gland.

Adrenal medulla

The secretory cells of the adrenal medulla derive from neural tissue during embryonic development and are the main source of **adrenaline** (epinephrine) and **noradrenaline** (norepinephrine). These hormones are chemically similar and are derived from amino acids.

These hormones increase the amount of energy available for immediate use and are secreted in response to stress. They increase the rate of glycogen breakdown in the liver and skeletal muscles, increase heart rate, promote glucose release by liver cells and dilate the bronchioles in the lungs to increase oxygen uptake. This prepares the body for action and is sometimes called the 'fight or flight' response. The hormones can increase metabolic rate by 100%.

Noradrenaline is the same substances as secreted by neurons in the sympathetic nervous system and some neurons in the CNS to be a neurotransmitter. Its effect on the body is similar but longer lasting as it is only slowly removed from the body.

Regulation of the adrenal glands

The adrenal cortex is stimulated from the pituitary gland. Adrenocorticotropin (ACTH) stimulates the adrenal cortex to produce cortisol. The adrenal medulla is stimulated by the sympathetic nervous system. The sympathetic nervous system increases cardiac output, increases blood pressure and increases ventilation rate.

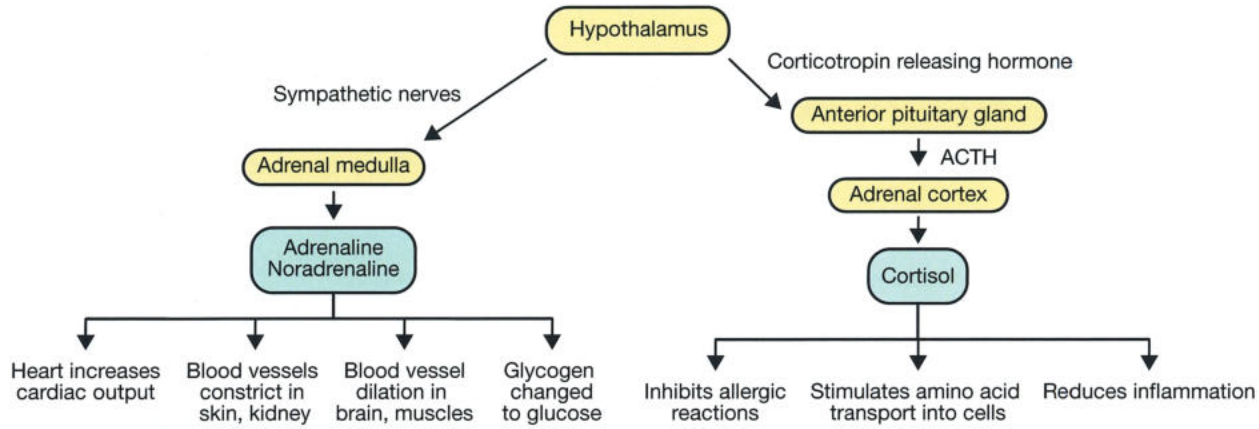


Figure 6.3 Action of adrenal gland hormones.

QUESTIONS

1. Describe the location of the adrenal glands.
2. Identify the two areas of the adrenal gland.
3. Distinguish between glucocorticoids and mineralocorticoids.
4. Outline the signal that causes the release of adrenocorticotrophic hormones.
5. Why is cholesterol important for normal functioning of the adrenal gland?
6. Outline the functions of:
   (a) Cortisol.
   (b) Aldosterone.
7. Name the main two hormones secreted by the adrenal medulla.
8. Outline the function of the main hormones of the adrenal medulla.
9. Construct a table to summarise the hormones produced by the adrenal gland.
10. Describe Addison's disease.
11. Compare the regulation of the adrenal cortex and the adrenal medulla.
12. Describe Cushing's syndrome.
13. Draw a flow chart to show the steps involved when a stress stimulus causes the hypothalamus to release CRF.
14. Which hormone is secreted by the adrenal medulla?
    (A) Aldosterone.
    (B) Cortisol.
    (C) Insulin.
    (D) Adrenaline.
15. About 95% of activity due to glucocorticoid activity is due to the hormone cortisol. What of the following is due to cortisol?
    (A) Development of follicles in the ovary.
    (B) Formation of glucose from proteins.
    (C) Build up of muscle tissue.
    (D) Clotting of blood.
16. The diagram shows the glands of the endocrine system for males and females.
    Which part is the adrenal gland?
    (A) Part B
    (B) Part C
    (C) Part G
    (D) Part D

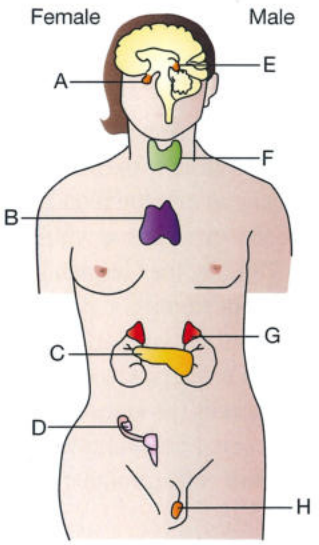


Figure 6.4 Endocrine system for males and females.

17. The diagram shows a flow chart for the release of some hormones.

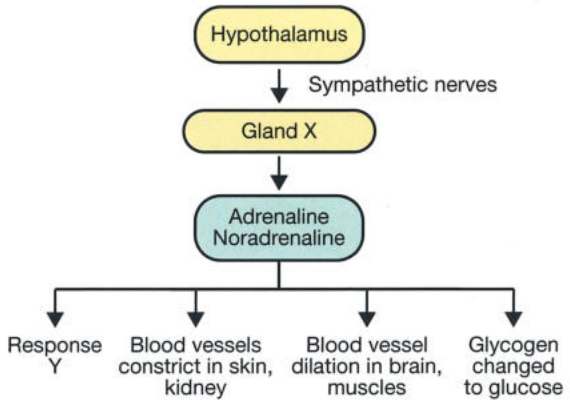


Figure 6.5 Release of some hormones.

Identify gland X and response Y.

	Gland X	Response Y
(A)	Adrenal medulla	Heart rate rises
(B)	Adrenal cortex	Fat metabolises
(C)	Thyroid	Basal metabolic rate rises
(D)	Pituitary	Blood sugar rises

7 The Pancreas

The pancreas is a gland located behind and parallel to the stomach. The pancreas has both endocrine and exocrine tissue.

Pancreas as an exocrine gland

Exocrine glands secrete their products into ducts. The **exocrine section** of the pancreas is involved in digestion and secreting enzymes in an alkaline solution into the small intestine. The acinar cells produce the digestive enzymes and the duct cells produce the bicarbonate ions and both products are released into the small ducts that empty into the pancreatic duct. The pancreatic duct empties into the duodenum.

Pancreas as an endocrine gland

The **endocrine section** of the pancreas is involved in homeostasis with the secretion of the hormones insulin and glucagon into the blood. The islets of Langerhans are patches of endocrine tissue within the pancreas. They were discovered in 1869 by the German histologist Paul Langerhans. There are about a million islets in an adult human pancreas making up 1% to 2% of pancreatic mass. The islets are composed of **alpha cells** (produce glucagon), **beta cells** (produce insulin), **delta cells** (produce somatostatin), **gamma cells** (produce pancreatic polypeptide and sometimes called PP cells) and **epsilon cells** (produce ghrelin).

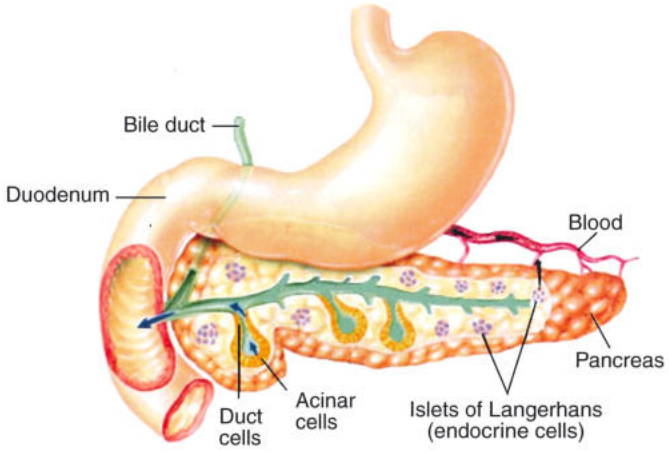


Figure 7.1 Pancreas.

Blood glucose

In humans blood glucose concentration should be 70 to110 mg/100 mL for optimal metabolism. Glucose is needed for cellular respiration and biosynthesis uses the carbon from glucose to build macromolecules. Blood glucose levels are closely regulated by homeostatic processes. Glucose levels in most humans is lowest in the morning before the first meal of the day.

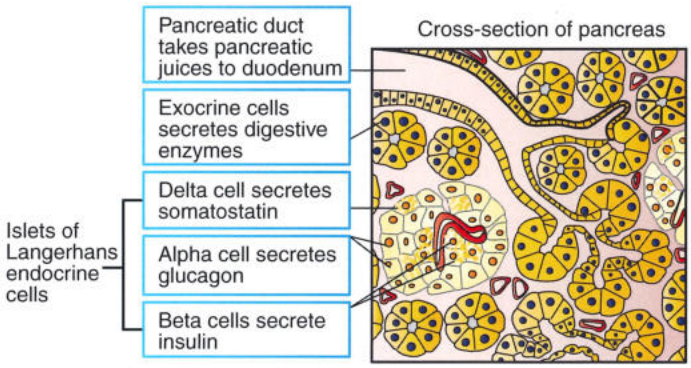


Figure 7.2 Cross-section of the pancreas.

Pancreatic hormones

Insulin and glucagon are two opposing (antagonistic) hormones that control blood glucose concentration. Insulin causes glucose in the blood to be taken up by body cells to lower blood glucose concentration while glucagon causes the release of glucose into the blood from energy stores, e.g. glycogen from the liver.

Insulin is a small protein composed of two amino acid chains. It acts by slowing down glycogen breakdown in the liver and stopping the conversion of glycerol and amino acids to glucose. It stimulates nearly all body cells to take up glucose. Within seconds after insulin binds with the membrane receptors of a target cell there is a marked increase in the uptake of glucose. Large amounts of insulin are secreted when there is a high intake of energy-giving foods, e.g. carbohydrates and proteins.

Glucagon is a polypeptide consisting of a chain of 29 amino acids. It acts mainly on target cells in the liver signalling an increase in glycogen hydrolysis with the conversion of glycerol and amino acids to glucose. Skeletal muscle cells and adipose tissues that store glycogen will also respond to glucagon.

Somatostatin is a small polypeptide consisting of 14 amino acids with a very short half-life of only 3 minutes in the circulating blood. It is secreted after eating food and its secretion is stimulated by increased blood glucose, increased amino acids and increased fatty acids. It extends the period of time that nutrients are assimilated into the blood and has several inhibitory actions, e.g. it acts locally in the pancreas to inhibit the secretion of both insulin and glucagon.

Regulation of blood glucose levels

If blood glucose levels are too high, the beta cells of the islets of Langerhans are stimulated to increase insulin secretion. The insulin stimulates cells, e.g. skeletal and fat cells to take up glucose from the blood. In muscle cells the glucose is either used immediately in respiration or stored as glycogen.

If blood glucose levels are too low the alpha cells of the islets of Langerhans are stimulated to increase glucagon secretion. The glucagon causes the liver to release stored glucose and mobilises fats and amino acids.

The regulation is a fast acting mechanism.

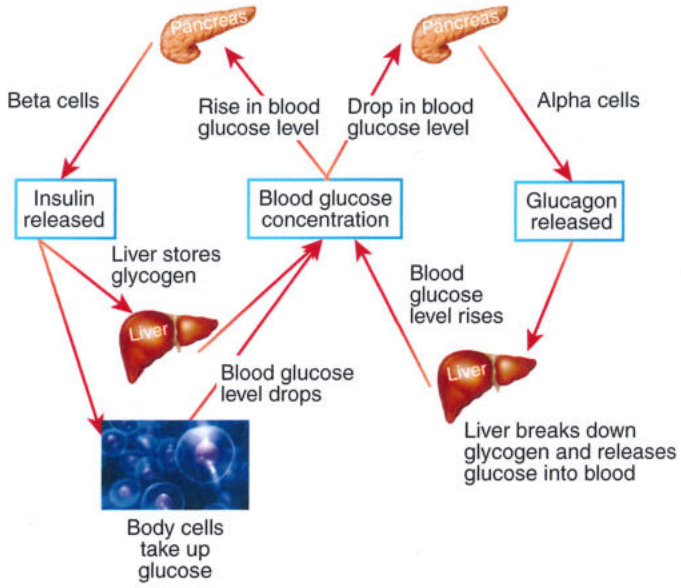


Figure 7.3 Regulation of blood glucose levels.

Diabetes mellitus

Diabetes mellitus is caused by a deficiency of insulin or a reduced response of target cells to insulin. There are two types of diabetes with both having a high blood glucose level.

Type 1 diabetes is insulin dependent diabetes and appears in childhood and is an autoimmune disease where the immune system destroys the beta cells of the pancreas.

Type 2 diabetes is non-insulin dependent where target cells do not correctly respond to insulin and usually appears in older people or people who are overweight and live a sedentary lifestyle.

QUESTIONS

1. Describe the location of the pancreas.
2. Explain why the pancreas is an exocrine gland.
3. Explain why the pancreas is an endocrine gland.
4. What are the islets of Langerhans?
5. Construct a table to summarise the five types of cells found in the islets of Langerhans and identify the substance produced by each type of cell.
6. Identify some uses of glucose in the body.
7. For most humans, when is their blood glucose level the lowest?
8. Discuss why blood glucose concentration is part of the homeostatic mechanism.
9. Explain why insulin and glucagon are antagonistic hormones.
10. How is the pancreas involved in homeostasis?
11. (a) Which cells produce insulin?
    (b) Which cells produce glucagon?
12. Explain why insulin secretion is linked with energy abundance.
13. Draw a flow chart to show the sequence of events when blood glucose levels become too low.
14. Draw a flow chart to show the sequence of events when blood glucose levels become too high.
15. Outline the function of somatostatin.
16. (a) What is diabetes mellitus?
    (b) Construct a table to compare type 1 diabetes and type 2 diabetes.
17. What is secreted by the alpha cells of the islets of Langerhans?
    (A) Pancreatic protease.
    (B) Insulin.
    (C) Glucagon.
    (D) Glucose.
18. The diagram shows the glands of the endocrine system for males and females. Which part is the pancreas?
    (A) Part B
    (B) Part C
    (C) Part G
    (D) Part D

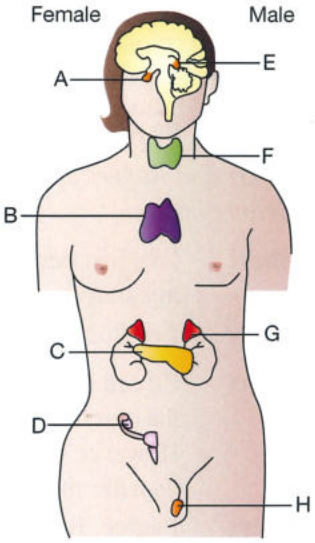


Figure 7.4 Endocrine system for males and females.

19. Which of the following gives the correct function of insulin and glucagon?

	Insulin	Glucagon
(A)	Inhibits secretion of glucagon	Inhibits secretion of insulin
(B)	Decreases absorption of amino acids	Increases absorption of amino acids
(C)	Increases blood glucose	Decreases blood glucose
(D)	Reduces blood glucose	Increases blood glucose

20. What will happen when glucose concentration in the blood falls?
    (A) Glucagon is produced to cause the release of glucose from the liver.
    (B) Insulin is produced to cause the release of glucose from the liver.
    (C) More glucose will be absorbed by the villi.
    (D) Liver cells spontaneously release glucose.

8 Play – Keeping Up With the Hormones

Dramatic personae:
Robert Red Bloodcell
Rupert Red Bloodcell
William White Bloodcell
Wally White Bloodcell
Mervin Mast cell
Owen Oxygen
Bertie Bacteria
Betty Bacteria
Penny Platelet
Pauline Platelet
Cain Sugar
Tim Testosterone
Adrian Adrenaline
Hannah Hormone

SCENE 1 – END OF A LONG BONE IN A LEG

(Robert and Rupert Red Bloodcell are talking in the end of a long bone)

Robert I've just been made in the marrow and I love my shape; what do you call this?

(Rupert studies Robert)

Rupert I'd call it disc shape and personally I think that the thin section in the middle looks a bit pale. However, the donut shape is more fashionable than the shape of those hormone macromolecules.

Robert I agree. Just because hormones are involved in long distance signalling they think they are the most important molecules in the body. We are far more important. And I also like our pale centre. It gives us character. Everyone can identify a red blood cell under a microscope. We are more important than the hormones.

Rupert And we have haemoglobin. It gives us a nice colour and my size is just perfect. How big are we?

Robert We're about 8 μm in diameter. Just perfect for fitting through little capillaries. Our shape and size is much better than those large white blood cells. Everyone should try to be like us. Keep up with the red blood cells.

Robert Stop being so vain, it's time to swim and see where this blood vessel goes.

Rupert Okay. Oh no! I just lost my nucleus. What does this mean?

Robert It means you're going to die and can't reproduce!

Rupert If that's the case, then let's enjoy ourselves before we go to that big red spleen in the abdomen!

William Excuse me, excuse me, out of the way. I'm on my way to an accident. There's a cut in the foot.

Wally Hurry up, move your red corpuscle, the white warriors are coming through and we have priority passage.

(Robert and Rupert watch the two white blood cells pushing their way along the blood vessel)

Rupert White blood cells are so big and pushy!

Adrian Out of my way! I'm important and need a clear path to my target cell.

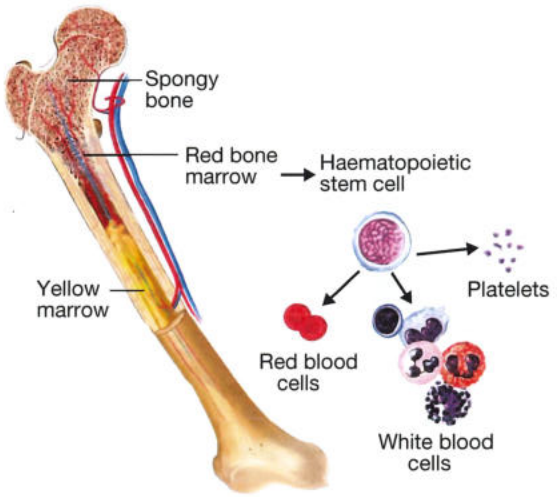


Figure 8.1 End of a long bone.

Robert Who are you?

Adrian I'm Adrian Adrenaline. I'm a hormone secreted by the adrenal medulla and this person was exercising before he tripped and cut his foot on the edge of the step.

Rupert Why are you so important?

Adrian I cause glycogen to be broken down to glucose in the liver to increase the amount of blood glucose.

Robert You like to control others?

Adrian I control changes in blood flow patterns and I am an important hormone. Only small amounts of hormone are needed so we are a very select group.

Rupert If your target cell is in the liver, why are you in the blood with us going to the foot?

Adrian Hormones travel everywhere in the blood. I'm taking the scenic route to the liver to have a look at the infection in the foot before I get to work.

(Adrian rushes forward pushing past the other components of the blood vessel)

Rupert Let's follow Adrian and those white blood cells. I love a good fight. It sounds as though there's trouble in the foot. Do you think they need us?

Robert I want to go to the lungs as I'm feeling a little blue, so I'll meet you later. How about we meet at the hepatic portal vein for afternoon coffee and biscuits?

Rupert Sounds good. See you later at the hepatic portal vein. By the way, where is this vein?

Robert The hepatic portal vein takes digested food, like glucose and amino acids, to the liver.

Rupert Oh THAT vein! I'll meet you later and tell you all about the fight in the foot.

Questions Scene 1

1. Where do red blood cells form?
2. Describe the shape and size of red cells.
3. Why do red cells have a limited life span?
4. Where is adrenaline produced?
5. Give a function of adrenaline.
6. Why would a red blood cell be blue?
7. Explain why they go to the hepatic portal vein for afternoon tea.

SCENE 2 – THE FOOT

(Bacteria are attacking through a cut in the foot)

Bertie Charge the foot, mates! We've got to infect quickly!

Betty I'll give this foot gangrene, and then it'll have to be amputated. It's fun being a bacterium.

William Stop! In the name of the immune system! I'm a law and order officer of the white cell squadron!

Wally I'll get Betty and engulf her!

Betty No you won't! You oversized phagocyte!

Wally Phagocytes are specialised white blood cells. We are the elite and we engulf horrible bacteria, like you.

Betty No! Go away! Don't come near me!

Bertie Keep fighting, Betty! Don't give in! Bacteria can win if we stick together.

William I'll get Bertie!

(William and Wally, the white blood cells, fight Betty and Bertie Bacteria. A savage battle follows, with valiant efforts made by both sides.)

Betty I may be the smallest cell around this place but I could've taken over the foot.

Wally But I'm here to see that you don't, even if I die also.

Betty I've killed 100 skin cells, so I can die happy.

Wally And I'm dying too, so what does that make us?

William Pus. Dead cells, bacteria and white cells make pus.

Rupert There's a lot of people at this scene.

Mervin We need a lot of people to fight this infection. I saw invaders and sent out the signal.

Rupert Who are you?

Mervin I'm a mast cell that is found in the connective tissue surrounding most blood vessels and nerves. I produce and contain histamine and other molecules that trigger inflammation in response to infection.

Rupert How do you trigger the inflammation response?

Mervin You weren't listening very carefully were you!

Rupert I've lost my nucleus.

Mervin You poor thing! Only 120 days to live your life! I release my stored histamine which increases the permeability of the capillaries to white blood cells and some proteins. Histamine also causes the capillaries to dilate.

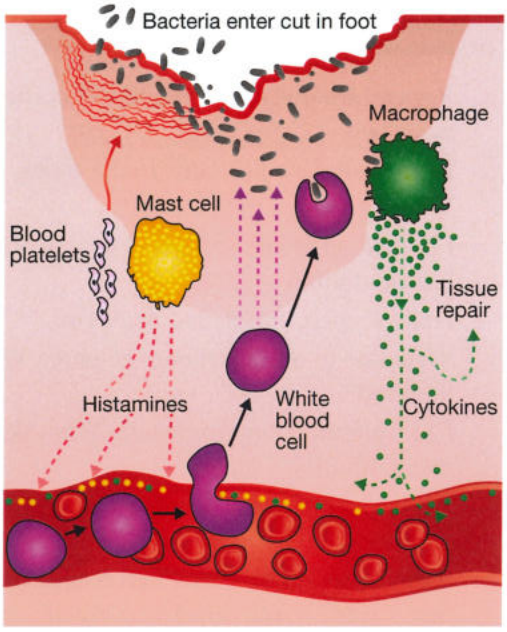


Figure 8.2 Mast cells release histamine.

Wally Is that why it was easy for me to leave the capillary and get to the bacteria?

Mervin Histamine is an inflammatory signalling molecule.

Rupert I'm glad you released the histamine. The dilated capillaries have increased the amount of blood flowing to this area and made it easy for me to see the battle first hand.

Pauline Hurry up Penny. If we don't hurry up and get to work, this foot is going to bleed to death.

Penny I'm ready to clot! What about all that pus in the way?

Pauline That's not our problem. Our job is to clot the blood.

Rupert If you two platelets are going to clot the blood, I'm leaving. Your union may allow scabs, but I'm not working at this site. This is no place for a red blood cell.

Penny If this gash doesn't seal up soon it'll need a toe-truck instead of a band-aid.

Pauline I love being a platelet and clotting blood. It's a pity that Rupert escaped. He would have made a nice part of this scab.

Questions Scene 2

1. What is the function of phagocytes?
2. What happens when bacteria invades the human body?
3. Where are mast cells found?
4. Outline the function of histamine.
5. What is the function of platelets?
6. What forms 'pus'?

SCENE 3 – THE HEPATIC PORTAL VEIN

(Rupert arrives at the hepatic portal vein)

Rupert Robert, are you here? I'm here for afternoon tea.

Robert Here I am. I've just been to the lungs.

Rupert What happened in the lungs?

Robert I picked up Owen Oxygen. He attached to my haemoglobin.

Owen Hello Rupert. Pleased to meet you.

Rupert Welcome to our afternoon tea party. Where's the food?

Cain I'm Cain the Sugar molecule. What do you mean afternoon tea?

Robert We heard there was free food here in the afternoon.

Cain Well, I used to be starch in a potato chip, and then I was broken down into glucose in the digestive system. Then I diffused across the wall of a villus in the small intestine, but I'm not food for you. I came to the liver for a respiration reaction.

Tim Hello. Who's interested in a reaction?

Rupert Who are you?

Tim I'm an important hormone. I'm testosterone.

Robert He's one of the steroids!

Tim That's right. I'm an anabolic steroid that is involved in muscle building.

Cain Why are you in the liver?

Tim This person takes steroid supplements to increase muscle mass and bone density and strength. The person is developing liver toxicity and I'm here to see what's happening.

Rupert We're talking about a respiration reaction.

Tim How boring. I'm off to somewhere more interesting. I heard there's some action down in the foot.

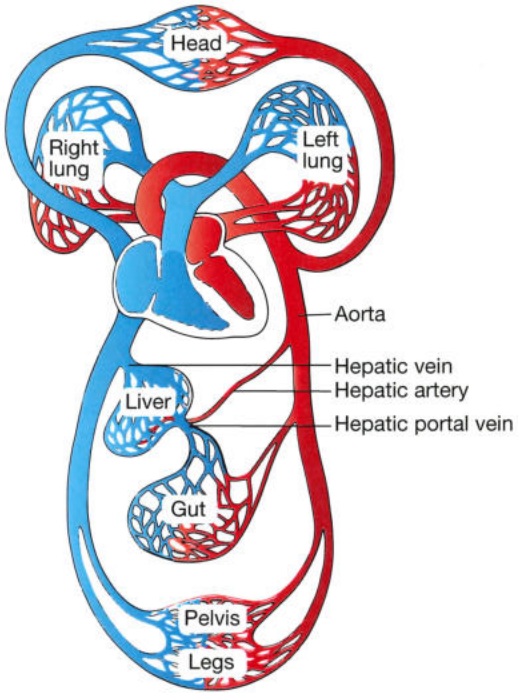


Figure 8.3 Hepatic portal vein in circulatory system.

Rupert Too late. The white blood cells killed the bacteria and the platelets are at work.

Tim Forming scabs is not my scene either. Goodbye. I'm off.

Robert Steroids! They think they are the most important chemicals in the body!

Robert What about this respiration reaction? Owen Oxygen and I had a reaction in the lungs on my way here. It was fun picking up oxygen. My haemoglobin became oxyhaemoglobin and was a brighter red.

Rupert Sounds like a hitchhiker to me.

Robert Don't be so boring. Another reaction sounds fun.

Cain The reaction is called respiration and sugar is the most important reactant.

Owen I beg your pardon, but it's the oxygen that's the most important reactant.

Rupert How am I involved?

Cain You're the transport but we can't have the reaction here. This is not the right place.

Robert The right place? What's the matter with this place?

Owen What's the matter with the hepatic portal vein?

Rupert Is this vein too wide?

Cain Yes it is. Respiration is needed in every living cell and we have to go to a needy cell for our reaction.

Owen We'll have to dissociate.

Robert You want to WHAT?

Owen Dissociate. When the oxygen concentration gets low, oxygen dissociates from the haemoglobin. How long will it take to get to a needy cell?

Cain It only takes 45 seconds for a red blood cell to complete one circuit of the circulatory system.

Rupert Well if that's the case, let's start this respiration reaction. Are there any side effects I should know about?

Cain You'll lose your oxygen.

Rupert What! That's not fair. I went all the way to the lungs to find Owen.

Robert You can get more oxygen in the alveoli.

Rupert But it won't be Owen.

(They all travel together for awhile and then suddenly Owen breaks away from Rupert)

Rupert What's happening?

Owen Help! Help! The diffusion gradient is too strong.

Cain Goodbye Rupert, I'm taking your oxygen and I'm diffusing out of this capillary into the tissues.

Owen Help me!

Rupert Come back here, thief!

Cain Ha! Ha! You're too late. I've got your oxygen.

Robert That glucose just diffused away and you lost Owen! He was such a nice chap.

Rupert Cain's a THIEF! They should lock him up in a dark cell.

Robert Their reaction is probably very boring.

Rupert He said I'd be part of the reaction. Why should they be the only ones to release energy?

Robert Well, if we meet again, we can pay him back with some mean trick of our own.

Questions Scene 3

1. How does sugar get into the circulatory system?
2. What is testosterone?
3. Identify a response when a person takes steroid supplements.
4. Write the word equation for the respiration reaction.
5. Where does the chemical reaction occur?
6. What happens to red blood cells in the lungs?
7. How long does it take for blood to complete one circuit of the system?
8. When does oxygen dissociate from haemoglobin?

SCENE 4 – A SMALL CAPILLARY

(The two friends Rupert and Robert meet again)

Robert Have you found that sweet sugar?

Rupert Sweet? That Cain Sugar stole Owen away from me. Cain was not sweet.

Robert They both diffused through the wall of the capillary, but we can't follow them and leave the capillary. We can't get out that way.

Rupert I don't want to go out there anyway. There's lymph out there and it has no colour at all. Very boring.

Robert Does this mean we're stuck between these walls forever?

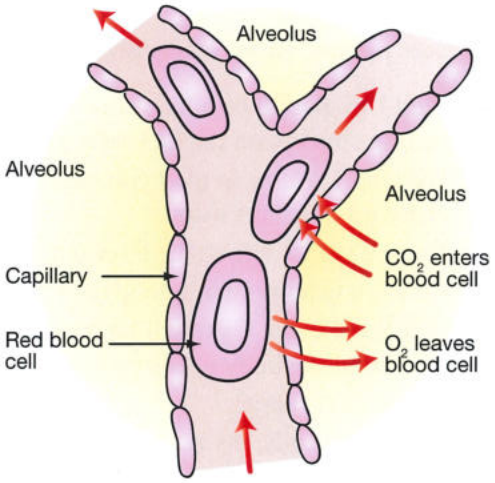


Figure 8.4 A capillary – red blood cells flow in single file.

Hannah Hello red men. What's going on here?

Rupert Nothing.

Robert He had a friend called Owen and was looking forward to a special reaction and then Owen diffused off with Cain.

Rupert It's true. I lost my oxygen and now I'm blue.

Hannah Well I'm a hormone, and I can't help you personally, but I'll keep you company on the journey.

Robert Most kind of you. You are very friendly for a hormone. And which hormone are you in particular?

Hannah Actually, I'm insulin. I was made in the pancreas and my job is to stop the formation of new glucose.

Rupert and **Robert** What!!!!

Rupert Did you say STOP the formation of glucose?

Hannah Yes. I make the liver and muscle cells change the glucose from the intestine into glycogen for storage.

Robert Fantastic!

Rupert It certainly is.

Robert We were conned and led up the garden path, or hepatic portal vein to be precise, by a certain sugar molecule.

Rupert And we would just love to escort a hormone that's going to put sugar into storage for a long time.

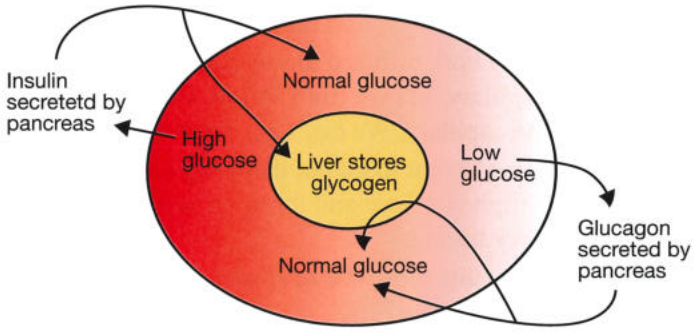


Figure 8.5 Insulin released when glucose levels are high.

Hannah Well I'm glad to be a help. Just don't go into insulin shock with the lack of sugar.

Robert We won't. We're so glad you came along to cheer up our short lives.

Hannah Do you want to go to the liver with me?

Rupert Can you change Cain into glycogen?

Hannah Of course I can. Cain will be trapped in the liver until the body needs more glucose.

Robert Fantastic revenge!

Rupert Let's go friends, my revenge will be sweet.

Robert It's not so hard to keep up with the hormones when you have insulin on your side.

Questions Scene 4

1. How did the glucose molecules move through the wall?
2. What makes red blood cells become 'blue'?
3. Where is insulin made?
4. What is the function of insulin?
5. When do people suffer 'insulin shock'?
6. What is found in blood?
7. Copy out Table 8.1 and complete the table by outlining the function of each part.

Table 8.1 Function of parts in this play.

Part	Function
Red blood cell	
White blood cell	
Platelet	
Testosterone	
Adrenaline	
Insulin	
Glucose	
Oxygen	
Mast cell	
Histamine	

8. The diagram shows an arteriole and what happens when insulin is secreted.

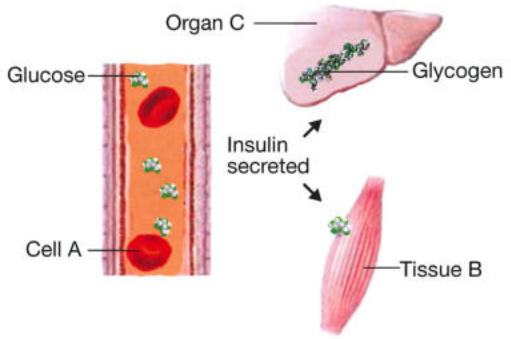


Figure 8.6 Arteriole and insulin secretion.

(a) Identify cell A, tissue B and organ C.
(b) Explain what is happening in this diagram.

9. Identify each of the parts in the diagram of the circulatory system.

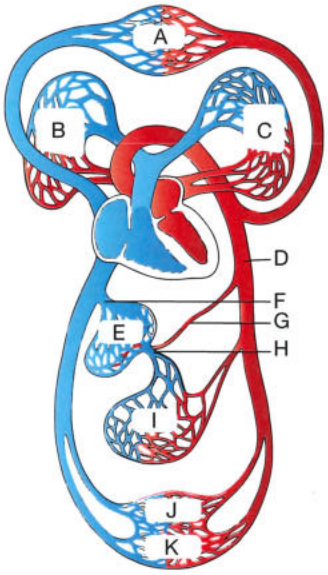


Figure 8.7 Circulatory system to label.

10. The diagram shows the anterior pituitary gland and some of its secretions.

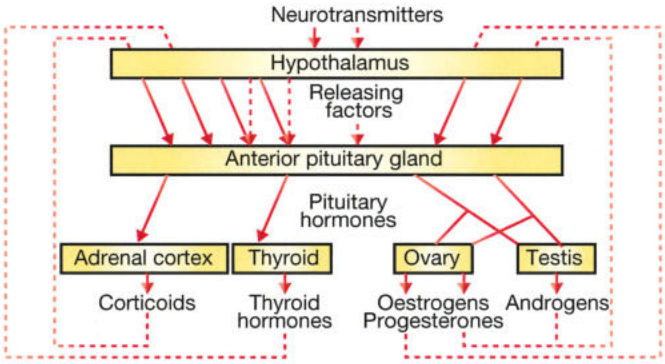


Figure 8.8 Anterior pituitary gland secretions.

How is negative feedback involved in hormone secretion and shown in the diagram?

9 Hormones and Feedback Regulation

There are two main types of feedback regulation for the control of hormone secretion – the simple endocrine pathway and the simple neuroendocrine pathway.

Simple endocrine pathway

Many hormones are part of a **simple endocrine pathway** where a stimulus, either internal or external leads to the secretion of the hormone. The hormone travels in the bloodstream to the target cell where specific receptors interact with the hormone to cause the cell to produce and release the hormone. For example, when low blood glucose level is detected by alpha cells in the pancreas, glucagon, a peptide hormone produced in the alpha cells in the pancreas is secreted into the bloodstream. Glucagon causes liver cells to convert glycogen into glucose molecules which will lead to a rise in blood glucose levels. Similarly when a low pH is detected in the duodenum the S cells of the intestinal glands in the duodenum secrete the hormone secretin. Secretin regulates water balance and will inhibit secretion of gastric acid by the parietal cells of the stomach and stimulates cells in the pancreas to produce hydrogen carbonate.

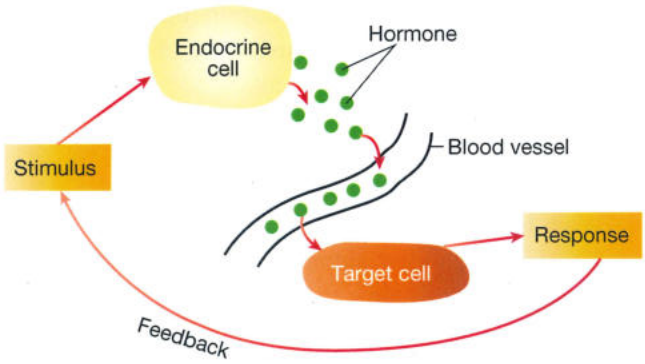


Figure 9.1 Simple endocrine pathway.

Simple neuroendocrine pathway

In a **simple neurosecretory pathway** the stimulus is detected by a sensory neuron which stimulates a neurosecretory cell to produce a neurohormone. For example, the neurosecretory cells of the hypothalamus make antidiuretic hormone (ADH) which is stored in the posterior pituitary gland. ADH is released from the posterior pituitary gland when there is a reduction in plasma volume and increases water retention in the kidneys causing less urine to be released. Oxytocin is another neurohormone produced by the hypothalamus and stored in the posterior pituitary gland. Oxytocin is involved in birth causing dilation of the cervix, labour contractions and is also involved in lactation.

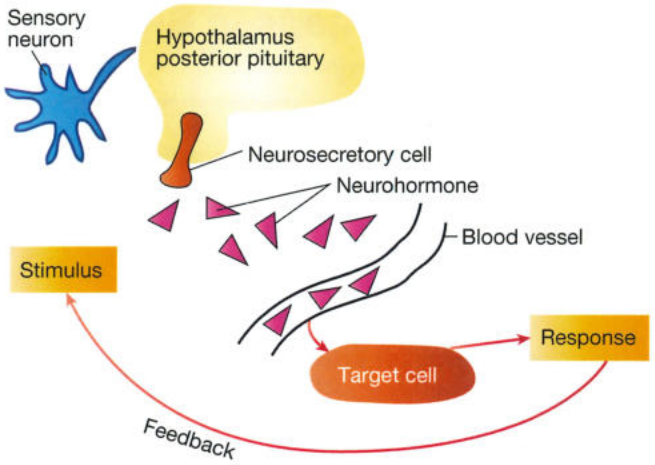


Figure 9.2 Simple neuroendocrine pathway.

In **negative feedback** the effector response reduces the initial stimulus and the response stops thus preventing overreaction by the system to the initial stimulus.

In **positive feedback** the stimulus is reinforced and leads to a greater response, e.g. suckling stimulates sensory nerves in the nipple that sends messages to the hypothalamus which releases oxytocin from the posterior pituitary gland to cause the mammary glands to secrete more milk.

QUESTIONS

1. Identify the two types of feedback regulation for the control of hormone secretion.
2. Draw a flow chart to show the sequence of events when low blood glucose levels are detected.
3. Distinguish between positive and negative feedback.
4. Explain why the release of oxytocin is an example of positive feedback.
5. Describe how the release of the hormone secretin is an example of negative feedback.
6. In humans the release of the hormone cortisol from the adrenal gland occurs when the body is under stress and produces the following effects:
   - Breakdown of proteins separating amino acids.
   - Conversion of amino acids to glucose in the liver.
   - Reduced glucose uptake by many body cells except those in the brain.

   (a) How does cortisol reach its target cells?
   (b) What would happen in negative feedback in this situation?
   (c) Suggest why the body benefits from the release of cortisol when under stress.
7. Which of the following is a neurohormone?
   (A) Oxytocin.
   (B) Secretin.
   (C) Insulin.
   (D) Glucagon.

10 The Three Steps in Signal Transduction

The process of signal transduction is a pathway that can be divided into three main steps – reception, transduction, and cellular response.

Reception

In reception the receptor detects a chemical signal, e.g. an integral plasma protein is activated by a signal molecule. For intracellular receptor proteins to be activated the signalling molecule must pass through the plasma membrane which means that the signalling molecule is either hydrophobic or very small, e.g. steroid hormones such as testosterone can pass through the plasma membrane. Signalling molecules can act like a **ligand** which is a molecule that binds specifically to another molecule, usually a large molecule.

The malfunction of many cell-surface receptors is associated with many human diseases, e.g. asthma, heart disease and cancer.

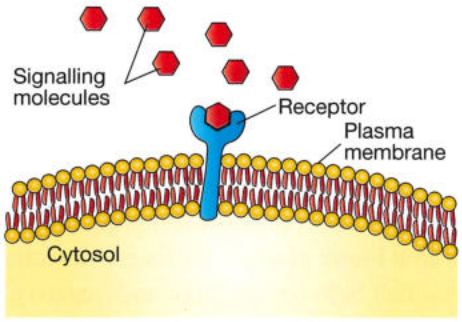


Figure 10.1 Reception.

Transduction

Activation of the receptor causes a change that stimulates other changes including the signal transduction pathways. If a signal molecule binds to a receptor protein in the plasma membrane there can be a conformational change in the protein so that the changed shape of the protein affects another part of the receptor molecule or the change directly transmits to information-relaying molecules called **secondary messengers** in the cytoplasm, e.g. calcium ions and cyclic AMP (cAMP) are secondary messengers. This leads to a series of reactions which can cause gene expression or gene repression.

There are several benefits in using a multistep pathway:

- The signal can be amplified with molecules in the pathway able to signal many molecules in the next step.
- It allows a fine tuned and coordinated response to a signal.

Cellular response

The cellular response can be a variety of cellular activities.

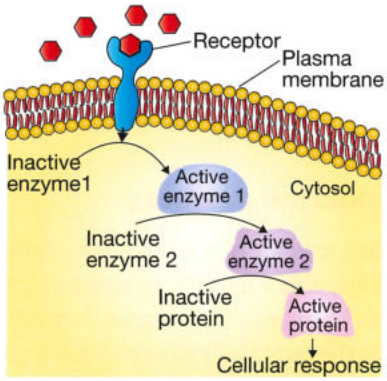


Figure 10.2 Transduction pathway.

QUESTIONS

1. List the three steps in signal transduction.
2. Where are the receptors for signalling molecules?
3. What types of signalling molecules are detected by intracellular receptors?
4. Retinitis pigmentosa (RP) is a degenerative eye disease caused by a mutation in the photoreceptors in the retina. There are several forms of retinitis pigmentosa with the most common mutation causing misfolding of the light sensitive protein rhodopsin found in the rods of the retina. Explain how misfolding of rhodopsin can affect the transduction pathway.
5. Outline some benefits of the transduction pathway being a multistep process.
6. What are secondary messengers?
7. What is a ligand?
8. The diagram shows signal transduction pathway that involves a G protein.

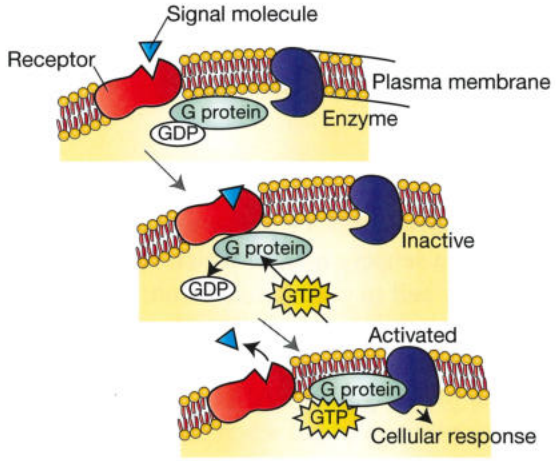


Figure 10.3 Signal transduction with G protein.

In this pathway, what is acting as a ligand?

(A) Signal molecule. (B) Enzyme.
(C) Receptor. (D) G protein.

11 The Nervous System

The nervous system is involved in coordination – sending messages when stimuli are detected to bring about appropriate responses. For example, the nervous system coordinates the activities of the heart, lungs and legs muscles. The coordination of body functions occurs through electrical signals and the release of regulatory molecules, e.g. neurotransmitters.

The nervous system and endocrine system act together in coordination and communication maintaining homeostasis.

The nervous system responds to changes that occur in the internal environment, e.g. changes in blood pressure or digestive activity or to changes that occur in the external environment, e.g. controlling voluntary actions and responding to information picked up by the sense organs.

The nervous system can be divided into two sections.

- The **central nervous system** (CNS) consists of the brain and spinal cord. The brain and spinal cord are completely surrounded by the **meninges** (three layers of membranes) with the spaces between the membranes containing cerebrospinal fluid.
- The **peripheral nervous system** (PNS) consists of all nerves outside the brain and spinal cord.

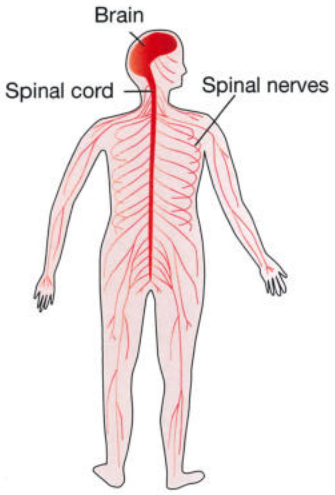


Figure 11.1 Human nervous system.

Neurons

The nervous system consists of excitable nerve cells called **neurons**. **Sensory neurons** carry an electrochemical impulse from a receptor in a sense organ to the CNS. **Motor neurons** carry the electrochemical impulse from the CNS to an effector which can be a muscle or a gland. **Multipolar neurons** (or connector neurons) make connections to other neurons inside the CNS.

Supporting cells

Glial cells are supporting cells found in the nervous system and outnumber neurons by 10 to 50 to one. They are essential in maintaining the structure of the nervous system for the normal functioning of neurons. Glial cells have different functions, e.g. provide physical support, direct growth, repair and development of neurons, give metabolic support to neuron functioning and help provide barriers and insulation to neurons.

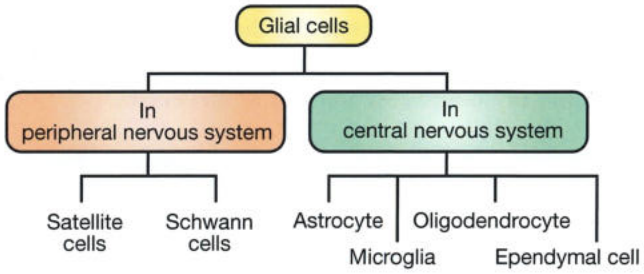


Figure 11.2 Supporting cells of nervous system.

Central nervous system

The central nervous system contains neurons that are supported by four main types of non-excitable **glial cells**. **Astrocytes** are the main supporting tissue of the CNS and are star shaped and found near blood vessels. **Oligodendrocytes** are smaller than astrocytes and cluster around nerve cell bodies and along the length of myelinated nerve fibres. **Microglia** are derived from monocytes and become phagocytic if necessary. **Ependymal cells** form the epithelial lining of ventricles of the brain and central canal of the spinal cord.

The ventricles of the brain contain cerebrospinal fluid which helps support and protect the brain and spinal cord, acts as a shock absorber, keeps the brain and spinal cord moist and maintains a uniform pressure around the brain and spinal cord.

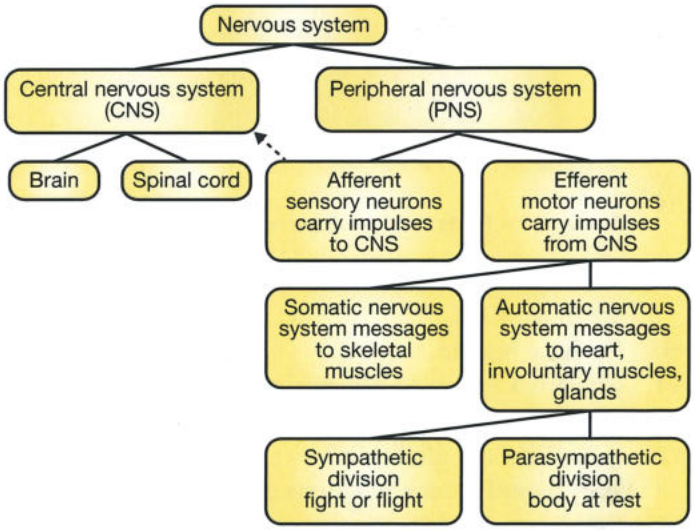


Figure 11.3 Divisions of the nervous system.

Peripheral nervous system

The peripheral nervous system is divided into the **afferent system** which takes information to the CNS and the **efferent system** which carries information from the CNS.

The efferent system can by subdivided into the **somatic system** which controls skeletal muscle and the **autonomic system** which controls smooth and cardiac muscle taking messages to the heart, involuntary muscles and glands. The autonomic system is the involuntary part of the nervous system and can be further subdivided into the **sympathetic system** and the **parasympathetic system**. The responses of the autonomic nervous system are usually rapid and essential for homeostasis.

QUESTIONS

1. What is the function of the nervous system?
2. What is involved in coordination?
3. Name the two body systems that act together in coordination.
4. Identify the location of stimuli recognised by the nervous system.
5. Identify the two sections of the nervous system.
6. What are the meninges?
7. Name the components of the central nervous system.
8. What components are in the peripheral nervous system?
9. The diagram shows the nervous system.

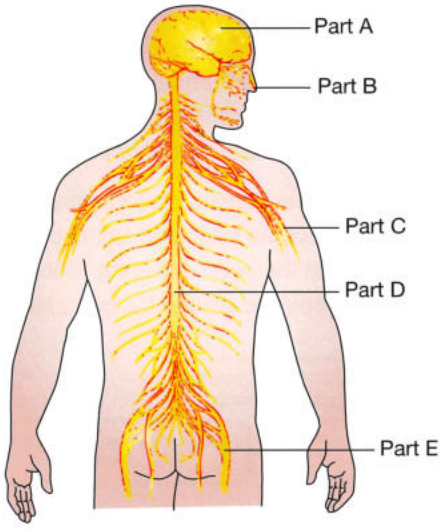


Figure 11.4 Parts in the human nervous system.

Identify which parts are in the CNS and which parts are in the PNS.

10. Define a neuron.
11. Construct a table to summarise the three types of neurons.
12. (a) What are glial cells?
    (b) Identify some functions of glial cells.
13. Construct a table to summarise the four types of glial cells found in the CNS.
14. State the functions of cerebrospinal fluid.
15. Describe the two divisions of the PNS.
16. The diagram shows structures in the arm and in the hand.

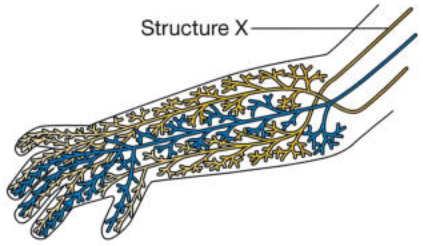


Figure 11.5 Structures in the arm and hand.

    (a) What part of the nervous system is shown?
    (b) What is the difference in the function of the sensory neurons and motor neurons in the hand?
17. Identify the divisions of the efferent nervous system.
18. The diagram shows the division of the autonomic nervous system.

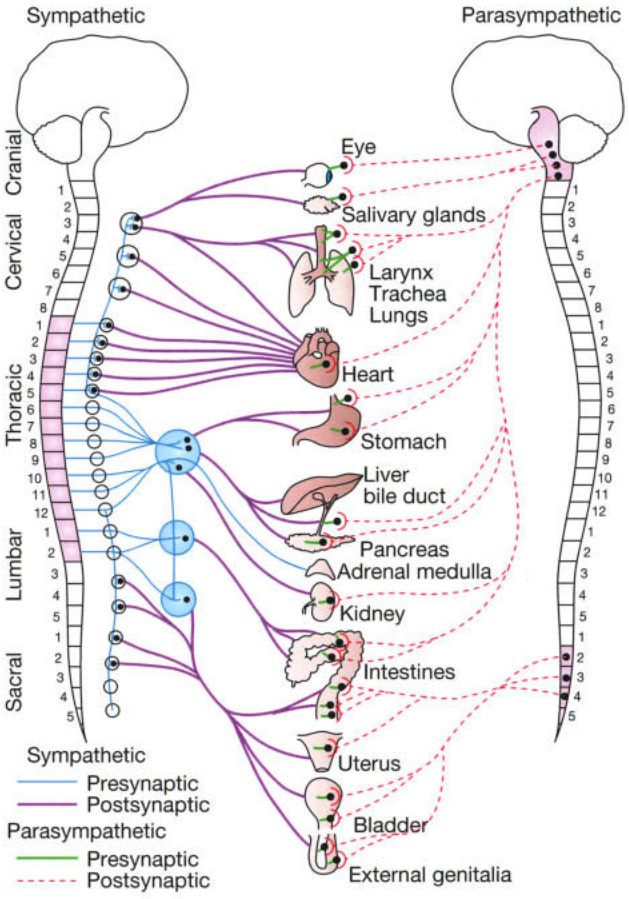


Figure 11.6 Division of the autonomic nervous system.

    (a) What are the two divisions of the autonomic nervous system?
    (b) From the diagram how many connections of the sympathetic system go from vertebrae to the heart?
19. Which cells take electrochemical impulses from a receptor to the central nervous system?
    (A) Astrocyte. (B) Motor neuron.
    (C) Sensory neuron. (D) Multipolar neuron.

12 The Role of the Nervous System

The nervous system is involved in coordination and carries out a vital role in detecting and responding to environmental changes.

In embryonic development the **ectoderm** differentiates to form a hollow **neural tube** which becomes the brain and spinal cord and **neural crests** which become the sensory nerve fibres and the autonomic nervous system.

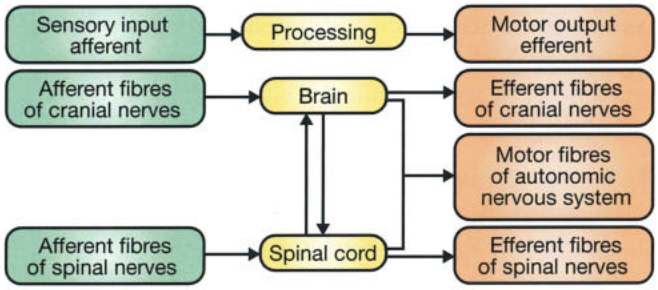


Figure 12.1 Organisation of the nervous system.

Detecting environmental changes

Sense organs contain **receptors** that detect changes in the environment. Sensory nerve endings in the skin respond to touch, pain, pressure, and heat and cold.

Taste buds on the tongue recognise four classes of chemicals – sweet, sour, salty and bitter.

Olfactory receptors are chemical receptors in the nasal cavity that determine the 'flavour' of food and detect smell.

The **retina** in the eye has photoreceptor cells called rods and cones. Cones respond to bright light and determine colour while rods respond to dim light.

The **cochlea** in the inner ear has sensory nerve endings that detect vibrations and messages sent to the brain are interpreted as sound.

Receptor → sensory neuron → connector neuron → motor neuron → effector

Figure 12.2 Path of nervous message.

Responding to environmental change

Some responses of the CNS involve reflex arcs which are **involuntary reactions**. Reflex arcs are important in protecting the body, e.g. blinking if dirt flicks in your eye, coughing if particles touch the lining of the windpipe. You cannot stop this reflex and often are not aware it is happening, e.g. iris dilation in dim light.

The brain determines **voluntary actions** in response to stimuli. The brain sends a message along motor neurons to instruct the effector to carry out a particular action, e.g. put on a jumper because you 'feel' cold.

QUESTIONS

1. What is the main role of the nervous system?
2. Trace the pathway from receptor to effector.
3. From which cells in the embryo does the nervous system develop?
4. The diagram shows a simple spinal reflex arc.

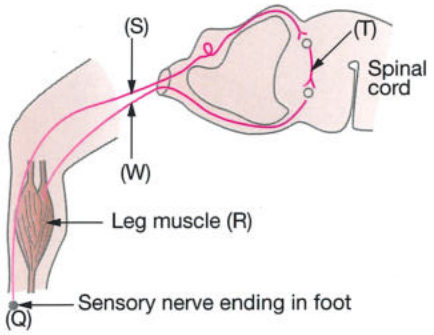


Figure 12.3 Spinal reflex arc.

Identify the type of neuron labelled (T) and explain its function.

5. Use examples to show how the tongue can be both a receptor and an effector.
6. The diagram shows a flow chart for the coordination of body temperature in response to changes in environmental conditions.

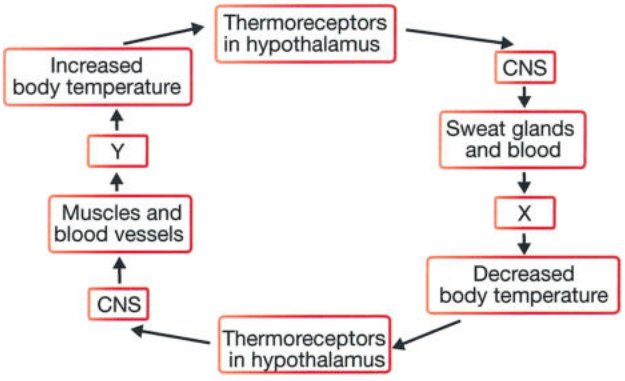


Figure 12.4 Coordination of body temperature.

Suggest suitable responses for part X and part Y.

7. The following steps are involved in the pathway of a nervous message.
   Step I – detection of change in environment.
   Step II – reaction by body to stimulus.
   Step III – electrochemical message sent from receptor.
   Step IV – motor neuron transmits nerve impulse.
   What is the correct order of these steps?
   (A) I, III, IV, II (B) I, II, III, IV
   (C) I, IV, III, II (D) III, I, II, IV

13 The Brain

The human brain is one of the largest organs in the body. Each neuron makes hundreds of synaptic contacts with other neurons. The brain is protected by the bones of the cranium, by being suspended in the cerebrospinal fluid and by the three meninges.

During embryo development the neural tube develops three bulges which become the forebrain, midbrain and hindbrain. The hindbrain develops into the cerebellum and part of the brainstem and the forebrain becomes the cerebrum.

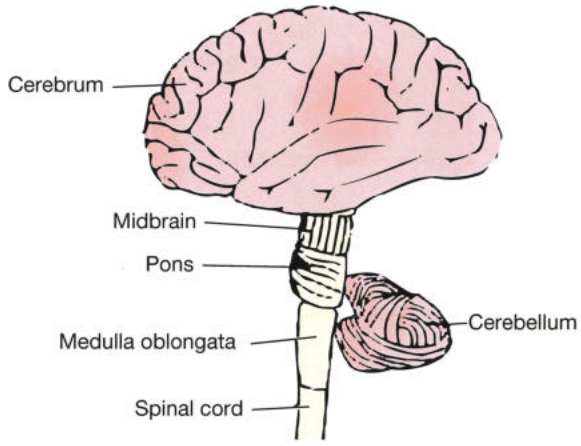


Figure 13.1 Parts of central nervous system.

Meninges

The brain and spinal cord are completely surrounded by three meninges – the dura mater, arachnoid mater and pia mater.

Dura mater is the outermost layer close to the skull consisting of two layers of dense tough fibrous tissue and provides a protective coating for the brain. Spinal dura mater forms a loose sheath around the spinal cord.

Arachnoid mater is a delicate transparent membrane between the dura and pia maters. In the brain it passes over the convolutions.

Pia mater is fine connective tissue with many small blood vessels that sticks to the brain completely covering the convolutions and fissures.

Cerebrum

The largest part of the human brain is the **cerebrum** where all conscious processes occur – it allows us to think, reason and remember. It is divided into two hemispheres and each hemisphere is divided into four lobes named after the bones of the cranium under which they lie – frontal, parietal, temporal and occipital.

There is a sensory area where signals from sense receptors are received and a motor area that sends out orders to muscles all over the body. The surface of the cerebrum is highly folded and covered by a layer of grey matter about 3 mm deep, known as the **cerebral cortex**.

Different parts of the cerebrum do different jobs. For example, the visual cortex is at the back and detects simple visual stimuli; the visual association area is above the visual cortex and is involved in the complex processing of visual information; and the auditory cortex is in front of these sections and is involved in the detection of sound quality (e.g. loudness and tone).

Association fibres connect different parts of a cerebral hemisphere. The left half of the cerebrum controls the right side of the body while the right half controls the left side of the body.

The **hypothalamus** is below and in front of the thalamus and linked to the **pituitary gland**. The hypothalamus controls the release of hormones from the pituitary gland, is involved in appetite, maintaining homeostasis. thirst and water balance, body temperature, emotional reactions, e.g. pleasure, fear, anger, sexual behaviour and the autonomic nervous system.

Figure 13.2 Cross-section through a human skull.

Corpus callosum

The corpus callosum is a thick band of nerve fibres in the central region of the brain that connect the right and left cerebral hemispheres.

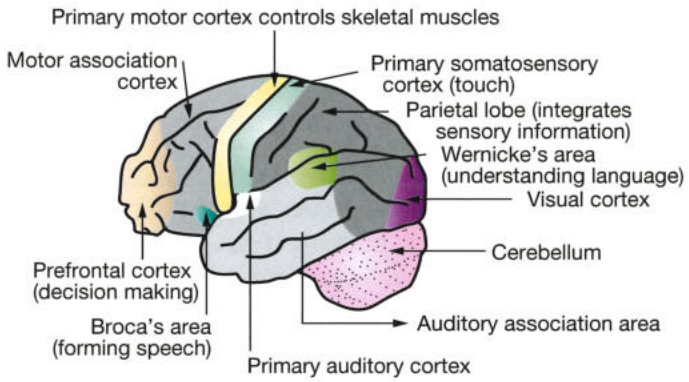


Figure 13.3 Functional regions of the human brain.

Cerebellum

The **cerebellum** is located behind the brainstem and is involved in the coordination of movement, balance and posture. The cerebellum, pons and medulla oblongata form from the hindbrain. The cerebellum is the second largest division of the brain and consists of two lateral hemispheres and receives sensory information about the positions of muscles and joints and from the visual and auditory systems. By monitoring messages from the cerebrum it coordinates responses, e.g. eye-hand responses and is the reflex centre for muscular coordination.

All activities of the cerebellum are unconscious and automatic and many are inhibitory as it adjusts motor impulses according to received sensory information. If injured movements become clumsy and uncoordinated.

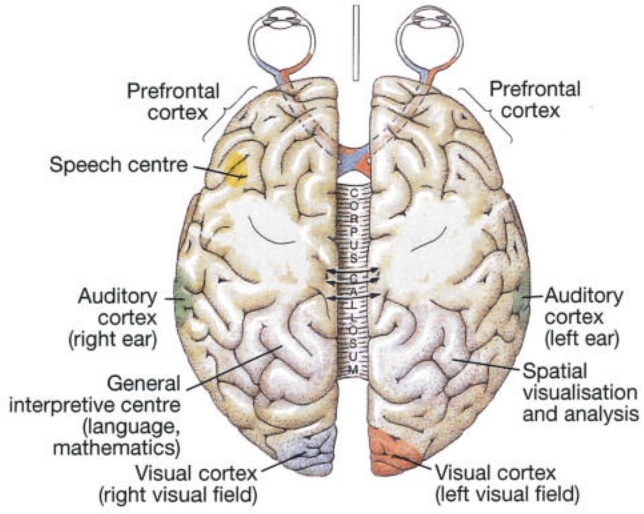


Figure 13.4 Corpus callosum connects left and right cerebral hemispheres.

Medulla oblongata

The **medulla oblongata** is continuous with the spinal cord and controls several visceral (autonomic, homeostatic) functions including breathing, heart and blood vessel activity, blood pressure, swallowing, vomiting and digestion. It extends from the pons down to the spinal cord transmitting information between the PNS and the forebrain and midbrain. The thick walls of the medulla consist mainly of nerve tracts that connect the cerebral cortex and spinal cord with many tracts crossing the midline to the opposite side of the body in the medulla region called the **pyramids**. One of the largest fibres in the pyramidal tract are myleinated fibres with a mean diameter of 16 micrometres which originate from giant pyramidal cells called **Betz cells** in the primary motor cortex.

The **pons** is the bulge above the medulla and is a relay centre for information between the cerebellum and cerebrum. It also acts with the medulla in coordinating breathing.

QUESTIONS

1. Describe the structures that protect the brain.
2. Distinguish between the cerebrum, cerebellum and medulla oblongata.
3. The diagram shows a cross-section of a human brain.

Figure 13.5 Brain cross-section.

Label the cerebrum, cerebellum, corpus callosum, spinal cord, medulla oblongata and the pituitary gland.

4. Some year 12 Human Biology students dissected a fresh sheep's brain purchased from the local butcher. They described it as a large pinkish-white, soft mass.
   (a) Identify two safety procedures they should have followed during their dissection.
   (b) Outline one way in which the sheep's brain is similar to a human brain and one way in which it is different.
   (c) In the brain, where would the students have found the white matter and the grey matter as compared with their respective locations in the spinal cord?
5. (a) Where is the hypothalamus?
   (b) Outline the function of the hypothalamus.
6. What is the corpus callosum?
7. Explain why the cerebellum can be called the reflex centre of muscular coordination.
8. Describe the function of the medulla oblongata.
9. What is the pons?
10. The diagram shows the rear view of the brain.

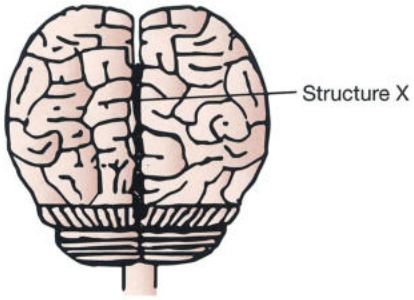


Figure 13.6 Rear view of brain.

What is structure X?

(A) Cerebrum. (B) Cerebellum.
(C) Corpus callosum. (D) Medulla oblongata.

14 The Brain Interprets Signals

The cerebrum is involved in mental activities, e.g. memory, intelligence, thinking, learning, reasoning, consciousness, sensory perception and initiation and control of voluntary muscle (e.g. skeletal muscle contraction. To bring about a coordinated response to stimuli such as light and sound and other signals from the environment, the brain needs to interpret the information correctly. **Perception** is the brain's interpretation of sensory information and the perceived stimulus may be very different from the actual stimulus. The decision about the type of voluntary action to take is decided in the cerebrum; the coordination of voluntary movements is controlled by the cerebellum.

The cerebrum is divided into two cerebral hemispheres. The right and left hemispheres consist of millions of nerve fibres and are connected by a bundle of fibres called the corpus callosum, which acts as a communication bridge between the two hemispheres. Each hemisphere is responsible for a different kind of thinking. For most people, reasoning, logic, scientific skills, mathematics, written and spoken language are controlled by the left hemisphere; while art, imagination, artistic ability, insight, intuition, music appreciation are controlled by the right hemisphere.

Mapping the brain

When studying areas of the brain, the size of the body part is *not* proportional to the nerve supply to that body part. Figure 14.1, a map of the brain by Wilder Penfield, shows the sensory areas on the left hand side of the diagram and motor areas on the right hand side of the diagram with each body part drawn to a scale proportional to the amount of brain space that that part occupies. This map shows the large amount of sensory information from the face and the large amount of motor connections to the hand.

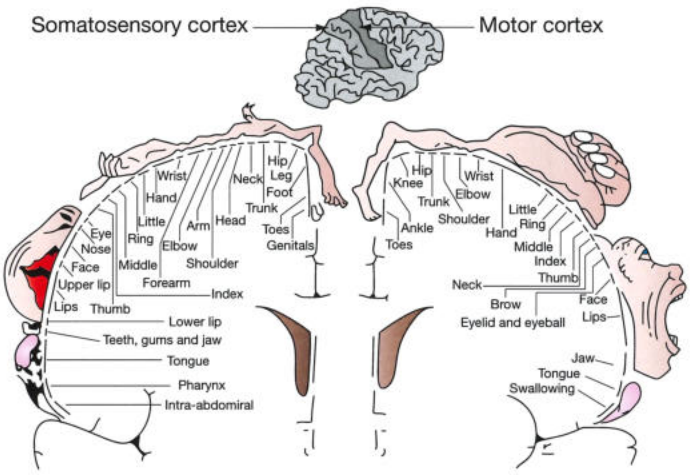


Figure 14.1 Map of the brain.

New technologies

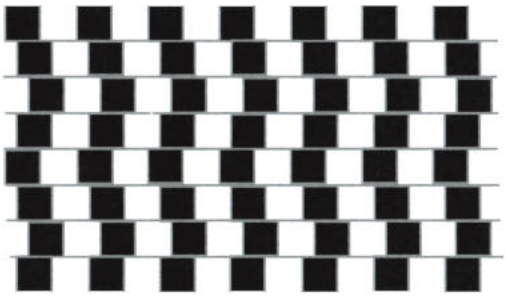
New imaging techniques e.g. **positron emission tomography** (PET) and **functional magnetic resonance imaging** (fMRI) allow scientists to observe how the brain functions. In PET scans, the volunteer is injected with radioactive water and the brain activity, which is detected as an increase in blood flow and more radioactive water, shows up on the PET scanner. The newer fMRI technique shows deep activity in the brain with high spatial resolution. It does not use radioactive materials, but surrounds the person's head with a giant magnet, which detects changes in the level of blood oxygen in the brain. Higher oxygen levels indicate more active levels, which are detected by fMRI as increased radio signals from the brain, as changes in the direction of the magnetic field cause hydrogen atoms in the brain to emit radio signals.

Magnetoencephalography (MEG) and **electroencephalography** (EEG) use an array of sensors or electrodes to map the active areas of the brain when different functions are being performed. These methods are able to track brain messages in real time – at a faster rate than PET and fMRI machines.

In more recent developments there has been an integration of MRI and fMRI with MEG and EEG in functional neuroimaging. This links the good spatial resolution of fMRI with the speed of the EEG to give an accurate sequence of events and a way to observe and measure brain activity in humans.

Optical illusions

Optical illusions occur when objects are drawn in such a way as to 'trick' the brain. Study Figure 14.2 and decide how the optical illusion is achieved.



Are the horizontal lines parallel or do they slope?

Figure 14.2 Optical illusion.

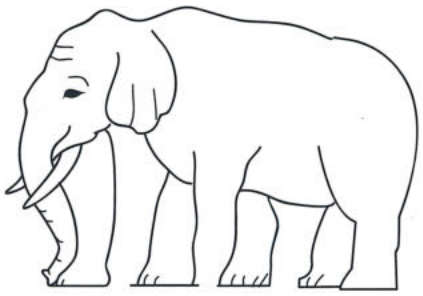
When given information the brain will use past experience to fill in missing information. For example, when given a two-dimensional drawing, the brain will translate the drawing into a three-dimensional image. Clothing designers use patterns and shapes that give an optical illusion to help a person emphasise or obscure a particular body feature.

Behaviour

Behaviour is determined by decisions made by the brain about the information received by the senses. If the received information seems faulty or unreliable (e.g. optical illusions) or if it is contradictory (e.g. different senses giving different information), behaviour will be affected. For example, if a human spins around and around on the spot, and then suddenly stops still, the eyes give information that the body is still, but the semicircular canals provide information that the body is still spinning. The brain is given conflicting information and frequently decides that the solution is to 'feel sick' forcing the body to stay still until all sensory information agrees.

QUESTIONS

1. Identify the three main functions of the cerebrum.
2. Define perception.
3. (a) Outline the structure of the cerebrum.
   (b) Compare the amount of brain space concerned with sensory receptors on the fingertip compared to sensory receptors on the hip.
4. Describe the different types of thinking that have been allocated, for most people, to each cerebral hemisphere.
5. Outline how PET has provided information about the functioning of the brain.
6. Describe how a PET scan is different from fMRI.
7. Outline why newer instruments have increased information about the functioning of the brain.
8. (a) What is a more recent development?
   (b) Outline a benefit of this development.
9. What is an optical illusion?
10. Study Figure 14.3 and outline what is 'wrong' with this optical illusion.



How many legs does this elephant have?

Figure 14.3 Optical illusion problem.

11. What determines behaviour?
12. Use an example to show how conflicting sensory information can sometimes cause a person to 'feel sick'.
13. If a person loses a body part, e.g. a limb, suggest why they sometimes feel a phantom pain in that body part.
14. In many instances of visual perception, the brain adds information from memory and makes an interpretation that goes beyond the sensory information. Study Figure 14.4 based on a triangle drawn by the Italian psychologist Gaetano Kanizsa.

Figure 14.4 Seeing what is not there.

   (a) Outline how Figure 14.4 shows an interpretation of the brain that goes beyond the sensory information being received.
   (b) If you block out part of Figure 14.4, what happens to the triangle? Suggest why this occurs.
15. The feeling of pain is an interesting example where the brain needs to interpret the signals correctly so that the animal can alter its behaviour to relieve the pain. Pain is only felt when the brain interprets the signal, although the brain itself is insensitive to pain. Choose an example to show how our response to pain can be modified by past experience.
16. The drawing shows the drawing of two men of equal size and shape.

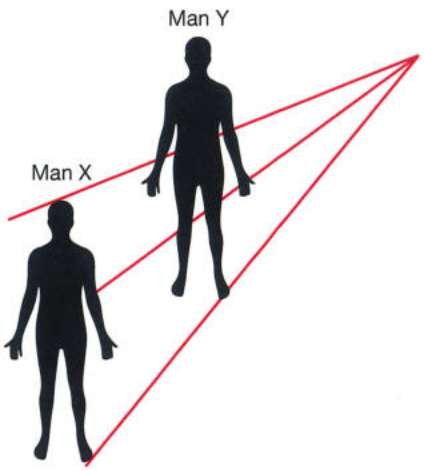


Figure 14.5 Optical illusion problem.

Explain why man Y appears larger and taller than man X even though they are the same size.

17. Which of the following would have the greatest area in the primary somatosensory cortex of the brain?
   (A) Lips. (B) Neck.
   (C) Thumb. (D) Abdominal organs.

15 The Spinal Cord

The spinal cord runs lengthwise inside the vertebral column from the base of the brain and is completely surrounded by three meninges – the dura mater, arachnoid mater and pia mater and cerebrospinal fluid. It is cylindrical with a small central canal and connects to the medulla oblongata of the brain. In an adult male it is approximately 45 centimetres long and its diameter varies at different levels, e.g. enlarged in cervical and lumbar regions.

Grey matter

The central part of the cord is grey matter with cell bodies, synapses and non-myelinated connector neurons. When looking at a cross-section of the spinal cord the grey matter seems to form the letter 'H'. It is grey as it lacks myelin and appears grey in unstained preparations.

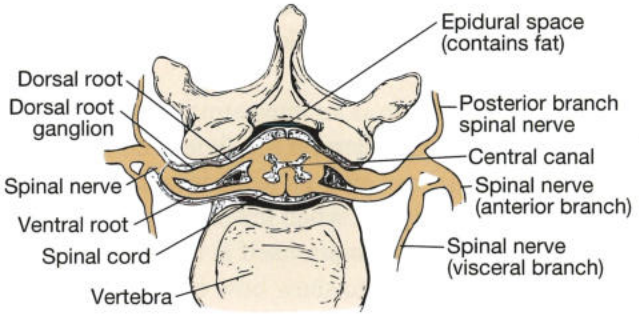


Figure 15.1 Cross-section spinal cord.

The **dorsal nerve root** contains sensory axons with incoming messages and the cell bodies of these neurons are in the **dorsal root ganglion**. The **connector neurons** are short neurons in the grey matter of the spinal cord and synapse with the sensory neurons, other connector neurons and motor neurons. The **ventral nerve root** contains the axons of motor neurons.

White matter

The outer part is white matter with myelinated fibres running longitudinally. The white matter has **ascending tracts** taking sensory information to the brain and **descending tracts** which bring motor impulses down from the brain.

Spinal nerves

Spinal nerves are formed by the union of both the dorsal and ventral roots after they leave the spinal cord. Spinal nerves are called **mixed nerves** as they carry both sensory and motor impulses. Each spinal nerve divides into a posterior and anterior branch and are named and numbered according to the vertebrae to which they connect, e.g. nerve C2 passes between the atlas and the axis vertebrae.

QUESTIONS

1. Where is the spinal cord?
2. What protects the spinal cord?
3. How long is the spinal cord in an average adult male?
4. Figure 15.2 shows a cross-section of a spinal cord.

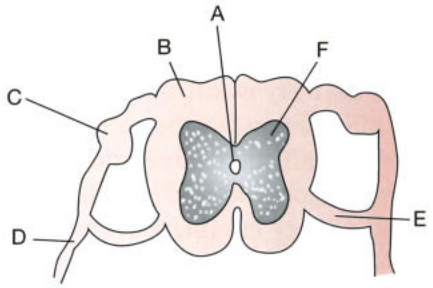


Figure 15.2 Student drawing of spinal cord.

Identify parts A to F.

5. In the spinal cord, where is the grey matter and the white matter?
6. (a) What are the components present in grey matter?
   (b) Why is it called grey matter?
7. What is the dorsal nerve root?
8. Outline the relationship between the dorsal nerve root and the dorsal root ganglion.
9. (a) Where are the connector neurons in the spinal cord?
   (b) List the connections of the connector neurons.
10. What is the ventral nerve root?
11. What is the white matter of the spinal cord?
12. (a) What are the two types of nerve tracts in the white matter of the spinal cord?
    (b) Outline the function of these two tracts.
13. What are spinal nerves?
14. Explain why spinal nerves are called mixed nerves.
15. Outline what happens to the spinal nerves as they leave the spinal area.
16. The diagram shows a reflex arc involving the spinal cord.

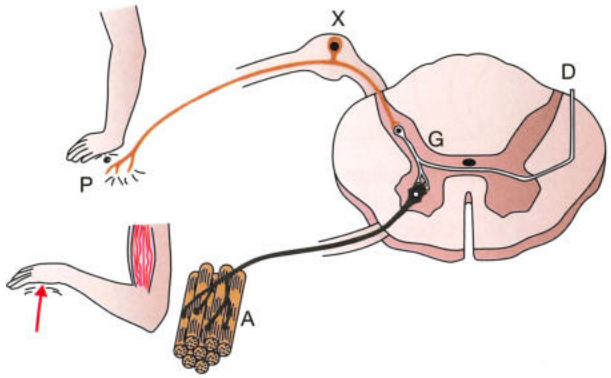


Figure 15.3 Reflex arc.

Use this diagram to write the correct sequence events in a reflex arc and identify the parts A, D, G, P and X.

16 The Autonomic Nervous System

The autonomic nervous system is the **motor division** of the **peripheral nervous system**. It is the **involuntary system** controlling functions that are carried out automatically, e.g. controlling the internal environment and homeostasis.

The **effector organs** are smooth muscle, cardiac muscle and the glands. The main control of the autonomic system is from **centres in the hindbrain**, e.g. cardiac centre, respiratory centre and thermoregulatory centre which are below the level of the cerebrum.

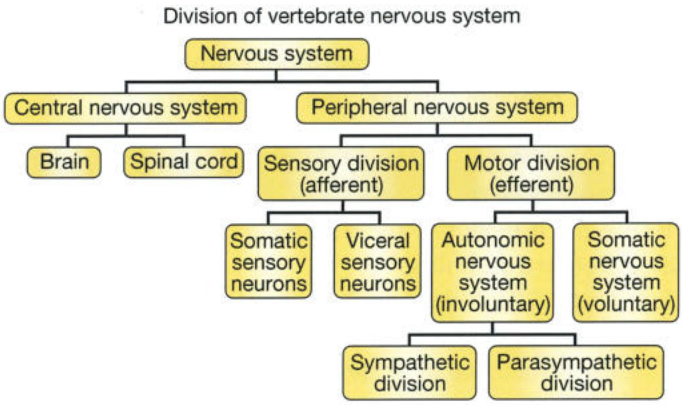


Figure 16.1 Structure of the nervous system.

Although control of the autonomic nervous system is involuntary a person is conscious of its effects, e.g. change in heart rate and some parts can be put under voluntary control, e.g. anal and bladder sphincter muscles.

The two divisions of the autonomic nervous system are the **sympathetic system** and the **parasympathetic system**. Although the two systems work in opposition and are **antagonistic,** e.g. sympathetic increases ventilation rate and the parasympathetic decreases ventilation rate, the two systems need to work together to control involuntary responses involving smooth muscle, cardiac muscle and glandular secretion.

Sympathetic nervous system

The sympathetic nervous system prepares the body for immediate action, e.g. flight or fight and increases heart rate. It mobilises the body's resources under stress to induce the flight or fight response. It causes an increase heart rate, increase in blood pressure, dilation of the iris and decreases the flow of blood in the gut. The neurotransmitter acetylcholine is released from preganglionic neurons and noradrenaline is released from postganglionic neurons.

Parasympathetic nervous system

The parasympathetic nervous system is involved with rest and digestion and decreases heart rate. It is involved in functions that do not require immediate reaction, e.g. for rest, for digestion, salivation, urination, lacrimation and defecation. It decreases heart rate, causes constriction of the iris, reduces blood pressure and increases flow of blood in the gut. The neurotransmitter acetylcholine is released from both pre and postganglionic neurons.

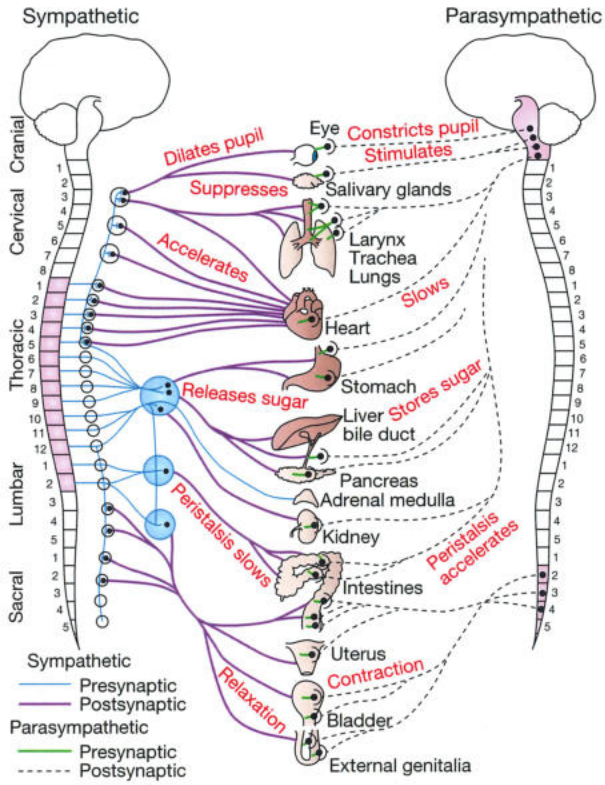


Figure 16.2 Connections of the autonomic nervous system.

QUESTIONS

1. What is the autonomic nervous system?
2. What is the main role of the autonomic nervous system?
3. Name the two divisions of the autonomic nervous system.
4. Why can the sympathetic and parasympathetic nervous systems be considered to work in opposition to each other?
5. Why do the sympathetic and parasympathetic nervous systems need to work together?
6. Construct a table to compare the sympathetic nervous system and the parasympathetic nervous system.
7. What are the effector organs of the autonomic nervous system?
8. Where is the main control of the autonomic nervous system?
9. What action would the sympathetic nervous system cause to the salivary glands and liver?
10. What action would the parasympathetic nervous system cause to the bladder and intestines?

17 The Stimulus-Response Model

The stimulus-response model gives the basic pathway for a nervous impulse where a receptor picks up a stimulus, the message is interpreted and another message is transmitted to an effector which carries out the response.

A **stimulus** is a change in the environment of an organism that is detected.

Stimulus → receptor → nervous system → effector → response

Figure 17.1 Simple stimulus-response model.

The three main parts of the stimulus-response model are:

- **Receptor** – there are many types of receptors in the body that can detect specific stimuli both internal and external. The receptor converts the stimulus into a nerve impulse which is transmitted by a sensory neuron to the central nervous system (CNS).
- **CNS** – connector neurons in the spinal cord and brain take the message to a control centre where the information is processed.
- **Effector** – motor neurons that connect to the CNS transmit a resultant nervous impulse from the CNS to an effector organ which is a muscle or a gland. The effector carries out the response.

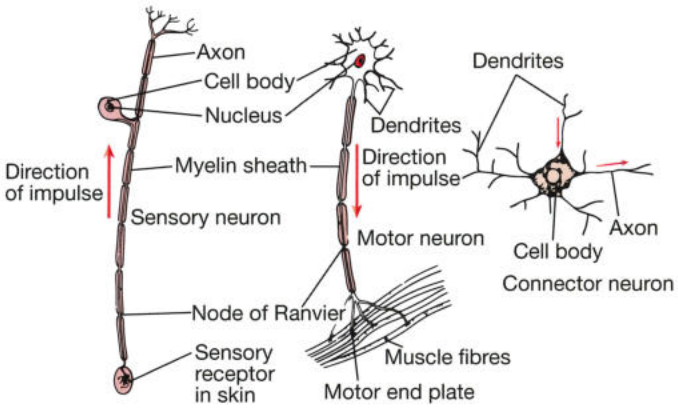


Figure 17.2 Three types of neurons.

Types of neurons

There are three main types of neuron which have specific functions – sensory, motor and connector neurons.

Sensory neurons carry messages from receptors in the sense organs, or in the skin to the brain and spinal cord.

Motor neurons carry messages from the brain and spinal cord to the effectors. A motor neuron to a muscle has many branches in the muscle fibre forming a motor end plate.

Connector neurons are located in the brain and spinal cord. These neurons are also called relay neurons, association neurons or interneurons.

A **nerve** is a bundle of nerve fibres, e.g. axons held together by connective tissue.

A **synapse** is a small gap at the junction between adjacent neurons.

A **neuromuscular junction** is the small gap between the end of an axon and a skeletal muscle cell.

Neurotransmitters are substances that diffuse across a chemical synapse transmitting the message from the synaptic terminal of a neuron to the postsynaptic cell triggering a response, e.g. from the axon to dendrite or from axon to cell body. There are many neurotransmitters, e.g. acetylcholine, adrenaline and dopamine. Acetylcholine is the neurotransmitter at neuromuscular junctions between a neuron and a skeletal muscle.

QUESTIONS

1. What is meant by the stimulus-response model?
2. Define a stimulus.
3. Draw a flow chart to show the steps in the stimulus-response model.
4. Construct a table to summarise the three main parts of the nervous system.
5. Distinguish between a motor neuron and a sensory neuron.
6. What is a nerve?
7. What is the synapse?
8. Outline the function of neurotransmitters.
9. Identify the following cell.

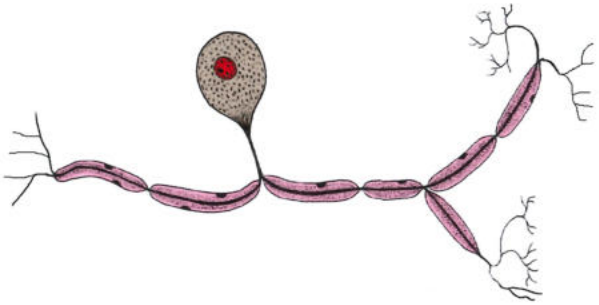


Figure 17.3 Type of cell.

(A) Motor neuron.
(B) Sensory neuron.
(C) Connector neuron.
(D) Muscle cell.

10. Which neurons are never myelinated?
(A) Motor neurons and sensory neurons.
(B) Sensory neurons.
(C) Motor neurons.
(D) Connector neurons.

18 Receptors

One of the characteristics of living things is that they can respond to stimuli. A **stimulus** may come from the external environment (e.g. light intensity, sound) or may come from the internal environment (e.g. hormone levels, arrival of food). **Receptors** detect stimuli.

There are many types of receptors and each type of receptor is highly sensitive to one type of stimulus. Central receptors are found in the **sensory organs**, e.g. the **cochlea** has hair cells that detect pressure waves in the ear.

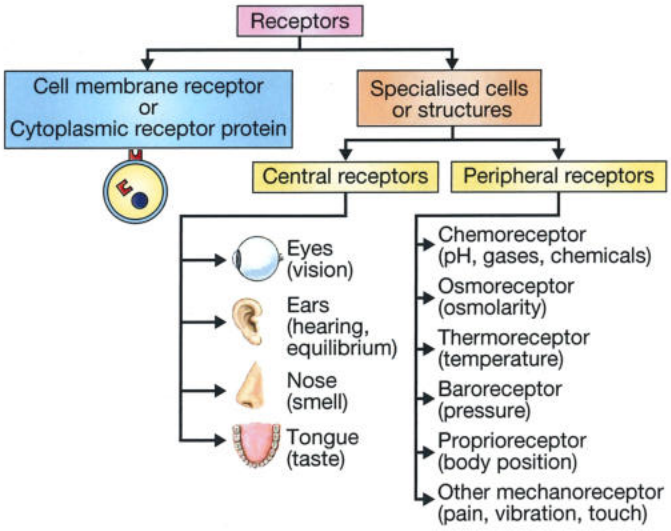


Figure 18.1 Types of receptors.

Proprioceptors are position receptors with one type receiving data about static position and another type receiving data about rate of movement, e.g. in muscles, tendons and joints.

Mechanoreceptors detect mechanical compression or stretching of the receptor or of adjacent tissue, e.g. respond to stretching, movement, touch, pressure, gravity and pain receptors in the skin. The number of mechanoreceptors varies in different parts of the body and varies between types of receptors, e.g. pain receptors are nearly 27 times more abundant than cold receptors.

Chemoreceptors respond to and are activated by the binding to a particular chemical, e.g. detecting taste in the mouth, smell in the nose, oxygen levels in arterial blood, osmolarity of body fluids. There are **olfactory receptors** in the nasal cavity and **gustatory receptors** in the taste buds on the tongue. **Osmoreceptors** are in the anterior hypothalamus and detect changes in plasma sodium concentrations. An increase in sodium causes the osmoreceptor cells to shrink which in turn causes the cells to send nerve signals to the pituitary gland to stimulate the release of antidiuretic hormone (ADH). ADH stimulates the nephrons in the kidney to increase water reabsorption.

Photoreceptors are sensory receptors in the retina of the eye that detect light. There are two types of photoreceptors – rods which are responsible for monochromatic night time vision and work in low light levels and cones which give colour vision during the daytime and high acuity vision.

Thermoreceptors detect heat and cold and changes in temperature.

Internal receptors are found in many locations. For example, **chemoreceptors** in the **medulla** of the brain monitor carbon dioxide levels in the blood. If the carbon dioxide level rises, impulses are sent to the inspiratory centre of the hindbrain, which in turn sends impulses to the diaphragm and intercostal muscles to increase the rate of breathing. Receptors in the hypothalamus detect levels of blood glucose, amino acids and fatty acids.

When a stimulus excites a receptor the membrane potential of the receptor is changed. This is the **receptor potential**. When the receptor potential rises above the threshold an action potential is initiated in the nerve fibre attached to the receptor.

The messenger, usually nerve impulses, takes the message from the receptor to the effector (e.g. muscles and glands in multicellular organisms), which carries out the response. The response to the stimuli helps the organism survive.

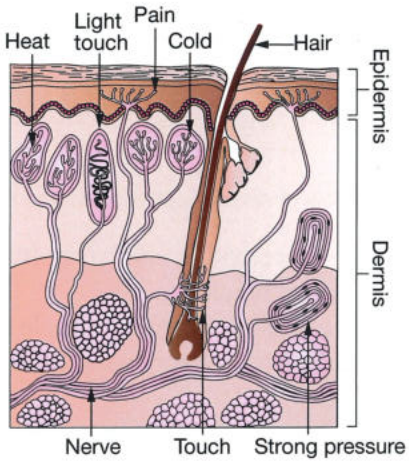


Figure 18.2 Senses in the skin. Most sense receptors in the dermis have capsules around them. Those in the epidermis and those wrapped around the hairs have naked nerve endings.

The receptors for **water soluble hormones** are found on cell membranes while the receptors for **lipid soluble hormones** are found inside the cell.

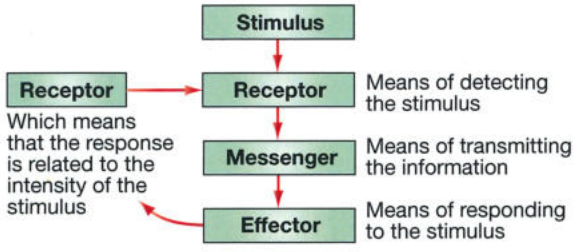


Figure 18.3 Main features of a coordinating system.

QUESTIONS

1. Create a table to show the relationship between the stimulus, the receptor and the sense organ.
2. Distinguish between proprioceptors, mechanoreceptors, chemoreceptors, photoreceptors, thermoreceptors and electroreceptors.
3. Identify the steps in the stimulus-response pathway.
4. Choose one type of sensory receptor found in the skin and briefly describe how it operates.
5. The flow chart shows the control of breathing in humans.

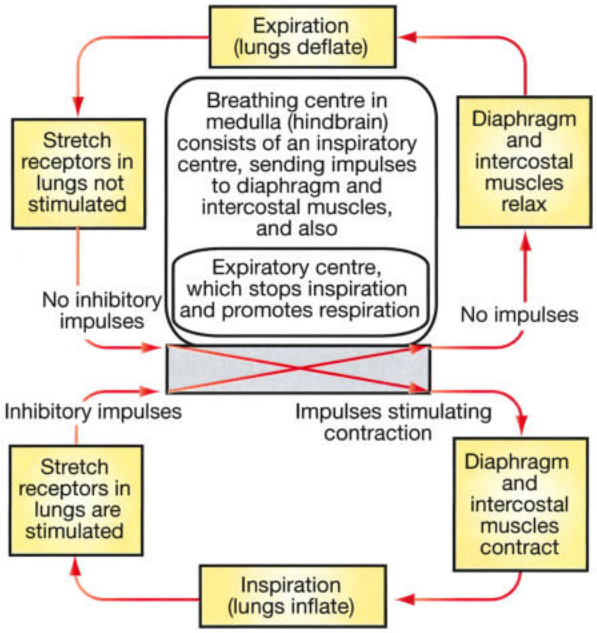


Figure 18.4 Flow chart of breathing in humans.

(a) Identify the receptors involved with the control of the rate of breathing.
(b) Explain a response in this example to show the stimulus, receptor, messenger, effector and response.
(c) Explain how a feedback mechanism could turn off the response you described in part (b).
(d) Use an example to show how the rate of breathing can be influenced by conscious thought.

6. Osmoreceptors in the hypothalamus detect changes in the concentration of solutes in body fluids. For example, the osmoreceptors detect an increase in the salt levels if a person has drunk little water or has been sweating heavily or has eaten large amounts of salty food. The osmoreceptors generate an impulse to the posterior pituitary gland to trigger the release of antidiuretic hormone (ADH) into the bloodstream. The target organ for the hormone is the kidney, where the ADH causes increased permeability of the walls of the collecting ducts so that more water is reabsorbed and the urine becomes more concentrated. Draw a flow chart to show the sequence of events in this stimulus-response example.
7. The diagram shows some different types of receptors.

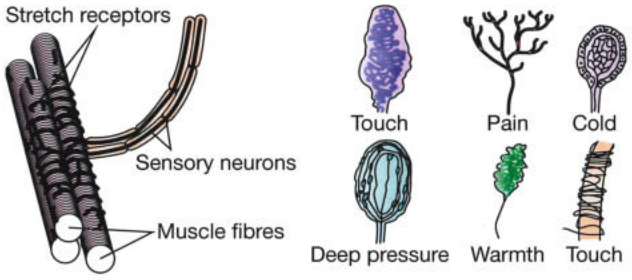


Figure 18.5 Different types of receptors.

(a) Outline the function of the stretch receptors.
(b) From the diagram name some of the different types of receptors found in the skin and comment on their abundance and distribution.

8. Identify the range of senses used in communication.
9. Suggest why it is important for people to be able to detect changes in their environment.
10. Give an example of how humans use touch as a form of communication.
11. The diagram shows the stimulus-response model in a human.

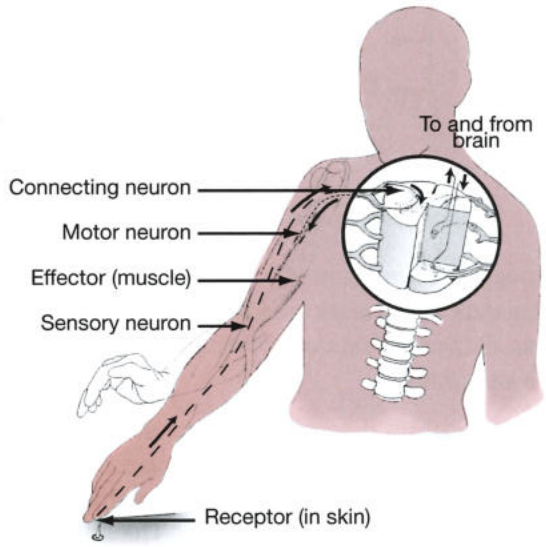


Figure 18.6 Stimulus-response model.

(a) In this diagram, what is the stimulus?
(b) What is the response?
(c) Identify the type of receptor cell.
(d) Draw a flow chart to show the pathway from receptor to response.
(e) The knee jerk stimulus-response is an example of a reflex arc. Compare this example with the stimulus of a ball coming towards you and you deciding to dodge away, rather than catching it.

12. What is the function of proprioceptors?
(A) Senses light.
(B) Senses sound.
(C) Senses temperature.
(D) Senses movement, position of the body.

19 Reflex Arcs

A **reflex** is a fast response that is involuntary and does not involve the brain in the decision making process. Reflexes help maintain homeostasis. In some reflexes the message from the receptor enters the spinal cord and is passed to motor neurons at the same level in the cord or may travel a few segments up the cord before travelling out through a motor neuron. Once the response has been made a person becomes consciously aware of the situation.

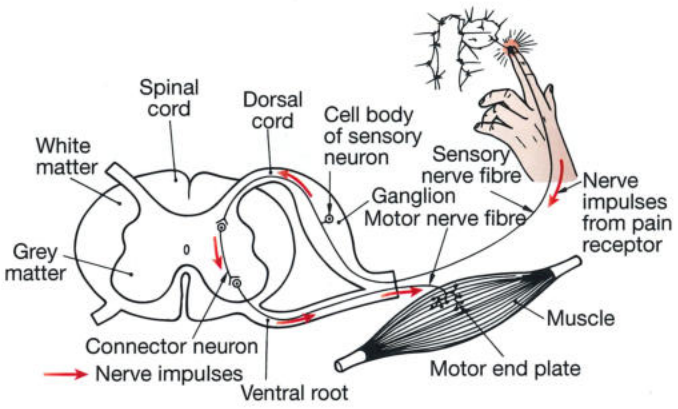


Figure 19.1 Reflex arc.

The brain and spinal cord contain grey matter and white matter. **Grey matter** is mainly the cell bodies of neurons, dendrites and unmyelinated axons. **White matter** consists of axons with myelin sheaths. In the spinal cord the white matter is on the outside to allow linking of the CNS to sensory and motor neurons of the PNS. The grey matter in the spinal cord is roughly in the shape of the letter H. Most of the grey matter in the brain is on the outside surrounding the white matter.

In a **monosynaptic reflex** there are only two neurons – a sensory neuron and a motor neuron, e.g. peripheral muscle reflexes such as the Achilles reflex. In a polysynaptic reflex arc there are one or more connector neurons in the circuit. Most reflexes are **polysynaptic**.

Autonomic nervous system

The autonomic nervous system is a subdivision of the motor nervous system of vertebrates that controls the internal environment. It controls involuntary muscle, e.g. it is the nerves that control cardiac muscle, glands and the smooth muscle that is found in the walls of blood vessels and in the digestive, respiratory, excretory and reproductive tracts. The autonomic nervous system is subdivided into the sympathetic nervous system and the parasympathetic nervous system. The **parasympathetic nervous system** generally produces responses that maintain the body during quiet conditions while the **sympathetic nervous system** usually produces response that prepare the body for activity, e.g. fight or flight responses.

QUESTIONS

1. What is a reflex?
2. When does a person know a reflex has happened?
3. Distinguish between grey matter and white matter.
4. Identify the location of grey matter and white matter:
   (a) In the brain.
   (b) In the spinal cord.
5. Distinguish between a monosynaptic reflex arc and a polysynaptic reflex arc.
6. The diagram shows a reflex arc.

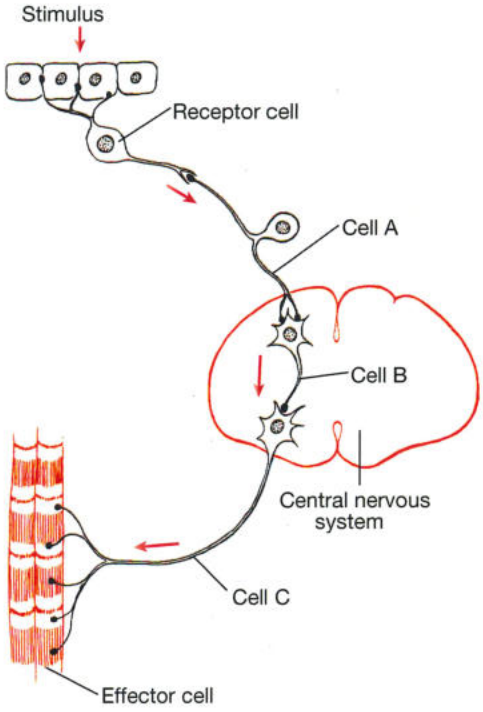


Figure 19.2 Reflex arc.

Identify the cells labelled A, B and C and explain your reasoning.

7. What is the autonomic nervous system?
8. The change in the size of the pupil of the eye is controlled by the autonomic nervous system. Explain why the size of the pupil needs to be a reflex caused by involuntary muscles.
9. Which of the following would *not* be part of the autonomic nervous system?
   (A) Muscle layer in wall of arteries.
   (B) Cardiac muscle.
   (C) Skeletal muscle.
   (D) Muscle layer in wall of intestines.
10. A flexor reflex is a spinal reflex that moves a limb away from a stimulus detected by a skin receptor. Which of the following would cause a flexor reflex?
    (A) Your hand touches a hot object.
    (B) You smell some pepper.
    (C) You see a ball coming towards you.
    (D) You hear a loud noise behind you.

20 The Structure of a Neuron

The function of the nervous system is to coordinate the activities of body organs so that they work together in harmony. The central nervous system contains more than 100 billion neurons.

Neurons or nerve cells are the functional unit of the nervous system. They are specialised cells that transmit signals from one location in the body to another location by electrochemical changes in their membranes.

A **nerve** is a bundle of neuronal fibres. In the brain each neuron may communicate with thousands of other neurons.

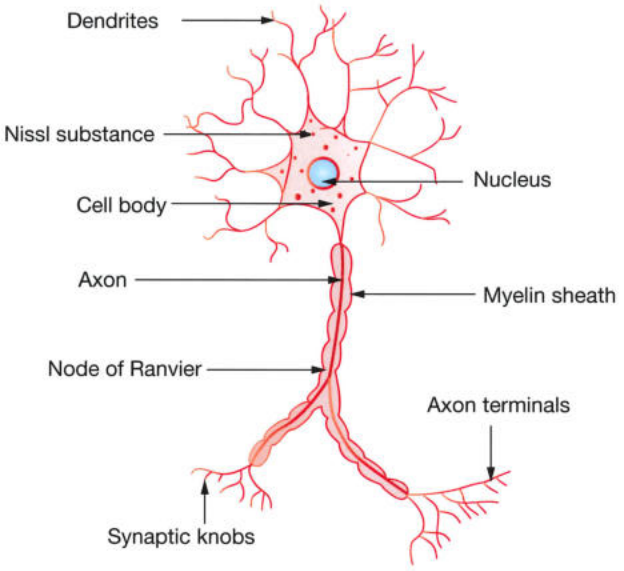


Figure 20.1 Typical neuron.

All nerve cells contain a cell body, dendrites and an axon.

- The **cell body** contains the nucleus and other cell organelles. The high level of cellular activity in neurons means there is usually a large amount of rough endoplasmic reticulum, which secretes proteins – visible as Nissl substance in the cytoplasm.
- The **dendrites** pick up the message using their extensive branches to increase the surface area of the 'receiving end' of the nerve cell.
- The **axon** conducts the message away from the cell body. The axons of many vertebrates are surrounded by **Schwann cells**, which are supporting cells that form an insulating layer called the **myelin sheath** and a myelinated axon. The **nodes of Ranvier** are the intervals along the axon where there is no myelin sheath. The myelin sheath acts as an insulator, protects the axon from damage and increases the speed of the nervous impulse along the axon. The conical region where the axon meets the cell body is the **axon hillock**.

Synapse

The **synaptic terminal** is at the end of the axon. The synaptic terminal releases chemical messengers called **neurotransmitters**, e.g. acetylcholine and noradrenaline. The neurotransmitters are held in many small vesicles in the synaptic knob at the end of the axon.

The **synapse** is the cell junction between a synaptic terminal and a target cell, which can be another neuron or an effector cell, such as a muscle cell. The synapse can be electrical or chemical. Synapses act as a relay and control point and ensure the message is in one direction.

The complexity in the shape of each neuron is related to the number of synapses it has with other neurons.

QUESTIONS

1. What is a nerve?
2. What is a neuron?
3. Identify the three main parts of a neuron and briefly describe each part.
4. Describe the synapse.
5. The diagram shows a photomicrograph of a neuron.

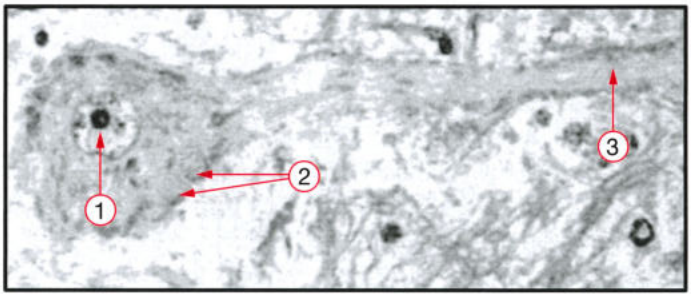


Figure 20.2 Photomicrograph of a neuron.

   (a) Identify the parts labelled (1), (2) and (3).
   (b) Neurons often contain a large amount of endoplasmic reticulum. State where you would expect to find this endoplasmic reticulum and discuss why it is necessary in neurons.
6. What are the nodes of Ranvier?
7. Suggest some functions of the myelin sheath.
8. Almost all axons that are more than 2 micrometres in diameter are myelinated while axons with a smaller diameter are usually unmyelinated. Suggest a reason why large axons need to be myelinated.
9. Outline what happens when a nerve impulse reaches the synaptic terminal.
10. Compare the three types of neurons.
11. Many drugs, such as alcohol, sedative hypnotics, opiate analgesics and hallucinogens, alter neurotransmitter functioning. Outline how these drugs influence perception by the brain.

21 The Action Potential

An **action potential** is a rapid and short change in the membrane potential of an excitable cell. In nerve cells it causes a nerve impulse while in muscle cells is causes contraction.

There is a difference in electrical charge across the plasma membrane – this voltage across the membrane is called the **membrane potential**.

The **resting potential**, which is usually around −70 millivolts (symbol is mV), is the normal condition in most animal cells and means that the inside of the cell is **negative** compared to outside the cell. In the resting state the membrane is said to be **polarised**.

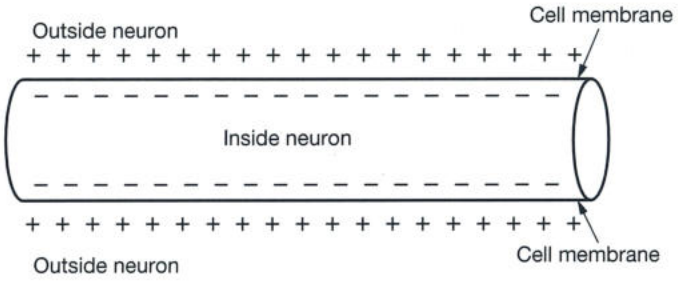


Figure 21.1 Parts of the action potential.

This difference in charge is due to differences in the concentration of ions inside and outside the cell. The inside of the cell has more potassium ions (K^+) than sodium ions (Na^+) while the outside has more sodium ions than potassium ions. This difference is maintained by a sodium-potassium pump which uses energy (ATP) to actively transport Na^+ out of the cell and K^+ into the cell.

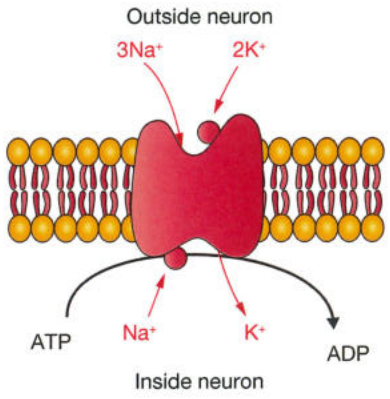


Figure 21.2 Sodium-potassium pump.

The sodium-potassium pump is an electrogenic pump as more positive charges are pumped to the outside than to the inside. An **electrogenic pump** is an ion transport protein that generates a voltage across a membrane. In the sodium-potassium pump three sodium ions are pumped to the outside and only two potassium ions are pumped to the inside thus making the inside negative compared to the outside.

A nerve impulse is an action potential where there is a temporary and local change in the resting potential. This change moves along a neuron and is the way a nerve transmits a message.

Stages of an action potential

- **Depolarisation stage** – At the beginning of a nerve impulse Na^+ ions move into the cell making the inside of the cell positive. This change in electrical potential is called **depolarisation**. The change occurs because ion channels or pores in the membrane open and allow sodium ions to move into the cell. Ion channels consist of a protein molecule which spans the lipid bilayer and the pore down the centre allows specific ions to pass. One type of protein is permeable to Na^+ while another type of protein is permeable to K^+.
- **Repolarisation stage** – The change in permeability of the membrane allowing Na^+ ions to enter stops and K^+ ions channels open allowing K^+ ions to move outside. This causes the inside to become negative again and is **repolarisation**.

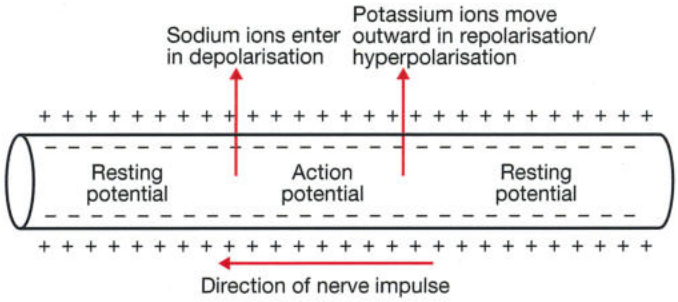


Figure 21.3 Propagation of an action potential.

- **Hyperpolarisation stage** – After repolarisation the inside of the cell becomes more negative than the resting potential. This is called **hyperpolarisation** and occurs when K^+ ions continue to move out of the cell.

An action potential can be recorded using one electrode inside the cell and one just outside the cell membrane.

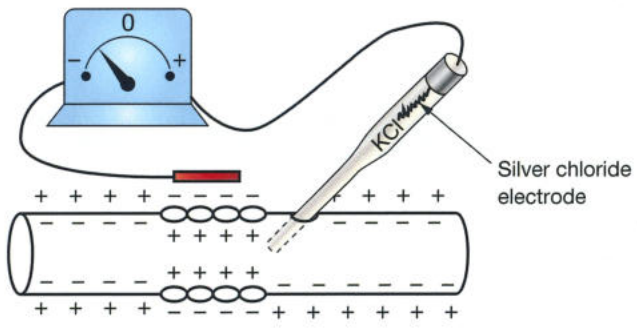


Figure 21.4 Measuring an action potential.

Graphing an action potential

Electrodes in a neuron while it transmits an action potential show the change in voltage across the membrane. The action potential can then be plotted on a graph.

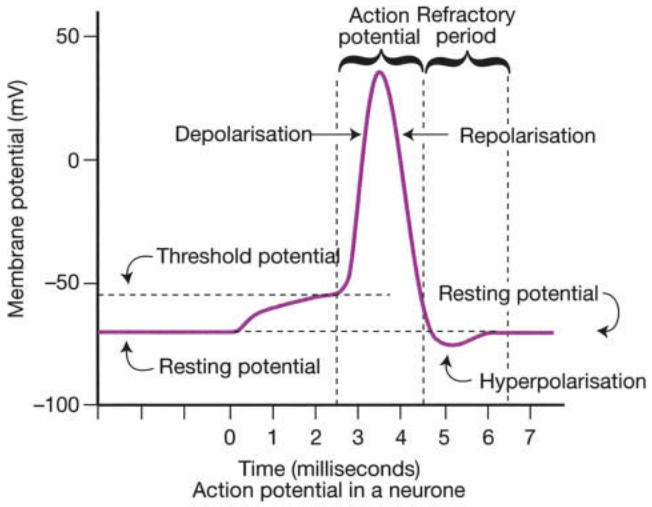


Figure 21.5 An action potential graph.

The **absolute refractory period** is the time after an action potential when the axon is recovering and cannot send another impulse.

The **relative refractory period** occurs after the absolute refractory period and the axon can transmit an impulse if it is more intense than normal.

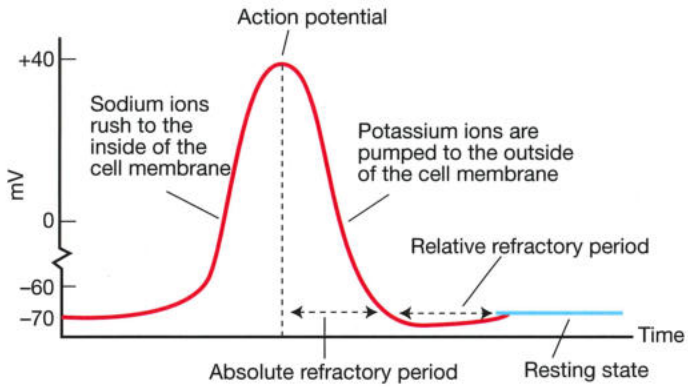


Figure 21.6 Parts of the action potential.

QUESTIONS

1. Define an action potential.
2. What is the membrane potential?
3. Define resting potential and state its value for most animal cells.
4. Why is a membrane in the resting state said to be 'polarised'?
5. Explain how the resting potential is maintained.
6. Define an electrogenic pump.
7. Explain why the sodium-potassium pump is an electrogenic pump.
8. Explain why the nerve impulse is an action potential.
9. What happens at the beginning of a nerve impulse?
10. Describe what happens during depolarisation.
11. Describe what happens during repolarisation.
12. Outline how you can measure voltage changes during an action potential.
13. What is the refractory period?
14. Construct a table to show the characteristics of resting potential, depolarisation, repolarisation and hyperpolarisation.
15. Study the following graph of an action potential.

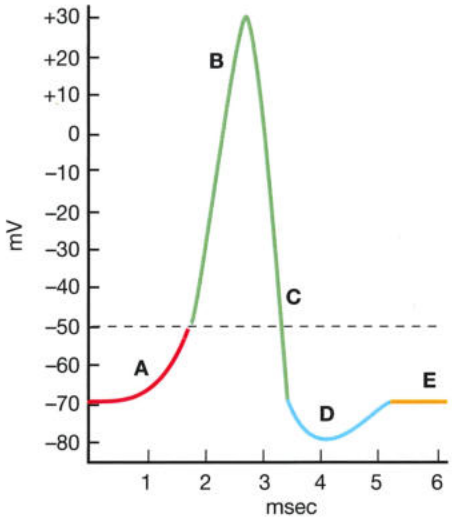


Figure 21.7 Graph of action potential.

Identify each part of the graph and explain what is happening at each part.

16. The diagram below shows part of an axon during a nervous impulse transmission.

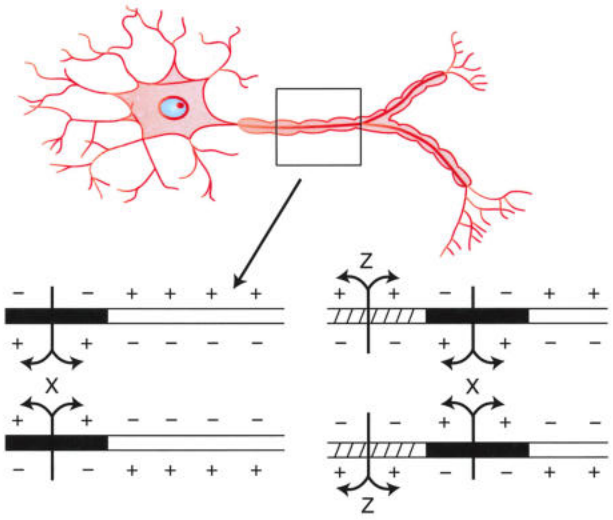


Figure 21.8 Nerve impulse.

(a) Identify the parts labelled X and Z.
(b) Which way is the impulse travelling? Explain your reasoning.

22 Crossing the Synapse

When the nerve impulse reaches the end of the axon there is a space between this neuron and the next neuron. This space is called the **synapse**. Synapses occur between the synaptic knobs of axon terminals of one neuron (presynaptic neuron) and the dendrite or cell body of another neuron (postsynaptic neuron). The site where motor neurons synapse with skeletal muscle cells is called a **neuromuscular junction**.

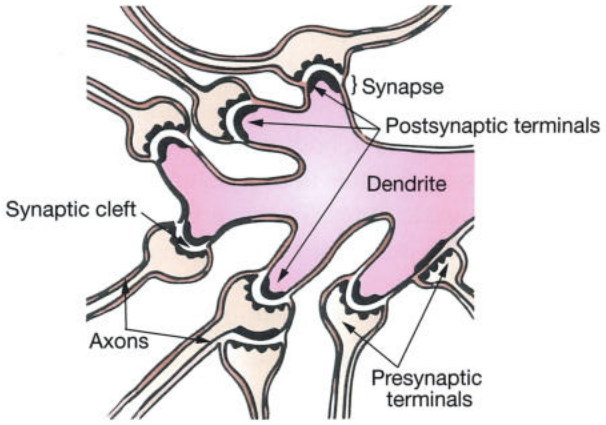


Figure 22.1 The synapse.

There are two types of synapses – electrical and chemical. At **electrical synapses** the action potential moves directly from the presynaptic terminal to the postsynaptic terminal. Electrical synapses are found where there is only a short distance between neurons called **gap junctions**. The nerve impulse is quickly passed across the gap junction and is usually responsible for fast, unvarying behaviours such as defensive reflexes, e.g. escape responses of lobsters and giant squids. There are many electrical synapses in the vertebrate brain. Chemical synapses are more common and involve the release of neurotransmitters.

Neurotransmitters

At a chemical synapse the action potential stimulates the release of a chemical called a **neurotransmitter**, e.g. acetylcholine, adrenaline and noradrenaline. There are more than 100 identified neurotransmitters and a single neurotransmitter may bind specifically to several different types of receptors.

The neurotransmitter diffuses across the synapse and stimulates the postsynaptic terminal which either starts another action potential in the next cell or initiates a response in an effector, e.g. muscle or gland. The diffusion time for a neurotransmitter to cross the synapse is usually short.

Because synaptic vesicles containing neurotransmitters are only found at the end of the axon, the synapse ensures that the nerve impulse can only flow in one direction along a neuron pathway.

Presynaptic terminals

Electron micrographs show that there is a variety of forms of presynaptic terminals with the end of most axons having small oval or round knobs called terminal knobs or synaptic knobs. The cell membrane of the presynaptic terminals have many voltage gated calcium channels. The amount of calcium ions released into the terminal is directly related to the amount of transmitter substance released into the synapse.

Steps in chemical transmission

- When the **impulse arrives** at the synaptic knob it depolarises the terminal membrane causing calcium ion channels to open. The open channels allow calcium ions to flow into the terminal from the synaptic cleft.
- The **rise in calcium ion concentration** causes a few vesicles containing the neurotransmitter to fuse with the presynaptic membrane. The contents of the vesicle are then released into the synaptic cleft by exocytosis.
- The **neurotransmitter diffuses** across the synaptic cleft and the effect on the postsynaptic cell can be either direct or indirect.
- In **direct synaptic transmission** the neurotransmitter binds directly to an ion channel causing the channel to open. In **indirect synaptic transmission** the neurotransmitter binds to a receptor that is not part of an ion channel. This activates a signal transduction pathway which is slower but more long lasting than direct transmission. When the neurotransmitter binds with a receptor protein of an ion channel in direct synaptic transmission the channel opens allowing one or more types of ion to pass, e.g. sodium, potassium, chloride ions. There are receptor sites specific for each type of neurotransmitter.
- An inflow of sodium ions through a sodium ion channel will depolarise the postsynaptic membrane and if the threshold is passed an action potential is generated.
- The neurotransmitter releases from the receptors and the channel closes.
- There are various ways in which the effect of the neurotransmitter is stopped. Enzymes can remove the neurotransmitter from the synaptic cleft to stop continuous firing of impulses in the postsynaptic neuron. For example, the enzyme acetylcholinesterase hydrolyses acetylcholine to choline and ethanoic acid.
- The inactivated forms of neurotransmitters re-enter the presynaptic knob and are resynthesised as neurotransmitter and stored in the vesicles.

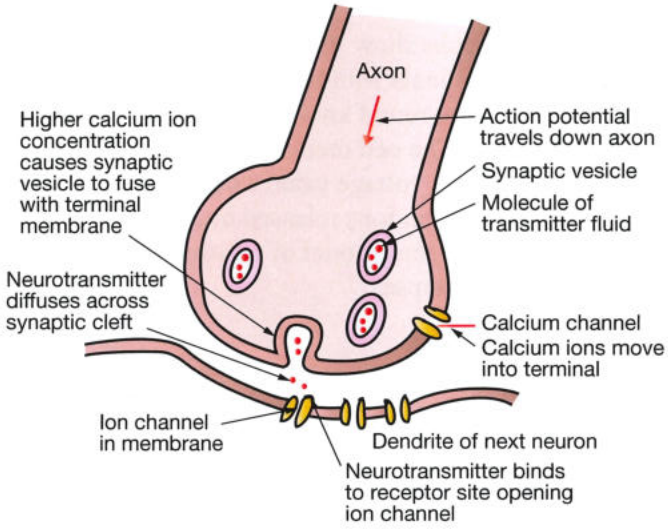


Figure 22.2 The synapse.

In an **excitatory synapse**, the receptor binds to a channel that allows both sodium ions and potassium ions to diffuse. This causes depolarisation and an action potential is initiated in the postsynaptic neuron. In an **inhibitory synapse**, the neurotransmitter only binds to the potassium ion channels and the action potential is inhibited.

Table 22.1 Comparing excitatory and inhibitory presynaptic neurons.

Feature	Excitatory presynaptic neuron	Inhibitory presynaptic neuron
Effect on postsynaptic membrane	Depolarises postsynaptic membrane and generates a new action potential.	Hyperpolarises postsynaptic membrane and makes it more difficult for an impulse to cross the synapse.
Effect on membrane potential	Influx of sodium ions causes increase in membrane potential of postsynaptic membrane.	Influx of chloride ions or loss of potassium ions causes the membrane potential of the postsynaptic membrane to become more negative – lowers resting potential.

QUESTIONS

1. What is a synapse?
2. Distinguish between a presynaptic neuron and a postsynaptic neuron.
3. Identify two types of synapses.
4. Define neurotransmitter.
5. Name some neurotransmitters.
6. Outline the function of neurotransmitters.
7. Where are neurotransmitters stored?
8. Explain why neurotransmitters are important in making the direction of a nervous impulse only go in one direction.
9. Explain how ions can cause a change in the electrical voltage across the plasma membrane of an axon.
10. The diagram shows two neurons in a neuron pathway.

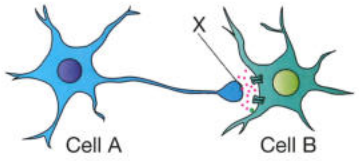


Figure 22.3 Two neurons.

(a) Explain what is happening at point X.
(b) In what direction is the nervous impulse travelling in this diagram?

11. Distinguish between direct and indirect synaptic transmission.
12. Summarise the steps in direct synaptic transmission.
13. Acetylcholine is one of the most common neurotransmitters in both vertebrates and invertebrates. It binds to acetylcholine receptors on skeletal muscle fibres. The diagram shows where acetylcholine is released in many vertebrates.

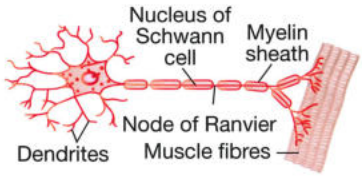


Figure 22.4 Neuron and muscle fibres.

(a) Name the type of neuron shown in this diagram.
(b) What is the name of the site between the neuron and the muscle fibres?
(c) Outline the function of acetylcholine at this site.
(d) Outline what happens to the acetylcholine after it has activated the receptors.

14. The diagram shows a sodium ion channel.

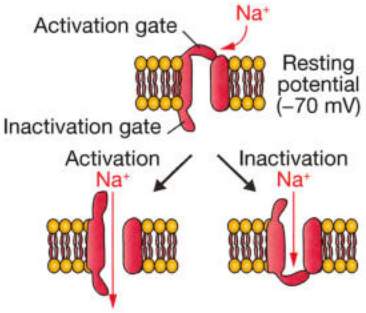


Figure 22.5 Sodium ion channel.

Distinguish between the activated state and the inactivated state of a sodium ion channel.

15. When an action potential reaches a synaptic terminal the terminal membrane is depolarised. Which ion channels are opened to allow ions into the terminal to cause the release of neurotransmitter from the synaptic vesicles?
(A) Iodine ion channels.
(B) Calcium ion channels.
(C) Sodium ion channels.
(D) Potassium ion channels.

23 Types of Neurotransmitters

There are five main groups of neurotransmitters.

- **Acetylcholine** is usually excitatory and used by the spinal cord neurons to control muscles. Also used by many neurons in the brain to regulate memory and learning.
- **Amino acids**, e.g. GABA (gamma-aminobutyric acid) which is a major inhibitory neurotransmitter in the brain and glutamate which has an excitatory effect on postsynaptic cells and is involved in the formation of long term memory. Glycine is an amino acid that acts as an inhibitory neurotransmitter used mainly by neurons in the spinal cord.
- **Biogenic amines**, e.g. **dopamine** which is usually inhibitory and produces feelings of pleasure when released by the brain reward system. Dopamine has multiple functions depending on where in the brain it acts. Noradrenaline is usually excitatory but is inhibitory in a few brain areas. **Noradrenaline** acts as a neurotransmitter and a hormone and in the PNS it is part of the fight or flight response while in the brain acts as a neurotransmitter regulating normal brain processes. **Serotonin** is an inhibitory neurotransmitter in pain pathways and is involved in many functions, e.g. mood, appetite and sensory perception.
- **Neuropeptides** are relatively short chains of amino acids, e.g. **substance P** and endorphins. **Endorphins** act as natural analgesics being released in times of stress decreasing pain perception, releasing sex hormones and enhancing the immune response.
- **Gases**, e.g. **nitric oxide** (NO) acts as a local regulator, e.g. in vascular smooth muscle relaxation and vasodilation relaxation of gastrointestinal smooth muscle increasing the ability of the stomach to store food. It also stimulates enzymes to synthesise a second messenger to alter cellular metabolism.

Neurotransmitters of sympathetic and parasympathetic nervous systems

Both the sympathetic and parasympathetic nervous systems are part of the autonomic nervous system which controls involuntary responses. The sympathetic nervous system prepares the body for immediate action, e.g. flight or fight, while the parasympathetic nervous system is involved with rest and digestion. The two systems can be considered to work in opposition, e.g. the sympathetic NS increases heart rate with acetylcholine released from preganglionic neurons and noradrenaline released from postganglionic neurons while the parasympathetic NS decreases heart rate with acetylcholine released from both pre and postganglionic neurons.

The two systems need to work together to control smooth muscle, cardiac muscle and glandular secretion to control involuntary responses.

QUESTIONS

1. Define neurotransmitter.
2. Name some neurotransmitters.
3. Construct a table to summarise the effect of six neurotransmitters.
4. The diagram shows neurotransmitters crossing the synapse.

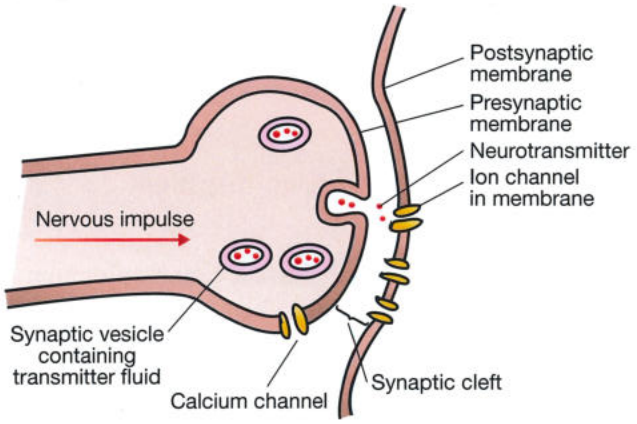


Figure 23.1 Neurotransmitters cross the synapse.

Outline the steps in synaptic transmission.

5. What are the sympathetic and parasympathetic nervous systems?
6. Compare the neurotransmitters of the sympathetic and parasympathetic systems.
7. The diagram shows parts of two neurons and the action of an inhibitory neurotransmitter.

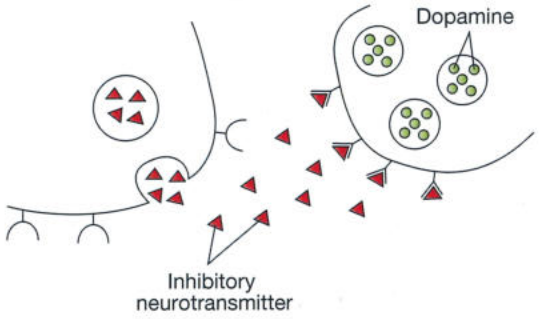


Figure 23.2 Action of an inhibitory neurotransmitter.

Which of the following correctly states what is happening in this diagram?

(A) Dopamine secretion has been blocked.
(B) Dopamine secretion has been stimulated.
(C) Secretion of the inhibitory neurotransmitter is blocked.
(D) A nervous impulse has been initiated.

24 Transmission of a Nervous Impulse

Any cell which can have an action potential generated in it is said to be **excitable** and is able to be made active, e.g. neurons are excitable and will rapidly transmit the signal while muscle cells are excitable and when stimulated will contract.

An action potential can be stimulated in excitable cells. Different sensory neurons are fired by different stimuli. Different excitable cells have different thresholds for firing. The **threshold** is the particular value of the membrane voltage that needs to be reached for an action potential to be started.

All or none law of nerve impulse transmission

The stimulus needs to be above a certain minimum strength, e.g. it often needs to be at least 15 mV more positive than the resting potential for an action potential to begin. If the normal resting potential is −70 mV then in this instance the threshold would be −55 mV.

If depolarisation is below this level an action potential will *not* be initiated. This means that there was not enough Na^+ ions moving into the cell. Once the stimulus is at or above the threshold the full action potential occurs. The magnitude of the action potential is independent of the triggering stimulus. Hence there is a full response or no response at all – the all or none law of nerve impulse transmission.

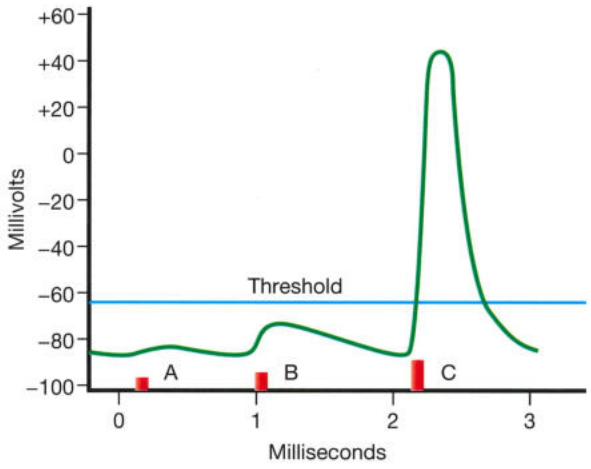


Figure 24.1 Threshold – if the stimulus is too small and there is insufficient depolarisation (e.g. stimulus A and B) then an action potential will not occur. If the stimulus causes a sufficient depolarisation (e.g. stimulus C) then an action potential is generated.

Axon diameter

The diameter of the axon is very important in determining the speed of the axon potential. Resistance to electrical current is inversely proportional to the cross-sectional area of the conductor. Thus a wide axon has less resistance and has a greater speed of the action potential. Giant axons in some arthropods and molluscs that are up to 1 mm in diameter have speeds up to 30 m/s which assists in rapid behavioural responses for defence or feeding.

The axons in most vertebrates are narrow and have other features that increase the speed of transmission of a nervous impulse. Small cells have a large surface area to volume ratio and ion leakage can weaken the membrane. Myelin stops ion leakage, therefore diameter is only important for non-myelinated neurons.

Myelin sheath

Myelin is an electrically insulating material that forms a layer around the axon of a neuron. Myelin is produced by two types of glial cells. Schwann cells supply the myelin for neurons in the peripheral nervous system and oligodendrocytes myelinate the axons in the central nervous system. Myelinated axons appear white in colour and make up 'white matter' in the brain and spinal cord. Myelin mostly consists of lipids, e.g. cholesterol. During development the glia wrap the axon in a spiral fashion in many layers of membrane to form the myelin sheath. Each myelinating cell makes myelin for only one segment of the axon. The myelin sheaths provide electrical insulation of the axon.

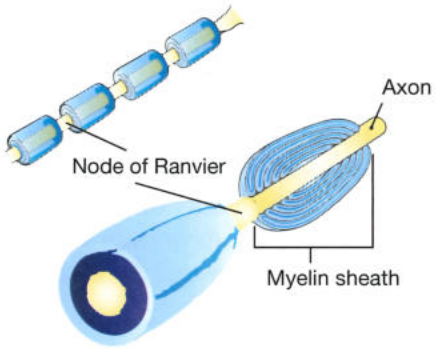


Figure 24.2 Myelin sheath.

Myelin increases the speed of transmission of a nervous impulse. In myelinated axons the voltage gated sodium channels are confined to the gaps in the myelin sheath at the nodes of Ranvier. The nerve impulse jumps from node to node in **saltatory conduction**. In unmyelinated axons the step by step depolarisation is continuous conduction. And is much slower.

The disease multiple sclerosis involves the gradual destruction of the myelin sheaths causing progressive loss of coordination as the nerve impulse becomes disrupted.

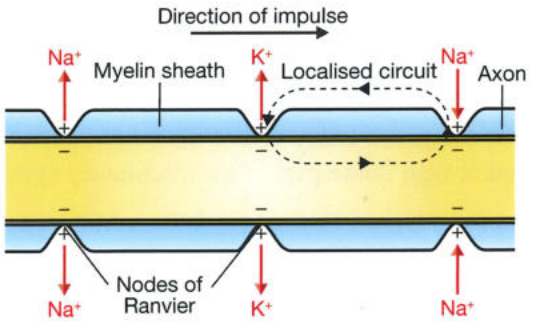


Figure 24.3 Saltatory and continuous conduction.

QUESTIONS

1. Define an excitable cell.
2. What is meant by the threshold of a nervous impulse?
3. State the all or none law of nerve impulse transmission.
4. If a stimulus causes either a full action potential or no action potential, how do we distinguish between a large and small stimulus?
5. There are two main structural features that affect the speed of conduction of an action potential – the diameter of the axon and the presence of myelin sheaths.

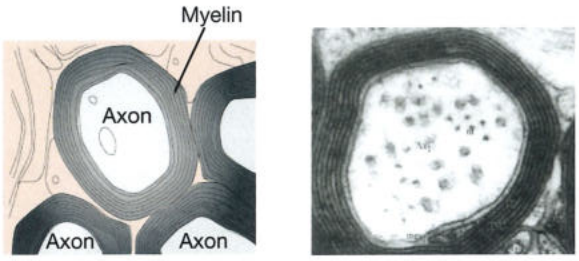


Figure 24.4 Myelin sheath around an axon.

   (a) Suggest why axons with a large diameter can conduct faster action potentials.
   (b) Suggest why axons with myelin sheaths can conduct faster impulses.
6. Multiple sclerosis (MS) is an inflammatory disease that damages the myelin insulation of nerve cells in the brain and spinal cord. T cells identify myelin as non-self and attack causing inflammation and the activation of macrophages and destructive proteins such as cytokines. Scientific research suggests that both genetic and environmental factors affect the development of MS in a person. Discuss why demyelination of neurons will affect the health of a person.
7. Giant squids up to 18 metres long have been found.

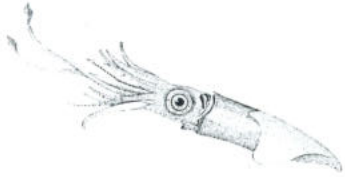


Figure 24.5 Giant squid.

AL Hodgkin and AF Huxley in 1940 experimented on the giant squid axon, worked out the mechanism of action potential and received a Nobel prize for their work. Suggest why they chose the giant squid as their experimental animal.

8. *Mimosa pudica* is a small herb with compound leaves that show rapid leaf movements with leaves folding inwards and drooping when the leaves are touched or shaken. Scientists have identified mechanoreceptor cells that can generate an action potential following a mechanical stimulus. The stimulus causes the cells to expel potassium ions (K^+) and chloride ions (Cl^-) and take up calcium ions (Ca^{2+}). The change in the osmotic gradient causes water to leave the cell and as the action potential passes along the phloem there are quick changes in turgor pressure so that the cells lose turgidity and the petioles droop. The leaves will reopen within a few minutes.
   Discuss how the action potential found in the *M. pudica* plant is similar to an action potential in neurons.
9. The speeds at which impulses travel along a neuron are different in different neurons and for different organisms.

Table 24.1 Speeds of nerve impulses.

Nerve type	Speed
Cat muscle fibre with myelin sheath	80 to 120 m/s
Cat muscle fibre without myelin sheath	0.5 to 2.5 m/s
Crayfish claw 'opener' neuron	8 m/s
Crayfish claw 'closer' neuron	20 m/s
Squid giant axon	35 m/s
Cockroach giant fibre	10 m/s

What can be concluded from this information about impulse speed in different nerve fibres?
   (A) Neurons in invertebrates are not myelinated.
   (B) Cockroaches have faster responses than squids.
   (C) Myelin sheaths increase impulse speeds in cat muscle fibres.
   (D) Myelin sheaths decrease impulse speeds in cat muscle fibres.
10. How do higher calcium levels in an axon terminal affect synaptic vesicles?
   (A) Causes vesicles to degrade.
   (B) Causes vesicles to depolarise.
   (C) Causes vesicles to release potassium.
   (D) Causes vesicles to fuse with terminal membrane.

25 Comparing the Endocrine System and Nervous System

Coordination and communication within the bodies of vertebrates is brought about by both the endocrine system and the nervous system which can work independently of each other and in conjunction.

When excited or alarmed the increase in heart rate and the sensation of 'butterflies in the stomach' are a result of both nervous messages from the brain to various body organs and the secretion of hormones from the adrenal gland.

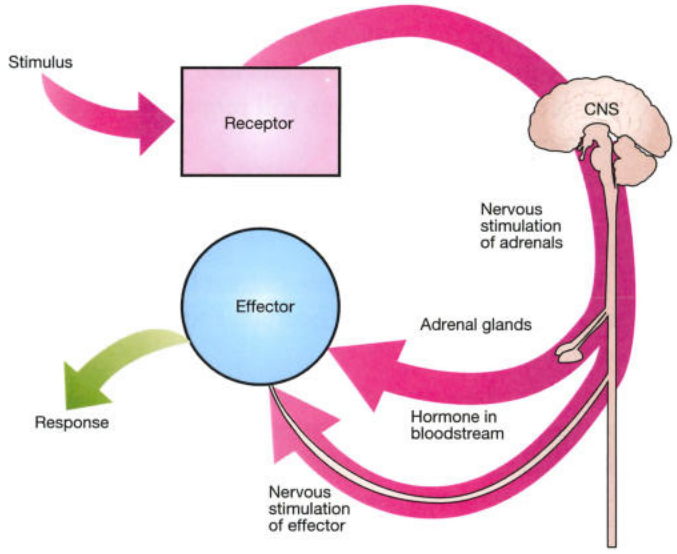


Figure 25.1 Both the nervous system and endocrine system cause 'butterflies in the stomach'.

Nature of message

A nervous impulse is an electrochemical impulse, while the messenger in the endocrine system is a chemical.

Mode of transmission

A nervous impulse is conducted along a neuron with chemical or electrical transmission at the synapse. Hormones travel in the bloodstream and body fluids.

Speed of transmission

Nervous impulses travel quite quickly with the response often occurring very soon after the receptor detects the stimulus and change of state. While the effect of hormones is often delayed as time is needed for the release of hormones from the gland and the transport of the hormone in the bloodstream to the target cell.

Cells contacted

A nervous impulse will only cause a response in muscles or glands that are part of the nervous pathway. While hormones can reach every cell of the body that is contacted or near some part of the circulatory system.

Time for effect

Impulses produce their effect within a few milliseconds while many hormones produce effects over many minutes or over several hours or even longer.

Duration of effect

The effects of nervous stimulation is short lived and usually reversible while the effects of endocrine stimulation are fairly long.

QUESTIONS

1. Construct a table to compare the endocrine and nervous systems for nature of message, mode of transmission, effect, time for effects to occur and duration of effects.
2. The diagram shows the divisions of the nervous system.

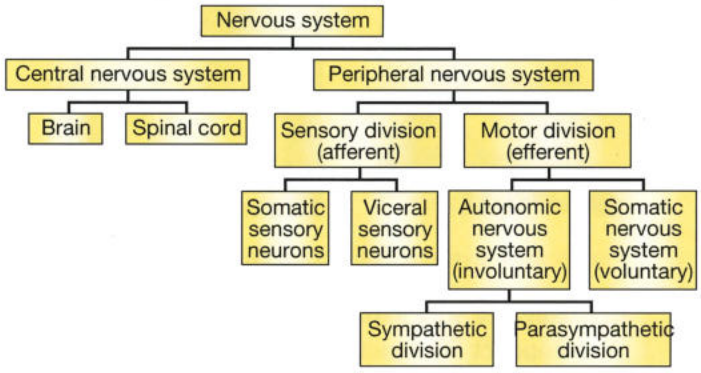


Figure 25.2 Divisions of the nervous system.

Give two examples to show how the endocrine system interacts with the peripheral nervous system.

3. Which of the following correctly distinguishes between a hormonal response and a nervous response?

	Hormonal response	Nervous response
(A)	Electrochemical signal	Chemical signal
(B)	Travels specific path	Travels non-specific path
(C)	Voluntary responses	Non-voluntary response
(D)	Slow acting to take effect	Fast acting to take effect

4. What are the major division of the peripheral nervous system?
   (A) Sympathetic and parasympathetic.
   (B) Brain and spinal cord.
   (C) Efferent and afferent.
   (D) Endocrine and exocrine.

26 Homeostasis

Homeostasis is the process by which organisms maintain a relatively stable internal environment within narrow limits. To maintain homeostasis, organisms need to detect stimuli from both their internal environment and from the external environment. Homeostasis is vital for the survival of all cells as all the chemical processes of life can only function within a narrow range of conditions. It maintains optimal metabolic efficiency.

Homeostasis consists of two stages:

- Detecting changes.
- Responding to the change.

Receptors detect the change from the stable state and the **effector** counteracts the change from the stable state with an appropriate response.

Humans, like other mammals and birds are endotherms and have the greatest ability to control their internal environment. **Endotherms** use metabolic heat to regulate body temperature, e.g. birds and mammals. Their body temperature can vary quite markedly from ambient temperature. In humans, homeostasis maintains body temperature at approximately 37°C, maintains pH of interstitial fluid within a tenth of 7.4 and maintains the sugar concentration of the blood at approximately 0.1%. The homeostatic control is carried out by the nervous and endocrine systems.

The nervous system has receptors in the sense organs to detect changes in the external environment. For example, the retina at the back of the eye has rods and cones that can detect light; the cochlea in the ear has hair cells that detect pressure waves in the cochlear fluid; the tongue has taste buds; the nose has olfactory receptors and the skin has a range of mechanoreceptors, thermoreceptors and pain receptors.

A message being carried by the nervous system usually follows the stimulus-response model.

Stimulus → receptor → sensory neuron → interneuron → motor neuron → effector → response

Figure 26.1 Stimulus-response model.

QUESTIONS

1. Define homeostasis.
2. Explain why a constant internal environment is necessary.
3. Identify the two stages of homeostasis.
4. Identify some variables in 'higher' vertebrates that are controlled by homeostasis.
5. Explain why endotherms require the greatest capacity to control their internal environment.
6. Outline the conditions of homeostasis maintained by the human body.
7. Construct a table to show the sense organ, its receptor and the stimuli it detects.
8. Draw a flow diagram to show the steps in the stimulus-response model.
9. The diagram shows a control system.

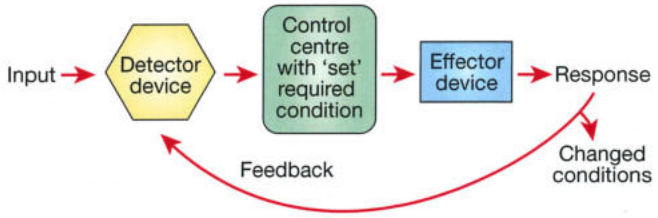


Figure 26.2 Control system.

Outline the functioning of a control system.

10. Define an endotherm.
11. Which of the following best defines homeostasis?
    (A) Inputs such as food and oxygen equal outputs such as faeces and carbon dioxide.
    (B) Metabolic processes are stationary.
    (C) Maintaining constant body temperature.
    (D) Internal environment is kept constant.
12. What are the two stages of homeostasis?
    (A) Detecting and responding to changes.
    (B) Positive and negative feedback.
    (C) Sensory stimuli and nervous impulse.
    (D) Nervous messages and endocrine hormones.
13. Which of the following correctly identifies the stimulus-response pathway?
    (A) Stimulus → control centre → effector → receptor → response
    (B) Stimulus → effector → control centre → receptor → response
    (C) Stimulus → receptor → control centre → effector → response
    (D) Stimulus → effector → receptor → control centre → response
14. Which of the following correctly identifies the stimulus, type of receptor and organ?

	Stimulus	Receptor	Organ
(A)	Light	Photoreceptor	Eye
(B)	Heat, cold	Chemoreceptor	Skin
(C)	Sound	Olfactory receptor	Ear
(D)	Pressure	Thermoreceptor	Nose

15. What is meant by ambient temperature?
    (A) Maximum temperature reached during the day.
    (B) Lowest temperature reached during the day.
    (C) Temperature of surroundings.
    (D) Range of temperatures over one day.

27 Feedback Mechanisms

Homeostasis usually involves a feedback mechanism. In a feedback mechanism the response is monitored.

Situation monitored	→	changes made	→	situation reassessed	→	changes made

Figure 27.1 Feedback mechanism.

The response becomes the new stimulus.

In a **negative feedback system** a specific change results in a response opposite to the initial situation. For example, in humans if the body temperature becomes too low (stimulus), the person may start shivering (response). The shivering generates heat and the body becomes warmer (new stimulus). The feedback mechanism monitors the rise in temperature and will cause the **opposite** to the initial response – it will **stop** the shivering.

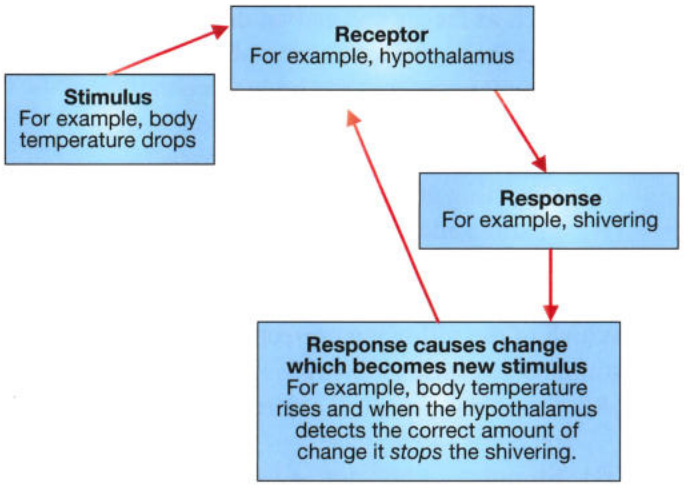


Figure 27.2 Negative feedback mechanism.

In a **positive feedback system** the monitoring will reinforce and amplify the situation, causing more of the same situation to take place. For example, during childbirth oxytocin is released to cause contractions of the uterine muscles and the pressure of the baby's head against the opening of the uterus causes more oxytocin and more contractions. As the baby's head continues to push against the uterus, sensors near the cervix stimulate even more contractions so that the pressure to keep dilating the opening to the uterus increases until the baby is born. Positive feedback is not as common as negative feedback.

QUESTIONS

1. Define feedback mechanism.
2. Compare negative and positive feedback, explaining an example for each.
3. The diagram shows how an air conditioner controls the temperature of a room

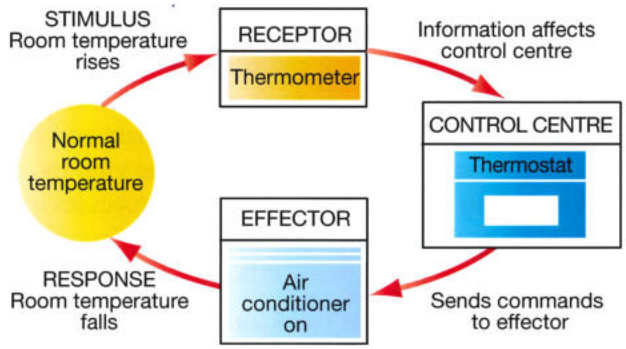


Figure 27.3 How an air conditioner works.

Compare the functioning of the air conditioner with thermoregulation in humans.

4. The diagram shows thermoregulation in humans.

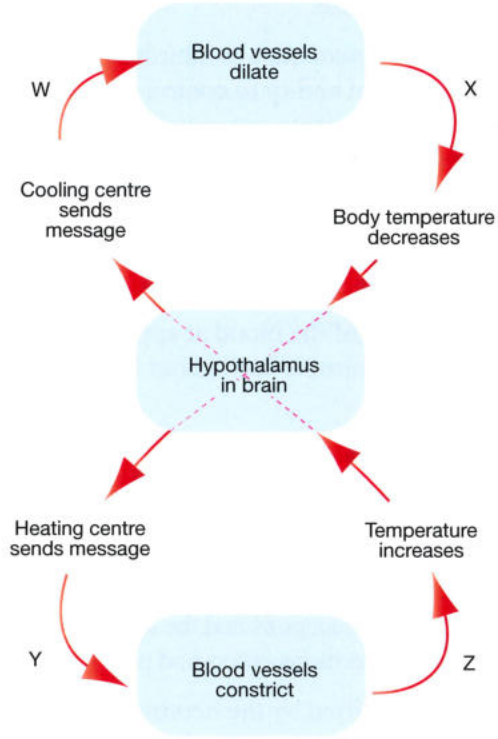


Figure 27.4 Thermoregulation in humans.

Which arrow(s) show negative feedback?

(A) W only. (B) W and Y.
(C) Y and Z. (D) X and Z.

5. When the environmental temperature around a human decreases, receptors in the skin detect the changes and send messages to the hypothalamus which, in turn, initiates a message to the hair erector muscles to contract so the hair can stand up. Which of the following most correctly identifies the type of system involved in this an example?

(A) Positive feedback system.
(B) Negative feedback system.
(C) Enantiostasis.
(D) Stimulus-response system.

28 Homeostasis and Temperature Control

Endotherms maintain a constant internal temperature because of internal processes using metabolic heat, e.g. birds and mammals while **ectotherms** use the energy from their environment and behavioural adaptations to regulate their body temperature, e.g. fish, amphibians, reptiles. The term 'warm blooded' is not used to describe temperature control as many animals such as reptiles are 'warm' when basking in the sun to absorb external heat.

Core body temperature

Deep tissues in the core of the human body maintain a constant temperature ±0.6°C except during an illness. The average core temperature is 37°C. When temperature is measured orally there is a small temperature range that is considered normal depending on the activity and surroundings of the person. A naked person can maintain body core temperature in dry air when exposed to temperatures from 12°C to 54°C. Body temperature measured rectally is usually around 0.5°C higher than an oral measurement.

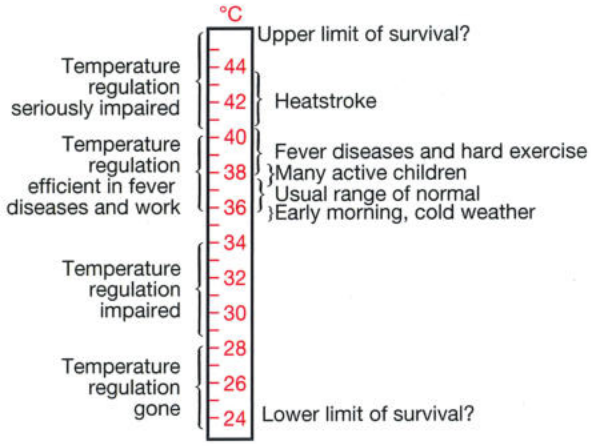


Figure 28.1 Body temperature range.

Core body temperature rises during exercise and will vary if the person is exposed to extreme ambient temperatures. The heat produced by hard exercise can raise core temperature to 39°C to 40°C or drop to 35°C if exposed to extreme cold.

Skin temperature

Skin temperature can rise or fall depending on the surroundings. It is highly important in losing heat to the surroundings to help maintain core body temperature.

Heat transfer

Heat is a form of energy and can be transferred by conduction, convection, radiation or evaporation. **Conduction** is the transfer of heat by direct contact.

Convection is the movement of heated particles in a liquid or gas with warmer particles rising and being replaced by cooler particles.

Radiation is the movement of energy in the form of parts of the electromagnetic spectrum, e.g. infra-red radiation (heat) without direct contact between heat source and receiver. Infra-red heat from the body is electromagnetic radiation with wavelengths 5 to 20 micrometres which radiates in all directions. All objects not at absolute zero radiate this heat. If room temperature is lower than body temperature then more heat is being radiated from the body than radiated to the body.

Evaporation is the transfer of energy to particles in the liquid state so that the liquid becomes a gas which is removed. This is often called **evaporative cooling** as the removal of heat from the body causes cooling.

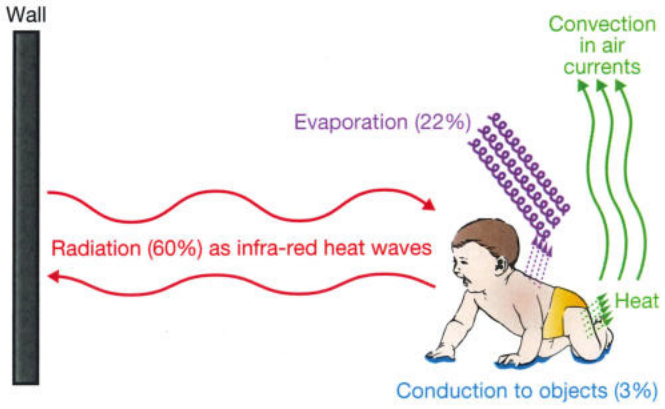


Figure 28.2 Heat transfer.

Heat production

In the body most heat is produced as a **by-product of metabolism**. The amount of heat produced is related to the basal metabolic rate (BMR), muscle activity, additional hormone activity, e.g. thyroxine, growth hormone and additional effects of adrenaline and sympathetic stimulation of cells. Most heat is produced in the brain, liver, heart and exercising skeletal muscles.

Heat loss

Heat loss requires heat to move from its source of production to the skin from where it is transferred to the surroundings. The layer of insulating fat under the skin restricts heat loss as fat conducts heat around one third as easily as other tissues.

The skin gives off heat to passing air currents causing heat loss by **convection**. When skin touches cold objects heat is lost by **conduction**.

Vasoconstriction and vasodilation – vasoconstriction is the narrowing of blood vessels, e.g. arterioles while vasodilation is the widening of blood vessels, e.g. arterioles. Since arterioles provide the main resistance to blood flow a change in the diameter of an arteriole will greatly influence blood pressure and the amount of blood flowing to an area. As a person becomes hotter vasodilation allows more blood to flow to the capillaries in the skin, the person looks flushed and red and there is an increase in the amount of heat lost by radiation, conduction and convection. In vasoconstriction the arteriole narrows restricting blood flow to the skin surface and heat is retained.

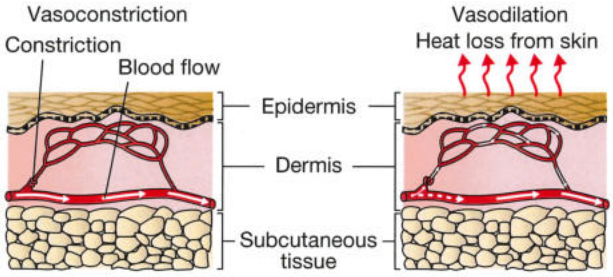


Figure 28.3 Vasoconstriction and vasodilation.

Evaporative cooling occurs when sweat lying on the surface of the skin evaporates taking large amounts of heat from the body to convert the liquid into a gas (evaporation). All sweat glands are simple exocrine glands that consist of a coiled tube that empties into a duct that releases sweat from a pore in the skin. If you exercise too strenuously, stop while covered in sweat and do not put on a jacket, evaporative cooling will continue and your body can become too cold.

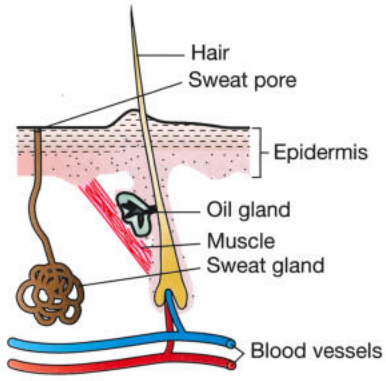


Figure 28.4 Sweat gland.

QUESTIONS

1. Explain why human temperature regulation is called endothermic and not warm blooded.
2. (a) What is the average core body temperature?
   (b) Explain why humans need to maintain body temperature within narrow limits.
3. How can body temperature be measured?
4. In what temperature range can a naked person maintain normal body core temperature?
5. What conditions can cause a change to core body temperature and still be considered normal?
6. Construct a table to summarise the different ways heat can be transferred.
7. Why is the skin important in controlling body temperature?
8. What is evaporative cooling?
9. What causes the production of most heat in the body?
10. Distinguish between vasoconstriction and vasodilation.
11. Explain why vasoconstriction and vasodilation assist thermoregulation.
12. What is the relationship between heat and the electromagnetic spectrum of radiation?
13. The diagram shows the formation of a 'blackhead' which is the beginning stage of acne.

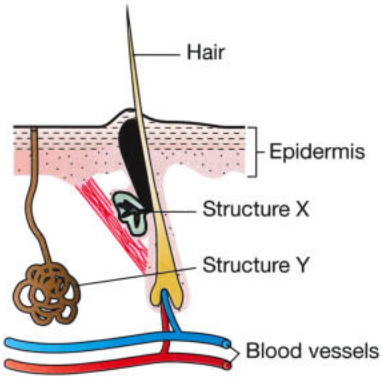


Figure 28.5 Formation of a blackhead.

What causes the blackhead in this diagram to form?

(A) Blockage of a secretory duct from a sweat gland structure X.
(B) Blockage of a secretory duct from a sweat gland structure Y.
(C) Blockage of a secretory duct from an oil gland structure X.
(D) Blockage of a secretory duct from an oil gland structure Y.

14. The diagram shows a section of human skin as seen on a cold day.

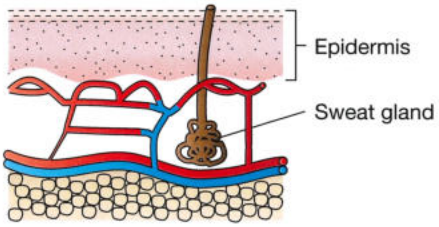


Figure 28.6 Section of human skin.

Which thermoregulation mechanism is occurring in this diagram?

(A) Shivering. (B) Vasodilation.
(C) Vasoconstriction. (D) Evaporative cooling.

29 Modelling Human Thermoregulation

Normal human core body temperature is 37°C. If the thermometer is placed under the tongue the accepted normal range is 36.8°C ±0.4°C. Skin temperature, not like core temperature, can rise and tall with the temperature of the surroundings.

Body temperature is balanced by controlling heat production with heat loss. If core body temperature falls below 35°C a person will suffer hypothermia. The low temperature affects normal metabolism and body functioning. If core body temperature rises above the normal set point, without a change in the temperature set point the person has hyperthermia, e.g. heat stroke. A fever occurs when the core temperature is set higher by the heat control centre and hypothalamus in response to an infection.

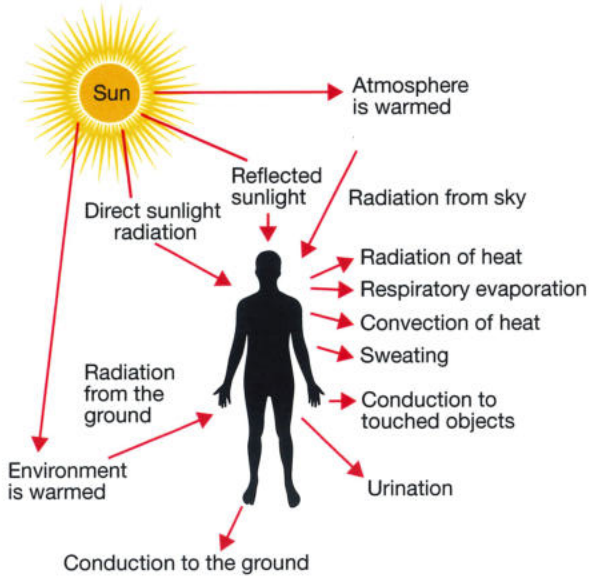


Figure 29.1 Heat transfer with the environment.

Heat is produced as a by-product of metabolism. The rate of heat production depends on the basal chemical reactions occurring in cells and any additional reactions, e.g. due to muscle activity and/or the action of some hormones such as thyroxine and growth hormone. Most heat is produced in the deep organs – the liver, brain and heart with skeletal muscles producing heat during exercise.

Computer models

Computer models of human thermoregulation responses have been developed for use in the design of clothing and environments that aim to protect humans from hyper- and hypothermia. The models have data about heat transfer, perspiration, respiration and blood flow and produce designs of military chemical suits, industrial protective clothing and space suits.

Computer models using data about human thermoregulation are also used to design and monitor environments such as space stations, aircraft, vehicles and buildings.

Computer simulations using different values for required conditions aid designers and enable scientists to study and predict the effects of extreme environments on the human body. This assists in establishing essential safety regulations for people working in extreme environments such as firefighters, pilots, foundry workers and soldiers.

QUESTIONS

1. What is normal human core body temperature?
2. What is hypothermia?
3. Construct a table to show the heat transfer with the environment for a person.
4. What produces heat in the body?
5. What determines the rate of heat production?
6. Distinguish between a fever and hyperthermia.
7. Explain why firefighters need special clothing that needs to be correctly designed.
8. List some clothing designs that have been improved by using computer models of human thermoregulation responses.
9. Why do scientists and engineers need to run computer simulations when designing buildings for people who will be living and working in extreme environments?
10. Firefighters wear overtrousers made of synthetic fabric with high temperature resistance for increased protection. How does the loose fit of the overtrousers over the other pair of trousers aid thermal protection?
    (A) Air between the layers of trousers helps stabilise temperature.
    (B) Convection currents in the layers of trousers remove heat from the firefighter's body.
    (C) The air layer increases radiation of heat from the firefighter.
    (D) Loose clothing increases the mobility of the firefighter.
11. What human responses are considered in the computer modelling for thermoregulation?
    (A) Heat transfer.
    (B) Perspiration rates.
    (C) Blood flow rates.
    (D) All of the above.
12. What will happen if an animal cannot maintain a constant internal environment for a period of time?
    (A) Metabolic efficiency will increase.
    (B) Metabolic efficiency will decrease.
    (C) The animal will go to sleep.
    (D) The animal will immediately die.

30 Homeostasis and Blood Composition

Water is essential for life. Many compounds and substances dissolve in water and most of the chemical reactions in organisms involve solutes dissolved in water. Water has a high specific heat and helps maintain the temperature of organisms. Water also has a high heat of vaporisation which takes heat from an organism in evaporative cooling.

Water is constantly lost from organisms, e.g. urine, sweat and homeostatic mechanisms are needed to maintain water balance. When the water is lost from the body fluids the dissolved solutes become more concentrated.

The blood circulating through blood vessels transports needed materials to cells and transports wastes from the cells to the excretory organs. As substances are delivered or picked up, the composition of blood changes as it travels around the body and the concentration of different solutes is different in different parts of the body. Homeostasis maintains the composition of fluids within the body within specific ranges.

Small intestine

Glucose, mineral salts, vitamins and some of the products of fat digestion pass through the walls of the villi of the small intestine into the capillaries. Most movement into the capillaries is by diffusion (passive transport) as the concentration of the food molecules is higher in the small intestine than in the capillaries. Sometimes active transport is needed if blood sugar levels are high. The capillaries join veins and the digested food molecules are transported in the blood to the liver in the hepatic portal vein.

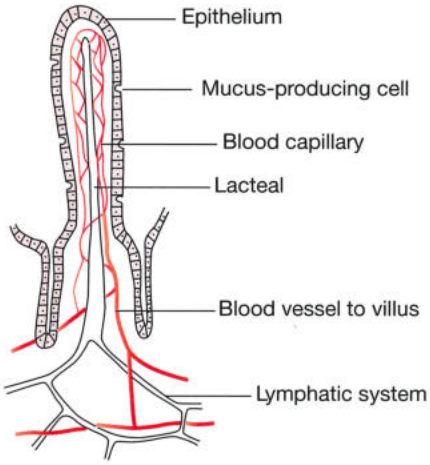


Figure 30.1 Structure of a villus.

A large proportion of the products of lipid digestion, e.g. fatty acids and glycerol move into the lacteal and are transported in the lymphatic system.

Liver

The liver has many functions. It receives the products of digestion and adjusts the blood concentration for each substance. Four functions of the liver include: control of suger levels; deamination; detoxification; and storage of iron.

Control of sugar levels – The liver removes excess glucose from the blood and converts it to glycogen in a process called **glycogenesis**. Glycogen cannot be used by the cells so that when the blood is low in glucose the liver will convert the stored glycogen back to glucose or other monosaccharides in a process called **glycogenolysis**. The glucose is released into the bloodstream.

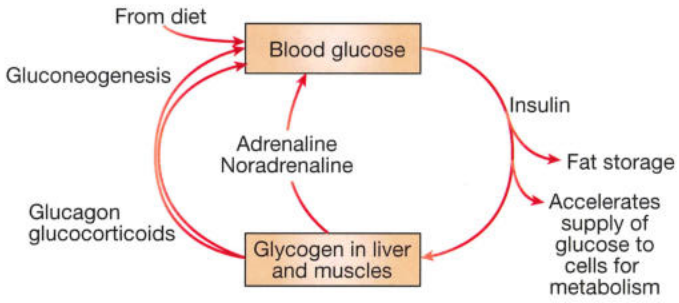


Figure 30.2 Homeostasis of sugar levels in the blood.

Deamination – If there are surplus amino acids, the liver will remove the amino section (NH_2) of the amino acid. This nitrogen-containing section is changed to urea to be excreted by the kidney. The remaining section of the amino acid is converted to glycogen to be stored in the liver.

Detoxification – Many toxic compounds that are ingested and absorbed into the blood are made harmless in the liver.

Storage of iron – Old red blood cells are broken down in the liver and the iron from the haemoglobin is stored in the liver.

Lungs

Oxygen dissolves in the moist lining of the alveoli in the lungs and diffuses into surrounding capillaries, while carbon dioxide in the capillaries diffuses out of the blood and into the alveoli. When air enters the alveolus some of the moisture lining the sac evaporates and saturates the air with water vapour.

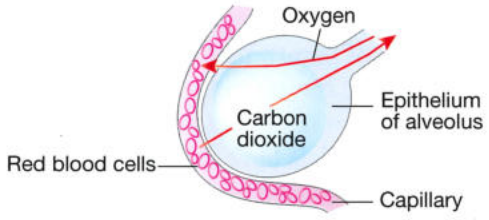


Figure 30.3 Gas exchange in an alveolus.

Kidney

The renal artery delivers blood to the kidneys. The kidneys are bean shaped organs that are at the back of the abdomen and filter the blood removing nitrogenous waste and excess salts. The kidneys are also involved in osmoregulation controlling the amount of water in the blood and thus the osmotic strength of the blood.

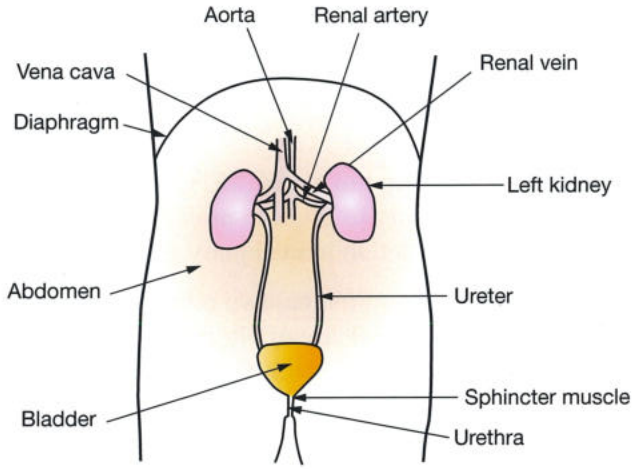


Figure 30.4 Kidney excretory system.

QUESTIONS

1. Compare blood composition in arteries going to the small intestine and veins leaving the small intestine and explain why any differences occur.
2. The table shows the difference between inhaled and exhaled air.

Substance	Inhaled (%)	Exhaled (%)
Oxygen	21	16
Carbon dioxide	0.04	4
Water vapour	Variable	Saturated

   Explain how the difference in the concentration of inhaled and exhaled air reflects changes in blood composition in the lungs.
3. Discuss why both the lungs and the kidneys are considered to be organs of excretion.
4. A healthy diet for a human requires carbohydrates. The main types of carbohydrates are cellulose, starch and sugar.
   (a) How does the body use each of these carbohydrates?
   (b) Which of these carbohydrates cause a change in blood composition?
5. Osmoregulation is concerned with controlling the water balance in the body. How can water be lost from the body?
6. The diagram shows the cycle of a red blood cell around the body.

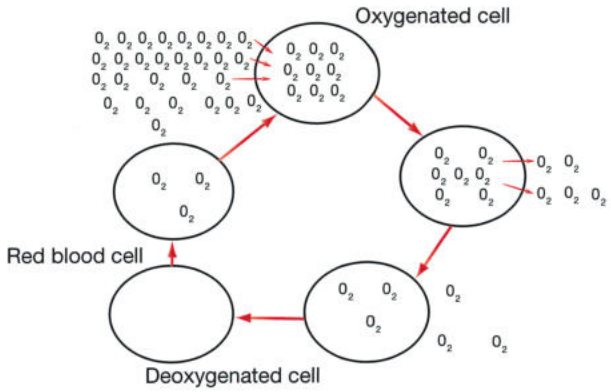


Figure 30.5 Cycle of a red blood cell.

   Identify the section on the diagram that relates to:
   (a) Lungs.
   (b) Body cell.
   (c) Area of high concentration of oxygen.
   (d) Area of low concentration of oxygen.
7. In which blood vessel would you expect to find the highest concentration of each of the following substances?
   (a) Oxygen.
   (b) Carbon dioxide.
   (c) Glucose.
   (d) Urea.
8. Complete the table to show where substances enter the circulatory system and where they travel to.

Substance	From	To
Oxygen		
Carbon dioxide		
Glucose		
Urea		

9. If an athlete was strenuously exercising what change in blood composition would you expect in the muscles of the athlete?
   (A) Increase in oxygen and decrease in carbon dioxide.
   (B) Decrease in oxygen and increase in carbon dioxide.
   (C) Increase in oxygen and increase in carbon dioxide.
   (D) Decrease in oxygen and decrease in carbon dioxide.
10. Hormones are chemicals produced by endocrine glands and travel in the blood to their target organ. Insulin is a hormone produced in the pancreas and stimulates the conversion of glucose to glycogen. What is the target organ for insulin?
    (A) Kidney.
    (B) Lungs.
    (C) Heart.
    (D) Liver.

31 The Hypothalamus and Homeostasis

The hypothalamus is a part of the brain below the thalamus that links to both the nervous system and the endocrine system. The thalamus and hypothalamus form the major coordinating centre of the brain.

The hypothalamus consists of cells with distinct small nuclei that process and sort sensory information and is responsible for activities associated with sex, hunger, thirst, pleasure, pain and anger. The hypothalamus contains the sensors for thermoregulation with warm receptors and cold receptors and it helps regulate arterial blood pressure.

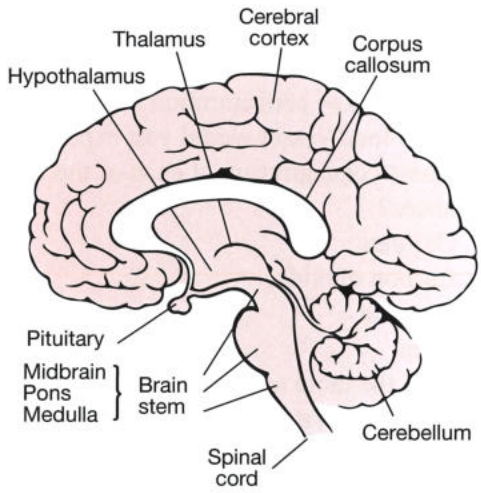


Figure 31.1 Parts of the brain.

Hormones of the hypothalamus

The hypothalamus produces many hormones. For example, it produces two hormones that are released from the hypothalamus and are stored in the posterior pituitary gland:

- **Antidiuretic hormone** (ADH) which increases water absorption into the blood by the kidneys. It is part of a complex feedback system that helps maintain suitable osmolarity of the blood.
- **Oxytocin** which is involved in a variety of processes such as contraction of the uterine muscles during labour, and causing the mammary glands to eject milk during lactation.

The hypothalamus also produces releasing hormones that control the secretion of hormones from the anterior pituitary gland. It produces:

- **Corticotropin releasing hormone** (CRH) which is produced in response to stress causing the anterior pituitary gland to secrete adrenocorticotropic hormone (ACTH) and beta-endorphins. ACTH stimulates the adrenal glands for the synthesis of cortisol, glucocorticoids and mineralocorticoids.
- **Thyrotropin releasing hormone** (TRH) which stimulates the release of thyroid stimulating hormone (TSH) and prolactin from the anterior pituitary gland.
- **Gonadotropin releasing hormone** (GnRH) which stimulates the anterior pituitary to release follicle stimulating hormone (FSH) and luteinising hormone (LH), which work together to ensure the normal functioning of the ovaries and testes.
- **Growth hormone releasing hormone** (GHRH) which stimulates the anterior pituitary gland to secrete growth hormone (GH). It also promotes slow wave sleep (deep sleep).
- **Prolactin releasing hormone** (PRH) which causes the anterior pituitary to produce prolactin which causes milk production in mammary glands.

The hypothalamus also produces two releasing factors that actually inhibit the release of pituitary hormones which are sometimes called **inhibiting hormones**.

- **Dopamine** functions both as a neurotransmitter in the brain affecting sleep, mood, attention and learning and as a chemical messenger, e.g. inhibits the release of prolactin which in turn leads to the inhibition of milk production. Dopamine is a biogenic amine closely related to adrenaline and noradrenaline.
- **Somatostatin** or growth hormone inhibiting hormone affects both neurotransmission and cell proliferation, and inhibits the release of several hormones, e.g. GH, TSH, insulin, glucagon. It is a peptide hormone that decreases both secretion and absorption in the gastrointestinal tract.

The releasing and inhibiting hormones are secreted near capillaries at the base of the hypothalamus. The capillaries drain into portal vessels which subdivide into capillaries in the anterior pituitary gland.

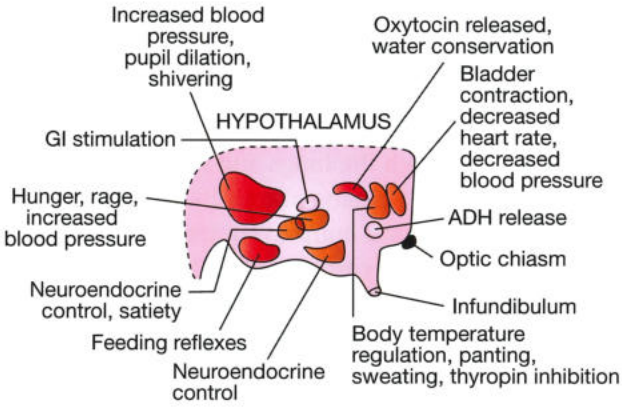


Figure 31.2 Parts of the hypothalamus.

Various nuclei in the hypothalamus secrete releasing hormones and inhibitory hormones into the blood. Each hormone made in the anterior pituitary is secreted in response to a specific releasing hormone. Secretion from the posterior pituitary is controlled by nerve signals from the hypothalamus.

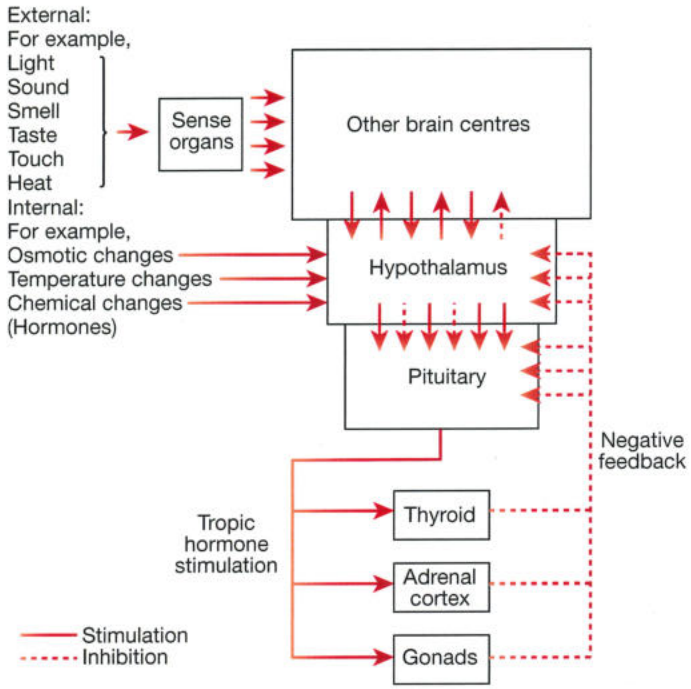


Figure 31.3 Functions of the hypothalamus.

QUESTIONS

1. Where is the hypothalamus in the body?
2. Identify some functions of the hypothalamus.
3. Name two hormones produced by the hypothalamus.
4. Explain why the hypothalamus needs to be supplied with blood vessels.
5. What is the relationship between the hypothalamus and the pituitary gland?
6. Copy and complete the following table to summarise the main hormones of the hypothalamus.

Type	Hormone	Function
Hormone	ADH	
	Oxytocin	
Releasing hormone	CRH	
	TRH	
	GnRH	
	GHRH	
	PRH	
Inhibiting hormone	Dopamine	
	Somatostatin	

7. Identify where releasing and inhibiting hormones are secreted in the hypothalamus.
8. The diagram shows a stimulus-response that involves the hypothalamus.

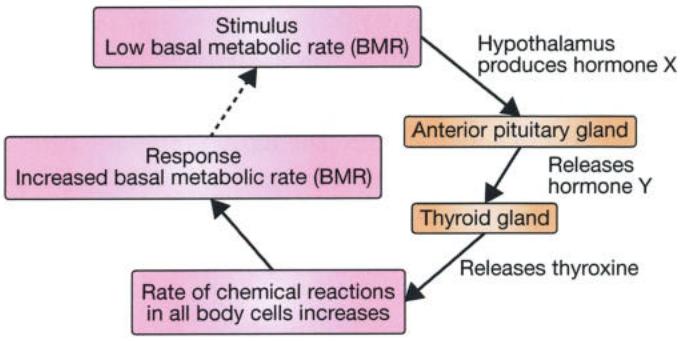


Figure 31.4 Stimulus-response involving hypothalamus.

Identify hormones X and Y.

9. The diagram shows the structure of the skull and brain.

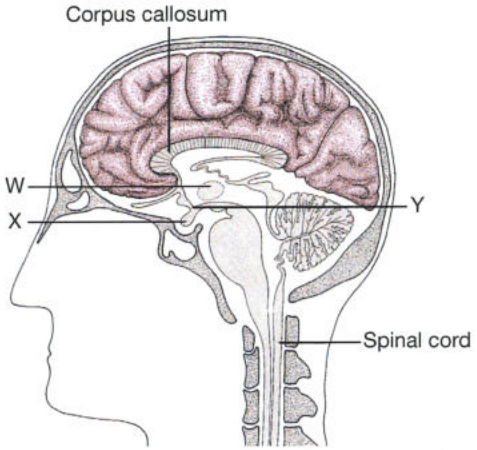


Figure 31.5 Skull and brain.

What are structures W, X and Y?

	W	X	Y
(A)	Pituitary	Hypothalamus	Thalamus
(B)	Hypothalamus	Pituitary	Thalamus
(C)	Thalamus	Pituitary	Hypothalamus
(D)	Pituitary	Thalamus	Hypothalamus

10. Which of the following hormones will cause water absorption into the blood by the kidneys to maintain homeostasis?
    (A) Dopamine.
    (B) Thyrotropin releasing hormone.
    (C) Somatostatin.
    (D) Antidiuretic hormone.
11. In which part of the brain would you find the sensors for heat and cold for thermoregulation?
    (A) Hypothalamus. (B) Cerebrum.
    (C) Medulla oblongata. (D) Cerebral cortex.
12. Which hormone would increase water absorption into the blood by the kidneys?
    (A) Oxytocin. (B) ADH.
    (C) Dopamine. (D) Somatostatin.

32 Changing Metabolic Activity

Metabolism is the sum of all chemical reactions within an organism. Metabolism occurs in all living cells.

The **metabolic rate** is the total amount of energy an animal uses in a unit of time. The **basal metabolic rate** (BMR) is the metabolic rate of a resting, fasting and non-stressed endotherm. The BMR accounts for about 50% to 70% of daily energy expenditure in most sedentary individuals.

Energy is needed for many functions, e.g. for muscle activity, secretion by glands, maintaining membrane potentials of nerve and muscle fibres, synthesis of substances in cells and the absorption of foods from the gastrointestinal tracts. Heat is the end product of nearly every reaction that releases energy in an organism. All energy foods, e.g. carbohydrates, proteins and lipids can be oxidised in cells releasing large amounts of energy.

ATP

Energy for use in the cell is in the form of **adenosine triphosphate** (ATP). ATP is present in the cytoplasm and nucleoplasm of all cells. ATP consists of a ribose sugar, adenine and three phosphate groups that are attached to the ribose. There is a high energy bond holding the last phosphate group to the ribose. When this bond is broken the energy is released and ATP reverts to ADP. There is high synthesis of ATP in the last stages of aerobic respiration in mitochondria.

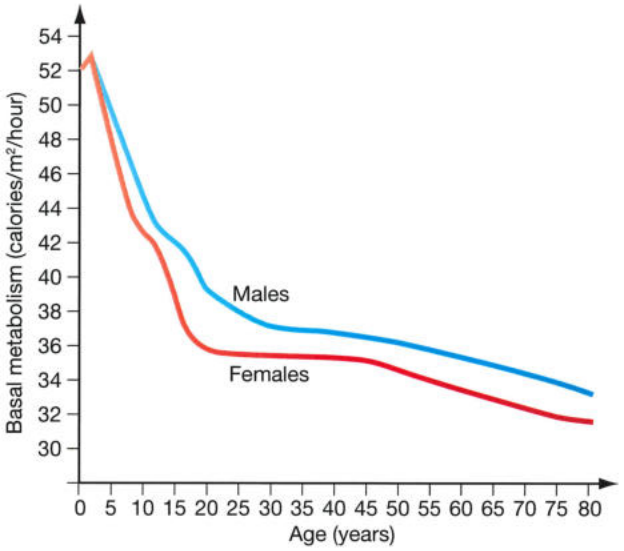


Figure 32.1 Basal metabolic rate for males and females.

The **metabolic rate** can be affected by the following.

- **Thyroxine** secreted by the thyroid gland, e.g. if maximum amounts of thyroxine are secreted the metabolic rate can increase by 50% to 100% or a total lack of thyroxine can decrease the metabolic rate by 40% to 60%.
- **Growth hormone** (GH) is produced and secreted by the anterior pituitary gland. Growth hormone can increase metabolic rate 15% to 20% due to stimulation of cell metabolism. GH acts on a wide range of target tissues with both tropic and non-tropic effects.
- **Sex hormones** – male sex hormone, e.g. testosterone can increase metabolic activity by 10% to 15% due to anabolic pathways that synthesise new proteins increasing skeletal muscle mass. Female sex hormones, e.g. oestrogen and progesterone slightly increase metabolic activity but not significantly.
- **Sleep** – during sleep the metabolic rate drops about 10% to 15% below the normal rate due to less activity of the central nervous system and decreased skeletal muscle activity.
- **Thermogenic effect of food** – after eating a meal metabolic rate increases for digestion, absorption and storage of food which require energy and involve the release of energy.

QUESTIONS

1. What is metabolism?
2. What is the metabolic rate?
3. What is the basal metabolic rate?
4. List some processes in humans that require energy.
5. What is the main form of energy for use in cells?
6. Explain how thyroxine affects metabolic rate.
7. How does growth hormone affect metabolic rate?
8. Suggest why male sex hormones affect metabolic rate.
9. (a) Describe the structure of ATP.
   (b) What happens when ATP releases energy?
10. When is there a high synthesis of ATP?
11. What happens during sleep?
12. What is meant by the thermogenic effect of food?
13. What is the main form of energy for use in cells?
    (A) ATP (B) ADP
    (C) Glucose (D) Glycogen
14. What is the sum of all chemical reactions within a body?
    (A) Metabolic rate. (B) Basal metabolic rate.
    (C) Respiration. (D) Metabolism.
15. How would an increase in thyroxine affect cells?
    (A) Decrease metabolism 40% to 60%.
    (B) Increase metabolism 50% to 100%.
    (C) Induce cell division.
    (D) Will not affect cells.

33 Homeostasis, Excretion and Osmoregulation

For proper cell functioning the extracellular fluid must maintain a relatively constant concentration of water, electrolytes and other solutes. **Osmoregulation** refers to the way organisms regulate solute concentrations and balance the gain and loss of water.

The **kidney** is the main excretory organ for most vertebrates. The kidney is bean shaped in mammals and has the dual functions of excretion and osmoregulation. In humans the kidneys are on the posterior wall of the abdomen and in adults each weighs around 150 grams and is about the size of a clenched fist.

The blood flow to the kidneys is unusual as it has two capillary beds which are positioned in series. This helps regulate the hydrostatic pressures needed for rapid filtration.

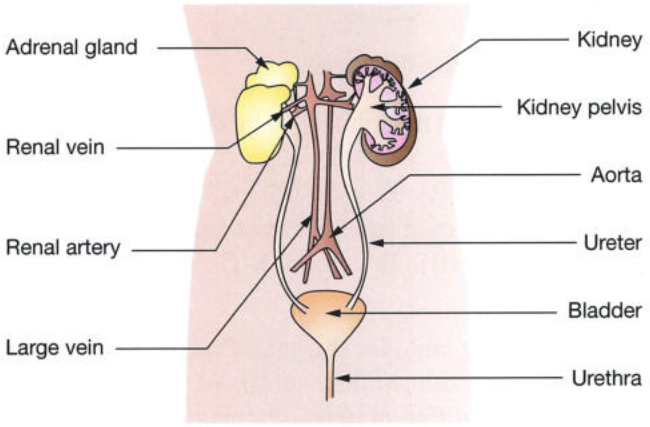


Figure 33.1 Excretory system of mammals.

Nephron

The functional unit of the kidney is the **nephron**. Each nephron has three major parts – Bowman's capsule, the glomerulus and a long thin tubule. Urine produced by normal kidneys can be four times more concentrated than the waste products in blood. The kidney needs to be able to concentrate wastes to reduce water loss and hence dehydration. Excretion from the kidney involves three main stages – filtration, reabsorption and secretion.

Filtration occurs in the glomerulus, where blood pressure forces water, salts, and other small solutes from the blood into Bowman's capsule. The larger blood components, like red blood cells and proteins, are too large to filter through the tiny pores in Bowman's capsule.

Reabsorption occurs in the tubules (proximal and distal tubules and the loop of Henle) and controls the concentration of substances in body fluid, e.g. amino acids, glucose, vitamins; some water and some salts are reabsorbed. Reabsorption involves both passive and active transport.

Most glucose, amino acids and salt (sodium chloride) move by active transport using energy, while water moves passively by osmosis. Salt moving out of the tubules changes the concentration gradient and the water will passively follow.

Secretion occurs in the proximal and distal tubules and adds substances, e.g. H^+, K^+, NH_4^+ and some drugs, to the filtrate. It is highly selective. It may involve either active or passive transport, e.g. movement of H^+ involves active transport and maintains the pH of blood and urine, while secretion of NH_4^+ stops the pH of the filtrate from becoming too low.

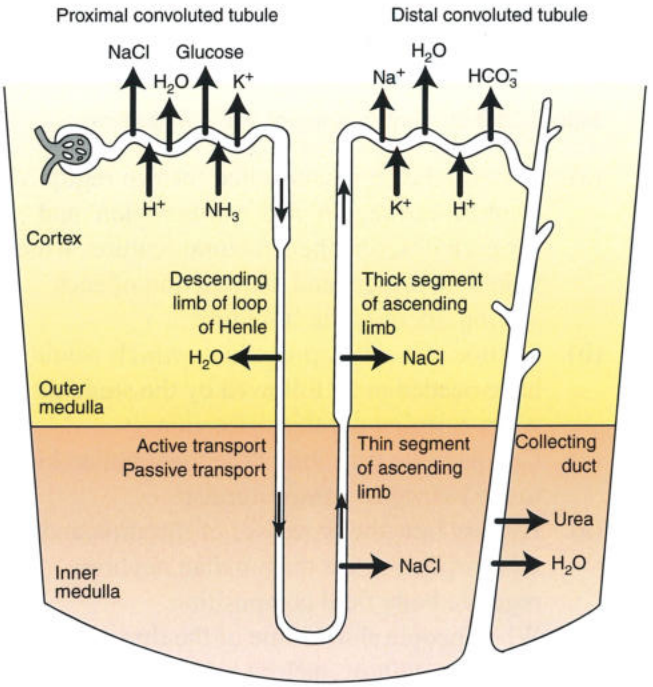


Figure 33.2 Functional activities in the nephron.

QUESTIONS

1. Define osmoregulation.
2. Define the terms.

Nephron	Filtration	Reabsorption
Secretion	Filtrate	Osmoregulation
Solute	Ureter	Urethra

3. Compare active and passive transport using examples of these processes occurring in the mammalian kidney.
4. The kidney is one of the main excretory organs of the body. One of its functions is homeostasis.
   (a) Define homeostasis.
   (b) Draw a flow diagram to show the pathway of urea from the blood in the renal artery to leaving the body through the urethra.
   (c) Predict how the kidney will respond if a person drinks no fluids on a very hot day.
   (d) Explain why the presence of red blood cells in urine indicates a serious problem.

5. Students dissected a mammalian kidney and made the following drawing.

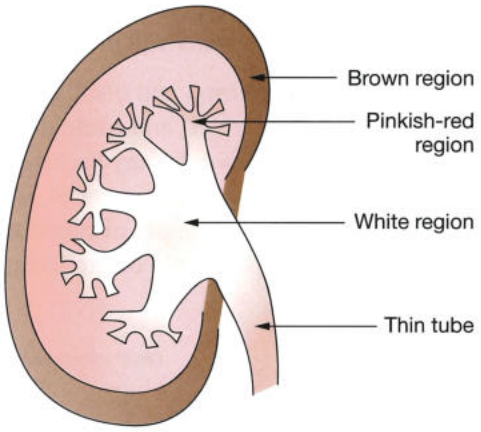


Figure 33.3 Student drawing of a dissected kidney.

(a) Identify the regions labelled 'brown region', 'pinkish-red region' and 'white region' and for each describe the structural features which would be present and the function of each section. Identify the 'thin tube'.
(b) Outline two safety procedures which would have needed to be followed by the students when carrying out this dissection.
(c) Compare the function of the mammalian kidney to the kidney of freshwater fish.

6. (a) Explain how the processes of filtration and reabsorption in the mammalian nephron regulate body fluid composition.
(b) When people show some of the first symptoms of kidney failure, such as pale and yellow complexion, fatigueness, shortness of breath, frequent urination at night, doctors will often advise their patient to change their diet restricting the intake of proteins such as poultry, seafood and soya products. However, they have to be careful not to reduce this intake too low. Explain why this measure slows down the rate of renal failure and why the protein intake cannot become too low.

7. Explain why the removal of metabolic wastes, such as nitrogenous wastes, is essential for efficient metabolic activity.

8. Explain why the processes of diffusion and osmosis are inadequate for excretion in mammals.

9. What causes filtration from the glomerulus into Bowman's capsule?
(A) High blood pressure due to the narrowing of the arteriole diameter.
(B) Difference in protein concentration causes water to move by osmosis out of the glomerulus.
(C) Diffusion of small molecules into Bowman's capsule.
(D) All of the above.

10. Mammalian kidneys excrete toxic wastes. What is another function of their kidneys?
(A) Regulate salt concentrations.
(B) Make urea.
(C) Break down and recycle old red blood cells.
(D) Deamination of proteins.

Use the diagram for the next TWO questions. The diagram shows the structure of a nephron in the kidney.

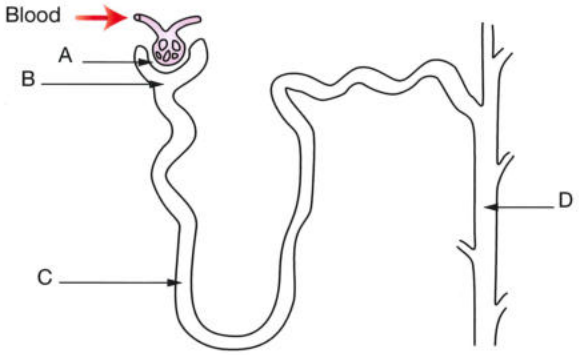


Figure 33.4 Nephron in the kidney.

11. Which point is most likely to be the site of active transport?
(A) A (B) B (C) C (D) D

12. Which site would have the least amount of glucose?
(A) A (B) B (C) C (D) D

Use the following table for the next TWO questions. The table shows the results of the analysis of samples of liquid taken from three different parts of a mammalian nephron.

Substance	Part A (%)	Part B (%)	Part C (%)
Water	91.35	99.00	96.20
Glucose	0.25	0.25	0.00
Proteins	8.15	0.00	0.00
Urea	0.25	0.75	3.80

13. Which sample(s) is most likely to have been taken from the collecting duct of the nephron?
(A) Sample A (B) Sample C
(C) Sample A and C (D) Sample B

14. Which of the following identifies the most likely process that is responsible for the difference in concentration of glucose between parts B and C?
(A) Filtration (B) Mitosis
(C) Reabsorption (D) Secretion

15. What is the correct link between metabolic waste and its organ of excretion for humans?
(A) Urea from the liver.
(B) Urea from the kidney.
(C) Carbon dioxide from the skin.
(D) Salt from the lungs.

34 Osmoreceptors and Water Balance

An **osmoreceptor** is a specialised sensory receptor that detects changes in osmotic pressure. Osmoreceptors are mainly found in the hypothalamus of endotherms. Osmoreceptors are very sensitive and can respond to a very small change, e.g. 1% to 2% increase in sodium ion concentration.

Water intake can greatly vary and there are **volume receptors** (low pressure **baroreceptors**), e.g. in walls of the right atria and large arteries and veins that respond to pressure changes.

The excretion of water and solutes operate in a way independently of each other when forming urine as concentrated urine has solutes in excess of water while dilute urine has water in excess of solutes.

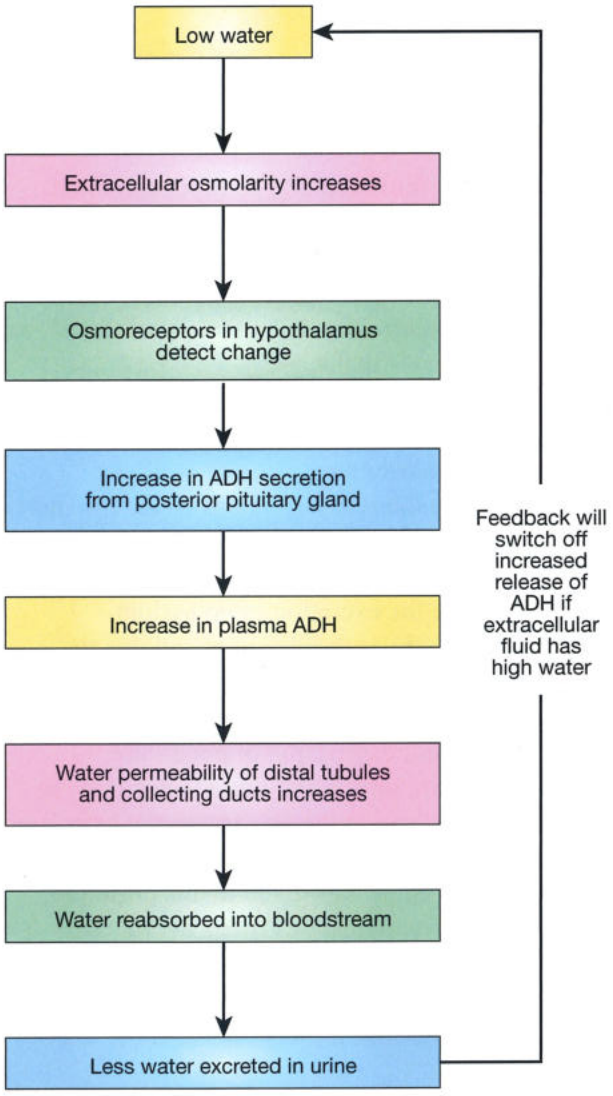


Figure 34.1 Release of ADH and feedback.

Release of ADH

When osmoreceptors in the hypothalamus detect an increase in the sodium ion concentration in the plasma (extracellular fluid osmolarity) above the normal range due to a lack of water, they send nerve signals to the posterior pituitary gland to release the hormone ADH (**antidiuretic hormone** or vasopressin). ADH is produced by the hypothalamus and stored by the posterior pituitary gland.

ADH enters the bloodstream and travels to the kidney where it changes the water permeability of the convoluted tubules and collecting duct walls so that more water is reabsorbed into the bloodstream from the ducts. Its also increases the permeability of the collecting duct walls to urea which diffuses in from the bloodstream. These two effects cause the urine concentration to increase. Thus ADH causes water to be conserved while sodium and other solutes continue to be excreted in the urine.

The release of ADH is also controlled by a decrease in blood pressure and/or blood volume. When arterial baroreceptors detect a drop in blood pressure and blood volume, e.g. due to haemorrhage then ADH is secreted and water is reabsorbed by the kidneys to help restore blood volume and pressure to normal.

If there is abnormal secretion of ADH, e.g. either too much or too little ADH then the kidneys will have difficulty concentrating or diluting urine correctly. The release of large volumes of dilute urine can lead to dehydration.

Thirst mechanism

The thirst mechanism is regulated by fluid intake and the osmoreceptor-ADH mechanism.

As water is lost from the body, e.g. in sweat, urine, faeces and exhaled breath the extracellular fluid becomes more concentrated and the reduction in plasma volume causes a feeling of thirst.

There is a **thirst centre** in the hypothalamus with osmoreceptors that activate the thirst mechanism.

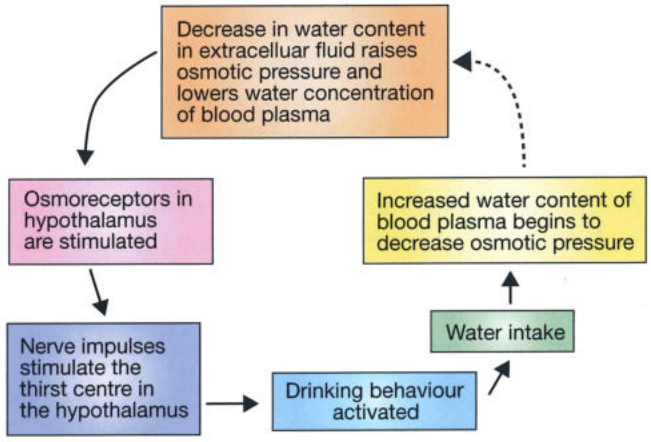


Figure 34.2 Thirst mechanism.

Release of aldosterone

Aldosterone is a hormone produced by the adrenal cortex in the adrenal gland, which is found just above the kidneys. Aldosterone causes the convoluted tubules to reabsorb more sodium ions and decrease the reabsorption of potassium ions. Because of osmosis, this leads to an increased absorption of chloride ions and water from the urine as they follow the sodium ions. This results in a rise in blood volume and blood pressure.

ADH and aldosterone work together to increase the water reabsorption, but they can solve slightly different problems. ADH release is a response to an increase in salt concentration in the blood, an increase in blood osmolarity while aldosterone is released as a response to a loss of volume in the blood or when salt and water are both lost from the blood.

Hormone replacement therapy

Hormone replacement therapy is a treatment given when a gland is not producing enough of a particular hormone. In Addison's disease when there is insufficient secretion of aldosterone there is insufficient secretion of potassium causing the extracellular fluid to then have dangerously high levels of potassium. Fludrocortisone is a synthetic hormone which usually has to be taken orally once a day for patients who need aldosterone replacement. These patients may be suffering from diabetes, a kidney disease or have a hereditary condition.

QUESTIONS

1. Define an osmoreceptor.
2. Where are most osmoreceptors in endotherms?
3. What are baroreceptors?
4. Where are baroreceptors located?
5. How can the extracellular fluid osmolarity increase?
6. Outline why the excretion of water and solutes operate somewhat independently of each other.
7. (a) Where is ADH produced?
   (b) Where is ADH stored?
8. What is the thirst mechanism?
9. Where is aldosterone produced?
10. Outline why the kidneys must continually excrete some fluid even in a dehydrated person.
11. What is replacement hormone therapy?
12. The functioning of the kidney is controlled by two hormones – aldosterone and antidiuretic hormone. Compare the role of these two hormones.
13. Outline the symptoms you would expect to find in a person who will require aldosterone replacement therapy.
14. People suffering from Addison's disease appear weak and thin, feel lethargic and often have a salt craving. Some require hormone replacement therapy of aldosterone. Explain how aldosterone would help these people.
15. The release of ADH follows a negative feedback pathway.

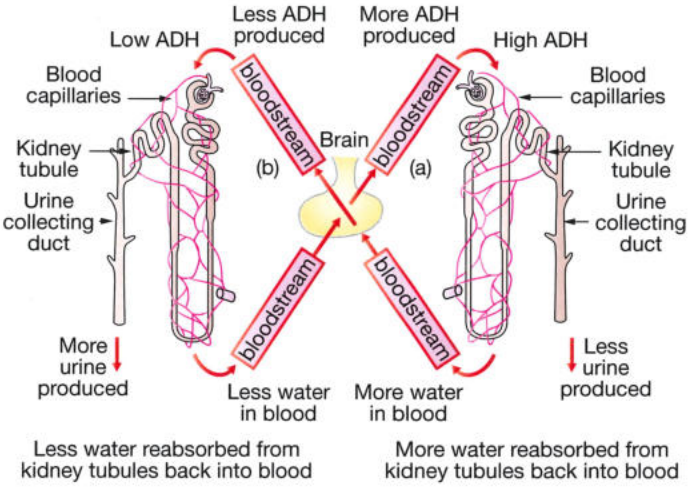


Figure 34.3 Negative feedback for ADH.

(a) Identify the receptors, control centre and effector which control the release of ADH.
(b) Outline the conditions which would cause very little ADH to be released.
(c) Describe how alcohol affects water levels and ADH levels in the body.

16. Aldosterone is a steroid hormone produced by the cortex of the adrenal gland. What is the main function of aldosterone?
   (A) Increase the reabsorption of water and increase the excretion of sodium.
   (B) Increase the reabsorption of sodium and increase the excretion of potassium.
   (C) Increase the reabsorption of sodium and increase the excretion of water.
   (D) Decrease the reabsorption of water and decrease the excretion of potassium.
17. High concentrations of antidiuretic hormone in many species lead to the constriction of arterioles and hence lead to increased arterial pressure. For this reason, what name was given to this hormone?
   (A) Osmoconstrictus.
   (B) Aldosterone.
   (C) Vasopressin.
   (D) Constricturetic protein.

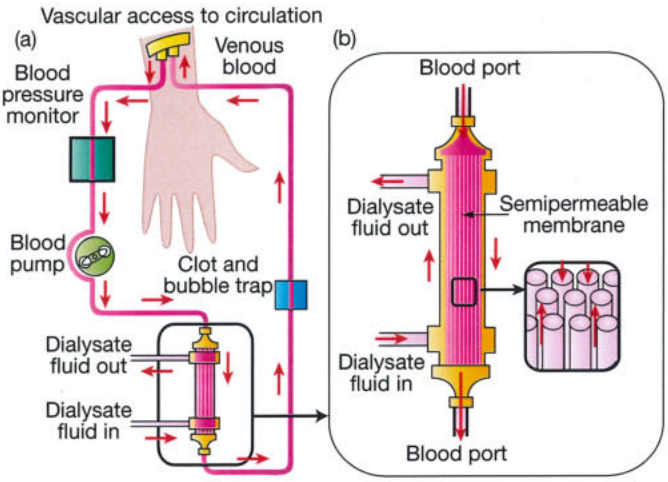
35 Renal Dialysis

Dialysis is a process that removes waste products from the blood.

Acute dialysis is needed in emergency situations; when potassium levels are high there is fluid overload and pulmonary oedema – fluid accumulates in the lungs. Acute dialysis is also required if there is a need to remove medications or toxins from the blood.

Chronic dialysis is needed for long term renal failure.

Haemodialysis is the most common method of dialysis and uses dialysers or artificial kidneys. Shunts are inserted into an artery and a vein so that blood can flow from the person into the dialysis machine, be filtered, and then flow back to the person and into their vein.



Haemodialysis system (a) Blood from an artery is pumped into (b) a dialyser, where it flows through the cellophane tubes that act as a semipermeable membrane (inset). The dialysate fluid has the same chemical composition as blood, except that it is free of urea and waste products. The blood flows down the tubes, bathed in the dialysate flowing in the reverse direction. Waste products in the blood diffuse through the semipermeable membrane into the dialysate.

Figure 35.1 Haemodialysis system.

In the artificial kidney, toxins and wastes move by diffusion and water moves by osmosis and ultrafiltration across an artificial membrane. The pores in the artificial membrane allow diffusion of small ions but prevent large plasma proteins leaving the blood. The body's buffer system, which maintains homeostasis, is maintained using a dialysate, which is a solution of all the extracellular ions such as hydrogen carbonate and acetate. Heparin, an anticoagulant, is administered to stop the blood clotting in the dialysis tubing.

QUESTIONS

1. (a) What is oedema?
   (b) Describe pulmonary oedema.
2. Outline some situations when renal dialysis may be needed.
3. Haemodialysis is the most common method of dialysis.
   (a) What is dialysis?
   (b) Describe one advantage and one disadvantage of haemodialysis.
   (c) Compare the process of renal dialysis with the function of the kidney.
   (d) During a football match, a player received a kick in his back. He began to pass urine with blood. Suggest one way his kidney could be damaged.
4. Explain why heparin is administered.
5. Outline how the body's buffer system is maintained during dialysis.
6. Urea, uric acid, potassium ions, phosphate ions and sulfate ions move into the dialysis fluid, while hydrogen carbonate ions and glucose pass into the blood from the dialysis fluid. Outline the importance of the direction of movement of these substances.
7. During dialysis, many small molecules can easily diffuse through a synthetic membrane. In what way does a cell membrane behave differently to synthetic membranes?
   (A) Cell membranes can transport large molecules by active transport.
   (B) Cell membranes prevent the movement of water against a concentration gradient.
   (C) Cell membranes allow the passive transport of very large molecules into cells.
   (D) Cell membranes maintain a permanent rigid structure with molecules locked into place.
8. The diagram shows the basic structure of the kidney dialysis machine.

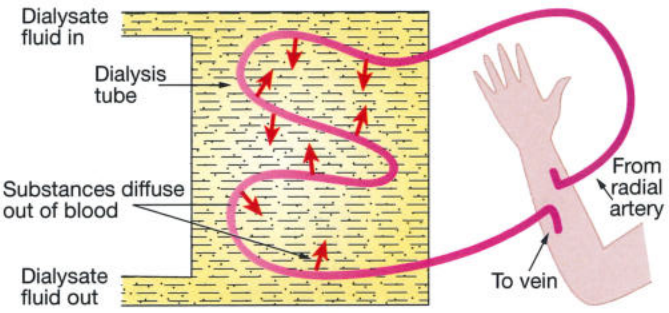


Figure 35.2 Kidney dialysis.

What comparison can be made between the dialysis machine and the human kidney?
(A) Both filter the same substances from blood.
(B) The kidney machine acts like Bowman's capsule, the renal tubules and collecting ducts.
(C) The kidney machine acts like Bowman's capsule and the renal tubules.
(D) Only the kidney has reabsorption.

36 Homeostasis and Gas Concentrations

Oxygen and carbon dioxide need to be transported in the body with concentrations kept within narrow limits to maintain homeostasis. Nitrogen is also transported around the body in the plasma and has no physiological function.

Gas exchange is the uptake of molecular oxygen from the environment and the release of carbon dioxide to the environment. Gas exchange occurs in the lungs between the alveoli and the blood with oxygen diffusing into the blood and carbon dioxide diffusing from the blood into the alveoli. Gas exchange also occurs between blood in the capillaries and body cells with oxygen leaving the blood and entering body cells and carbon dioxide leaving body cells and entering the capillaries. Gas exchange relies on a difference in the partial pressure across the cell membrane. Gas exchange occurs across the thin walls of capillaries. The walls of arteries are too thick for easy gas exchange.

Table 36.1 Partial pressures of gases.

Gas	Alveolar air kPa	Deoxygenated blood kPa	Oxygenated blood kPa
Oxygen	13.3	5.3	13.3
Carbon dioxide	5.3	5.8	5.3
Nitrogen and inert gases	76.4	76.4	76.4
Water vapour	6.3		

When blood arrives at tissues it has a higher partial pressure oxygen (P_{O_2}) and a lower partial pressure carbon dioxide (P_{CO_2}) than the tissues. Pressure is directly proportional to the concentration of the gas molecules. The difference in pressure creates a concentration gradient that causes diffusion and gas exchange.

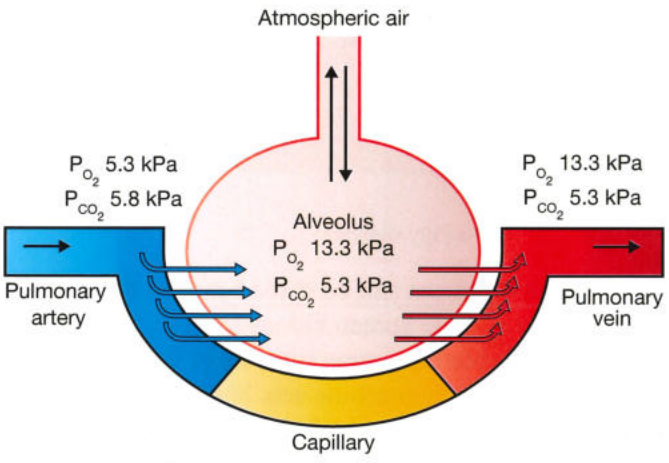


Figure 36.1 Gas exchange.

The **rate of diffusion** and gas exchange depends on thickness of the membrane, the surface area of the membrane, the blood supply and the concentration gradient (or pressure difference across the membrane).

Carbon dioxide

Carbon dioxide (CO_2) is a waste product of respiration and needs to be excreted. In water carbon dioxide forms carbonic acid (H_2CO_3). Large amounts of carbonic acid which forms hydrogen ions (H^+) and hydrogen carbonate ions (HCO_3^-) which lowers the pH of the blood and affects the ability of enzymes to function efficiently.

$$CO_2 + H_2O \rightleftharpoons H_2CO_3 \rightleftharpoons HCO_3^- + H^+$$

Figure 36.2 Carbon dioxide reacts with water.

In the blood carbon dioxide is 7% dissolved in plasma, 23% as carbaminohaemoglobin combined with haemoglobin in the red blood cells and 70% as the hydrogen carbonate ion in plasma.

Carbon dioxide can easily diffuse across cell membranes to enter/leave cells by diffusion.

Oxygen

Oxygen can easily diffuse across cell membranes to enter/leave cells and combines with haemoglobin in red blood cells to form **oxyhaemoglobin** with 98% of oxygen being transported as oxyhaemoglobin in the bloodstream. Oxygen also travels in solution in **plasma** water (2% dissolved in plasma).

Oxyhaemoglobin is unstable and easily dissociates to release the oxygen.

Controlling gas concentrations

Gas concentrations are controlled by balancing the intake of oxygen and the removal of carbon dioxide via the lungs in breathing. **Breathing** is an involuntary action though there is some conscious control, e.g. when speaking, singing, fearful, crying or holding your breath for a limited period of time.

The main control of gas concentration is by **medulla oblongata** which has the **respiratory centre** with neurons that control the rate and depth of breathing.

Chemoreceptors on the surface of the medulla oblongata called **central chemoreceptors** are surrounded by cerebrospinal fluid and detect an increase in arterial P_{O_2}. An increase in P_{CO_2} causes the central chemoreceptors to stimulate the respiratory centre which in turn stimulates an increase in lung ventilation, e.g. contraction of the diaphragm for increased inhalation, stimulation of the phrenic nerve to stimulate the diaphragm and stimulation of the intercostal muscles between the ribs to expand the rib cage for increased inhalation.

Peripheral chemoreceptors are in the arch of the aorta and carotid bodies and detect very small increases in arterial P_{CO_2}. An increase in P_{CO_2} causes peripheral chemoreceptors to send impulses to the respiratory centre which increases the rate and depth of breathing. These receptors also detect an increase in blood acidity (lower pH) which also increases breathing rate and increases CO_2 excretion.

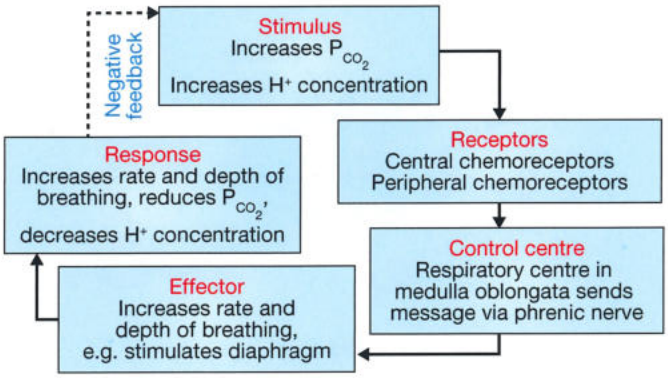


Figure 36.3 Control of breathing.

Temperature can also affect breathing, e.g. a person with a fever will have an increased metabolic rate and increased respiration while metabolism is lower when a person has hypothermia.

The body can **alter diffusing capacity for oxygen** to adjust for changing needs such as when exercising or at rest. To increase the oxygen diffusing capacity, e.g. with increased pulmonary blood flow and faster breathing rate during exercise, the pulmonary capillaries around the alveoli have additional dilation and dormant pulmonary capillaries become open.

The **autonomic nervous system** controls breathing and gas concentrations with the **sympathetic nervous system** increasing ventilation rates, increasing cardiac output and increasing blood pressure while the **parasympathetic system** decreases ventilation rate, decreases cardiac output and reduces blood pressure.

QUESTIONS

1. Which gases are transported in the blood?
2. (a) Define gas exchange.
   (b) Where does gas exchange occur?
3. Why is most gas exchange across capillary walls and not artery walls?
4. How is partial pressure of gases related to gas concentrations?
5. (a) What is the difference in partial pressures of oxygen and carbon dioxide when blood arrives in the tissues?
   (b) What is a result of this difference in gas concentration?
6. Identify the main factors that affect the rate of diffusion and gas exchange.
7. Some pulmonary diseases cause fibrosis of the lungs with a thickening of some respiratory membranes. How will this affect diffusion and gas exchange?
8. When a person has panacinar emphysema the walls between adjacent alveoli break down so that the air sacs merge. Why will this affect diffusion and gas exchange?
9. Explain why carbon dioxide needs to be excreted.
10. How does carbon dioxide travel in the blood?
11. How does oxygen travel in the blood?
12. Is breathing a voluntary or involuntary action? Explain your reasoning.
13. Where is the main control centre for gas concentrations?
14. Identify the receptors involved in detecting gas concentrations.
15. Outline what happens if an increase in P_{CO_2} is detected.
16. Use an example to show how body temperature can affect breathing.
17. Outline how the body can alter the diffusing capacity for oxygen, e.g. during exercise.

Use the graph which shows the oxygen dissociation curve for haemoglobin for the next TWO questions.

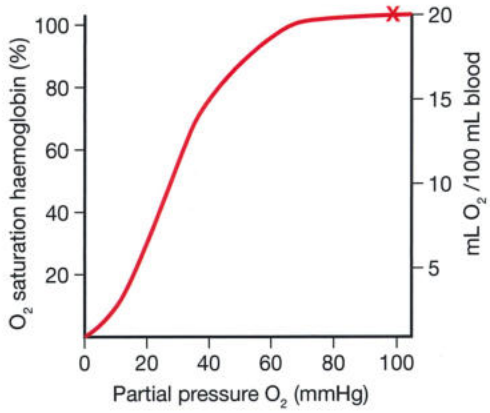


Figure 36.4 Oxygen dissociation curve for haemoglobin.

18. If the partial pressure in tissues at rest is usually around 40 mmHg, what is the per cent oxygen saturation in tissues at rest?
    (A) 30% (B) 50% (C) 70% (D) 90%
19. On the graph, where is the most likely location of part X?
    (A) Lungs
    (B) Brain
    (C) Renal tubules
    (D) Peripheral capillaries

37 Infectious Disease

A disease is any condition that impairs or interferes with the normal functioning of the body. Infectious diseases differ from other diseases, e.g. genetic and lifestyle diseases in that they are caused by the invasion by a pathogen and can be transmitted from one host to another. Infectious diseases are also called communicable or transmissible diseases.

Infectious diseases have dramatically affected life on Earth acting as a selective pressure in the evolution of different populations. The history of human civilisations through the ages shows how natural immunity and the ability to combat infectious diseases has affected politics, economics and public works and health. During the middle ages in the 14th century nearly one third of the world's population died from bubonic plague – the Black Death which is an infectious disease caused by the bacteria *Yersinia pestis*.

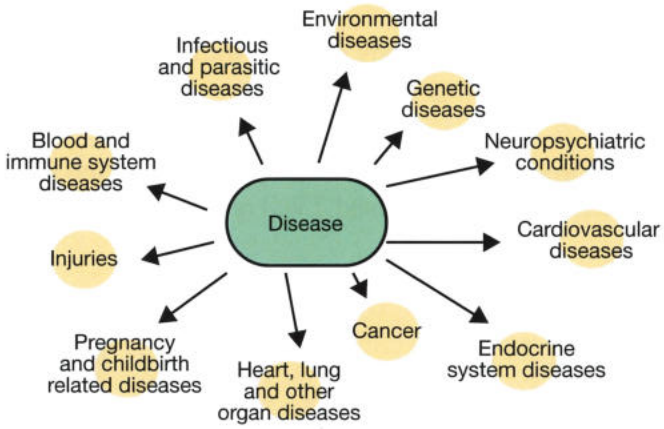


Figure 37.1 Types of disease.

Diseases can be classified in many ways, e.g. by the major categories of infectious disease and non-infectious disease or by the body part/system that is mainly affected, e.g. cardiovascular diseases, heart, lung and other organ diseases or endocrine system diseases or by the way the disease is acquired, e.g. genetic diseases, injuries and environmental diseases.

Technology and infectious disease

The invention of the light microscope and developments in its design to improve resolution and magnification enabled scientists to observe micro-organisms and then to link particular organisms to particular infectious diseases.

Anton van Leeuwenhoek was the first to describe 'little animals' in rainwater and his discovery led to many other scientists using the light microscope to look for micro-organisms in different substances.

Louis Pasteur carried out experiments using swan neck flasks that led to his germ theory of disease. Then in 1845-6 he showed that a mystery disease that threatened the silkworm industry was caused by micro-organisms that were only visible under a microscope and only found in the tissues of diseased silkworms, moths and eggs. This work refuted the theory of spontaneous generation which had been a common belief that living organisms could arise from non-living matter.

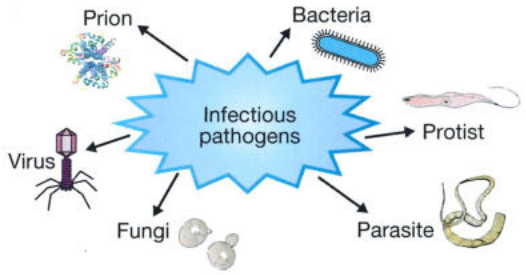


Figure 37.2 Types of pathogens.

Robert Koch built on the work of Pasteur developing improved ways to study micro-organisms. He solidified liquids using gelatine and agar to produce suitable mediums on which to grow bacteria. His assistant **Julius Petri** invented the Petri dish to aid the growth of pure cultures. Improved staining techniques meant that Koch was able to identify the causative bacteria for tuberculosis and cholera. This work led Koch to outline a systematic way, known as Koch's postulates, which identifies the causative pathogen for a particular disease.

New molecular technologies and computer assisted analysis are now speeding up the diagnosis process for infectious diseases which means specific treatments can be started earlier which hastens recovery times. New electronic tools and computer processing also assist in drug dosing and prescribing to improve the quality and efficiency of treatment procedures.

QUESTIONS

1. Define disease.
2. How are infectious diseases different from other diseases?
3. Use an example to show how infectious disease has affected human populations in the past.
4. How can diseases be classified?
5. How did the invention of the microscope increase scientific knowledge about infectious disease?
6. Construct a table to show at least *three* ways technology has helped aid our understanding of infectious disease.
7. Who was the first to describe 'little animals' under a microscope?
   (A) Van Leeuwenhoek. (B) Louis Pasteur.
   (C) Robert Brown. (D) Robert Koch.

38 Louis Pasteur

A **pathogen** is an organism, virus, viroid or prion that causes disease. Louis Pasteur made significant contributions to the scientific understanding of the relationship between micro-organisms and disease. His work laid many of the foundations for further investigations into the identification of specific pathogens, how they cause disease and how to produce vaccinations to prevent symptoms occurring in the host.

Early work

Pasteur was born in Ole and became a chemist, studying the shapes of organic crystals. He showed that crystals of paratartrate had one chemical composition but two crystal forms – one that rotated light to the left and one, its mirror image, rotated light to the right. His work led to the new science of stereochemistry.

Figure 38.1 Louis Pasteur (1822-1895).

Pasteur and the cause of disease

In 1856, M Bigot, the father of one of Pasteur's chemistry students, approached Pasteur asking for help with problems with his wine business. M Bigot manufactured alcohol by fermenting the juice from the roots of beet, yet sometimes the alcohol went sour and turned to vinegar, or if they were trying to make vinegar it turned to lactic acid.

At the time, the accepted scientific theory of fermentation was that it was a chemical breakdown of sugar into alcohol due to unstabilising vibrations. These unstabilising vibrations could be transferred to a new vat to start the process again. They knew yeast cells were living organisms but thought they were probably a catalyst or a product of fermentation. Biologists who had suggested yeast caused fermentation were ridiculed.

Using a microscope, Pasteur discovered that vats with good alcohol contained yeast cells that were round and budding. When lactic acid formed instead of alcohol, there were small rod shaped microbes (lactobacilli) mixed with the yeast. Chemical analysis showed the presence of several complex organic molecules such as amyl alcohol, which inferred that fermentation involved more steps than the simple breakdown of sugar. Pasteur also discovered that some of these compounds were asymmetric and rotated light, which he believed could only be formed from living things. He concluded that the yeast caused fermentation of sugar from the beet juice into alcohol and that contaminating micro-organisms turned the fermentations sour.

He showed that sugar without yeast will not form alcohol and that lactic acid forms if the rod shaped organisms are present. He referred to the spoilage of wine as a 'disease' so that infectious diseases caused by micro-organisms became known as the **germ theory of disease**.

Further study discovered the specific organisms that caused the fermentation or spoilage of wine, beer and vinegar. Pasteur suggested that makers first heat the wine, beer or milk to moderately high temperatures (50°C to 60°C) for a few minutes to kill the micro-organisms and sterilise their batches (pasteurise). They could then add the microbes they wanted and it would not go sour.

Pasteur's classic experiment

In the 1860s there was still debate about spontaneous generation, that is, many believed that life forms, such as beetles, maggots, rats suddenly came to life from decaying matter. Pasteur conducted a series of experiments that proved spontaneous generation did not occur. He placed broth in long neck flasks and then bent some of the necks into an S shape (swan neck flasks), while others were left straight to act as a control. He then boiled the broths to kill the micro-organisms. Germs in the air could reach the sterile broth in the straight neck flasks but the swan neck flasks trapped dust and germs in the film of moisture on the humid curve.

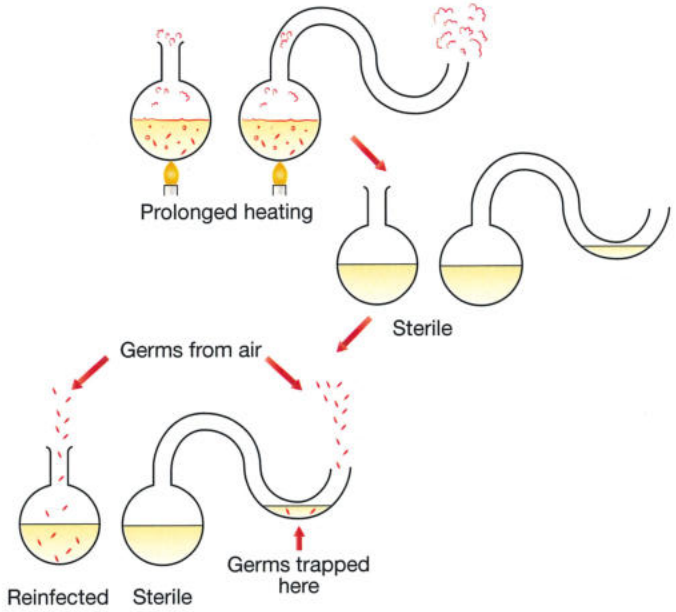


Figure 38.2 Pasteur's classic experiment.

After a while the control broth contained many organisms while the swan neck flask broth was still sterile. If the swan neck was broken off, the broth developed colonies of mould and bacteria. Since air alone did not cause the broth to develop bacteria and mould, the experiment showed that microbes cause decay and there is no spontaneous generation. Pasteur's flasks are on display at the Pasteur Institute in Paris – with the swan neck flask still free of micro-organisms.

Pasteur, vaccinations and the germ theory

The germ theory became firmly established with Pasteur's work with anthrax.

Anthrax is a deadly disease of sheep, horses and cattle. In 1876, Robert Koch isolated the anthrax bacillus (*Bacillus anthracis*) and suggested that sick animals be killed or burned as the bacterial spores survived for months in contaminated fields. Pasteur decided to work on anthrax and added one drop of blood from a sheep dying of anthrax to 50 mL of sterile culture and grew the culture. He repeated the dilution 100 times so that none of the original culture cells were present and found that the final culture was as active as the first in causing anthrax. This showed that the bacteria caused the disease as only the reproducing bacillus could escape the dilution.

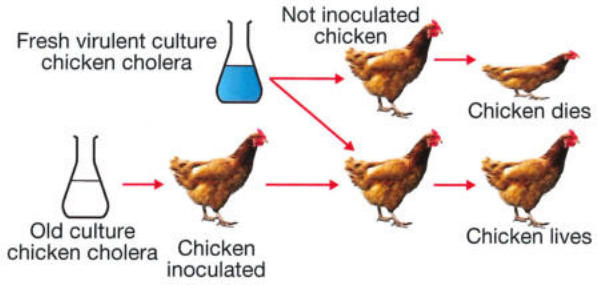


Figure 38.3 Louis Pasteur and chicken cholera.

His work with chicken cholera made Pasteur believe he could find a vaccine for anthrax. He tried several techniques such as oxidation and ageing, and eventually developed a vaccine. His work was challenged by a well known veterinarian, causing Pasteur to conduct a public test. Using 25 sheep as controls and 25 vaccinated, all sheep were given a lethal dose of anthrax. There was intense publicity. On 5 May 1882, two days after the final inoculation, all control sheep were dead and all vaccinated sheep were alive and healthy. The publicity educated the world about microbes and vaccination.

QUESTIONS

1. Although Pasteur is known as the father of microbiology and immunology, how did he begin his science career?
2. Why did Pasteur begin investigating the production of wine?
3. What was the current belief about fermentation when Pasteur began investigating wine production?
4. Describe the three clues that allowed Pasteur to solve the puzzle of fermentation.
5. What did Pasteur conclude?
6. State the 'germ theory of disease'.
7. Describe the role of Pasteur in identifying the causes of disease.
8. Describe pasteurisation and explain why it is used.
9. What is spontaneous generation?
10. In Pasteur's classic experiment, identify the control and explain how it is used.
11. Explain why Pasteur boiled each broth.
12. Explain why the flask with the swan neck did not develop mould or bacteria.
13. What is anthrax?
14. Who discovered the microbe that causes anthrax?
15. Describe Pasteur's experiment that confirmed the germ theory of disease.
16. How did Pasteur develop a vaccine?
17. Explain how microbes and vaccination came to the attention of the public.

The information is to be used for the next THREE questions. In 1864 Louis Pasteur carried out a series of experiments where he put broth into several special S shaped flasks.

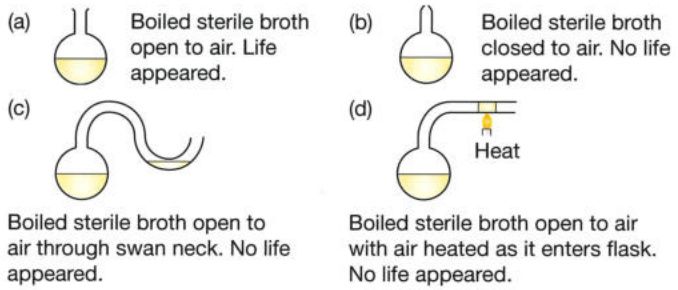


Figure 38.4 Louis Pasteur's flasks of broth.

18. Which theory was disproved by this experiment?
    (A) Spontaneous generation.
    (B) Microbes caused disease.
    (C) Life forms are made up of cells.
    (D) Decay is caused by micro-organisms.
19. If the swan neck is broken off the flask, what would you would expect to happen to the experimental broth?
    (A) Remain sterile.
    (B) Decay due to microbes from the air reaching the broth.
    (C) Decay due to microbes already present in the broth now multiplying.
    (D) Decay only when the contents of the swan neck are deposited into the flask.
20. In Pasteur's experiment, the broth in the flask with the swan neck did not decay because:
    (A) Boiling the broth in step one prevents decay.
    (B) The temperature changes along the length of the swan neck kill bacteria.
    (C) Bacteria were trapped in the swan neck.
    (D) The swan neck prevents oxygen reaching the broth.
21. In 1882 Louis Pasteur carried out a public test to show the benefit of vaccination. What did he use in this public test?
    (A) Dogs with cholera. (B) Dogs with anthrax.
    (C) Sheep with cholera. (D) Sheep with anthrax.

39 Robert Koch

A **pathogen** is an organism, virus, viroid or prion that causes disease. Robert Koch made significant contributions to scientific understanding making people aware of micro-organisms as causative agents in disease. He was a pioneering bacteriologist and established the rules of procedure for proving a particular micro-organism is the cause of a particular disease.

Robert Koch was a German physician who took his medical degree in 1866 and served as a surgeon in the Prussian army. He became a country physician in Wollstein where many cattle and sheep died of anthrax. Anthrax has the characteristics of boils and swelling of the spleen and oedema, and death occurs quickly. Learning of Pasteur's theory that micro-organisms caused disease, he investigated, and in 1876 discovered the anthrax bacillus. He isolated these bacteria from other bacteria present and injected them into healthy mice. The mice developed anthrax similar to the sick sheep. Koch's work was supported by Pasteur, and his study methods were so successful he was invited to the University of Breslau and then to Berlin.

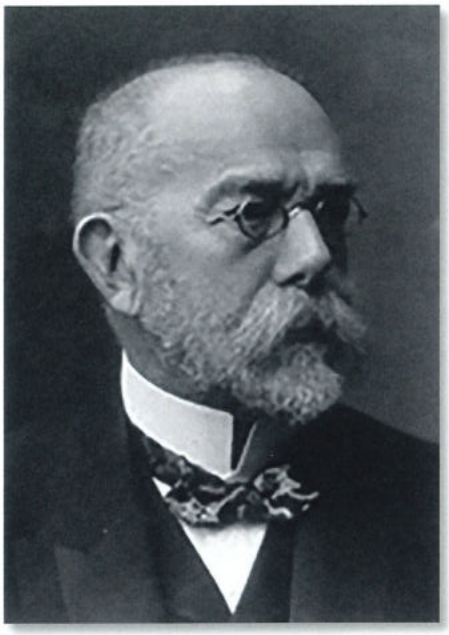


Figure 39.1 Robert Koch (1843-1910).

In 1877 Koch described techniques of fixing, staining and photographing bacteria. He used dyes to stain bacteria and began using solid gels such as gelatin instead of sticky liquid mediums. The bacteria formed colonies on the agar and were easier to isolate.

Richard J Petri (1852-1921) invented a special glass dish with lid to hold the agar media. The gelatin was replaced with agar.

In 1884, Koch made a list of criteria needed to prove that a particular organism causes a particular disease. These are known as **Koch's postulates**.

In 1891, Koch was appointed Director of the Institute of Infectious Diseases in Berlin and by 1892 he and his pupils had found the agents causing diphtheria, typhoid, tetanus and tuberculosis.

Koch pioneered bacteriology and was awarded the Nobel Prize in Physiology or Medicine in 1905 for developing tuberculin to test for tuberculosis.

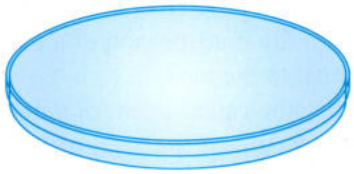


Figure 39.2 Petri dish invented by Richard J Petri.

Koch's postulates

1. The organism believed to be the cause of the disease must always be present when the disease occurs.
2. The organism must be isolated from the host and grown in pure culture.
3. Organisms from the pure culture, when inoculated into healthy, suitable, susceptible hosts must produce the disease.
4. The organism must be reisolated, grown in pure culture and compared with the organism first injected.

This basic bacteriological technique is still used today to identify the pathogen of an infectious disease.

QUESTIONS

1. What is anthrax?
2. Describe Robert Koch's contribution to our understanding of anthrax.
3. Outline some of the techniques developed by Koch to study micro-organisms.
4. How did Richard J Petri assist bacteriology?
5. Outline Koch's postulates.
6. How was Koch rewarded for his work?
7. Explain how Koch's postulates can be used to identify the causative organism of an infectious disease.
8. No one has been yet able to culture on artificial media the bacteria (*Treponema pallidum*) that causes syphilis.
   (a) How could they decide that *Treponema pallidum* causes syphilis?
   (b) Does this mean that Koch's postulates are invalid?
9. Describe the circumstances that led to a named micro-organism being identified as the causal agent of a particular disease.

10. Which of the following identifies Koch's postulates?
    (A) Procedures used to establish the cause of a disease.
    (B) Steps to follow when culturing micro-organisms.
    (C) Rules for cleanliness and hygiene when dealing with pathogens.
    (D) Procedures to follow to detect pathogens in blood samples.
11. Which of the following shows a practical use of Koch's postulates?
    (A) Determine the evolutionary tree of several closely related species.
    (B) Estimate the effectiveness of a new vaccine.
    (C) Explain the transmission of a pathogen from one host to the next.
    (D) Identify the causal agent of a disease.
12. When using Koch's postulates, which of the following is an essential requirement for the suspected pathogen?
    (A) Is present in all organisms with the disease.
    (B) Can be cultured on an agar plate.
    (C) Is tested on a wide variety of organisms.
    (D) Can reproduce sexually within the human body.
13. When using Koch's postulates, which of the following features of the suspected pathogen needs to be investigated?
    (A) Discover the effect of the pathogen on a wide variety of organisms.
    (B) Cultivate the pathogen in pure culture.
    (C) Have many specimen samples of the pathogen.
    (D) Classify the pathogen to genus species level.
14. What does a colony on an agar plate represent?
    (A) A single bacterium.
    (B) A community of different organisms clumped together to help each other.
    (C) A population produced by a single bacterium.
    (D) One species of virus.
15. While using Koch's postulates, scientists took a sample of a bacteria suspected of being the pathogen which caused a particular disease. After several days the Petri dish of nutrient agar had many colonies that were white, circular and convex. There was also one round, furry patch. Suggest the most likely identification of the furry patch.
    (A) A colony of mutated bacteria.
    (B) Contamination by a mould fungi.
    (C) A yeast infection.
    (D) A parasitic virus growth.
16. Why is the use of a solid medium for culturing micro-organisms preferred to the use of a liquid broth?
    (A) Yeast cells cannot multiply in a broth.
    (B) Viruses prefer to detach from each other on a solid medium.
    (C) Fungal mycelium causes the broth to solidify.
    (D) Bacteria form separate colonies on solids.
17. A bacterial pathogen which lives in human blood was isolated and cultivated on agar plates at 37.5°C. After 5 days the plates had no bacterial colonies. Identify the most likely explanation for the lack of bacterial colonies.
    (A) Agar inhibited the growth of the bacteria due to its antiseptic properties.
    (B) Bacteria grew but did not reproduce.
    (C) Agar lacked an essential food necessary for the growth of the bacteria.
    (D) Temperature was unsuitable for this particular bacteria.
18. In 1876, Robert Koch discovered that the bacteria, *Bacillus anthracis,* caused the disease anthrax in sheep and other animals. Louis Pasteur also became interested in anthrax and found that a weakened culture injected into sheep prevented the sheep dying when they were later infected with bacterial spores. Which of the following shows how our understanding of the cause and prevention of disease develops?
    (A) One person does the work and others copy.
    (B) Rivalry prevents progress for many years.
    (C) Each scientist builds on the work of others.
    (D) Each scientist works alone guarding their own discoveries.
19. In 1854 Louis Pasteur was appointed to the University of Lille where he studied samples of wine under a microscope. He found that the micro-organisms in good fermentations were different from those in bad fermentations. He stated that good and bad fermentation is due to the activities of different micro-organisms. What type of statement is this?
    (A) Hypothesis. (B) Theory.
    (C) Inference. (D) Conclusion.
20. The diagram shows the anthrax bacillus.

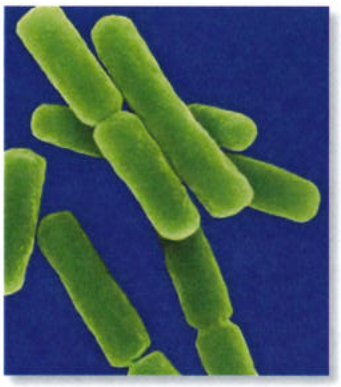


Figure 39.3 Anthrax bacillus.

What was Robert Koch's contribution to the study of anthrax?
    (A) He first described the symptoms of anthrax.
    (B) He discovered the anthrax bacillus.
    (C) He first suggested this bacillus caused anthrax.
    (D) He proved the bacillus caused anthrax.

40 Pathogen Transmission

A **pathogen** is an organism, virus, viroid or prion that causes disease. Most pathogens are micro-organisms such as bacteria, viruses, protozoans or fungi. All viruses are potential pathogens as they can only metabolise and reproduce inside a host cell.

A **parasite** benefits by living on or in a host which is harmed.

Pathogens can come from the external environment, e.g. entering the body from the air, with ingested food and water, through wounds or cuts in the skin or they can be endogenous are originate from within an organism, tissue or cell.

The transmission or communicability of an infectious disease depends on the way the pathogen leaves its host and how it enters a susceptible person.

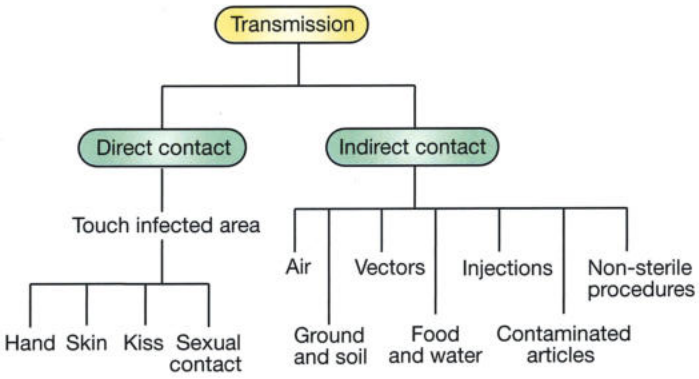


Figure 40.1 Disease transmission.

Direct contact

Direct contact involves touching the affected area of the infected person with the contagious person passing the pathogen to another person.

The communicable person may be showing symptoms of the disease or may be a symptomless carrier. There are many diseases that can be transmitted by both direct and indirect contact, e.g. measles is caused by the measles virus and is spread by contact with saliva or nasal secretions of an infected person but can also be spread through contact with surfaces that have droplets from coughs and sneezes from an infected person.

Direct contact can involve touching skin, handshaking, kissing and sexual contact.

Diseases include *Herpes simplex* infection, chickenpox, warts, ringworm, scabies, head lice, *Staphylococcus aureus* infection, syphilis, HIV infection, gonorrhoea, pubic lice and conjunctivitis.

Indirect contact

There are many ways a pathogen can be transmitted from person to person through indirect contact.

- **Airborne diseases** – involves inhalation of bacterial spores, or droplets containing pathogens from coughing, sneezing or speaking. Airborne diseases include chickenpox, common cold, diphtheria, influenza, measles, meningitis (bacterial), mumps, rubella, tuberculosis and whooping cough.
- **Food and water** – involves eating or drinking contaminated food which can be caused by lack of proper sewage treatment of water supplies, poor hygiene, use of sewage as a fertiliser. Diseases include *Cryptosporidium* infection, *Giardia* infection, hepatitis A, meningitis (viral), worms, *Salmonella* infection, cholera, typhoid and botulism.

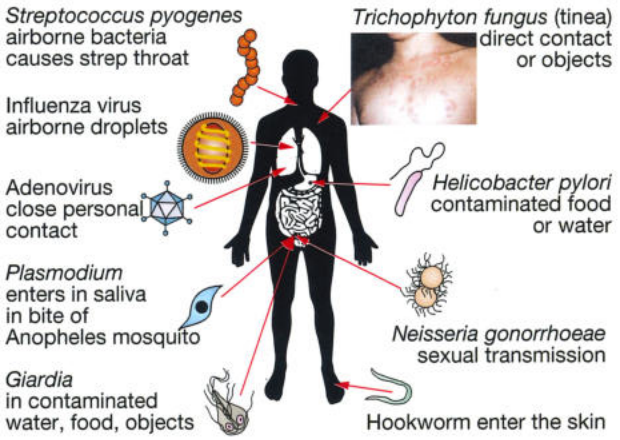


Figure 40.2 Pathogens in humans.

- **Vectors** – involve transmission through other organisms that carry the pathogen from one person to another or from an infected animal to another person. The vector may or may not be an intermediate host of the pathogen. Parasitic life cycles can be complex involving more than one host. The intermediate or secondary host is usually the organism in which the parasite goes through the larval stage or undergoes asexual reproduction. Many different organisms act as vectors, e.g. arthropods, mammals and birds.

  Arthropods – e.g. houseflies, cockroaches carry pathogens on their feet and bodies and contaminate food and water, e.g. houseflies transmit anthrax, typhoid and cholera. While mosquitoes, fleas, lice and tick carry pathogens internally and transmit them while feeding, e.g. Lyme disease is transmitted in the bite of a tick.

  Mammals – many different mammals can transmit pathogens to infect humans, e.g. dogs transmit rabies and hydatid tapeworms and pigs transmit pork tapeworm.

  Birds – certain birds can transmit avian influenza, e.g. water birds and parrots can transmit psittacosis.

- **Contaminated articles** – many articles can become contaminated, e.g. eating and drinking utensils, cooking equipment and chopping boards, clothing, toys, books, doorknobs. Faecal-oral transmission occurs when pathogens in the stool of one person or animal is swallowed by another person. This often occurs via contaminated articles and shows the importance of safe and careful food handling practices. Diseases include hepatitis A, typhoid, poliomyelitis, salmonella and cholera.
- **Non-sterile procedures** – there are many procedures that require sterile conditions. Sterile means free from germs. All surgeries, caring for wounds, having injections, dental work and tattooing require sterile conditions. Sterile techniques include washing with soap and running water, keeping work surfaces dry and clean, only touching the outside wrapper with bare hands, wearing gloves, wearing a mask over nose, mouth and hair and covering your mouth when you cough or sneeze.
- **Across the placenta** – some pathogens can move across the placenta from the mother to enter the foetal circulation. Diseases include rubella and syphilis.

Pattern of infection

Infectious diseases tend to follow the pattern:

- Pathogen enters the body.
- Incubation period when the pathogen reproduces increasing in numbers but the host does not show symptoms.
- Host shows symptoms of infectious disease, e.g. fever, headache, pain, rash.
- Crisis – the effect on the host is at its most severe stage from which the host either recovers or dies.
- Convalescence when the host recovers from the infection and the symptoms disappear.

QUESTIONS

1. What is a pathogen?
2. What is a parasite?
3. From where do pathogens come?
4. Identify the two main factors that determine the communicability of an infectious disease.
5. Distinguish between transmission by direct and indirect contact.
6. Use an example to show how a pathogen can be transferred by both direct and indirect contact.
7. Outline the process of airborne transmission.
8. Explain how pathogen transmission can occur in food or water.
9. What is a disease vector?
10. Use examples to show how each of the following can act as disease vectors.
    (a) Arthropods.
    (b) Mammals other than humans.
    (c) Birds.
11. Construct a table to show the pattern of infection for most infectious diseases.
12. Identify some sterile techniques that are used to prevent disease transmission.

Use the diagrams that show different ways of disease transmission for the next TWO questions.

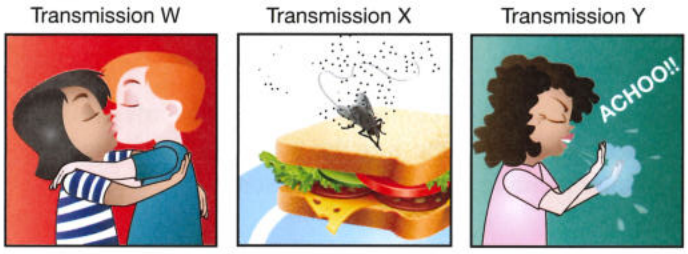


Figure 40.3 Disease transmission.

13. Which of the following shows possible modes of transmission for each disease?

	Transmission W	Transmission X	Transmission Y
(A)	Malaria	HIV	Tuberculosis
(B)	Tuberculosis	Malaria	Anthrax
(C)	HIV	Tuberculosis	Malaria
(D)	Gonorrhoea	Anthrax	Tuberculosis

14. Botulism is an infectious disease caused by the bacteria *Clostridium botulinum* and the botulism toxins are one of the most lethal substances known. The bacterial spores can be spread in soil and water. In which of the modes of transmission is botulism not likely to occur?
    (A) Transmission W.
    (B) Transmission X.
    (C) Transmission Y.
    (D) Transmissions X and Y.
15. Which of the following applies to all pathogens?
    (A) Are viruses.
    (B) Able to live in human blood.
    (C) Microscopic.
    (D) Able to cause disease.
16. What is meant by the incubation period?
    (A) The time the pathogen spends outside the host.
    (B) The time from the moment of exposure to an infectious agent until symptoms appear.
    (C) The time during recovery of health and strength in the later stage of an infectious disease.
    (D) When the host is an infectious carrier.

41 Experiment – Microbes in Food and Water

Several Human Biology students wished to discover if there were microbes in food and water. They decided to investigate food bought from their school canteen, food from the local shop and food brought from home.

Student actions

They took water samples from the school bubblers, from a tap in the school biology laboratory and from bottled water.

To culture the microbes they prepared agar plates with high protein nutrients, being careful not to expose the plates to the air. One plate was left unexposed to be used as a control.

For each of the 'water plates', the students inoculated the plates with 0.5 mL of the particular water sample and gently rocked the plate so that the water spread evenly over the agar surface.

For the food sample the students chose the bread from a cheese sandwich. They mashed the bread with distilled water and streaked the plate with a loopful of bread solution.

All plates were incubated at 25°C for 7 days.

The plates were examined and the number and types of different colonies visible were recorded. The lids were not removed from the Petri dishes.

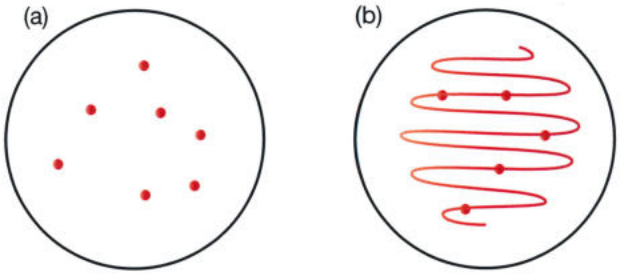
Problems encountered

The students encountered problems when they tried to identify and classify the different colonies on their plates.

Most identification tests for different bacterial species are biochemical, physiological or immunological. Bacteriologists have a systematic identification procedure which allows them to quickly identify a pathogen, particularly useful in hospitals. For example, Gram stain with crystal violet, Jensen's iodine, alcohol and carbol fuchsin identifies if the bacteria is Gram positive (purple) or Gram negative (pink). Another identification procedure is growth tests on different food sources, for example sugars, proteins, nitrates and sulfur compounds. In the school laboratory, the students decided they would simply examine the size (mm), shape (circular, filamentous, irregular or rhizoid), texture (smooth, rough, shiny, matt, moist or dry) and colour of the bacteria. Fungi have a different appearance, being larger and cottony.

QUESTIONS

1. Explain why it is essential to use aseptic techniques in microbiology and give an example of one technique.



Tap water from biology laboratory

Bread from homemade sandwich

Figure 41.1 Experiment results.

Table 41.1 Student results.

Source	Bacteria		Fungi	
	Number of different types	Total number of colonies	Number of different types	Total number of colonies
Control	1	1	0	0
Bubbler water	3	17	0	0
Lab water	2	8	0	0
Bottled water	2	3	0	0
Canteen sandwich	3	25	1	1
Homemade sandwich	2	6	0	0
Local shop sandwich	2	22	0	0

2. Define the following terms – pure culture, colony, streaking.
3. Explain why the students were careful not to expose the plates to the air.
4. How could the control be used in this experiment?
5. Explain why a full nutrient agar was prepared and used.
6. From the student results, which sandwich was the most 'healthy'? Explain your results.
7. Suggest some possible ways in which the sandwich was contaminated.
8. Which 'water' was the most 'healthy'? Explain your answer.
9. Suggest some possible ways in which the water became contaminated.
10. Describe another way in which the students could present their results.
11. Suggest a possible reason why the students did not pursue the identification of their bacterial colonies.

12. Identify two safe work practices needed during this investigation.
13. Water samples are often tested for bacteria that indicate faecal contamination, for example *Escherichia coli* and *Enterobacter aerogenes*. These bacteria live in large numbers in the intestines of humans and other animals. Suggest possible reasons why it is important for cities such as Sydney to know the amount of *E. coli* in the water off Bondi Beach.
14. Some bacteria are always present in milk. They may be present on the skin of the cow or they may be in the milk ducts and are flushed out with the milk, or they may come from animal hair, milking equipment, manure, dirt or human workers. Some of these bacteria are harmless, some are helpful (e.g. in forming cheese or yoghurt) and some are pathogenic (e.g. causing tuberculosis, brucellosis and mastitis). Explain how the pasteurisation of milk prolongs the life of the milk.
15. Water treatment depends on the cleanliness of the water at its source. A large, undisturbed catchment area needs little treatment. A river system with industrial run-off, or farming or human waste requires extensive treatment. Primary treatment includes screening, degritting, flocculation, sedimentation and sludge dredging. Secondary treatment involves filtration by passing the water through sand beds or charcoal. Tertiary treatment involves chlorination, fluoridation or the addition of other chemicals. Describe three methods to treat drinking water and the effectiveness of each method.
16. Discuss the importance of water treatment.
17. Cryptosporidiosis is caused by the parasite *Cryptosporidium*. It is a notifiable infectious disease in Western Australia, e.g. 254 cases reported in 2015, 454 cases reported in 2011. The pathogen lives in the intestine, causing diarrhoea, abdominal pain and sometimes vomiting and slight fever. Suggest some ways in which people can prevent the spread of cryptosporidiosis.
18. In July and September 1998, Sydney had a series of problems with its water supply. The National Health and Medical Research Council and the Agricultural and Resource Management Council of Australia and New Zealand have set the guidelines for drinking water. Two pathogens, cryptosporidium and giardia, are transmitted in contaminated water. Explain why it is important to have a 'closed' water catchment.
19. Leeuwenhoek described unicellular organisms in 1676 but it was nearly two centuries before antiseptics were used by surgeons to kill microbes. Which of the following suggests a reason for this lapse in time?
    (A) Inability to classify micro-organisms.
    (B) Difficulties in observing microbes.
    (C) Lack of suitable antiseptics.
    (D) Failure to link micro-organisms with disease.
20. Louis Pasteur investigated fermentation of sugar to produce alcohol. Which of the following causes fermentation?
    (A) Fungi.
    (B) Virus.
    (C) Bacteria.
    (D) Prions.
21. Cryptosporidiosis is an infectious disease caused by the parasite *Cryptosporidium* which infects the human intestine, producing watery diarrhoea and abdominal pain. The disease has been reportable in Western Australia. Since the disease can be spread by contaminated water and through person to person contact, identify one way to prevent the spread of the disease.
    (A) Kill all mosquitoes.
    (B) Provide antibiotics to anyone who comes in contact with a person with the disease.
    (C) Stop people with diarrhoea swimming in pools and recreational swimming areas.
    (D) Filter all drinking water.
22. The diagram shows Brewer's yeast (*Saccharomyces cerevisiae)*.

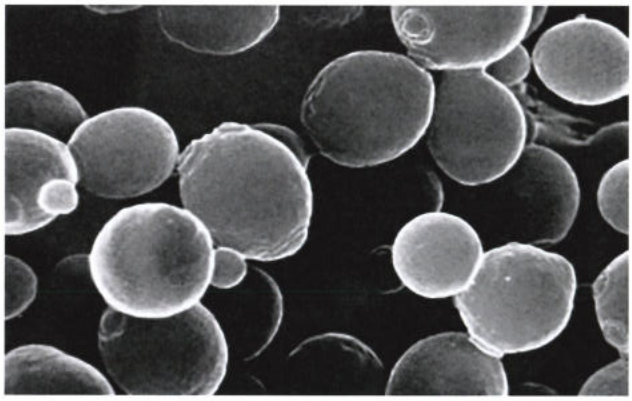


Figure 41.2 Brewer's yeast.

Brewer's yeast is the yeast remaining after beer making. It is inactive and is often used in the health industry as a vitamin B supplement. It is also used to make certain foodstuffs, e.g. Vegemite and Marmite. Which of the following statements is true about *Saccharomyces cerevisiae*?
    (A) It is a beneficial microbe used for leavening bread and beer making.
    (B) It is a beneficial microbe used to supply protein, minerals and B vitamins.
    (C) It is a beneficial microbe best consumed in its active state to provide nutrients.
    (D) It is a harmful microbe that can cause infectious disease.

42 Water Treatment

Water treatment is very important in preventing disease and maintaining a healthy population.

Many pathogens are transmitted through water, especially those that affect the alimentary canal, e.g. *Vibrio cholerae* (cholera), *Salmonella typhi* (typhoid) and *Shigella* spp (dysentery).

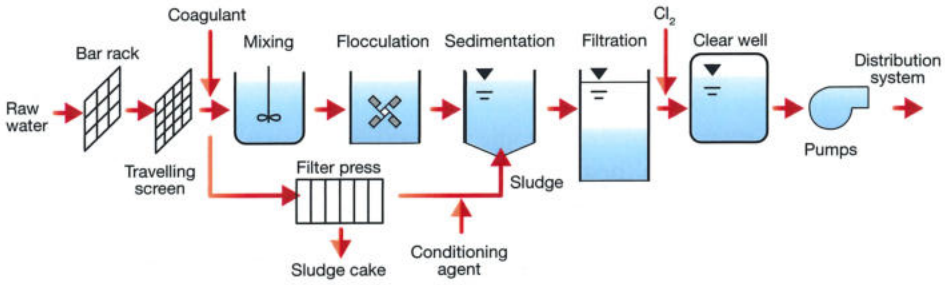


Figure 42.1 Water treatment process.

In water treatment plants the water is purified, filtered and disinfected to make it safe to drink. The type of treatment given to any water depends on the source of the water and its flow from the source to the final tap. Treatment can be divided into three main stages.

Primary treatment

- **Flocculation** – Chemicals, e.g. polyelectrolytes are added to the water to cause particles of dirt, leaves etc to stick together to form a large clumps (flocs). The clump is easier to filter or settle from water.
- **Coagulation** – aluminium sulfate is added to help sink flocs and make sedimentation faster.
- **Sedimentation** – clumps of dirt and particles sink to the bottom in calm, still water to form sludge.
- **Sludge processing** – solids and liquids are collected from the settling tanks.

Secondary treatment

- **Filtration** – water is filtered, e.g. through sand beds so that leaves, sticks, dirt, algae etc are trapped and the water is clean.
- **Backwashing** – occurs when clean water is pumped backwards through the filters to flush them clean of sludge.

Tertiary treatment

Involves the addition of chemicals, e.g. for disinfection to destroy pathogens.

- **Chlorination** – chlorine gas can be injected into the water.
- **Fluoridation** – sodium fluorosilicate is added to the water, e.g. in a ratio of 1 in a million fluoride, to reduce teeth decay.
- **Other ways to disinfect water** – (a) Heat – boiling water for a few minutes. (b) UV light – strong UV lamps or sunlight radiation. (c) Ozone – injected as gas into water. (d) Other chemicals – e.g. iodine, potassium permanganate, hypochlorite.
- **Changing pH** – pure water has a neutral pH and impurities can make the water acidic or alkali. Soda ash is often added so the pH is suitable for the best action of chlorine.

QUESTIONS

1. Name some diseases that are caused by the transmission of pathogen through water supplies.
2. The diagram shows the bacteria *E. coli*.

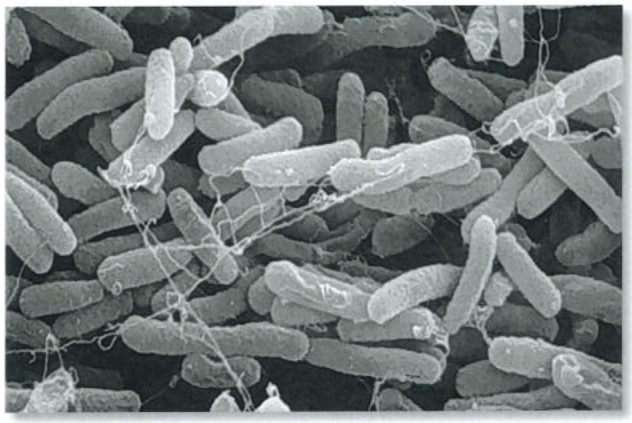


Figure 42.2 *Escherichia coli*.

E. coli is found in the intestines of many animals, e.g. humans, cattle, dogs. It can cause severe illness due to its release of toxins in the alimentary canal. Symptoms may include bloody diarrhoea, abdominal cramps and fever. In severe cases, especially children under 5 years old, it can cause acute kidney failure. Explain why a study of *E. coli* abundance needs to be a part of a water treatment program.

3. Which of the following processes would be most effective in removing pathogens from a water supply?
   (A) Flocculation. (B) Sedimentation.
   (C) Filtration. (D) Chlorination.
4. What is the main purpose of sedimentation in water treatment?
   (A) Disinfection to destroy pathogens.
   (B) Clumping of particles of dirt and leaves.
   (C) Allowing flocs to drop to the bottom.
   (D) Trapping leaves, dirt and algae in filters.
5. Which process is involved in causing particles of dirt and leaves to stick together in water treatment?
   (A) Flocculation. (B) Sedimentation.
   (C) Filtration. (D) Chlorination.

43 Health, Disease and Cleanliness

Health is often defined as a state of physical, mental and social well being. There are difficulties in applying this definition, however, as different people have different standards about what is a 'healthy' level of physical activity, application of mental ability or socialising with friends, co-workers, or neighbours. Health involves the effective functioning of the body and is not just the absence of disease or infirmity.

Disease is any condition that impairs or interferes with the normal functioning of the body.

Infectious disease is caused by pathogens which invade the body. Infectious diseases are contagious and can often be passed from one person to another.

Non-infectious disease is not caused by a pathogen. It can, for example, be caused by heredity, nutrition, physiological malfunction, environment and chemicals. Non-infectious diseases cannot be easily passed from one person to another (except by heredity).

Pathogens are disease-causing parasites that live on or in a host. They can be microscopic or macroscopic and are found in air, water, soil and any contaminated articles. Transfer from one person to another can be by direct or indirect contact.

History of cleanliness – Many ancient civilisations recognised the importance of cleanliness in maintaining health. The Romans built public baths, sewers and aqueducts. Over 3000 years ago, the Chinese and Hebrews supported cleanliness in food, water and personal hygiene.

Notifiable diseases – The Australian Government Department of Health lists the national notifiable diseases and case definitions. Occurrences need to be reported to the Commonwealth's National Notifiable Diseases Surveillance System.

Maintaining health

Function of genes – Genes consist of DNA with the nitrogenous bases forming **codons** that relate to the production of specific amino acids. Since amino acids are the functional unit of proteins such as enzymes, the gene message often translates into a particular enzyme that controls a specific reaction. **Gene expression** means that information coded in the DNA in the gene directs the production of particular polypeptide and the gene is 'switched on'.

Mitosis – Mitosis is part of the cell cycle and is important in the maintenance and repair of body tissues. Two types of genes control the cell cycle — the **proto-oncogenes** stimulate cell growth and cell division while **tumour suppressor genes** slow down or stop cell growth and cell division. If either of these types of genes are damaged or mutated, cancers can form due to the uncontrolled cell replication.

Cell differentiation and specialisation – Multicellular organisms need specialised cells to carry out particular functions to allow the whole body to survive. Specialised cells include red blood cells, white blood cells, neurons, muscle cells, bone and epithelium.

Controlling infectious disease

As infectious diseases are caused by pathogens, control measures need to be able to maintain cleanliness to prevent pathogens from entering and then multiplying in food, water or on/in your body and to prevent the transmission of the pathogen from one person to another. Examples of control measures include washing hands before eating food, cooking food properly, effective sewage treatment, providing uncontaminated water, not coughing over food, and covering hair and open sores when preparing food.

QUESTIONS

1. Define health and disease.
2. Discuss one difficulty in defining health.
3. Outline how each of the following assists in the maintenance of health.
   (a) Gene expression.
   (b) Mitosis.
   (c) Cell differentiation.
4. Use an example to show that the need for cleanliness in food, water and personal hygiene was recognised by ancient cultures.
5. (a) Distinguish between infectious and non-infectious disease.
   (b) Distinguish between direct and indirect contact.
6. Explain why cleanliness in food, water and personal hygiene practices assist in the control of disease. Use examples.
7. Define pathogen.
8. Describe where pathogens can be found and how they can transfer from one person to another.
9. Using the example of a mutation in the allele in the beta-haemoglobin gene, show how health is determined by gene expression.
10. Explain how a mutation or damage to genes can lead to the growth of a cancer.
11. Use an example to show why multicellular organisms need differentiation.

12. Leukaemias are a group of conditions involving the increase in numbers of leucocytes in bone marrow, bloodstream and other organs. Discuss how leukaemia is related to health and mitosis.
13. The diagram shows four different types of cells. Identify each type of cell and describe a disease which involves that particular cell type.

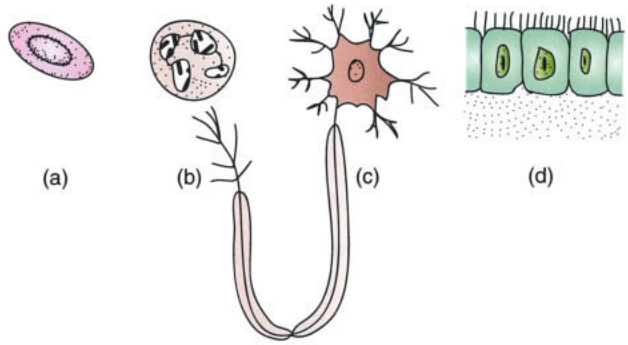


Figure 43.1 Different types of human cells.

14. Classify each of these diseases as either infectious or non-infectious.
    (a) Cholera.
    (b) Chickenpox.
    (c) Down syndrome.
    (d) Scurvy.
    (e) Scoliosis.
    (f) Rheumatic fever.
    (g) Measles.
    (h) Meningitis.
    (i) Glaucoma.
    (j) Anorexia nervosa.
15. The bacteria which causes cholera has a direct life cycle and infects people when ingested with contaminated food or water. Suggest how sewage treatment helps prevent a cholera epidemic.
16. If there is a change in the sequence of bases, gene expression is affected and the health of the organism can be impaired. For example, if the allele in the beta-haemoglobin gene has a mutation with thymine instead of guanine at one particular location, the person suffers from the disease thalassemia major which results in abnormalities such as severe anaemia and growth retardation. Use this example to evaluate how major advances in scientific understanding and/or technology have influenced scientific thinking.
17. (a) Identify the cause of some cancers.
    (b) Each body cell with a nucleus contains the full number of chromosomes. Outline the degree of activity of each gene in any one cell.
18. Using the understanding of the causes of disease, assess the impact of implications of biology on society and the environment since the beginnings of recorded history.
19. Which is the best definition of health?
    (A) Physical, mental and social wellbeing.
    (B) Correct gene expression to maintain and repair body tissues.
    (C) Not suffering from a disease.
    (D) Absence of infirmity with mental alertness.
20. Which of the following describes how gene expression affects the health of the organism?
    (A) Directly monitors homeostasis.
    (B) Prevents entry of pathogens into the body.
    (C) Controls the manufacture of antibiotics.
    (D) Controls protein synthesis.
21. Uncontrollable mitosis leads to:
    (A) A large healthy body.
    (B) Repair of body tissues.
    (C) Cancer.
    (D) Formation of differentiated tissues.
22. Which of the following describes an infectious disease?
    (A) Regulated by a strict diet.
    (B) Transmitted from one host to another.
    (C) The result of a genetic abnormality.
    (D) Caused by a micro-organism.
23. What is a common sign of disease in mammals?
    (A) Gain in body weight.
    (B) Decrease in body hair.
    (C) Increase in antagonistic behaviour.
    (D) Increase in body temperature.
24. Which of the following relates to a non-infectious disease?
    (A) Cured using antibiotics.
    (B) Transmitted from one host to another.
    (C) The result of a genetic abnormality.
    (D) Caused by a micro-organism.
25. Scientists are now looking at historical literature to give a modern diagnosis for diseases that were observed in ancient times but not understood. For example, a hieroglyph from Memphis in ancient Egypt dated 3700 BCE shows a temple priest with the typical clinical signs of paralytic poliomyelitis. Which successful procedure was used by the Chinese and Hebrews over 3000 years ago to reduce the incidence of disease?
    (A) Addition of the mineral chlorite to their water wells.
    (B) Practice of cleanliness in food, water and personal hygiene.
    (C) Wearing of leather suits sewn together once on the body.
    (D) Vaccination against diseases by using animal extracts.

44 Prion Pathogens

The term prion was first used in 1982 by Stanley Prusiner as a combination from *protein* and *infection*. He discovered and identified prions as a class of infectious self-reproducing pathogens that are only made of protein and won the Nobel Prize in Physiology or Medicine in 1997 for 'the discovery of Prions – a new biological principle of infection'.

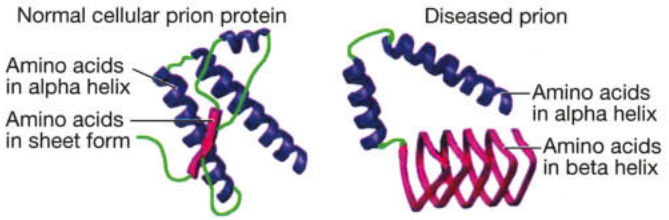


Figure 44.1 Normal protein and diseased prion.

Prions are infective proteins and do not have a nucleic acid genome.

The **normal form** of the protein is called the cellular prion protein (PrP^C). The **prion** is a misfolded version of the normal cellular protein.

The prions increase in number by converting correctly folded versions of the protein to more prions with the altered abnormal shape.

Comparisons of the normal protein and the prion protein show that normal prion proteins consist of about 40% alpha-helices while prion proteins are about 55% beta-pleated sheets and 20% alpha-helices. It is believed that the pathogenic features of the misfolded prion proteins relates to the high amounts of beta-pleated sheets.

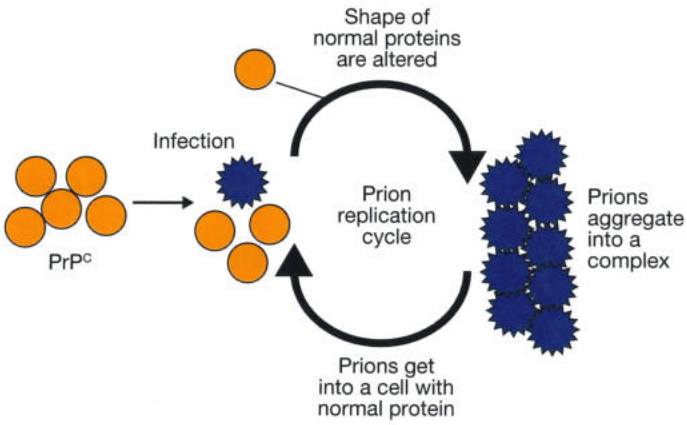


Figure 44.2 Prion replication.

The cellular prion proteins are mainly located on the surface of cells in the central nervous system but can occur in other body tissues in mammals.

Transmissable spongiform encephalopathies (TSEs)

TSEs are a group of progressive conditions that affect the brain and nervous system of many animals. When prions accumulate in tissues they cause large vacuoles so that the tissue begins to resemble a sponge. Tissues commonly affected are the cortex and cerebellum of the brain with degeneration of brain tissue. All prion diseases show a severe loss of neurons. This causes loss of full control of bodily movements and changes in behaviour. In humans the symptoms can involve personality changes, psychiatric problems, insomnia, progressive loss of intellectual capacity and the loss of the ability to move or speak.

TSEs in sheep and goats cause scrapie, in humans it causes CJD (Creutzfeld-Jakob disease), FFI (fatal familial insomnia) and kuru; in cattle it causes BSE (bovine spongiform encephalopathy or mad cow disease) and in cats, pumas and cheetahs causes feline spongiform encephalopathy.

Infection

Humans may be infected in three ways: acquired infection, for example through food or medical procedures; or through apparent hereditary transmission or sporadically due to a spontaneous change into the infectious form of the protein which begins a chain reaction altering other proteins in the cell. This means that prion diseases are unusual as they can occur as *both* infectious and hereditary diseases.

Many prion diseases have a very long incubation period, e.g. at least ten years and at first act very slowly. The long incubation period means that sources of infection may not be identified until long after the first cases appear which means there may have been many infections from the original source or from later infected individuals.

After the long incubation there is often a very aggressive growth phase which causes fatal loss of brain and nervous functionality.

Many prions are not affected by many measures taken to destroy pathogens, e.g. the prions are not destroyed or deactivated by normal cooking temperatures or by normal sterilisation techniques such as boiling or irradiation.

Treatment

Prion diseases are always fatal. When there is rapid development death can occur within a few months to a few years. Most treatments are aimed at alleviating symptoms to make the patient as comfortable as possible.

There are several research projects, e.g. some researchers are developing therapeutic strategies against the diseased prion protein while others are trialling stem cell transplants to restore lost tissue and recolonise damaged areas.

Mad cow disease epidemic

The first confirmed death of an animal due to BSE occurred in 1986 in the United Kingdom. It is believed, but not proven that a scrapie agent jumped species when cattle were fed sheep offal as a protein supplement. The epidemic peaked in the early 1990s with 1000 cases of infected cattle being reported each week in 1993.

In 1996 the British Health Secretary announced to the British House of Commons that mad cow disease is the 'most likely explanation at present' for '10 cases of CJD in people under 42'. There has been over 160 cases of the new human TSE named variant Creutzfeldt-Jakob disease (vCJD) since 1995.

After the announcement in the British parliament about 4.5 million cattle were destroyed and many countries banned the importation of UK beef and beef products. The export ban on British beef was lifted in 1999 but a cow born in 2000 in the UK was diagnosed with BSE. In 2003 a cow born in Washington State, USA and a cow born in Canada tested positive for BSE.

Australia has a national monitoring program checking cattle and sheep populations for evidence of TSE. There is also a complete ban on the importation of live cattle from all countries that have reported cases of BSE. The Ruminant Feed Ban prohibits the feeding to ruminant (animals that chew cud) of any material, tissue or blood taken from an animal.

QUESTIONS

1. How was the word 'prion' invented?
2. How was Stanley Prusiner rewarded for his work on prions?
3. What is a prion?

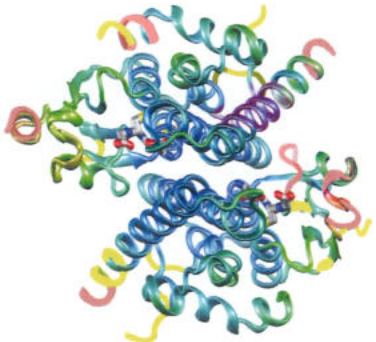


Figure 44.3 Prion.

4. What is PrP^{C}?
5. Identify one of the main differences between the structure of the normal form of the protein and the form of the prion protein.
6. Where are most cellular prion proteins located?
7. Suggest why many prion diseases are classified as spongiform encephalopathies.
8. Outline the effects of prion diseases.
9. Name a prion disease found in:
   (a) Goats and sheep.
   (b) Humans.
   (c) Cattle.
   (d) Cats.
10. Construct a table to summarise the three ways human TSEs can occur.
11. Explain why the transmission of prions is different to the transmission of other pathogens.
12. Discuss why prion diseases can become widely spread before they are detected.
13. Discuss the resilience of the prion protein.
14. What is the mortality rate for prion diseases?
15. How are researchers trying to overcome prion diseases?
16. (a) What is the believed cause of the outbreak of BSE in the United Kingdom in 1986?
    (b) Explain why many countries were concerned when there was an outbreak of BSE in the United Kingdom in 1986.
    (c) What measures have been introduced in Australia to prevent the spread of BSE?
17. The diagram shows three structures.

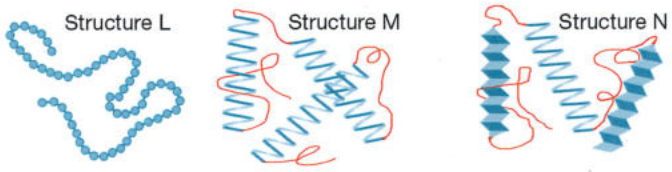


Figure 44.4 Three structures.

Which of the following correctly identifies the normal prion protein and the diseased prion protein?

	Normal cellular prion protein	Diseased prion protein
(A)	Structure L	Structure M
(B)	Structure L	Structure N
(C)	Structure N	Structure M
(D)	Structure M	Structure N

18. Which group lists diseases caused by prions?
    (A) AIDS, influenza, poliomyelitis.
    (B) Scrapie, kuru, Creutzfeldt-Jakob disease.
    (C) Creutzfeldt-Jakob disease, BSE, AIDS.
    (D) Fatal familial insomnia, CJD, malaria.
19. What is a prion?
    (A) Type of nucleic acid which can reproduce in brain tissue.
    (B) Protein particle that causes brain degeneration.
    (C) Lipid which can modify nucleic acids.
    (D) Amino acid which can transmit an infectious disease.

45 Case Study – Creutzfeldt-Jakob Disease

Creutzfeldt-Jakob disease (CJD) is a rapidly progressive degenerative brain disorder that is a transmissible spongiform encephalopathy (TSE). It is called spongiform as the infected tissue resembles a sponge due to the disintegration of the tissue by the misfolded prions. An encephalopathy is a disease that affects the functioning of the brain due to some agent or condition. CJD causes brain deterioration and dementia.

There are several variants of the disease, e.g. vCJD was first found in the United Kingdom. It affects younger individuals who show psychiatric symptoms and it has a longer duration from onset of symptoms to death.

CJD affects many areas of the brain with cortical shrinkage and the loss of neurons.

Figure 45.1 Creutzfeldt-Jakob disease.

Transmission and cause of disease

CJD is a prion disease. Once the prion is present it alters other normal cellular prion proteins to take on the misshaped form of the prion protein. The abnormal prion proteins aggregate into a clump that leads to the loss of neurons and brain damage.

Like other TSEs CJD can be caused by:

- Sporadic appearance even though the person has no risk factors for the disease. This spontaneous change into the infectious form of the protein begins a chain reaction altering other proteins in the cell. Sporadic CJD accounts for more than 85% of cases of CJD.
- Hereditary transmission where there is a family history of the disease and/or the genetic mutation is identified in the person's genome. Hereditary CJD accounts for 5% to 10% of cases of CJD.
- Acquired infection, for example through food or medical procedures. Acquired CJD accounts for less than 1% cases of CJD.

CJD is often difficult to diagnose as the early symptoms are vague and similar to other brain disorders.

Symptoms of disease

The symptoms usually begin to show around the age of 60 and most people die within one year from the onset of symptoms.

Symptoms can include:

- Confusion, disorientation and failing memory which develops into dementia.
- Personality changes with impaired judgement and thinking.
- Behavioural changes.
- Weakness or loss of balance and muscle control causing problems with coordination and movement.
- Muscle spasms and paralysis.
- Vision problems, e.g. double vision or blindness.
- Coma.
- Incontinence.
- Other infections can occur, e.g. pneumonia which can lead to death.

Treatment of disease

There is no cure for CJD. Medical treatments try to ease the symptoms and aid the patient with the difficulties they have in coping with day to day activities. There are several research projects that are investigating prions, their nature and behaviour and details of their mode of transmission. Increased knowledge of prions will aid the development of new treatments for CJD.

QUESTIONS

1. What is CJD and what causes CJD?
2. What is the most common form of CJD?
3. What is vCJD?
4. How do prion diseases act in the body?
5. Why is CJD classified as a transmissible spongiform encephalopathy (TSE)?
6. (a) Outline the three ways CJD can be transmitted.
   (b) Give the approximate percentages for infection by the three different ways of transmission of CJD.
7. Why is CJD difficult to diagnose?
8. Identify some symptoms of CJD.
9. When do symptoms usually begin for CJD?
10. How is CJD cured and treated?
11. Which of the following diseases has a similar cause and transmission to CJD?
    (A) Measles (B) Malaria.
    (C) Pneumonia. (D) Kuru.

46 Virus Pathogens

Viruses are very small (approximately 30 to 300 nm) and reproduce by taking over a host cell. They are infectious particles with either DNA or RNA genome surrounded by a protein coat called a **capsid**.

A **phage** (bacteriophage) is a virus that infects bacteria. Phages are used as antibacterial agents.

A **viroid** is a plant pathogen consisting of a molecule of naked circular RNA with only a few hundred nucleotides.

Virus infections have been known since ancient times. One of the first written records of a virus infection is a hieroglyph from Memphis, the capital of ancient Egypt dated about 3700 BCE which shows typical paralytic poliomyelitis.

Living or non-living?

There has been debate as to whether they are living or non-living – they are not cellular and hence do not comply with the cell theory that states that all living things consist of cells. They also cannot metabolise or reproduce on their own. But viruses contain nucleic acid (either DNA or RNA) which is a requirement for life. Viruses can be crystallised.

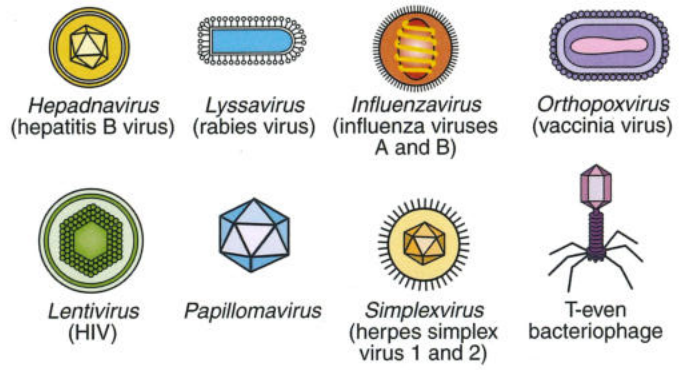


Figure 46.1 Viruses.

Types of viruses

There are four main types of viruses – icosahedral (capsid has 20 flat sides giving it a special shape), helical, (capsid is rod shaped), enveloped (capsid is encased in a baggy membrane) and complex (no capsid).

Features of viruses

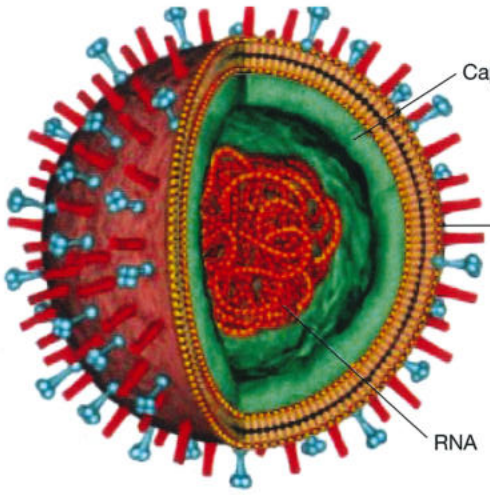
Features include:

- Genome can be DNA or RNA as single linear or circular molecule.
- Capsid is a protein shell with the shape dependent on the type of virus, e.g. rod shape, polyhedral or more complex.
- Viral envelope occurs in some viruses and is derived from the host cell phosopholipids and membrane proteins.
- Viruses can be crystallised and remain viable and infectious.
- They lack metabolic enzymes and ribosomes for making proteins but can express their genes and do chemical synthesis when they are inside a living host cell using the host cell's metabolic equipment and chemical pathways to make more virus proteins and nucleic acid.
- Host range is the limited number of host species that each particular virus can infect. The host specificity is due to a 'lock and key' fit between viral surface proteins and specific receptor molecules on the outside of host cells.

Viral diseases

At present there are no cures for viral diseases but vaccination reduces their incidence. Viral diseases include measles, mumps, poliomyelitis, chickenpox, AIDS, influenza and glandular fever. Some viruses are used to control pest populations, for example myxomatosis in the control of rabbits.

Influenza A virus – H1N1



Causes influenza in birds and some mammals. There are many variants and subtypes and are RNA viruses with a viral envelope. Strains of H1N1 cause flu in humans, pigs and birds. It has caused pandemics, e.g. 1918 Spanish flu and 2009 swine flu and epidemics, e.g. 1978 Russian flu.

Figure 46.2 Influenza A virus – H1N1.

Viral vaccines

The first viral vaccines were based on attenuated or weaker viruses, e.g. smallpox vaccine used cowpox virus.

Vaccines using live attenuated viruses include measles, mumps, rubella and chickenpox.

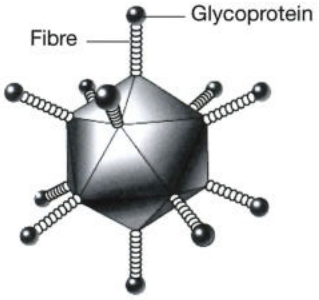
Vaccines using inactivated or killed viruses include polio and hepatitis A. A vaccine that uses segments or conjugates include hepatitis B and the influenza vaccine.

Some viruses keep mutating from one person to the next and vaccination is difficult for these viruses as the virus has already changed its format by the time the vaccine is developed, e.g. common cold.

QUESTIONS

1. What is a virus?
2. Distinguish between a phage and a viroid.
3. Explain why some people consider viruses to be living while others consider them to be non-living.
4. Identify the four main types of viruses.
5. What is a capsid?
6. Describe a viral envelope.
7. Explain why a virus can only metabolise inside a host cell.
8. Outline how viruses cause disease.
9. Outline the disease caused by H1N1 virus and why the disease is closely monitored.
10. The box of information provides data about viral conjunctivitis.

Viral conjunctivitis



Caused by an adenovirus with no viral envelope. The capsid has an icosahedral shape with a glycoprotein spike at each vertex. It is highly contagious and if eyes are touched can be transmitted by shaking hands, sharing towels, contaminated swimming pools.

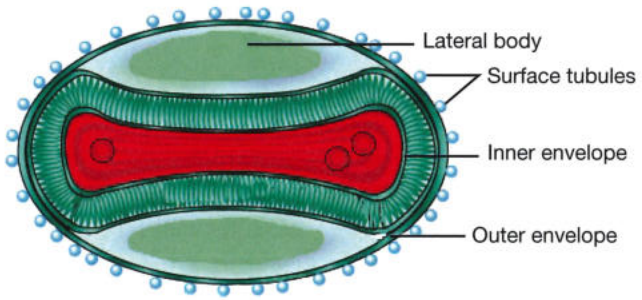
Figure 46.3 Viral conjunctivitis.

(a) How is viral conjunctivitis spread from person to person?
(b) Describe the conjunctivitis virus.

11. What was used in the first viral vaccines?
12. Explain why it is difficult to produce a long term vaccine against some viruses.
13. The vaccine for hepatitis B has been available in Australia since 1982 and the hepatitis B vaccination program started in 1988. Hepatitis B is a life threatening liver infection caused by the hepatitis B virus (HBV) and spread through contact with blood or other body fluids. HBV can survive outside the body for at least 7 days. Suggest why the World Health Organisation recommends vaccination for hepatitis B in the first day of life.

The box of information provides data about smallpox. Use this information for the next TWO questions.

Variola virus and smallpox



Variola virus is a double stranded DNA virus with two envelopes with the outer envelope only present in the extracellular state. The core has a dumbbell shape with two lateral bodies. It is transmitted by direct contact or airborne droplets.

Figure 46.4 Smallpox.

14. Smallpox is a highly contagious disease. The WHO certified the eradication of smallpox in 1979. Which of the following could be used as a vaccine for smallpox?
    (A) Cowpox virus.
    (B) H1N1 influenza virus.
    (C) *Salmonella* bacteria.
    (D) Human immunodeficiency virus.
15. What is the variola virus?
    (A) Single stranded DNA virus.
    (B) Double stranded RNA virus.
    (C) Single stranded DNA virus.
    (D) Double stranded DNA virus.
16. In 1892, Dmitri Iwanowsky discovered that the disease which caused the leaves of tobacco plants to become mottled and wrinkled was caused by an agent that was so small it could pass through a porcelain filter and could not be seen under the microscope. In 1935, WM Stanley isolated and crystallised the agent for this disease. When the crystals were injected into healthy tobacco plants, mould appeared. These experiments show that the tobacco plant disease is caused by a:
    (A) Prion. (B) Virus.
    (C) Bacterium. (D) Fungus.

47 Case Study – Human Immunodeficiency Virus

Human immunodeficiency virus (HIV) is a type of retrovirus that causes acquired immunodeficiency syndrome (AIDS). HIV weakens the immune system allowing other infections and cancers to affect the host,

HIV has a roughly spherical shape about 120 nm in diameter with two copies of single stranded RNA that contains its genetic information. The RNA is enclosed in a conical shaped capsid made of protein. Necessary enzymes, e.g. reverse transcriptase bond to the RNA inside the capsid.

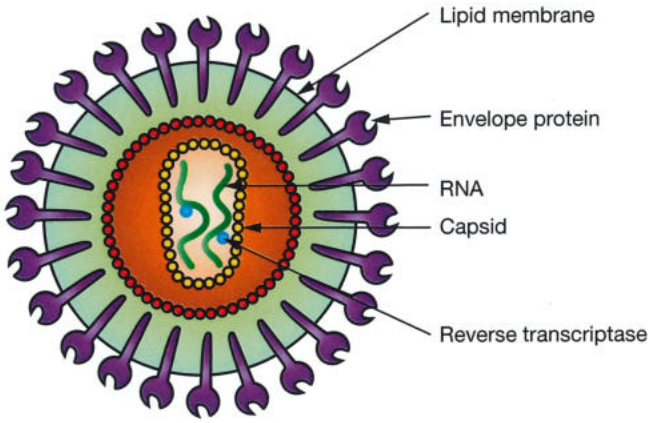


Figure 47.1 Human immunodeficiency virus.

A person becomes HIV positive when they have been infected with the virus and an infected person may develop acquired immunodeficiency syndrome (AIDS) when the virus causes progressive failure of their immune system.

AIDS is a syndrome which means it is diagnosed from a set of symptoms and is not inheritable.

Transmission and cause of disease

HIV can have a long incubation period as the viral genome upon entry into a host cell is converted into a double stranded DNA by the reverse transcriptase.

The viral DNA then moves into the cell nucleus of the host and integrates itself into the host DNA.

Once part of the host DNA the virus can stay latent and undetected by the host immune system or it can cause gene expression to produce new RNA genomes and viral proteins that are released from the cell as new virus particles.

HIV targets and infects helper T cell (CD4 cells), macrophages and dendritic cells. CD4 is a naturally occurring protein found on the surface of helper T cells. When the HIV virus is taken into the cell it begins replicating. Helper T cells are an important part of the immune response.

They secrete cytokines that influence other cells, e.g. they secrete gamma interferon which activates macrophages and interleukins that activate antibody production.

There are several ways that HIV can be transmitted.

- **Sexual contact** is the most common transmission of HIV, e.g. sexual contact with an infected partner. The virus can enter through the lining of the vagina, vulva, penis, rectum or mouth during sex.
- **Contaminated equipment** can transmit HIV, e.g. when sharing contaminated syringes or needles, e.g. drug users, tattoos, piercings.
- **Mother to child** when HIV crosses the placenta during pregnancy or the child is infected during birth or when drinking infected breast milk.
- **Blood** – at one stage HIV was transmitted through blood transfusions or blood products that used infected blood. However, there is now blood screening for evidence of HIV infection and treatments to destroy HIV in blood products.

Symptoms of disease

The symptoms of AIDS can be the same as the symptoms for other diseases, e.g. flu-like symptoms, extreme tiredness, fevers, chills and night sweats, swollen lymph glands, skin blemishes or bumps, a dry cough and white spots or unusual marks in the mouth.

Blood tests can determine the presence of HIV antibodies which are detectable between 6 to 12 weeks after infection.

The destruction of the CD4 cells means that the body cannot fight other infections which leads to the final stage of HIV infection which is AIDS. HIV infection can be classified into three categories.

Category A – mild symptoms similar to glandular fever with swollen lymph nodes mainly on the neck, headaches, fever and dry cough. At this stage the amount of virus in the blood can be high and there is greater chance of transmission than in the later stages of infection. After the initial reaction many people feel better and do not show other symptoms for several years.

Category B – advanced symptoms with persistent generalised lymphadenopathy – swelling of lymph nodes, weight loss, diarrhoea, fatigue, fever, unexplained bleeding and memory loss. Many people get thrush – oral yeast infections or shingles (herpes infections) at this stage.

Category C – final stage with severe symptoms and AIDS when other serious diseases develop such as pneumonia, cancer and brain disorders.

Without treatment an HIV infection will progress to AIDS in about 10 years.

Treatment of disease

At present there is no cure for AIDS.

There are several medications that help control the virus and slow the progression of the disease and also help the person maintain their health for a longer period of time.

Antiretroviral medications used in antiretroviral therapy (ART) target different stages of the virus replication life cycle. However, the medications do not work well for all people and the virus can become resistant to the medication.

There are several treatments that reduce AIDS-related illnesses to help maintain the quality of life for the affected person and those who care for them.

There is pre-test counselling for those who request an HIV test and there is also post-test counselling for people, regardless of the outcome of the test.

Control and management of disease

The main control is safe sex practices, e.g. using a condom during sexual intercourse. Other control measures include not sharing needles, syringes and other injecting equipment.

In Australia there were 1236 newly diagnosed cases of HIV infection in 2013 with about 70% of new diagnoses occurring due to sexual contact between men.

Ethical issues

Worldwide issue – AIDS is a pandemic that has killed more than 39 million people since 1981. It is estimated that 1.5 million people died of AIDS in 2013 which is 22% fewer than in 2009 and 35% fewer than the number in 2005. About 35 million people were living with HIV at the end of 2013. The global nature of HIV infection means everyone needs to be aware of how HIV is transmitted and take precautions against infection.

Uneven regional distribution – At the end of 2013, of the 35 million living with HIV/AIDS, about 69% were living in Sub-Saharan Africa.

Economic productivity – young people with HIV/AIDS may be unable to work, their health care may be expensive and families lose their primary income person or child care person. The country loses skilled labour and there is a collapse in the economy and social structure. If both parents die, society needs to care for the children now orphaned.

Discrimination – there are many ways different groups in society can discriminate against people living with HIV/AIDS, e.g. ostracism, loss of previous social network, work restrictions, travel restrictions, denial of hospital facilities or be subject to violence.

Social stigma – the forms of transmission can lead to prejudices and people living with HIV/AIDS have issues with loss of reputation, feelings of worthlessness, withdrawal from society, e.g. some people associate AIDS with other areas they consider a stigma – homosexuality, promiscuity, prostitution, intravenous drug use.

Education programs can help reduce the incidence of HIV/AIDS and reduce the negative social implications.

QUESTIONS

1. Distinguish between HIV and AIDS.
2. Describe HIV.
3. Explain how HIV can evade the immune system.
4. Which cells are infected by HIV?
5. Identify some ways in which HIV can be transmitted.
6. What are some initial symptoms of HIV infection?
7. Describe the three categories of HIV infection.
8. Outline the cure and treatment for patients that are HIV positive.
9. The diagram shows part of the life cycle of HIV.

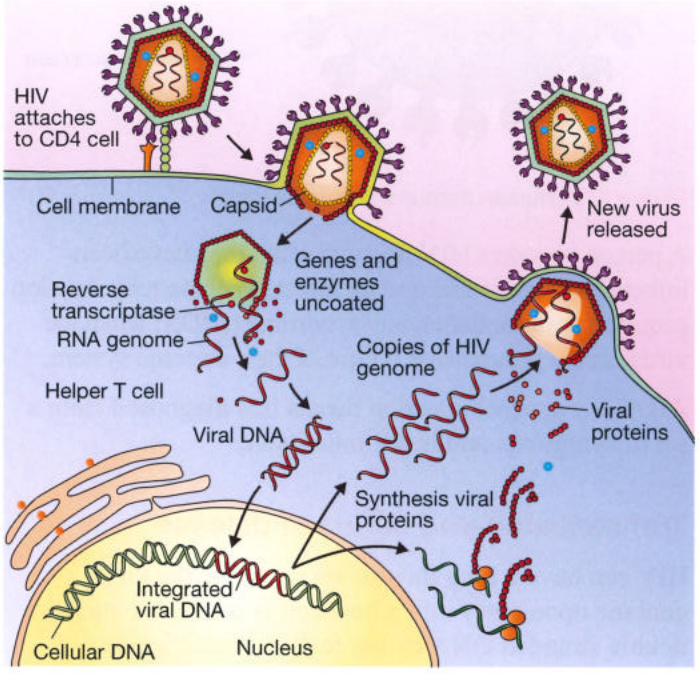


Figure 47.2 Part of HIV life cycle.

Use the diagram to draw a flow chart to show how HIV uses the host cell for replication.

10. Explain why HIV/AIDS is a worldwide issue.
11. Outline why HIV infection can lead to discrimination.
12. What is the best description of HIV?
    (A) Retrovirus with RNA genome.
    (B) Retrovirus with DNA genome.
    (C) Single stranded DNA virus.
    (D) Double stranded DNA virus.

48 Bacteria Pathogens

Bacteria are prokaryotes (no membrane bound organelles or nucleus) ranging in size from 0.5 to 5.0 μm, with a single strand of DNA. Bacteria are present in the soil, air and water and in many human body parts such as skin, throat, mouth, nose and digestive tract. Human armpits have over a million bacteria per square centimetre, although many do not cause disease.

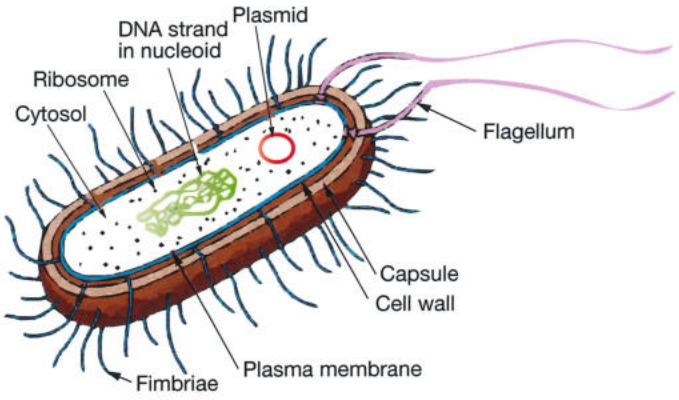


Figure 48.1 Generalised bacterium.

Features of bacteria

Structural features of bacteria include the following.

- A **cell wall** is a rigid structure which can be surrounded by a capsule. It is a protective layer outside the cell membrane.
- The **capsule** is a sticky layer of polysaccharide or protein and if less well organised is called a slime layer. It is a virulence factor as it aids the disease causing ability and protects against phagocytosis by macrophages.
- **Fibriae** are hairlike appendages that help the bacteria stick to their substrate or to one another.
- **Flagella** may be concentrated at one or both ends or can be scattered over the entire surface. They are used for locomotion.
- **Nucleoid** region where the bacterial double stranded circular DNA chromosome is located.
- **Plasmids** are small rings of independently replicating DNA with only a few genes. The genes in the plasmids often provided genetic advantages, e.g. antibiotic resistance.
- **Endospores** – in harsh conditions the bacterium surrounds a copy of its chromosome with a tough multilayered structure forming an endospore and metabolism stops. The original cell lyses and releases the endospores. Endospores can survive boiling and high pressure and can lie dormant for long periods of time.

Bacterial reproduction

Bacteria reproduce asexually by binary fission so that within a short time the host can contain thousands of bacteria. Bacteria produce toxins, often as waste products, which harm the host.

A bacterium can also directly transfer DNA to another bacterium by **conjugation** when two cells are temporarily joined.

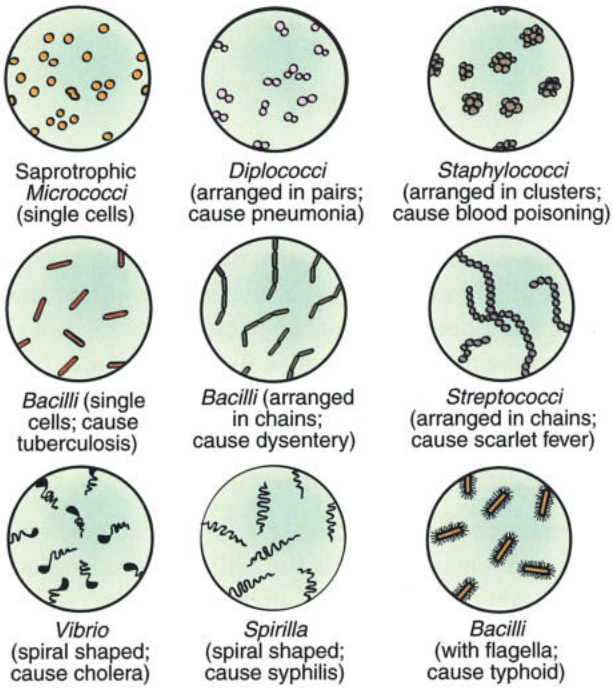


Figure 48.2 Bacteria are classified by their shape.

Bacterial classification

Bacteria can be classified and named by shape, e.g. cocci, bacilli, spirilla, vibrio, rickettsiae and others as shown in Figure 48.2.

Bacteria are also subdivided on Gram staining. Gram staining distinguishes bacteria on the properties of their cell wall. Gram positive bacteria keep the crystal violet dye colour identifying the peptidoglycan in the cell wall while Gram negative stain red or pink. The Gram stain is usually the first step in bacteria identification.

Bacterial diseases

Bacteria cause about half of all human diseases. Bacterial diseases include bubonic plague, diphtheria, gonorrhoea, leprosy, scarlet fever, syphilis, tetanus, tonsillitis, typhoid and whooping cough.

Cholera is caused by an exotoxin secreted by the bacterium *Vibrio cholerae*. The exotoxin causes the intestinal cells to release chloride ions into the gut and water follows by osmosis. This leads to severe diarrhoea, dehydration and an electrolyte imbalance.

In plants, bacteria are found in the conducting and leaf tissues. They cause some leaf spot diseases, bacterial canker on tomatoes, bacterial soft rot of lettuce and carrots and some rot diseases such as stem rot of potatoes, although most plant diseases are caused by fungi. Since many bacteria can survive in the soil, crop debris, seeds, plant parts and in weeds they can remain dormant until suitable conditions allow them to multiply. Favourable conditions include high humidity, crowding, poor air circulation and warm, wet weather.

Bacteria and antibiotics

Antibiotics are chemicals that kill or inhibit the growth of bacteria and are used to treat bacterial infections. Antibiotics are produced by soil bacteria and fungi. Antibiotics can inhibit cell wall synthesis, inhibit protein synthesis (e.g. translation), alter the cell membrane and its permeability or inhibit nucleic acid synthesis. For example, penicillins affect cell wall synthesis and the tetracyclines are protein synthesis inhibitors.

Bacterial vaccines and toxoids

A bacterial **exotoxin** is a toxin, e.g. protein secreted by bacteria and released outside the cell. They are produced by both Gram negative and Gram positive bacteria. There are many types of exotoxins based on structure and function. Exotoxins may be secreted or like endotoxins are released during lysis of the cell.

Bacterial **endotoxins** are part of the outer membrane of the cell wall of Gram negative bacteria. They are lipopolysaccharides and only one type of endotoxin.

Vaccines contain killed or attenuated bacteria that activate the immune system. Antibodies are created against that particular bacteria, e.g. whooping cough vaccine has killed bacteria while measles vaccine has living, attenuated bacteria.

Toxoids are modified toxins that have the antigen but it cannot destroy tissues, e.g. toxoids for tetanus and diphtheria. Booster injections for vaccines and toxoids increase the number of antibodies and the number of circulating memory cells to sustain immunity.

QUESTIONS

1. Define a prokaryote.
2. How large are bacteria?
3. Where are bacteria found?
4. Construct a table to summarise the function of different parts of a bacterium.
5. How do bacteria reproduce?
6. Distinguish between cocci, bacilli and spirilla bacteria.
7. How is the Gram stain used in bacteria identification?
8. (a) Outline the cause of cholera.
   (b) Identify some symptoms of cholera.
9. The diagram shows a generalised bacterium.

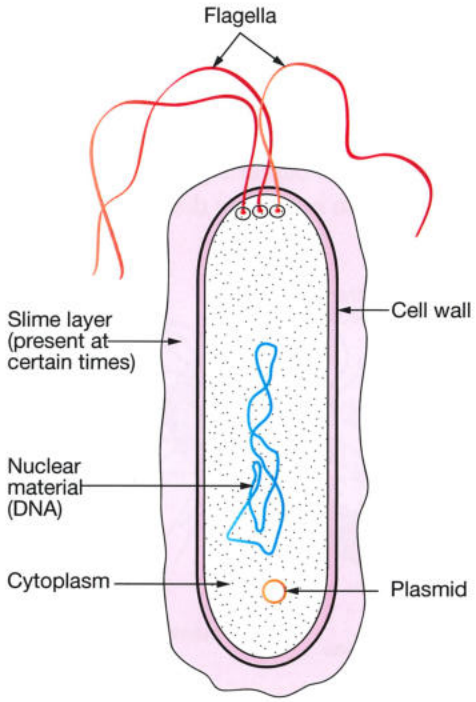


Figure 48.3 A bacterium.

Explain why numbers of a bacterial pathogen can rapidly increase.

10. Describe how many bacteria are named and classified.
11. Outline why bacteria cause disease.
12. Distinguish between an exotoxin and an endotoxin.
13. (a) What is an antibiotic?
    (b) How are antibiotics used?
    (c) Outline some ways in which antibiotics work.
    (d) Name an antibiotic and state how it works.
14. How do vaccines work against pathogenic bacteria?
15. What is a toxoid?
16. Why are booster injections needed in vaccination programs?
17. Which choice correctly matches the infectious disease, pathogen and classification of pathogen?

	Infectious disease	Pathogen	Classification of pathogen
(A)	AIDS	HIV	Virus
(B)	HIV	HIV	Bacteria
(C)	HIV	AIDS virus	Virus
(D)	AIDS	*Streptococcus* HIV	Bacteria

18. Which group lists diseases caused by bacteria?
    (A) AIDS, influenza, poliomyelitis.
    (B) Pneumonia, cholera, tetanus.
    (C) Ringworm, tinea, thrush.
    (D) Malaria, amoebic dysentery, sleeping sickness.

49 Case Study – Cholera

Cholera is caused by the vibrio bacteria – *Vibrio cholerae*. The bacteria has a direct life cycle; it is ingested in contaminated food or water, infecting the bowel of the human and is then egested with the faeces. Incubation may be a few hours to five days.

For at least 2000 years, cholera was endemic to the southern parts of India, spreading to the rest of the world along the trade routes. In 1854, John Snow established the link between cholera and its source – sewage-contaminated water. Also in 1854, Filippo Pacini first clearly described the cholera vibrio. *Vibrio cholerae* can live in both fresh and salt water and survives outside the human body with no need for an animal host or insect vector. Due to the severe diarrhoea it causes, anything fouled can transmit the disease, for example bedding, clothing and unprotected water sources. The European cholera epidemic of 1831-32 claimed many lives.

In 2010 there was an outbreak of cholera in Haiti ten months after an earthquake hit the island. In 2012 during the rainy season the epidemic returned and it is estimated that more than 5% of the population were infected with over 7000 deaths.

Figure 49.1 *Vibrio cholerae*.

Major symptoms and host response

The bacteria produce a toxin which interrupts the sodium and potassium transport systems in the intestine. The bacteria attaches to the mucous lining of the microvilli, causing excessive secretion of fluids and electrolytes.

The lining of the intestine is broken down by bacterial enzymes. Reabsorption of electrolytes, nutrients and fluid is prevented. Muscle cramps are experienced. The bowel becomes inflamed (enteritis), causing severe 'rice water stool' diarrhoea, and loss of water and salts results in dehydration; death can occur within 12 hours. Cholera victims can lose 10 litres of water a day.

Victims develop cold, shrivelled skin and sunken eyes, blood pressure falls and the pulse becomes faint.The toxin causes antibodies to be produced. These antibodies provide resistance to the same strain of bacteria.

Treatment

It is essential to replace the fluids and salts. Antibiotics are effective only in the early stages and if given during the first day reduce the period of diarrhoea and the need for fluid replacement. If treated with the proper amount of glucose, cholera patients can still reabsorb fluid. Oral rehydration solution (ORS) containing sodium and potassium chloride and glucose and sodium bicarbonate prevents dehydration. A simple recipe for treatment is 1 litre of water to 1 teaspoon of salt and 8 teaspoons of sugar.

Control

The control of a disease can involve many factors, such as laws, infrastructure and economics, necessary to create an environment where the pathogen cannot survive or be transmitted. For cholera, it is important to prevent contaminated faeces infecting the water supply. Chlorination of water and an effective sewerage system can guard against this. Cholera is a notifiable disease and all outbreaks of cholera in cholera-free zones are reported to the WHO which has established regulations to prevent travellers spreading the disease to other areas.

Prevention

Prevention is a type of control which prevents people from becoming infected. For example, travellers who wish to visit areas where cholera is found can have vaccines that are effective for six months. They should avoid drinking water or eating food that may be contaminated.

Cholera in Australia

The European epidemic of cholera in 1831 led the Bourke Government to introduce Australia's first Quarantine Act in 1832. The area within a quarter mile of the high water mark of Spring Cove in Sydney Harbour was established as a Quarantine Area in 1833 and four years later the area was extended to the whole of the North Head Peninsula.

QUESTIONS

1. Identify the cause and transmission of cholera.
2. Describe the host response to *Vibrio cholerae* infection.
3. Outline the major symptoms of cholera.
4. Describe oral rehydration therapy.
5. How can cholera be prevented?

6. On the diagram identify the point of entry for the pathogen, the site of colonisation and the point of exit from the body.
7. Distinguish between prevention and control.
8. How did the European cholera epidemic in 1831 affect Australia?
9. Cholera is a bacterial infectious disease common in tropical countries. Identify the most likely method of transmission.
   (A) Ingestion of contaminated food or water.
   (B) Inhalation of airborne droplets.
   (C) Direct contact.
   (D) Vector.

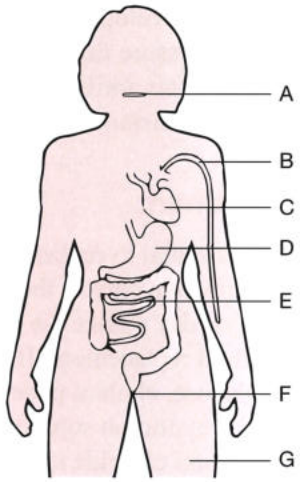


Figure 49.2 *Vibrio cholerae* in the body.

10. The bacterium *Vibrio cholerae* infects the human intestine disrupting the salt and electrolyte transport in the lining of the intestine. The bacteria also produce an enzyme that damages the intestine. Identify the most likely effect on the human.
   (A) Anaemia.
   (B) Severe diarrhoea.
   (C) Pink spots on the skin of the abdomen that become itchy.
   (D) Fever with regular periodic shivering.
11. Drugs such as tetracycline and chloramphenicol can be used to treat cholera. Which of the following gives the most probable reason why drugs can be used against the spread of cholera?
   (A) It is a genetic disease.
   (B) It is a disease with a long incubation period.
   (C) It is caused by a virus.
   (D) It is caused by a bacteria.
12. Cholera was endemic in southern India for at least 2000 years before it gained world attention in 1817 when travellers reported the disease spreading along the trade routes. It had reached Moscow by 1831. People feared cholera as it claimed the lives of so many regardless of age, wealth, or fitness. Which of the following gives one of the reasons why cholera is such a threatening disease?
   (A) A protozoan parasite has both a human host stage and an insect vector host stage.
   (B) A bacterial parasite can be an intracellular parasite on other bacteria.
   (C) Vibrio bacteria can survive outside the human body with no animal host or insect vector.
   (D) Virus is easily transmitted by close contact with blood and vomit.
13. It has been suggested that the periodic outbreaks of cholera are related to the climatic conditions of southern India, where the disease is endemic. The vibrio bacteria survives and increases in numbers in water storage areas such as wells and cisterns. When the monsoon rains come, the bacteria is flushed out of these storage places and spread over wide areas causing an epidemic. Some parts of Australia have similar climatic conditions to southern India. Which of the following is a possible method to control the spread of cholera in Australia?
   (A) Cover water storage areas to prevent the bacteria entering the water.
   (B) Continually flush the water storage areas. through the dry season with salt compounds.
   (C) Prevent contaminated faeces infecting the water supply.
   (D) Adjust the pH of water in storage areas so it lies within a range of pH 2.0 to pH 8.5.
14. In 1854, John Snow studied the spread of cholera through London districts, charting maps to show its occurrence. An analysis of his results convinced him that a common factor involved the water from the Broad Street pump. He persuaded the aldermen to remove the hand of the Broad Street pump. What link was established by the work of John Snow?
   (A) The cause of cholera and the life cycle of the pathogen.
   (B) A cholera epidemic and its source.
   (C) Host response to cholera and bacterial toxins.
   (D) The vector and the source of cholera.
15. The worldwide cholera epidemic of 1831 led to Australia's first Quarantine Act being established by the Bourke Government in 1832. Which of the following resulted from this Act?
   (A) The Federal Government assumed responsibility for quarantine.
   (B) The establishment of a Board of Health.
   (C) Statewide immunisation program against cholera.
   (D) The establishment of a quarantine area at North Head Peninsula.
16. Which of the following is the most suitable control to prevent the spread of cholera?
   (A) Provide continual medication to the population of drugs such as tetracycline.
   (B) Chlorination of the water supply.
   (C) Rapid fluids and salt doses to people.
   (D) Avoid sexual contact with infected people.

50 Antibiotics and Bacterial Infections

Antibiotics are chemicals which inhibit or destroy micro-organisms such as bacteria. They are drugs that have a toxic effect on microbial cells, inhibiting the growth or destroying pathogens. They affect the micro-organism but not the host.

Discovery of antibiotics

In 1928 Alexander Fleming (1881-1955) discovered the first antibiotic when he observed the effect of *Penicillium* mould on the growth of bacterial colonies on an agar plate. He realised the mould produced a substance that killed nearby bacteria.

A decade later an Australian, Howard Florey, and Ernst Chain developed methods to isolate and produce antibiotics on a large scale. Fleming, Florey and Chain received the 1945 Nobel Prize for medicine for their work.

Many antibiotics have been discovered but only a few are safe for human use. The first antibacterial drug used on humans was sulfonamide in 1935. During the 1940s research produced a variety of drugs derived from penicillin. In the 1950s, two tetracyclines and chloramphenicol were also used for infectious diseases.

How antibiotics work

Antibiotics work in several ways. Penicillin disrupts the cell wall of bacteria. Amphotericin destroys cell membranes. Streptomycin disrupts protein synthesis. Broad spectrum antibiotics act against a range of bacteria, including non-pathogenic forms. Some patients have side effects from the use of antibiotics ranging from vomiting, nausea, diarrhoea and skin rashes to more serious effects such as deafness, loss of balance, damage to bone marrow and serious allergy.

Resistance to antibiotics

Natural selection has led to the evolution of some strains of bacteria resistant to antibiotics. In order to minimise the number of resistant strains, it is important that people take the complete course of their antibiotic and that antibiotics are not overprescribed.

The evolution of antibiotic resistant bacteria is accelerated by the following.

- The overuse of antibiotics, e.g. in the treatment of many diseases such as viral infections – 'cough' and 'colds'.
- When symptoms disappear and people stop taking the antibiotics some resistant forms of the bacteria may survive and cause a future infection.
- Overuse of antibiotics for food-producing animals, e.g. chickens and pigs are fed antiobiotics to prevent infection and to promote growth.
- Producing cleaning products that contain antibacterial ingredients adds a selective pressure into the environment that favours the survival of antibiotic resistant strains of bacteria. In most instances the quantities of these products are not enough to kill all the bacteria.

QUESTIONS

1. What is an antibiotic?
2. Describe the discovery of penicillin.
3. Describe how penicillin functions.
4. Identify the role of antibiotics in controlling bacterial infections.
5. Explain why many people cannot use penicillin.
6. Explain why, in some instances, penicillin does not cure diseases that it once cured.
7. Assess two possible implications of the widespread use of antibiotics on society and the environment.
8. Antibiotics are effective against many urinary tract infections, most ear infections and some sinus infections but are ineffective against all colds and flu, most coughs and most sore throats. Outline why antibiotics do not cure everything.
9. It has been discovered that some pathogenic bacteria, e.g. *Shigella* spp which cause dysentery, and *Yersinia enterocolitica* which is related to the plague bacteria have injectisomes on their surfaces. Injectisomes are tiny structures that act like a 'molecular syringe' to help the bacteria to get into the cells or to inject effectors to deactivate immune systems.

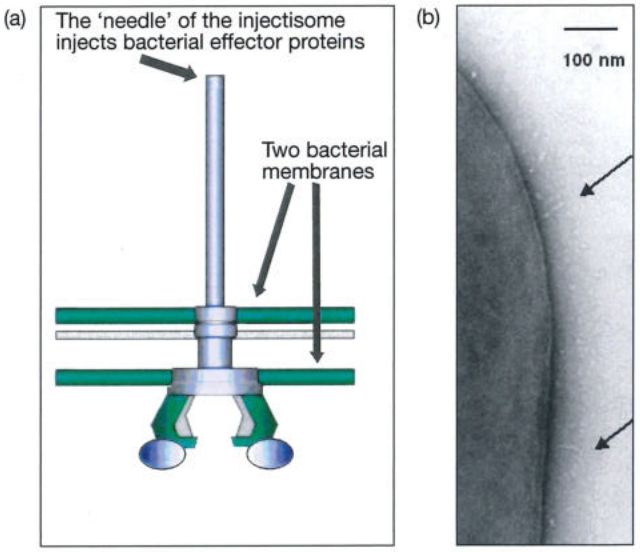


Figure 50.1 An injectisome of *Yersinia*. (a) A drawing of the injectisome. (b) Electron micrograph of needles extending from the bacterial surface.

The bacteria also has a lipopolysaccharide on its surface that hides the needles from antibodies. These features have been called the 'sword and shield' mechanism. Discuss why research into the ways in which bacteria attack cells is important in the development of new antibiotics.

10. Research the diseases listed in the table and complete the table.

Table 50.1 Diseases.

Type of pathogen	Disease	Pathogen	Symptoms	Treatment/ prevention
Bacteria	Cholera	*Vibrio cholerea*		
	Tetanus	*Clostridium tetani*		
	Syphilis	*Treponema pallidum*		
	Whooping cough	*Bordetella pertussis*		
Virus	AIDS	HIV		
	Poliomyelitis	Poliovirus		
	Smallpox	*Variola major*		
	Influenza	Influenza A, B, C viruses		
	Rubella	Rubella virus		
Fungi	Tinea	Three genera: *Microsporum*, *Erichophyton*, *Epidermophyton*		
	Thrush	*Candida albicans*		
Protozoa	Malaria	Four *Plasmodium* species		
	Amoebic dysentery	*Entamoeba histolytica*		
	African sleeping sickness	*Trypanosoma gambiense*		
Macro-parasite	Human tapeworm	*Taenia saginata*		
	Threadworms	*Enterobius vermicularis*		
	Lice	*Pediculus humanus capitus*		
	Scabies	*Sarcoptes scabei*		

11. Antibiotics are often prescribed by a doctor when you have a viral infection such as a cold because:
    (A) Antibiotics are extremely effective against viruses.
    (B) Colds are often accompanied by secondary bacterial infections.
    (C) Antibiotics stimulate the production of antigens.
    (D) The cell membrane of viruses is destroyed by antibiotics.

12. Which is *not* a means of transmission of an infectious disease?
    (A) Airborne droplets.
    (B) Water supply.
    (C) Malnutrition.
    (D) By a vector.

13. The diagram shows testing drug sensitivity of *Staphylococcus aureus*. The agar plate was completely covered with *Staphylococcus aureus* colonies. The antibiotic discs A, B, C and D were then added. The antibiotics that were effective destroyed the bacteria around them, leaving a clear ring.

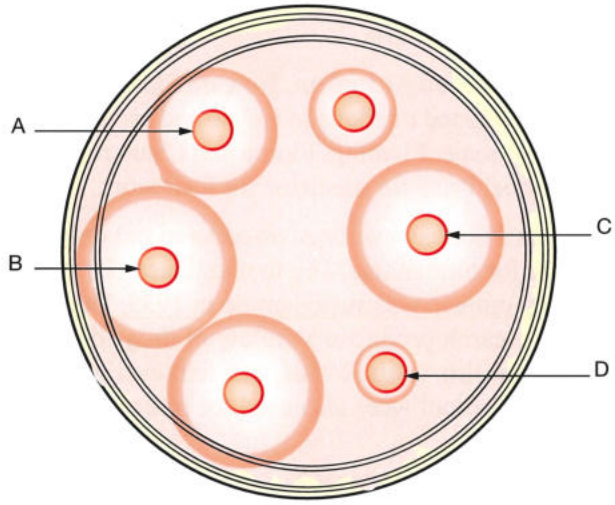


Figure 50.2 Drug sensitivity testing of *Staphylococcus aureus*.

Which drug would be least effective against *Staphylococcus aureus*?
    (A) A
    (B) B
    (C) C
    (D) D

14. The lower section of the male and female urethra can be colonised by *Mycobacterium smegmatis*, *Corynebacterium* spp, *Streptococcus* spp and *Staphylococcus epidermidis*. Identify the specific conditions needed by commensal bacteria which inhabit a specific region of the human body.
    (A) Nutritional conditions.
    (B) pH.
    (C) Temperature.
    (D) All of the above.

51 Protist Pathogens

The **protists** are an extremely diverse group of eukaryotes. Most are unicellular but some are colonial and some are multicellular.

As a group protists have relatively simple levels of organisation and few features common to all, e.g. they have different life cycles, different specialised structures, different modes of nutrition, different reproduction methods and they occupy diverse habitats.

At one stage there was the kingdom Protista but the diverse nature of the group and eukaryotic systematics has led to reclassification.

Protozoans

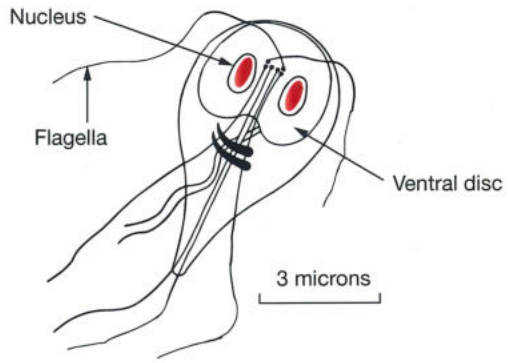
The **protozoans** are a diverse group of mostly motile unicellular, heterotrophic eukaryotic organisms that were in the Protista kingdom. Protozoans range in size from 2 to 1000 μm and many are easily seen under a school light microscope. The protozoans were classified by their form of locomotion, e.g. flagella, cilia, pseudopods or sporozoa. The term protozoa is **no longer** used in taxonomy.

Diplomonads and parabasalids

The diplomonads and parabasalids are two clades of protists that lack plastids and have mitochondria that lack DNA for enzymes in the citric acid cycle. They are anaerobic.

Diplomonads have two equal sized nuclei and multiple flagella, e.g. *Giardia lamblia*.

Giardia



Giardia is a genus of anaerobic flagellated protozoan parasites, e.g. *G. lamblia* that live and reproduce in the small intestines of several vertebrates, e.g. dogs, cats, sheep, cattle. They cause the disease giardiasis with stomach cramps, diarrhoea and excessive gas. They have two nuclei each with four associated flagella and lack mitochondria and Golgi apparatus. Giardia spreads when hands, objects or food is contaminated with faeces of infected people or animals or by contaminated drinking water, e.g. contaminated streams, ponds, swimming pools.

Figure 51.1 *Giardia* is a genus of diplomonad parasites.

Parabasalids are flagellates with one or more clusters of flagella near the anterior of the cell. They include the trichomonads which have 4 to 6 flagella, e.g. *Trichomonas vaginalis*.

Kinetoplastids

Kinetoplastids are flagellated protists with a kinetoplast which is an organised mass of DNA in a single large mitochondrion. The kinetoplast can contain many copies of the mitochondrial genome.

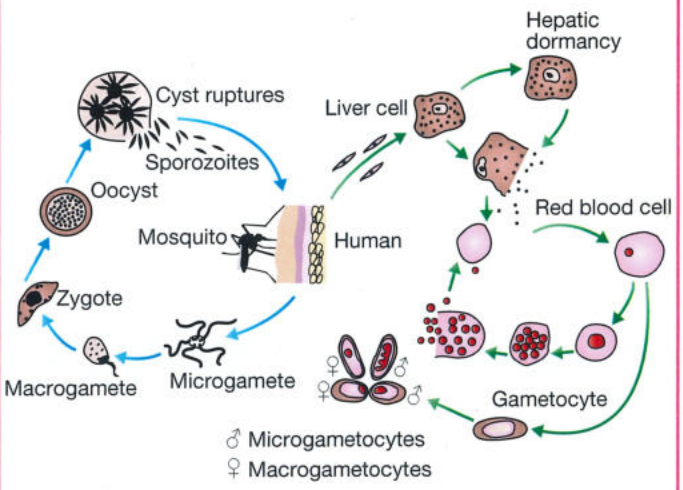
The kinetoplastids include free living and parasitic species, e.g. the trypanosomes include the genus *Trypanosoma* with *T. brucei* which causes human sleeping sickness and *T. cruzi* which causes Chagas' disease.

Apicomplexans

All apicomplexans are protist parasites of animals with most having life cycle stages that help them survive in several different environments. The life cycles often involve two or more different species for completion and have both sexual and asexual stages.

The sporozoite is the cell form that infects a new host and it has a complex of organelles at the cell apex that aids penetration into the host cells and tissues.

Malaria



Caused by *Plasmodium* which is transmitted to humans in the saliva of a bite of an *Anopheles* mosquito. The sporozoites enter the liver and divide and become merozoites.

Merozoites enter red blood cells, multiply and break out releasing toxins that cause the chills and fever. The incidence of malaria decreased in 1960s with the use of insecticides against *Anopheles* mosquito populations and antimalarial drugs but there are now mosquito varieties resistant to insecticides and *Plasmodium* varieties resistant to drugs. Thus there is a resurgence of malaria.

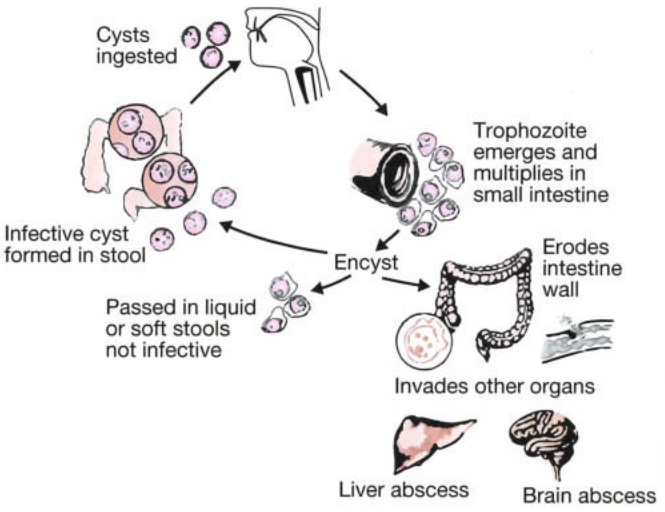
Figure 51.2 *Plasmodium* is an apicomplexan parasite.

Amoebozoans

Amoeboid protists include many amoebas and the slime moulds. They all have an endoplasm which is a granular central mass and an ectoplasm. **Amoeboid movement** involves the formation of pseudopodia with the cytoplasm sliding forward in front of the cell and the cell changes shape.

The genus ***Entamoeba*** are all parasites that infect many different animals. There are at least six species that infect humans but only *E. histolytica* is pathogenic.

Amoebic dysentery



Amoebic dysentery is caused by the amoeba *Entamoeba histolytica* which is transmitted through contaminated food and water.

The amoeba form infective cysts which are found in faeces. Free amoeba known as trophozoites are without cysts and do not survive for long outside the human gastrointestinal tract. The amoeba can burrow through the intestinal wall and spread through the bloodstream to infect other organs, e.g. liver, lungs and brain.

There is no vaccine and the amoeba are destroyed by antimicrobial drugs, e.g. metronidazole.

Figure 51.3 *Entamoeba histolytica.*

QUESTIONS

1. What are the protists?
2. Why was the kingdom Protista abandoned?
3. Identify some ways in which the protists are dissimilar.
4. What is a protozoan?
5. How were protozoans classified?
6. Why is it unlikely you will find 'protozoan' in a classification system?
7. (a) Describe *Giardia lamblia.*
   (b) Describe the disease caused by *G. lamblia.*
   (c) How is *G. lamblia* transmitted to a new host?
8. What is a kinetoplast?
9. The diagram shows a diagram of *Trichomonas vaginalis.*

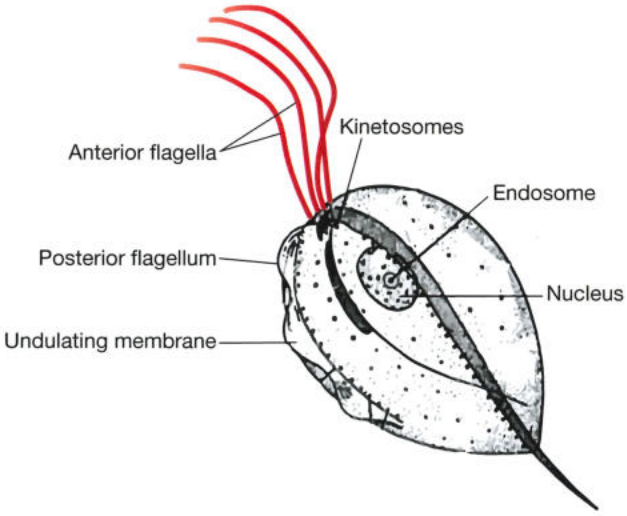


Figure 51.4 *Trichomonas vaginalis.*

What features show that *T. vaginalis* is a parabasalid?

10. What are apicomplexans?
11. What is a sporozoite?
12. Use an example to show the sporozoite and complex life cycle of an apicomplexan protist parasite.
13. Explain why there was a drop in the incidence of malaria in the 1960s and then a resurgence of the disease.
14. Which organisms are amoeboid protists?
15. Describe amoeboid movement.
16. (a) Describe amoebic dysentery.
    (b) How can the amoeba be destroyed?
17. In microbiology, sterility means the absence of all living organisms. Heating to a temperature of 100°C causes protein to break down, yet this high temperature does not guarantee sterility. Why is sterility not guaranteed?
    (A) Viruses do not contain protein.
    (B) Unicellular fungi can crystallise.
    (C) Some bacteria form resistant spores.
    (D) Prions have a protective cell wall resistant to high temperatures.
18. A micro-organism causing a disease is described as being cellular, heterotrophic, eukaryotic and with no rigid cell wall. What is the most likely classification of this organism?
    (A) Protist. (B) Prion.
    (C) Bacteria (D) Virus.
19. What is the name of the infectious cell in an apicomplexan life cycle?
    (A) Pseudopod. (B) Gametocyte.
    (C) Kinetosome. (D) Sporozoite.

52 Case Study – Cryptosporidium

Cryptosporidium is a genus of apicomplexan protists that cause the disease cryptosporidiosis. Several species infect mammals with *C. parvum* and *C. hominis* being the main species that infect humans. They cause the disease **cryptosporidiosis**.

Cryptosporidiosis is a group B disease and must be notified in writing within 5 days of diagnosis. School exclusions apply.

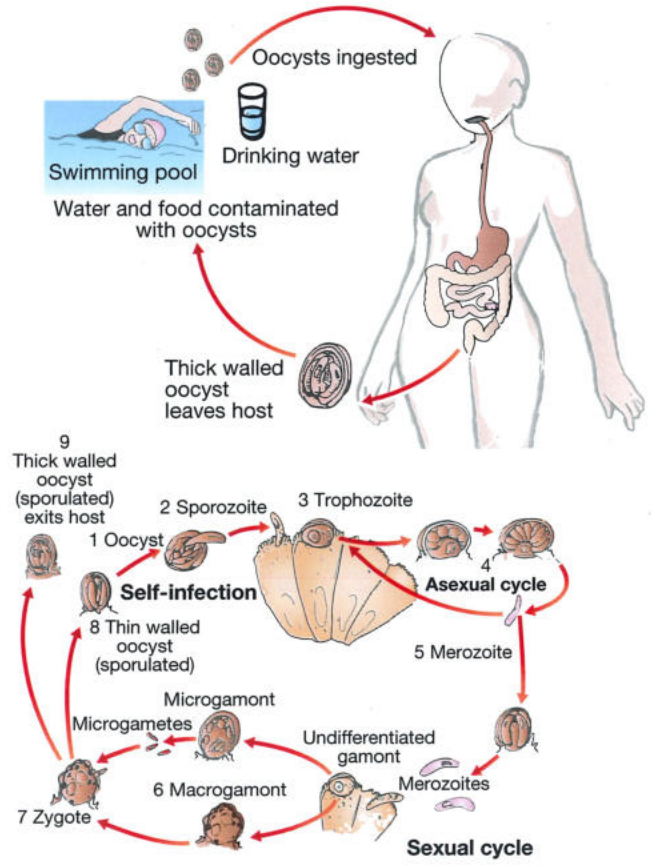


Figure 52.1 *Cryptosporidium* life cycle.

Transmission and cause of disease

The oocysts of *Cryptosporidium* are thick walled and present in faeces of infected hosts. The **oocyst** is a spore about 4 to 6 micrometres in diameter that can be present in food and water and survive for long periods outside a host.

Transmission occurs when the contaminated food or water is ingested. **Self-infection** can occur when a person has poor personal hygiene practices.

The thick wall of the oocyte is resistant to many cleaning chemicals and disinfectants and can survive in chlorinated swimming pools for short time spans but long exposure to high levels of chlorine destroys the oocyst.

Transmission is often linked with child care centres, public swimming pools and contaminated water supplies.

A person becomes infectious as soon as the symptoms develop and they can stay infectious for several weeks after the symptoms disappear.

In March 2013 several swimming pools in Melbourne were temporarily closed and others cleaned with a 'superchlorination' due to a 300% increase in gastroenteritis caused by *Cryptosporidium*.

In 1998 there was a water crisis in Sydney when *Cryptosporidium* and giardia were found in the water supply leading to a 'boil tap water' alert. In 1999 the Sydney Catchment Authority was created to take control of the catchment areas and dams as the contamination was caused by low quality raw water entering dams.

Life cycle of cryptosporidium

Like other apicomplexan protists *Cryptosporidium* forms sporozoites. The sporozoites are released from the ingested oocyst. The sporozoites invade the epithelial lining of the intestines and in these cells undergo asexual reproduction.

Asexual reproduction will produce merozoites that are identical daughter cells. The sexual cycle produces a zygote.

Symptoms of disease

Cryptosporidium is a gastrointestinal disease that causes watery diarrhoea, stomach cramps and bloating. Ingested oocysts travel to the small intestine and invade the epithelial cells lining the intestine wall.

There is usually an incubation period of around 7 days before symptoms appear after infection but symptoms can sometimes appear within one day or as late as 12 days after infection.

Most people are not seriously affected with symptoms lasing 1 to 2 weeks. Though children and adults with poor immune systems, e.g. people with AIDS can become more seriously ill with vomiting and fever and they will be ill for a longer period of time with severe prolonged diarrhoea.

Treatment of disease

Cryptosporidium can be identified from a stool sample viewed under a light microscope. *Cryptosporidium* can also be identified in a polymerase chain reaction (PCR) test in a pathology laboratory.

Treatment involves care for the individual to make them as comfortable as possible. Like other types of gastroenteritis diseases people are advised to take plenty of fluids to prevent dehydration.

It is also advisable to avoid foods high in sugar or fat. In severe cases fluid and electrolyte replacements may need to be administered and if blood is present in the diarrhoea medical advice should be immediately sought.

Control and management of disease

Cryptosporidiosis is a nationally notifiable disease within Australia. Only confirmed cases need to be notified with laboratory definitive evidence of the presence of *Cryptosporidium*.

Water treatment plants using filtration and coagulation steps remove nearly all *Cryptosporidium* oocysts from the water. Filtering through a reverse osmosis filter, an 'absolute one micron ' filter or cyst removal filter will also remove *Cryptosporidium* from water. High levels of chlorine and exposure to UV light will inactivate *Cryptosporidium*.

Good person hygiene practices, e.g. washing hands with soap and water after going to the toilet prevents the spread of the pathogen. Hand washing is also needed when changing nappies and handling food or animals. *Cryptosporidium* is not killed by alcohol gels and hand sanitisers.

If a person becomes infected, e.g. a young child it is recommended that they stay home from school or from the child care centre until the diarrhoea has stopped and that they do not swim in swimming pools for two weeks after the diarrhoea has stopped.

When travelling in countries with minimal water treatment and sanitation systems it is advised to avoid drinking tap water or consuming untreated ice. It is also advisable not to swallow recreational water, e.g. in swimming pools, hot tubs, lakes or streams.

QUESTIONS

1. What disease is caused by *Cryptosporidium*?
2. What is *Cryptosporidium*?
3. Which species of *Cryptosporidium* infect humans?
4. What part of the life cycle involves the oocyst?
5. How can self-infection occur?
6. What happened in the 1998 Sydney water crisis?
7. When is a person infectious?
8. Identify some symptoms of cryptosporidiosis.
9. Give an example of direct transmission and an example of indirect transmission of cryptosporidium.
10. How long do the symptoms normally last?
11. How is cryptosporidiosis diagnosed?
12. What is a legal requirement if a person has cryptosporidiosis?
13. What advise would you give to people travelling overseas about possible infection of *Cryptosporidium*?
14. Describe some good hygiene practices that prevent the spread of *Cryptosporidium*.
15. On 3 August 1998 the Sydney Water Corporation announced that the city's tap water was safe to drink. *Cryptosporidium* and *Giardia* had been detected in Sydney's water supply on 21 July 1998 and on 30 July all Sydney residents were advised to boil tap water before consumption. Explain why a 'boil alert' was issued to Sydney residents.
16. The diagram shows steps in water treatment.

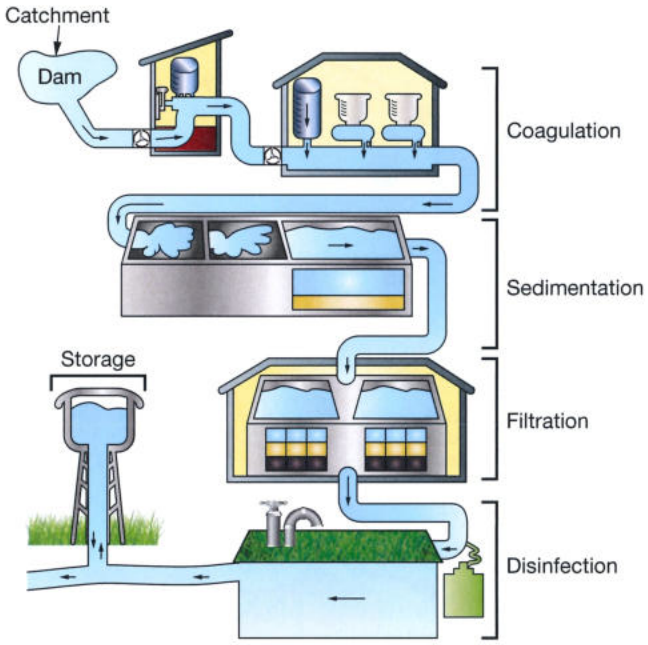


Figure 52.2 Water treatment.

Which steps remove nearly all *Cryptosporidium* during water treatment?
(A) Disinfection.
(B) Coagulation and storage.
(C) Sedimentation and filtration.
(D) Coagulation and disinfection.

17. What type of organism is *Cryptosporidium*?
(A) Apicomplexan protist.
(B) Amoebaean protist.
(C) Virus.
(D) Bacteria.
18. In November 2010 there was an outbreak of cryptosporidiosis in Ostersund, Sweden caused by *Cryptosporidium hominis* that was present in drinking water. What advice would be suitable to give to the people of Ostersund while the drinking water was contaminated?
(A) Add chlorine to all drinking water.
(B) Boil drinking water.
(C) Use alcohol gels and wipes before and after handling any foods.
(D) Do not go swimming.

53 Fungi Pathogens

Fungi are eukaryotes, do not contain chlorophyll, are heterotrophs and can be unicellular, such as yeasts, or multicellular, such as mushrooms. Fungal cell walls contain chitin and they digest their food by secreting exoenzymes that break down complex molecules into smaller molecules the fungi can absorb. They are widespread and diverse.

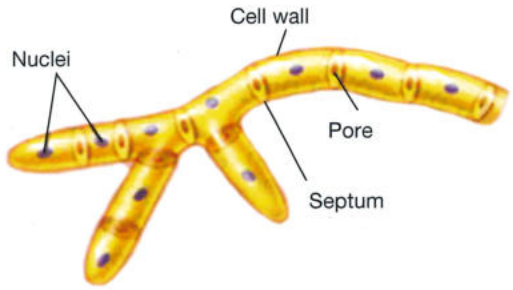


Figure 53.1 Fungi hyphae.

Fungi can be saprophytic (living on dead matter) or parasitic (living on a host). Some fungi form symbiotic relationships with other organisms, e.g. lichens are a symbiotic relationship of photosynthetic micro-organisms such as cyanobacteria or filamentous green algae held in a mass of fungal hyphae.

Many fungal diseases are caused by fungi that are common in the environment.

Pathogenic fungi plant diseases

Around 30% of fungi are parasites with many causing plant disease. About 80% of plant diseases are caused by fungi, for example leaf curl, rot, smut, rust, powdery mildew and black spot.

Epidemics that destroy crops are often fungal diseases, e.g. the Irish potato famine that started in 1845 and lasted until 1852 was due to potato blight *Phytophthora infestans*.

Pathogenic fungi animal diseases

A fungal infection is known as mycosis with fungal diseases on animals usually being less serious, affecting the skin, hair and lungs. In many animals the skin and mucous membranes act as preventative barriers to fungal penetration.

In **amphibians** the chytrid fungus *Batrachochytrium dendrobatidis* has spread globally adding to the other problems of habitat destruction, loss of breeding sites by drainage of wetlands, use of insecticides and herbicides and other human activities which have seen a worldwide decline in frog population numbers. The fungus can cause sporadic individual deaths though there is no agreement about the main cause of the decline in frog populations.

In **humans** several superficial infections are caused by fungi, e.g. infections of the skin, hair or nails such as ringworm and athlete's foot with these fungi feeding on keratin and preferring warm, moist areas. Different species cause different infections, e.g. *Tinea pedis* causes athletes foot, *Tinea corporis* infects the arms, legs and trunk, *Tinea capitis* infects the scalp and hair and *Tinea cruris* infects the groin region.

Thrush (candidiasis) is a yeast infection caused by *Candida albicans*. It naturally occurs in the intestines and is usually harmless but a disruption to the natural microflora in the bowel can lead to an overgrowth, e.g. vaginal thrush involves a white discharge and itching/burning sensation. *Candida* can infect the mouth causing oral thrush and can cause diaper rash in babies. Thrush is one of the most common fungal infections.

Figure 53.2 *Candida albicans*.

Some fungi can infect deep tissues and organs and spread to other regions of the body. **Histoplasmosis** is a fungal disease caused by breathing in spores of the fungus *Histoplasma capsulatum*, e.g. from bird or bat droppings. The lungs are usually affected and the fungus can spread to multiple organ systems. Since most pathogenic fungi are opportunists infections can take hold when the immune system is lowered, e.g. histoplasmosis is common among AIDS patients.

QUESTIONS

1. What are the main features of fungi?
2. Identify the main ways in which fungi obtain their food.
3. Explain why agriculturalists are concerned about fungal infestations.
4. Explain why the spread of the chytrid fungus *Batrachochytrium dendrobatidis* is being monitored.
5. Identify some fungal diseases found in humans.
6. Explain why AIDS patients often have fungal infections that spread throughout their body.

54 Case Study – Tinea

Tinea is a highly contagious fungal skin infection. Tinea reproduce rapidly in hot, moist areas and if there is poor hygiene. Tinea infections occur in specific areas:

- Athletes foot – *Tinea pedis* affects the feet.
- Jock itch – *Tinea cruris* affects the groin.
- Scalp ringworm – *Tinea capitis* affects the head.
- Body ringworm – *Tinea corporis* affects the body.
- Onychomycosis – *Tinea unguium* affects the nails, e.g. toe or fingernails.

Tinea is a dermatophyte fungus which means it requires keratin for growth. Keratin is the major structural protein in skin, nails and hair.

Figure 54.1 Tinea infection.

Transmission and cause of disease

Tinea has a direct life cycle. The dermatophyte fungus usually lives in epidermal tissues or under the nails without causing symptoms. The fungus can very occasionally invade deeper tissues getting into the hair follicle and the dermis. Transmission occurs by direct contact with infected people or with other infected animals such as cats and dogs. Transmission can also be indirect with contact of contaminated articles or from the soil.

Symptoms of disease

Tinea infections can occur in people who are otherwise healthy but have compromised immune systems. Each type of tinea infection has particular clinical features, e.g. *Tinea pedis* affects the skin webs between the toes and because the fungus can survive under toenails the infection reoccurs with itching following by the development of small blisters which release a thin fluid on rupturing. The skin softens and peels, cracks can appear and bacteria can infect the open wound. An allergic reaction to the fungal antigens can lead to a rash or itch in other parts of the body, e.g. hands.

The main symptoms of tinea infections include:

- Itches and stinging sensation.
- Red scaly rash that is shaped like a ring.
- Cracking, splitting and peeling in the toe web spaces.
- Blisters.
- Yellow or white discolouration of the nails.
- Bald spots on the scalp.

Host response

The fungus has to attach to the skin and penetrate into the epidermal layer. If it overcomes this barrier host enzymes, e.g. keratinases are activated and the inflammatory response is initiated. Most humans have a certain level of resistance to dermatophyte fungi related to antifungal factors in their blood. There is a certain level of innate immunity and infection will cause acquired immunity with the production of antibodies and T cells.

Tinea pedis is not usually transmitted between marriage partners and usually occurs in adults rather than children. The difference in susceptibility of different people to tinea infection may be due to the difference in production of certain fatty acids in adults and children.

Treatment of disease

Antifungal creams reduce the symptoms but some infections require oral antifungal medication. Infected and dead skin is often removed.

For *Tinea pedis* antifungal creams are usually used at night while antifungal powders are used during the day to help minimise the moisture between the toes during daily activities.

QUESTIONS

1. What is tinea?
2. What conditions favour the growth of tinea?
3. Construct a table to summarise some of the different types of tinea infections.
4. How is tinea transmitted from person to person?
5. Describe some of the symptoms of tinea infection.
6. Outline the host response to a tinea infection.
7. What treatments are used for tinea infections?
8. Which part of the body is most likely to be infected by *Tinea pedis*?
   (A) Scalp.
   (B) Feet.
   (C) Blood.
   (D) Heart.

55 Macroparasites

Macroparasites can be seen with the naked eye and live either internally in the host (endoparasites) or externally (ectoparasites).

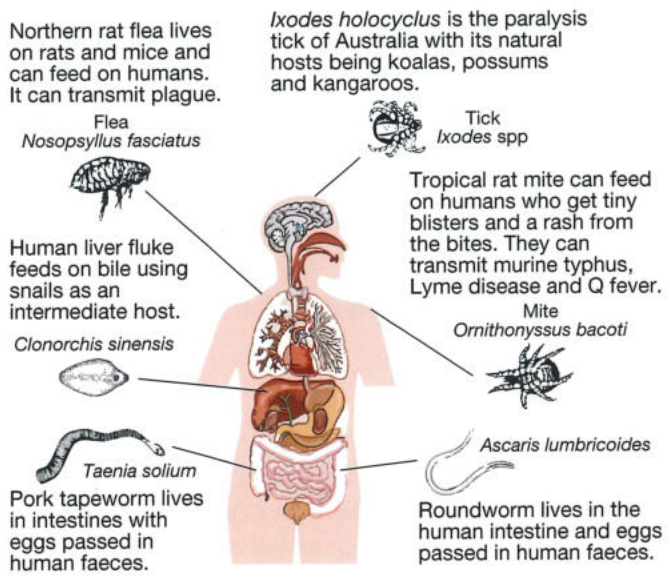


Figure 55.1 Macroparasites.

Endoparasites

Most endoparasites are flatworms or roundworms and have a long association with the host.

Trichinella spiralis is a roundworm normally parasitic in pigs and rats and infects humans when eaten in raw or uncooked pork. Small *filarias* nematode roundworms live in the human bloodstream causing filariasis and are passed from human to human by blood sucking vectors such as mosquitoes. The hookworm *Ankylostoma* inhabits the human gut with *A. duodenale* penetrating intact skin usually through the feet and travelling in the blood to alveoli where they move into the bronchi and trachea to be coughed up and reswallowed to eventually mature in the small intestine. *Ascaris* is a roundworm that can grow to 40 cm in length and infects the gut of pigs, cattle and sheep but can infect humans who eat uncooked vegetables that have been contaminated with faeces of infected animals.

Flukes are parasitic in or on the bodies of a wide variety of animals. *Fasciola hepatica* is the liverfluke that lives in the liver of sheep and the larvae in water snails.

Tapeworms have a long ribbon-like body with a small head with suckers and hooks for attachment in the intestines of its host. They have no mouth or alimentary canal as they absorb food through their body wall from the host's alimentary canal. *Echinococcus granulosus* is the hydatid tapeworm with the adult living in the intestines of dogs and the bladderworm invading any tissue in sheep, humans or other mammals.

Ectoparasites

Most ectoparasites are arthropods (e.g. lice, fleas, mosquitoes, ticks, leeches and mites) and have a brief association with the host.

Plant macroparasites are considered to be pests, for example aphids, mites, scale insect and borers. The dodder plant can be considered to be a parasite on its host tree. It wraps around the host and inserts haustoria into the vascular system of the host to take sugars and water from the host.

QUESTIONS

1. Define macroparasite.
2. Distinguish between ectoparasites and endoparasites.
3. The diagram shows the life cycle of *Echinococcus granulosus.*

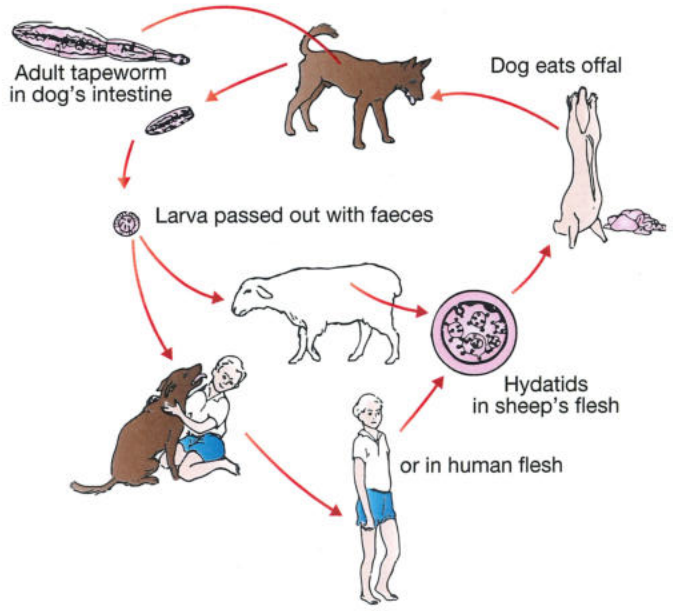


Figure 55.2 Life cycle of *Echinococcus granulosus.*

   (a) What type of parasite is *Echinococcus granulosus?*
   (b) Explain how a human can become infected by this parasite.
4. Use an example to show how an ectoparasite can act as a vector for another pathogen.
5. Explain why tapeworms do not need to have a mouth and an alimentary canal.
6. Endoparasites encounter specific problems. What feature of the endoparasite needs to be well developed for it to be successful in the human gut?
   (A) Digestive system.
   (B) Method of locomotion.
   (C) Exoskeleton.
   (D) Method of anchorage.

56 Case Study – Pork Tapeworm

Pork tapeworm, *Taenia solium* is a platyhelminth that can infect humans. It causes the disease taeniasis (tapeworm disease).

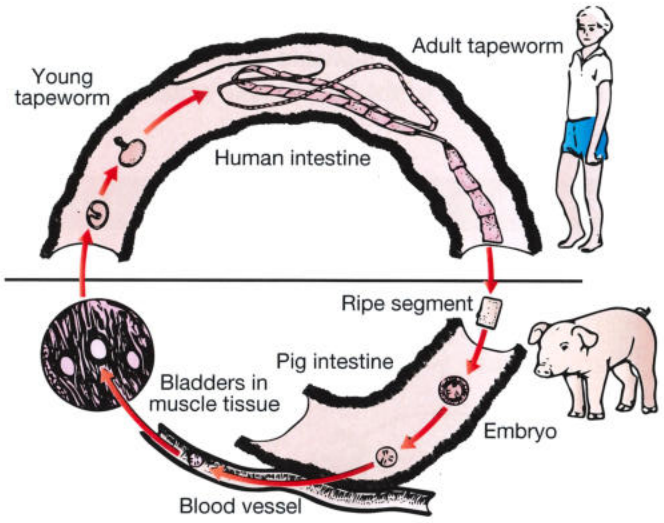


Figure 56.1 Life cycle of *Taenia solium*.

Transmission and cause of disease

The tapeworm has an indirect life cycle with human as the final host and the pig as the intermediate host. When a person eats undercooked pork that contains infective live bladderworms the larvae are released in the human intestine and develop into young tapeworms.

Life cycle *of T. solium*

The adult *Taenia solium* is the ribbon-like tapeworm with a scolex (head) with suckers and a double row of curved hooks. Most of the tapeworm is a linear series of segments called proglottids. The worms absorb food digested by the host. Behind the scolex is a narrow neck region where the proglottids are formed.

The body of the tapeworm is covered in mucus which is secreted in large amounts to defend itself against the digestive enzymes of the host. The worms also secrete inhibitor substances onto the surface of the proglottids to cause local inactivation of digestive enzymes.

The tapeworm is a hermaphrodite that reproduces by self-fertilisation. They produce large numbers of fertilised eggs that are released in the human faeces.

Contaminated vegetation is eaten by pigs and the eggs hatch in the intestines of the pig. They move through the intestinal wall and enter the blood and lymph and travel to various organs, e.g. muscle, liver, brain. They form cysts in these tissues and organs. When a person eats undercooked or raw pork they become infected and the worm breaks free from the cyst.

Symptoms of disease

Tapeworm infection in the intestines causes malnutrition and loss of weight in the human. If the worm becomes very large and long it can cause intestinal blockage and abdominal pain in the host.

The most visible symptom is the passing of the tapeworm segments in faeces.

If a person ingests the fertilised eggs then the larvae will hatch and move into the circulatory and lymphatic systems as in pigs and cysts can form in human tissues and organs.

Host response

In the intestine the tapeworm usually evades the host's immune response. If the macroparasite enters the bloodstream it has a complex and diverse set of antigens which will activate the immune response. The host response can harm the host depending on the location of the cyst, e.g. the inflammation response in the brain can affect cognitive functioning of the brain.

Treatment and control of disease

After diagnosis people with taeniasis are given medications such as praziquatel or niclosamide to clear the infestation.

To control the spread of the disease it is important to always correctly cook pork at suitable temperatures and not consume undercooked pork. Correct disposal of human waste away from pig areas is also needed to interrupt the life cycle of the tapeworm.

Taeniasis is not a notifiable disease.

QUESTIONS

1. What is taeniasis?
2. Outline the cause of taeniasis.
3. What happens when a person becomes infected with pork tapeworm cysts?
4. What is a proglottid?
5. How does the tapeworm defend itself against the host digestive enzymes?
6. How does the tapeworm stay attached to the host so it does not pass out with the faeces?
7. Explain why tapeworms do not need complex mating and courtship rituals.
8. Explain the benefit of many cultures banning the eating of pork.

57 Disease With an Insect Vector – Malaria

History of our understanding of malaria

Symptoms of malaria have been recorded as early as 18 BCE but it was not until 1880 that **Charles Laveran** linked the cause of the disease with micro-organisms when he observed the malaria parasite in blood. In 1885 **Celli** and **Marchiafara** observed that malaria is transmitted through infected blood and in 1886 **Golgi** observed asexual reproduction of the protozoan in infected blood. In 1898 **Bastianelli** and **Grassi** suggested the *Anopheles* mosquito as the vector and in 1899 **Ronald Ross** established the *Plasmodium* parasite as the cause of malaria. Ross became interested in malaria while stationed as a medical officer with the British army in India. He found that human volunteers bitten by mosquitoes infected with malaria later had the malarial parasite in their red blood cells. Ross eventually described the complete life cycle of *Plasmodium* and was awarded the Nobel Prize in 1907.

Malaria in humans is caused by four species of the protozoan *Plasmodium*: *P. falciparum, P. vivax, P. malariae* and *P. ovale*. The pathogen has a complex indirect life cyle with an intermediate host (humans) and a final host (female *Anopheles* mosquito). As the mosquito transfers the pathogen to humans, the mosquito is a **vector**. The disease is found in tropical climates.

Life cycle and transmission of *Plasmodium*

When a mosquito feeds on human blood, it injects saliva containing an anticoagulent which prevents the blood clotting in its feeding tube (proboscis). If the saliva contains *Plasmodium* sporozoites, the human will become infected. The sporozoites invade the liver cells and reproduce asexually. After about 2 to 4 weeks, merozoites of the parasite burst out with toxins and invade the red blood cells. Merozoites then develop into gametocytes which are sucked up by a mosquito.

In the gut of the female *Anopheles* mosquito the gameteocytes become gametes which fertilise to become zygotes (sexual reproduction). Meiosis occurs, forming sporozoites that migrate to the salivary glands where they may be injected into a human, along with saliva.

Host symptoms

When the merozoites burst out of the red blood cells, toxins are also released. The toxins cause fever that may reach 41.1°C, shivering fits, headache, nausea and then profuse sweating. This occurs every 48 hours in tertian forms (*P. falciparum, P. vivax* and *P. ovale*), and every 72 hours in quartan forms (*P. malariae*).

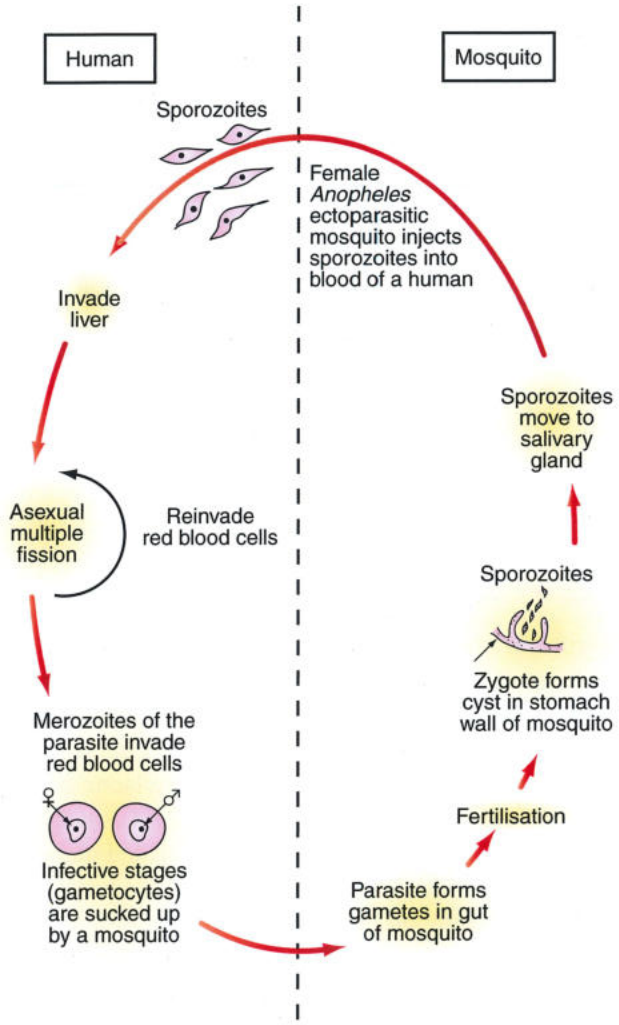


Figure 57.1 Life cycle of *Plasmodium*.

Reinvasion of the liver may occur. A malarial attack lasts two or more hours; after the body temperature returns to normal the host feels better but is weak, and suffers anaemia from loss of red blood cells and haemoglobin. The most serious malaria is caused by *P. falciparum* as the host becomes weaker with each attack and most die if untreated.

Host response

When the *Plasmodium* is in the red blood cells, the host produces antibodies. However, the surface antigen of the cell membrane of *Plasmodium* changes periodically, which means the new surface antigen is not recognised by the antibodies. While in the liver, the *Plasmodium* is isolated from the host's immune system. People with the sickle cell trait have a low susceptibility to malaria.

Treatment

Antimalarial drugs such as chloroquine, primaquinine, mefloquine or quinine reduce temperature or cure the infection. However, drug resistant strains of *Plasmodium* have evolved.

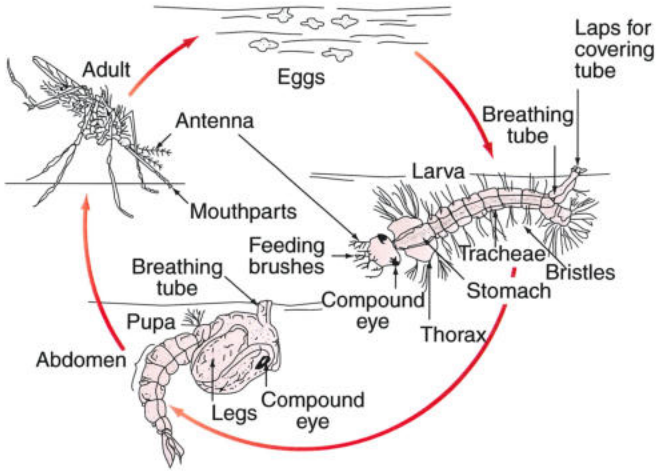


Figure 57.2 Life cycle of mosquito.

Control

Control can involve protecting people directly or destroying the mosquito vector. For example, people in malarial areas should wear protective clothing, use insect repellent and use mosquito nets over beds. People who are about to visit malaria areas can take prophylactic drugs such as chloroquine, which prevents malaria as well as curing it, and people who have recently visited malarial areas should not donate blood. The adult mosquito can be destroyed by spraying insecticides; swamps and standing water can be drained, covered with a lid or a layer of oil to destroy the larvae and pupa; and ships, planes and vehicles should be sprayed before arrival in malaria-free locations.

QUESTIONS

1. Discuss the importance of Ronald Ross's work with malaria.
2. What is meant by vector?
3. Identify the causative agent of malaria.
4. For malaria, what is the intermediate host and the final host?
5. Describe some of the symptoms of malaria.
6. Distinguish between treatment and control of a disease.
7. Discuss the methods for control and prevention of malaria.
8. Explain when control or prevention is most effective.
9. Early in the 20th century, there were outbreaks of malaria across the tropical areas of the north of Australia. Discuss how quarantine and eradication of the *Anopheles* mosquito helped to eradicate malaria from Australia.
10. Describe two adaptations of *Plasmodium* which are an advantage for a parasitic existence.
11. In an infectious disease, what is the vector?
    (A) A disease-causing organism.
    (B) A disease-carrying organism.
    (C) Always a blood-sucking or biting organism.
    (D) Always the intermediate host.
12. People who are infected with *P. vivax, P. ovale* or *P. malariae* can be treated and have an excellent chance of recovery. Infection with *P. falciparum* is still a life threatening condition. *P. falciparum* is life threatening because:
    (A) *P. falciparum* invades the liver rather than red blood cells.
    (B) *P. vivax, P. ovale* or *P. malariae* do not cause anaemia while *P. falciparum* causes anaemia.
    (C) *P. vivax, P. ovale* or *P. malariae* can be treated with AZT, but *P. falciparum* is not responsive to AZT.
    (D) Most *P. falciparum* are chloroquine resistant.

The diagram below is to be used for the next THREE questions. It shows a simplified flow chart for the life cycle of the protozoan *Plasmodium vivax* that causes malaria in humans.

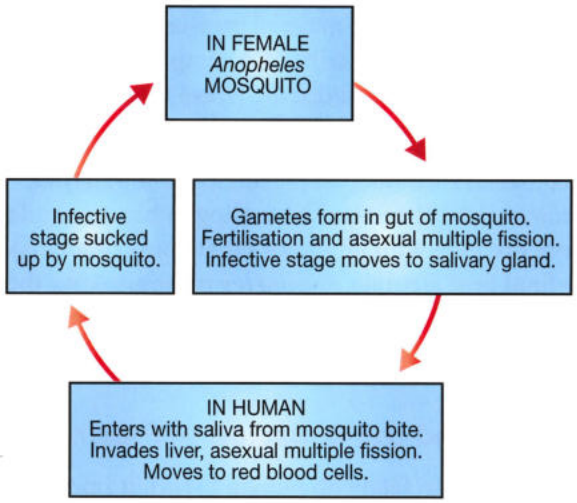


Figure 57.3 Life cycle of *Plasmodium*.

13. In this disease, what is the vector?
    (a) Mosquito.
    (b) *Plasmodium.*
    (c) Human.
    (d) Both the mosquito and *Plasmodium.*
14. Identify the intermediate host in this disease.
    (A) Mosquito.
    (B) *Plasmodium.*
    (C) Human.
    (D) Both the mosquito and *Plasmodium.*
15. An analysis of the life cycle of the pathogen shows prevention measures for a disease. Which of the following would not be a preventive measure for this disease?
    (A) Destroying adult mosquitoes by spraying insecticides.
    (B) Draining standing water.
    (C) Humans wearing protective clothing and sleeping under mosquito nets.
    (D) Removing game animals from infected areas.

16. The disease myxomatosis caused a dramatic reduction in the numbers of rabbits when the virus was first introduced. The virus is transmitted from one rabbit to another by a mosquito. How would the mosquito be described?
(A) Vector. (B) Parasite.
(C) Pathogen. (D) Antigen.

17. The disease trypanosomiasis, or African sleeping sickness, is caused by two flagellates of the genus *Trypanosoma*. While studying this disease, a scientist discovered *Trypanosoma* in the salivary glands of the tsetse fly, both male and female. The diagram shows the protozoan *Trypanosoma*.

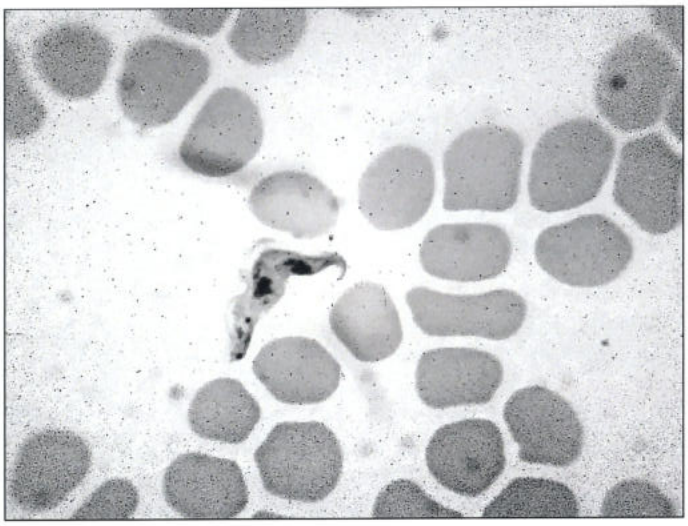


Figure 57.4 *Trypanosoma*.

Which of the following is a possible means of transmission of this disease?
(A) An uninfected fly bites an infected human.
(B) An infected fly bites an infected human.
(C) A fly bites an infected human and then feeds on a healthy human.
(D) A fly bites a healthy human and then feeds on an infected human.

18. Malaria is endemic in the tropics and subtropics. Which of the following gives the most likely reason for its movement into temperate zones?
(A) People moving from tropical areas to temperate areas.
(B) Airborne *Plasmodium* are carried in air currents to higher latitudes.
(C) Importing cattle without suitable quarantine measures.
(D) Using blood from overseas donors for medical purposes.

19. Most fevers are due to viral illnesses. If a person is suffering from a persistent and unexplained fever that lasts for more than three weeks, accompanied by sweating, nausea and abdominal pain, which of the following would confirm the disease as malaria?
(A) Low $CD4^+$ count.
(B) Blood film with protozoans present.
(C) Low blood pressure.
(D) Chest X-ray.

20. Thalassemia is an inherited condition found in specific populations . For example, beta thalassemia is found in people in China, the Mediterranean, the Middle East and India. Each person has two beta genes – one on each chromosome 11. A homozygous abnormality in the beta globin gene often leads to death in childhood. Thalassemia provides an advantage in infection with *P. falciparum* malaria and there is a high occurrence of this lethal gene; approximately 2.5% to 15% of the population in the 'thalassemia belt' have it. Away from this belt, you would expect the incidence of this gene and this disease to:
(A) Decrease.
(B) Increase.
(C) Stay the same.
(D) Only change for recent immigrants.

21. Many people have been involved in the development of our understanding of the cause and prevention of malaria. Which of the following correctly associates the person with their contribution?

	Person	Contribution
(A)	Ross	Synthesised chloroquine to treat soldiers in the war
(B)	Golgi	Identified *P. vivax* and *P. malariae*
(C)	Laveran	Malaria is only transmitted by *Anopheles* mosquitoes
(D)	Grassi and Bastianelli	Introduction of DDT to kill the mosquitoes

22. Who demonstrated that there were malarial oocysts in the gut tissue of the female *Anopheles* mosquito and thus proved these mosquitoes were the vectors for malaria?
(A) Charles Laveran.
(B) Camillo Golgi.
(C) Macfarlaner Burnet.
(D) Ronald Ross.

58 Barriers – The First Line of Defence

Natural resistance or **immunity** refers to the defence mechanisms present at birth. The first line of defence of the human body involves barriers that prevent the entry of pathogens, for example skin, mucous membranes, cilia, chemical barriers and other body secretions.

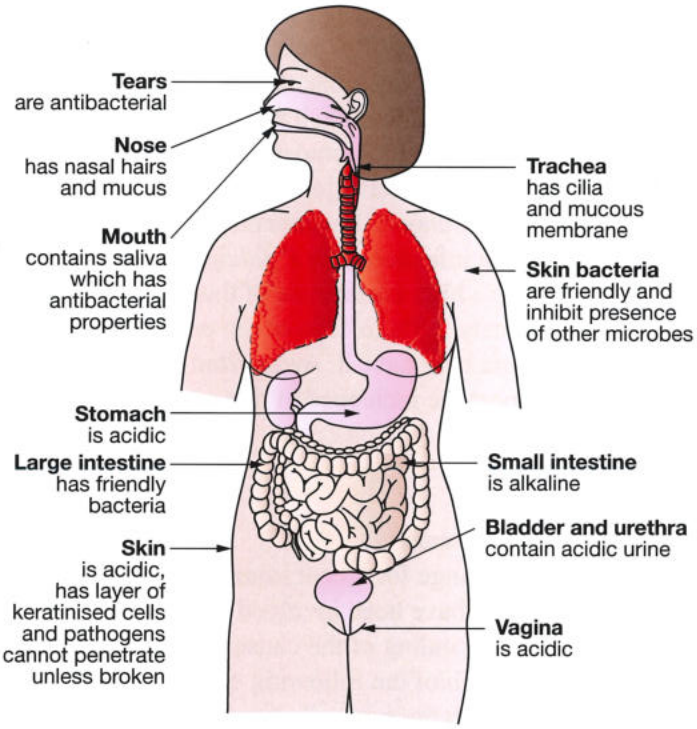


Figure 58.1 Barriers to microbe entry.

Skin

Intact, outer tough layer with keratin protein prevents penetration. Dryness prevents growth. Glands secrete antibacterial and antifungal chemicals. Pathogens overcome this barrier by direct touch of infected area, through abrasions, cuts, openings, or by active penetration. Infection through the skin occurs with mites, hookworm larvae, tetanus, chickenpox, cold sores, staphylococcal infections and ringworm.

Mucous membranes

These cover the digestive, respiratory, reproductive and urinary ducts with a thick, viscous mucus which traps pathogens until they can be removed, for example by cilia, or through coughing or sneezing. Secretions containing lysozyme which can break down the cell wall of some bacteria help flush out pathogens trapped in mucus (saliva, nasal secretion, tears etc). Mucus also can contain the antibody IgA which prevents the attachment of viruses and bacteria to epithelial surfaces.

Pathogens overcome this barrier as many spores are resistant to heat and chemicals, such as bacterial spores. Diseases include cholera, gonorrhoea and influenza.

Cilia

Cilia are very small microtubules which usually occur in large numbers on the cell surface and are used for locomotion. Cilia beat like oars, creating a force at right angles to the axis of the cilium. The movement sweeps the mucus, with trapped pathogens, along the tract. The upper respiratory system is lined with cilia (e.g. along the trachea and bronchi) preventing microbes entering the lungs. Coughing, sneezing or swallowing finally clears the tract.

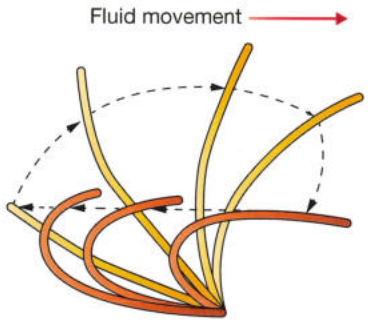


Figure 58.2 Beating of cilia creates fluid movement.

Chemical barriers

There are several antimicrobial secretions which inhibit the entry of pathogens, such as mucous membranes and skin secretions. The stomach wall secretes gastric juices that contain hydrochloric acid and protein-digesting enzymes which destroy pathogens in the stomach. The acid mantle of the vagina inhibits growth of fungi and bacteria in the female reproductive tract.

Other secretions

Urine cleans and flushes the lower urinary tract; it is sterile and its acidity inhibits bacterial growth. Sebaceous (oil) glands in the skin secrete fatty acids which inhibit fungal and bacterial growth.

Imbalance of microflora

If there is an imbalance of microflora, disease can result. For example, the skin and membranes of the mouth, throat, intestines and vagina normally have an array of microscopic organisms of which some are harmful and some are not. The use of antibiotics often destroys all microbes, including the useful varieties. When this happens, the environment changes and the fungus *Candida albicans* can invade normal healthy tissue. The fungus multiplies and the infection is known as candidiasis, or thrush.

QUESTIONS

1. What is meant by the 'first line of defence'?
2. What is natural immunity?
3. Explain how the skin prevents the entry of pathogens.
4. Describe how secretions in the stomach act as a defence barrier.
5. Outline the role of urine in preventing infection.
6. Use an example to show how one disease results from an imbalance of microflora in humans.
7. Describe how cilia and mucous membranes work together to prevent the entry of pathogens.
8. Complete the following table.

Table 58.1 Summary of barriers.

Barrier	How it protects	How pathogens overcome barrier	Diseases
Skin			
Mucous membranes			
Cilia			
Chemical barriers			
Body secretions			

9. Name one opening in the body which can act as a site for entry by a pathogen and describe the natural protection provided at this site to reduce the ability of the pathogen to infect the body.
10. There are many defence mechanisms present in the human body to prevent the entry of pathogens and hence prevent disease. Which of the following identifies the type of defence mechanism and how it works?

	Trapping	Killing	Washing away	Expelling
(A)	Mucus	Acidic mantle	Tears	Sneezing
(B)	Hairs	Urine	Gastric juice	Cilia
(C)	Cilia	Mucus	Perspiration	Peristalsis
(D)	Macrophage	Phagocytosis	Urine	Pus

11. Which mechanism is *not* considered to be part of the natural resistance of the human body?
    (A) Cilia in the respiratory tract.
    (B) Acidity on the skin.
    (C) Mucous membranes in the urinogential tract.
    (D) B lymphocytes in the intestine
12. Which of the following examples demonstrates protection against a pathogen at the point of entry?
    (A) Antibodies produced by B cells.
    (B) Killer T cells.
    (C) Acidity in the stomach.
    (D) Phagocytosis.
13. Which of the following describes why unbroken skin provides an effective barrier at the point of entry?
    (A) Beating of cilia prevents pathogens touching the skin surface.
    (B) Dead cells with keratin cell walls form a hardened outermost layer.
    (C) Dry conditions prevent pathogen growth.
    (D) Oil secretions are alkaline to inhibit pathogen replication.
14. Mucous membranes are part of the first line of defence to prevent the entry of pathogens. Which of the following best describes mucus?
    (A) Is acidic and kills micro-organisms on contact.
    (B) Contains lysozyme which digests and destroys pathogens.
    (C) Triggers the inflammatory response when pathogens are identified.
    (D) Traps pathogens and foreign substances.
15. The diagram shows the body as a castle fighting against the entry of pathogens.

Figure 58.3 Entry of pathogens.

Which of the following describes protection at point A?
    (A) Tears containing lysozyme.
    (B) Acidic mantle.
    (C) Cilia trap pathogen.
    (D) Intact layer with keratin.
16. *Lactobacillus* spp inhabit the vaginas of women during their reproductive years. They help to maintain a low pH by fermenting the glycogen present in the lining cells. If a woman undergoes antibiotic therapy it is likely that:
    (A) pH rises, causing an increase in body temperature.
    (B) pH rises and resistant microbes will colonise the area.
    (C) pH drops and vaginal infections occur.
    (D) pH drops and oestrogen levels rise accordingly.

59 Defence by Specific Body Parts

Different parts of the body, e.g. skin, digestive tract, urogenital tract, respiratory system, the ear and the eye have specific external defence mechanisms.

Skin

Pathogens can enter the skin through abrasions or openings, by active invasions, e.g. by mites, hookworm larvae, cercariae of blood flukes or through scratching, e.g. in rickettsia or trench fever.

The skin is protected by the fibrous protein **keratin** in the epidermis which forms an impervious, hardened outer protective mat on the surface of the skin, by the secretion of oil and sweat which contain **antibacterial** and **antifungal secretions** on to the skin surface and by the **normal microflora** on the skin. Skin secretions, e.g. organic acids can give the skin a pH from 3 to 5 which prevents the growth of many microbes. Most skin has little surface moisture which hinders microbe growth.

Digestive tract

Pathogens enter the digestive tract with food or drink, from dirty fingers, soiled objects, from touching animals and lack of hygienic practices. Mucous membranes lining the digestive system provide an intact barrier against pathogens. Some cells produce mucus which traps microbes and other particles and the mucus contains the enzyme **lysozyme** which digests the walls of many bacteria, e.g. in saliva in the mouth. The constant production of **saliva** helps flush microbes from the mouth and the production of **gastric juice** by the stomach with pH = 2 kills many microbes that reach the stomach. **Peristalsis** keeps substances moving along the alimentary canal. The large intestines have a resident microflora that actively retain their territory. The anus has mucus that traps pathogens. Body temperature is unfavourable for many micro-organsims.

Urinogenital tract

Microbes can enter the urinogenital tract through unsafe sexual practices. Mucous membranes lining the urinogenital tract provide an intact barrier against pathogens. Some cells produce **mucus** which traps microbes and other particles and the presence of lysozyme enzyme helps break down the bacterial cell walls. The vagina is **acidic** and normal flora inhibit pathogen growth. **Urine** from the bladder flushes pathogens from the urethra. Body temperature is unfavourable for the growth of many microbes.

Respiratory system

Transmission of pathogens occurs by inhalation of fungal/bacterial spores with pathogens contained in droplets from coughs and sneezes. **Mucous membranes** lining the respiratory system provide an intact barrier against pathogens. Some cells produce **mucus** which traps microbes and other particles and contains **lysozyme** which destroys some types of bacteria. **Cilia** line the respiratory tract, e.g. in the trachea that sweep the mucus containing the microbes, dirt and debris out of the tract, e.g. to be swallowed, sneezed or coughed out. The nasal cavity has **hairs** that help trap larger particles.

The ear

Airborne pathogens can enter the ear, e.g. *Streptococcus pneumoniae* causes ear infections and sinus infections. The ear produces **cerumen** (ear wax) that has antibacterial properties. Ear wax prevents desiccation of the ear canal and protects the skin in the passageway by lubricating the cells and provides a barrier against infection. Ear wax is pushed out of the ear by skin growth.

The eye

The eye has many defence mechanisms against infection. The eyelid and eyelashes are used to prevent particles contacting the surface of the eyes with the **blink reflex** pushing particles from the cornea and conjunctiva which also provide an intact protective barrier. The tear glands produce **tears** that contain lysozyme and salts that inhibit or destroy pathogens and flush the cornea removing pathogens. The different layers of the cornea increase the impermeability of substances into the eyes.

QUESTIONS

1. Draw the following table and write notes to summarise the protection found at different points of entry into the body.

Point of entry	Transmission	Protection at point of entry
Skin		
Digestive tract		
Urogenital tract		
Respiratory system		
The ear		

60 Antigens

An **antigen** is a unique molecule or part of a molecule that initiates an immune response, for example a virus, part of a bacterium, pollen or part of a toxic molecule. Antigens bind to receptors of B cells, antibodies or T cells.

Antigens are usually foreign and large molecules such as proteins or polysaccharides. The antigens can be on the surface of the foreign cell or virus or bacterium or can be a toxin that has been secreted into the extracellular fluid.

A **non-self antigen** originates outside the body. They are markers on cells, tissues or organs that enter the organism, e.g. viral, bacterial, fungal, prion infection or a transplant that the body's immune system recognises as foreign.

Self antigens are marker molecules on the surface of cells that are recognised as being part of that organism.

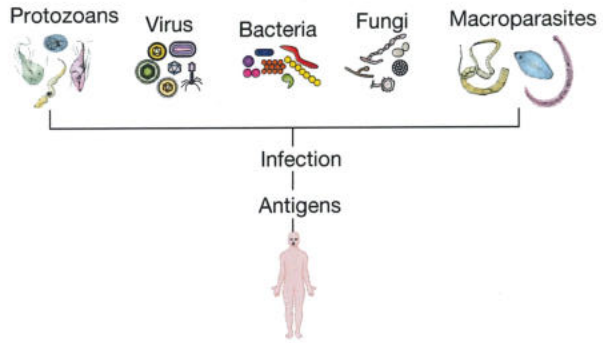


Figure 60.1 Antigens from infection agents.

Adaptive immunity

Adaptive immunity involves B cells or T cells recognising an antigen and binding with it via their antigen receptor. It is the antigen specific immune response (also called acquired immunity).

Antibodies (immunoglobulins) are proteins produced by B plasma cells in response to specific antigens and have the capacity to react against antigens.

An **allergen** is an antigen that triggers an exaggerated immune response leading to an allergy. There are many sources of allergens, e.g. different foods especially different nut types, seafood, shellfish and eggs, pet dander, insect stings or excretion, pollen and some drugs, e.g. penicillin, salicylates and sulfonamides.

The prevalence of food allergens seems to be increasing with the most common allergens of small children being milk and egg. Many of these children outgrow this allergy by the ages of 5 to 7 years. However, seafood and nut allergies can develop at older ages and stay with the person for the rest of their life. About 90% of all food allergies are caused by the following nine categories – peanuts, tree nuts, soy, milk, egg, cereals, seafood, fish and sesame.

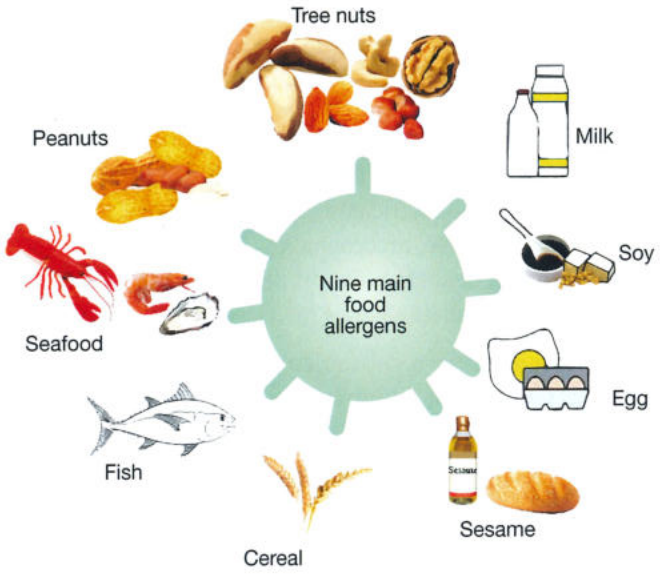


Figure 60.2 Most common food allergens.

Food allergy symptoms affect the mouth, stomach and intestines causing vomiting, hives, coughing, tight hoarse throat, weak pulse, swelling of the tongue and dizziness. Allergy symptoms for airborne allergens affect the eyes, nose and lungs causing sneezing, blocked, itchy or runny nose, red weeping eyes and headache.

QUESTIONS

1. Define an antigen.
2. What can be an antigen?
3. Distinguish between a self antigen and a non-self antigen.
4. What is meant by adaptive immunity?
5. What is an antibody?
6. Define an allergen.
7. Name some sources of allergens.
8. What are the most common food allergens?
9. Describe some symptoms for an allergic reaction to a food that is eaten.
10. Describe some symptoms for an allergic reaction to an airborne allergen.
11. There are four different human blood types O, AB, A and B. The allele I^B leads to the production of the glycoprotein, self antigen B, the allele I^A leads to the production of the glycoprotein, self antigen A while the allele I is recessive and does not code for an antigen. Explain why a person with blood type A will produce antibodies Anti-B and a person with blood type B will produce antibodies Anti-A.
12. What do you call a unique molecule or part of a molecule that originates outside the body and initiates an immune response?
    (A) Self antigen. (B) Non-self antigen.
    (C) Antibody. (D) Anticodon.

61 The Immune Response – The Second Line of Defence

If a pathogen penetrates the barriers of the first line of defence, it triggers the second line of defence. This group of responses is non-specific and includes phagocytosis, inflammation response, macrophages, lymph system response and cell death (to seal off pathogens) as well as antimicrobial chemicals such as interferons and complement.

Antigen is a term for a substance that triggers the immune response, for example a virus, part of a bacteria or part of a toxic molecule.

The **lymph system** is a system of vessels and lymph nodes that returns fluid and proteins to the blood. The lymphatic system is important in controlling tissue fluid balance, lipid transport and defence against disease.

Lymph is a colourless fluid in the lymphatic system. Lymph drains from the interstitial spaces into lymph capillaries, blind-end tubes with valves which prevent lymph flowing back into the tissues. The fluid moves with the contraction of nearby skeletal muscles and smooth muscle in the walls of larger lymph vessels. The lymph passes through lymph nodes and eventually joins up with the circulatory system at the heart.

Lymph nodes are located along the lymphatic vessels and act as filters, removing microbes, foreign particles, tissue debris and dead cells from circulation. Antibodies and memory cells formed in an activated lymph node travel to the bloodstream so they can circulate throughout the whole body.

Phagocytosis is a non-specific process where white blood cells, called **phagocytes**, attack foreign substances, and engulf and destroy them. Phagocytes actively move into the tissues and destroy and ingest microbes, toxins and abnormal cells. Phagocytosis is not always successful as some pathogens can repel phagocytes; some bacteria have special capsules which the phagocyte cannot grasp, and some pathogens escape before being completely destroyed. Pus is a mixture of dead phagocytes, bacteria, tissue fluid and damaged body cells.

Macrophages are the largest phagocytic cells and are mobile, long living and mononucleated. Macrophages are involved in wound healing, inflammation and the immune response when activated by lymphokines released by T cells. Macrophages extend pseudopodia and engulf dead and damaged cells, debris, antibody coated microbes and damaged fatty particles. They then release digestive enzymes and lytic enzymes to destroy the particles. There are fixed macrophages in some locations, such as lymph nodes and alveoli, and wandering macrophages. After a macrophage has destroyed the pathogen, parts of the partially digested antigen are displayed on the surface of the macrophage. Contact with T cells stimulates the production of more helper T cells for that particular antigen. Thus, macrophages are part of the non-specific response but are also involved in the production of specific defence cells.

Inflammation response. Tissue damage followed by the invasion of a pathogen causes the inflammatory response which includes the release of histamine. The histamine causes the dilation of blood vessels, increased permeability of the blood vessel walls and the release of chemicals to attract phagocytes. This results in increased fluid leaving the blood flow The area becomes red, hot, swollen and painful with increased phagocytosis, and in more severe cases causes raised body temperature (fever).

Cell death to seal off pathogen. A cluster of cells may surround the pathogen and damaged tissue. A cyst may even form. The area is sealed off. Other cells such as white blood cells and healthy cells may be sacrificed to make sure the pathogen is sealed off.

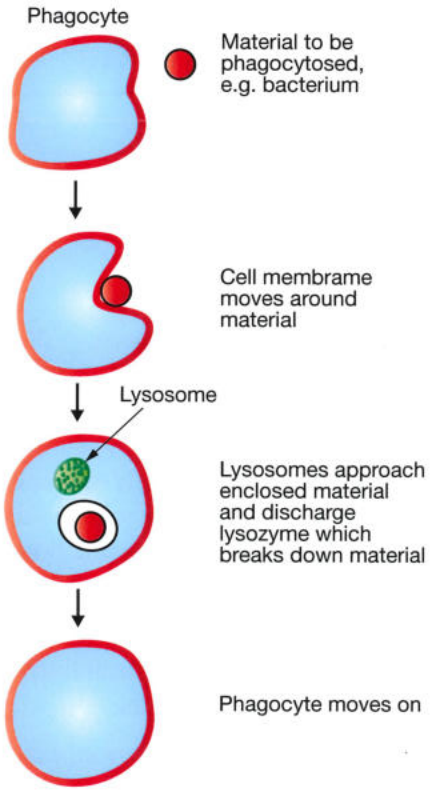


Figure 61.1 Phagocytosis.

Other secretions. The body also secretes special proteins which act against pathogens. **Interferons**, first identified in 1957, are produced in virus-infected cells and induce nearby non-infected cells to produce antiviral enzymes. These enzymes block the translation of the viral messenger RNA to protein. Viral replication is thus inhibited. Interferons are most effective in short term viral infections such as colds and influenza. There are several types of interferons, some of which are mass produced using recombinant DNA technology and antiviral therapy.

Complement proteins are a group of at least 20 proteins which assist other defence mechanisms. They enhance phagocytosis and cause cell lysis, that is, the cell membrane is disrupted and water and ions enter through holes and intensify the inflammatory response.

Organ transplants. A body's immune system will identify the transplant organ as a foreign body (antigen) and thus stimulate the non-specific responses such as inflammatory response and phagocytosis, as well as the specific responses of B and T lymphocytes. This can cause tissue rejection. Immunosuppressive drugs are often administered to prevent tissue rejection.

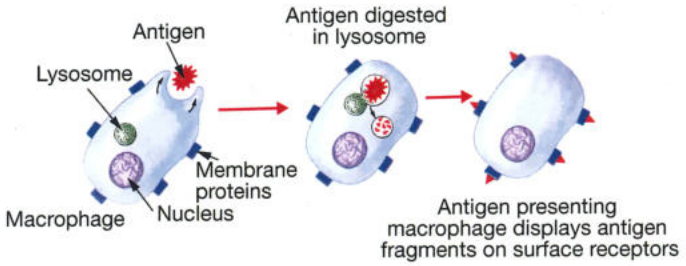


Figure 61.2 Phagocytosis.

QUESTIONS

1. Define antigen.
2. What is the lymph system?
3. Describe the function of a lymph node.
4. Distinguish between blood and lymph.
5. Explain how the inflammatory response fights infection.
6. Explain why macrophages are both part of the specific and the non-specific response.
7. Describe the action of interferons.
8. Explain why organ transplants trigger an immune response.
9. Compare the first line of defence with the second line of defence.
10. Explain why phagocytosis is considered to be a non-specific immune response.
11. Plasma and tissue fluids contain substances referred to as the C1, C2, C3 … C9 components of complement. When activated, these components can have important effects, for example C2-kinin stimulates vasodilation and C3-opsonin increases phagocytosis. Discuss why complement is considered to be a link between the non-specific and specific defence systems.
12. The diagrams show a process that occurs in many parts of the human body.

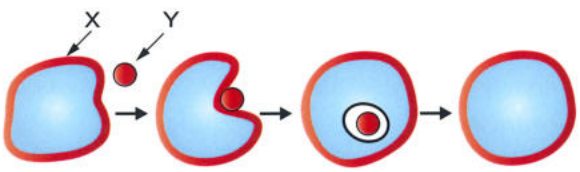


Figure 61.3 Process in body.

The parts labelled X and Y are respectively:
(A) Phagocyte and macrophage.
(B) Antigen and antibody.
(C) Macrophage and bacteria.
(D) Antigen and phagocyte.

13. What are phagocytes?
(A) A type of B lymphocyte.
(B) Cells which engulf and destroy foreign bodies.
(C) Cells which produce antibodies.
(D) Chemicals produced in response to pathogens entering the body.

14. Which of the following best describes the process of phagocytosis?
(A) Is always effective against pathogens as it causes the inflammatory response.
(B) Produces phagocytes which are chemicals designed to react with specific antigens.
(C) Protects the body by the manufacture of interferons to combat viral infection.
(D) Is the engulfing and digestion of substances recognised as foreign to the body.

15. The inflammatory response, caused by cutting your skin or injuring cells, involves the release of histamine. What will be caused by histamine?
(A) Dilation of blood vessels and increased vascular permeability.
(B) Decrease in the ingestion of microbes by phagocytes.
(C) Reduction in body temperature.
(D) Interferons to bind with bacteria to form an inactive complex.

16. The normal flora on a healthy body includes commensal bacteria that live on the skin and mucous membranes. These bacteria do not normally invade the body, but under some circumstances they can become opportunistic pathogens and invade tissues. This can occur:
(A) After exercising when the body temperature is raised.
(B) When the body's defences are impaired.
(C) Before a meal.
(D) Within 24 hours of a vaccination injection.

17. Which of the following does *not* apply to lysozyme?
(A) An enzyme which speeds up DNA replication.
(B) Found free in sweat, tears, nasal secretions, saliva and tissue fluids.
(C) An oily substance secreted by sebaceous glands.
(D) Able to break down a large variety of microbes.

62 Play – Inside the Immune System

Dramatic personae:
Tetani – the antigen (tetanus bacteria, *Clostridium tetani*)
Histo – the antigen (amoeboid protozean for dysentery, *Entamoeba histolytica*)
Diphtheriae – the antigen (diphtheria bacteria, *Corynebacterium diphtheriae*)
Macro – a macrophage
Macruu – another macrophage, a constable in the White Force
Macraa – another macrophage, a sergeant in the White Force
Letty – the lymph node Commandant
B Memory Cell No. 2834 – plasma cell from B lymphocyte
B Memory Cell No. 2836 – plasma cell from B lymphocyte

SCENE 1 – OUTSIDE THE BODY

(Three diabolical friends, Tetani, Histo and Diphtheriae, are outside a human body discussing ways of infecting the human)

Tetani I'm tired of living anaerobically in the soil. Let's enter that human body and make it sick.

Histo Good idea, Tetani. I love making people ill. It's a very stimulating experience.

Diphtheriae How will we enter this human?

Tetani There are lots of ways to enter.

Histo Well, let's choose a path that doesn't have too many defences. I'm a fairly virulent type but I can still be destroyed.

Tetani You have to be careful to enter the correct way or you have no effect.

Diphtheriae What do you mean Tetani?

Tetani My identical twin went in the wrong entry point and hasn't been seen since.

Histo What do you mean, your identical twin?

Tetani Binary fission causes two identical daughter cells to form

Histo Is that how bacteria reproduce?

Tetani Well, you weren't found in a cabbage patch, unless you're a plant bacteria.

Diphtheriae Never mind about all this reproduction, which isn't even sexual, what happened to your identical twin Tetani?

Tetani My twin was a tetanus bacteria like me and decided to go in the mouth.

Diphtheriae So what happened?

Tetani Nothing happened when she was ingested. It was very sad. Tetanus bacteria have to enter the bloodstream via a wound.

Diphtheriae I'm very sorry.

Tetani We think she must have mutated. No one else in our family has tried to change the conventional methods of our ancestors.

Diphtheriae So how are you going to enter?

Tetani I'm going to sit on that rusty nail and when the human stands on it, I'll enter through the wound and give it the disease lockjaw.

Histo Lockjaw? Is that it's proper name?

Tetani Some people now call lockjaw 'tetanus'.

Histo Well, I have to be ingested. I CAN'T go through the skin.

Tetani How do you infect people, Histo?

Histo I'm a cyst of the parasite and I'm in the faeces of an infected person.

Tetani You're in the faeces? How horrible for you!

Diphtheriae So what disease do you cause?

Histo I cause dysentery.

Diphtheriae And how do you get in?

Histo I'm going to be eaten with food or drunk with contaminated water.

Tetani This person will actually EAT FAECES?

Histo Well, I'm a bit more clever than that. I hide on dirty hands and they don't know I'm there.

Tetani How are you going to enter Diphtheriae?

Diphtheriae I cause diphtheria and I'm an airborne droplet from a sneeze.

Histo So how are you going to get in?

Diphtheriae I'm going to be breathed in and I'll infect the larynx.

Tetani The larynx?

Diphtheriae The larynx is the voice box. You may know about infecting the skin Tetani, but you don't know much about the respiratory systems.

Tetani OK, OK, let's just go and infect this person and see who does the most damage.

(The three friends separate and each enter the human in their own way)

Questions Scene 1

1. What causes:
   (a) Tetanus?
   (b) Diphtheria?
   (c) Dysentery?
2. How do each of these pathogens enter the human body?
3. How do bacteria reproduce?
4. Explain why Tetani has an identical twin

SCENE 2 – IN THE BLOOD

(The three friends meet again in the blood)

Tetani Isn't this wonderful fun? Causing disease is SO, SO sick.

Histo How are you making this person sick?

Tetani I've produced a neurotoxin that binds to the synapses and interferes with nerve impulses to cause muscular spasms.

Diphtheriae Speak English, please.

Histo She means that she will try to make the jaw muscles lock up.

Diphtheriae Does that mean she's won our contest and will cause the most damage?

Histo Never. I've been very active and will cause terrible pain to this person.

Tetani Do tell. What have you been doing, Histo?

Histo I was licked off a finger with sugar from a donut and then I hatched in the intestine, invaded the intestine lining and fed on the blood and caused abscesses.

Diphtheriae That doesn't sound very impressive.

Histo I caused severe, continuous diarrhoea, and THAT was impressive.

Diphtheriae OK, you can have a few points for that. Are you going to do anything else?

Histo To get some points, I thought I may invade the liver and cause amoebic hepatitis.

Tetani What have you done, Diphtheriae?

Diphtheriae I released a toxin into the blood and will cause a fever, headache and vomiting all within 5 days.

Tetani Excellent, you certainly work fast.

Diphtheriae I sure do.

Macro Invaders, Invaders !!! Help ! Help !

Diphtheriae Who's this?

Macro I'm Macro, a macrophage.

Tetani Well, diffuse off Macro. We are in the middle of a contest.

Macruu I came as fast as I could, Macro. What seems to be the problem?

Macro We have been invaded by these three antigens and the immune system must come to the rescue.

Histo Leave me alone Macro. What are you doing?

Macro Come on Histo, I have you now. I'm taking you to the nearest lymph node.

(Macro transports Histo away)

Macraa Hello, Hello. What do we have here, Constable Macruu?

Macruu Antigen invaders, Sir.

Macraa Well, what are you waiting for? You take the bacteria and I'll get the protozoan.

Macruu Right, Boss.

(Macruu and Macraa engage Tetani and Diphtheriae in a fierce struggle, but eventually the two antigens are taken into custody and taken to the nearest lymph node)

Questions Scene 2

1. Describe the symptoms caused by:
   (a) Tetanus bacteria.
   (b) Diphtheria bacteria.
   (c) Dysentery protozoan.
2. What happens when an antigen enters the body?
3. Where do macrophages take the antigens?

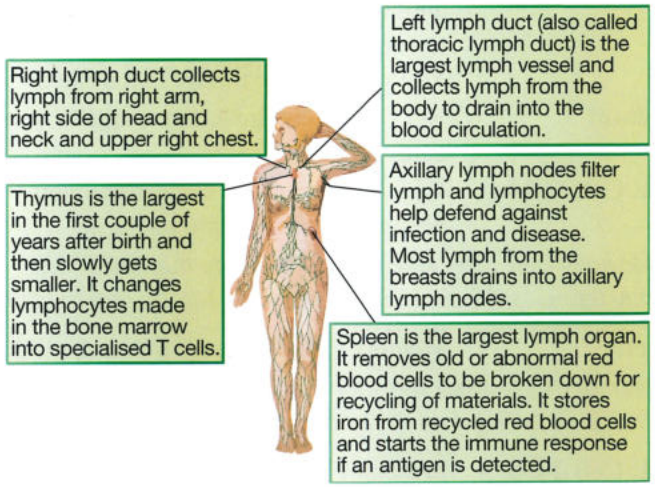


Figure 62.1 Lymphatic system.

SCENE 3 – THE LYMPH NODE

(The three friends meet again, this time in a lymph node. They are brought before Letty, the lymph node Commandant)

Histo They got you two also, did they?

Tetani They certainly did. They are very efficient, these white knights.

Letty Prisoners, be QUIET!

Tetani Yes, Madam.

Letty Step forward, prisoners, and be identified.

Diphtheriae How can you identify us? You've never met us before.

Letty We have our ways.

Histo I don't like the sound of this.

Letty And we've met your kind before. Sadists that like seeing pain in others.

(The three antigens are checked over)

Letty Tetani and Diphtheriae step forward.

(Tetani and Diphtheriae move one pace forward.)

Letty You two have been identified by our memory cells.

Tetani You have a computer?

Letty Quiet. And we know who you are. Memory Cell, Number 2834, step forward, please.

(Memory Cell 2834 steps forward from the ranks of plasma cells lined up in the lymph node)

M. Cell 2834 Present, Commandant Letty. What job can I do?

Letty Do you recognise this prisoner here?

(Letty points to Tetani, the tetanus bacteria)

M. Cell 2834 Certainly, I was made to identify that particular rodent.

Tetani What are you going to do to me?

M. Cell 2834 I'm going to initiate the large number of plasma cells which will secrete antibodies that will combine with you to form an antigen-antibody complex. You will then be totally inactivated.

Tetani NO ! NO! Not the antigen-antibody complex!

M. Cell 2834 YES ! The immune system has you beaten, my pathogenic bacteria.

(Memory Cell 2834 releases an antibody that combines with Tetani and Tetani falls to the floor, totally inactivated)

Letty Memory Cell 2836, step forward, please.

(Memory Cell 2836 steps forward from the ranks of plasma cells lined up in the lymph node)

M. Cell 2836 Present, Commandant Letty. What job can I do?

Letty Do you recognise this prisoner here?

M. Cell 2836 Certainly, I was made to recognise that particular prokaryote.

Diphtheriae NO! Don't let it happen to me also.

M. Cell 2836 I'm afraid so.

(Memory Cell 2836 causes the release of an antibody that combines with Diphtheriae, the antigen and an antigen-antibody complex forms)

Diphtheriae What do we do now that we're inactivated?

M. Cell 2836 You do nothing. WE have stimulated phagocytosis and YOU are soon to be gobbled up by a phagocyte.

Tetani Tell me, how come you could identify us, when we've never been here before?

M. Cell 2836 This human is very clever and has been immunised with the triple antigen.

Diphtheriae The triple antigen?

M. Cell 2836 When this human was 2 months old, 4 months old and 6 months old, it received an injection that immunised it against diphtheria, whooping cough and tetanus.

Tetani So this human body is protected against us?

M. Cell 2836 That's correct.

Diphtheriae And is that why you're 2834?

(Histo points to Memory Cell 2834)

Diphtheriae And you're 2836?

(Histo points to Memory Cell 2836)

M. Cell 2836 That's correct.

Diphtheriae And I presume that Memory Cell 2835 is the whooping cough memory cell?

M. Cell 2834 Correct ! You're fairly smart for a prokaryote, it's just a pity that you weren't smart enough to fool us.

(The two antigens Tetani and Diphtheriae disappear as phagocytosis takes over and they are never seen again)

Histo Well, it looks like I've won the contest.

Letty We have antibodies for you as well.

Histo Those antibodies can't protect this host from me.

Letty You may be able to avoid the immune system, but I warn you, this human looks after itself and at this very instance is taking the medicinal drug 'emetine' to kill you.

Histo Oh No ! Help me, help me. I'm being poisoned !!!

(Histo staggers, than drops down dead)

Letty Well done, troops.

Macruu Thank you, Commandant.

Macro Thank you, Commandant.

Macraa It's just another exciting episode in the journal of a soldier in the WHITE FORCE of the IMMUNE SYSTEM.

Questions Scene 3

1. What happens to the antigens in the lymph node?
2. What produces antibodies?
3. Why were Tetani and Diphtheriae identified?
4. How did the human treat the dysentery?
5. Who won the contest?

63 Leucocytes

The three main cellular components of blood are red blood cells (erythrocytes), white blood cells (leucocytes) and platelets. Since the red blood cells and platelets do not have nuclei, the leucocytes are the only fully functional cells in circulation. Leucocytes can be divided into **granulocytes** which have secretory granules in their cytoplasm and **agranulocytes** which do not have granules in their cytoplasm and have a one-lobed nucleus. Under a microscope leucocytes can be distinguished from each other by the size and shape of the nucleus, the staining features of the cytoplasm, the regularity of the cell border and the cytoplasmic inclusions.

All blood cells come from **pluripotent haematopoietic stem cells** in bone marrow. In adults only the spine, ribs and pelvis and proximal ends of long bones make new blood cells. Active bone marrow is a red colour while inactive bone marrow is a yellow colour and contains many fat cells. Cytokines, e.g. interleukins and colony stimulating factors stimulate the production and development of blood cells. **Leucopoiesis** is the process of white blood cell production and development and once a leucocyte matures it cannot undergo mitosis and cell division. **Leukaemias** are a group of diseases caused by an excess or lack of white blood cells.

Leucocytes play a major role in defence against foreign invaders with most of their activity carried out in body tissues rather than in the bloodstream.

There are five types of mature leucocytes – eosinophils, basophils, neutrophils, monocytes and lymphocytes. Monocytes that leave the bloodstream develop into macrophages in body tissues.

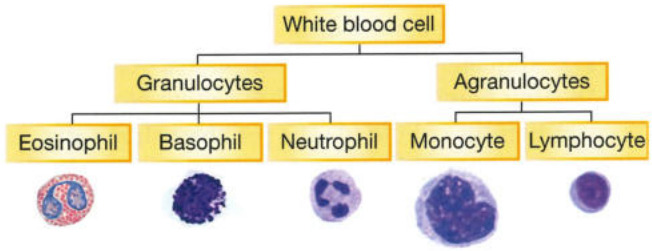


Figure 63.1 Types of leucocytes.

Phagocytes

Granulocytes and monocytes are formed only in bone marrow and are phagocytic ingesting invading organisms. Of these white blood cells it is mainly the neutrophils and macrophages that attack and destroy invading bacteria, viruses and other agents identified as foreign.

Chemotactic substances cause both monocytes and neutrophils to move to the source of the substance, e.g. they move up the concentration gradient of toxins produced by the invading virus or bacteria, towards the degenerative products produced by the inflamed tissue, towards the source of complement proteins and towards the source of reaction products caused by the clotting in an inflamed area.

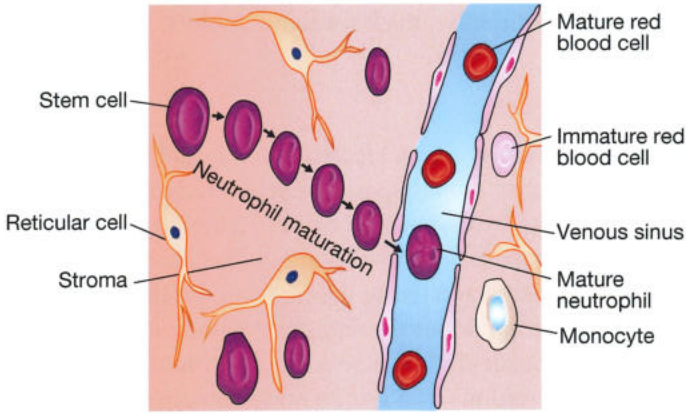


Figure 63.2 Neutrophil development in red bone marrow.

Neutrophils are the most abundant type of granulocyte and mature neutrophils can attack and destroy bacteria in both the bloodstream and tissues. They can move through the pores of blood vessels and chemotaxis causes them to move to an area of tissue damage. After formation mature neutrophils usually spend 4 to 8 hours circulating in the bloodstream and then 4 to 5 days in tissues. A neutrophil is one of the first cells to respond and move to the site of inflammation. They attach to a particle and send pseudopodia around the particle to enclose the particle in a phagocytic vesicle. Lysosomes and other cytoplasmic granules move to the vesicle and begin digestion of and breakdown of the phagocytised particle. One neutrophil can phagocytise between 3 to 10 bacteria before it is inactivated and dies.

Monocytes in the bloodstream are immature cells and when they move into tissues they swell in size to become mature macrophages. Macrophages are more powerful phagocytes than neutrophils with a longer life span, able to phagocytise up to 100 bacteria and can engulf larger particles than neutrophils. After phagocytising particles a macrophage can extrude any residual product and keep functioning for several months. Some macrophages become attached to tissues, e.g. fixed tissue macrophages are found in the spleen, lymph nodes and other tissues of the lymphatic system as well as in the kidney, bone, liver, peritoneal cavity, placenta, connective tissues and skin. In the lymph nodes macrophages trap bacteria, viruses and other foreign agents preventing them from spreading to other parts of the body. During an infection the lymph nodes in the neck and armpit can become swollen due to the increased activity of the immune response. Macrophages are also important in muscle repair, growth and regeneration and in wound healing.

Dendritic cells are a type of phagocyte that become antigen presenting cells as part of the immune response.

Lymphocytes

Lymphocytes have a non-granular cytoplasm and are the main cells involved in the acquired immune response. Some lymphocytes are specialised to produce antibodies while others directly attack foreign invaders.

Lymph nodes

Lymph nodes are organs located along lymph vessels. They are a mass of spongy tissue separated into compartments by connective tissue. Lymph nodes filter lymph and trap and remove foreign particles such as micro-organisms and tissue debris from the fluid. There are single lymph nodes throughout the body and clusters of nodes in particular areas, e.g. neck, armpits and groin. The cortical regions of a lymph node contain reticulum cells that are the germinating centres of lymphocytes.

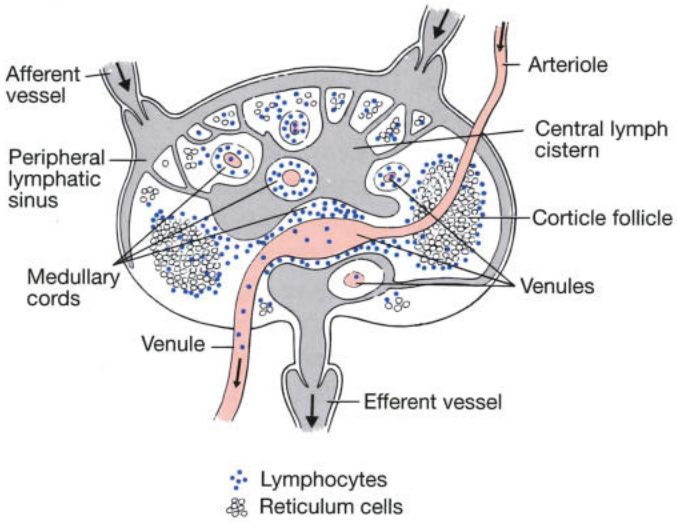


Figure 63.3 Lymph node.

QUESTIONS

1. Name the three main cellular components of blood.
2. Distinguish between granulocytes and agranulocytes.
3. What features distinguish the leucocytes from each other?
4. (a) What is the source of all blood cells?
   (b) What is the location of these sources in adult humans?
5. Distinguish between red and yellow bone marrow.
6. What is the main function of leucocytes?
7. Where is most of the activity of leucocytes carried out?
8. What will stimulate the production and development of blood cells?
9. Define leucopoiesis.
10. What is leukaemia?
11. Name the five main types of leucocytes.
12. (a) What is phagocytosis?
    (b) Which leucocytes are phagocytic?
13. (a) Define chemotaxis.
    (b) Identify the source of some chemotactic substances that attract phagocytes.
14. Explain how a neutrophil acts against bacteria and viral particles.
15. What is the life expectancy of a neutrophil?
16. Construct a table to compare neutrophils and macrophages.
17. What are dendritic cells?
18. What are lymphocytes?
19. What are lymph nodes?
20. Which of the following all have granular cytoplasm?
    (A) Neutrophil, eosinophil, basophil.
    (B) Monocyte, neutrophil, lymphocyte.
    (C) Lymphocyte, eosinophil, macrophage.
    (D) Basophil, macrophage, monocyte.
21. If you observed a stained smear of blood under high power light microscope, what would be the likely identity of a very large cell with a large non-lobed nucleus that was over twice the size of the red blood cells?
    (A) Neutrophil. (B) Macrophage.
    (C) Lymphocyte. (D) Basophil.
22. Which of the following leucocytes is not phagocytic?
    (A) Neutrophil. (B) Macrophage.
    (C) Lymphocyte. (D) Eosinophil.
23. Tissue macrophages are found at lymph nodes. What is the function of these macrophages?
    (A) Increase the rate of respiration.
    (B) Produce antibodies against foreign particles.
    (C) Storage of antibodies.
    (D) Trap and phagocytise foreign particles.
24. An adult human has about 7000 white blood cells per microlitre of blood. The pie chart shows the percentage of different leucocytes found in blood.

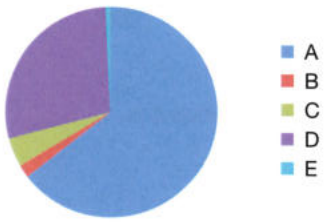


Figure 63.4 Leucocytes in blood.

Which of the sections refers to neutrophils?
(A) Section A.
(B) Section B.
(C) Section C.
(D) Section D.

64 Mast Cells and Dendritic Cells

Dendritic cells and mast cells are important components of the innate (non-specific) immune response to an antigen. They have roles in the inflammatory response.

Dendritic cells

Dendritic cells are a type of phagocyte that become antigen presenting cells as part of the immune response. They are mainly in lymphatic tissues and skin and are particularly efficient in presenting antigens to naive T helper cells as part of the primary immune response. They link the innate response to the acquired immunity response.

Specialised Langerhans cells first describe by Paul Langerhans are found on the surface of the skin in the dermis and epidermis and come from cellular differentiation of monocytes. These cells take up and process microbial antigens to become functional antigen presenting cells.

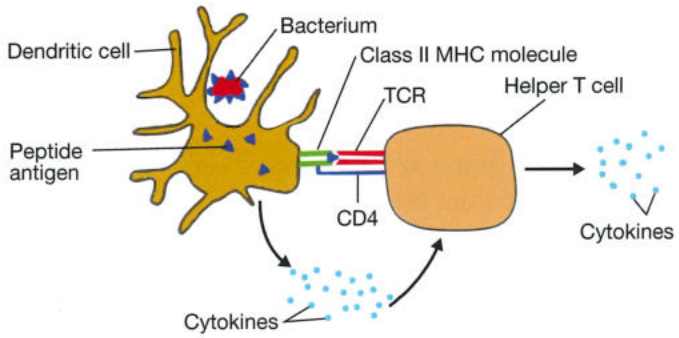


Figure 64.1 Dendritic cell engulfs bacterium and presents antigen.

After a dendritic cell phagocytises a bacterium it will move a fragment of the protein (peptide) antigen to the cell surface in **antigen presentation**. It will then display the antigen fragments with a class II **major histocompatibility complex (MHC) molecule**. T cells can recognise displayed antigens bound to MHC molecules. Most helper T cells have a surface protein called **CD4** which recognises and binds with the class II MHC molecule aiding the **T cell receptor** (TCR) on the helper T cell. The helper T cell now activated will secrete several different cytokines that stimulate other lymphocytes and help begin the acquired immune response.

Mast cells

Mast cells are a type of body cell in vertebrates that produce histamine and other molecules that trigger the inflammatory response. Mast cells are found in connective tissue and under mucous membranes that line the digestive tract and airways so when an injury occurs the mast cells release the stored histamine triggering dilation and increased permeability of the nearby capillaries.

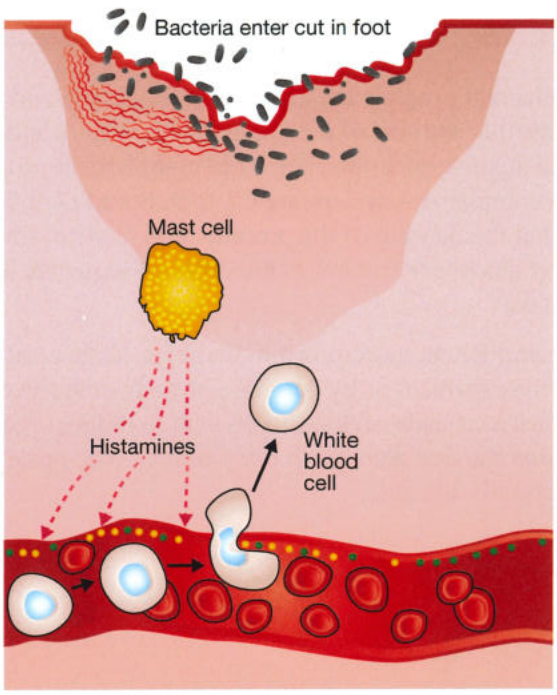


Figure 64.2 Mast cell releases histamine when tissue is damaged.

Histamine is an organic nitrogenous compound that is stored in the granules in mast cells and is the active molecule that initiates the inflammatory response. It is released when mast cells degranulate and it causes the opening of pores in capillaries and the dilation of blood vessels. This causes plasma proteins to escape into the tissue leading to swelling and oedema of the tissue. In nasal passages the inflammatory response causes a stuffy, runny nose and antihistamine drugs are used to treat this symptom.

QUESTIONS

1. Identify the importance of dendritic cells and mast cells.
2. What is a dendritic cell?
3. What is the location of most dendritic cells?
4. Define an antigen presenting cell.
5. What are MHC molecules?
6. What is CD4?
7. What is a mast cell?
8. Where are most mast cells found?
9. Outline the function of mast cells.
10. What is histamine?
11. (a) How is histamine involved in the inflammatory response?
    (b) What is the function of antihistamines?
12. Which chemical plays a significant role in anaphylactic shock?
    (A) Penicillin.
    (B) Histamine.
    (C) Heparin.
    (D) Glucose.

65 Complement Proteins

Complement proteins are around 30 different blood proteins that are part of the complement system and defend against infection. The main complement proteins in the complement system are C1 to 9, B and D, though note that the naming of the proteins was made in the order of their discovery and not in the order of sequence in the reactions.

The complement system is initiated by a surface antigen or micro-organism or by antigen-antibody complexes and becomes a cascade of steps to prevent invading organisms or toxins causing damage to body tissues. The proteins are normally inactive.

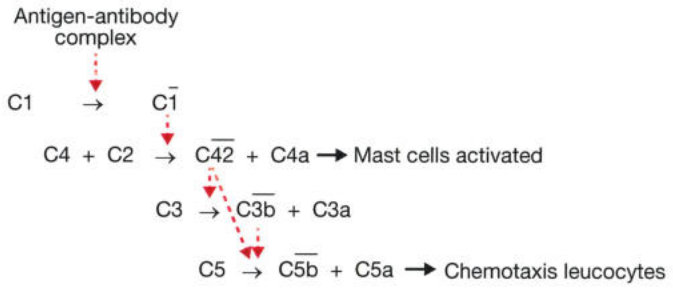


Figure 65.1 Complement system with part of a cascade. Note that a reaction product is needed to activate the next reaction as drawn by a dotted line.

Function of complement proteins

Most of the sequence of reactions that make up the cascade occurs on the surface of the pathogen leading to the destruction of the pathogen and end of the infection. Complement proteins can do the following.

- Amplify the inflammatory response, e.g. by causing increased blood flow to the area, causing increased leakage of capillary proteins into the area and causing the coagulation of interstitial fluid proteins to prevent movement of invading organisms to other areas.
- Enhance phagocytosis.
- Cause **chemotaxis** by attracting macrophages and neutrophils.
- Directly lyse pathogens, e.g. complement C5b triggers the assembly of a **membrane attack complex** (MAC) that can rupture bacterial and other cell membranes.
- Activate mast cells and neutrophils.
- Agglutinate invading organisms by changing the surface structures of organisms so that the organisms adhere to each other in clumps which reduces their virulence.
- Neutralise viruses, e.g. by attacking viral structures.

Many of these proteins are enzyme precursors and they react with one another to opsonise pathogens and start the inflammatory response. An **opsonin** is any molecule that marks an antigen for an immune response making it a more attractive target for phagocytosis. Opsonisation is a molecular mechanism that chemically modifies bacteria to make them more attractive to phagocytes. Complement proteins C3b and C4b are opsonising proteins and this aids both phagocytosis and the inflammatory response.

Classical and alternate pathway

The **classic complement pathway** is activated when the first protein (C1q) in the complement cascade binds directly to the pathogen surface, though it can also be triggered during an adaptive immune response due to an antigen-antibody reaction. In some of the steps proteolysis occurs with a complement protein being cleaved into a smaller protein and a peptide. The smaller protein remains bound to the complex at the surface of the micro-organism while the peptide diffuses away. Figure 77.1 shows complement C3 being cleaved into C3b and 3Ca. The complement protein 3Cb is an important opsonin and strongly activates phagocytosis attracting both macrophages and neutrophils to engulf the organism to which it is attached.

The **alternate pathway** starts when a large polysaccharide from the membrane of an invading micro-organism combines with complement proteins B and D. This pathway also forms an amplification loop for the classic pathway.

QUESTIONS

1. What are complement proteins?
2. Identify the main complement proteins.
3. What initiates and activates the complement system?
4. Explain what is meant by the complement system being a cascade of steps.
5. Where do most of the steps in the cascade occur?
6. In what ways can complement proteins amplify the inflammatory response?
7. Use an example to show how complement proteins are involved in chemotaxis.
8. List some of the ways complement proteins act in defence of the body.
9. Define an opsonin.
10. What is the classic complement pathway?
11. How do complement proteins increase defence against pathogens?
    (A) Opsonisation.
    (B) Enhanced phagocytosis.
    (C) Cause chemotaxis.
    (D) All of the above.

66 The Inflammatory Response

The inflammatory response is a localised innate immune response that occurs when there is physical injury or infection of a tissue. It is a protective mechanism that involves several activities. The clinical characteristics of inflammation are redness, heat, oedema and pain.

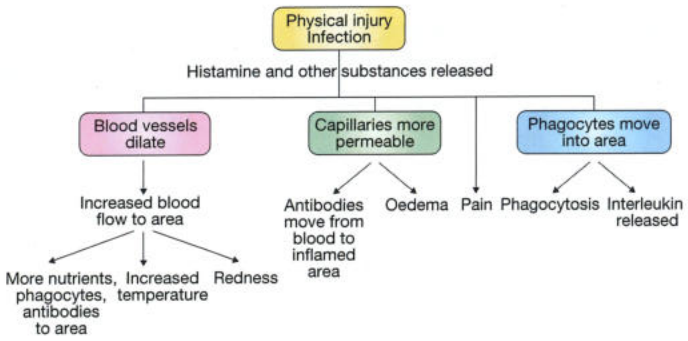


Figure 66.1 Inflammatory response.

Actions in the Inflammatory response

After the injury and/or infection the following changes occur as part of the inflammatory response.

- Histamines and other substances released, e.g. serotonin, prostaglandins, reaction products of the complement system and lymphokines. Histamine is stored in mast cells, e.g. in connective tissue and when injured they degranulate and release the histamine.
- Histamines cause dilation of blood vessels in the affected area increasing blood flow to the area.
- Increased blood flow makes the skin look red and feel warm.
- Histamines make the walls of nearby capillaries more permeable allowing more fluid to leave circulation and enter the tissues causing oedma and swelling of the tissue.
- Increased permeability of the capillary walls allows blood proteins and antimicrobial proteins, e.g. immunoglobulins (antibodies) to leave the bloodstream and enter the tissues.
- Clotting elements move into the affected area aiding in tissue repair and blocking the movement of the pathogen into other areas.
- The injured area is 'walled off' from other sections by fibrinogen clots reducing fluid movement through the spaces.
- Increased blood flow brings more phagocytes to the area, e.g. first neutrophils and then monocytes. Increased phagocytosis is the major factor in destroying and removing the invading pathogen.
- White blood cells release cytokines and other signals, e.g. interleukins that affect the hypothalamus to help adjust and regulate body temperature and help mediated the inflammation response.
- If the whole body becomes affected rather than the localised area, then fever can occur. **Fever** is an abnormally high body temperature and is usually accompanied by shivering, sweating and headache. Fever affects viral activity, decreases circulating iron which decreases metabolic activity of viruses, interrupts the reproduction of some bacteria and viruses and promotes the production of some T lymphocytes and antibodies.

QUESTIONS

1. What is the inflammatory response?
2. What are the clinical characteristics of inflammation?
3. Outline the importance of histamines in the inflammation response.
4. When staphylococci bacteria invade tissues they release large amounts of lethal cellular toxins. Explain how the 'walling off' of the affected area is an important part of the defence mechanism.
5. Where is histamine stored?
6. Explain why redness is a typical sign of infection and the inflammatory response.
7. Explain why an infected area, e.g. pinprick on a finger can become hotter than other parts of the finger.
8. Explain why there is swelling in an affected area.
9. Explain why a thick whitish to green pus can appear in an infected area.
10. Explain how fever can help fight an infection.
11. Draw a flow chart to show some of the changes that occur during the inflammation response.
12. The diagram shows several stages of the inflammation response.

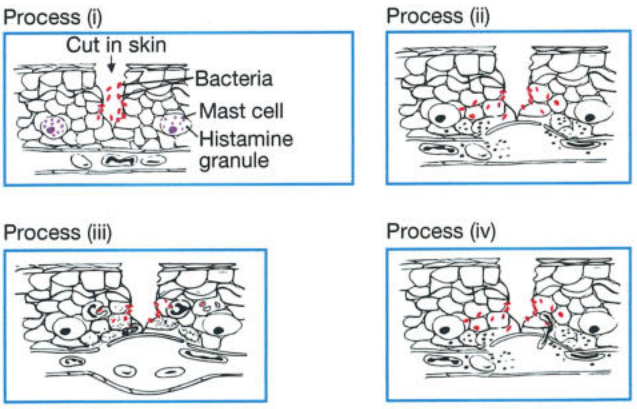


Figure 66.2 Steps in the inflammatory response.

What is the correct order of these steps?

(A) i, ii, iii, iv
(B) i, iii, ii, iv
(C) i, ii, iv, iii
(D) i, iv, ii, iii

67 The Third Line of Defence

The third line of defence involves the production of specific cells to combat specific antigens: B cells, T cells and antibodies. **Innate immunity** is present before exposure to any pathogens and works from the time of birth. **Acquired immunity** (also called adaptive immunity) occurs after exposure to inducing agents, e.g. pathogens, toxins, abnormal body cells and other foreign substances.

An important part of acquired immunity is immunological memory. **Immunological memory** allows the immune system to remember the antigens on a particular pathogen, tumour cell or other foreign substance and to respond more rapidly and more effectively to that particular encountered antigen.

Acquired immunity is very specific and combats specific antigens. **Lymphocytes** are a particular type of white blood cell that are involved in acquired immunity. Lymphocytes produce two main types of specific immune response.

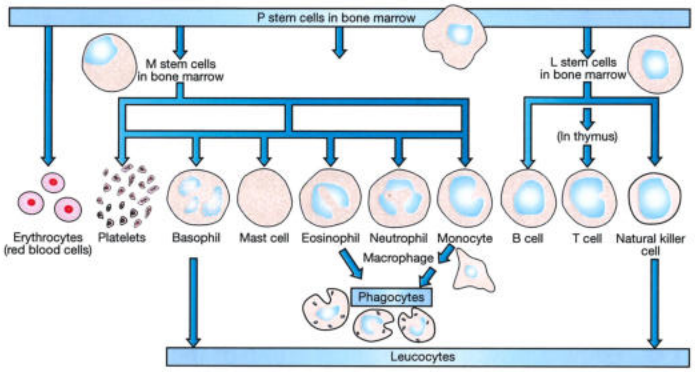


Figure 67.1 Types of leucocytes (white blood cells).

There are two main types of specific immune response – humoral (or antibody mediated) immunity and cell mediated immunity.

Humoral immunity

In humoral immunity the lymphocytes produce specific antibodies that are designed to destroy the pathogen. **Antibodies** (immunoglobulins) are proteins that mark a particular antigen for elimination. The type of lymphocyte that produces antibodies is the B lymphocyte (B cell). B cells are formed in bone marrow and complete their development in bone marrow. B cells were actually named after the bursa of Fabricius where B cells mature in birds. B memory cells retain immunological memory of the antigen and the antibody that needs to be produced to combat that antigen.

Cell mediated immunity

In cell mediated immunity the lymphocytes directly attack the invading pathogen. The type of lymphocyte involved in cell mediated immunity is a range of T lymphocytes (T cells). T cells are formed in the bone marrow and complete their development in the thymus. T memory cells retain immunological memory and can quickly respond to the encountered antigen.

QUESTIONS

1. Distinguish between innate immunity and acquired immunity.
2. Outline the importance of immunological memory.
3. Which cells are involved in adaptive immunity?
4. Compare T cells and B cells.
5. What are antibodies?
6. Identify the components of the third line of defence.
7. The body has three main types of defence mechanisms. Identify the three main types and give an example to show how each can maintain a line of defence against infection.
8. Complete the flow chart to show what happens once an antigen enters the body.

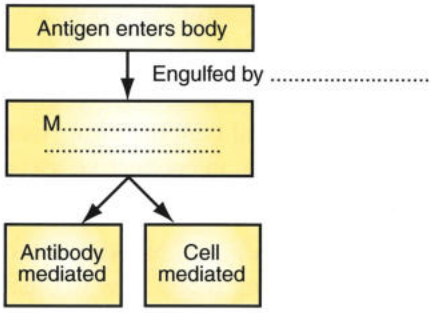


Figure 67.2 Antigen flow chart.

9. Antibodies are proteins that are produced in response to an antigen. What produces antibodies?
   (A) Phagocytes.
   (B) Lymphocytes.
   (C) Neutrophils.
   (D) Macrophages.
10. What is the function of B memory cells?
   (A) Produce antibodies on a daily basis.
   (B) Increase production of antigens after next exposure to pathogen.
   (C) Storage of antibodies.
   (D) Recognise antigen if it invades again.
11. Cells which are part of the body's defence mechanism, produced in the bone marrow and which mature in the thymus gland are called:
   (A) Phagocytes.
   (B) Macrophages.
   (C) T lymphocytes.
   (D) B lymphocytes.

68 B Lymphocytes

B lymphocytes (B cells) are white blood cells, formed in the bone marrow like all white blood cells from pluripotent haematopoietic stem cells. They mature in the bone marrow and are released into the bloodstream where they circulate through blood and lymph. The lymphocytes are found concentrated in lymphoid tissue such as lymph nodes, tonsils, Peyer's patches and spleen. B cells provide antibody mediated (humoral) immunity.

Humoral immunity does not develop until after invasion by a pathogen or toxin. Most invading organisms are first phagocytosed and partially digested by **macrophages**. The macrophages become **antigen presenting cells** and also secrete cytokines, e.g. interleukin to promote growth and reproduction of specific lymphocytes. The antigen is bound and displayed on the **major histocompatibility complex** (MHC) molecule which is a cell surface molecule.

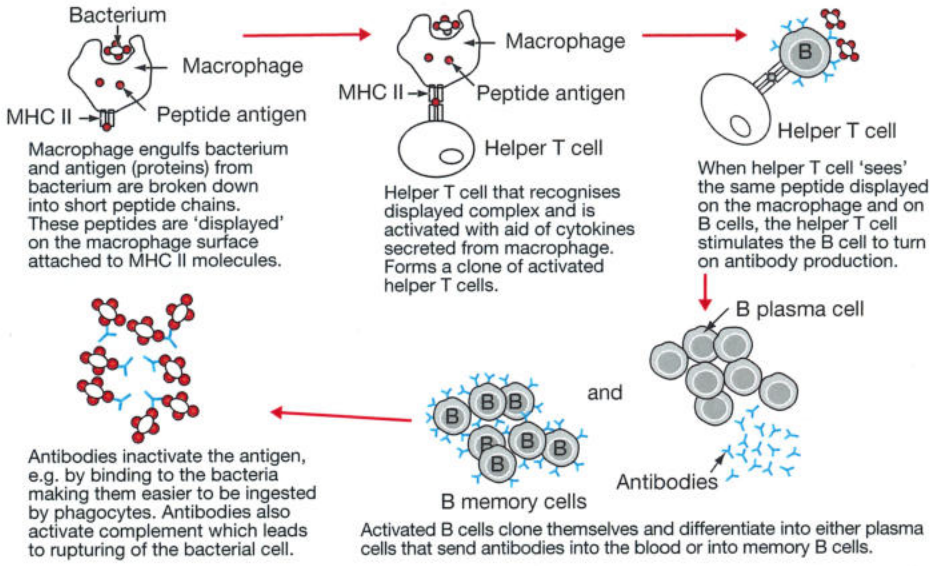


Figure 68.1 Antibody production.

B lymphocyte receptor

A B lymphocyte has antigen receptors for only one type of antigen and all the receptors recognise the same epitope. The **epitope** (also called the antigenic determinant) is the small accessible region of an antigen to which the antibody or antigen receptor binds. A single B lymphocyte can have about 100 000 antigen receptors.

The B cell receptor is a Y shaped molecule with the long tail of the Y shape consisting of two heavy chains that are anchored in the plasma membrane of the cell. Each B cell receptor has two antigen binding sites – one at each end of the short arms of the Y shape. Secreted antibodies are similar to the B cell receptors having the same Y shape and same antigen binding sites but antibodies lack the transmembrane regions that are present in the B cell receptors.

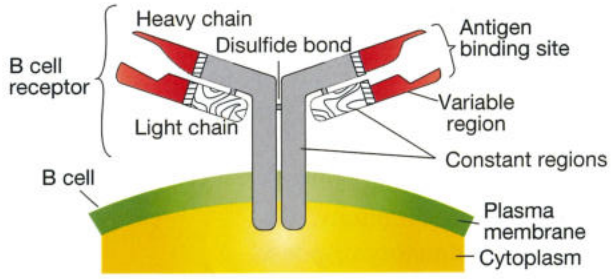


Figure 68.2 B cell receptor.

B lymphocyte activation

The presence of an antigen activates B cells to divide and differentiate into either plasma cells or memory cells. **Plasma cells** secrete immunoglobulins called antibodies with a shape compatible to that of the antigen. The antibody can bind with its particular antigen to form an antibody-antigen complex. This binding destroys the antigen by either immobilising it, or by blocking and neutralising an active binding site of the antigen, or by causing the antibody-antigen complexes to agglutinate and clump which makes them easier to eliminate by phagocytosis.

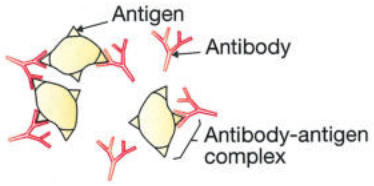


Figure 68.3 The antigen binding site at the end of the short arm of the antibody binds with the antigen to form an antibody-antigen complex.

Immunoglobulins

There are five classes of immunoglobulins (antibodies).

- **IgM** antibodies are the first circulating antibodies to appear after an infection and last a short time. They cause the antigens to agglutinate.
- **IgG** are the most abundant antibodies and activate complement proteins. They easily leave the blood vessels and enter tissues. They pass immunity from the mother to the foetus across the placenta, and from the mother to the baby in breast milk.

- **IgA** are present in external secretions such as tears, saliva and milk. They are found in mucous membranes and prevent the attachment of viruses and bacteria to the epithelium.
- **IgD** are found on the surface of antibody-forming B cells, probably acting as antigen receptors.
- **IgE** cause the release of histamine to begin the inflammatory response and play a role in the expulsion of parasites such as intestinal worms. The release of histamine and other chemicals can cause an allergic reaction.

B memory cells

The B memory cells remain in the body for a long time and can identify that particular antigen. A later infection of the same antigen will be detected by the memory cells, which cause a rapid, large scale production of the needed antibody. Memory cells provide immunity.

Interaction between B and T lymphocytes

B cells cannot function properly without T cells. Helper T cells induce B cells to divide and produce large numbers of clones (clonal selection), resulting in millions of B cells dedicated to destroying a particular antigen. T cells stimulate the production of antibodies by B cells. Interaction occurs between the B and T lymphocytes either through cell contact between the two cell types, or the production of chemicals by the T cell (such as interleukins) which induce activated B cells to divide.

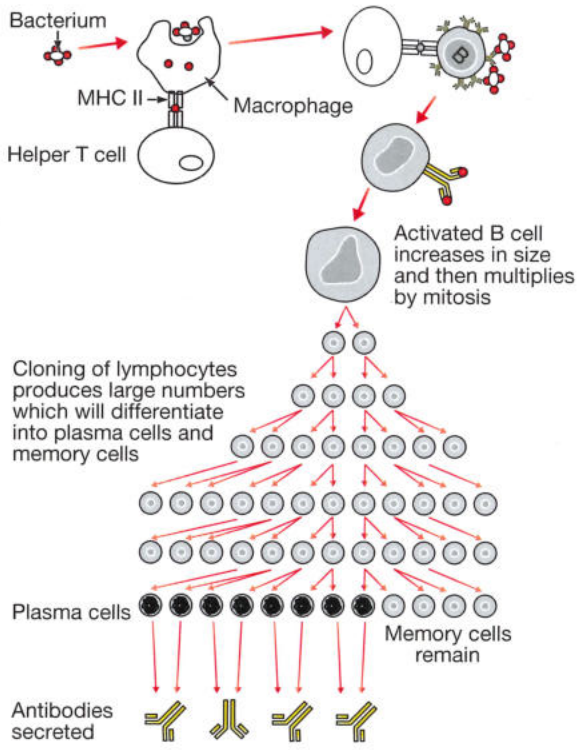


Figure 68.4 Clonal selection of lymphocytes.

Clonal selection of lymphocytes

Clonal selection is a process in which an antigen binds to and activates only the lymphocytes that have the specific receptor for that antigen. Since many lymphocytes are needed these specific lymphocytes are then cloned to produce great numbers. The clones will differentiate into effector cells (e.g. plasma cells for B lymphocytes) and memory cells that are specific for that antigen.

QUESTIONS

1. Where are B cells formed and where do they mature?
2. When does a specific humoral immunity develop?
3. Outline the first step that usually occurs when a new pathogen or toxin enters the body.
4. What is the major histocompatibility complex?
5. What is the epitope?
6. Describe the B lymphocyte receptor.
7. Compare a B lymphocyte receptor to an antibody.
8. What happens when B cells are activated?
9. Outline the function of plasma cells
10. Explain how B lymphocytes act against an antigen.
11. Construct a table to summarise the five classes of immunoglobulins.
12. What is the function of B memory cells?
13. What is clonal selection?
14. Describe how B and T lymphocytes interact.
15. Describe a possible mechanism which allows the interaction between B and T cells.
16. Which of the following best fits the formation of antibodies?
    (A) When B cells stimulate T cells to begin manufacture.
    (B) When T cells stimulate B cells to begin manufacture.
    (C) Only in response to a malfunction in the immune system.
    (D) After a full course of antibodies in an immunisation program.
17. What are B Lymphocytes?
    (A) White blood cells, mainly effective against viruses.
    (B) White blood cells which can recognise antigens.
    (C) A type of phagocyte which can engulf and destroy pathogens.
    (D) Red blood cells, mainly effective against bacteria.
18. Which of the following is caused by the production of antibodies?
    (A) Acquired immunity.
    (B) Passive immunity.
    (C) Inflammatory response.
    (D) An individual to suppress tissue rejection.

69 The Structure of Antibodies

Antibodies (also called immunoglobulins) are globular proteins secreted by plasma cells that bind to particular antigens and mark it for elimination.

All antibodies have the same **Y shape** and in their monomer form consist of two identical heavy polypeptide chains and two identical light chains joined by disulfide bridges. The shape is the same as the shape of the B cell receptor without the transmembrane region that holds the receptor protein in the plasma membrane. Antibodies have molecular weights between 160 000 and 970 000 and make up around 20% of all plasma proteins.

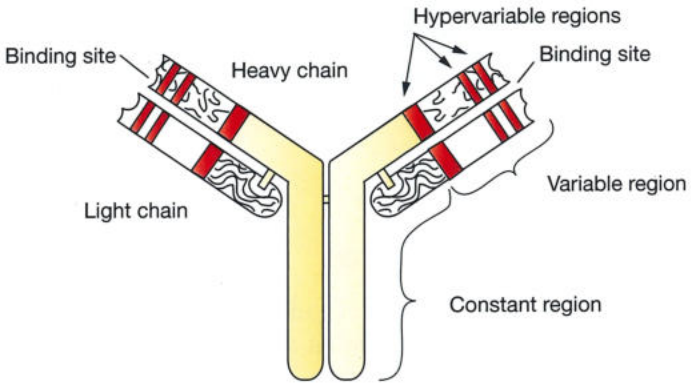


Figure 69.1 Y shaped antibody.

The **variable region** at the end of the heavy and light chains determines the specificity of the antibody and has the antigen binding site. The organisation of the amino acids in this region gives the variable portion different shapes which allows the antibody to become specific for a specific antigen. The high specificity is due to many bonding sites between the antigen and antibody with the complex held together by hydrophobic bonding, hydrogen bonding and ionic attraction. The Y shape allows each antibody to bind with two antigen molecules.

The **constant region** determines other properties such as its ability to diffuse into tissues, its ability to pass through membranes, its ability to adhere to specific structures within the tissues, ability to attach to the complement complex and other biological properties.

Classes of antibodies

There are five general classes of antibodies – IgA, IgD, IgE, IgG and IgM. In a normal person about 70% to 75% of antibodies are IgG. **IgG** is abundant in blood and in tissue fluid and is the only class of antibody that can cross the placenta giving passive immunity to the embryo/foetus. IgG is important as it promotes opsonisation, neutralisation and agglutination of antigens.

IgE is only a small proportion of antibodies in a person but triggers the release of histamines from mast cells and basophils and is involved in allergies.

Antibody functioning

Antibodies can directly attack an invader and can activate the complement system.

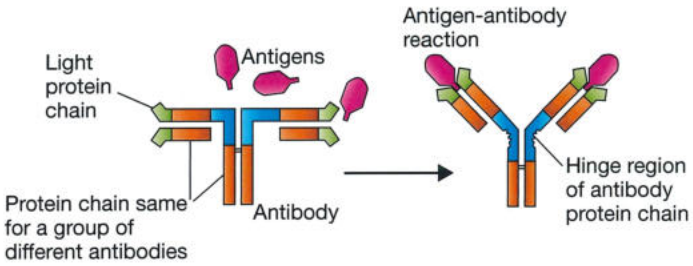


Figure 69.2 Antibody functioning.

When antibodies directly attack an invader they combine to form an antigen-antibody complex. This can cause:

- **Agglutination** binding everything into a clump.
- **Precipitation** of the clump which has become insoluble.
- **Neutralisation** with the binding sites to the host blocked by the antibody.
- **Lysis** with the cell membrane of invader being ruptured.
- **Opsonisation** and increased phagocytosis.

The activation of the complement system causes amplification of the effects of the immune system and enhances phagocytosis.

QUESTIONS

1. What is an antibody?
2. Describe the shape of an antibody.
3. What are the main components of an antibody?
4. Explain why each antibody can be specific to a specific antigen.
5. What is the benefit of the constant region of an antibody?
6. Name the five classes of antibodies.
7. (a) What is the most abundant antibody in the blood?
   (b) What is the function of this antibody?
8. Which antibody gives passive immunity to a foetus?
9. Explain how antibodies act against an antigen.
10. What is the maximum number of antigens which can be bound by one antibody?
    (A) One (B) Two
    (C) Three (D) Four

70 T Lymphocytes

T lymphocytes (T cells) form in the bone marrow but mature in the thymus gland where they are 'programmed' for a specific antigen. The thymus is a gland above the heart in the thoracic cavity.

T cells provide cell mediated immunity, being particularly effective against body cells which have been infected by a virus or other pathogen. There are several different types of T cells; killer T cells (cytotoxic T cells), helper T cells and suppressor T cells. Memory T cells recognise the antigen if it is reintroduced.

Note that **natural killer (NK) cells** are part of the **innate** defences. They attach to cells that are affected by virus or cancer and once bound to the cell cause apoptosis. Natural killer cells act more rapidly than other lymphocytes and secrete antiviral cytokines.

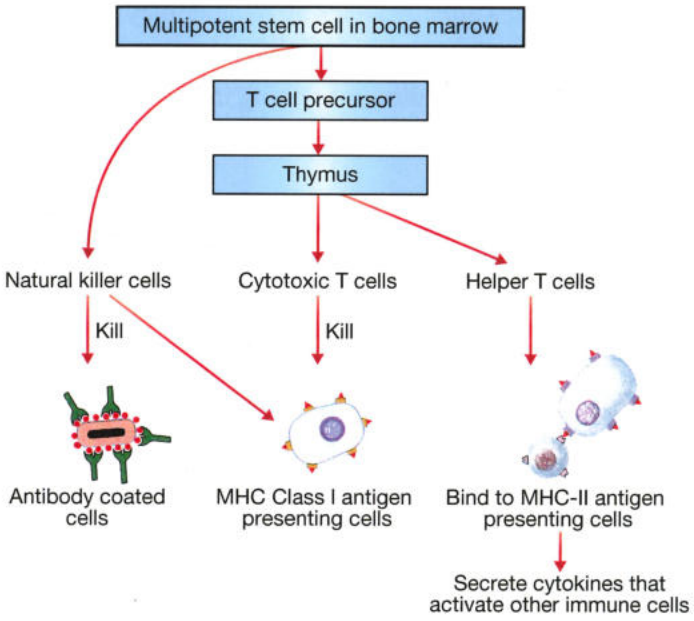


Figure 70.1 Types of leucocytes.

T lymphocyte receptors

The receptors for T lymphocytes are two polypeptide chains linked by a disulfide bridge that have a transmembrane region that locks the receptor into the plasma membrane of the cell. The outer tip of the molecule has an antigen binding site. The T lymphocyte receptors can detect antigens presented by **major histocompatibility complex** (MHC) molecules. These molecules are found on many types of cells, e.g. class I MHC molecules are on almost all nucleated body cells and if cancerous will attract cytotoxic T cells signalling the cell is to be destroyed while class II MHC molecules are mainly on macrophages, dendritic cells and B cells which become antigen presenting cells attracting T helper cells.

The T cell receptors are most important as the sites where the antigen binds to the T cell. There can be as many as 100 000 receptors sites on a single T cell.

T cells recognise cells infected with a virus as the viral proteins are moved to the cell surface and displayed by MHC molecules. The immune system recognises such cells as non-self and T cells are sent to destroy it.

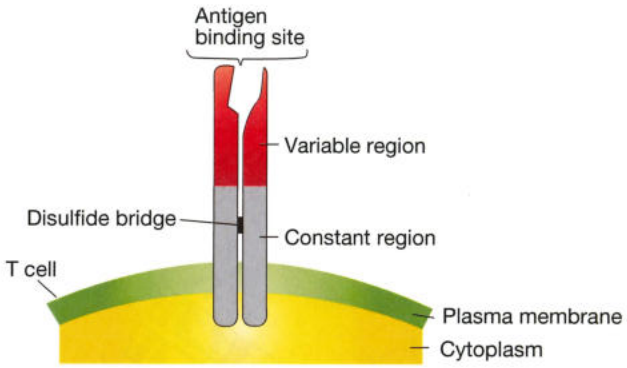


Figure 70.2 T cell receptor.

Types of T lymphocytes

Cytotoxic T cells destroy cells identified as non-self. They are a direct attack cell that can kill some micro-organisms and the body's own cells. They are formed when either a macrophage or an infected cell displays the antigen. The T cell binds with the infected cell and inserts a toxic chemical called **perforin**, a protein which causes the cell membrane to rupture, and the cell lyses. They also release **granzymes** which are digestive enzymes that enter the cell through perforin channels. They also target cancer cells and foreign graft cells.

Helper T cells (T4 cells) stimulate the production of plasma cells and help activate B cells to produce antibodies. Many B cells cannot go into production unless T4 cells for that antigen have first acted. Helper T cells secrete chemicals, such as lymphokines and interleukins, which cause activated B cells or T cells to divide and also stimulate macrophages for phagocytosis. The production of killer T cells is stimulated by T4 cells.

Suppressor T cells (T8 cells) turn off the immune response and suppress the production of antibodies. Some believe they are actually a form of helper T cell. The ratio of T4 to T8 cells is important in the diagnosis of AIDS.

Memory T cells have immunological memory allowing a faster and stronger response to a previously encountered antigen. When a T cell clone is activated by an antigen some of the new clones move to lymphoid tissue throughout the body to be preserved in the tissues. This means that re-exposure to that particular antigen, anywhere in the body will lead to an immediate recognition of the antigen and activation of the adaptive immune response.

Malfunctions with T lymphocytes

Acquired immunodeficiency syndrome (AIDS) – Most antigens activate both T cells and B cells at the same time. Helper T cells are very numerous and act as a major regulator of the immune response and thus a virus like **HIV** which infects helper T cells has a severe effect on the immune response. After an antigen is identified the helper T cells secrete lymphokines that activate specific B cells. The helper T cells proliferate and differentiate into a clone of activated helper T cells and memory T cells. When the HIV virus infects helper T cells it causes the destruction of the helper T cells and nullifies the immune response causing AIDS.

Autoimmune diseases – In an autoimmune disease the body loses tolerance for self and turns against certain molecules in the body. In **multiple sclerosis** T cells destroy the myelin sheath on neurons in the central nervous system.

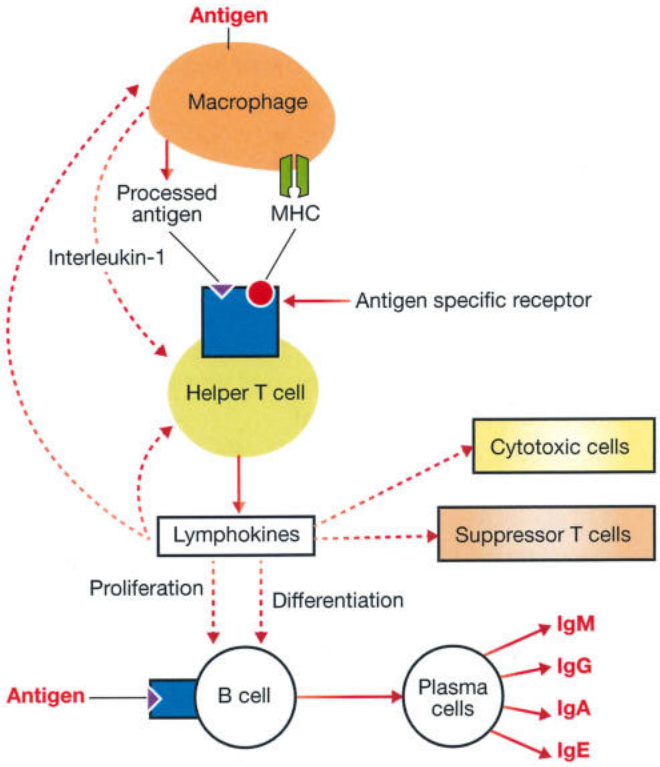


Figure 70.3 Actions of T cells.

QUESTIONS

1. Where are T cells formed and where do they mature?
2. What types of cells are attacked by T cells?
3. Describe natural killer cells.
4. Identify different types of T lymphocytes and explain the difference in their roles.
5. Describe T cell receptors.
6. What is needed for a T cell to detect an antigen?
7. Distinguish between the function of class I MHC molecules and class II MHC molecules.
8. How do T cells act against viral infection?
9. What is the function of memory T cells?
10. Outline how helper T cells help mediate the immune response.

Use the following flow chart which compares the cell mediated and humoral immunity for the next TWO questions.

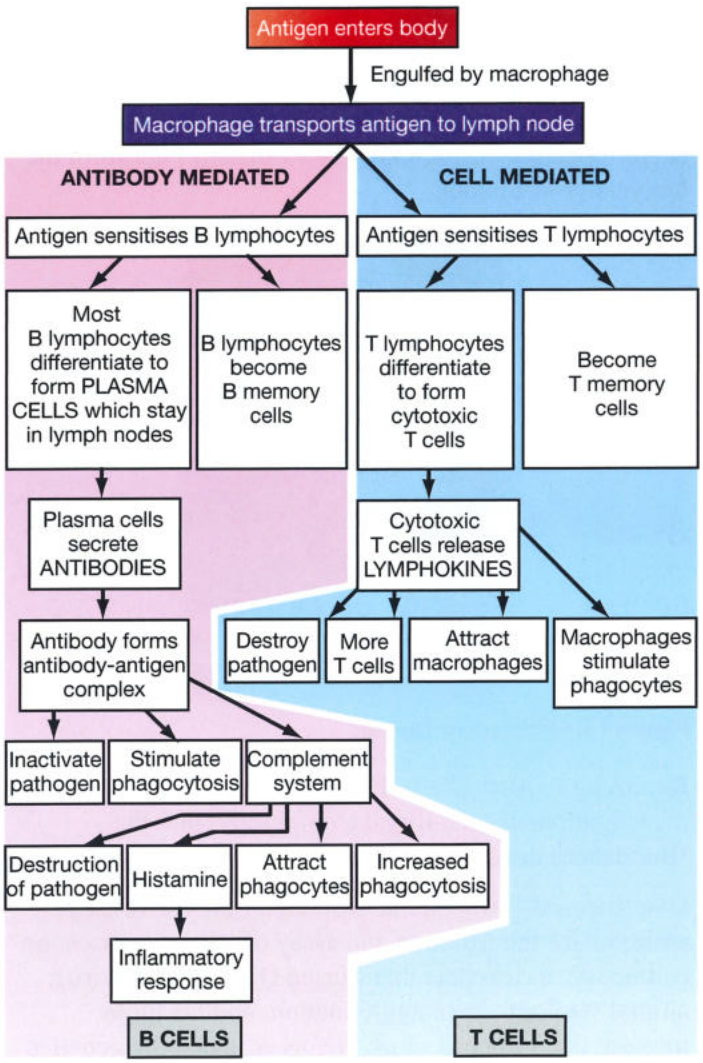


Figure 70.4 Third line of defence.

11. Which of the following is *not* performed by T cells?
    (A) Attract macrophages.
    (B) Directly destroy pathogen.
    (C) Release lymphokines.
    (D) Produce antibodies.
12. What are the main cells involved in cell mediated immunity?
    (A) T lymphocytes. (B) B lymphocytes.
    (C) R lymphocytes. (D) Macrophages.

71 Macfarlane Burnet

Frank Macfarlane Burnet (1899-1985) was born in Traralgon, eastern Victoria and is considered by many to be the greatest Australian scientist of his era. He graduated from the University of Melbourne (MB, BS) and then worked from 1922 to 1924 at Melbourne Hospital as a resident medical officer. In 1924 he graduated as MD and worked at Melbourne Hospital as pathological registrar, and then as senior resident pathologist. In June 1925 he was a ship's surgeon while en route to England. From 1925 to 1927 he worked at the Lister Institute, London, as assistant to the curator of the National Collection of Type Cultures which allowed about two thirds of his time for research on bacteriophages. In 1928 he was awarded a PhD from the University of London.

Figure 71.1 Macfarlane Burnet.

Returning to Australia in 1928, Burnet carried out investigations for the Royal Commission into the 'Bundaberg disaster' while still studying bacteriophages.

Over the next few years he worked on the use of chick embryos for the isolation and assay of animal viruses; on psittacosis, a rickettsia that caused Q fever; poliovirus; animal virology; haemagglutination; and his major interest, the influenza virus. He investigated the genetics of the influenza virus, revealing its high frequency recombination. He was knighted knight bachelor in 1951.

In 1960 he formulated what he regarded to be his major contribution to science, the clonal selection theory of antibody production. This theory helped explain how the body learns to distinguish between 'self' and 'non-self', which is vital in understanding why rejection occurs following organ transplantation. He shared the Nobel Prize in Physiology or Medicine with Peter Medawar in 1960 for his work on immunological tolerance. To culture viruses, he developed an important methodology using living egg embryos which was used to develop a vaccine.

He also made the discovery that antibody production occurs during foetal life and is not inherited. In 1981 he was awarded 'Australian of the Year'. Other honours included an Order of Merit in 1958; Second Order of the Rising Sun (with double rays) in 1961; President of the Australian Academy of Science in 1965; Knight Commander of the Order of the British Empire in 1969; and Knight of the Order of Australia in 1978.

Besides his research discoveries, whilst Director of the Walter and Eliza Hall Institute of Medical Research in Melbourne, Burnet helped train a generation of researchers. Some people felt that he had a socially awkward personality and his public statements on controversial issues (against smoking and nuclear weapons, and in favour of euthanasia, population control and solar energy) created dispute. He did not follow orthodox views on science, religion or politics.

QUESTIONS

1. Outline how the work of Sir Frank Macfarlane Burnet led to a better understanding of the immune response.
2. How was he rewarded for his work?
3. Explain the importance of the discovery that viruses could be cultivated in living chicken eggs.
4. What is the clonal selection theory?
5. Peter Medawar shared the 1960 Nobel Prize in Physiology or Medicine with Macfarlane Burnet. Medawar once described viruses as a piece of nucleic acid surrounded by bad news. Explain what he meant by this statement.
6. In 1900 the German physician Paul Ehrlich proposed that antibodies exist as specific receptors on cell surfaces. Explain why Ehrlich has been called the 'conceptual father' of the clonal selection theory.
7. When studying the structures of various antibodies, Norbert Hilschmann found that amino acid sequence in the protein chains was only different in the first half of each chain, and that all antibodies had the same sequence in the remainder of the chain. Suggest a reason why the presence of both variation and constancy is important in the structure of antibodies.
8. Which of the following cannot be used to culture viruses?
   (A) Chicken eggs. (B) Mice brains.
   (C) Nutrient agar. (D) Bacterial broth.
9. What is Macfarlane Burnet most remembered for?
   (A) Clonal selection theory.
   (B) Naming the chemicals that appear after infection *Antikörper* or antibodies.
   (C) Devising a technique to measure antibodies.
   (D) Using chicken eggs to grow bacteria.

72 Immunity

Immunity involves the body activating the immune response against an invading pathogen. This can occur naturally in **natural immunity** without human intervention or after human intervention in **artificial immunity** when the person has been given antigens or antibodies, e.g. by injection.

Acquired resistance describes the defence mechanisms a person builds up during their lifetime.

Passive acquired immunity can occur naturally for a short time, such as when antibodies cross the placenta to the foetus, or when the presence of antibodies in breast milk allows a baby to acquire the resistance of the mother. Passive immunity can also be artificially induced. For example, injection of immune serum globulin or 'gamma globulin' containing antibodies is sometimes given to protect travellers to countries where hepatitis is widespread. Passive immunity typically lasts only a few weeks, does not involve the production of memory cells and the person cannot make their own antibodies.

Active acquired immunity occurs after the person is exposed to the antigen. This can occur naturally when the person catches the disease and recovers, or it can be artificially induced by vaccination (e.g. tetanus injections). In active acquired immunity, memory cells are produced and the person is able to produce antibodies for that particular antigen.

Immunisation is a process which stimulates the immune system to produce lymphocytes or antibodies to fight an infection, giving immunity to that disease.

Vaccination is a method of providing immunisation, in which weakened or dead pathogens are introduced into the body.

Vaccines can contain either the killed bacteria or viruses (e.g. cholera or typhoid); living attenuated forms of the pathogen (e.g. rubella or poliomyelitis); or a toxoid (e.g. tetanus or diphtheria). Through genetic engineering, scientists can insert specific genes into DNA of certain microbes or mammalian cells, making them into living factories, mass producing a desired antigen. In this way, vaccines are created for diseases such as hepatitis B and malaria.

How vaccination prevents infection. The vaccine acts as an antigen, inducing the immune response; plasma cells and memory cells are produced. The vaccine does not cause the disease or the symptoms so the plasma cells are not really needed. The memory cells are important as they give the body immunological memory for a particular pathogen.

If that pathogen enters the body, the circulating memory cells can respond very quickly and in greater numbers so that the person does not suffer the symptoms or the disease. As the number of memory cells can decrease over time, booster injections are sometimes needed to maintain immunity.

Graph of immune responses to successive exposure to antigen. A graph of the antibody titre in the body against time shows how the immune response after initial exposure compares with the response after a second exposure; the second response is quicker, higher and longer.

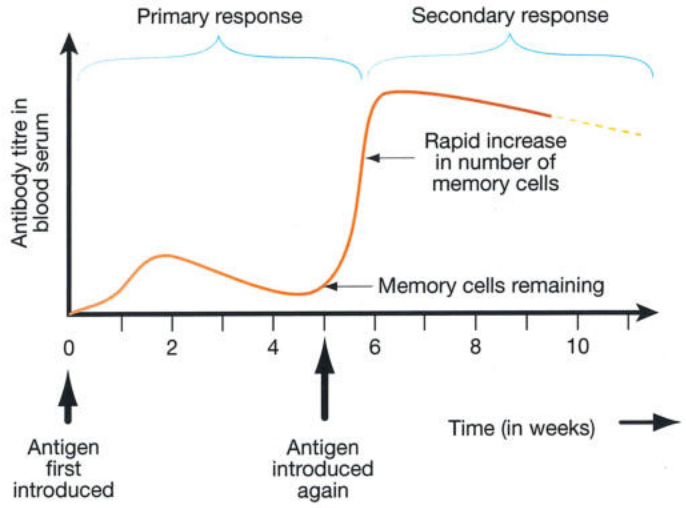


Figure 72.1 Immune response.

QUESTIONS

1. Distinguish between natural and artificial immunity.
2. Compare passive and active acquired immunity.
3. Explain how immediate protection can be given to someone who has been exposed to a particular disease such as hepatitis.
4. What is the difference between immunisation and vaccination?
5. What is a vaccine?
6. Explain how vaccination prevents infection.
7. On the First Fleet in 1788, John Hunter, captain of the Supply, recorded in his journals that they carried small vials containing smallpox scabs. It was common practice to use these scabs to prevent smallpox. Use current understanding about the immune response to explain why the use of smallpox scabs was successful.
8. Which of the following best describes the process of immunisation?
   (A) Injecting antibodies to fight an infection.
   (B) Using vaccines to stimulate natural immunity.
   (C) A process which gives immunity.
   (D) Giving gamma globulins to stimulate the body's immune system.

73 Vaccination Programs

Vaccination programs are important as they help maintain herd immunity. **Herd immunity** involves the majority of individuals in a population being vaccinated against a specific disease. This decreases the chance of transmission of the disease to the few individuals in the population who are susceptible and are not immune, e.g. herd immunity against smallpox led to its eradication and people no longer have to be immunised against smallpox.

The National Health and Medical Research Council has issued guidelines for childhood immunisation. These include the triple antigen (diphtheria, tetanus and whooping cough) at 2, 4, 6, and 18 months, and the Sabin polio oral vaccine at 2, 4, and 6 months and 5 years. It is recommended that people update their diphtheria and tetanus immunisation every 10 years and that people over 65 years of age have the influenza vaccine.

Even though the vaccination programs are in place, whooping cough has returned. In 1991, 342 Australians contracted whooping cough, in 1993 there were 4452 cases and in 2003 there were 5106 cases of whooping cough. In 2015 there was a 70% spike in the number of cases of whooping cough with 2324 cases reported between January 1 and June 10.

The Western Australia State Government provides free vaccines under the National Immunisation program and also funds some other vaccines with each vaccine having criteria for eligibility. The Western Australia immunisation schedule is posted on the government website.

Vaccination and smallpox

Smallpox is a highly contagious viral disease with a high death rate. Symptoms include high temperature and skin blisters. Edward Jenner (1749-1823) developed the vaccine against smallpox using scrapings from the sore of a dairymaid who had cowpox. A global vaccination program for smallpox was introduced by the World Health Organisation in the 1960s, resulting in its eradication by1980. The world's last reported case was the death of a British laboratory worker who became infected with a live smallpox virus kept at a research institution. The only sources of the virus are stored in two high-containment laboratories. There has been debate about the fate of these stores. In 1986 a specialist committee from the WHO suggested the stores be destroyed, but others wish to study the DNA of the virus.

Vaccination and diphtheria

Diphtheria is a highly contagious bacterial disease that begins with a sore throat, fever and rapid pulse. Before the development of a vaccine, mortality was high – up to 90% in diphtheria of the larynx – and two thirds of those who died were under five years old. In 1923 a vaccine was introduced. Diphtheria vaccination is part of the triple antigen, which is part of the childhood immunisation program.

Vaccination and polio

Poliomyelitis is also called infantile paralysis and is a viral disease with no cure. Poliomyelitis has been around since ancient times. The virus attacks the nerve cells and sometimes the central nervous system, with muscle wasting, paralysis and sometimes death following. Jonas Salk began investigating the poliovirus in 1947 and in 1954 began a nationwide testing of his vaccine with 60% to 70% prevention. However, 200 people contracted the disease from the vaccine and 11 of them died. All testing was halted until investigators found that the disease-causing vaccine all came from one batch that had been poorly made. Albert Sabin experimented with monkeys and chimpanzees and isolated a rare form of poliovirus that would reproduce in the intestinal tract but not in the central nervous system. This vaccine could be swallowed, was cheaper to make and became the vaccine of choice. Poliomyelitis is now extremely rare in industrialised nations with most cases being caused by the Sabin vaccine itself as the live virus can mutate to stronger forms. The WHO aims to eradicate polio completely in the first decade of the 21st century.

Organ transplants and the immune response

Tissue rejection after a transplant operation is a possibility, even though tissue proteins have been matched with the donor. To prevent tissue rejection the patient is given immunosuppressive drugs such as antilymphocyte globulin and cyclosporin. Most rejection is due to T cells. Once the immune system is suppressed, the patient is more susceptible to other infections like pneumonia. The implantation of artificial devices such as a pacemaker does not stimulate the production of antibodies.

QUESTIONS

1. Why are vaccination programs important?
2. What is herd immunity?
3. Choose one infection which is treated by the childhood immunisation program and describe the procedure and its success rate.

4. The graph shows the amount of antibodies in the blood of a person who was exposed to a particular pathogen. If the same person was exposed to the same pathogen two months later, draw in the expected graph to show how the number of antibodies in the blood would change. Explain your reasoning.

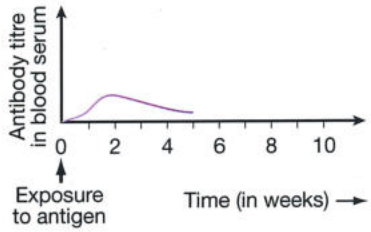


Figure 73.1 Graph of person exposed to particular antigen.

5. There has been a decline in the number of children being immunised. In 1989 to 1990, 73% of children were immunised against whooping cough while in 1995, 60% were immunised. There has been a corresponding rise in the number of cases of whooping cough, with the death of several babies in 1997. Suggest two reasons why people are not having their children immunised.
6. Present one argument for and one argument against the destruction of the last stores of the smallpox virus.
7. Outline reasons for the suppression of the immune response in organ transplant patients.
8. For the year 1988 there were 35 251 reported cases of poliomyelitis on a global basis. This number had dropped to 3228 reported cases in 1998. Outline a reason for the dramatic change in reported cases.
9. Smallpox has been eradicated since the 1980s. Poliomyelitis is now very rare in industrialised nations and the WHO hopes to eradicate it early in the 21st century. What control measures are needed to eradicate an infectious disease?
   (A) Protect susceptible people, prevent transmission and eliminate the reservoir of the pathogen.
   (B) Cure all symptoms and educate people about the correct methods of personal hygiene.
   (C) Build the infrastructure to make sure there are proper sewage treatment plants and chlorination of the water supply.
   (D) Change people's diet making sure they cook food properly and suitably dispose of wastes.
10. In Australia, more people die from non-infectious diseases such as cardiovascular diseases and cancers than from infectious diseases such as tuberculosis and poliomyelitis. The decrease in deaths due to infectious diseases is a result of:
    (A) Higher standard of living (e.g. electricity, motor vehicles, the internet).
    (B) Eradication of pathogens.
    (C) Discovery of antibodies.
    (D) Public sanitation, personal hygiene and immunisation programs.
11. During organ transplant operations, immunosuppressive drugs restrict tissue rejection. Why does this occur?
    (A) They prevent the patient sensitising T cells to the transplant tissues.
    (B) They inhibit the donor tissue from producing either T or B cells.
    (C) They suppress the recipient's production of antigens.
    (D) They attack any foreign particles which cause infection of the transplant tissues.
12. The graph shows the incidence of whooping cough over fifty years.

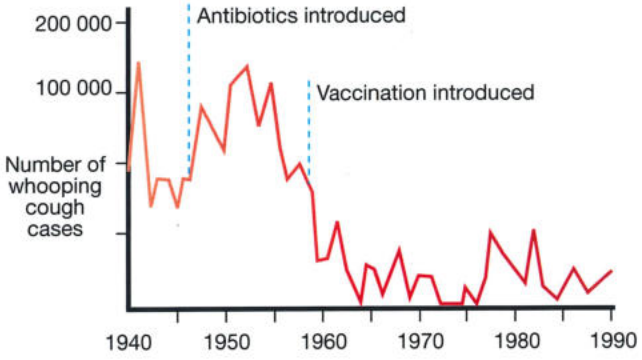


Figure 73.2 Graph of whooping cough.

What does the graph suggest?
   (A) Antibiotics were unnecessary once the vaccine had been discovered.
   (B) In the mid 1970s all children were vaccinated and were immune to whooping cough.
   (C) The introduction of antibiotics caused a decrease in whooping cough.
   (D) Parents in the 1980s were not as concerned about whooping cough as previous generations.
13. Organ transplants from one human to another can have the adverse effect of organ rejection by the recipient. What causes rejection?
    (A) The recipient is unable to produce lymphocytes.
    (B) The transplanted organ acts as an antigen.
    (C) Antigens produced by the recipient attack the antibodies present in the donor organ.
    (D) Antibiotics produced by the recipient recognise the donor tissue as non-self.
14. What is the aim of long term vaccination?
    (A) Produce constant, suitable levels of antibodies in case of infection.
    (B) Maintain the level of circulating memory cells.
    (C) Enable the body to manufacture antigens when infection occurs.
    (D) Provide autoimmunity.

74 Epidemiology

Epidemiology is the study of disease and its prevalence in the community. **Epidemics** occur when there is a rapid spread of the pathogen and large numbers of people are infected. Epidemics can be **pandemic**, spread to many countries of the world; **endemic**, always present in particular parts of the world; or **sporadic,** occurring in different places at different times.

To understand the reasons for an epidemic, several factors need to be investigated: the cause of the disease, how it is transmitted, who is at risk, method of treatment and control, long term measures of prevention, virulence of the strain, and whether it changes over time. To study these issues, scientists need to collect and analyse statistical data for whole populations. Only then can suitable procedures be set in place. Often this involves the government and the administrative departments which deal with relevant areas, for example construction of sewage treatment works, freshwater supplies and immunisation programs.

Case study – cholera

In 1830 and 1831 a massive cholera epidemic swept through Europe. The people living in the crowded, underserviced slums were often hardest hit. Since the cholera would disappear just as quickly as it appeared, quarantine was not particularly effective. There was considerable dispute about the cause of the disease. Some felt it was contagious, caused by a micro-organism, while others felt it was environmental and moral with the clean, godly and prudent being saved.

In 1854, Filippo Pacini identified the vibrio bacteria that causes cholera after studying the bodies of cholera victims.

In England, John Snow studied the epidemic. He made maps of the areas which had been affected by cholera. He interviewed families to find out where they obtained their water for drinking and washing. He worked in one of the slum areas of London and had a belief that cholera was spread through the water supply. Two weeks after the beginning of the epidemic, on 7 September 1854, he persuaded the local aldermen to remove the pump handle from a public pump on Broad Street. However the most serious phase of the epidemic had already passed.

In 1866, New York followed John Snow's theory and clean water was brought to people in a cholera infected area. Houses were also disinfected. Study of the history of the epidemic had helped to stop this cholera outbreak, even though scientists did not fully understand the cause of the disease or why this therapy was effective.

QUESTIONS

1. What is epidemiology?
2. Distinguish between epidemics that are pandemic, endemic and sporadic.
3. Describe the main features of epidemiology.
4. Use a case study to show how epidemiology can save lives.
5. Why is the work of John Snow significant in our understanding of disease?
6. In the 19th century yellow fever, or yellow jack, was one of the most feared diseases in the Americas. The disease was endemic to West Africa and was brought to America on the slave ships. Most black Africans did not die of the disease but it killed many Caucasians, American Indians and Asians. Suggest a possible reason why black Africans were not as susceptible to the disease.
7. In February 2015 there was an outbreak of hepatitis A in Australia. Frozen berry products from overseas markets were recalled from supermarkets. Explain how this epidemic would have started.
8. In April 2000 there was an outbreak of Legionnaires' disease in Melbourne and northern Victoria. Four people died from the disease. How can authorities find the source of an outbreak such as Legionnaire's disease?
9. The outbreaks of yellow fever epidemics in the 19th century are well recorded. Many people mapped the spread of yellow fever in attempts to find its cause and method of transmission. In 1927 the life cycle of the virus was investigated and shown to involve both the *Aedes aegypti* mosquito and monkeys. Unlike smallpox, yellow fever will probably never be completely eradicated. Which of the following gives a possible reason for this?
   (A) Yellow fever is caused by a virus while smallpox is a bacterial disease.
   (B) The virus remains in jungles in the monkey-mosquito life cycle.
   (C) Smallpox has a vaccine while it is impossible to make a vaccine for yellow fever.
   (D) Yellow fever has a longer incubation time than smallpox.
10. The ebola virus is endemic to Africa and is highly infectious, causing haemorrhagic fever where blood oozes from the skin and mouth and out of the rectum. In the recent outbreak in Africa there were over eleven thousand deaths by the end of 2015. Why did this outbreak reach such an epidemic size?
    (A) It spread to densely populated cities.
    (B) Increased people movement between areas.
    (C) Dysfunctional health care system.
    (D) All of the above.

75 Pandemics – Influenza

Influenza is an infectious disease caused by a variety of strains of a RNA virus. The virus has a direct life cycle and is usually transmitted when the host coughs or sneezes without using a handkerchief and the airborne droplets carry the virus to another host. Although the influenza virus is relatively 'hardy' and can exist independently outside a host body for quite a while it can be inactivated by sunlight, disinfectant and detergents.

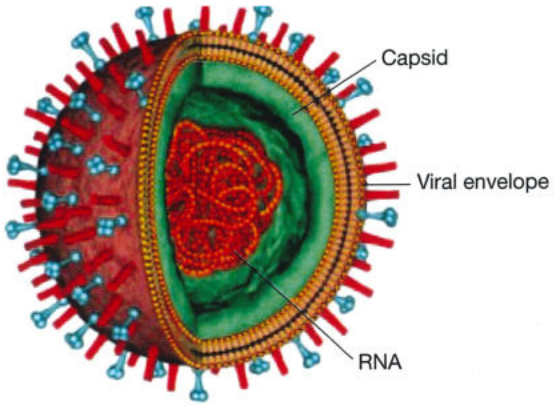


Figure 75.1 H1N1 virus.

Flu symptoms

Typical 'flu' symptoms include – sore throat with inflammation of the upper respiratory tract, fever, chills, congested nose, pain in the back and limbs, headache and total exhaustion. Complications can occur when the virus settles in parts of the respiratory system, e.g. adults can get pneumonia in the lungs or children can get croup in the larynx or when there are secondary bacterial infections, e.g. causing bronchitis.

Strains of the influenza virus

There are three main strains of influenza virus: A, B and C. Type A strains cause severe illness and have caused human pandemics. Type B causes sporadic human cases and small scale outbreaks. Type C strains rarely cause human infection and have not caused outbreaks.

Influenza A (H1N1) virus is a subtype of influenza A. It was the most common cause of human influenza in 2009. Some strains of H1N1 are endemic in pigs (swine influenza), in birds (avian influenza) and in humans. If transmission of the swine flu virus causes human influenza, it is called zoonotic swine flu. The outbreak of H1N1 was declared a pandemic by the WHO in June 2009.

The host produces antibodies against each specific strain of influenza to which they have been exposed. Once a particular strain has passed through a population there is a change in its structure and it becomes a new strain. The antibodies produced for the previous strain are often ineffective against the new strain and people can thus get the 'flu' many times. Influenza cannot be cured and treatment involves relieving the symptoms, e.g. bed rest, extra fluids, aspirin (not for children) or paracetamol to reduce fever and muscle aches.

Control and prevention involve the use of the 'flu vaccine'. Each vaccine is specific to particular strain(s) and people who rely on the vaccine for prevention need to have the injection yearly before the beginning of winter. A new version of the vaccine is developed twice a year due to the rapid changing nature of the virus.

To prevent the spread of infected airborne droplets, people should cover their mouth and nose when coughing or sneezing. Some people wear face masks in winter as a preventive measure. Prevention of H1N1 involves prevention in swine, prevention of transmission to humans and prevention of its spread among humans.

1918 influenza epidemic

The 1918 influenza epidemic known as the Spanish flu killed more people in one year than the Black Death plague killed from 1347 to 1351. About one fifth of the world's population was infected with the Spanish flu. It is one of the worst pandemics recorded in history.

A common ditty sung during the 1918 outbreak was –

There was little bird called enza
I opened the window and in flew enza.

QUESTIONS

1. What is influenza?
2. Describe the influenza pathogen.
3. Describe influenza symptoms.
4. When can complications arise with influenza?
5. Compare the three main strains of the influenza virus.
6. Outline how influenza H1N1 has been involved in pandemics.
7. How does the host respond to H1N1?
8. What often happens to a particular strain of the virus after it has passed through a population and how does this affect humans?
9. What is the treatment and cure for influenza?
10. Describe the flu vaccine.
11. What actions can people take to prevent the spread of H1N1?
12. Why are antibiotics ineffective against H1N1 influenza?
    (A) Antibiotics do not work against viruses.
    (B) H1N1 keeps changing its surface structure.
    (C) Viruses produce cytokines that destroy antibiotics.
    (D) H1N1 is a bacterial pathogen.

76 Modelling Disease Outbreak and Spread

Scientific models are used to show an idea or system giving a representation that makes it easier to understand or see what is happening.

Theoretical models are usually simplified descriptions that can be used to make predictions. In many instances experiments can be carried out to test the accuracy and reliability of the model. These experiments either confirm or disprove the model.

Modelling for disease and life expectancy began in the 19th century.

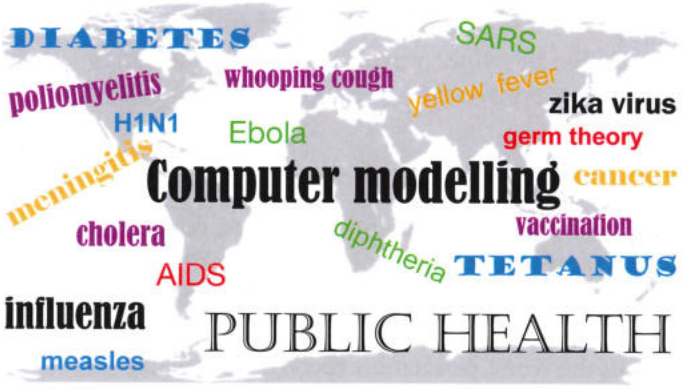


Figure 76.1 Disease modelling.

Daniel Bernoulli

Daniel Bernoulli (1700-1782) is known for Bernoulli's principle on the conservation of energy, but he also used statistics in 1766 to analyse smallpox **morbidity** (presence of the disease in patients) and **mortality** (number of deaths) data in an attempt to work out the efficiency of vaccination. He used **statistical censoring** which means only some of the values, measurements or observations are known. This was one of the first times that censored data was used in a statistical analysis.

A kind of smallpox vaccination, **variolation** using material taken from a patient, e.g. rubbing powered smallpox scab was introduced into England by **Lady Mary Wortley Montagu** between 1716 and 1718. Intentional infection was first practised in China with powdered scabs being inhaled as snuff. Inoculation became a common practice in the 18th century but there were serious risks. Smallpox caused by variolation was just as contagious as normally contracted smallpox and is responsible for some epidemics.

Edward Jenner carrier out his famous smallpox vaccination experiment in 1796 using pus from blisters caused by cowpox after learning that smallpox never infected milkmaids. Before his famous experiment Jenner tried variolation on milkmaids who had had cowpox and could not induce the usual infection.

Contemporary models

Developments in computer software with increased processing capacity and data storage have enabled the development of more complex models with computers manipulating large amounts of data to simulate events. Some contemporary models investigate how diseases will progress and spread,

Contemporary models can simulate the effects of possible interventions such as mass immunisation in public health programs. Models are also used to investigate new variables and new relationships such as the relationship between epidemic frequency and location. Computer programs that simulate the relationship between environmental changes, population size and the spread of particular diseases can show what would happen in different scenarios. Knowing the likelihood of diseases spreading to certain areas due to climate change and global warming can aid future development plans and the construction of new preventive strategies.

Case study – zika fever and congenital microcephaly

Zika fever is caused by the zika virus (ZIKV) and is related to dengue and yellow fever with milder symptoms such as mild headache, macropapular rash, fever, conjunctivitis and joint pains. About one in five people who are infected with the zika virus show symptoms.

The **zika virus** is a single stranded RNA molecule. Like other flaviviruses the zika virion attaches to the host cell membrane receptors and induces virion endocytosis. Inside the host cell it replicates in the host cytoplasm in the endoplasmic reticulum and new virions are transported to the Golgi apparatus to be secreted from the host cell. The virus is transmitted to humans by the *Aedes* mosquito which means the spread of the disease is related to the distribution of the vector. Incubation time of the virus in the mosquito vector is around 10 days.

An outbreak in 2015 reached pandemic proportions with global interest especially across the Americas due to the linking of zika virus during pregnancy with the birth of a child with congenital microcephaly. A child born with microcephaly will have an abnormally small head compared to babies of the same sex and age and incomplete brain development. In severe cases of microcephaly the child can have seizures, hearing and vision problems, intellectual disability with developmental delay and decreased ability to learn and function in daily life.

In 2016 the World Health Organisation (WHO) designated the situation with the zika virus as a Public Health Emergency of International Concern (PHEIC). An epidemic trajectory was plotted using computer models considering human demographics, e.g. annual numbers of pregnancies, the economic index of areas, the occurrence and abundance of the *Aedes* mosquito and temperature conditions.

The models estimated human-mosquito ratios, the attack rate projections and the potential numbers of infections. These models quantified the potential magnitude of the zika epidemic, e.g. they predicted that Brazil would have the largest number of infections.

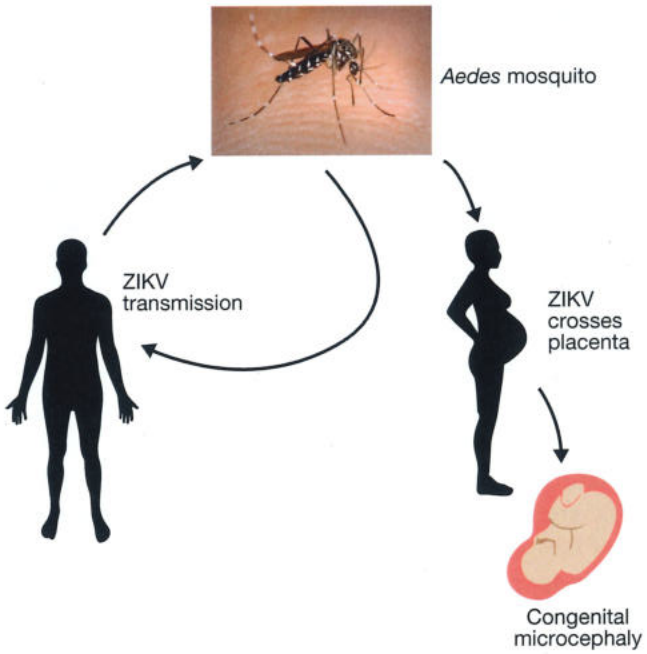


Figure 76.2 Transmission of zika virus.

The models also projected the numbers of children born with zika-induced microcephaly.

QUESTIONS

1. Why do scientists create models?
2. For what is Daniel Bernoulli most famous?
3. Outline the milestone achievement of Bernoulli in the understanding of biology and disease.
4. Distinguish between morbidity and mortality.
5. What is meant by statistical censoring?
6. Outline how the procedure of variolation is a type of inoculation.
7. What type of vaccination procedure was used at the time that Bernoulli carried out his statistical analysis of the efficacy of smallpox vaccination?
8. When did Edward Jenner carry out his famous vaccination experiment using pus from a cowpox pustule as the vaccine for smallpox?
9. What developments have improved contemporary models?
10. Explain why models about the spread of disease made in the 18th century were not very reliable.
11. Describe zika fever.
12. Describe the zika virus.
13. Identify the arthropod vector of ZIKV and outline how this affects the incidence of zika fever.
14. What is the incubation time of ZIKV in the vector?
15. Zika fever has mild symptoms compared to similar diseases such as yellow fever and dengue fever. Explain why there was global concern in 2015 when there was an outbreak of this disease across the Americas.
16. What is congenital microcephaly?
17. How did the World Health Organisation respond to the outbreak of zika fever in 2015 across the Americas?
18. How were computer models used in the outbreak of zika fever?
19. What factors were programmed into the computer model for zika fever and congenital microcephaly?
20. Outline some of the calculations that were worked out by the computer models for zika fever and zika-induced microcephaly.
21. Severe acute respiratory syndrome (SARS) is a type of pneumonia caused by a virus. It can spread through the air in infected droplets from sneezes and coughs and can exist on surfaces such as hands, tissues, and utensils. The death rate from SARS is approximately 9% to 12% of those diagnosed but the death rate is much higher for the elderly. There were outbreaks of SARS in 2002, 2003 and 2004. A computer model was made to show a SARS epidemic. Which of the following would *least* be shown by a computer model of a disease epidemic?
    (A) How the epidemic progresses.
    (B) Effectiveness of different health controls.
    (C) Probability of mortality level.
    (D) Time and location of next outbreak.
22. Which of the following is linked to the zika virus?
    (A) Down syndrome.
    (B) Measles.
    (C) Congenital microcephaly.
    (D) Turner syndrome.
23. The diagram shows factors that were put into a computer model.

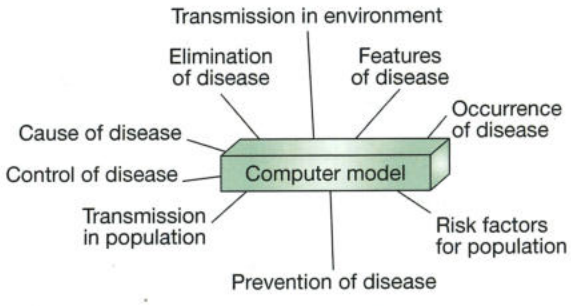


Figure 76.3 Computer modelling.

What is the best description of the function of this model?
(A) Investigate a biochemical process.
(B) Investigate an epidemic.
(C) Summarise an non-infectious disease.
(D) Summarise an effect of global climate change.

77 Managing Pandemics in the Asia Region

Epidemics occur when there is a rapid spread of a pathogen and large numbers of people are infected. The epidemic becomes **pandemic** when the disease spreads around the world to many countries. Epidemics and pandemics that are global or regional are becoming more prevalent.

Diseases that are a concern globally include HIV/AIDS, diphtheria, malaria, measles and swine flu. Diseases that are of concern in the Asia region include severe acute respiratory syndrome (SARS) and avian flu.

The Australian Government, National Health and Medical Research Council (NHMRC) has guidelines that can be downloaded or viewed on their website on '*Outbreak investigation and management*' of reportable diseases. The Australian Government also provides management plans for particular diseases, e.g. *Australian Health Management Plan for Pandemic Influenza*.

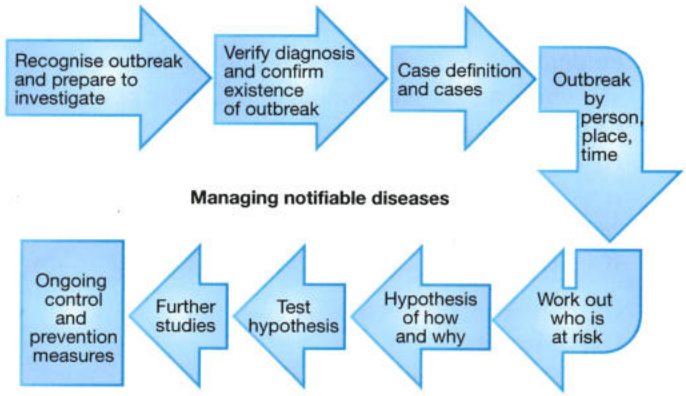


Figure 77.1 Managing notifiable diseases following NHMRC guidelines.

Increasing migration and global travel means that diseases are spreading more rapidly across the world. With high population density in urban areas and underdeveloped health care systems in some countries the introduction of a disease can quickly lead to an epidemic.

If funding for drugs and vaccines is limited the epidemic will affect a larger number of people. In some instances people mistrust vaccination reducing herd immunity below recommended safe levels. Recommendations from international businesses support collaboration between business, governments and international organisations to help prevent the spread of infectious diseases among poor populations and communities. This includes increasing drug and vaccine availability and pandemic preparedness planning.

HIV/AIDS

The UNAIDS (Joint United Nations Program on HIV/AIDS) report 2013 for HIV in Asia and the Pacific reported that there had been a 26% reduction in new HIV infections since 2001 but many people eligible for antiretroviral treatment were not accessing it. To manage the AIDS/HIV epidemic the report advised that HIV testing, and counselling should be expanded and sustained, there should be stronger political leadership and national ownership of programs, key populations and affected communities at highest risk should be targeted and involved in all stages of the AIDS response, funding for programs needs to be maintained and there needs to be an increased push both in the media and legally to reduce discrimination and to ensure human rights and gender equality.

Data shows that HIV infections in South and South-East Asia are decreasing while increasing in East Asia. Although HIV prevalence is only 0.3% in India, the large population of this country means that over 2.1 million people are infected with HIV. Groups with the highest risk of new infection are men who have sex with men and transgender people, people who inject drugs and female and male sex workers and their clients.

Diabetes

The World Health Day 2016 took the slogan Prevent. Treat. Beat diabetes. Globally there is an increase in diabetes with a high increase in incidence in low and middle income countries. It is estimated that over 90 million adults in the South-East Asia region have diabetes with half undiagnosed.

QUESTIONS

1. What is an epidemic?
2. What is meant by pandemic?
3. Identify some diseases that are of concern:
   (a) Globally. (b) In the Asia region.
4. Outline one way in which he Australian Government assists in managing pandemics.
5. List the steps that are in the guidelines from NHMRC for 'Outbreak investigation and management'.
6. What is UNAIDS?
7. Outline some of the suggestions to manage HIV/AIDS epidemic made in the UNAIDS report 2013.
8. Which groups are at most risk of HIV infection?
9. What was the slogan for World Health Day 2016 and explain why this disease was chosen.
10. What will help manage the spread of a pandemic?
    (A) Increased vaccine availability.
    (B) Free immunisation programs.
    (C) Disease information education programs.
    (D) All of the above.

78 Immune Deficiency Diseases

Immune deficiency diseases occur when the immune system fails to protect the body from pathogens or cancer cells. These diseases can be an inborn or **primary immunodeficiency** caused by a genetic or developmental defect or an acquired or **secondary immunodeficiency** after exposure to several chemical or biological agents.

Primary immunodeficiency

Most primary immunodeficiency diseases are diagnosed before the child is one year old though in some people who have less severe cases the disorder is not picked up until they are older. These diseases are usually caused by a defect in the ability to produce specific proteins, e.g. IgA or specific complement proteins and also cause reduced numbers or dysfunctional B cells and T cells.

A newborn infant has antibodies, e.g. IgG that have diffused across the placenta from the mother and in the first month the number of antibodies in the baby's blood dramatically decreases as it had passive immunity and cannot make its own antibodies. If a child is premature they can show signs of immunodeficiency as their organ development can be incomplete including an underdeveloped immune system.

Severe combined immunodeficiency (SCID) is a group of disorders caused by genetic mutations that reduce the ability of the immune system to fight infection. The mutations lead to non-functional helper T cells and/or defective B cells and problems with antibody production or ineffective complement proteins. One form is due to a mutation on the X chromosome which means that female carriers with one copy of this gene will not show symptoms while males show symptoms if their one X chromosome has the mutated gene. Children with SCID have a high risk of developing severe infections and often have frequent, severe respiratory infections, pneumonia, sepsis blood infections or meningitis. Early diagnosis is important so that treatments can begin and the child has less time to get a life-threatening infection. Diagnosis uses TREC (T cell receptor excision circle) assay to identify most forms of SCID and white blood cell counts are also used. Treatments include using bone marrow transplants so that the child can make new white blood cells or gene therapy, e.g. with the child's own bone marrow cells genetically engineered with the needed functional gene or enzyme replacement therapy. One of the forms of SCID is due to a lack of the enzyme adenosine deaminase (ADA) which leads to an accumulation of substances toxic to both B and T cells. Enzyme replacement therapy and gene therapy where a functional adenosine deaminase gene is inserted into the patient's own bone marrow cells have helped children with this form of SCID.

Secondary immunodeficiency

Secondary immunodeficiency develops later in life and can be due to several factors, e.g. the use of immunosuppressive drugs in transplant surgery, the presence of certain cancers, e.g. Hodgkin's disease or pathogen infection, e.g. AIDS is caused by HIV infection.

Acquired immunodeficiency syndrome (AIDS) is caused by infection with the human immunodeficiency virus (HIV). HIV is a retrovirus. Like all viruses, when HIV locates its target cell it binds to the surface molecules in this case the CD4 receptor molecule on helper T cells and injects its genes into the interior of the cell.

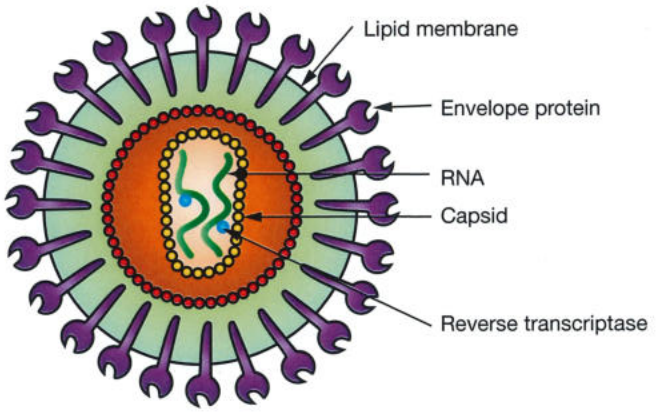


Figure 78.1 Human immunodeficiency virus.

Once **inside the cell** HIV is uncoated and the viral RNA is copied to DNA (it is a retrovirus). The viral DNA integrates with the DNA of the helper T cell cell and becomes part of the genome. The lymphocyte uses the viral DNA as a template for reproducing the HIV RNA genome and synthesises viral proteins. The viral proteins assemble and new viral particles bud from the lymphocyte and move to infect another helper T cell cell. A depletion of helper T lymphocytes weakens the immune system and allows life-threatening diseases called opportunistic infections to occur. Helper T cells activate plasma B cells to produce antibodies and thus antibody production decreases with fewer helper T cells. Fewer T lymphocytes also means that cell mediated immunity is reduced.

Diagnosis requires a test for HIV infection. Early symptoms of HIV infection include fever, swollen glands, sore throat, rash, fatigue, muscle and joint aches and pains and headache. Late stage infection when the immune system is weakening symptoms include rapid weight loss, recurring fever and night sweats, extreme tiredness, prolonged swelling of lymph glands, pneumonia, purplish blotches on the skin, mouth, nose, eyes, memory loss and depression.

Transmission of HIV involves the transfer of body fluids containing infected cells and can occur by sexual contact with an infected partner, the use of HIV contaminated equipment, from an infected mother to child across the placenta during pregnancy or during birth, through blood transfusions or contaminated blood products.

Because HIV weakens the immune system **opportunistic pathogens,** e.g. fungi, bacteria, viruses and parasites can take hold more easily in an AIDS patient. **Viral-induced cancers**, e.g. Kaposi sarcoma, cervical cancer, and Burkitt's lymphoma are also likely to develop due to reduced numbers of functional helper T cells.

Treatment involves anti-HIV drugs which interfere with the replication cycle of the virus and measures to help AIDS-related illnesses to help maintain the quality of life for the patient. There are different types of antiviral drugs and they work in different ways. Treatment in antiviral therapy (AVT) often consists of taking three or more antiviral drugs at the same time.

Since mutations occur at each round of virus reproduction new strains of the virus are being generated which can be drug resistant. This means that the use of drugs keeps putting a selection pressure on the virus which is continually evolving and changing. At the present time there is no cure for AIDS. The main control measures are safe sex practices and not sharing needles, syringes or other injecting equipment.

QUESTIONS

1. What happens in an immune deficiency disease?
2. What are the two types of immunodeficiency diseases?
3. When are most primary immunodeficiency diseases diagnosed?
4. Why can premature babies show signs of immunodeficiency?
5. How do many primary immunodeficiency diseases affect the child with the disease?
6. What is SCID?
7. Construct a table to summarise the cause, symptoms, diagnosis and treatment for SCID.
8. Explain why early diagnosis is important for infants born with SCID.
9. When do secondary immunodeficiency diseases develop?
10. What does AIDS stand for and what causes this disease?
11. Construct a table to summarise the cause, symptoms, diagnosis, transmission, treatment and control for AIDS.
12. Describe HIV and how it enters a cell.
13. Explain why some drugs used in the treatment of HIV do not stay effective over many years.
14. Discuss how the HIV virus acts inside a helper T cell and why this weakens the immune system.
15. It was estimated that in 2014 there were 27 150 people in Australia who were HIV positive with about 73% receiving retroviral treatment. What is the benefit in taking drugs that will interfere with the replication cycle of the virus?

Use the diagram for the next FOUR questions.

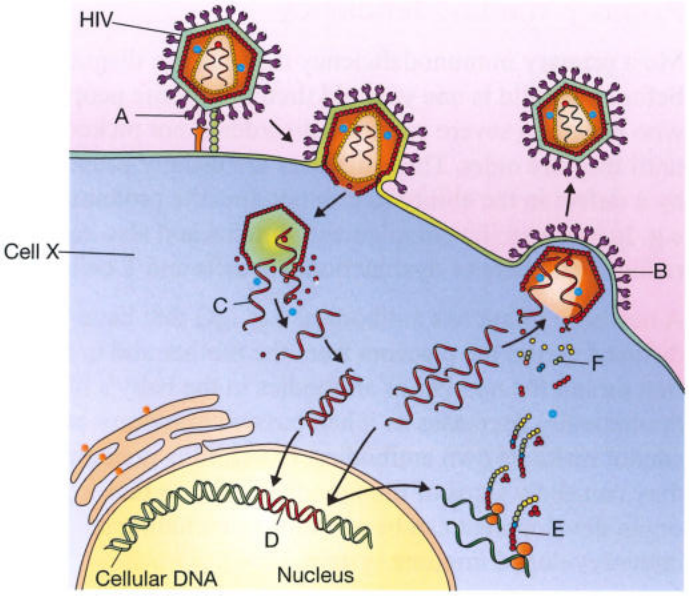


Figure 78.2 Infection of a helper T cell.

16. What is the most likely identity of cell X?
    (A) Cytotoxic T cell. (B) Helper T cell.
    (C) Suppressor T cell. (D) Plasma B cell.
17. Which arrow points to the CD4 receptor?
    (A) Arrow A. (B) Arrow B.
    (C) Arrow C. (D) Arrow E.
18. What is happening at part D?
    (A) The viral genome is being replicated.
    (B) The viral genome is incorporated into the host DNA.
    (C) The viral enzyme reverse transcriptase is generating DNA from RNA.
    (D) Viral proteins are being synthesised.
19. What is happening at part F?
    (A) The viral genome is being replicated.
    (B) The viral genome is incorporated into the host DNA.
    (C) The viral enzyme reverse transcriptase is generating DNA from RNA.
    (D) Viral proteins are released in the cytoplasm.
20. Which of the following diseases is a primary immunodeficiency disease?
    (A) AIDS (B) Measles
    (C) SCID (D) Chickenpox

79 Allergic Reactions

An **allergy** is a hypersensitivity to a non-pathogenic antigen causing an antigen-antibody reaction in some people to substances that do not cause the same response in other people. An **allergen** is an antigen that triggers an exaggerated immune response leading to an allergy. Common types of allergies include hay fever, asthma, urticarial, childhood eczema and some gastrointestinal disorders. Allergic inflammatory responses range from mild tissue damage to fatal reactions.

Antigens that are likely to cause allergies are on pollen grains, fungal spores, dust, certain foods, bites of insects, animal fur or feathers, and plant substances.

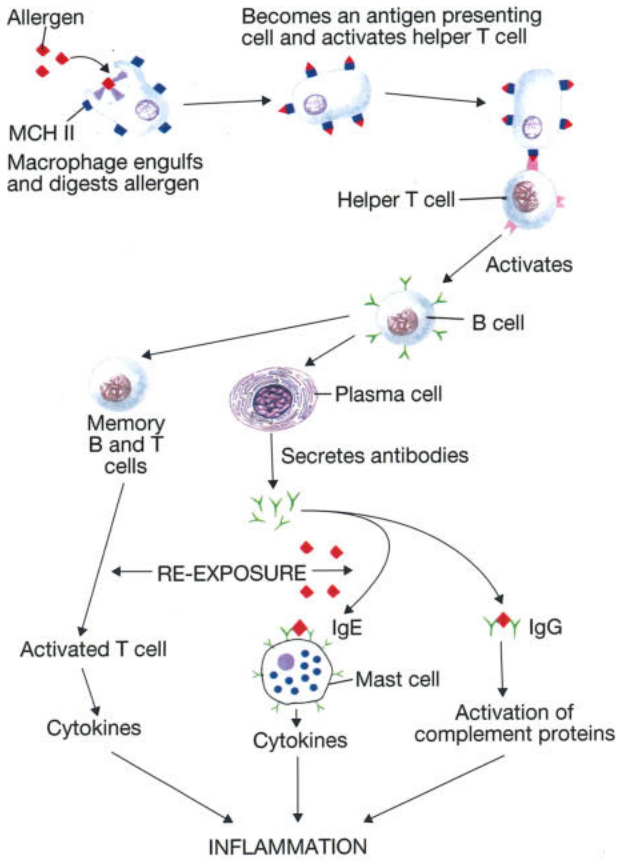


Figure 79.1 Flow chart for an allergic reaction.

In an allergic reaction histamines are released by basophils and mast cells causing dilation of blood vessels and an increase in the permeability of capillary walls. This leads to the inflammation response with skin rash and swelling. The number of eosinophils increases to relieve the symptoms and remove the histamine. Food allergy symptoms affect the mouth, stomach and intestines causing vomiting, hives, coughing, tight hoarse throat, weak pulse, swelling of the tongue and dizziness. Allergy symptoms for airborne allergens affect the eyes, nose and lungs causing sneezing, blocked, itchy or runny nose, red weeping eyes and headache.

Hay fever

Hay fever is an allergic reaction that occurs in the nose. Histamine is released causing dilation of intranasal blood vessels and an increased permeability of capillary walls. This causes fluid to leak into the nasal cavities and into other tissues of the nose. The nasal linings become swollen and secretory. The irritation can cause sneezing and treatment is taking antihistamine drugs.

Anaphylaxis

Anaphylaxis is a dangerous allergic condition where a widespread reaction takes place very quickly with cardiac output and arterial pressure dramatically decreasing. When a person with a severe allergy is exposed to the antigen basophils in the blood and mast cells in tissues immediately outside the blood vessels release large amounts of histamine which causes extreme dilation of blood vessels and increased permeability of the walls of capillaries. This in turn causes a very large loss of fluid and proteins from the bloodstream into tissues. So much plasma from the blood can cause circulatory shock and death can occur within a few minutes. Adrenaline is used to oppose the effects of the histamine.

The Western Australia Government provides Anaphylaxis Management Guidelines for WA schools. By law each school must have a School Anaphylaxis Management Policy and inform and educate students and staff about the policy and how to deal with an anaphylactic situation. Students who are anaphylactic need to have spare adrenaline autoinjection devices and staff need regular training and updates on how to recognise and respond to situations. It is important for students to have the back-up device as in severe situations the first injection of adrenaline may not be sufficient to lower the symptoms from a dangerous level and a second injection needs to be given. Information can be found on the government website with links to other resources such as Health Direct and ASCIA anaphylaxis e-training for schools.

Pollen allergy

Many people suffer from pollen allergies. For each person the allergy is caused by hypersensitivity to a particular pollen from a specific plant species or to a group of plant species. Pollen is produced by flowering plants and many plants are wind pollinated, e.g. grasses. These plants produce large amounts of light pollen that can be carried long distances. Common pollen allergies occur with pellitory wee (often called asthma weed), Paterson's curse *(Echium plantagineum)*, ragweed, white cypress (Murray) pine and parthenium weed. Allergic rhinitis (hay fever) and asthma can be caused by pollen.

For **diagnosis** of a pollen allergy it is important to take careful note of the timing of symptoms and to identify local plants and trees. Allergy tests using skin prick tests and/or blood allergen specific IgE tests will confirm the allergy.

When a person inhales the pollen the allergen stimulates the release of IgE from plasma cells that have been already sensitised to that antigen. The IgE attaches to the surface receptors on mast cells. This activates the mast cell to degranulate. When the granules are in the extracellular fluid they release the stored histamine and other chemicals, e.g. serotonin that cause inflammation. Dilation of blood vessels in the nose cause the area to become red and swollen and the increase in permeability of capillary membranes makes the nose run, eyes water and the irritation can cause sneezing. If the skin becomes infected and mast cells release histamines into the skin these areas will also become red and swollen (hives). Other symptoms include sinus pressure, swollen areas under the eyes and the eyelids, difficulty breathing and wheezing.

Treatment involves the use of antihistamine tables to reduce the runny nose and watery eyes. Nasal sprays containing corticosteroids act against inflammation. Decongestant sprays clear blocked noses. Desensitisation therapy (also called allergen immunotherapy) involves giving the patient very small amounts of the allergen over a period of months or years to stimulate the production of IgG antibodies against the antigen. The IgG antibodies combine with the allergen blocking the action of IgE.

Control measures for people with pollen allergies include staying inside on windy days in the pollen season, not being near lawn mowing, keeping windows closed when airborne pollen counts are high and planting suitable garden plants.

QUESTIONS

1. What is an allergy?
2. Define an allergen.
3. Name some common types of allergies.
4. What is the range of allergic inflammatory response?
5. Which cells release histamine in the inflammatory response?
6. Construct a table to summarise the cause, symptoms, diagnosis, how the inflammation response is involved, treatment and control for pollen allergy.
7. Name some plants that have pollen which causes allergic reaction in some people.
8. Compare allergic reactions for food allergens and airborne allergens.
9. Explain why eosinophil counts increase in the inflammation response.
10. The diagram shows a pollen allergic reaction.

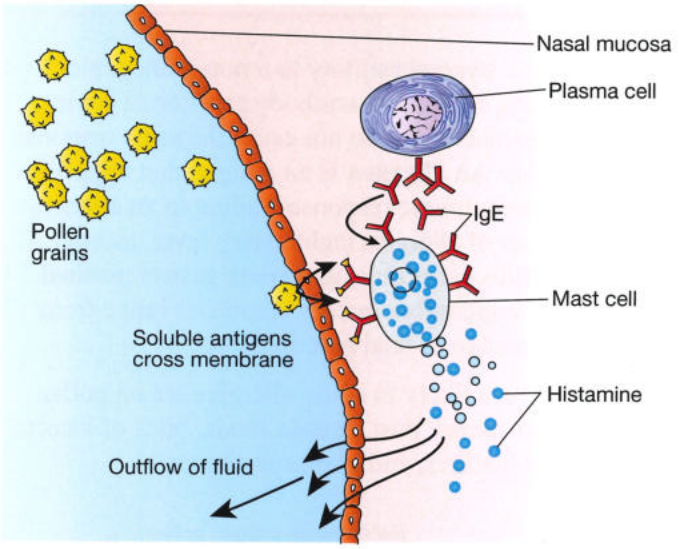


Figure 79.2 Pollen allergic reaction.

Outline the sequence of events that occurred to cause the outflow of fluid into the nasal cavity.

11. What is hay fever?
12. What is anaphylaxis?
13. Explain the principle of desensitisation therapy.
14. (a) Explain why anaphylaxis can be life threatening.
    (b) How do you treat a person who is having an anaphylactic attack?
15. Outline the legal requirements on anaphylaxis for schools.
16. It is estimated that about 20% of the current Australian population has an allergic disease. Which of the following is not an allergic condition?
    (A) Asthma.
    (B) Hay fever.
    (C) Eczema.
    (D) Mumps.
17. What can act as an allergen?
    (A) Insect stings.
    (B) Mould.
    (C) Dust mites.
    (D) All of the above.
18. An epigenetic study has found evidence that food allergies could begin before a baby is born. How could an epigenetic study provide this information?
    (A) Epigenetics studies show how the environment can switch genes on and off.
    (B) Epigenetics investigates biomarkers in the genome for hereditary diseases.
    (C) Epigenetics is the large scale study of proteins produced by genes.
    (D) Epigenetics is the science of collecting and analysing complex biological data such as genetic codes.

80 Snake Antivenom Production

Every year hundreds of people die of snakebite, with most of these people being in developing countries.

In Australia there are around 2000 recorded incidents of snakebite with only a couple leading to death even though Australia has a large number of snake species, many of which are venomous.

In Australia there are more venomous snakes than non-venomous snakes – of the 172 species, 115 are venomous.

Snakes belong to the order Squamata – the same order as lizards. However, snakes are legless, have an elongated body and a modified backbone.

Table 80.1 Some venomous Australian snakes.

Snake	Features
Red-bellied black snake	Very common on east coast, not hugely venomous, causing blood clotting with muscle and nerve damage.
Tiger snake	Common in SW Australia from NSW, Victoria, Tasmania and corner of South Australia. Will attack and bite. Is fatal if untreated.
Eastern brown snake	Common in inland and rural areas, very venomous, fast moving and aggressive.
Western brown snake	Widespread except for wetter areas of eastern Australian and SW Western Australia. The venom is not as toxic as the eastern brown snake and they are less aggressive.
Inland taipan	Widespread in dry inland areas. Reclusive and rare. Very toxic venom considered most potent of all land snakes.
Coastal taipan	Found on east coast from Brisbane to northern NSW and northern Western Australia, e.g. in canefields. Alert and fast with multiple attacks.
King brown snake	Widespread except in Victoria, Tasmania and southern Western Australia. Also called mulga snake with large venom output per bite.
Copperhead	Inland areas of Victoria and NSW in cooler areas.
Death adder	Found in eastern Australia, southern South Australia and Western Australia. Lies motionless twitching end of tail to attract prey. Neurotoxin venom.

How venom works

Snakes will bite and inject venom into food animals to immobilise their prey and assist digestion, but will also bite and inject the venom into animals that are not food as a means of self-defence. Snakes have to swallow their prey whole as their teeth cannot tear food apart.

More than half the land snakes in Australia, about 85 species, are **elapids**. All elapids are venomous.

There are several types of venom – **neurotoxins** to paralyse voluntary muscles, **haemotoxins** to destroy red blood cells, **coagulants** to cause blood clotting, **anticoagulants** to stop blood clotting, **cytotoxins** to destroy tissue and **hyaluronbidase enzymes** to increase the spreading factor of the venom.

Venom of Australian snakes

The **dugite**, native to Western Australia has a very toxic venom that causes progressive paralysis and stops the blood from clotting. Victims can collapse within a few minutes and many doses of antivenom have been used to reverse the effects.

The venom of the **tiger snake** causes pain in the feet and neck, tingling, numbness and sweating, followed by breathing difficulties and paralysis. The venom damages the blood and muscles leading to renal failure.

The **coastal taipan** has the third most toxic venom of land snakes with very long fangs up to 12 mm long. The venom is specialised for warm blooded mammals affecting the nervous system and blood causing nausea, convulsions, internal bleeding, destruction of muscles and kidney damage. Death can occur within 30 minutes.

Conventional antivenom

An **antivenom** is an antidote that counteracts the venom neutralising its effects. There are specific antivenoms for specific species of snakes. A general broad spectrum antivenom (polyvalent) is used when the snake species is not known. Red-bellied black snake antivenom is used for several closely related species.

Traditionally antivenom is produced by 'milking' venomous animals and immunising large animals, e.g. horse with small quantities of the collected venom. The dose of venom is increased until the animal has a large number of antibodies in its blood. The antibodies are then collected to produce the antivenom.

There are several problems with the conventional method:

- Process is risky.
- Process is labour intensive.
- Products often cause allergic reactions.
- Products highly expensive.
- Products difficult to transport and store.

Synthetic antivenom

There have been attempts by some organisations for global cooperation and investment by science, business and government bodies to increase the availability of antivenoms in the developing world. To increase the supplies of antivenoms there is new research into the production of synthetic DNA that is injected into cells to generate an antibody response to the protein that closely resembles the most toxic parts of the actual venom.

QUESTIONS

1. What is the incidence of snakebite and death by snakebite in Australia per year?
2. What proportion of Australian snakes are venomous?
3. To which classification order do snakes belong?
4. How are snakes different to lizards?
5. Name three venomous snakes that are found in Western Australia.
6. The inland taipan, *Oxyuranus microlepidotus*, is considered to be the most venomous snake in the world. Its venom consists of neurotoxins, haemotoxins, myotoxins and hyaluronidase enzymes that cause a high mortality rate in untreated cases. Suggest why there are few people currently receiving fatal bites from the inland taipan.
7. Define antivenom.
8. The elapidae is a family of venomous snakes that have fangs used to inject venom from glands found near the back of the upper jaw. Many Australian snakes are elapids, e.g. taipans, copperheads, death adders. The diagram shows the fangs of an elapid.

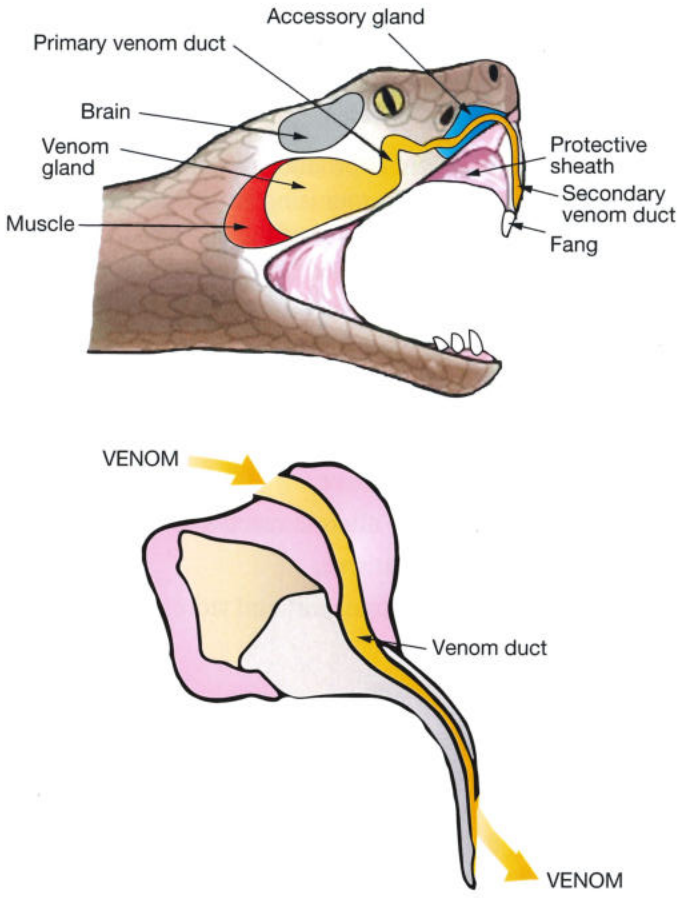


Figure 80.1 Elapid snake fangs.

Using the diagram outline how venom is injected into prey by an elapid snake.

9. In 2006, Simon Wagstaff and his colleagues at the Liverpool School of Tropical Medicine in the UK reported that they had made an antivenom serum using snake venom. They used DNA from the carpet viper which is found in West Africa and is the cause of many deaths. They began by identifying the genes that are active when the snake is refilling its venom sacs. Extracting and using these genes they created a single length of synthetic DNA that would code and produce the protein molecules that cause haemorrhaging and the destruction of blood vessels. They then injected this synthetic DNA into mice and the mice made antibodies to act against the poison. The serum extracted from these mice was tested on other mice and was found to be more powerful that the traditional antivenom produced from carpet viper venom and the new serum also acted against venom from other vipers, e.g. a viper from North Africa. However, the use of large animals, e.g. horses would be needed to produce antivenom for human use using the Wagstaff technique.
   (a) Draw a flow chart to show the steps used by Simon Wagstaff and his colleagues to produce the new antivenom.
   (b) Suggest why the production of synthetic antivenom is an important development in bioinformatics.
10. The diagram shows an activity carried out by producers of antivenom.

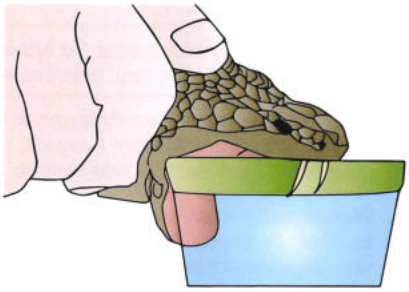


Figure 80.2 Snake activity.

What is shown in this diagram?
(A) Feeding a venomous snake.
(B) Milking a snake for venom.
(C) Measurement of fang length.
(D) Stimulating egg laying process.

11. Which of the following animals would be suitable to use to collect antibodies for an antivenom for human use?
(A) Mouse. (B) Rabbit.
(C) Carpet snake. (D) Horse.
12. Many venomous snakes use constriction as well as venom to kill their prey, e.g. brown snakes, tiger snakes, whipsnakes and rainforest crowned snakes. What would be the main benefit of this strategy?
(A) Prey cannot bite or scratch while being swallowed.
(B) Constriction softens the flesh making it tastier.
(C) Snake expends less energy.
(D) Ensures live flesh is not swallowed.

81 Monoclonal Antibodies and Cancer

Clonal selection was proposed by **Macfarlane Burnet** in his 1959 book *The Clonal Selection Theory of Acquired Immunity*. He proposed that immunological memory was due to one clone acting immediately to produce a specific type of antibody while the other clone remained in the immune system providing long term immunity to that antigen. These cells proliferated making many clones against a particular antigen.

In 1975 **Georges Kohler** and **Cesar Milstein** worked out a way to produce monoclonal antibodies and in 1984 they received the Nobel Prize in Physiology or Medicine 'for theories concerning the specificity in development and control of the immune system and the discovery of the principle for production of monoclonal antibodies'. They injected a protein antigen into a mouse in two doses – a primer dose and a booster dose. The immune system of the mouse identified the antigen as non-self and B cells produced antibodies specific for this antigen. These antibody producing lymphocytes were added to a culture of myeloma cancer cells and mouse hybrid cells were produced using cell fusion techniques that had previously been used in chromosome mapping investigations. The **hybridoma** cells were screened to find which made pure monoclonal antibodies. Since the hybridoma has the properties of a cancer cell it will not die in the tissue culture like other cells.

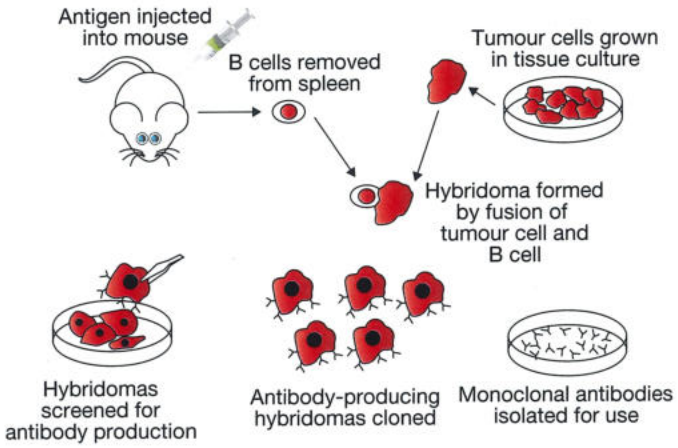


Figure 81.1 Monoclonal antibody production.

Specificity of monoclonal antibodies

A line of monoclonal antibodies come from a single ancestral B cell clone. Monoclonal antibodies are highly specific as an antibody is only capable of binding to the particular epitope of the antigen to which it was sensitised. To produce a particular type of antibody only the epitope is needed to bind to the antibody and not the presence of the whole antigen.

Modern production of monoclonal antibodies

New methods to streamline quality, reduce timelines, increase production and decrease costs are constantly being worked out in the production of monoclonal antibodies.

The method devised by Kohler and Milstein is still the basic procedure with an antigen being injected into a host followed by fusion and selection. One problem had been the presence of mouse antibodies which caused a reaction by the human immune system. The creation of 'chimeric' antibodies increases the amount of human protein involved in the fusion process.

Use of monoclonal antibodies

Large amounts of particular antibodies are needed for diagnosis and treatment of particular diseases and by researchers in academic, medical and commercial fields. Use of monoclonal antibodies include the following.

- **Radioimmunodetection** which used radiolabelled antibodies for diagnostic imaging of neoplasms and cancers and **radioimmunotherapy** which combines radiation therapy and immunotherapy to treat different types of cancers.
- **Diagnostic testing**, e.g. pregnancy tests checking for human chorionic gonadotropin (hCG) and ELISA testing for AIDS. In pregnancy tests the monoclonal antibodies bind to the hormone hCG which is produced in the placenta in the early stages of pregnancy with small amounts of the hormone being passed out of the body in urine. ELISA is the enzyme linked immunosorbent assay which is a diagnostic tool that uses antibodies to identify a particular antigen and besides testing for HIV is used to test for food allergens, hepatitis B markers and to screen for certain types of drugs.
- **Treatment of viral diseases**, e.g. emergency treatment for a rabies infection which is caused by a number of lyssaviruses including the rabies virus and the Australian bat lyssavirus. Monoclonal antibodies are also used in the therapeutic control of hepatitis C virus and hepatitis B virus.
- **Classification** of strains of single pathogens, e.g. *Neisseria gonorrhoeae* which causes gonorrhoea and to detect *Plasmodium* which causes malaria.
- **Identify and trace** specific cells or molecules in an organism, e.g. which proteins are responsible for cell differentiation in the respiratory system.
- Aid **the problem of organ rejection** in patients. The antibodies are antigen specific immunosuppressants that reduce the immune response to the graft.

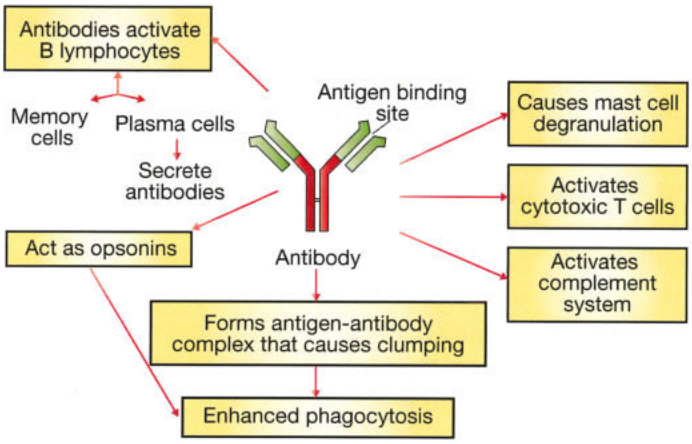


Figure 81.2 Functions of antibodies.

Monoclonal antibodies and cancer

New cancer drugs are based on monoclonal antibody technology. The laboratory-produced monoclonal antibodies are engineered to attach to specific parts of cancer cells. They can tag particular cells and toxin molecules can be bound to tumour-specific monoclonal antibodies to carry out a specialised search and destroy action against specific tumour cells.

The laboratory-produced antibodies act by:

- **Opsonisation** attracting lymphocytes and macrophages to the cancer cell, e.g. the drug rituximab is a chimeric monoclonal antibody that attaches the CD20 proteins found on B cells from which certain lymphomas and leukaemias are generated. The drug is also used for people with rheumatoid arthritis. However, the drug can cause other adverse effects, e.g. cardiac arrest or cytokine release syndrome.
- **Blocking growth factor receptors** which restricts the fast growth of cancer cells which often grow faster than normal cells.
- **Reducing the formation of blood vessels** as additional blood vessels are needed to form to supply the tumour growth with nutrients and remove wastes.

Monoclonal antibodies are also used in combination with radiotherapy and chemotherapy treatments. A radioactive particle can be combined with a monoclonal antibody to deliver the radiation to the cancer cell and similarly chemotherapy drugs can be combined with monoclonal antibodies to directly deliver the drug to specific cancer cells. For example, chemotherapy drugs are used with monoclonal antibodies to treat some types of breast cancer.

Some new methods using antibodies can target cell membranes of cancerous cells.

There is a great advantage in using monoclonal antibodies in the treatment of cancer as the antibodies are chosen to combine with specific cells and healthy cells are not affected. However, like all drugs there have been side effects which means further research still needs to be carried out.

QUESTIONS

1. What was Macfarlane Burnet's contribution to clonal selection?
2. (a) What was the contribution of Georges Kohler and Cesar Milstein to our understanding of monoclonal antibody production?
   (b) Describe how Kohler and Milstein produced monoclonal antibodies.
3. Define a hybridoma.
4. Explain the specificity of monoclonal antibodies.
5. Define an epitope.
6. Explain why researchers investigate new ways to produce monoclonal antibodies.
7. Outline one problem that occurs with the traditional method of monoclonal antibody production and how it has been addressed.
8. Outline why large amounts of monoclonal antibodies are needed.
9. Distinguish between a neoplasm and a cancer.
10. Distinguish between radioimmunodetection and radioimmunotherapy in their used of monoclonal antibodies and cancer.
11. Summarise some of the uses of monoclonal antibodies.
12. Outline a benefit of using laboratory-produced monoclonal antibodies.
13. Use an example to show how laboratory-produced monoclonal antibodies can be used to aid a particular cancer disease.
14. Who proposed the theory of clonal selection in 1959?
    (A) Macfarlane Burnet.
    (B) Georges Kohler and Cesar Milstein.
    (C) Camillo Golgi.
    (D) James Watson.
15. What is used to form a hybridoma?
    (A) UV radiation for a short period on a B cell.
    (B) Fusion of a macrophage with a melanocyte.
    (C) Enzymic catalysis of a complement protein with a tumour cell.
    (D) Fusion of an antibody-producing B cell and a tumour cell
16. What is a neoplasm?
    (A) A disease of newborn children.
    (B) A type of connective tissue.
    (C) An abnormal new growth of cells.
    (D) The intracellular fluid part of a neutrophil.

82 Rational Drug Design

Rational drug design is a process that looks for new medications based on a biological target. The drug is usually a small organic molecule that is complementary in shape and charge to its target which it activates or inhibits, e.g. a target protein such as an enzyme. The action of the drug has a therapeutic effect aiding the patient. This method of investigation is used by many pharmaceutical companies as it reduces the need of 'chance' findings as occurred with Alexander Fleming discovering the action of penicillin.

Each rational drug design starts with the **proteomics** which is the study of the proteome – the full protein set encoded by the genome. The proteome of normal cells are compared with the proteome of cells from a particular disease state.

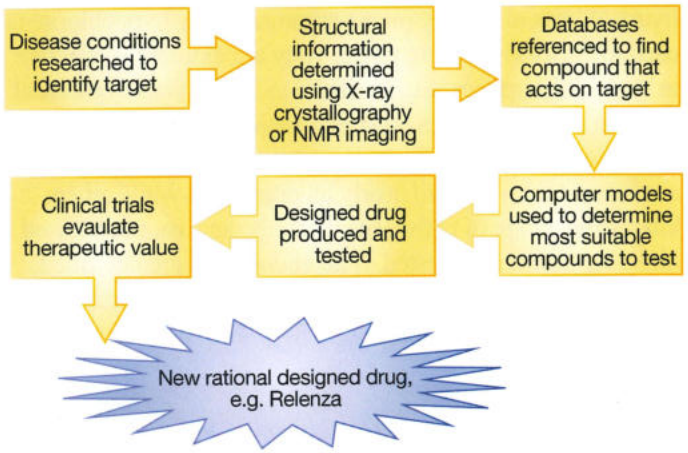


Figure 82.1 Rational drug design.

Computer modelling

Drug design often involves computer modelling techniques in **computer aided drug design**. Research into the structure and activity of the specific target molecule allows researchers to modify the target molecule to see if the changes have therapeutic value. The 3-D shape of the target molecule gives the shape to which the potential drug needs to attach and physical models in **structure based drug design** are used to predict the functionality of changes in the structure of the molecule. Different parameters can be set in the computer modelling to predict properties of the changed molecule, e.g. binding ability, electronic properties before the molecule is produced. Once suitable forms have been found possible forms can be produced reducing the time, effort and cost of research.

Steps in rational drug design

There are three basic steps in drug design.

1. **Target**, e.g. receptor or enzyme is identified relating to a particular disease state. There are many macromolecules involved in disease and many have specific signalling pathways. The disease can cause a disruption or imbalance in the signalling pathway leading to specific symptoms. With knowledge about differences between normal cell proteins and diseased cell proteins, a suitable target can be chosen. The target needs to be disease modifying, e.g. enzyme activator or inhibitor. The target must not affect any other 'off target' molecules which will cause side effects of the potential drug.
2. **Target is fully characterised** giving structural information about the given target. In this process the target is usually cloned and produced and purified to establish a screening assay which is used to determine the 3-D structure. Magnetic resonance spectroscopy and X-ray crystallography are used to find the structure of molecules.
3. **Molecule chosen** – screening libraries that contain information about the structure and functionality of biomolecules are searched to find possible molecules that will suitably interact with the target. The chosen molecule is then designed so that it binds to the target to either restore the normal signalling pathway or inhibit a dysfunctional biomolecule. The design should make a compound that can be taken as a drug, e.g. form an oral medicine with suitable chemical and metabolic stability and have few toxic effects.

Relenza

Relenza is an antiviral drug developed by the Australian biotech firm Biota Holdings. It is a neuraminidase inhibitor used in the treatment of influenza caused by influenza A and B viruses. Neuraminidase is a viral enzyme that is part of the mechanism that causes newly formed virions in an infected host cell to be released from the host cell. Relenza acts by blocking the active site of neuraminidase using competitive inhibition to stop the host cell releasing the virions and spreading the infection.

Critics of rational drug design

Critics argue that the most significant discoveries have been by chance and this formula approach to discovery may limit development and restrict initiatives.

QUESTIONS

1. What is rational drug design?
2. Identify the main features of rational drugs.
3. What is the expected effect of a rational drug?
4. How is computer modelling involved?
5. Outline the three main steps in rational drug design.
6. (a) What is Relenza?
   (b) How does Relenza work?

83 Antiviral Drugs and Antibiotics

Antibiotics

Antibiotics are antimicrobials that kill or inhibit the growth of bacteria. Antibiotics can be naturally produced by other micro-organisms or can be semisynthetic or synthetic.

Broad spectrum antibiotics are effective for killing many types of bacteria while **narrow spectrum antibiotics** target disease-causing bacteria.

Bacteriostatic antibiotics do not kill the bacteria, but inhibit their growth allowing the immune system to overcome the infection. Many bacteriostatic antibiotics interfere with protein synthesis, e.g. tetracycline, chloramphenicol, macrolides and spectinomycin.

Bactericidal antibiotics destroy bacterial pathogens. Bactericidal antibiotics can target outer cell walls, inner cell membranes or metabolic pathways of bacteria, e.g. penicillins and polymixins.

Antiviral drugs

Antiviral drugs are antimicrobials that are used to cure or control virus infections. Most antiviral drugs do not kill viruses; they inhibit their replication. Viruses can only replicate and metabolise inside a host cell. Some antiviral drugs prevent the virus entering the cell. Others prevent the virus releasing its DNA or RNA into the cell nucleus or prevent it splicing its genetic material into the host cell DNA. Antiviral drugs include anti-herpetic agents, e.g. Helpin, Famvir and anti-influenza agents, e.g. Relenza, Tamiflu.

Antiseptics

Antiseptics are antimicrobial chemicals that are applied to living tissue or skin to reduce the likelihood of infection, sepsis or putrefaction, e.g. alcohols are used to disinfect skin before injections. Chlorhexidine is a skin antiseptic or is used to treat inflammation of the gums. Antiseptics kill and/or prevent microbial growth and prevent infection.

Antiseptics are mild chemicals less effective than disinfectants but not damaging to skin and mucous membranes. Toxicity means they cannot be ingested. Since many are mild chemicals all endospores may not be killed; the resistant forms multiply leading to a population that is not affected by the antiseptic.

Disinfectants

Disinfectants are antimicrobial agents that are applied to non-living objects to destroy micro-organisms, e.g. hypochlorite is used to purify swimming pools. They are extremely effective in sterilising surfaces, e.g. sterilisation in hospitals of medical tools, in dental surgeries, kitchens, bathrooms and floors. Choice of disinfectant depends upon the need, e.g. some have a wide spectrum and kill nearly all organisms while others kill a small range of pathogens. Disinfectants cannot be used on living tissue, or ingested. If all endospores are not killed, the resistant forms multiply leading to a population that is not affected by the disinfectant.

QUESTIONS

1. Distinguish between antibiotics and antiviral drugs.
2. Distinguish between broad spectrum and narrow spectrum antibiotics.
3. Distinguish between bacteriostatic and bactericidal antibiotics.
4. Construct a table to evaluate the use of antiseptics and disinfectants giving advantages and disadvantages of each and an evaluation.

Table 83.1 Mode of action for some antibiotics.

Mechanism	Description	Examples
Inhibition of cell wall synthesis	Antibiotics, e.g. penicillin inhibit the formation of peptioglycan cross-links for the cell walls of bacteria. This weakens the cell walls and without strong cells walls, the new bacterium is unable to survive.	Penicillin used to treat syphilis and *Staphylococcus* infections Cephalosporin is used to treat typhoid fever.
Inhibition of protein synthesis	Antibiotics block some stage of protein synthesis by attacking the bacterial ribosomes. They can bind to the bacterial ribosomes (70S) but do not attack the ribosomes in human cells (80S). Some cause the misreading of the genetic code which leads to the synthesis of abnormal proteins.	Erythromycin is used for respiratory tract infections and for people who have an allergy to penicillin, Chloramphenicol is a broad band antimicrobial active against Gram positive and Gram negative bacteria and anaerobes usually used topically, e.g. eye infections. Neomycin is in many topical medications, e.g. creams, ointments.
Inhibition of nucleic acid synthesis	Rifampicin inhibits DNA-dependent RNA polymerase preventing initiation of transcription, especially in Gram positive bacteria. Antibiotics affect the synthesis of DNA/RNA or attach to DNA/RNA so they cannot be read. Anthracyclines inhibit DNA and RNA synthesis by intercalating between base pairs of the DNA/RNA strand and prevent cell replication.	Rifampicin (semisynthetic) used to treat the meningitis form of tuberculosis. Anthracyclines can kill or inhibit bacteria but due to toxicity to humans are never used to treat infections but are used to treat a wide range of cancers.

84 Synthetic Hormones

Synthetic hormones may be developed to control or treat endocrine dysfunction, e.g. for diabetes mellitus, hypothyroidism and hyperthyroidism, to improve the quality of life for individuals. A **synthetic hormone** is a hormone that acts on target cells and is produced artificially. The shape of the synthetic hormone must be very similar (bioidentical) to the 'natural' hormone as it must 'fit' with the receptor sites on the cell membrane or inside the cell.

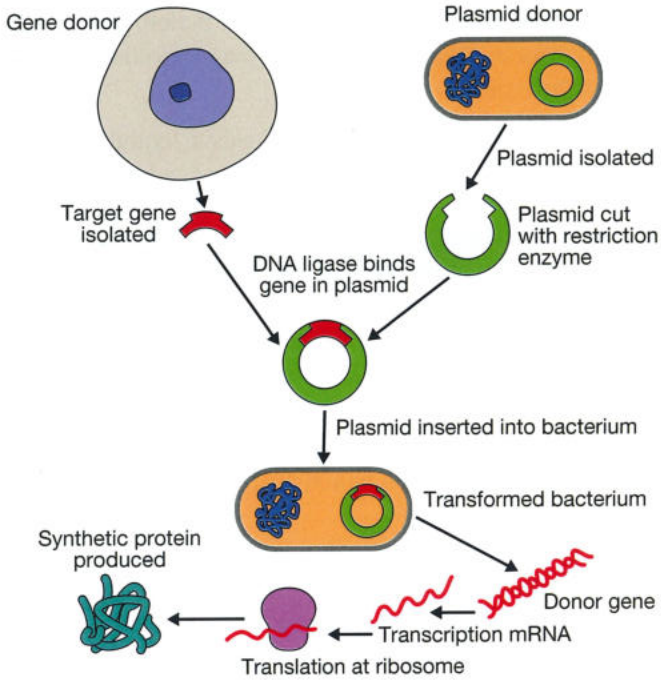


Figure 84.1 Production of synthetic hormone.

Recombinant DNA technology

Synthetic hormones can be produced using recombinant DNA technology. This technology reduces the possibility of rejection or allergic reaction, is cheap to produce, can be produced in quantity and reduces the need to extract particular hormones from farm animals for the treatment of particular human medical disorders.

Recombinant DNA technology involves the following.

- Identifying the human gene for a particular hormone.
- Isolating this gene.
- Inserting the gene into a bacterial plasmid, e.g. insertion of the human gene into *Escherichia coli* plasmids (Ti plasmids).
- Culturing the transformed bacteria in optimum growing conditions so that the bacteria produce large quantities of the gene product (required hormone).
- Extracting the human hormone and packaging for medical treatments for particular disorders.

Hormone replacement therapy (HRT)

HRT is medical treatment that involves giving hormones to a person to replace hormones which are malfunctioning or insufficient.

Diabetes mellitus

Diabetes mellitus is an endocrine disorder where the body cannot maintain glucose homeostasis. **Diabetes mellitus type 1** is a type of diabetes mellitus when not enough insulin is produced leading to high blood glucose levels. **Diabetes mellitus type 2** is a long term progressive metabolic disorder with high blood glucose levels, insulin resistance and the body loses its ability to produce enough insulin. The symptoms of diabetes include thirst due to high glucose levels, high glucose levels in urine, and problems with the eye, kidney and circulatory system. The main treatment for diabetes is controlling blood glucose levels with insulin medications.

Insulin is a hormone produced by beta cells in the islets of Langerhans in the pancreas. Insulin consists of two polypeptide chains that are linked by disulfide bonds. It is released when there is increased blood glucose levels and causes increased glycogen synthesis (**glycogenesis**). Glycogenesis is the production of glycogen from glucose for storage.

Traditional treatment of diabetes required the injection of insulin gathered from the pancreases of pigs and calves. There were several problems with the use of animal insulin, e.g. it was difficult to extract in required quantities for the number of diabetics and was expensive. Animal insulin is also slightly different to human insulin which means it is detected by the immune system and can potentially cause an allergic response and using animal insulin creates the possibility of the transmission of animal viruses into the human population.

The production of synthetic insulin also needs to be linked to an efficient and beneficial method of hormone delivery. Patients using synthetic hormones may still have hypoglycaemic attacks. There are several devices that test and monitor blood glucose levels which people use throughout the day providing information about their need for insulin.

Hypothyroidism

The thyroid gland secretes two hormones – thyroxine (T4) and triiodothyronine (T3). Both these hormones affect metabolic rate, are produced by follicular cells of the thyroid gland and are regulated by thyroid stimulating hormone (TSH) from the anterior pituitary gland. Most T4 travels in the blood bound to transport proteins. Many receptors for the thyroid hormones are near DNA genetic strands and the hormones activate transcription causing an increase in protein production, e.g. enzyme production which increases metabolic activity.

Hypothyroidism is a condition in which the body lacks sufficient T4 and T3 which means body metabolism slows with reduced growth, fatigue and low energy levels. Hypothyroidism requires lifelong hormone replacement, e.g. with thyroxine. Low levels of thyroxine can be due to low iodine levels or an autoimmune disease where the body attacks the thyroid gland as it would attack a foreign antigen.

The main symptoms of hypothyroidism include weakness, slow heart rate, weight gain or difficulty regulating weight, intolerance to cold, muscle cramps, depression, slowing of mental capabilities, constipation, dry or rough skin, deepening voice, memory loss, swollen tongue and face and goitre. It can cause cretinism in children.

When hypothyroidism is caused by iodine deficiency there is limited production of T4 and T3 and for this type of condition treatment involves increasing iodine intake with iodised salt or iodine-rich foods. Hormone replacement therapy is used if there is failure or damage to the thyroid gland, pituitary gland or hypothalamus.

After an injection of a large quantity of thyroxine the change in the metabolic rate of the person will not be noticed for 2 to 3 days as it has a long latent period before activity and activity slowly increases reaching a maximum after 10 to 12 days whereas triiodothyronine acts four times as rapidly as thyroxine. T3 has a latent period of 6 to 12 hours with maximum cellular activity occurring within 2 to 3 days. When given HRT an excess of T3 can occur so that patients can become hyperthyroid for part of the day while synthetic T4 causes a slower rise and fall of metabolism.

Sources of synthetic thyroid hormones come from desiccated animal thyroid glands, e.g. from pigs (porcine) and there are many available brands, e.g. levothyroxine and liothyronine.

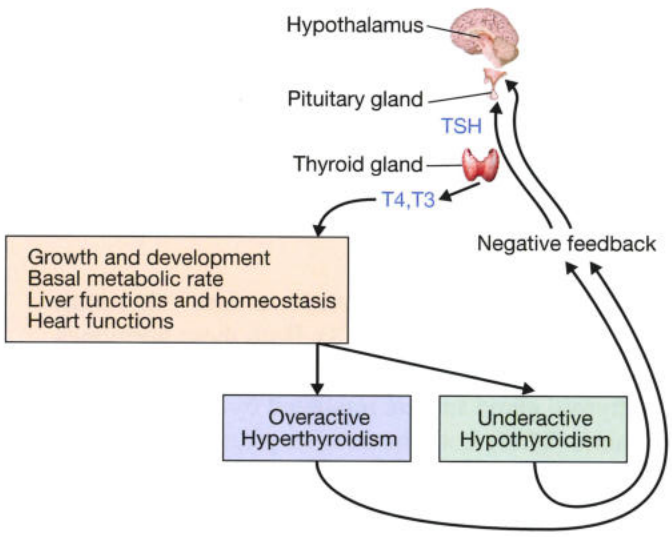


Figure 84.2 Hyperthyroidism and hypothyroidism.

Hyperthyroidism

Hyperthyroidism is the overactivity of the thyroid gland. Excess production of thyroid hormones T3 and T4 (thyroxine) cause rapid heartbeat and an increased rate of metabolism. Symptoms include increased rate of metabolism, nervousness, anxiety, irritability, weight loss, rapid heart rate or palpitations, hand tremors, sleeping problems, excessive sweating and feeling hot and many people develop some degree of eyeball protrusion that stretches the optic nerve affecting vision. Treatment can involve antithyroid drugs which interfere with the ability of the thyroid gland to use iodine, surgery which removes part or all of the thyroid gland or radiotherapy using radioactive iodine which is taken up by active follicular cells and kills these cells.

Treatment for hyperthyroidism often leads to hypothyroidism which is then treated with hormone replacement therapy.

QUESTIONS

1. Why are synthetic hormones being developed?
2. Define a synthetic hormone.
3. Why does the synthetic hormone need to be bioidentical to the natural hormone?
4. What technology is used to produce synthetic hormones?
5. Outline some of the benefits of using synthetic hormones.
6. Outline the main steps in the production of synthetic hormones.
7. What is hormone replacement therapy (HRT)?
8. What is diabetes mellitus?
9. Distinguish between type 1 and type 2 diabetes mellitus.
10. Describe some symptoms for diabetes mellitus.
11. Identify the main treatment for diabetes mellitus.
12. What is insulin?
13. Define glycogenesis.
14. Compare the traditional production of insulin for treatment for diabetes mellitus with the use of synthetic insulin.
15. Outline why there is still further research into synthetic insulin.
16. What are T4 and T3?
17. Distinguish between hypothyroidism and hyperthyroidism.
18. Construct a table to compare the symptoms and treatment of hypothyroidism and hyperthyroidism.
19. A person with hypothyroidism is given hormone replacement therapy. Which of the following would you expect to happen?
    (A) Fat metabolism is inhibited.
    (B) Decrease in protein production in a cell.
    (C) Reduction in ion transport across membranes.
    (D) Number and size of mitochondria increase in the body cells.

85 Gene Therapy

Gene therapy is a broad area which aims to correct genes that cause disease. The problem gene could be switched on/off, replaced, swapped for a normal gene or repaired through selective reverse mutation. Most gene therapy is still at the experimental stage, e.g. clinical trials for different types of cancer, treating children with X-SCID (severe combined immunodeficiency) or the 'bubble' children. Some people are concerned that gene therapy could be used to enhance 'desired' characteristics which could lead to discrimination against particular groups.

Somatic gene therapy treats the diseased body cells of the person, lasting for their lifetime but cannot be inherited. **Germ line gene therapy** corrects the gene in the gamete, zygote or embryo so that the change lasts not just their lifetime but can be inherited. Many countries forbid germ line gene therapy.

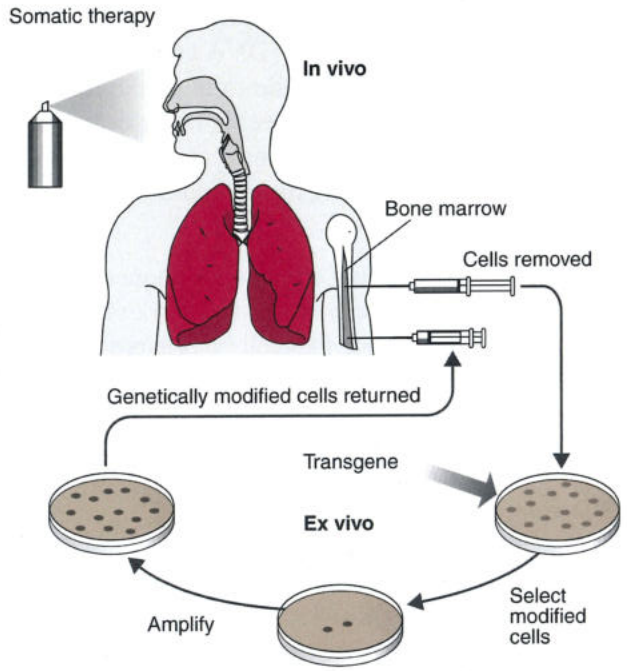


Figure 85.1 Somatic gene therapy.

One part of gene therapy is using gene tests to diagnose genetic disorders even if the person at present has no symptoms. This can help the person by providing information about the course of the disease and the risk of future disease in their children. There are many genetic tests available, e.g. for Duchenne muscular dystrophy, cystic fibrosis, sickle-cell anaemia and Huntington's disease. However, there are many issues arising from gene testing, e.g. the emotional and psychological impact on the person; even if the disease has been diagnosed some diseases have no effective treatment or cure; privacy issues related to healthcare cover and employment. Genetic counsellors are available to help explain disorders and provide information about future options.

Most methods of gene delivery involve using virus vectors. Nasal sprays have been used to introduce new genes into the lungs. The target lung cells are invaded with the viral vector (e.g. common cold – an adenovirus with a double stranded DNA genome). Non-viral delivery includes direct introduction of therapeutic DNA into target cells or using **liposomes**. Liposomes carry the therapeutic DNA inside an artificial lipid sphere and it fuses with the target cell's membrane. Some researchers are trialling the introduction of a 47th chromosome into target cells. This introduced chromosome holds the therapeutic gene.

QUESTIONS

1. Define gene therapy.
2. Identify some areas where gene therapy is being explored.
3. Construct a table to distinguish between somatic gene therapy and germ line gene therapy.
4. Describe two different ways therapeutic genes can be delivered.
5. In March 2005, Professor Thrasher and Dr Gaspar from Great Ormond Street Hospital in London announced they had successfully used gene therapy to treat a four year old boy called Mustaf. Mustaf suffered from SCID (severe combined immunodeficiency) as he had a faulty gene for the enzyme adenosine deaminase (ADA). Mustaf entered the program because there was no matching bone marrow donor available and he did not respond well to drugs providing the missing enzyme.
   (a) Explain why children with SCID are often referred to as 'bubble children'.
   (b) Why is it important to provide these children with early treatment?
   (c) To treat Mustaf, a sample of bone marrow was removed. Stem cells were purified from the bone marrow and mixed with a mouse virus that contained a normal copy of the ADA gene. The virus was introduced to Mustaf and the virus implanted the normal ADA gene in his bone marrow. Mustaf had chemotherapy to suppress his bone marrow and the transfused GM bone marrow cells were allowed to multiply. Identify the vector in this example and explain its function.
   (d) Another type of SCID is X-linked (X-SCID). Gene therapy trials on X-SCID were halted in 2003 after a second child developed a leukaemia-like condition after treatment with a retroviral blood stem cell. Discuss why it is important for government agencies to monitor clinical trials.
6. Bone marrow cells which include stem cells that give rise to all cells of the blood and immune systems have been used in gene therapy trials. Outline a benefit of using stem cells in gene therapy.
7. Evaluate gene testing as a genetic procedure.

86 Gene Cloning and Cell Replacement

Gene cloning is a process which isolates a single gene so that multiple copies can be made, e.g. by inserting the gene into a vector such as a bacterial plasmid and then producing many copies through replication. The terms 'recombinant DNA technology', 'DNA cloning', 'molecular cloning' and 'gene cloning' all apply to the same process.

Vectors

Bacteria are the most common vector and host cell for recombinant DNA molecules. **Bacterial plasmids** are used to produce multiple copies of the same gene. Plasmids are circular DNA molecules additional to the normal bacterial chromosome. The plasmid usually stays separate from the bacterial chromosome and will replicate and be passed on to daughter cells. Plasmids are highly infective and are easily taken up by bacteria. The bacteria are often treated with calcium salts to make their membranes more permeable and allow plasmids to be taken up more easily. **Bacterial artificial chromosomes** (BACs) use the F-factor plasmid found in *E. coli* to carry DNA inserts. **Yeast artificial chromosomes** (YACs) are also used as vectors and can carry foreign DNA. Other cloning vectors include viruses and mammalian cells.

Cloning a gene

The first step in cloning a gene involves using a restriction enzyme to isolate the required gene from chromosomal DNA. The same restriction enzyme needs to be used on the plasmid so that the sticky ends are complementary and can unite. The joining of the desired gene with the cloning vector forms the recombinant DNA molecule which can then be cloned with the host cell DNA.

Cloning tissues and organs

One aim of gene cloning is to produce whole tissues and then organs from single cells. This means that damaged cells, e.g. in Alzheimer's or Parkinson's disease could be replaced by healthy cells. Stem cell research is investigating how to produce these tissues and organs. Another area of research in producing organ transplants involves genetically modified pigs. Current technology seems more suitable for the cloning of some species than others and the species most similar to humans that has responded to methods is the pig. To prevent the human immune system rejecting the implanted pig organ, scientists need to switch off the involved gene.

In 2002 it was reported a 'double knock-out' pig, that lacked both copies of a gene involved in transplant rejection, had been engineered.

Therapeutic proteins

Inserting the human gene which codes for a particular protein into bacteria means that these proteins can be produced in large quantities. For example, recombinant DNA in *E. coli* have been used to manufacture insulin, growth hormone and antihaemophilic globulin (the clotting factor missing in haemophilia).

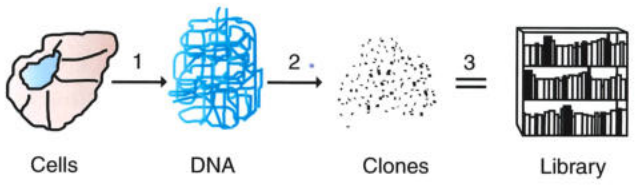


Figure 86.1 Making a DNA library.

DNA libraries

Cloning genes can lead to the formation of DNA libraries. A DNA library is a collection of DNA genes. There are genomic libraries which are made from total nuclear DNA of an organism; chromosome-specific libraries with DNA from purified isolated chromosomes; and cDNA libraries which are made from mRNA for particular tissues, e.g. heart, liver, leaf, root.

A collection of overlapping clones (e.g. sequenced tagged site STS clones) can be used to work out a long continuous DNA sequence that can be ordered to create a physical genetic map of a chromosome.

QUESTIONS

1. Define gene cloning.
2. What is a bacterial plasmid?
3. Outline how a desired gene is isolated from chromosomal DNA.
4. Name some vectors for gene cloning.
5. Describe how researchers increase the chance that the host bacteria will take up the plasmid.
6. Why are researchers investigating tissue and organ cloning?
7. Describe how therapeutic proteins can be produced using gene cloning.
8. What is a DNA library?
9. Gene cloning and selective breeding are two different processes that lead to a change in the gene pool of a population. Compare these two processes and use examples to show how each changes the gene pool in a population.
10. Distinguish between gene cloning and whole organism cloning.
11. Outline two benefits of a clone library.

87 Alzheimer's Disease and Parkinson's Disease

There are several diseases that affect the nervous system, particularly the brain. **Dementia** is a progressive, irreversible degeneration of the cerebral cortex that affects memory, intellect and reasoning. **Alzheimer's disease** is one of the most common forms of dementia. Parkinson's disease affects the neurons especially the dopamine releasing neurons in the middle area of the brain that are involved in muscle movement.

Alzheimer's disease

In Australia around two thirds of dementia cases involve Alzheimer's disease with females affected around twice as often as males. Alzheimer's disease is sometimes referred to as premature ageing of the brain. In the disease amyloid plaques (fibrous patches) ranging in diameter from 10 micrometres to several hundred micrometres and neurofibrillary tangles (protein filament strands) form on brain cells disrupting their functioning. The affected brain cells die leading to shrinkage (atrophy) of the brain which eventually affects the ability of the brain to function correctly causing memory loss, decreased ability to plan and problem solve, vagueness in conversation, becoming disoriented and taking longer to do routine tasks. Some people lose the ability to recognise people, including family members.

Brain imaging techniques are used to investigate the development of the amyloid plaque and proteins and to see changes in brain structure. There is no cure for Alzheimer's disease. There are medications that can help slow down the loss of memory and thinking abilities, manage behavioural symptoms or to treat other aspects of the disease, e.g. depression and sleep disorders. Researchers are investigating ways to stop amyloid proteins forming clumps and then plaques in the brain, using monoclonal antibodies designed to lower amyloid in the brain, ways to stop the protein fibres from collapsing and twisting into tangles, vaccines to stimulate the immune system to destroy protein fibre tangles in the brain and drugs that slow down or prevent symptoms in people who have the genetic mutations for Alzheimer's disease.

Parkinson's disease

Parkinson's disease causes a progressive degeneration of neurons in the circuitry of the basal ganglia. The **basal ganglia** are the nuclei surrounding the thalamus that help with planning movement. The loss of control and coordination leads to fixed muscle tone which causes an expressionless face, stooping posture, shuffling gait and rigid muscles. There is often muscle tremor of the extremities which makes simple movements difficult and high mental effort and concentration is needed to perform simple tasks or speak clearly. These symptoms often create distress and anxiety.

In many cases there is no clearly identifiable cause for Parkinson's disease, though most researchers believe it is a combination of genetic and environmental factors.

There is currently no cure for Parkinson's disease. Treatments help patients manage their conditions and medications can include dopamine replacement therapy, the use of anticholinergics which block the effect of acetylcholine transmitters which function in conjunction with dopamine or inhibitors that block the metabolism of dopamine in the brain. Some patients have surgery to stop the trembling.

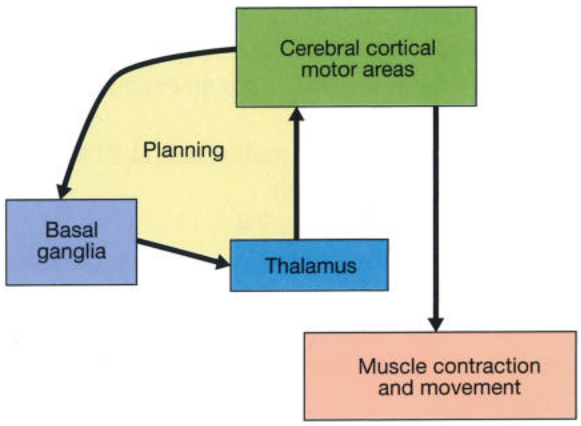


Figure 87.1 Pathway interrupted by Parkinson's disease.

QUESTIONS

1. Define dementia.
2. (a) Outline what happens to the brain when a person has Alzheimer's disease.
   (b) List some of the symptoms of Alzheimer's disease.
3. How is Alzheimer's disease treated?
4. What are the basal ganglia?
5. What is Parkinson's disease?
6. Describe some of the symptoms of Parkinson's disease.
7. What causes Parkinson's disease?
8. A suggested potential cure for Parkinson's disease is the implantation of dopamine secreting neurons into the basal ganglia. How would this aid a person with Parkinson's disease?
9. Scientists have found that embryonic stem cells can be stimulated or genetically engineered to develop into dopamine-secreting neurons. How could this aid a person with Parkinson's disease?
10. Which cells are most affected in Parkinson's disease?
    (A) Neural stem cells.
    (B) Neurons in the cerebellum.
    (C) Dopamine-secreting neurons.
    (D) Neurons in the temporal lobe.

Topic Test

Section A – Multiple Choice (35 marks)

1. What is the definition of homeostasis?
   (A) Homeostasis is the maintenance of body sugar levels and temperature.
   (B) Homeostasis is the process where the internal and external environment is controlled.
   (C) Homeostasis is the maintenance of constant internal environment in a limited range.
   (D) Homeostasis is the process in plants and animals to control the internal environment.
2. Which of the following is *not* an example of homeostasis?
   (A) Body temperature maintained at 37°C.
   (B) Eating a large lunch.
   (C) pH of blood kept at 7.4.
   (D) Breathing faster when carbon dioxide levels in the blood rise.
3. There are two stages in homeostasis. One stage is detecting changes from the stable state, what is the other stage?
   (A) Negative feedback to turn off a response no longer needed.
   (B) Sending a message along a sensory neuron to the brain.
   (C) The central nervous system interprets the message from receptors.
   (D) Counteracting changes from the stable state.
4. What is meant by cell metabolism?
   (A) All the chemical reactions that take place in a cell.
   (B) The rate of biochemical reactions in the cell.
   (C) A particular chemical pathway where the product of one reaction becomes the substrate of the next reaction.
   (D) A process that releases energy for use in the cell.
5. What will happen if a person cannot maintain a constant internal environment for a period of time?
   (A) Metabolic efficiency will increase.
   (B) Metabolic efficiency will decrease.
   (C) The person will go to sleep.
   (D) The person will immediately die.
6. What is the correct order of the stimulus-response pathway?
   (A) Stimulus → central nervous system → sensory neuron → effector → response.
   (B) Stimulus → motor neuron → central nervous system → effector → response.
   (C) Stimulus → receptor → central nervous system → effector → response.
   (D) Stimulus → receptor → central nervous system → sensory neuron → response.

The diagram shows a section of the nervous system that controls a reflex arc in the leg. Use this diagram for the next TWO questions.

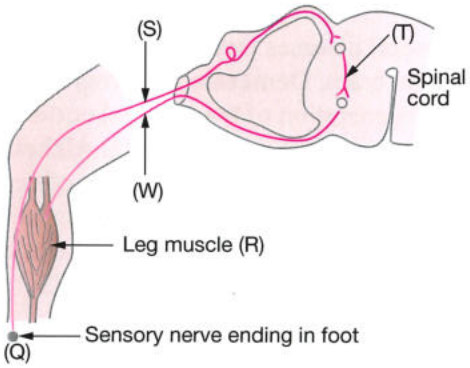


Figure TT.1 Reflex arc.

7. What is the pathway from sensory nerve to effector?
   (A) Q, W, T, S, R
   (B) Q, T, W, S, R
   (C) Q, S, T, W, R
   (D) Q, W, S, T, R
8. What is the advantage of a reflex arc in detecting and responding to a change in the environment?
   (A) It connects the nervous system to the endocrine system and enables a hormonal response.
   (B) It means the body can respond very quickly without waiting for the brain to interpret the information.
   (C) It protects the legs from changes in temperature.
   (D) It enables sensory neurons to connect to other parts of the body to inform each part about the stimulus.
9. What is the best definition of osmosis?
   (A) Diffusion of water across a semipermeable membrane from a region of high concentration of water to a region of low concentration of water.
   (B) Diffusion of water across a semipermeable membrane from a region of low concentration of water to a region of high concentration of water.
   (C) Movement of water across a semipermeable membrane from high salt concentration to low salt concentration.
   (D) Movement of particles from an area of high concentration to an area of low concentration.
10. After aerobic respiration in a cell in your right hand, a newly formed carbon dioxide molecule will travel in the circulatory system to be excreted. In which of the following structures will it *not* travel?
    (A) Bronchus.
    (B) Right ventricle.
    (C) Pulmonary vein.
    (D) Pulmonary artery.

11. The graph shows the change in metabolic rate of a person with hypothyroidism after the injection of a large dose of a thyroid hormone.

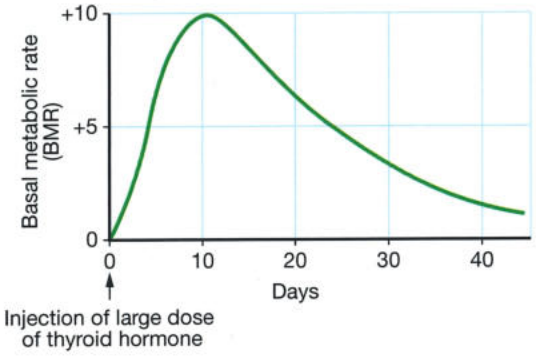


Figure TT.2 Hyperthyroidism and hypothyroidism.

What is the most likely cause of this change in metabolism?

(A) Hormone was thyroxine as maximum activity is around 11 days after injection.
(B) Hormone was triiodothyronine as maximum activity is around 11 days after injection.
(C) Hormone was thyroxine as it denatures after 11 days.
(D) Hormone was triiodothyronine as it denatures after 11 days.

12. The graph shows the effect of a change in diet and exercise on the body sugar levels of a young person.

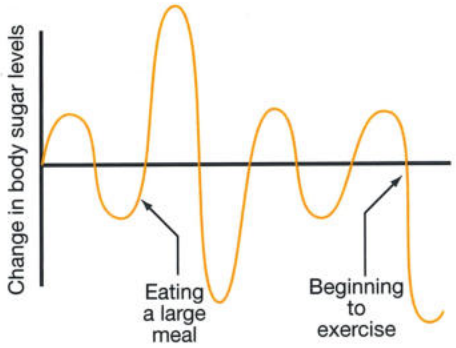


Figure TT.3 Blood sugar levels of an endotherm over time.

What does this graph represent?

(A) A feedback system.
(B) A nerve impulse.
(C) The digestive enzymes.
(D) The rate of respiration.

13. Why would a diabetic inject glucagon?

(A) Lower blood glucose levels by promoting gluconeogenesis.
(B) Raise blood glucose levels by promoting gluconeogenesis.
(C) Lower blood glucose levels by inhibiting gluconeogenesis.
(D) Raise blood glucose levels by inhibiting gluconeogenesis.

14. What is hypothyroidism?

(A) Underproduction of insulin.
(B) Excessive production of insulin.
(C) Underproduction of T3 and T4 hormones.
(D) Excessive production of T3 and T4 hormones.

15. Which of the following is a physiological response of humans to decreased oxygen concentrations at high altitudes?

(A) Using oxygen breathing masks.
(B) Formation of more red blood cells.
(C) Decrease in respiration and heart rate.
(D) Increase in body weight to change body mass index.

16. The diagram shows the nerves that leave the spinal cord.

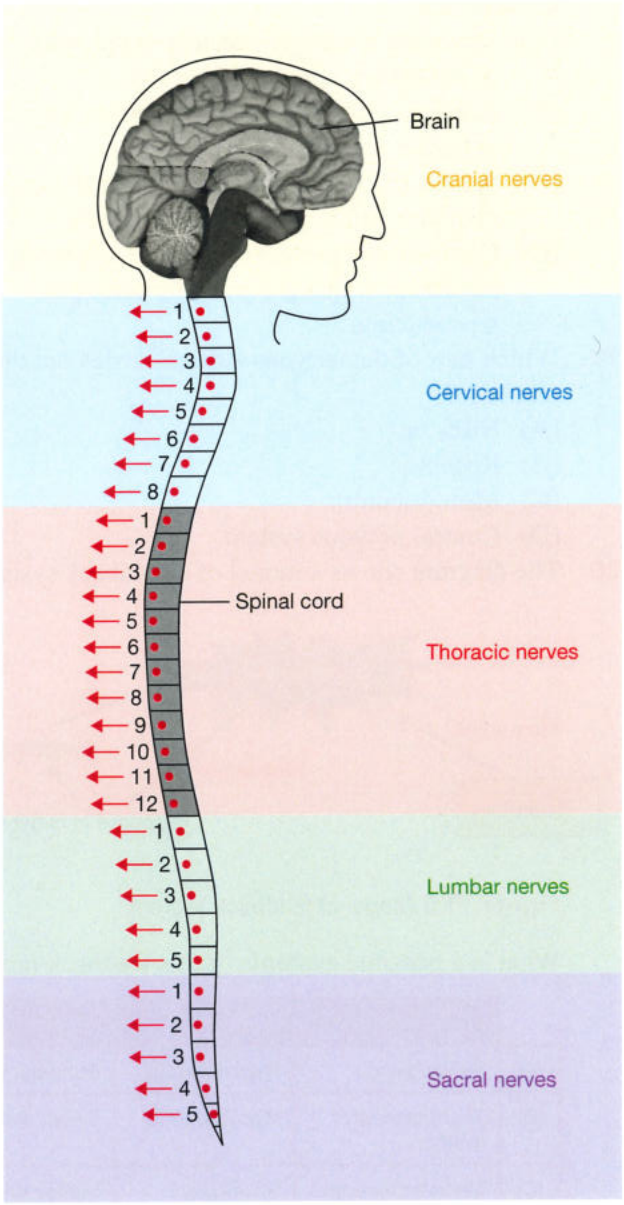


Figure TT.4 Nerves leaving the spinal cord.

If a person has normal feeling in their arms and body but feels no pain when their big toe is touched with a pin, which of the following spinal nerves could be damaged?

(A) C2 (B) C6 (C) S4 (D) T5

17. In 1862, a scientist showed that heating for 30 minutes at 55°C killed the micro-organisms that soured wine. Identify this scientist.
    (A) Robert Koch.
    (B) Edward Jenner.
    (C) Louis Pasteur.
    (D) Ronald Ross.
18. Many ancient cultures recognised the relationship between the survival from a disease and that person's immunity for that disease. For example, after a typhus epidemic in Athens, Thucydides (460-400 BCE) wrote 'none was ever attacked [by the pathogen] a second time or with a fatal result'. Which of the following gives the current understanding of this relationship?
    (A) Exposure to antigens results in the production of antibodies and memory cells.
    (B) Antigens entering the body induce phagocytosis at higher levels of activity.
    (C) Antibodies of pathogens stimulate the immune response and increased phagocytosis.
    (D) Complement proteins cause the release of histamine which prevents the pathogen reproducing.
19. Which part of the nervous system carries out the response?
    (A) Effector.
    (B) Receptor.
    (C) Motor neuron.
    (D) Central nervous system.
20. The diagram shows a model of a feedback system.

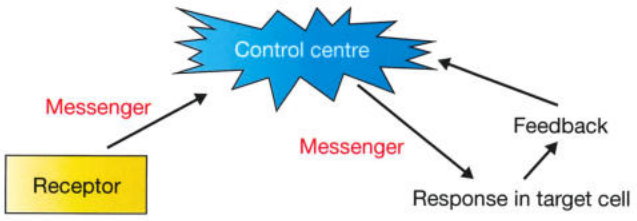


Figure TT.5 Model of feedback system.

What is a possible example of this feedback model?

	Receptor	Control centre	Target cell
(A)	Sweat gland	Hypothalamus	Stomach wall
(B)	Thermoreceptor in skin	Hypothalamus	Sweat glands
(C)	Mechanoreceptor	Spinal cord	Kidney tubules
(D)	Sensory neuron	Brain	Liver

21. Many bacteria are found on the skin. The genera *Staphylococcus, Micrococcus, Corynebacterium* and *Propionibacterium* are considered 'normal' flora. Which of the following could be used as a safety measure to temporarily dispose of this normal flora to prevent infection as is needed during an operation or surgery?
    (A) Expose all vital areas to gamma radiation.
    (B) Wash all areas with antimicrobial preparation such as an iodine-containing preparation.
    (C) Cover all areas with petroleum jelly to form an artificial impenetrable covering.
    (D) Ingest an antibiotic such as penicillin.
22. The diagram shows a cross-section of a bronchiole in the human respiratory tract.

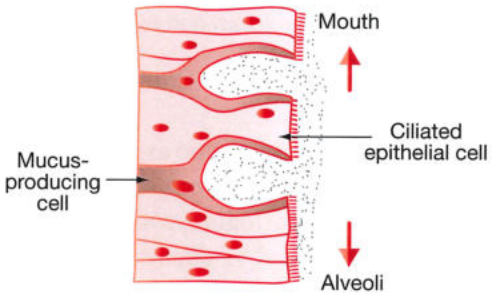


Figure TT.6 Lining of a bronchiole.

What is the function of cilia?
    (A) To move mucus in a particular direction.
    (B) To beat and trap pathogens.
    (C) To slow down the movement of particles into the body along open tracts.
    (D) To begin the immune response by detecting foreign particles.
23. Comparing the blood of sheep that had died of anthrax with the blood of healthy sheep, Robert Koch (1843-1910) found that the blood of the diseased sheep contained small rod shaped organisms. This discovery became the first step in Koch's postulates. What is this first step in Koch' postulates?
    (A) The organism must be grown in pure culture.
    (B) The organism must be seen under the microscope in all organisms.
    (C) The microbe causing the disease must be isolated and grown in pure culture.
    (D) The microbe causing the disease must be found in each diseased organism.
24. Asbestosis, sometimes called mesothelioma, is a disease caused by the accumulation of asbestos fibres in the lung, heart or abdominal cavity. It can take between 15 and 50 years to develop. Identify the most suitable classification for this disease.
    (A) Inherited non-infectious disease.
    (B) Nutritional non-infectious disease.
    (C) Environmental non-infectious disease.
    (D) Infectious disease.

25. Which of the following correctly identifies a disease that is transmitted by an insect vector, the name of the vector and the organism infected by the disease?

	Disease	Insect vector	Organism infected by disease
(A)	Malaria	Mosquito	Human
(B)	Psittacosis	Flea	Human
(C)	Rabies	Cockroach	Dog
(D)	Sleeping sickness	Tsetse fly	Bat

26. The diagram shows the defences against infection.

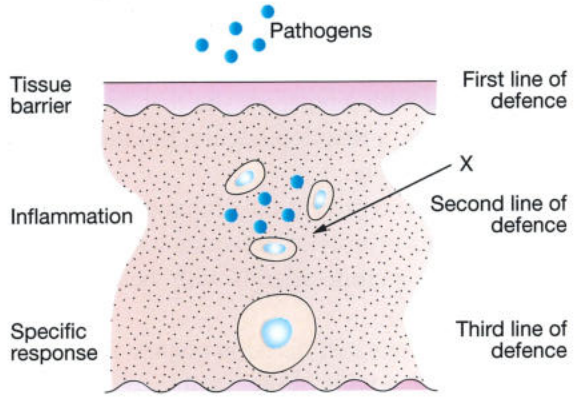


Figure TT.7 Defences against infection.

Which of the following is most likely to occur at the point labelled 'X'?

(A) Secretion of mucus.
(B) Lysosomes of phagocytes release enzymes toxic to micro-organisms.
(C) Killer T cells secrete interleukins which modulate the function of the immune system.
(D) Plasma cells synthesise antibodies which move into the bloodstream and tissue spaces.

27. One of the first recorded incidences of a transplant operation was in the 3rd century CE when Saint Cosmas and Saint Damian transplanted a leg onto a white recipient from a dead Ethiopian. Modern transplant operations often use drugs, such as corticosteroids, to suppress the immune response. Which type of response leads to rejection of grafted tissue?

(A) Inflammatory response.
(B) Cell-mediated response.
(C) Antibody-mediated response.
(D) Phagocytic response.

28. Creutzfeldt-Jakob disease (CJD) was independently observed in the 1920s by two German physicians, Hans Creutzfeldt and Alfons Jakob. This disease is a mental degenerative disease where the patient gradually loses memory and intellect, and has increasing difficulty performing simple tasks. The disease causes microscopic vacuoles to form in neurons so that they appear 'sponge-like', hence the name of spongiform encephalopathy disease.
What type of pathogen is believed to be the cause of CJD?

(A) Virus.
(B) Protozoan.
(C) Prion.
(D) Bacteria.

29. The table shows some vitamins, their source and why they are required by the body.

Vitamin	Source	Why required
A Retinol	Green and yellow vegetables, fish liver oils, carrots, tomatoes, kidneys	Epithelial tissues, e.g. skin, digestive tract, urinary tract
B1 Thiamine	Whole grain products, seafood, meat	Carbohydrate metabolism, nerve and muscle functions
B2 Riboflavin	Meat, eggs, green vegetables, mushrooms	Cell respiration, protein metabolism, synthesis of hormones
C Ascorbic acid	Citrus fruits, tomatoes, leafy vegetables	Teeth and gum development, wound healing, absorption of iron, connective tissues
D Calciferol	Liver, eggs, produced in skin by sunlight	Bone and teeth development by aiding absorption of calcium and phosphorus
E Tocopherols	Wheat germ, whole grain products, green leafy vegetables	Maintenance of reproductive system, prevents damage to cell membrane

If a person was suffering from easy bruising and bleeding, sore gums and loose teeth, which of the following is most likely to be recommended?

(A) Increase retinol intake to repair epithelial tissue.
(B) Eat citrus fruits to increase vitamin C levels.
(C) Increase tocopherols by eating more meat and eggs.
(D) Take ascorbic acid tablets to increase vitamin A levels.

30. Rheumatoid arthritis is a disease caused by the body producing antibodies to the synovial membrane which lines the joints. How does this affect the joint?
(A) Antibodies and the immune system cause joint inflammation and shape distortion.
(B) The circulatory system increases blood flow into the joint causing joint distortion.
(C) Increased formation of articular cartilage increases joint flexibility.
(D) Causes sodium urate crystals deposition in the joint and tendons distorting the joint.

31. An excess of growth hormone can lead to an excessive growth of bones, enlargement of internal organs, e.g. heart, a rise in blood pressure and growth of excess connective tissue.

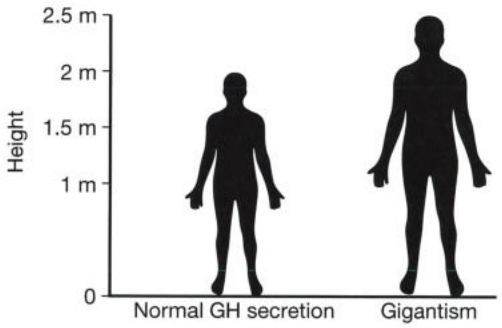


Figure TT.9 Normal and excessive GH secretion.

Which endocrine gland is likely to be malfunctioning causing gigantism?
(A) Adrenal gland.
(B) Thyroid gland.
(C) Posterior pituitary gland.
(D) Anterior pituitary gland.

32. A year 10 science student carried out a survey for a research project. She interviewed 125 year 9 students, asking them about the decision to participate in immunisation programs. She tabulated her results in the graph shown below for the question 'Is whooping cough vaccination related to mental retardation?'

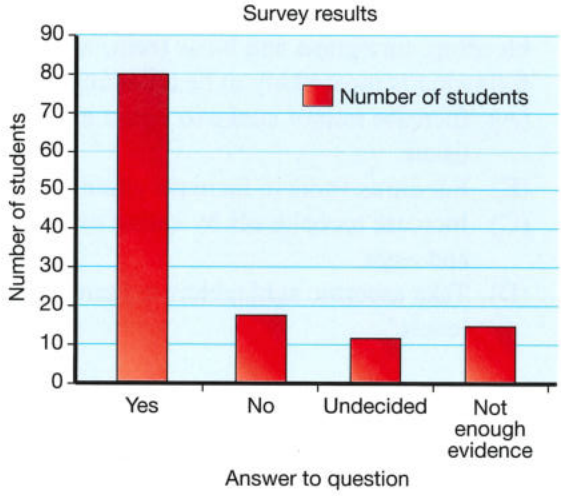


Figure TT.9 Student graph.

She concluded that whooping cough vaccination is not related to mental retardation Which of the following is a correct assessment of the validity of her conclusion?
(A) Very valid as she interviewed 125 students which is a large sample size.
(B) Very valid as she took into account a range of answers.
(C) Not very valid as her sample involved a limited age group.
(D) Totally invalid as she did not give them any information about vaccination programs.

33. It is currently believed that when an antigen first enters the body it circulates and comes into contact with many lymphocytes. When it meets a lymphocyte with a receptor site that best matches its own contours, a reaction occurs, binding the antigen to the lymphocyte.
What is the next step in the clonal selection theory?
(A) A series of mitotic divisions is triggered, making a clone of cells with the same antigen specificity.
(B) Continuous cell division occurs, making multiple copies of the combined antigen and lymphocyte.
(C) The antigen acts as a template for multiple replication, forming many cloned antibodies.
(D) Antibodies are secreted from macrophages, initiating the formation of more cloned lymphocytes.

34. The diagram shows the structure of an antibody.

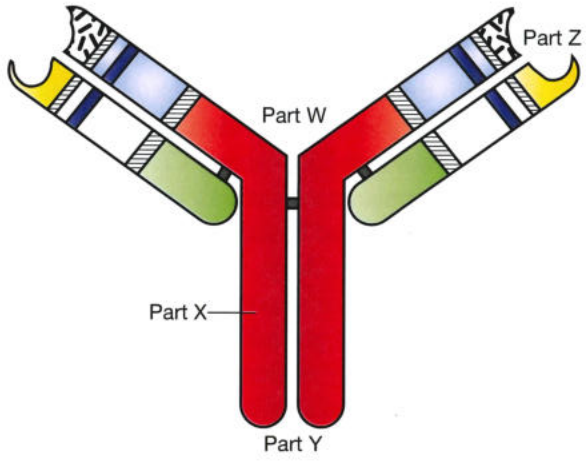


Figure TT.10 Antibody.

Where is the antigen binding site on an antibody?
(A) Part W
(B) Part X
(C) Part Y
(D) Part Z

35. The diagram shows a sweat gland in the skin.

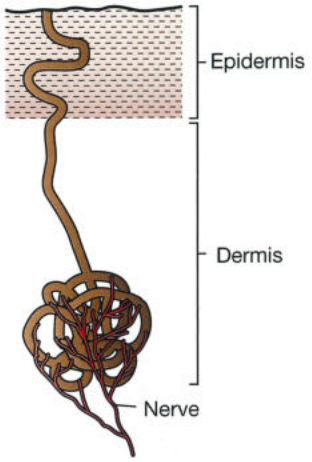


Figure TT.11 Sweat gland.

What type of nerve would stimulate secretion by the sweat gland?

(A) Sensory neurons.
(B) Sympathetic nerve.
(C) Parasympathetic nerve.
(D) Cervical C4 nerve.

Section B – Written Response (65 marks)

36. (a) Outline one advantage of homeostasis. (1 mark)
(b) List at least four factors that need to be kept constant in homeostasis. (3 marks)

37. The flow chart shows a feedback mechanism.

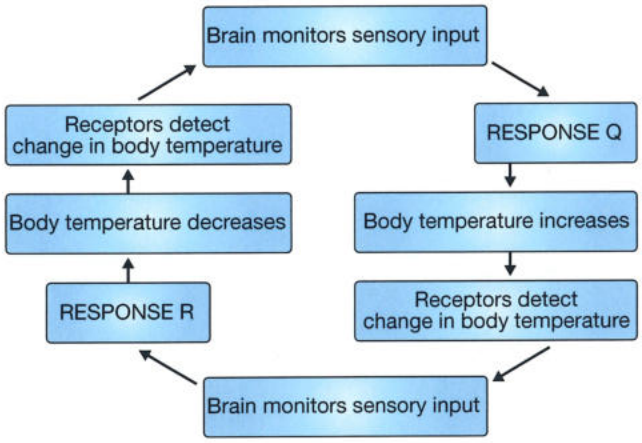


Figure TT.12 Feedback mechanism.

What would be response R and response Q respectively? (2 marks)

38. Complete the table to show the difference between the three types of hormones. (6 marks)

Type of hormone	Structure	Example
Peptide		
Steroid		
Amine		

39. The kidney is an organ involved in osmoregulation. Draw a cross-section of a kidney and label, the cortex, the medulla, a pyramid of the medulla, the pelvis of the kidney, the renal vein, the renal artery and the ureter. (4 marks)

40. During this topic, you performed a first-hand investigation to gather information about microbes in food or water.
(a) Explain the procedure you followed. (4 marks)
(b) Identify two safety procedures you needed to follow during this investigation. (2 marks)

41. The diagram shows how glucose levels change over time.

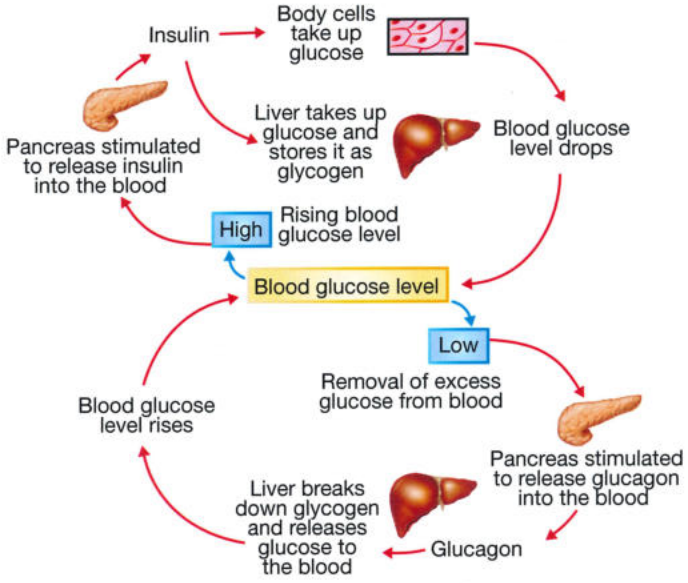


Figure TT.13 Changing blood glucose levels.

(a) From the diagram what is the function of insulin? (2 marks)
(b) Explain why insulin and glucagon can be considered to be antagonistic hormones. (2 marks)

42. The diagram shows the control of carbon dioxide levels in the blood.

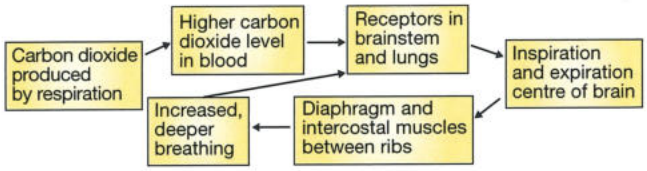


Figure TT.14 Changing carbon dioxide levels.

(a) From the chart, what causes an increase in carbon dioxide levels in the blood and what detects this change? (2 marks)
(b) Explain why high carbon dioxide levels in the blood can seriously affect metabolic processes. (3 marks)

43. Discuss why the immune response poses a problem for people undergoing gene therapy. (3 marks)

44. (a) Identify two chemicals that are removed from blood by the kidney? (2 marks)
    (b) Identify two chemicals that are reabsorbed into the blood by the kidney. (2 marks)
    (c) Explain the main steps involved in the formation of urine as part of homeostasis. (5 marks)

45. Outline how one disease results from an imbalance of microflora in humans. (4 marks)

46. Copy and complete the table to compare natural and acquired immunity. (4 marks)

Feature	Natural immunity	Acquired immunity
How it develops		
Short duration immunity		

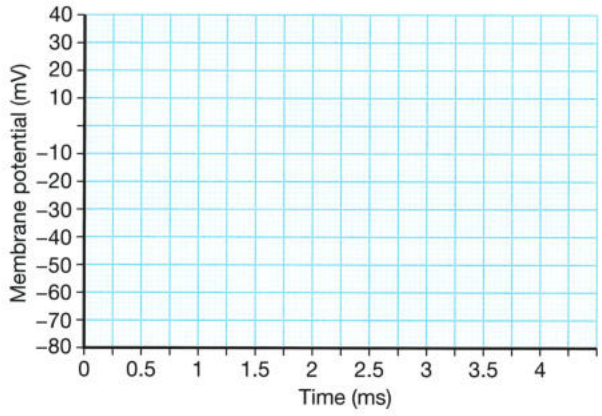
47. Complete the table by identifying the three types of neurons and outlining their function. (6 marks)

Neuron	Description
	Name
	Function
	Name
	Function
	Name
	Function

48. The following data was collected when a microelectrode was placed inside a nerve fibre to measure the membrane potential as impulses travelled past the microelectrode.

Time (ms)	Membrane potential (mV)
0	–70
0.25	–70
0.5	–65
0.75	–50
1	10
1.25	34
1.5	–50
1.75	–60
2	–80
2.25	–75
2.5	–73
2.75	–71
3	–70
3.25	–70
3.5	–70
3.75	–70
4	–70

(a) Plot these points on the graph. (3 marks)



(b) Distinguish between depolarisation and repolarisation and mark them on the graph. (3 marks)

(c) Explain why some stimuli do not result in an action potential. (2 marks)

Answers

1 Assumed Knowledge

1. Metabolism is the sum of all the chemical reactions occurring within a cell or other parts of an organism.
2. The environment is everything, both living and non-living, around an organism. The external environment refers to anything outside the body; the internal environment refers to anything inside the body.
3. A receptor is a cell or organ that can detect variations of some kind in an organism's environment. An effector is a structure that causes a response to counteract changes from the stable state.
4. Five senses are hearing, sight, touch, taste and smell.
5. Five sense organs are eyes, ears, tongue, nose and skin.
6. Skin has heat and cold receptors, light touch, pain and strong pressure receptors.
7. Interstitial fluid is the intercellular fluid which fills the spaces between the cells.
8. Endotherms, e.g. humans and other mammals maintain a constant internal environment because of internal processes.
9. The nervous system coordinates sensory information with the body's responses.
10. Both the CNS and PNS are composed of neurons and together make up the nervous system. The CNS consists of the brain and spinal cord and is mainly composed of interneurons, while the PNS consists of the nerves branching from the CNS and passing to all other body parts.
11. Ambient temperature is the temperature of surroundings.
12. The circulatory system transports materials around the body.
13. Red blood cells transport oxygen from the lungs to needy tissues in the body. White blood cells are part of the body's defence against infection and platelets are involved in clotting the blood.
14. Arteries are vessels which carry blood away from the heart; veins are vessels which carry blood towards the heart; capillaries are vessels which connect arteries to veins.
15. Diffusion is the movement of particles from an area of high concentration of particles to an area of low concentration of particles.
16. A control is part of an experiment that has the identical situation but often without the variable, and is used for comparison.
17. The lymph system is a network of nodes, veins and blind-end capillaries which carry lymph from body tissues and drain into blood vessels.
18. The cerebrum controls conscious thought, memory, sensory reception and motor activities. The spinal cord is involved in reflex actions involving body structures below the neck and sending sensory impulses to the brain and carrying motor impulses from the brain.
19. Carbon dioxide diffuses out of the capillary into the alveoli and oxygen diffuses into the capillary from the alveoli. Both move down a concentration gradient.
20. Water is an important molecule because of its unique properties as a solvent; many solutes dissolve in water. It also enters such reactions as photosynthesis, respiration and digestion. It can also act as a cooling agent, e.g. water evaporates from the skin and removes heat from the body.
21. A parasite lives on or in a host, gaining food, shelter and protection and causing harm to the host.
22. Bacteria are prokaryotes; they are microscopic with a cell wall and a single coiled strand of DNA.
23. Viruses are very small, are not cellular, consist of nucleic acid (either RNA or DNA) and are enclosed in a protein coat.
24. A pathogen is an organism capable of producing a disease.
25. A hormone is an organic chemical produced by one part of the body and transported to another part where it affects the metabolism of the target cells.

2 The Endocrine System

1. The endocrine system is a body system composed of different endocrine glands that are ductless glands which secrete hormones directly into the bloodstream or body fluids.
2. Hormones are chemical signals that travel in the blood and body fluids and act on specific target cells to cause a specific response for internal communication and regulation and maintaining homeostasis.
3.

Feature	Nervous system	Endocrine system
Speed of transmission	Rapid	Slower
Mode of transmission	Electrochemical impulse along nerve fibres	Hormones travelling in bloodstream and tissue fluids
Response time	Immediate	Slower acting
Response duration	Short time	Long time – up to years
Specificity	One effector	Several organs or target cells

4. A = pituitary gland
   B = thyroid gland
   C = ovaries
   D = pineal gland
   E = parathyroid glands
   F = thymus gland
   G = adrenal glands
   H = pancreas – islets of Langerhans
5. (a) The diagram shows how the hypothalamus (releasing TRH) and anterior pituitary gland (releasing TSH) control the secretion of the thyroid hormones. The release of hormones from several glands is controlled by the release of other hormones. The thyroid hormones are important in controlling many body functions, e.g. maintaining normal blood pressure, heart rate, muscle tone and digestion. The diagram also shows how hormone control is a feedback mechanism.
   (b) To reach their target organ, e.g. travelling from the anterior pituitary gland to the thyroid gland, the hormone travels in the blood.
6. Insulin and glucagon are antagonistic hormones as they both regulate the concentration of glucose in the blood but have opposite effects. Low blood glucose levels will cause the alpha cells of the pancreas to release glucagon into the blood which will cause liver cells to break down glycogen and release glucose into the blood. High blood glucose levels will cause the beta cells of the pancreas to release insulin into the blood which will cause the liver to take up glucose and store it as glycogen as well as body cells to take up more glucose.
7. The pituitary gland is often called the 'master gland' as its hormones stimulate other endocrine glands to secrete hormones, e.g. anterior pituitary produces thyroid stimulating hormone that stimulates the thyroid gland to produce thyroxine.
8. Adrenaline and noradrenaline are the flight or fight hormones.
9. (a) Neurotransmitters (e.g. when metabolic rate is slow) trigger the hypothalamus to secrete releasing factors (e.g. thyroid releasing factor – TRF) which will stimulate the pituitary gland to secrete thyroid stimulating hormone (TSH) that will then stimulate the thyroid gland to produce the hormone thyroxine.
   (b) Negative feedback is a type of regulation where a change is monitored and there is a response that will counteract the initial change. The dotted lines show the negative feedback. For hormone secretion negative feedback will inhibit the production of the hormone if the hormone concentration becomes too high, e.g. by controlling the amount of releasing factor and secretions of the pituitary gland.
   (c) The hypothalamus produces hormones and releasing factors which stimulate the pituitary gland to produce hormones.
10. There are three main types of hormones – the peptide hormones, the steroid hormones and the amine hormones.
11. The structure of the hormone determines if it is water soluble (hydrophilic) or fat soluble (hydrophobic), e.g. polypeptide hormones such as insulin are water soluble while steroid hormones such as cortisol are lipid soluble.
12. (a) TSH is a tropic hormone as it stimulates the thyroid gland.
    (b) Prolactin is a non-tropic hormone as it stimulates milk production and secretion.
    (c) ACTH is a tropic hormone as it stimulates the adrenal cortex to secrete glucocorticoids.
13. D
14. A
15. C

3 Types of Hormones

1. The endocrine system consists of endocrine glands which are ductless glands that release chemical messengers called hormones into the bloodstream.
2. A hormone is a chemical messenger, e.g. protein, amine or steroid molecule formed in specialised cells that travel in body fluids to act on specific target cells to change their functioning.
3. In vertebrates the endocrine glands can include the pituitary, sex glands, adrenal gland, thyroid gland, parathyroid gland, pancreas, islets of Langerhans and pineal gland.
4. Growth hormone causes growth of many parts of the body and thyroxine increases the rate of many chemical reactions in nearly all body cells.
5. The three main types of hormones are proteins and peptides (small polypeptides with up to 30 amino acids), amines derived from amino acids and steroids.
6. Steroid hormones are lipid soluble and can easily pass through cell membranes. They are usually concerned with long term responses of the body, e.g. oestrogen, testosterone and aldosterone.
7. As steroid hormones are lipid soluble they need to travel around the body bound to carrier proteins.
8. The receptors for steroid hormones can be in the cytoplasm or in the nucleus.
9. Many steroid hormones, e.g. oestrogens and cortisol act by directly regulating gene expression of target genes.
10. The amine hormones are derivatives of the amino acid tyrosine and include hormones secreted by the thyroid gland, e.g. thyroxine and triiodothyronine and the adrenal medulla (adrenaline and noradrenaline).
11. The protein/peptide groups include hormones secreted by the anterior and posterior pituitary gland, e.g. thyroid stimulating hormone (TSH) and antidiuretic hormone ADH), the pancreas (insulin and glucagon) and the parathyroid gland, e.g. parathyroid hormone (PTH).
12. In general the response time is much faster for amine and protein/peptide hormones than for steroid hormones. The amine/protein/peptide hormones such as ADH, TSH and insulin bring about a response within a few minutes and their actions are usually short lived while steroid hormones such as testosterone and oestrogen have long term responses that are longer lived.
13. Both steroid hormones and amine/protein/peptide hormones travel in the blood in the circulatory system around the body. Steroid hormones are formed from cholesterol and need to be bound to plasma proteins whereas amine/protein/peptide hormones can dissolve in the plasma.
14. A prehormone is a chemical produced by a gland that is converted into an active hormone in a mature form in peripheral tissues.
15. Signal transduction is the transmission of a molecular signal from the exterior of the cell into its interior using cell-surface receptors.
16. Hormone secretion is mainly controlled by a negative feedback system that will stop hormone secretion to prevent overactivity of the target cells. Hormone secretion is also affected by other factors such as daily rhythms, seasonal changes, stages of development and ageing and sleep patterns, e.g. growth hormone has higher secretion rates during early sleep time than in later sleep time.
17. The hypothalamus is part of the ventral section of the forebrain.
18. The hypothalamus plays an important role in the functioning of the endocrine system, e.g. secreting the hormones oxytocin and ADH which are stored in the posterior pituitary gland and secreting the releasing factors that regulate the anterior pituitary gland. The hypothalamus is also important as it maintains homeostasis by coordinating the endocrine system with the nervous system.
19. Stimulus → transcription → translation → synthesis of peptide hormone at ribosome on endoplasmic reticulum → packaging of hormone by Golgi apparatus → storage in vesicle → secretion by exocytosis from cell.
20. C
21. A

4 The Pituitary Gland

1. The pituitary gland lies below the hypothalamus.
2. The pituitary gland is often called the 'master gland' as it regulates the activity of other endocrine glands.
3. The hypothalamus controls the release of many of the hormones from the pituitary gland.
4.

Anterior pituitary hormone	Function
Growth hormone	Acts on body tissues especially during childhood and adolescence increasing the rate at which amino acids are taken up by cells and built into proteins.
Prolactin	Helps initiate and maintain milk secretion in females and promotes development of female breasts.
Luteinising hormone	Works with FSH in females to bring about ovulation and form the corpus luteum after ovulation. In males stimulates interstitial cells in testes to secrete male sex hormones.
Thyroid stimulating hormone	Stimulates the production and release of hormones from the thyroid gland.
Adrenocorticotrophic hormone	Controls the production and release of some of the hormones from the cortex of the adrenal glands.
Posterior pituitary gland	**Functions**
ADH	Causes the kidneys to reabsorb water to maintain body water balance.
Oxytocin	Stimulates the contraction of muscles of the uterus, e.g. during childbirth and stimulates the contraction of cells in mammary glands in the release of milk while suckling.

5. (a) Stimulation hypothalamus → secretion of growth hormone releasing hormone (GHRH) → anterior pituitary secretes growth hormone → increase in uptake of amino acids by cells.
   (b) Stimulation of hypothalamus → secretion of prolactin releasing hormone (PRH) → anterior pituitary secretes prolactin → mammary glands stimulated to produce milk.
   (c) Stimulation of hypothalamus → secretion of corticotropin releasing hormone (CRH) → anterior pituitary gland secretes adrenocorticotropic hormone (ACTH) → adrenal glands synthesise cortisol.
6. Hyposecretion involves reduced amounts of hormone produced whereas hypersecretion involves increased amount of hormone produced.
7. A secretory tumour of the pituitary gland leads to hypersecretion of a particular hormone. Overproduction of a hormone will interfere with normal body functioning and if the tumour mass becomes large and presses against other areas it can cause headaches and vision problems.
8. D
9. A
10. B
11. C

5 The Thyroid Gland

1. The thyroid gland lies below the larynx on each side and anterior to the trachea.
2. The thyroid gland produces thyroxine (T_4), tiiodothyronine (T_3) and calcitonin.
3. Thyroxine is an amino acid with four iodine atoms.
4. T_4 and T_3 affect the metabolic rate of the body and bring about the release of energy when complex molecules are synthesised from simple molecules. They are also necessary for cellular differentiation.
5. Low concentration of thyroid hormones in the blood → anterior pituitary gland is stimulates to secrete more TSH → thyroid gland is stimulated to secrete more thyroid hormones → concentration of thyroid hormones in blood increases.

6. (a) Hyperthyroidism is the overproduction of thyroxine.
   (b) Hyperthyroidism causes nervousness, insomnia and excessive excitability; heat intolerance and excessive sweating; increased heart rate and blood pressure; and weight loss.
7. (a) Hypothyroidism is the underproduction of thyroxine.
   (b) Hypothyroidism during infancy affects the development of brain cells and can lead to permanent mental deficiency and dwarfism. In adults it causes dry skin, intolerance to cold and lack of energy.
8. TSH (thyroid stimulating hormone) from the anterior pituitary gland stimulates the release of thyroxine from the thyroid gland.
9. Common salt is iodised to make sure people do not have an iodine deficiency which would lead to an inability to make thyroxine and hypothyroidism. This can lead to a constant production of TSH which causes the thyroid gland to grow and form a goitre.
10. Cretinism is caused by extreme hypothyroidism during foetal development, infancy or childhood. The low levels of thyroid hormones cause failure of body growth and mental retardation and can be due to lack of iodine in the diet or a genetic defect of the gland. A child with cretinism will be stocky, short and obese.
11. The thyroid hormones activate gene expression and the transcription of a large number of genes, e.g. to cause the synthesis of protein enzymes, structural proteins and transport proteins.
12. A
13. C
14. B
15. D
16. A

6 The Adrenal Glands

1. The adrenal glands are on either side of the spinal column above each kidney.
2. The adrenal gland consists of the adrenal cortex and the adrenal medulla.
3. Both corticosteroids are secreted by the adrenal cortex with glucocorticoids affecting glucose metabolism while mineralocorticoids affect mineral metabolism maintaining salt and water balance.
4. Stressful signals cause the hypothalamus to secrete a releasing hormone that stimulates the anterior pituitary to release adrenocorticotropic hormone (ACTH). ACTH causes the adrenal cortex to secrete the corticosteroids.
5. Cholesterol is needed for normal functioning of the adrenal glands as the adrenocortical hormones are steroids derived from cholesterol.
6. (a) Cortisol has multiple metabolic functions and is involved in the metabolism of proteins, carbohydrates and fats. It also has anti-inflammatory effects.
   (b) Aldosterone increases the reabsorption of sodium in the kidney nephrons and increases the secretion of potassium and hydrogen ions from the kidney.
7. The main two hormones of the adrenal medulla are adrenaline (epinephrine) and noradrenaline (norepinephrine).
8. The main hormones of the adrenal medulla are secreted in response to stress and increase the rate of glycogen breakdown in the liver and skeletal muscles, increase heart rate, promote glucose release by liver cells and dilate the bronchioles in the lungs to increase oxygen uptake. This prepares the body for action and is sometimes called the 'fight or flight' response.
9.

Adrenal cortex hormone	Function
Cortisol	Metabolism of proteins, carbohydrates and fats; has anti-inflammatory effects.
Aldosterone	Increases the reabsorption of sodium in the kidney nephrons and increases the secretion of potassium and hydrogen ions from the kidney.
Adrenal medulla hormone	**Function**
Adrenaline and noradrenaline	Increase the rate of glycogen breakdown in the liver and skeletal muscles, increase heart rate, promote glucose release by liver cells and dilate the bronchioles in the lungs to increase oxygen uptake.

10. Addison's disease is caused by insufficient production of adrenocortical hormones, e.g. lack of aldosterone. This causes a loss of sodium ions, chloride ions and water in urine and a drop in plasma volume. Cardiac output decreases and the person can die of shock within two weeks from the time when mineralocorticoid production stopped.
11. The adrenal cortex is stimulated from the pituitary gland, e.g. adrenocorticotropin (ACTH) stimulates the adrenal cortex to produce cortisol. Whereas the adrenal medulla is stimulated by the sympathetic nervous system, e.g. to increase cardiac output, increase blood pressure and increase ventilation rate.
12. Cushing's syndrome occurs when there is hypersecretion of the adrenal cortex leading to excess cortisol in the blood. It is often caused by excess ACTH secretion. It causes fat from the lower body to mobilise and deposit in the upper abdomen and thorax causing a buffalo torso appearance. The excess steroids also cause increased facial hair and fat deposits on the face.
13. Stress stimulus → hypothalamus secretes CRF → anterior pituitary secretes ACTH → adrenal cortex secretes cortisol → increased glucose and fuel supplies in the blood for body cells under stress.
14. D
15. C
16. C
17. B

7 The Pancreas

1. The pancreas is located behind and parallel to the stomach.
2. Exocrine glands secrete their products into ducts. The pancreas has acinar cells that produce digestive enzymes and duct cells that secrete sodium bicarbonate solution into the ducts that empty into the pancreatic duct. Thus the acinar cells and duct cells are part of the exocrine section of the pancreas.
3. Endocrine glands are ductless glands that secrete hormones into extracellular spaces and then diffuse into the circulatory system. The pancreas secretes the hormones insulin and glucagon and this section of the pancreas is thus an endocrine gland.
4. The islets of Langerhans are patches of endocrine tissue within the pancreas that were first discovered by Paul Langerhans in 1869.
5. Islets of Langerhans.

Cell type	Product
Alpha cells	Glucagon
Beta cells	Insulin
Delta cells	Somatostatin
Gamma cells	Pancreatic polypeptide
Epsilon cells	Ghrelin

6. Glucose is needed for cellular respiration and biosynthesis of macromolecules.
7. For most humans blood glucose levels are the lowest first thing in the morning before breakfast.
8. Homeostasis is the maintenance of a constant internal environment within narrow limits. Blood glucose levels need to be maintained within narrow limits so that normal metabolic processes can function properly. Changes in solute concentrations will affect diffusion rates, osmotic pressure and the need for active transport across membranes.
9. Glucose and glucagon are antagonistic hormones as they have opposing effect – insulin causes glucose in the blood to be taken up by body cells to lower blood glucose concentration while glucagon causes the release of glucose into the blood from energy stores, e.g. glycogen from the liver.
10. Homeostasis is the maintenance of a constant internal environment within narrow limits. The hormones produced by the pancreas – insulin and glucagon control blood glucose concentration keeping the concentration within narrow limits for optimal metabolism. Thus the pancreas is important in maintaining homeostasis.
11. (a) Insulin is produced by the beta cells within the islets of Langerhans.
    (b) Glucagon is produced by the alpha cells within the islets of Langerhans.

12. Insulin acts by slowing down glycogen breakdown in the liver and stopping the conversion of glycerol and amino acids to glucose. It stimulates nearly all body cells to take up glucose. If the diet contains high amounts of energy giving food, e.g. carbohydrates and protein then the body will make large amounts of insulin.
13. Blood glucose levels too low → alpha cells of islets of Langerhans secrete glucagon → liver cells release stored glucose and fats and amino acids mobilised → blood glucose levels rise.
14. Blood glucose levels too high → beta cells of islets of Langerhans secrete insulin → body cells take in glucose, e.g. muscle cells store glucose as glycogen → blood glucose levels drop.
15. Somatostatin extends the period of time that nutrients are assimilated into the blood and has several inhibitory actions, e.g. it acts locally in the pancreas to inhibit the secretion of both insulin and glucagon
16. (a) Diabetes mellitus is caused by a deficiency of insulin or a reduced response of target cells to insulin.

(b)

Type 1 diabetes	Type 2 diabetes
Is insulin dependent diabetes.	Is non-insulin dependent.
Usually appears in childhood.	Usually appears in older people or people who are overweight and live a sedentary lifestyle.
Is an autoimmune disease where the immune system destroys the beta cells of the pancreas.	Is where target cells do not correctly respond to insulin.

17. C
18. B
19. D
20. A

8 Play – Keeping Up With the Hormones

Scene 1 – End of a Long Bone in a Leg

1. RBCs form in bone marrow.
2. RBCs are biconcave disc shape, approximately 8 μm in diameter.
3. Mature RBCs do not have a nucleus and hence have limited life span.
4. Adrenaline is produced by the adrenal medulla of the adrenal gland.
5. Adrenaline causes glycogen to be broken down to glucose in the liver to increase the amount of blood glucose. Adrenaline controls changes in blood flow patterns.
6. RBC feeling 'blue' as it is not carrying any oxygen.
7. The hepatic portal vein takes products of digestion from the small intestine to the liver and hence they can have afternoon tea there as it is full of 'food'.

Scene 2 – The Foot

1. Phagocytes are specialised WBCs which engulf and destroy bacteria.
2. When bacteria enter, phagocytes rush to the area to engulf and destroy bacteria.
3. Mast cells are found in the connective tissue surrounding most blood vessels and nerves.
4. Histamine is an inflammatory signalling molecule. It increases the permeability of the capillaries to white blood cells and some proteins and causes the capillaries to dilate.
5. Platelets are involved in clotting blood.
6. Pus is dead cells, bacteria and WBCs.

Scene 3 – The Hepatic Portal Vein

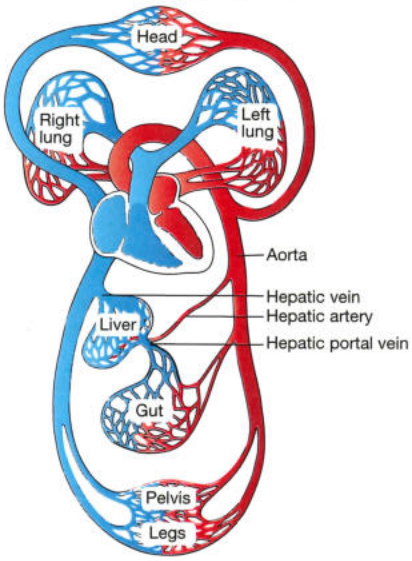
1. Sugar-containing foods are broken down into glucose in the digestive system. Then the glucose diffuses across the wall of a villus in the small intestine into the bloodstream.
2. Testosterone is an anabolic steroid hormone.
3. Taking steroid supplements increases muscle mass and bone density and strength.
4. Respiration reaction:
   Glucose + oxygen → carbon dioxide + water + energy
5. Respiration occurs in all living cells.
6. In the lungs RBCs pick up oxygen.
7. 45 seconds.
8. Oxygen dissociates from haemoglobin when there is low oxygen and there is a high diffusion gradient.

Scene 4 – A Small Capillary

1. Glucose moved through the wall by diffusion.
2. Losing the oxygen made the RBC blue – it is deoxygenated.
3. Insulin is made in the pancreas.
4. Insulin causes glucose in the bloodstream to be converted into glycogen for storage.
5. Lack of sugar causes insulin shock.
6. Blood contains RBCs, WBCs, platelets, hormones, glucose.
7.

Part	Function
Red blood cell	Picks up oxygen in the lungs and transports to a needy cell.
White blood cell	Fights infection.
Platelet	Aids clotting.
Testosterone	Increases muscle mass and bone density and strength. Maintains male characteristics of males, supports spermatogenesis.
Adrenaline	Increases blood sugar, dilates some blood vessels, increases rate of heart beat.
Insulin	Lowers blood sugar, increases storage of glycogen.
Glucose	Used for cellular respiration and biosynthesis of macromolecules.
Oxygen	Used in aerobic respiration.
Mast cell	Releases histamine and other molecules that trigger inflammation response to infection.
Histamine	Increases permeability of the capillaries to white blood cells and some proteins. Also causes capillaries to dilate.

8. (a) Cell A is a red blood cell, tissue B is skeletal muscle.
   (b) The diagram shows glucose molecules in the arteriole. When insulin is secreted it causes the liver to take up the glucose and convert it to glycogen. It also causes skeletal muscle to take up glucose which is either used in respiration or converted to glycogen for storage.
9.



A = head
B = right lung
C = left lung
D = aorta
E = liver
F = hepatic vein
G = hepatic artery
H = hepatic portal vein
I = gut
J = pelvis
K = legs

10. Negative feedback is a type of regulation where a change is monitored and there is a response that will counteract the initial change. The dotted lines show the negative feedback. For hormone secretion negative feedback will inhibit the production of the hormone if the hormone concentration becomes too high, e.g. by controlling the amount of releasing factor and secretions of the pituitary gland.

9 Hormones and Feedback Regulation

1. Two types of feedback regulation for the control of hormone secretion are the simple endocrine pathway and the simple neuroendocrine pathway.
2. Stimulus (low blood glucose level) → endocrine cell (alpha cell of pancreas) → hormone (glucagon) secreted → hormone enters blood vessel and travels in bloodstream → target cell (liver cell) → response (liver cell converts glycogen to glucose molecules which raises blood glucose level).
3. In negative feedback the effector response reduces the initial stimulus and the response stops thus preventing overreaction by the system to the initial stimulus whereas in positive feedback the stimulus is reinforced and leads to a greater response.
4. Oxytocin can be an example of positive feedback when suckling stimulates sensory nerves in the nipple that sends messages to the hypothalamus which releases oxytocin from the posterior pituitary gland to cause the mammary glands to secrete more milk. More suckling causes more oxytocin and more milk flowing to feed the baby.
5. The release of secretin is an example of negative feedback when a low pH is detected in the duodenum and the S cells of the intestinal glands in the duodenum secrete the hormone secretin. Secretin regulates water balance and will inhibit secretion of gastric acid by the parietal cells of the stomach and stimulates cells in the pancreas to product hydrogen carbonate. This response reduces the pH and when the change is detected negative feedback will stop the production of secretin.
6. (a) Like other hormones cortisol is released directly into the bloodstream and travels in blood vessels to target cells.
   (b) In negative feedback cortisol would stop being released from the adrenal gland.
   (c) The effect of the breakdown of proteins and amino acids is to increase the level of blood glucose. Glucose uptake is also reduced. This means that the body can increase the rate of respiration providing energy for thought processes to improve preparedness and alertness to cope with the stress.
7. A

10 The Three Steps in Signal Transduction

1. The three steps in signal transduction are reception, transduction and cellular response.
2. Receptors for signalling molecules can be integral proteins in the plasma membrane or intracellular receptors.
3. Intracellular receptors detect signalling molecules that are either hydrophobic or very small, e.g. steroid hormones such as testosterone as these signalling molecules need to pass through the plasma membrane to reach the receptor.
4. Rhodopsin is a light sensitive protein found in rods of the retina and is an important part of the photoreceptor. A mutation that causes misfolding of rhodopsin means that the shape of the protein changes in turn which means it will no longer be able to function in responding to light. This means that the transduction pathway will not be activated reducing the cellular response and vision will be reduced.
5. There are several benefits in using a multistep pathway: 1. The signal can be amplified with molecules in the pathway able to signal many molecules in the next step. 2. Allows fine-tuned and coordinated response to a signal.
6. Secondary messengers are information relaying molecules that take the signal received at the receptor on the cell surface to the target molecules in the cytosol or nucleus, e.g. calcium ions or cAMP.
7. A ligand is a molecule that binds specifically to another molecule, usually a large molecule.
8. A

11 The Nervous System

1. The nervous system is involved in coordination – sending messages when stimuli are detected to bring about appropriate responses.
2. The coordination of body functions occurs through electrical signals and the release of regulatory molecules, e.g. neurotransmitters.
3. The nervous system and endocrine system act together in coordination and communication maintaining homeostasis.
4. The nervous system responds to stimuli from the internal environment, e.g. changes in blood pressure or digestive activity or to stimuli from the external environment, e.g. to information picked up by the sense organs.
5. The nervous system is divided into the central nervous system (CNS) and the peripheral nervous system (PNS).
6. The meninges are three layers of membrane between the spinal cord and vertebrae or between the brain and the skull.
7. The CNS consists of the brain and spinal cord.
8. The PNS consists of all nerves outside the brain and spinal cord.
9. Parts A and D are part of the CNS while parts B, C and E are part of the PNS.
10. A neuron is an excitable nerve cell.
11.

Type of neuron	Function
Sensory neuron	Carries an electrical impulse from a receptor to the CNS.
Motor neuron	Carries an electrochemical impulse from CNS to effector, e.g. muscle or gland.
Multipolar neuron	Makes connections to other neurons inside the CNS.

12. (a) Glial cells are supporting cells found in the nervous system that are essential in maintaining the structure of the nervous system for the normal functioning of neurons.
    (b) Functions of glial cells include: provide physical support, direct growth, repair and development of neurons, give metabolic support to neuron functioning and help provide barriers and insulation to neurons.
13.

Glial cell in CNS	Function
Astrocyte	Main supporting tissue found near blood vessels.
Oligodendrocyte	Cluster around nerve cell bodies and length of myelinated nerve fibre.
Microglia	Derived from monocytes and become phagocytic if necessary.
Ependymal cells	Form epithelial lining of ventricles of brain and central canal of the spinal cord.

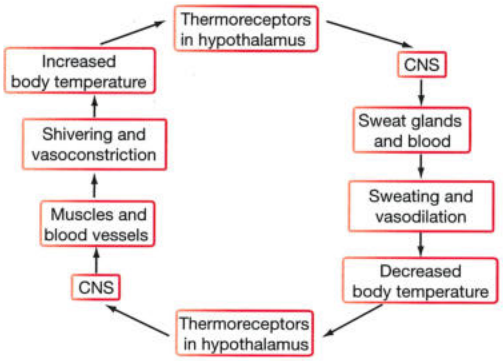
14. Cerebrospinal fluid – 1. Helps support and protect the brain and spinal cord. 2. Acts as a shock absorber. 3. Keeps the brain and spinal cord moist. 4. Maintains a uniform pressure around the brain and spinal cord.
15. The peripheral nervous system is divided into the afferent system which takes information to the CNS and the efferent system which carries information from the CNS.
16. (a) The diagram with structure X shows part of the PNS.
    (b) The sensory neurons would be taking messages from the receptors in the hand to the CNS while the motor neurons would be bringing messages from the CNS, e.g. to muscle cells.
17. The efferent nervous system can by subdivided into the somatic system which controls skeletal muscle and the autonomic system which controls smooth and cardiac muscle.
18. (a) Two divisions of the autonomic nervous system are the sympathetic system and the parasympathetic system.
    (b) The diagram shows eight connections from vertebrae to the heart in the sympathetic system.
19. C

12 The Role of the Nervous System

1. The main role of the nervous system is coordination with the detection and response to environmental stimuli.
2. Receptor → sensory neuron → connector neuron → motor neuron → effector.
3. The nervous system develops from the ectoderm with differentiation to form a hollow neural tube which becomes the brain and spinal cord and neural crests which become the sensory nerve fibres and the autonomic nervous system.
4. Neuron T is as a connector neuron in the CNS. Its function is to relay the impulse from the sensory neuron to the motor neuron to enable a fast reflex to occur without waiting for the message to be interpreted by the brain.

5. The tongue acts as a receptor when the receptor cells in taste buds detect sweet, salty, sour or bitter in food. The tongue can act as an effector when it is used for speech to respond to a comment or when it is used to spit something out of the mouth.

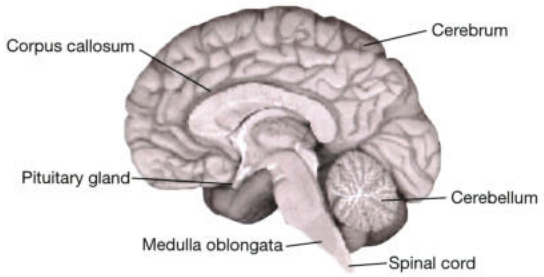
6.



7. A

13 The Brain

1. The brain is protected by: 1. The bones of the cranium, which provide an outer strong layer. 2. The meninges, which are three protective layers of connective tissue. 3. Cerebrospinal fluid, which fills the spaces in the brain and acts as a shock absorber to cushion the brain.
2. The cerebrum controls all conscious processes (e.g. thinking, reasoning and remembering) as well as receiving sensory signals from sense receptors and sending messages to motor areas to carry out orders. The cerebellum is involved in the coordination of movement, balance and posture. The medulla oblongata controls visceral functions including breathing, heart rate and swallowing. They each occupy different regions of the brain. The cerebrum is in the forebrain, and the cerebellum and medulla oblongata are part of the hindbrain.
3. Human brain.



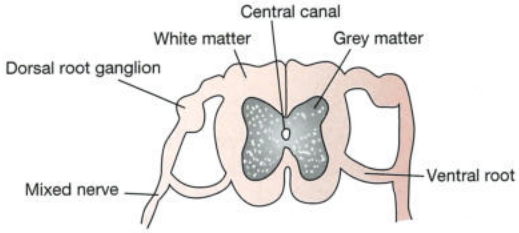
4. (a) Students should wear rubber gloves to protect their skin from any substances that may be present. Students should wear safety goggles in case some materials squirts up into their eyes.
   (b) Both human and sheep brains have a cerebrum, cerebellum, medulla oblongata and are surrounded by meninges. The human brain has a larger cerebral cortex than the sheep's brain.
   (c) In the brain the grey matter is the outer layer with the white matter on the inside; in the spinal cord the white matter is on the outside and the grey matter is on the inside.
5. (a) The hypothalamus is below and in front of the thalamus and linked to the pituitary gland.
   (b) The hypothalamus controls the release of hormones from the pituitary gland, is involved in appetite, maintaining homeostasis, thirst and water balance, body temperature, emotional reactions, e.g. pleasure, fear, anger, sexual behaviour and the autonomic nervous system.
6. The corpus callosum is a thick band of nerve fibres in the central region of the brain that connects the right and left cerebral hemispheres.
7. The cerebellum receives sensory information about the positions of muscles and joints, e.g. from the visual and auditory systems and by monitoring messages from the cerebrum it coordinates responses, e.g. eye-hand responses. Thus it can be called the reflex centre for muscular coordination.
8. The medulla oblongata controls several visceral (autonomic, homeostatic) functions including breathing, heart and blood vessel activity, blood pressure, swallowing, vomiting and digestion.
9. The pons is the bulge above the medulla and is a relay centre for information between the cerebellum and cerebrum. It also acts with the medulla in coordinating breathing.
10. C

14 The Brain Interprets Signals

1. Three main functions of the cerebrum are – 1. Mental activities, e.g. memory, intelligence, thinking, learning, reasoning, consciousness. 2. Sensory perception. 3. Initiation and control of voluntary muscle, e.g. skeletal muscle contraction.
2. Perception is the brain's interpretation of sensory information and the perceived stimulus may be very different from the actual stimulus.
3. (a) The cerebrum is divided into the left and right cerebral hemispheres connected by a bundle of fibres called the corpus callosum.
   (b) Both the fingertip and the hip have specific areas in the brain for interpretation of information. The fingertip has a much larger area and is more sensitive than the hip, which has a smaller area.
4. For most people, reasoning, logic, scientific skills, mathematics, written and spoken language are controlled by the left hemisphere; while art, imagination, artistic ability, insight, intuition, music appreciation are controlled by the right hemisphere.
5. In positron emission tomography (PET) scans, the person is injected with radioactive water and the brain activity is detected as an increase in blood flow and more radioactive water shows up on the PET scanner.
6. PET uses radioactive water while functional magnetic resonance imaging (fMRI) does not use radioactive water. PET scans are considered invasive while fMRI is not.
7. Newer instruments – magnetoencephalography (MEG) and electroencephalography (EEG) track brain messages in real time – at a faster rate than PET and fMRI machines as the older imaging machines cannot keep up with the speed of the messages from the senses.
8. (a) In more recent developments there has been an integration of MRI and fMRI with MEG and EEG in functional neuroimaging.
   (b) This links the good spatial resolution of fMRI with the speed of the EEG to give an accurate sequence of events and a way to observe and measure brain activity in humans.
9. An optical illusion is usually a drawing that seems to make sense but upon analysis contradicts reality.
10. The legs of the elephant have been incorrectly joined at the base to create confusion about the actual number of legs present. The brain initially identifies the diagram and pre-learning tells us that the elephant has four legs. It is only when you try to count the legs and work out how each connects to the body that the problem becomes evident.
11. Behaviour is determined by decisions made by the brain about the information received by the senses.
12. If a human spins around and around on the spot, and then suddenly stops still, the eyes give information that the body is still, but the semicircular canals provide information that the body is still spinning. The brain is given conflicting information and frequently decides that the solution is to 'feel sick' forcing the body to stay still until all sensory information agrees.
13. It is possible that the nerve endings in the stump have been stimulated so that the brain interprets the message as coming from a region that has been lost, e.g. amputated and a phantom pain is perceived.

14. (a) The brain remembers the shape of a triangle and sees this shape in the space, due to the way the circles have been cut even though there are no lines defining a triangle.
    (b) If you block out part of the diagram, the triangle disappears. The brain is deprived of some of the clues it needs to make a conclusion about the missing piece and hence it does not perceive a triangle.
15. The approach of a nurse with an injection is an example where past experience can modify our response. If a person has had a painful experience with injections, they may flinch, break out into a sweat or their skin may go pale, just at the sight of the nurse with the needle, even though their skin has not yet been pierced.
16. Even though the two men are the same size the brain translates a two-dimensional drawing into a three-dimensional image. The lines show how the size of objects changes with distance to the horizon and thus man Y appears much larger as the brain interprets him as being further away and must be larger than man X.
17. A

15 The Spinal Cord

1. The spinal cord runs lengthwise inside the vertebrae from the base of the brain.
2. The spinal cord is protected by the vertebrae, by three meninges – the dura mater, arachnoid mater and pia mater and by cerebrospinal fluid.
3. In an adult male it is approximately 45 centimetres long.
4. 

5. In the spinal cord the grey matter is on the inside forming a letter 'H' and the white matter is on the outside.
6. (a) Grey matter consists of cell bodies, synapses and non-myelinated connector neurons.
   (b) It is grey as it lacks myelin and appears grey in unstained preparations.
7. The dorsal nerve root is the nerve that enters the back of the spinal cord and contains the axons of the sensory neurons.
8. The dorsal root ganglion contains the cell bodies of the sensory neurons that are in the dorsal nerve root.
9. (a) The connector neurons of the spinal cord are in the grey matter (letter 'H') of the spinal cord.
   (b) The connector neurons of the spinal cord synapse with the sensory neurons, other connector neurons and motor neurons.
10. The ventral nerve root comes out the front of the spinal cord and contains the axons of motor neurons.
11. The white matter of the spinal cord has myelinated fibres running longitudinally.
12. (a) The two types of tracts of white matter in the spinal cord are the ascending tracts and the descending tracts.
    (b) The ascending tracts take sensory information to the brain and descending tracts bring motor impulses down from the brain.
13. Spinal nerves are formed by the union of both the dorsal and ventral roots after they leave the spinal cord.
14. They are called mixed nerves as they carry both sensory and motor impulses.
15. Each spinal nerve divides into a posterior and anterior branch and are named and numbered according to the vertebrae to which they connect.
16. Part P shows a receptor in the skin on the hand. A nervous impulse will travel in the sensory neuron which has its cell body in part X, the dorsal nerve ganglion. The synapse is in the spinal cord and the message is passed to part G, a connector neuron. part G passes the message to motor neurons which take the message up to the brain (via part D) and the effector (part A) which is a muscle in the arm which moves the arm as the response to the stimulus.

16 The Autonomic Nervous System

1. The autonomic nervous system is the motor division of the peripheral nervous system.
2. The autonomic nervous system is the involuntary system controlling functions that are carried out automatically, e.g. controlling the internal environment and homeostasis.
3. The two divisions of the autonomic nervous system are the sympathetic system and the parasympathetic system.
4. Since the sympathetic nervous system prepares the body for immediate action, e.g. flight or fight, while the parasympathetic nervous system is involved with rest and digestion, the two systems can be considered to work in opposition. The sympathetic NS increases heart rate while the parasympathetic NS decreases heart rate.
5. The two systems need to work together to control smooth muscle, cardiac muscle and glandular secretion to control involuntary responses.
6.

Feature	Sympathetic nervous system	Parasympathetic nervous system
Subdivision of nervous system	Part of the autonomic nervous system.	Part of the autonomic nervous system.
General action	Mobilises body's resources under stress to induce the flight or fight response.	For functions that do not require immediate reaction, e.g. for rest, for digestion, salivation, urination, lacrimation and defecation.
Heart rate	Increases heart rate.	Decreases heart rate.
Movement of iris	Causes dilation.	Causes constriction.
Flow of blood in gut	Decreases flow of blood in gut.	Increases flow of blood in gut.
Neurotransmitter at synapse	Acetylcholine released from preganglionic neuron. Noradrenaline released from postganglionic neuron.	Acetylcholine released from both pre and postganglionic neurons.

7. The effector organs are smooth muscle, cardiac muscle and the glands.
8. The main control of the autonomic system is from centres in the hindbrain, e.g. cardiac centre, respiratory centre and thermoregulatory centre which are below the level of the cerebrum.
9. The sympathetic nervous system would cause suppression of secretion from the salivary glands and increase of release of glucose (conversion of glycogen to glucose) in the liver.
10. The parasympathetic nervous system would cause contraction of the walls of the bladder and the sphincter relaxed and an increase in peristalsis.

17 The Stimulus-Response Model

1. The stimulus-response model gives the basic pathway for a nervous impulse where a receptor picks up a stimulus, the message is interpreted and another message is transmitted to an effector which carries out the response.
2. A stimulus is a change in the environment of an organism that is detected.
3. Stimulus → receptor → nervous → effector → response system.
4.

Part of stimulus-response model	Description
Receptor	Detects stimulus and converts it into a nerve impulse which is transmitted by a sensory neuron to CNS.
CNS	Connector neurons in spinal cord and brain take the message to a control centre where the information is processed.
Effector	Motor neurons connect to the CNS and transmit a resultant nervous impulse from CNS to effector organ, e.g. muscle or gland to carry out the response.

5. A motor neuron takes the nerve impulse from the CNS to an effector whereas a sensory neuron takes the nerve impulse from the receptor to the CNS. The motor neuron usually has a long axon that is myelinated while the axon of the sensory neuron may or may not be myelinated.

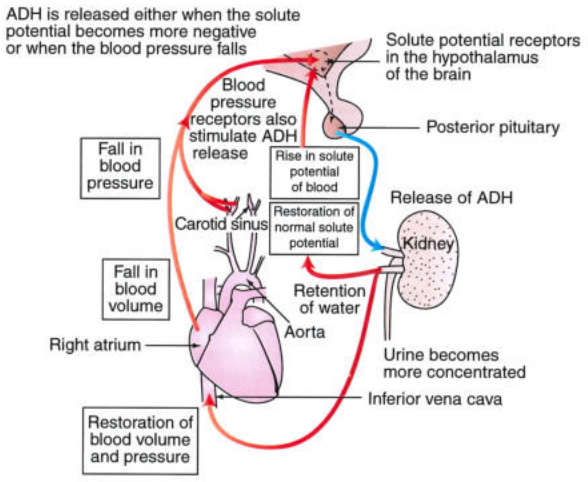
6. A nerve is a bundle of nerve fibres, e.g. axons held together by connective tissue.
7. The synapse is the gap at the junction between adjacent neurons.
8. Neurotransmitters are substances that diffuse across a chemical synapse transmitting the message from the synaptic terminal of a neuron to the postsynaptic cell triggering a response.
9. B
10. D

18 Receptors

1.

Sense organ	Receptor	Stimuli
Eyes	Retina has rods and cones	Light
Ears	Cochlea has hair cells	Sound
Tongue	Gustatory receptors in taste buds	Chemicals
Nose	Olfactory receptors in nasal cavity	Chemicals
Skin	Mechanoreceptors detect pressure, touch, stretch	Pressure, touch
Skin	Thermoreceptors detect heat and cold	Heat, cold
Skin	Pain receptors	Pain

2. Proprioceptors are in muscles, tendons and joints; mechanoreceptors respond to energy such as stretching, movement, touch, pressure, gravity; chemoreceptors respond to chemicals; photoreceptors detect light; thermoreceptors detect heat and cold and some animals (e.g. fish have electroreceptors that detect electrical energy).
3. Stimulus → receptor → sensory neuron → interneuron → motor neuron → effector → response.
4. Example of touch response – the strong pressure receptor (Pacinian corpuscle) is found deep in the skin and around joints and is deformed when the skin experiences firm pressure. The sense cell converts the stimulation into a nerve impulse.
5. (a) Receptors include chemoreceptors in medulla of brain and stretch receptors in lungs.
   (b) The stimulus could be an increase in the carbon dioxide levels in the blood that is detected by the chemoreceptors in the medulla. This causes a messenger (nerve impulse) to be sent to the inspiratory centre, which in turn sends nerve impulses to the effectors – the diaphragm and intercostal muscles – to increase the rate of contraction. The response is faster and deeper breathing, and an increase in the rate of inspiration.
   (c) The stretch receptors in the lungs are stimulated by the faster breathing rate and send messages back to the expiratory centre of the hindbrain. The inspiratory centre is switched off, followed by the expiratory centre.
   (d) When a person holds their breath or is playing a wind instrument, they can override the impulses that control inspiration for a short time.
6. Stimulus-response model for antidiuretic hormone (ADH).



7. (a) Stretch receptors (proprioceptors) are in muscles and tendons and send impulses to the CNS to inform the brain about the position and movement of various parts of the body.
   (b) There are several types of receptors in the skin, e.g. touch, pain, cold, deep pressure, warmth and touch receptors. The relative abundance and distribution of each is different and different on different parts of the body, e.g. pain receptors are nearly 27 times more abundant than cold receptors.
8. Senses include sense of hearing, sight, touch, pressure, detection of chemicals by taste and smell.
9. People need to be able to detect external climate changes, presence of food, predators, danger, to ensure survival of the individual.
10. Humans use the handshake as a form of greeting communication. Some people use the handshake to assess the quality of the other person (e.g. on the firmness, length of grip, angle of the palm).
11. (a) Finger touching pin.
    (b) Arm muscle causes the finger to jerk away.
    (c) Pain receptor.
    (d) Stimulus → receptor → sensory neuron → interneuron → motor neuron → effector → response.
    (e) Both the knee jerk and dodging the ball are examples of stimulus-response where the body is subjected to an external stimulus. However, the knee jerk is a pressure stimulus detected by pressure receptors in the skin, while the ball approaching is a light stimulus detected by receptors in the eye. The knee jerk is a reflex, occurring without conscious decision by the brain while the ball response is a conscious response decided by the brain.
12. D

19 Reflex Arcs

1. A reflex is a fast response that is involuntary and does not involve the brain in the decision making process.
2. A person becomes aware a reflex has happened after it has occurred when the message reaches the brain and is interpreted.
3. Grey matter is mainly the cell bodies of neurons, dendrites and unmyelinated axons. Whereas white matter consists of axons with myelin sheaths.
4. (a) Most of the grey matter in the brain is on the outside surrounding the white matter.
   (b) In the spinal cord the white matter is on the outside to allow linking of the CNS to sensory and motor neurons and the grey matter is on the inside roughly forming a shape like the letter H.
5. In a monosynaptic reflex there are only two neurons – a sensory neuron and a motor neuron, e.g. peripheral muscle reflexes such as the Achilles reflex. Whereas in a polysynaptic reflex arc there are one or more connector neurons in the circuit.
6. Cell A is a sensory neuron as it connects to the receptor cell and takes the message to the CNS. Cell B is in the CNS (spinal cord) and is therefore a connector neuron. Cell C is a motor neuron as it takes the message from the CNS to the effector cell.
7. The autonomic nervous system is part of the motor nervous system that controls the internal environment, e.g. involuntary muscles such as cardiac muscle and the smooth muscle in the walls of blood vessels and in the digestive tract.
8. The change in size of the pupil of the eye is a response to light stimuli. In bright light the pupil will constrict to reduce the amount of light entering the eye to avoid damage to the retina. In dim light the pupil will dilate to let more light through to the retina to aid vision and help the person see in darker conditions. Pupil size needs to be a reflex so that there is a fast and continuous response that adjusts to changing light conditions and does not need conscious thought cluttering the brain with constant decision making about pupil size.
9. C
10. A

20 The Structure of a Neuron

1. A nerve is a bundle of neuronal fibres.
2. A neuron is a nerve cell that transmits signals by electrochemical changes in their membranes.
3. The three main parts are: 1. The cell body, which contains the nucleus and other cell organelles for maintaining the cell. 2. The dendrites, which pick up the message using their extensive branches to increase the surface area of the 'receiving end' of the nerve cell and connecting to the axons of other neurons. 3. The axon, which conducts the message away from the cell body.
4. The synapse is the cell junction between a synaptic terminal and a target cell, which can be another neuron or an effector cell, such as a muscle cell.
5. (a) Part 1 = nucleus, part 2 = Nissl substance, part 3 = axon.
   (b) ER is found in the cell body. It is needed for protein synthesis as all neurons have a high level of metabolic activity. The high level is due to the massive surface area to volume ratio, which requires a large energy input and the need to produce electrochemical gradients, which also requires a large energy input.
6. Nodes of Ranvier are intervals along the axon where there is no myelin sheath.
7. The myelin sheath acts as an insulating layer against any leakage of the electrochemical impulse and also increases the electrical capacitance of neurons, and the faster the nerve impulse will travel.
8. Axons take the message to the next neuron. A large axon would be long and the presence of a myelin sheath will help speed the nervous impulse along the axon. A small axon would be a small neuron which would be able to quickly pass the message along the short distance to the next cell.
9. When the nerve impulse reaches the end of the axon, a neurotransmitter is released into the synapse. The neurotransmitter diffuses across the space and binds to receptors in the membranes of the dendrites of the next cell.
10. All three types of neurons transmit impulses in one direction. Sensory neurons transmit information about the internal and external environment to the CNS. Motor neurons transmit the impulse from the CNS to effector cells such as muscles or glands. Interneurons are located within the CNS and connect with other neurons.
11. Alcohol gives relaxation but intoxication affects motor performance and judgement. Sedative-hypnotics (e.g. barbiturates such as Valium) are brain depressants producing relaxation, euphoria and tranquillity. and are often used as sleeping pills and as anti-anxiety agents. Opiate analgesics (e.g. morphine, heroin) give pain relief and produce euphoria. Hallucinogens (e.g. LSD, mescaline, marijuana) alter thinking patterns and perception.

21 The Action Potential

1. An action potential is a rapid and short change in the membrane of an excitable cell.
2. The membrane potential is the difference in electrical charge (voltage) across the plasma membrane.
3. The usual resting potential for most animal cells is around 70 mV and is the normal condition of the plasma membrane. Inside of the cell is negative compared to the outside of the cell.
4. A membrane in the resting state is said to be polarised as the inside is negative compared to the outside.
5. The resting potential is maintained by a sodium-potassium pump which uses energy (ATP) to actively transport Na^+ out of the cell and K^+ into the cell.
6. An electrogenic pump is an ion transport protein that generates a voltage across a membrane.
7. The sodium-potassium pump is an electrogenic pump as it generates a voltage across the membrane. It pumps three sodium ions outside and two potassium ions inside thus making the inside negative compared to the outside creating the voltage across the membrane.
8. The nerve impulse is an action potential as it involves a temporary and local change in the resting potential.
9. At the beginning of a nerve impulse sodium ions move into the cell making the inside of the cell positive.
10. During depolarisation ion channels in the membrane open and allow Na^+ to move into the cell making the inside of the cell positive.
11. During repolarisation the change in permeability of the membrane allowing Na^+ to enter stops and K^+ channels open allowing K^+ to move outside. This causes the inside to become negative again.
12. To measure voltage changes during an action potential you can use one electrode inside the cell and one just outside the cell membrane.
13. The refractory period is the time after an action potential which lasts from 5 to 10 milliseconds when the axon is recovering and cannot send another impulse.
14.

Stage	Characteristic
Resting potential	More K^+ inside and Na^+ outside maintained by sodium-potassium pump and active transport.
Depolarisation	Na^+ channels open and Na^+ move inside the cell.
Repolarisation	K^+ channels open and K^+ moves outside cell.
Hyperpolarisation	K^+ continue to move out making the inside more negative than resting potential.

15. Part A on the graph shows the resting potential of around −70 mV and is the period just before the beginning of the action potential. Part B is depolarisation when the sodium ions are moving inside the cell.
    Part C is repolarisation when the sodium channels have closed and the potassium channels are open and the potassium ions are moving outside the cell. By the end of part C, some potassium channels will be closing.
    Part D is hyperpolarisation of the membrane when potassium ions continue to move out and the inside becomes more negative than the resting potential.
16. (a) X is sodium ions (moving inside the cell) and Z is potassium ions (moving outside the cell).
    (b) The impulse is travelling from left to right as the first action is sodium ions moving inside and then potassium ions move outside.

22 Crossing the Synapse

1. The synapse is the specialised junction between two neurons where the activity in one influences the activity in another.
2. The presynaptic neuron is the transmitting cell before a synapse and the postsynaptic neuron is the target cell at a synapse.
3. There are electrical and chemical synapses.
4. A neurotransmitter is a chemical released from the synaptic terminal at a chemical synapse that diffuses across the synapse to stimulate the postsynaptic terminal.
5. Neurotransmitters include acetylcholine, adrenaline and noradrenaline.
6. After a neurotransmitter has diffused across the synaptic cleft it will either start another action potential in the next neuron in the pathway or initiate a response in an effector.
7. Neurotransmitters are stored in vesicles at the end of the axon in the synaptic knob.
8. The nerve impulse can only go in one direction as the neurotransmitters are stored in the axon terminal and when released will diffuse to the postsynaptic cell.
9. The plasma membranes have ion channels that cross the bilipid layer. An action potential causes changes in the permeability of the membrane and these ion channels open allowing ions, e.g. Na^+ or K^+ to cross the membrane. The movement of these ions changes the electrical potential – voltage.
10. (a) At point X exocytosis is occurring with the release of a neurotransmitter from the terminal knob of the presynaptic neuron into the synaptic cleft.
    (b) The direction of the impulse would be from cell A to cell B (from left to right) in the diagram.

11. In direct synaptic transmission the neurotransmitter binds directly to an ion channel causing the channel to open. Whereas in indirect synaptic transmission the neurotransmitter binds to a receptor that is not part of an ion channel. This activates a signal transduction pathway which is slower but more long lasting than direct transmission.
12. Steps are: 1. When the impulse arrives at the synaptic knob it depolarises the terminal membrane causing calcium ion channels to open and calcium ions to flow into the terminal. 2. The rise in calcium ion concentration causes a few vesicles containing the neurotransmitter to fuse with the presynaptic membrane and the contents are released by exocytosis. 3. The neurotransmitter diffuses across the synaptic cleft and binds with a receptor protein of an ion channel in the postsynaptic membrane. 4. The channel opens and an inflow of sodium depolarises the postsynaptic membrane and if the threshold is passed an action potential is generated. 5. The neurotransmitter releases from the receptors and the channel closes. 6. Enzymes can remove the neurotransmitter from the synaptic cleft to stop continuous firing of impulses in the postsynaptic neuron. For example, the enzyme cholinesterase hydrolyses acetylcholine to choline and ethanoic acid. 7. The inactivated forms of neurotransmitters re-enter the presynaptic knob and are resynthesised as neurotransmitter and stored in the vesicles.
13. (a) Diagram shows a motor neuron.
    (b) It is a neuromuscular junction.
    (c) At a neuromuscular junction acetylcholine is released from the axon terminal of the presynaptic neuron by exocytosis. The acetylcholine receptors in the muscle membrane are activated by the acetylcholine and cause an ion channel to open. Sodium ions flow into the muscle fibre making the end plate potential more positive. The action potential spreads along the muscle membrane causing muscle contraction.
    (d) After the acetylcholine has activated the receptors it is rapidly destroyed by the enzyme acetylcholinesterase.
14. In the activated state the activation gate and inactivation gate of the sodium channel are both open and this allows sodium ions to flow inside the cell. Whereas in the inactivated state the inactivation gate is closed which prevents the sodium ions flowing into the cell.
15. B

23 Types of Neurotransmitters

1. Neurotransmitters are chemicals which diffuse across a synapse and bind to the postsynaptic cell to transmit the nerve impulse and trigger a response.
2. There are many different neurotransmitters, e.g. acetylcholine, epinephrine, norepinephrine, dopamine, serotonin and some amino acids act as neurotransmitters, e.g. glycine, glutamate and aspartate.
3.

Neurotransmitter	Effect
Acetylcholine	Usually excitatory and used by the spinal cord neurons to control muscles. Also used by many neurons in the brain to regulate memory.
Gamma-aminobutyric acid (GABA)	Major inhibitory neurotransmitter in the brain.
Dopamine	Usually inhibitory and produces feelings of pleasure when released by the brain reward system. Has multiple functions depending on where in the brain it acts.
Glycine	Inhibitory neurotransmitter used mainly by neurons in the spinal cord.
Noradrenaline	Usually excitatory bit is inhibitory in a few brain areas. Acts as a neurotransmitter and a hormone. In the PNS it is part of the fight or flight response while in the brain acts as a neurotransmitter regulating normal brain processes.
Serotonin	Is inhibitory in pain pathways and is involved in many functions, e.g. mood, appetite and sensory perception.

4. Steps in synaptic transmission: 1. An action potential travels down the axon to the synaptic terminal where neurotransmitter has been synthesised and stored in synaptic vesicles. The structure of the synaptic terminal varies with the type of postsynaptic cell (e.g. a synaptic knob where one neuron synapses another is simpler than a neuromuscular junction). The action potential depolarises the terminal membrane. 2. Depolarisation opens calcium channels and the presynaptic membrane and calcium ions diffuse into the terminal. 3. The increase in calcium concentration causes the synaptic vesicles to fuse with the terminal membrane. 4. Exocytosis releases the neurotransmitter into the synapse. 5. The neurotransmitter diffuses across the synaptic cleft and binds with the receptor site in the postsynaptic membrane. 6. (a) In an excitatory synapse, the receptor binds to a channel that allows both sodium ions and potassium ions to diffuse. This causes depolarisation and an action potential is initiated in the postsynaptic neuron. 6. (b) In an inhibitory synapse, the neurotransmitter only binds to the potassium ion channels and the action potential is inhibited.
5. Both the sympathetic and parasympathetic nervous systems are part of the autonomic nervous system which controls involuntary responses. The sympathetic nervous system prepares the body for immediate action, e.g. flight or fight, while the parasympathetic nervous system is involved with rest and digestion.
6. The sympathetic NS increases heart rate with acetylcholine released from preganglionic neurons and noradrenaline released from postganglionic neurons while the parasympathetic NS decreases heart rate with acetylcholine released from both pre- and postganglionic neurons.
7. A

24 Transmission of a Nervous Impulse

1. An excitable cell is any cell which can have an action potential generated and is able to be made active.
2. The threshold of a nervous impulse is the particular value of the membrane voltage that needs to be reached for an action potential to be started.
3. The all or none law states that if the stimulus is at or above the threshold it will start a full action potential. If the stimulus is below the threshold there will be no response.
4. The way we distinguish between a small and large stimulus is by the number and frequency of action potentials sent to the brain. If the brain receives many impulses along one nerve or a number of neurons are sending impulses, the brain interprets the information as a large stimulus. While few impulses are interpreted as a small stimulus.
5. (a) The diameter of the axon determines the amount of resistance it offers against the conduction of an action impulse. The narrower the axon the greater its resistance and hence the slower the transmission of the nerve impulse.
    (b) Myelin mainly consists of lipid and has a high electrical resistance. Along a myelinated axon there are points between the sheath cells known as nodes of Ranvier. The nerve impulse jumps from node to node along a myelinated axon rather than depolarising every bit of axon along the way and is thus faster than an action potential on an unmyelinated axon.
6. Demyelination of neurons means that there will be difficulty in transmitting a nervous impulse along a long axon and the neuron will become susceptible to damage. The action of T lymphocytes and destructive proteins such as cytokines against body cells can cause the death of the neuron which will stop nervous messages and impair movement, e.g. people with MS can have difficulty walking.
7. The giant squid was a suitable experimental animal as it has one of the largest and longest known axons. Hodgkin and Huxley needed large axons as they had to insert electrodes so they could measure the electrical potential across the membrane.

8. The action potential of the *M. pudica* plant is similar to an action potential of a neuron as both are generated by a stimulus. Mechanoreceptors in the plant generate the action potential and some action potentials in neurons are initiated by mechanoreceptors. Both action potentials involve the movement of ions from cells although the *Mimosa* involves the outflow of potassium and chloride ions while a neuron involves the outflow of sodium ions followed by an inflow of potassium ions.
9. C
10. D

25 Comparing the Endocrine System and the Nervous System

1.

Feature	Nervous system	Endocrine system
Nature of message	Electrochemical impulse.	Chemical messenger (hormone).
Mode of transmission	Conduction along a neuron and chemical transmission at synapse.	In the bloodstream and body fluids.
Effect	Changes membrane potential of affected cells, carries sensory and motor signals, is involved in consciousness, learning, memory, emotion and control of movements. Causes muscles to contract or glands to secrete.	Changes the rate of metabolic activity of affected cells. The hormone will reach every cell in the body that is contacted or near some part of the circulatory system.
Time for effect	Impulses produce their effect within a few milliseconds.	Hormones mostly have their effects over many minutes or over several hours or even longer.
Duration of effect	Effects of nervous stimulation are short lived and reversible.	Effects of endocrine stimulation are fairly long lasting.

2. Two interactions of endocrine glands and PNS can involve the autonomic nervous system (ANS): 1. Sympathetic nervous system controls glands, e.g. for the flight or fight responses which include the release of adrenaline from the adrenal glands. 2. Parasympathetic nervous system influences the functioning of the pancreas, e.g. insulin secretion.
3. D
4. C

26 Homeostasis

1. Homeostasis is the maintenance of a constant internal environment within narrow limits.
2. Homeostasis is necessary to maintain optimal metabolic efficiency.
3. The two stages of homeostasis are detecting changes from the stable state and counteracting or responding to changes from the stable state.
4. Variables controlled by homeostasis in 'higher' vertebrates include blood composition, e.g. blood sugar level, oxygen and carbon dioxide concentrations, pH and water potential; blood pressure, and in endotherms body temperature.
5. Endotherms use internal processes to keep a constant body temperature and hence they require extremely efficient homeostasis mechanisms to maintain this temperature.
6. In humans, homeostasis maintains body temperature at approximately 37°C, maintains pH of interstitial fluid within a tenth of 7.4 and maintains the sugar concentration of the blood at approximately 0.1%.
7.

Sense organ	Receptor	Stimuli detected
Eye	Rods and cones in retina	Light
Ear	Hair cells in cochlea	Sound
Tongue	Gustatory receptors in taste buds	Chemicals
Nose	Olfactory receptors in nasal cavity	Chemicals
Skin	Mechanoreceptor	Pressure, touch
Skin	Thermoreceptor	Heat, cold
Skin	Pain receptor	pain

8. Stimulus → receptor → sensory neuron → interneuron → motor neuron → effector → response.
9. In a control system there is: 1. A detector device that picks up the input, e.g. detects a change in temperature. 2. A control centre that has a 'set' required condition, e.g. a certain temperature such as body temperature. 3. An effector device functions to counteract the change to return the system to the set condition, e.g. to return the system to body temperature. 4. The return to the set condition is detected by a feedback mechanism and the new change becomes the input.
10. Endotherms use metabolic heat to regulate body temperature, e.g. birds and mammals.
11. D
12. A
13. C
14. A
15. C

27 Feedback Mechanisms

1. Feedback mechanism is when information about a change is monitored by the control centre, e.g. brain, which then causes another response.
2. Both negative and positive feedback monitor the changes and respond to these changes. They differ in that negative feedback will initiate a new response that is the opposite, counteracting the change, e.g. if the responses have caused the body temperature to rise with hair erector muscles causing hair to stand up, then negative feedback will cause the body temperature to drop with a response to cause the hair erector muscles to relax, making the hair lie flat. Positive feedback will reinforce and amplify the situation, e.g. during childbirth oxytocin is released to cause contractions of the uterine muscles and the pressure of the baby's head against the opening of the uterus causes the release of more oxytocin, and more contractions.
3. Both the air conditioner and thermoregulation in humans keep the temperature within a certain range using a negative feedback system. They both have a receptor, control centre and effector. The differences are in the nature of these components. As a receptor, the air conditioner has a thermometer while humans have two sets of temperature receptors – one in the skin and the other in the hypothalamus. For a control centre, the air conditioner has a thermostat which will send a command to turn on the air conditioner when the temperature is too high, while the hypothalamus controls body temperature and will send commands to blood vessels to dilate to increase blood flow through surface vessels, and to sweat glands to increase their production of sweat.
4. D
5. D

28 Homeostasis and Temperature Control

1. The terms warm blooded and cold blooded do not accurately describe temperature regulation. Humans are called endothermic as an endotherm regulates its body temperature by internal mechanisms while many ectothermic organisms can appear to be warm blooded, e.g. while basking in the sun or living in a hot environment using behavioural adaptations to maintain body temperature.

2. (a) The average core temperature is 37°C.
   (b) A constant internal environment is important for optimal metabolic efficiency – enzymes require specific temperature, pH, salt concentration etc for maximum rate of reaction.
3. Body temperature can be measured with a thermometer either orally or rectally. Body temperature measured rectally is usually around 0.5°C higher than an oral measurement.
4. A naked person can maintain body core temperature in dry air when exposed to temperatures from 12°C to 54°C.
5. Core body temperature rises during exercise and will vary if the person is exposed to extreme ambient temperatures. The heat produced by hard exercise can raise core temperature to 39°C to 40°C or drop to 35°C if exposed to extreme cold.
6.

Type of heat transfer	Description
Conduction	Transfer by direct contact.
Convection	Movement of heated particles in a liquid or gas, e.g. heated particles rise and are replaced by cooler particles.
Radiation	Movement of energy in the form of parts of the electromagnetic spectrum, e.g. infra-red (heat) without direct contact between heat source and receiver.
Evaporation	Transfer of energy to particles in the liquid state so that the liquid becomes a gas which is removed.

7. The skin is highly important in losing heat to the surroundings to help maintain core body temperature.
8. Evaporative cooling is the transfer of energy to particles in the liquid state so that the liquid becomes a gas which is removed and heat is taken from the body causing cooling.
9. In the body most heat is produced as a by-product of metabolism.
10. Vasoconstriction is the narrowing of blood vessels, e.g. arterioles while vasodilation is the widening of blood vessels, e.g. arterioles.
11. Vasoconstriction and vasodilation control the diameter of blood vessels, e.g. arterioles. Since arterioles provide the main resistance to blood flow a change in the diameter of an arteriole will greatly influence blood pressure and the amount of blood flowing to an area. As a person becomes hotter vasodilation allows more blood to flow to the capillaries in the skin, the person looks flushed and red and there is an increase in the amount of heat lost by radiation, conduction and convection. In vasoconstriction the arteriole narrows, restricting blood flow to the skin surface and heat is retained.
12. Heat is infra-red radiation with a wavelength of 5 to 20 micrometres and is a part of the electromagnetic spectrum of radiation.
13. C
14. C

29 Modelling Human Thermoregulation

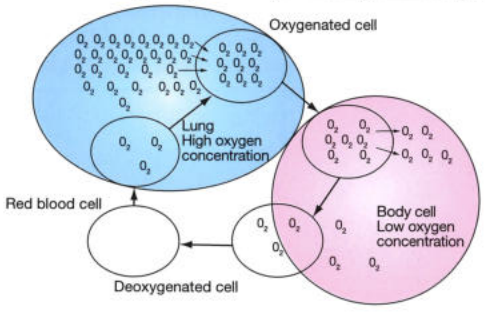
1. Normal human core body temperature is 37°C.
2. Hypothermia is a condition when the body's core temperature drops below the level needed for normal metabolism and body functioning, e.g. in humans the core body temperature drops below 35°C.
3.

Heat transfer with the environment	
Gain to human	**Loss from human**
Direct sunlight radiation. Reflected sunlight. Radiation from the sky. Radiation from the ground.	Radiation of heat. Respiratory evaporation/convection of heat. Sweating. Conduction to touched objects. Urination. Conduction to the ground.

4. Heat is produced as a by-product of metabolism, e.g. in deep organs such as the liver, brain and heart with skeletal muscles producing heat during exercise.
5. The rate of heat production is determined by the basal chemical reactions occurring in cells and any additional reactions, e.g. due to muscle activity and/or the action of some hormones such as thyroxine and growth hormone.
6. A fever involves a rise in core body temperature due to a change in the set point for core body temperature by the heat control centre of the brain and hypothalamus, e.g. in response to a bacterial or viral infection. Hyperthermia involves a rise in core body temperature, e.g. due to heat stroke with no change in the set point for core body temperature.
7. Firefighters are exposed to many dangers besides the heat of the fire, e.g. toxic fumes, sharp objects, uneven ground, slippery surfaces, biological fluids, spilt chemicals and electricity. This means their uniforms and equipment need to be correctly designed to protect the firefighter from injury or harm. Their tunic needs to be fire resistant, able to maintain its structural strength after fire exposure and able to resist cuts and tears. The tunic also needs to provide protection without increasing the metabolic heat stress of the firefighter. They also need flash hoods, boots, helmet, gloves, breathing apparatus, wet weather clothing, cold and extreme climate clothing and bushfire jacket.
8. Computer modelling has been used to design clothing for – military chemical suits, industrial protective clothing, space suits and clothing for firefighters.
9. When people have to work and live in extreme environments, e.g. space stations, submarines, extreme cold or extreme heat the scientists and engineers who design the buildings need to run computer simulations before construction begins to make sure the internal environment will protect and sustain human life. The scientists can run many simulations, changing the set value of variables to show how the human body will respond to the conditions and to determine the required safety regulations for each situation.
10. A
11. D
12. B

30 Homeostasis and Blood Composition

1. All blood, whether in arteries or veins is similar in that it contains red blood cells, white blood cells and plasma. A major difference is that the blood in arteries going to the small intestine carries less glucose, amino acids, mineral salts, vitamins and some products of fat digestion than the blood leaving the small intestine in veins. This is due to the absorption of the products of digestion in the small intestine from the gut to the capillaries in the villi.
2. The concentration of the blood entering the lungs is higher in carbon dioxide and lower in oxygen than the blood leaving the lungs. This difference means that oxygen in inhaled air is exchanged for carbon dioxide in exhaled air.
3. Excretion is the removal of metabolic wastes from the body. The lungs can be considered an organ of excretion as they remove carbon dioxide produced in respiration. The kidneys are an organ of excretion as they remove nitrogenous waste products from the blood.
4. (a) Cellulose is not easily digested by humans and is need for fibre or roughage. Both starch and sugar are digested into simple sugars which can be changed into glycogen for storage in the liver or the sugar is used in respiration to release energy.
   (b) Cellulose does not change blood composition as it is not absorbed into the bloodstream. Starch and sugar cause a change in blood composition as they are absorbed into the blood in the small intestine.
5. Water is lost by evaporation, sweating, urinating and exhalation.
6.



7. (a) Highest concentration of oxygen is in the pulmonary vein.
   (b) Highest concentration of carbon dioxide is in the pulmonary artery.
   (c) Highest concentration of glucose is in the hepatic portal vein.
   (d) Highest concentration of urea is in the renal arteries.
8.

Substance	From	To
Oxygen	Lungs	All body cells
Carbon dioxide	All body cells	Lungs
Glucose	Small intestine	All body cells
Urea	Liver	Kidneys

9. B
10. D

31 The Hypothalamus and Homeostasis

1. The hypothalamus is a part of the brain below the thalamus.
2. The hypothalamus processes and sorts sensory information and is responsible for activities associated with sex, hunger, thirst, pleasure, pain and anger. It has sensors for thermoregulation and produces the hormones ADH and oxytocin and releasing hormones.
3. The hypothalamus produces ADH and oxytocin.
4. The hypothalamus needs to be well supplied with blood vessels to supply oxygen and glucose for respiration and the constant metabolism within the cells of the hypothalamus, to remove waste products, to provide the stimulus material for the sensors within the hypothalamus, and to take the hormones and releasing factors to the pituitary gland.
5. The pituitary gland lies below the hypothalamus and works with the hypothalamus in controlling the release of hormones linking the nervous system and the endocrine system.
6.

Type	Hormone	Function
Hormone	ADH	Increases water absorption into the blood by the kidneys.
	Oxytocin	Variety of processes, e.g. contraction of uterine muscles during labour, causes mammary glands to eject milk during lactation.
Releasing hormone	CRH	Causes anterior pituitary gland to secrete ACTH and beta-endorphins.
	TRH	Stimulates anterior pituitary to release TSH and prolactin
	GnRH	Stimulates anterior pituitary to release FSH and LH.
	GHRH	Stimulates anterior pituitary to secrete GH.
	PRH	Stimulates anterior pituitary to produce prolactin.
Inhibiting hormone	Dopamine	Inhibits the release of prolactin.
	Somatostatin	Inhibits the release of several hormones, e.g. GH, TSH from anterior pituitary.

7. The releasing and inhibiting hormones are secreted near capillaries at the base of the hypothalamus.
8. Hormone X would be thyrotropin releasing hormone (TRH) released by the hypothalamus. Hormone Y would be thyroid stimulating hormone (TSH) released from the anterior pituitary.
9. C
10. D
11. A
12. B

32 Changing Metabolic Activity

1. Metabolism is the sum of all chemical reactions within a body.
2. The metabolic rate is the total amount of energy an animal uses in a unit of time.
3. The basal metabolic rate (BMR) is the metabolic rate of a resting, fasting and non-stressed endotherm.
4. Energy is needed for many functions, e.g. for muscle activity, secretion by glands, maintaining membrane potentials of nerve and muscle fibres, synthesis of substances in cells and the absorption of foods from the gastrointestinal tracts.
5. Energy for use in the cell is in the form of adenosine triphosphate (ATP).
6. Thyroxine secreted by the thyroid gland, e.g. high levels of thyroxine increase metabolic rate by up to 50% to 100% or a total lack of thyroxine can decrease the metabolic rate by 40% to 60%.
7. Growth hormone can increase metabolic rate 15% to 20% due to stimulation of cell metabolism.
8. Male sex hormone, e.g. testosterone can increase metabolic activity by 10% to 15% due to anabolic pathways that synthesise new proteins increasing skeletal muscle mass.
9. (a) ATP consists of a ribose sugar, adenine and three phosphate groups that are attached to the ribose.
   (b) When ATP releases energy the high energy bond holding the last phosphate group to the ribose is broken and ATP reverts to ADP.
10. There is a high synthesis of ATP during exercise and physical activity.
11. During sleep the metabolic rate drops slightly due to less activity of CNS and skeletal muscles.
12. The thermogenic effect of food refers to the need for energy and the release of heat when food is eaten and has to be digested, absorbed and energy reserves stored in the body.
13. A
14. D
15. B

33 Homeostasis, Excretion and Osmoregulation

1. Osmoregulation is the way organisms regulate solute concentrations and balance the gain and loss of water.
2. Nephron – functional unit of the kidney.
   Filtration – process which allows small particles to pass through filter but retains larger sized particles.
   Reabsorption – process in the kidney where particles which have been filtered out of the blood are returned to blood.
   Secretion – process where materials are released from one area to another.
   Filtrate – is the fraction that passes through the filter.
   Osmoregulation – the control of water and salt levels in the body to maintain homeostasis.
   Solute – the material that is dissolved in the solvent.
   Ureter – tube passing from the kidney to the bladder.
   Urethra – the tube passing from the bladder to the outside of the body.
3. Active transport uses energy to move materials across membranes, such as during the secretion of some ions, e.g. H^+, into the convoluted tubules of the kidney. Passive transport does not require an energy input, such as during the osmosis of water from the convoluted tubules back into the capillaries.
4. (a) Homeostasis is the maintenance of a constant internal environment.
   (b) Renal artery → glomerulus → Bowman's capsule → convoluted tubule → collecting duct → kidney pelvis → ureter → urinary bladder → urethra → outside body.
   (c) On a hot day with little fluid intake a person could dehydrate. The kidney will respond by increasing the amount of water reabsorbed in the renal tubules so that a concentrated urine is formed.
   (d) Red blood cells should not be present in urine as they are too large to pass through the pores in the epithelium of Bowman's capsule. If RBCs are present in the urine it indicates damage to the nephron and the person should seek immediate medical attention.
5. (a) Brown region is the renal cortex which consists of glomerulus and Bowman's capsule which is where the blood is filtered. Pinkish-red region is the medulla which consists of renal tubules and collecting tubules and is where reabsorption occurs. White region is the pelvis and consists of collecting ducts and is where urine collects to be passed to the ureter. The thin tube is the ureter.

(b) Students would need to wear gloves to protect their hands and prevent possible infection. Students would have to correctly dispose of the dissected kidneys, making sure to remove all pieces of kidney from the lab as the tissue will putrefy and could transmit disease if it became infected.

(c) The mammalian kidney is able to concentrate urine and conserve water when necessary, while the kidney of freshwater fish is mainly osmoregulatory. Freshwater fish cannot form a concentrated urine – they have cilia in the nephrons to push large amounts of very dilute urine from the body. Both freshwater fish and mammals are able to reabsorb salts from the filtrate.

6. (a) Filtration is the movement of small molecules such as glucose, amino acids, water, salts and urea from the glomerulus into Bowman's capsule. The movement is mainly due to increased blood pressure. The filtrate is the same as blood except it does not contain red blood cells or proteins. Reabsorption is the movement of needed nutrients back into the bloodstream from the renal tubules. All of the glucose and amino acids are reabsorbed and some salts and water is reabsorbed, depending on the needs of the body. Urea is not reabsorbed. In this way the kidney is regulating the amount of salt and water in the body, as well as excreting urea.

(b) Reducing the intake of proteins will help to reduce the pressure within the kidneys and hence protects the kidneys – if the blood has a high concentration of protein compared to the lack of protein in the filtrate, there will be a high 'suction pressure' drawing water by osmosis from the blood. Proteins are needed in the diet to provide amino acids, which are the building blocks for new cells and enzymes. If the protein intake is insufficient the patient will have difficulty maintaining cell growth, repair and maintenance.

7. Many of the waste products, especially the nitrogenous wastes such as ammonia and urea, are toxic and therefore the body needs a method to quickly remove these products so that normal chemical reactions can function correctly, catalysed by enzymes working at optimum efficiency.
8. Diffusion is slow and is effective over short distances where there is a large concentration gradient. Mammals are complex multicellular animals and the processes of diffusion and osmosis would not be able to remove waste products, especially the toxic nitrogenous waste products, quickly enough to prevent toxic poisoning. Therefore a circulatory system is needed to deliver waste products to the excretory system, which in turn removes the waste products from the blood.
9. D
10. A
11. C
12. D
13. B
14. C
15. B

34 Osmoreceptors and Water Balance

1. An osmoreceptor is a sensory receptor that detects changes in osmotic pressure.
2. Osmoreceptors are mainly found in the hypothalamus in endotherms.
3. Baroreceptors are volume receptors that respond to pressure changes.
4. Baroreceptors are located in the walls of the right atria and large arteries and veins.
5. The extracellular fluid osmolarity can increase due to an increase in the concentration of sodium ions in the plasma.
6. The excretion of water and solutes operate in a way independently of each other when forming urine as concentrated urine has solutes in excess of water while dilute urine has water in excess of solutes.
7. (a) ADH is produced by the hypothalamus.
   (b) ADH is stored in the posterior pituitary gland.
8. The thirst mechanism is the stimulation of the thirst centre in the hypothalamus for the need to drink water when there is decreased blood volume and an increase in the extracellular fluid osmolarity and sodium concentration.
9. Aldosterone is produced by the adrenal cortex in the adrenal gland.
10. Even in a dehydrated person there are excess solutes that are eaten in the food intake or produced by metabolic processes that need to be excreted. Thus the kidneys still need to produce urine and some water is lost in urine even though the person is dehydrated.
11. Hormone replacement therapy is a treatment given when a gland is not producing enough of a particular hormone.
12. Both aldosterone and ADH help regulate water and salt levels in blood. Aldosterone acts mainly on the loop of Henle and distal tubule, causing sodium to be reabsorbed and excreting potassium. ADH acts mainly on the collecting ducts, causing increased membrane permeability and increased water retention.
13. Symptoms include drop in blood pressure and low blood volume due to excessive losses of sodium and water in urine. This can lead to changes in the heart rhythm which can be fatal.
14. Aldosterone hormone replacement therapy would help counter osmoregulatory problems. It would stimulate the reabsorption of sodium, which addresses the salt craving, and the increased sodium level would cause more water to be reabsorbed due to osmosis. Sodium deficiencies lead to muscle cramps, mental apathy and reduced appetite which means these symptoms should decrease.
15. (a) Receptors are the osmoreceptors in the hypothalamus; the control centre is the pituitary gland; and the effector is the tubules and collecting ducts of the kidney.
    (b) Very little ADH would be released after drinking water and eating food as this would lower this blood osmolarity. Little water would need to be reabsorbed and a dilute urine would be released.
    (c) Alcohol dehydrates the body as it inhibits the release of ADH and causes an excessive loss of water in the urine.
16. B
17. C

35 Renal Dialysis

1. (a) Oedema occurs when tissues are swollen with fluid.
   (b) Pulmonary oedema is when fluid accumulates in the lungs.
2. Dialysis is used in emergency situations: when potassium levels are high and there is fluid overload and pulmonary oedema; if there is a need to remove medications or toxins from the blood; or if there is long term renal failure.
3. (a) Dialysis is a process which passes blood over a semipermeable membrane so the solutes, e.g. waste products, diffuse through the membrane and are removed from the blood.
   (b) One advantage of haemodialysis is that it prolongs life when the kidneys have failed. One disadvantage is that each treatment requires two needle sticks and for three times a week each patient must spend four hours at a dialysis session.
   (c) Both the kidney and renal dialysis filter urea from the bloodstream. The kidney filters all substances from the blood and then needed substances are reabsorbed. Thus not only is urea removed, but the kidney also regulates water levels and controls the concentration of many other substances, e.g. sodium, potassium, phosphate levels. The dialysis machine only filters substances with no reabsorption, so people using dialysis have to watch their diet, restricting their intake of high potassium foods such as leafy vegetables and fruits and high phosphate foods such as beans and peas, as dialysis does not control the level of these substances.
   (d) For blood to be entering his urine, the glomerulus and Bowman's capsule could be damaged so that red blood cells are now filtering across to the filtrate instead of staying in the bloodstream.
4. Heparin, an anticoagulant, is administered to stop the blood clotting in the dialysis tubing.

5. The body's buffer system, which maintains homeostasis, is maintained used a dialysate, which is a solution of all the extracellular ions such as bicarbonate and acetate.
6. Urea, uric acid and the potassium, phosphate and sulfate ions are considered wastes and need to be removed from the blood. The hydrogen carbonate is needed to maintain correct pH levels and the glucose is needed for energy requirements.
7. A
8. D

36 Homeostasis and Gas Concentrations

1. Oxygen, carbon dioxide and nitrogen are transported in the blood.
2. (a) Gas exchange is the uptake of molecular oxygen from the environment and the release of carbon dioxide to the environment.
   (b) Gas exchange occurs in the lungs in the alveoli.
3. Gas exchange occurs by diffusion. Capillary walls are thin allowing easy diffusion and gas exchange while artery walls are thick which hinders easy diffusion.
4. Pressure is directly proportional to the concentration of the gas molecules. The difference in pressure creates a concentration gradient that causes diffusion and gas exchange.
5. (a) When blood arrives at tissues it has a higher partial pressure oxygen (P_{O_2} = 13.3) and a lower partial pressure carbon dioxide (P_{CO_2} = 5.3) than the tissues (P_{O_2} = 5.3) and (P_{CO_2} = 5.8).
   (b) As a result of this difference in gas concentration gas exchange occurs with oxygen diffusing from the blood into the cells in the tissues and carbon dioxide diffusing out of the cells in the tissues into the bloodstream.
6. The rate of diffusion and gas exchange depends on – 1. Thickness of the membrane. 2. The surface area of the membrane. 3. The richness of the blood supply. 4. The concentration gradient (or pressure difference across the membrane).
7. The rate of diffusion and gas exchange depends on the distance to be travelled and the thickness of the membrane. A thickening of the respiratory membrane in a pulmonary disease will reduce gas exchange causing breathing difficulties and problems obtaining sufficient oxygen especially during exercise and problems excreting carbon dioxide.
8. When the alveoli walls break down in panacinar emphysema merging into large spaces there is a reduction in total surface area for diffusion and gas exchange. Since efficient diffusion and gas exchange require a large surface area, the reduction leads to problems obtaining oxygen and excreting carbon dioxide and maintaining efficient metabolism.
9. Carbon dioxide (CO_2) is a waste product of respiration and needs to be excreted as it forms carbonic acid (H_2CO_3) in water. Large amounts of carbonic acid form hydrogen ions (H^+) and hydrogen carbonate ions (HCO_3^-) which lowers the pH of the blood and affects the ability of enzymes to function efficiently.
10. In the blood carbon dioxide is 7% dissolved in plasma, 23% as carbaminohaemoglobin combined with haemoglobin in the red blood cells and 70% as the hydrogen carbonate ion in plasma.
11. Oxygen combines with haemoglobin in red blood cells to form oxyhaemoglobin with 98% oxygen being transported as oxyhaemoglobin in the bloodstream. Oxygen also travels in solution in plasma water (2% dissolved in plasma).
12. Breathing is an involuntary action though there is some conscious control, e.g. when speaking, singing, fearful, crying or holding your breath for a limited period of time.
13. The main control of gas concentration is by the medulla oblongata which has the respiratory centre with neurons that control the rate and depth of breathing.
14. Central chemoreceptors are on the surface of the medulla oblongata and peripheral chemoreceptors are in the arch of the aorta and carotid bodies.
15. An increase in P_{CO_2} causes peripheral chemoreceptors and central chemoreceptors to send impulses to the respiratory centre which increases the rate and depth of breathing.
16. Temperature can affect breathing, e.g. a person with a fever will have an increased metabolic rate and increased respiration while metabolism is lower when a person has hypothermia.
17. To increase the oxygen diffusing capacity, e.g. with increased pulmonary blood flow and faster breathing rate during exercise, the pulmonary capillaries around the alveoli have additional dilation and dormant pulmonary capillaries become open.
18. C
19. A

37 Infectious Disease

1. A disease is any condition that impairs or interferes with the normal functioning of the body.
2. Infectious diseases differ from other diseases, e.g. genetic and lifestyle diseases in that they are caused by the invasion by a pathogen and can be transmitted from one host to another.
3. In the past infectious diseases have affected human populations by causing the death of many individuals which can affect the politics and economics of an area, e.g. by the death of prominent citizens, leaders, insurgents. In the 14th century nearly one third of the world's population died from bubonic plague – the Black Death which is an infectious disease caused by the bacteria *Yersinia pestis*.
4. Diseases can be classified in many ways, e.g. by the major categories of infectious disease and non-infectious disease or by the body part/system that is mainly affected, e.g. cardiovascular diseases, heart, lung and other organ diseases or endocrine system diseases or by the way the disease is acquired, e.g. genetic diseases, injuries and environmental diseases.
5. The invention of the light microscope and developments in its design to improve resolution and magnification enabled scientists to observe micro-organisms and then to link particular organisms to particular infectious diseases. For example, in 1845-6 Louis Pasteur showed that a mystery disease that threatened the silkworm industry was caused by micro-organisms that were only visible under a microscope and only found in the tissues of diseased silkworms, moths and eggs.
6.

Technology	Used by	Scientific understanding of infectious disease
Light microscope	Anton van Leeuwenhoek	Observed micro-organisms under the microscope which led to other scientists searching for microbes in different substances.
Swan neck flasks	Louis Pasteur	Showed microbes caused decay giving rise to the germ theory and refuting the theory of spontaneous generation.
Petri dish with agar jelly	Robert Koch	Identified the causative pathogens for tuberculosis and cholera and proposed a systematic method that will identify the pathogen causing a particular infectious disease.

7. A

38 Louis Pasteur

1. Pasteur was a chemist who discovered that chemicals with the same composition could have mirror-image structures.
2. Pasteur was asked to help the wine maker who was the father of one of his chemistry students.
3. The current belief was that fermentation was a straightforward chemical breakdown of sugar to alcohol caused by destabilising vibrations.
4. The three clues are: 1. Normal wine has round, budding yeast cells while sour wine has additional rod-like micro-organisms. 2. Other organic compounds are present in wine. 3. Other compounds are asymmetrical.
5. Pasteur concluded the yeast caused fermentation and contaminating micro-organisms made the wine sour.
6. Germ theory states that infectious diseases are caused by germs (micro-organisms).
7. Pasteur discovered that micro-organisms caused disease.
8. Pasteurisation involves heating the substance to 50°C to 60°C and then rapidly cooling it. It kills microbes so the substance will not go sour.

9. Spontaneous generation is the sudden appearance of life forms from decaying matter.
10. The control is the flask with the straight neck. It is used for comparison to show what happens when air with dust and germs can enter the boiled broth.
11. The broths were boiled to kill any micro-organisms present before the experiment began.
12. The swan neck trapped the dust and germ laden air so that the germs could not reach the broth.
13. Anthrax is a fatal disease in sheep and cattle.
14. Robert Koch.
15. Pasteur diluted one drop of blood containing anthrax bacteria and tested the culture on healthy organisms. He then continued to dilute the culture and found that the final culture was just as active as the first. This showed that the bacteria could survive the dilution and must cause the disease.
16. Pasteur tried oxidation and ageing techniques to make a vaccine.
17. Pasteur's work with the anthrax vaccine was challenged by a well known veterinarian and so Pasteur conducted a public trial with 50 sheep. The media published the results.
18. A
19. B
20. C
21. D

39 Robert Koch

1. Anthrax is a disease in sheep and cattle causing boils, oedema and quick death.
2. Koch identified the bacteria which causes anthrax.
3. Koch developed fixing and staining techniques and grew pure cultures on a solid gel.
4. Petri invented the Petri dish on which cultures could be grown.
5. Koch's postulates: 1. The organism believed to be the cause of the disease must always be present when the disease occurs. 2. The organism must be isolated from the host and grown in pure culture. 3. Organisms from the pure culture, when inoculated into healthy, suitable, susceptible hosts must produce the disease. 4. The organism must be reisolated, grown in pure culture and compared with the organism first injected.
6. Nobel Prize in Physiology or Medicine 1905.
7. For any infectious disease, scientists must examine the affected hosts and find a pathogen present in each diseased individual. The pathogen is isolated, cultured and introduced to a healthy individual which should develop the same disease and symptoms. The pathogen is then reisolated, compared with the original and conclusions can be made about the cause of the disease.
8. (a) The volume of circumstantial evidence shows that *Treponema pallidum* causes syphilis. The microbe has been identified in every case of the disease.
   (b) No, the procedure still applies even if the pathogen cannot be grown on an agar plate.
9. Robert Koch was interested in finding the cause of anthrax, which killed many sheep and cattle. He isolated the bacteria from the blood of dying animals.
10. A
11. D
12. A
13. B
14. C
15. B
16. D
17. C
18. C
19. C
20. D

40 Pathogen Transmission

1. A pathogen is an organism, virus, viroid or prion that causes a disease.
2. A parasite is an organism that benefits by living on or in a host which is harmed.
3. Pathogens can come from the external environment, e.g. entering the body from the air, with ingested food and water, through wounds or cuts in the skin or they can be endogenous and originate from within an organism, tissue or cell.
4. The communicability of an infectious disease depends on: 1. The way the pathogen leaves its host. 2. How the pathogen enters a susceptible person.
5. Direct contact involves touching the affected area of an infected person, whereas indirect contact can be through: inhalation of bacterial spores; food and water; and vector transmission through other organisms that carry the pathogen from one person to another.
6. There are many diseases that can be transmitted by both direct and indirect contact, e.g. measles is caused by the measles virus and is spread by contact with saliva or nasal secretions of an infected person but can also be spread through contact with surfaces that have droplets from coughs and sneezes from an infected person.
7. Airborne transmission involves inhalation of bacterial spores, or droplets containing pathogens from coughing, sneezing or speaking. Airborne diseases include chickenpox, common cold and diphtheria.
8. Pathogens can be transmitted in food or water when a person eats or drinks contaminated food. This can occur with lack of proper sewage treatment of water supplies, poor hygiene or use of sewage as a fertiliser. Diseases include *Cryptosporidium* infection, *Giardia* infection, hepatitis A, meningitis (viral) and worms.
9. A disease vector is an organism that carries the pathogen from one person to another or from an infected animal to another person.
10. (a) Arthropods such as houseflies and cockroaches carry pathogens on their feet and bodies and contaminate food and water, e.g. houseflies transmit anthrax, typhoid and cholera.
    (b) Many different mammals can transmit pathogens to infect humans, e.g. dogs transmit rabies and hydatid tapeworms and pigs transmit pork tapeworm.
    (c) Birds – certain birds can transmit avian influenza, e.g. water birds and parrots can transmit psittacosis.
11.

Stage	Description
Infection	Pathogen enters the body.
Incubation	Pathogen reproduces increasing in numbers but the host does not show symptoms.
Host shows symptoms	Symptoms of infectious disease appear, e.g. fever, headache, pain, rash.
Crisis	The effect on the host is at its most severe stage from which the host either recovers or dies.
Convalescence	The host recovers from the infection and the symptoms disappear.

12. There are many procedures that require sterile conditions, e.g. all surgeries, caring for wounds, having injections, dental work and tattooing. Sterile techniques include washing with soap and running water, keeping work surfaces dry and clean, only touching the outside wrapper with bare hands, wearing gloves, wearing a mask over nose, mouth and hair and covering your mouth when you cough or sneeze.
13. D
14. A
15. D
16. B

41 Experiment – Microbes in Food and Water

1. Aseptic or sterile techniques involve the use of certain procedures to prevent contamination of cultures and to protect workers from infection. For example, all work areas are swabbed with alcohol before use and all specimens are properly labelled. When opening a Petri dish, the lid is only lifted at a small angle and you do not talk while the lid is up or breathe onto the dish.
2. Pure culture: growing only one type of organism. Colony: growth of one species of bacteria that is large enough to be seen with the naked eye. Streaking: method of inoculating a plate with micro-organisms.
3. Air contains many spores of bacteria and fungi that can enter when the plate is exposed.

4. The control would be used for comparison to show the conditions that could be expected for all plates. Since the control had one bacterial colony, it can be expected that one bacterial colony on each plate was due to experimental technique rather than due to the substance being tested.
5. Full nutrient agar should allow the maximum growth of different types and number of bacteria colonies.
6. The most 'healthy' sandwich was the homemade sandwich, as it had fewer different types of bacteria and fewer total numbers of bacteria.
7. Contamination comes from lack of cleanliness, for example lack of cleaning of the bread board, the knife, or the hands and coughing on the food. Cheese also has bacteria present.
8. The bottled water was the most healthy as it had fewer types of bacteria and fewer total numbers of bacteria.
9. The water may become contaminated due to direct contact with the pipes, for example people putting their mouths on bubblers, touching the nozzle of the tap in the laboratory, or from airborne particles from coughing or sneezing.
10. The students could draw a graph to show the differences between each source.
11. Identification involves techniques and equipment not necessarily available in school laboratories. Some of the bacteria may be pathogenic which would require additional precautions to prevent infection.
12. When working with micro-organisms it is essential to wash your hands well with soap, antiseptic and water before leaving the laboratory to prevent the spread of a pathogen and possible infection of other people. Another safe work practice is to wear a laboratory coat and closed-in footwear, which also helps prevent the spread of pathogens onto clothing or the body. Do not open the plate after it has grown micro-organisms.
13. *E. coli* indicates the amount of faeces pollution present in the water. Since faeces may also contain pathogens, such as those responsible for cholera, dysentery and typhoid, the *E. coli* count can indicate the likelihood of pathogens. *E. coli* is much hardier and very resistant which means its absence infers pathogens are not present.
14. Pasteurisation involves the rapid heating and then cooling of milk, destroying the majority of the bacteria. If the pasteurised milk is kept refrigerated it will last longer and contain less bacteria.
15. During secondary treatment, drinking water is filtered to remove particles of several sizes and the passing of the water through sand beds or charcoal is effective in removing nearly all remaining microbes. In tertiary treatment, chlorine is added to kill bacteria. The chlorine is more effective on bacteria than on viruses and protozoa. Water samples are tested in laboratories for the presence of micro-organisms to ensure the effectiveness of these treatments.
16. Treatment of drinking water is important as it decreases the spread of pathogens and infectious diseases.
17. To prevent the spread of cryptosporidiosis people should practise good personal hygiene and not drink untreated water from rivers and dams. Those with diarrhoea should not enter public swimming pools or prepare food for others.
18. A 'closed' water catchment is not open for public use and so protects the area from faeces contamination from humans, domestic animals, livestock, and agriculture. This protects people from an outbreak of infectious disease.
19. D
20. A
21. C
22. B

42 Water Treatment

1. Diseases caused by pathogens being transmitted through water include cholera (*Vibrio cholerae*), typhoid fever (*Salmonella typhi)* and dysentery (*Shigella* spp).
2. *E. coli* is found in the intestines of animals which means it is passed out of the body in stools. A study of *E. coli* abundance gives an indication of the amount of sewage pollution present in the water supply. It is important to monitor the amount of *E. coli* pollution as high levels can cause illness, e.g. bloody diarrhoea, and even acute kidney failure in children under 5 years old.
3. D
4. C
5. A

43 Health, Disease and Cleanliness

1. Health is defined as a state of physical, mental and social wellbeing. Disease is defined as any condition that impairs normal living processes.
2. Health is often considered to be the absence of disease. However, a physically fit person in their teens going through social traumas has issues which are affecting their health, hence the need to consider not only a person's physical state but also their mental and social wellbeing.
3. (a) Gene expression controls protein production, for example enzyme manufacture, which will then control the metabolic processes for growth, repair and maintenance of a healthy body.
   (b) Mitosis is cell division which is needed for growth, repair and maintenance. Uncontrollable cell division causes cancers.
   (c) Cell differentiation occurs when cells begin to specialise into muscle cells, blood cells and so on, allowing specialised tissues to carry out particular jobs in a multicellular organism.
4. Leviticus (third book of the Old Testament) sets down laws concerning health issues. For example, the law for skin diseases (leprosy was prevalent) dictated that those declared unclean, with boils or inflammation, be isolated for seven days. Regulations governed the purification ritual for people cured of skin diseases. There were also laws for cleaning houses to remove mildew, dealing with unclean bodily discharges and protection of water and food supplies.
5. (a) Infectious disease is caused by pathogens that invade the body. Infectious diseases are contagious and are passed from one person to another. Non-infectious diseases are not caused by a pathogen; they can be caused by heredity, nutrition, physiological malfunction, environment and chemicals. They cannot be easily passed from one person to another (except by heredity).
   (b) Direct contact involves touching the infected area, for example, sexual contact and kissing. Indirect contact can occur through means such as ingestion of food or water; inhalation of fungal or bacterial spores; contact with contaminated articles such as toys, drinking utensils, bedding, ground and soil; and non-sterile medical procedures.
6. As infectious diseases are caused by pathogens, control measures require cleanliness to prevent pathogens from entering and then multiplying in food, water or on or in your body, and to prevent the transmission of the pathogen fron one person to another. Examples include washing hands before eating food, washing and cooking food properly, effective sewage treatment, providing uncontaminated water, not coughing over food, and covering hair and open sores when preparing food.
7. A pathogen is a parasite that causes disease, living on or in a host. It can be microscopic or macroscopic.
8. Pathogens can be found in air, water, soil, contaminated articles, and non-sterile medical and nursing procedures or injections, and can transfer from one person to another by direct or indirect contact.
9. If the beta-haemoglobin gene has a mutation with thymine instead of guanine at one particular location, the person suffers from the disease thalassaemia major which results in abnormalities such as severe anaemia and growth retardation.
10. If either the proto-oncogenes (genes which stimulate cell growth and cell division) or tumour suppressor genes (which slow down or stop cell growth and cell division) are damaged or mutated, cancers can form due to the uncontrolled cell replication.
11. Multicellular organisms need specialised cells to carry out particular functions to allow the whole body to survive, for example, red blood cells to carry oxygen, white blood cells for defence against infection, and neurons to carry messages.
12. In leukaemia, uncontrolled mitosis leads to the formation of too many white blood cells, which crowd out the platelets, and red blood cells, which leads to lack of health (easy bruising, bleeding from gums, fever and skin infections).

13. A = red blood cell, disease is anaemia with haemoglobin deficiency so that insufficient oxygen can be carried by the blood.
    B = white blood cell, disease is leukaemia with too many white blood cells which are unable to fight infection.
    C = neuron (nerve cell), disease is multiple sclerosis where the myelin sheath degenerates and leads to damage of the nerve and reduces ability to move limbs.
    D = ciliated epithelium, disease is asthma with increased numbers of epithelial cells, mast cells and lymphocytes in bronchoalveolar fluids and increased airway reactivity.
14. (a) Infectious
    (b) Infectious
    (c) Non-infectious
    (d) Non-infectious
    (e) Non-infectious
    (f) Infectious
    (g) Infectious
    (h) Infectious
    (i) Non-infectious
    (j) Non-infectious
15. Preventing sewage from contaminating water supplies prevents faeces that contain the cholera bacteria from infecting drinking water. This stops the bacteria being transmitted to another person, which could lead to an epidemic if enough people were infected.
16. The scientific understanding of the structure of DNA and its base codes and how gene expression influences phenotype was a major scientific advance, along with the development of the technology which allows the base code sequences to be identified. This development led to a change in treatment practices and has directed future research into genetic engineering as a way to assist people with genetic diseases. Understanding the nature of the disease enables scientists to devise ways of curing the disease.
17. (a) Genetic mutation causes cancer. Some cancers occur when cells rapidly divide and cell differentiation does not occur properly.
    (b) Although each nucleus has each chromosome and hence contains each gene, not all genes are 'switched on'. At any one moment, a typical human cell only expresses about 3% to 5% of its genes.
18. Recorded history shows how people have always been concerned about disease and its cause. Leviticus (third book of the Old Testament) sets down hygienic practices that include personal cleanliness, protection of water and food supplies, and protection from the spread of infectious diseases, particularly the identification and treatment of leprosy. People recognised the relationship between cleanliness and lack of disease. The Minoans and Cretans had drainage systems and flushing toilets and the ancient Romans had public baths, sewers and carefully maintained water supplies. Written records from China dating from 2500 BCE show concern about health and medicine, with the belief that good health was the correct balance of the energy forces Yin and Yang. They treated an imbalance, or disease, with acupuncture and herbal treatments. As biological understanding developed, the impact on society was a reduction in disease-related deaths in humans. Biological understanding also led to a reduction in disease-related deaths of farm animals and growing of resistant strains of crops.
19. A
20. D
21. C
22. B
23. D
24. C
25. B

44 Prion Pathogens

1. In 1982 Stanley Prusiner invented the word prion as a combination from *protein* and *infection*.
2. Stanley Prusiner won the Nobel Prize in Physiology or Medicine in 1997 for 'the discovery of Prions – a new biological principle of infection'.
3. Prions are infective proteins that have been altered to an abnormal shape.
4. PrP^C stands for cellular prion protein and is the normal form of the protein that will become a prion.
5. The normal prion proteins consist of about 40% alpha-helices while prion proteins are about 55% beta-pleated sheets and 20% alpha-helices.
6. The cellular prior proteins are mainly located on the surface of cells in the central nervous system but can occur in other body tissues in mammals.
7. Many prion diseases are classified as spongiform encephalopathies because in a postmortem examination and viewed under a microscope the brain has large vacuoles in the cortex and cerebellum so that the tissue resembles a sponge.
8. Prions cause degeneration of brain tissue with severe loss of neurons. This causes loss of full control of bodily movements and changes in behaviour, e.g. in humans the symptoms can involve personality changes, psychiatric problems, insomnia, progressive loss of intellectual capacity and the loss of the ability to move or speak.
9. (a) TSEs in sheep and goats cause scrapie.
    (b) Human prion diseases include CJD (Creutzfeld-Jakob disease), FFI (fatal familial insomnia) and kuru.
    (c) In cattle a prion disease is BSE (bovine spongiform encephalopathy or mad cow disease).
    (d) A prion disease in cats, pumas and cheetahs is feline spongiform encephalopathy.
10.

Transmission	Description
Acquired infection	Contact with infected tissue, body fluids, ingestion of infected tissues, or during medical procedures with contact of contaminated medical instruments.
Hereditary transmission	A change or mutation in a prion protein gene can be transmitted to an individual's offspring.
Sporadic change	A spontaneous change in the normal prion protein begins a chain reaction that causes other normal prion protein to alter shape to the diseased prion form.

11. Pathogens cause infectious diseases. The transmission of prions is unusual as it can be transmitted through contact of infected materials, like other infectious diseases, but it can also be inherited. Other infectious diseases cannot be genetically inherited.
12. Prion diseases can become widely spread before they are detected as they have a very long incubation period, e.g. at least ten years and at first act very slowly. The long incubation period means that sources of infection may not be identified until long after the first cases appear which means there may have been many infections from the original source or from later infected individuals.
13. Prion proteins are highly resilient. They are not affected by many measures taken to destroy pathogens, e.g. the prions are not destroyed or deactivated by normal cooking temperatures or by normal sterilisation techniques such as boiling or irradiation.
14. The mortality rate or death rate for prion diseases is extremely high. Prion diseases are always fatal.
15. Researchers are developing therapeutic strategies against the diseased prion protein while other researchers are trialling stem cell transplants to restore lost tissue and recolonise areas damaged by the prion protein.
16. (a) It is believed, but not proven that the start of the outbreak of BSE in the United Kingdom in 1986 was caused by a scrapie agent jumping species when cattle were fed sheep offal as a protein supplement.
    (b) Many countries were concerned about the BSE outbreak as prion diseases are always fatal. When the British Health Secretary announced the death of 10 people due to vCJD millions of cattle were destroyed and import bans were put in place for British beef and beef products to protect people in other countries from contracting the disease.
    (c) Australia has introduced several measures to prevent the introduction and spread of BSE: 1. There is a national monitoring program checking cattle and sheep populations for evidence of TSE. 2. There is also a complete ban on the importation of live cattle from all countries that have reported cases of BSE. 3. The Ruminant Feed Ban prohibits the feeding to ruminant (animals that chew cud) of any material, tissue or blood taken from an animal.

17. D
18. B
19. B

45 Case Study – Creutzfeldt-Jakob Disease

1. CJD is Creutzfeldt-Jakob disease and is a type of transmissible spongiform encephalopathy (TSE). CJD is caused by a prion that can spontaneously appear as a sporadic occurrence, be genetically inherited or acquired through food or medical procedures.
2. Most cases of CJD, e.g. more than 85% of cases occur as the sporadic form. For an unknown reason there is a spontaneous change of the normal cellular prion protein into the misshaped form of the prion protein.
3. The vCJD is a variant of CJD that was first found in the United Kingdom. It affects younger individuals who show psychiatric symptoms and it has a longer duration from onset of symptoms to death.
4. In a prion disease the infectious prion protein causes other normal cellular prion proteins to take on the altered form. The prion proteins aggregate into a clump that leads to loss of neurons and brain damage.
5. CJD is a transmissible spongiform encephalopathy (TSE) as it is one of the progressive conditions that affect the brain and nervous system of many animals. It is called spongiform as the infected tissue resembles a sponge due to the disintegration of the tissue by the misfolded prions. An encephalopathy is a disease that affects the functioning of the brain due to some agent or condition.
6. (a) The three ways of transmission are: 1. Sporadic appearance even though the person has no risk factors for the disease. 2. Hereditary transmission where there is a family history of the disease and/or the genetic mutation is identified in the person's genome. 3. Acquired infection, for example through food or medical procedures.
   (b) Sporadic CJD accounts for more than 85% of cases of CJD. Hereditary CJD accounts for 5% to 10% of cases of CJD. Acquired CJD accounts for less than 1% of cases of CJD.
7. CJD is often difficult to diagnose as the early symptoms are vague and similar to other brain disorders.
8. Symptoms can include – confusion, disorientation and failing memory which develops into dementia, personality changes with impaired judgement and thinking, behavioural changes, weakness or loss of balance and muscle control causing problems with coordination and movement, muscle spasms and paralysis, vision problems, e.g. double vision or blindness, coma and incontinence.
9. Symptoms usually begin to show around the age of 60 and most people die within one year from the onset of symptoms.
10. There is no cure for CJD. Medical treatments try to ease the symptoms and aid the patient with the difficulties they have in coping with day to day activities.
11. D

46 Virus Pathogens

1. A virus is an infectious particle with either the DNA or RNA genome surrounded by a protein coat called a capsid.
2. A phage is a virus that infects bacteria while a viroid is a plant pathogen consisting of a molecule of naked circular RNA with only a few hundred nucleotides.
3. As viruses can be crystallised, many consider them non-living; however, they contain nucleic acid which indicates they are living.
4. Four main types of viruses: 1. Icosahedral – capsid has 20 flat sides giving it a special shape. 2. Helical – capsid is rod shaped. 3. Enveloped – capsid is encased in a baggy membrane. 4. Complex – no capsid.
5. A capsid is a protein shell with the shape dependent on the type of virus, e.g. rod shaped, polyhedral or more complex.
6. A viral envelope is derived from the host cell phospholipids and membrane proteins. It occurs in some viruses.
7. Viruses lack metabolic enzymes and ribosomes for making proteins and so they need a host cell to metabolise. They can express their genes and carry out chemical synthesis when they use the host cell's metabolic equipment and chemical pathways to make more virus proteins and nucleic acid.
8. Viruses inject their nucleic acid into the host and use the host's materials to reproduce and eventually burst the host cell, whereupon the released viruses can invade other cells.
9. H1N1 is a particular strain of the influenza A virus that causes influenza in birds, pigs and humans. It has caused pandemics, e.g. 1918 Spanish flu and 2009 swine flu. It is closely monitored because of its ability to cause pandemics.
10. (a) Conjunctivitis is spread from person to person if eyes are touched and the virus is transmitted by shaking hands, sharing towels, or being in a contaminated swimming pool.
    (b) The conjunctivitis virus is an adenovirus with no viral envelope. The capsid has an icosahedral shape with a glycoprotein spike at each vertex.
11. The first viral vaccines were based on attenuated or weaker viruses, e.g. smallpox vaccine used cowpox virus.
12. It is difficult to get a long term vaccine against some viruses as some viruses keep mutating from one person to the next and by the time the vaccine is in production the virus has already changed its format, e.g. common cold.
13. The World Health Organisation recommends vaccination for hepatitis B in the first day of life as it is a preventable disease that is easily transferred from person to person in blood or body fluids and the virus can survive for at least 7 days outside the body. As HBV affects the liver it can lead to cirrhosis and liver cancer which reduces life expectancy.
14. A
15. D
16. B

47 Case Study – Human Immunodeficiency Virus

1. HIV is the virus – human immunodeficiency virus whereas AIDS is the disease acquired immunodeficiency syndrome that is caused by the virus.
2. HIV is a retrovirus which has a roughly spherical shape with two copies of single stranded RNA that contains its genetic information. The RNA is enclosed in a conical shaped capsid made of protein.
3. HIV can evade the immune system as it can stay latent in the host DNA. Upon entry into a host cell the RNA of HIV is converted into a double stranded DNA by the reverse transcriptase. The viral DNA then moves into the cell nucleus of the host and integrates itself into the host DNA. Once part of the host DNA the virus can stay latent and undetected by the host immune system leading to a long incubation period.
4. HIV targets and infects helper T cell (T4 cells), macrophages and dendritic cells.
5. There are several ways that HIV can be transmitted: 1. Sexual contact is the most common way HIV is transmitted, e.g. sexual intercourse with an infected partner without using a condom. The virus can enter through the lining of the vagina, vulva, penis, rectum or mouth during sex. 2. Contaminated equipment – HIV is transmitted by sharing contaminated syringes or needles, e.g. drug users, tattoos, piercings. 3. Mother to child when HIV crosses the placenta during pregnancy or the child is infected during birth or by drinking infected breast milk. 4. Blood – at one stage HIV was transmitted through blood transfusions or blood products that used infected blood. However, there is now blood screening for evidence of HIV infection and treatments to destroy HIV in blood products.
6. The symptoms of AIDS can be the same as the symptoms for other diseases, e.g. flu-like symptoms, extreme tiredness, fevers, chills and night sweats, swollen lymph glands, skin blemishes or bumps, a dry cough and white spots or unusual marks in the mouth.
7. The three categories of HIV infection are: 1. Category A which is similar to glandular fever with swollen lymph nodes, headaches, fever and dry cough. 2. Category B with more advanced symptoms such as persistent generalised lymphadenopathy – swelling of lymph nodes, weight loss, diarrhoea, fatigue, fever, unexplained bleeding and memory loss. 3. Category C – fully developed AIDS with the development of other serious diseases such as pneumonia, cancer and brain disorders.
8. Once a person is HIV positive the virus cannot be removed and there is no current cure. There are several medications that help control the virus and slow the progression of the disease and also help the person maintain their health for a longer period of time. Antiretroviral medications used in antiretroviral therapy (ART) target different stages of the virus replication life cycle.

9. For viral replication: HIV attaches to CD4 cell → it enters through the cell membrane → the viral genes and enzymes are uncoated → reverse transcriptase changes the RNA genome into viral DNA → viral DNA enters the nucleus and integrates with the cellular DNA → the cell's protein synthesis machinery produces viral proteins and copies of HIV genome → virus leaves the cell as a new viral particle.
10. HIV/AIDS is a global issue as the virus is easily spread through unsafe sexual practices and sharing of needles and syringes. With modern international travel people can easily move to other areas and HIV has spread globally with an estimated 39 million deaths since 1981. Education about HIV/AIDS is needed in all countries to reduce its occurrence and transmission. Since there is no current cure an HIV infection will dramatically affect the quality of life for the person and reduce their life expectancy. Thus all countries need to maintain programs and public awareness about HIV/AIDS.
11. A person with HIV infection can be discriminated against in many ways, e.g. ostracism, loss of previous social network, work restrictions, travel restrictions, denial of hospital facilities or be subject to violence.
12. A

48 Bacteria Pathogens

1. A prokaryote has no membrane bound organelles, e.g. mitochondria, nucleus.
2. Bacteria range in size from 0.5 to 5.0 μm.
3. Bacteria are present in the soil, air and water and in many human body parts such as the skin, throat, mouth, nose and digestive tract.
4.

Part	Function
Cell wall	Protective rigid structure to help maintain cell shape.
Capsule	If present a sticky outside layer that aids disease causing ability and helps protects against phagocytosis by macrophages.
Fibriae	Help bacteria to sticky to their substrate or to one another.
Flagella	Used for locomotion.
Nucleoid region	Contains genetic information and main component of bacterial genome.
Plasmid	Accessory genes give genetic advantages, e.g. antibiotic resistance.
Endospore	When the bacterium is exposed to harsh conditions it forms thick-coated resistant cells which can lie dormant for long periods of time.

5. Bacteria reproduce asexually by binary fission and can produce endospores to survive harsh conditions. Bacteria can also conjugate and horizontally transfer genetic information when two cells are temporarily joined.
6. Cocci are any spherical or roughly spherical bacteria, a bacillus is a rod shaped bacterium while spirilla are bacteria with a rigid spiral structure.
7. The Gram stain is usually used as the first step in bacteria identification. Gram positive bacteria keep the crystal violet dye colour identifying the peptidoglycan in the cell wall while Gram negative stain red or pink.
8. (a) Cholera is caused by an exotoxin secreted by the bacterium *Vibrio cholerae*.
   (b) The exotoxin secreted by *V. cholerae* causes the intestinal cells to release chloride ions into the gut and water follows by osmosis. This leads to severe diarrhoea, dehydration and an electrolyte imbalance.
9. Bacteria reproduce by binary fission. This can occur as quickly as within 12 minutes if conditions are favourable (e.g. nutrients, moisture, temperature, pH, and availability of oxygen). This means that numbers can increase incredibly rapidly within a short time span.
10. Bacteria can be named and classified by shape, e.g. round cocci, rod shaped bacilli, spiral etc.
11. Bacteria cause disease when they release their waste products that may be toxic to the host. Rapid reproduction causes more toxins in the host.
12.

Feature	Exotoxin	Endotoxin
Released	Released outside cell.	Part of outer cell wall.
Type of bacteria	Produced by both Gram negative and Gram positive bacteria.	Only in Gram negative bacteria.
Structure	Toxin is a protein.	Lipopolysaccharide.
Number of types	Many types of exotoxins based on structure and function.	Only one type of endotoxin.
Cell lysis	Both released during lysis of the cell.	

13. (a) Antibiotics are chemicals that kill or inhibit the growth of bacteria.
    (b) Antibiotics are used to treat bacterial infections.
    (c) Antibiotics can: 1. Inhibit cell wall synthesis. 2. Inhibit protein synthesis (e.g. translation). 3. Alter the cell membrane and its permeability. 4. Inhibit nucleic acid synthesis.
    (d) Penicillins affect cell wall synthesis and the tetracyclines are protein synthesis inhibitors.
14. Vaccines contain killed or attenuated bacteria that activate the immune system, e.g. whooping cough vaccine has killed bacteria while measles vaccine has living, attenuated bacteria. The vaccine stimulates the production of antibodies that are specific against that particular bacteria.
15. Toxoids are modified toxins that have the antigen but it cannot destroy tissues, e.g. toxoids for tetanus and diphtheria.
16. Booster injections for vaccines and toxoids increase the number of antibodies and the number of circulating memory cells to sustain immunity.
17. A
18. B

49 Case Study – Cholera

1. Cholera is caused by the vibrio bacteria *Vibrio cholerae* and is transmitted by being ingested in contaminated food or water.
2. The toxin produced by the bacteria causes antibodies to be produced by the host.
3. Victims develop cold, shrivelled skin and sunken eyes, blood pressure falls and the pulse becomes faint. The lining of the intestine is broken down by bacterial enzymes; reabsorption of electrolytes, nutrients and fluid is prevented. The bowel becomes inflamed, causing severe 'rice water stool' diarrhoea with muscle cramps, dehydration and eventually death.
4. ORT involves the application of oral rehydration solution containing sodium and potassium chloride, glucose, and sodium bicarbonate to prevent dehydration.
5. Prevention involves vaccines and people in cholera areas should avoid drinking water or eating food that may be contaminated.
6. The point of entry is part A, the mouth. The site of colonisation is part E, the small intestines. The point of exit is F, the anus.
7. Control involves many factors, such as laws, infrastructure and economics, which create an environment where the pathogen cannot survive or be transmitted. Prevention is a type of control that prevents people from becoming infected.
8. It influenced the Bourke Government to introduce Australia's first Quarantine Act in 1832.
9. A
10. B
11. D
12. C
13. C
14. B
15. D
16. B

50 Antibiotics and Bacterial Infections

1. Antibiotics are drugs that have a toxic effect on microbial cells, inhibiting the growth or destroying pathogens.
2. Alexander Fleming discovered in 1928 that *Penicillium* mould prevented the growth of bacterial colonies on an agar plate.
3. Penicillin disrupts the cell wall of bacteria.
4. Antibiotics act in different ways to prevent the growth of bacteria and hence control infection.

5. Some patients have side effects such as vomiting, nausea, diarrhoea, skin rashes, deafness, loss of balance, damage to bone marrow and serious allergy.
6. Evolution resistant strains of bacteria have evolved.
7. The spread of bacterial infections can be controlled by the widespread use of antibiotics and hence the spread of disease to other countries or cities can be restricted. This can reduce the incidence of epidemics. However, indiscriminate or incorrect use of antibiotics can act as a selecting agent on the bacteria resulting in the evolution of multidrug resistant strains that are difficult to kill. This in turn could lead to a worldwide epidemic and the spread of disease, without a current treatment to prevent its spread.
8. Antibiotics are effective against bacteria, which are the pathogens causing the urinary tract infections, ear infections and sinus infections. However, all colds and flu, most coughs and most sore throats are caused by viruses and antibiotics are ineffective against viruses.
9. It is very important to research the ways in which bacteria attack cells as a knowledge of the mechanisms and chemicals involved in bacterial attack allows counteractive measures to be investigated. This can lead to new treatments or the development of vaccines which will prevent the pathogens causing disease in the host.
10. Students should make a complete table. Only the last three columns are given here.

Pathogen	Symptoms	Treatment/Prevention
Vibrio cholereae	Diarrhoea, inflamed bowel	Antibiotics in early stage, fluids
Clostridium tetani	Muscular spasms, lockjaw	Antitoxin, immunisation
Treponema pallidum	Ulcers, skin rash, blindness	Penicillin
Bordetella pertussis	Short, dry cough, catarrh, sneezing	Antibiotics, cough syrup
HIV	Destruction of immune system	Safe sex, no needle sharing, no cure as yet
Poliovirus	Affects muscles, nerves, paralysis	Immunisation
Variola major	Fever, headache, rash, vomiting	Immunisation
Influenza A, B, C viruses	Fever, headache, inflammation of windpipe and throat	Rest, warmth, paracetamol
Rubella virus	Headache, sore throat, fever	Paracetamol (temp), fluids
Three genera *Microsporum*, *Erichophyton*, *Epidermophyton*	Lesions expanding in all directions with raised borders	Antifungal cream, ointments
Candida albicans	White patches in mouth and pharynx	Antifungal cream
Four *Plasmodium* species	Shivering fits, fever, headache, nausea	Chloroquine
Entamoeba histolytica	Diarrhoea, fever, dehydration	Medicinal drugs, emetine
Trypanosoma gambiense	Headache, intermittent fever	Pentamidine, suramin
Taenia saginata	Malnutrition, weight loss	Drug – niclosamide
Enterobius vermicularis	Restlessness, irritability	Combantrin, Vermox
Pediculus humanus capitus	Red spots, itching, blisters	Insecticides – lindane
Sarcoptes scabei	Itching hands, wrists, elbows, eczema	Benzyl benzoate lotion

11. B
12. C
13. D
14. D

51 Protist Pathogens

1. The protists are an extremely diverse group of eukaryotes including unicells, colonial and multicellular organisms.
2. The diverse nature of the organism found in the kingdom Protista and the study of the diversification of living forms both past and present and their relationships in eukaryotic systematics has led to reclassification and the abandonment of the kingdom Protista. The protists have little in common except for a relatively simple level of organisation.
3. The protists are dissimilar as they: 1. Have different life cycles. 2. Have different specialised structures. 3. Have different modes of nutrition. 4. Have different reproduction methods. 5. Occupy diverse habitats.
4. The protozoans are a diverse group of mostly motile unicellular, heterotrophic eukaryotic organisms that were in the Protista kingdom.
5. Protozoans were classified by their means of locomotion.
6. The term protozoa is no longer used in taxonomy and thus it is unlikely you would find the term in a current classification system.
7. (a) *Giardia lamblia* is a diplomonad protist that is anaerobic, lacks plastids and has a mitochondrion that lacks DNA for enzymes in the citric acid cycle. They have two nuclei each with four associated flagella and lack Golgi apparatus.
   (b) *G. lamblia* causes the disease giardiasis with stomach cramps, diarrhoea and excessive gas.
   (c) *G. lamblia* is transmitted to a new host when hands, objects or food is contaminated with faeces of infected people or animals or by contaminated drinking water, e.g. contaminated streams, ponds, swimming pools.
8. A kinetoplast is an organised mass of DNA in a single large mitochondrion and can contain many copies of the mitochondrial genome.
9. *Trichomonas vaginalis* is a parabasalid as it is a protist flagellate with one or more clusters of flagella near the anterior of the cell.
10. Apicomplexans are protist parasites of animals with most having life cycle stages involving both asecual and sexual stages so they can survive in several different environments.
11. A sporozoite is the cell form of an apicomplexan protist parasite that infects a new host. It has a complex of organelles at the cell apex that aids penetration into the host cells and tissues.
12. *Plasmodium* is an apicomplexan protist parasite that is transmitted to humans in the saliva of a bite of an *Anopheles* mosquito. The sporozoites enter the liver and divide and become merozoites. Merozoites enter red blood cells, multiply and break out releasing toxins that cause the chills and fever.
13. The incidence of malaria decreased in the 1960s with the use of insecticides against *Anopheles* mosquito populations and antimalarial drugs but there are now mosquito varieties resistant to insecticides and *Plasmodium* varieties resistant to drugs. Thus there is a resurgence of malaria.
14. The amoeboid protists include many amoebas and the slime moulds.
15. Amoeboid movement involves the formation of pseudopodia with the cytoplasm sliding forward in front of the cell and the cell changing shape.
16. (a) Amoebic dysentery is caused by the amoeba *Entamoeba histolytica* which is transmitted through contaminated food and water. The amoeba form infective cysts which are found in faeces. Free amoebas known as trophozoites are without cysts and do not survive for long outside the human gastrointestinal tract. The amoeba can burrow through the intestinal wall and spread through the bloodstream to infect other organs, e.g. liver, lungs and brain.
    (b) The amoebas are destroyed by antimicrobial drugs, e.g. metronidazole.
17. C
18. A
19. D

52 Case Study – Cryptosporidium

1. *Cryptosporidium* causes the disease cryptosporidiosis.
2. *Cryptosporidium* is an apicomplexan protist.
3. *C. parvum* and *C. hominis* are the main species that infect humans.
4. The oocyst plays an important part of the life cycle as it is a thick walled spore that can survive for long periods outside the host and is the form that will be transmitted through infected food or water from one host to the next.

5. Self-infection can occur when a person has poor personal hygiene practices and the oocyst is transferred by faeces-oral transmission by the individual.
6. In 1998 low quality raw water entered the dams that supply Sydney with tap water. This caused contamination with cryptosporidium and giardia being detected in Sydney tap water. There was a 'boil tap water' alert and in 1999 the Sydney Catchment Authority was created to oversee dams and catchment areas to reduce the risk of such an event recurring.
7. A person becomes infectious as soon as the symptoms develop and they can stay infectious for several weeks after the symptoms disappear.
8. Cryptosporidiosis is a type of gastroenteritis that causes watery diarrhoea, stomach cramps, bloating, vomiting and fever.
9. Direct transmission can occur with an individual with poor hygiene not washing their hands after going to the toilet causing faeces-oral transmission. Indirect transmission can occur when objects or surfaces are contaminated with faeces from an infected person or animal and a person eats uncooked food that has been in contact with the surface or they touch the surface and use the unwashed hand to put food in their mouth.
10. For a person with a normal immune system the symptoms usually last 1 to 2 weeks. If a person has a weakened immune system the symptoms can last longer and be more severe.
11. Cryptosporidiosis is diagnosed when cryptosporidium is identified in a stool sample from the person with the symptoms when the sample is viewed under a light microscope.
12. Cryptosporidiosis is a nationally notifiable disease within Australia which means there is a legal requirement for authorities to be notified if there is a confirmed case with laboratory definitive evidence of the presence of *Cryptosporidium*.
13. If a person is travelling overseas to a country with minimal water treatment and sanitation systems it is advisable to avoid drinking tap water or consuming untreated ice as they could become infected with cryptosporidium and return home with the disease.
14. Good personal hygiene practices that prevent the spread of cryptosporidium include: 1. Washing hands with soap and water after going to the toilet. 2. Hand washing when changing nappies and handling food or animals. 3. An infected person does not swim in swimming pools for two weeks after the diarrhoea has stopped.
15. A 'boil alert' was issued for all Sydney residents as boiling water, e.g. a rolling boil for one minute will inactivate the cysts. The advice was a precautionary measure to prevent an epidemic with the infection of many of the population as the contamination occurred in the catchment area and dams.
16. C
17. A
18. B

53 Fungi Pathogens

1. Fungi are: 1. Eukaryotes. 2. Do not contain chlorophyll. 3. Are heterotrophs. 4. Fungal cell walls contain chitin. 5. They secrete exoenzymes to break down complex molecules into smaller molecules that can be absorbed. 6. Can be unicellular, such as yeasts, or multicellular, such as mushrooms.
2. Fungi are either saprophytes and feed off dead matter or parasites and live on or in a host. Some fungi are in a symbiotic relationship with other organisms.
3. Fungal infestations cause about 80% of plant diseases. These diseases can have a significant effect on crop yield and productivity. Thus agriculturalists investigate and are concerned about fungal infestations to protect food supplies and economic returns to farmers.
4. Scientists are monitoring the spread of the chytrid fungus as it is believed the disappearance and decline in abundance of some frog species is related to the spread of the fungus. The fungus attacks keratin and since frogs use their skin as a respiratory surface an infection affects gas exchange.
5. In humans several superficial infections are caused by fungi, e.g. infections of the skin, hair or nails such as ringworm and athlete's foot, while candidiasis infects the mouth, vagina and causes nappy rash. Histoplasmosis is caused by fungi infecting deeper organs and tissues such as the lungs.
6. The HIV virus that causes AIDS breaks down the immune system of its host. Since fungi are opportunistic feeders, a person with a compromised immune system such as a person with AIDS has little defence against the fungal pathogen which will spread throughout the host's body.

54 Case Study – Tinea

1. Tinea is a dermatophyte fungus that causes a highly contagious fungal skin infection.
2. Tinea reproduce rapidly in hot, moist areas and if there is poor hygiene.
3.

Tinea	Location of infection
Tinea pedis	Feet
Tinea cruris	Groin
Tinea capitis	Head and scalp
Tinea corporis	Body – arms, legs
Tinea unguium	Nails

4. Transmission occurs by direct contact with infected people or with other infected animals such as cats and dogs. Transmission can also be indirect with contact of contaminated articles or from the soil.
5. Each type of tinea infection has particular clinical features, e.g. *Tinea pedis* affects the skin webs between the toes and because the fungus can survive under toenails the infection reoccurs with itching following by the development of small blisters which release a thin fluid on rupturing. The skin softens and peels, cracks can appear and bacteria can infect the open wound. The main symptoms of tinea infections include – itches and stinging sensation, red scaly rash which is shaped like a ring, crackling, splitting and peeling in the toe web spaces, blisters, yellow or white discolouration of the nails, and bald spots on the scalp.
6. The fungus has to attach to the skin and penetrate into the epidermal layer. If it overcomes this barrier host enzymes, e.g. keratinases are activated and the inflammatory response is initiated. Most humans have a certain level of resistance to dermatophyte fungi related to antifungal factors in their blood. There is a certain level of innate immunity and infection will cause acquired immunity with the production of antibodies and T cells.
7. Antifungal creams reduce the symptoms but some infections require oral antifungal medication. Infected and dead skin is often removed.
8. B

55 Macroparasites

1. Macroparasites can be seen with the naked eye and live on a host which is harmed by the association.
2. Ectoparasites live on the host, while endoparasites live inside the host.
3. (a) *Echinococcus granulosus* is a parasitic tapeworm belonging to the phylum Playhelminth (flatworms). It is an endoparasite.
   (b) The adult tapeworm is in the intestines of dogs with larvae passing out in faeces. Contact with the faeces, e.g. due to poor hygiene can cause hydatid infection in a human where the bladderworm invades any tissue or organ and will form a cyst.
4. Tropical rat mites can feed on humans and transmit murine typhus, Lyme disease and Q fever. *Anopheles* mosquitoes can transmit *Plasmodium* spp from human to human with the pathogen causing malaria in the infected human.
5. Tapeworms do not need to have a mouth or alimentary canal as they absorb food through their body wall from the host's alimentary canal
6. D

56 Case Study – Pork Tapeworm

1. Taeniasis is human tapeworm disease.
2. Taeniasis is caused by a person eating undercooked pork that contains infective live bladderworms.
3. When a person becomes infected with live pork tapeworm cysts, the larvae are released in the human intestine and develop into young tapeworms.
4. A proglottid is one of the segments in the body of an adult tapeworm.

5. The body of the tapeworm is covered in mucus which is secreted in large amounts to defend itself against the digestive enzymes of the host. The worms also secrete inhibitor substances onto the surface of the proglottids to cause local inactivation of digestive enzymes.
6. The scolex (head) of the tapeworm has suckers and a double row of curved hooks that attach to the wall of the intestine and prevents the worm from being passed out in faeces.
7. Tapeworms are hermaphrodites with each proglottid having male and female organs so that they can self-fertilise. This means that the tapeworm does not need complex mating rituals to find another worm of a suitable sex in the intestines of the host.
8. The banning of the eating of pork by many cultures means that they are less likely to be infected by *Taenia solium* and will not get taeniasis.

57 Disease With an Insect Vector – Malaria

1. Ross established *Plasmodium* as the cause of malaria and described its complete life cycle.
2. A vector is an organism that transfers a pathogen to another organism.
3. Malaria in humans is caused by four species of the sporozoan protozoan *Plasmodium.*
4. The intermediate host is the human and the final host is the female *Anopheles* mosquito.
5. Symptoms include fever (up to 41.1°C), shivering fits, headache, nausea and profuse sweating.
6. Treatment involves dealing with those who have the disease; control involves preventing people from catching the disease.
7. Control and prevention are aimed at either humans, (e.g. use of prophylactic drugs, protective clothing and mosquito nets) or at the vector (e.g. insecticides on mosquitoes, draining swamps and spraying still bodies of water).
8. Control and prevention measures are most effective when applied early. Many people who die do not take suitable measures, such as the use of prophylactic drugs, protective clothing and mosquito nets, when the danger is greatest. Natural selection has also meant that many previous measures are no longer effective; there are now strains of *Anopheles* mosquito resistant to DDT and strains of *Plasmodia* resistant to drugs.
9. Authorities introduced the Quarantine Act 1908 which seized prohibited goods and organisms, and introduced measures such as eradication of the *Anopheles* mosquito. The Immigration Department now screens all new arrivals into Australia.
10. An indirect lifestyle increases its chances of survival; and while *Plasmodium* is in the liver cells it is isolated from the host's immune system.
11. B
12. D
13. A
14. C
15. D
16. A
17. C
18. A
19. B
20. A
21. B
22. D

58 Barriers – The First Line of Defence

1. The first line of defence of the human body involves barriers that prevent the entry of pathogens, such as skin, mucous membranes, cilia, chemical barriers and other body secretions.
2. Natural immunity refers to the defence mechanisms present at birth, for example mucous membranes.
3. Skin provides an intact, outer tough layer with keratin to stop penetration; its dryness prevents growth, and glands secrete antibacterial and antifungal chemicals.
4. The stomach wall secretes gastric juice (with HCl and proteases), which destroys pathogens.
5. Urine is sterile and acidic and flushes out microbes when it is released from the bladder.
6. Antibiotics can destroy microscopic organisms, causing an imbalance, so that the fungus *Candida albicans* can invade normal healthy tissue and multiply, causing candidiasis (thrush).
7. Beating of cilia sweeps the mucus, with trapped pathogens, along the tract and up to the nose.
8.

Barrier	How it protects	How pathogens overcome barrier	Diseases
Skin	Intact, outer tough layer with keratin protein prevents penetration. Dryness prevents growth. Glands secrete antibacterial and antifungal chemicals.	Direct touch, through abrasions, cuts, openings, or by active penetration.	Mites, hookworm larvae, tetanus.
Mucous membranes	Cover the digestive, respiratory, reproductive and urinary ducts with a thick mucus which traps pathogens and secretes inhibitory substances.	Pathogens overcome this barrier as many spores are resistant to heat and chemicals (e.g. bacterial spores).	Cholera, gonorrhoea, influenza.
Cilia	Beating movement sweeps the mucus, with trapped pathogens, along the tract.	Some bacteria infect upper respiratory tract.	Diphtheria.
Chemical barriers	Antimicrobial secretions, acidic gastric juice, acid mantle of the vagina inhibit growth of fungi and bacteria.	Resistant spores.	Syphilis.
Body secretions	Acidic urine flushes tract.	Use of antibiotics can destroy natural micro-organisms.	Thrush.

9. Skin acts as a site of entry. The intact, outer tough layer with keratin protein prevents penetration.
10. A
11. D
12. C
13. C
14. D
15. A
16. B

59 Defence by Specific Body Parts

1.

Point of entry	Transmission	Protection at point of entry
Skin	Direct transmission, e.g. touch affected area, through abrasions or openings or active penetration.	Hardening of outer layer by keratin. Organic acids, e.g. antifungal, antibacterial from sweat glands, surface bacteria. Little surface moisture.
Digestive tract	Ingestion of food and drinks, soiled fingers, toys, pets.	Mucous membranes. Saliva. Peristalsis. Gastrointestinal secretions. Temperature. Lysozyme enzyme.
Urinogenital tract	Direct sexual contact.	Mucous membranes. Urine. Temperature. Lysozyme enzyme.
Respiratory system	Inhalation of fungal or bacterial spores or pathogens in droplets from cough or sneeze.	Mucous membranes. Hairs and cilia. Coughing and sneezing. Temperature. Lysozyme enzyme.
The ear	Airborne pathogens enter ear.	Cerumen prevent dessication and lubricates skin in the ear canal. Cerebrum has antibacterial properties. Skin growth pushes out cerumen.
The eye	Airborne pathogens enter eye.	Eyelashes and eyelids with blink reflex. Tears flush cornea and conjunctiva. Lysozyme enzyme and salts in tears.

60 Antigens

1. An antigen is a unique molecule or part of a molecule that initiates an immune response,
2. Antigens can be a virus, part of a bacterium or part of a toxic molecule.
3. Self antigens are marker molecules on the surface of cells that are recognised as being part of that organism whereas a non-self antigen originates outside the body. Non-self antigens are markers on cells, tissues or organs that enter the organism, e.g. in a viral, bacterial, fungal, prion infection or in a transplant.
4. Adaptive immunity involves B cells or T cells recognising an antigen and binding with it via their antigen receptor. It is the antigen-specific immune response (also called acquired immunity).
5. Antibodies (immunoglobulins) are proteins produced by B plasma cells in response to specific antigens and have the capacity to react against antigens.
6. An allergen is an antigen that triggers an exaggerated immune response leading to an allergy.
7. There are many sources of allergens, e.g. different foods especially different nut types, seafood, shellfish and eggs, pet dander, insect stings or excretion, pollen and some drugs, e.g. penicillin, salicylates and sulfonamides.
8. About 90% of all food allergies are caused by the following nine categories – peanuts, tree nuts, soy, milk, egg, cereals, seafood, fish and sesame.
9. Food allergy symptoms affect the mouth, stomach and intestines causing vomiting, hives, coughing, tight hoarse throat, weak pulse, swelling of the tongue and dizziness.
10. Allergy symptoms for airborne allergens affect the eyes, nose and lungs causing sneezing, blocked, itchy or runny nose, red weeping eyes and headache.
11. Antibodies are proteins produced by B plasma cells in response to specific antigens and have the capacity to react against antigens. A person with blood type A will produce the self antigen A and recognise antigen B as foreign (non-self). As a foreign antigen it will make antibodies against antigen B and thus a person with blood type A will produce Anti-B antibodies. Similarly a person with blood type B will recognise antigen A as foreign and produce Anti-A antibodies.
12. B

61 The Immune Response – The Second Line of Defence

1. Antigens trigger the immune response. They are found on pathogens and foreign organs.
2. The lymph system is a series of capillaries and veins draining fluid from the tissues and returning it to the blood.
3. Lymph nodes act as filters, removing microbes, foreign particles, tissue debris and dead cells from circulation.
4. Blood and lymph have basically the same composition, except that blood has red blood cells.
5. The inflammatory response causes increased blood flow to the area (redness, heat and swelling) so that more phagocytes, antibodies, complement, and interferons can attack the pathogen.
6. Macrophages act as part of the non-specific response when they act as phagocytes. They act as part of the specific response when parts of the partially digested antigen are displayed on the surface of the macrophage and this activates the production of helper T cells for that particular antigen.
7. Interferons are produced in virus-infected cells and induce nearby non-infected cells to produce antiviral enzymes that inhibit viral replication.
8. The body's immune system identifies the transplant organ as a foreign body containing the antigen.
9. Both the first and second lines of defence are non-specific and act against all types of pathogens. The first line is the barrier to entry while the second line occurs after the pathogen has entered.
10. Phagocytes are non-specific as they attack any foreign substances, and engulf and destroy them.
11. Complement is part of the non-specific defence system as it increases phagocytosis and the inflammatory response, but it can be activated by antigen-antibody complexes which are part of the specific defence system.
12. C
13. B
14. D
15. A
16. B
17. C

62 Play – Inside the Immune System

Scene 1 – Outside the Body

1. (a) Tetanus is caused by a bacterium.
   (b) Diphtheria is caused by a bacterium.
   (c) Dysentery is caused by an amoeboid protozoan for dysentery.
2. Tetanus bacteria enter through skin (sits on rusty nail). Amoeboid protozoan of dysentery enters as a cyst from faeces of infected person (hides on dirty hands). Diphtheria bacteria enter as airborne droplets from a sneeze and are breathed in.
3. Bacteria reproduce by binary fission.
4. Bacteria have identical twins because binary fission produces two identical daughter cells from the parent cell.

Scene 2 – In the Blood

1. (a) Tetanus bacteria produce a neurotoxin that binds to the synapses and interferes with nerve impulses and causes muscular spasms.
   (b) Diphtheria bacteria release a toxin into the blood and will cause fever, headache and vomiting within 5 days.
   (c) Dysentery protozoan hatches in the intestine, invades the intestine lining and feeds on the blood. It causes abscesses accompanied by severe, continuous diarrhoea. May invade the liver and cause amoebic hepatitis.
2. Macrophages confront invading antigens.
3. Macrophages take antigens to a lymph node.

Scene 3 – The Lymph Node

1. Antigens are identified by memory cells.
2. Plasma cells produce antibodies.
3. The person was immunised with the triple antigen, which protects against tetanus, whooping cough and diphtheria.
4. Dysentery was treated by taking the medicinal drug 'emetine'.
5. The human defeated the pathogens.

63 Leucocytes

1. The three main cellular components of blood are red blood cells (erythrocytes), white blood cells (leucocytes) and platelets.
2. Granulocytes have secretory granules in their cytoplasm, e.g. eosinophils whereas agranulocytes do not have granules in their cytoplasm and have a one-lobed nucleus, e.g. monocytes.
3. Leucocytes can be distinguished from each other by: 1. The size and shape of the nucleus. 2. The staining features of the cytoplasm. 3. The regularity of the cell border. 4. The cytoplasmic inclusions.
4. (a) All blood cells come from pluripotent haematopoietic stem cells in bone marrow.
   (b) In adult humans the spine, ribs and pelvis and proximal ends of long bones make new blood cells.
5. Red bone marrow is active bone marrow producing blood cells while inactive bone marrow is a yellow colour and contains many fat cells.
6. Leucocytes play a major role in defence against foreign invaders.
7. Most of the activity of leucocytes is carried out in body tissues rather than in the bloodstream.
8. Cytokines, e.g. interleukins and colony stimulating factors stimulate the production and development of blood cells.
9. Leucopoiesis is the process of white blood cell production and development.
10. Leukaemias are a group of diseases caused by an excess or lack of white blood cells.
11. The five main types of mature leucocytes are eosinophils, basophils, neutrophils, monocytes and lymphocytes.
12. (a) Phagocytosis is a non-specific process where phagocytes attack foreign substances, and engulf and destroy them.
    (b) Granulocytes and monocytes are phagocytic.

13. (a) Chemotaxis involves movement of cells in response to a chemical stimulus.
    (b) Phagocytes are attracted and move up a concentration gradient to: 1. Toxins produced by the invading virus or bacteria. 2. Towards the degenerative products produced by the inflamed tissue. 3. Towards the source of complement proteins. 4. Towards the source of reaction products caused by the clotting in an inflamed area.
14. A neutrophil attaches to a particle and sends pseudopodia around the particle to enclose the particle to form a phagocytic vesicle. Lysosomes and other cytoplasmic granules move to the vesicle and begin digestion of and breakdown of the phagocytised particle. One neutrophil can phagocytise between 3 to 10 bacteria before it is inactivated and dies.
15. Neutrophils are relatively short-lived. After formation mature neutrophils usually spend 4 to 8 hours circulating in the bloodstream and then 4 to 5 days in tissues.
16.

Feature	Neutrophil	Macrophage
Life span	Short-lived, e.g. 4 to 5 days in tissues.	Longer lived, e.g. can be years in fixed tissue location.
Phagocytosis ability	Phagocytise 3 to 10 bacteria before dying – less powerful.	Phagocytoise up to 100 bacteria – more powerful.
Extrude residual particles	Cannot extrude residual particles.	Can extrude residual particles.
Antigen presenting cell	Do not become antigen presenting cells.	Can become antigen presenting cells.
Location	Can function in bloodstream and tissues.	Motile and patrol through tissues or become tissue fixed.
Formation	Formed in bone marrow.	Formed in bone marrow.

17. Dendritic cells are a type of phagocyte that become antigen presenting cells as part of the immune response.
18. Lymphocytes have a non-granular cytoplasm and are the main cells involved in the acquired immunity response.
19. Lymph nodes are organs located along lymph vessels that are a mass of spongy tissue separated into compartments by connective tissue which filter lymph and trap and remove foreign particles such as micro-organisms and tissue debris from the fluid.
20. A
21. B
22. C
23. D
24. A

64 Mast Cells and Dendritic Cells

1. Dendritic cells and mast cells are important components of the innate (non-specific) immune response to an antigen with roles in the inflammatory response.
2. Dendritic cells are a type of phagocyte that become antigen presenting cells as part of the immune response.
3. Most dendritic cells are in lymphatic tissues and skin.
4. An antigen presenting cell is a cell that ingests bacteria or viral particles, destroys them and then generates antigen fragments which are bound to class II MHC molecules and displayed on its cell surface, e.g. macrophages and dendritic cells.
5. MHC molecules are major histocompatibility complex molecules which can present antigens that will be recognised by T cells.
6. CD4 is a surface protein molecule found on most helper T cells. It recognises and binds with class II MHC molecules aiding the recognition of antigens by T cells and helping to start the acquired immunity response.
7. Mast cells are a type of body cell in vertebrates that produce histamine and other molecules that trigger the inflammatory response.
8. Most mast cells are found in connective tissue and under mucous membranes that line the digestive tract and airways.
9. When an injury occurs the mast cells release the stored histamine triggering dilation and increased permeability of the nearby capillaries.
10. Histamine is the active molecule that initiates the inflammatory response and causes the opening of pores in capillaries.
11. (a) When histamine is released the opening of capillary pores and dilation of blood vessels leads to plasma proteins escaping into the tissue and interstitial spaces leading to swelling and oedema of the tissue and most of the observable symptoms of inflammation.
    (b) Antihistamines reduce the permeability of membranes acting against the action of histamines and reduce symptoms such as runny nose and swelling.
12. B

65 Complement Proteins

1. Complement proteins are around 30 different blood proteins that are part of the complement system and defend against infection.
2. The main complement proteins in the complement system are C1 to 9, B and D with the proteins named in the order of their discovery and not in the order of sequence in their position in the cascade of reactions.
3. The complement system is initiated by a surface antigen or micro-organism or by antigen-antibody complexes.
4. The complement system being a cascade of steps means that it is a series of chemical reactions that will often occur spontaneously one after the other. Once initiated the product of the first reaction can be the enzyme that is needed to activate the second reaction (e.g. drawn as a dotted line in a diagram to show a reaction cascade).
5. Most of the sequence of reactions that make up the cascade occurs on the surface of the pathogen leading to the destruction of the pathogen and end of the infection.
6. Complement proteins can amplify the inflammatory response by causing increased blood flow to the area, causing increased leakage of capillary proteins into the area and causing the coagulation of interstitial fluid proteins to prevent movement of invading organisms to other areas.
7. The complement protein 3Cb is involved in chemotaxis as it strongly activates phagocytosis attracting both macrophages and neutrophils to engulf the organism to which it is attached.
8. Complement proteins can: 1. Amplify the inflammatory response. 2. Enhance phagocytosis. 3. Cause chemotaxis by attracting macrophages and neutrophils. 4. Directly lyse pathogens. 5. Activate mast cells and neutrophils. 6. Agglutinate invading organisms by changing the surface structures of organisms so that the organisms adhere to each other in clumps which reduces their virulence. 7. Neutralise viruses by attacking viral structures.
9. An opsonin is any molecule that marks an antigen for an immune response making it a more attractive target for phagocytosis.
10. The classic complement pathway is activated when the first protein (C1q) in the complement cascade binds directly to the pathogen surface, though it can also be triggered during an adaptive immune response due to an antigen-antibody reaction.
11. D

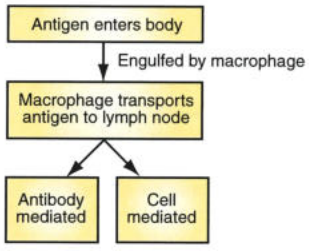
66 The Inflammatory Response

1. The inflammatory response is a localised innate immune response that occurs when there is physical injury or infection of a tissue. It is a protective mechanism that involves several activities.
2. The clinical characteristics of inflammation are redness, heat, oedema and pain.
3. Histamine causes dilation of blood vessels in the affected area which means that more blood is brought to the affected area. This is extremely important in the inflammation response as the increased blood flow brings more phagocytes and other white blood cells to the area, the higher temperature can affect the metabolism of the invading pathogen and the change in permeability of the capillary walls allows antimicrobial proteins and clotting elements to move into the tissue to destroy the pathogen and aid tissue repair.

4. 'Walling off' an area infected by staphylococci bacteria is an important defence mechanism as it prevents the bacteria spreading to other parts of the body. Since the bacterial toxins are destructive to cells and tissues it is imperative to confine the bacteria in as small an area as possible.
5. Histamine is stored in mast cells and when injured they degranulate and release the histamine.
6. Redness is caused by increased blood flow to an area due to the dilation of blood vessels and is one of the clinical characteristics of infection and the inflammation response.
7. One of the main functions of the circulatory system is to distribute heat around the body, especially to peripheral areas such as fingers and toes. The localised increased heat in an infected area is caused by the increased blood flow to that area due to dilated blood vessels.
8. Swelling is caused by more fluids and defensive substances moving into the tissues as the walls of the blood vessels become more permeable during the inflammation response.
9. Pus is an accumulation of dead and living phagocytes, cell debris and micro-organisms and forms as the remnants of the immune response as phagocytes attack the invading pathogen.
10. Fever is an abnormally high body temperature and is usually accompanied by shivering, sweating and headache. Fever affects viral activity, decreases circulating iron which decreases metabolic activity of viruses, interrupts the reproduction of some bacteria and viruses and promotes production of some T lymphocytes and antibodies.
11. Damage or infection → histamines and other chemicals released → local vasodilation and walls of blood vessels more permeable → increased flow of fluids into tissues → defensive chemicals and cells move to affected area → phagocytosis → pus may form → mitosis of tissue cells to replace damaged cells.
12. C

67 The Third Line of Defence

1. Innate immunity is present before exposure to any pathogens and works from the time of birth. Acquired immunity (also called adaptive immunity) occurs after exposure to inducing agents, e.g. pathogens, toxins, abnormal body cells and other foreign substances.
2. Immunological memory allows the immune system to remember the antigens on a particular pathogen, tumour cell or other foreign substance. This is very important as it allows the immune system to respond more rapidly and more effectively to that particular encountered antigen.
3. Lymphocytes which are a particular type of white blood cell are involved in adaptive immunity.
4. Both B and T lymphocytes are made in bone marrow. B cells are responsible for antibody immunity and T cells are responsible for cell mediated immunity.
5. Antibodies (immunoglobulins) are proteins that mark a particular antigen for elimination.
6. The third line of defence involves B cells, T cells and antibodies.
7. Defence mechanisms can be: 1. Barriers to entry, for example skin which provides an intact, outer tough layer with keratin protein to prevent penetration. 2. If the pathogen enters, phagocytosis can occur where phagocytes attack foreign substances and engulf and destroy them. 3. Adaptive immunity is a the third defence mechanism and is a specific defence where antibodies are produced by B lymphocytes in response to a specific antigen.
8. There are several ways in which the table could be completed.



9. B
10. D
11. C

68 B Lymphocytes

1. B lymphocytes (B cells) are white blood cells, formed in the bone marrow from pluripotent haematopoietic stem cells and they mature in the bone marrow before being released.
2. A specific humoral immunity does not develop until after invasion by that pathogen or toxin.
3. The first step after the invasion by a new pathogen or toxin is usually phagocytosis and partial digestion by macrophages. The macrophages become antigen presenting cells and also secrete cytokines, e.g. interleukin to promote growth and reproduction of specific lymphocytes.
4. The major histocompatibility complex (MHC) is a cell surface molecule that presents an antigen for identification by lymphocytes.
5. The epitope (also called the antigenic determinant) is the small accessible region of an antigen to which the antibody or antigen receptor binds.
6. The B cell receptor is a Y shaped molecule with the long tail of the Y shape consisting of two heavy chains that are anchored in the plasma membrane of the cell. Each B cell receptor has two antigen binding sites – one at each end of the short arms of the Y shape.
7. Secreted antibodies are similar to the B cell receptors having the same Y shape and same antigen binding sites but antibodies lack the transmembrane regions that are present in the B cell receptors at the end of the long tail of the Y shape.
8. The presence of an antigen activates B cells to divide and differentiate into either plasma cells or memory cells.
9. Plasma cells secrete immunoglobulins called antibodies with a shape compatible to that of the antigen.
10. B cells produce antibodies which bind with the antigen to form an antibody-antigen complex. This binding destroys the antigen by either immobilising the antigen, or blocking and neutralising an active binding site of the antigen, or by causing the antibody-antigen complexes to agglutinate and clump which makes them easier to eliminate by phagocytosis. Memory B cells remember antigens.
11.

Immunoglobulin	Function
IgM	Antibodies are the first circulating antibodies to appear after an infection and last a short time. They cause the antigens to agglutinate.
IgG	Are the most abundant antibodies and activate complement proteins. They easily leave the blood vessels and enter tissues. They pass immunity from the mother to the foetus across the placenta, and from the mother to the baby in breast milk.
IgA	Are present in external secretions such as tears, saliva and milk. They are found in mucous membranes and prevent the attachment of viruses and bacteria to the epithelium.
IgD	Are found on the surface of antibody-forming B cells, probably acting as antigen receptors.
IgE	Cause the release of histamine to begin the inflammatory response and play a role in the expulsion of parasites such as intestinal worms. The release of histamine and other chemicals can cause an allergic reaction.

12. The B memory cells remain in the body for a long time and can identify that particular antigen providing immunological memory. This means that a later infection of the same antigen can have a quicker response with the memory cells causing a more rapid, large scale production of the needed antibody to fight the invading pathogen or toxin.
13. Clonal selection is a process in which an antigen binds to and activates only the lymphocytes that have the specific receptor for that antigen. Since many lymphocytes are needed these specific lymphocytes are then cloned to produce great numbers.
14. Helper T cells induce B cells to divide, produce clones and produce antibodies.
15. One mechanism involves the production of chemicals, such as interleukins, by the T cell which induce activated B cells to divide.
16. B
17. B
18. A

69 The Structure of Antibodies

1. Antibodies are globular proteins secreted by plasma cells that bind to particular antigens and mark it for elimination.
2. All antibodies have the same Y shape in their monomer form.
3. Antibodies consist of two identical heavy polypeptide chains and two identical light chains joined by disulfide bridges.
4. Antibodies are specific as they have a variable region at the end of the heavy and light chains that determines the specificity of the antibody and has the antigen binding site. The organisation of the amino acids in this region gives the variable portion different shapes which allows the antibody to become specific for a specific antigen.
5. The constant region determines important properties of the antibody such as its ability to diffuse into tissues, its ability to pass through membranes, its ability to adhere to specific structures within the tissues, ability to attach to the complement complex and other biological properties.
6. The five general classes of antibodies – IgA, IgD, IgE, IgG and IgM.
7. (a) IgG is the most abundant antibody being around 70% to 75% of all antibodies in the blood.
   (b) IgG is important as it promotes opsonisation, neutralisation and agglutination of antigens.
8. IgG is the only antibody that can cross the placenta and thus give passive immunity to a foetus.
9. Antibodies combine with the antigen and can cause: 1. Agglutination binding everything into a clump. 2. Precipitation of the clump which has become insoluble. 3. Neutralisation with the binding sites to the host blocked by the antibody. 4. Lysis with the cell membrane of invader being ruptured. 5. Opsonisation and increased phagocytosis.
10. B

70 T Lymphocytes

1. T lymphocytes (T cells) form in the bone marrow but mature in the thymus gland where they are 'programmed' for a specific antigen.
2. T cells attack body cells which have been infected by a virus or other pathogen.
3. Natural killer (NK) cells are part of the innate defences. They attach to cells that are affected by virus or cancer and once bound to the cell cause apoptosis. Natural killer cells act more rapidly than other lymphocytes and secrete antiviral cytokines.
4. Cytotoxic T cells destroy cells identified as non-self (e.g. virus-infected cells, graft cells and tumour cells) by inserting perforin, which ruptures the cell membrane. Helper T cells stimulate the production of plasma cells and the production of antibodies. Helper T cells secrete chemicals such as lymphokines and interleukins. Suppressor T cells turn off the immune response and suppress the production of antibodies.
5. T lymphocyte receptors consist of two polypeptide chains linked by a disulfide bridge that have a transmembrane region that locks the receptor into the plasma membrane of the cell. The outer tip of the molecule has an antigen binding site. There can be as many as 100 000 receptors sites on a single T cell.
6. The T lymphocyte receptors can detect antigens presented by major histocompatibility complex (MHC) molecules.
7. MHC molecules are glycoprotein cell surface molecules that span the plasma membrane. MHC class I molecules bind antigens synthesised by the cell, e.g. due to cancer and present the antigen to cytotoxic T cells while MHC class II molecules bind antigens from bacteria or foreign particles that have been engulfed and digested and present the antigen to helper T cells. MHC class I molecules target the cell for destruction while MHC class II molecules attract immune cells to the area of infection.
8. T cells recognise cells infected with a virus as the viral proteins are moved to the cell surface and displayed by MHC molecules. The immune system recognises such cells as non-self and T cells are sent to destroy it.
9. Memory T cells have immunological memory allowing a faster and stronger response to a previously encountered antigen.
10. Helper T cells help mediate the immune response as they respond to a macrophage, dendritic cell or B cell displaying an antigen on a MHC class II molecule and will release lymphokines that will attract other lymphocytes to the area. After an antigen is identified the helper T cells secrete lymphokines that activate specific B cells. The helper T cells proliferate and differentiate into a clone of activated helper T cells and memory T cells increasing the effect of the immune response.
11. D
12. A

71 Macfarlane Burnet

1. Macfarlane Burnet was an immunologist who developed the clonal selection theory and helped to explain how the body identifies self and non-self.
2. He was awarded a knighthood in 1951, and in 1960 the Nobel Prize in Physiology or Medicine.
3. Viruses cannot reproduce outside a host cell and hence cannot be grown on agar plates. The discovery that they can be cultivated in living chicken eggs meant a high concentration of the virus could be produced and a vaccine developed.
4. Clonal selection theory is a model of how the immune system works to destroy an antigen by the selection of B and T lymphocytes for cloning and mass production.
5. Viruses contain either DNA or RNA as their genetic material, both of which are nucleic acids surrounded by a coat of protein. Since viruses cause diseases which often have no cure they are 'bad news'.
6. Ehrlich proposed his side chain theory of antibody formation in which the antigen was chemically bound to its receptor which in turn stimulated the cell to produce multiple copies of the bound receptor. The extra receptors (antibodies) were released into the blood. Although he did not correctly describe the actual procedure, Ehrlich's theory in which an antigen binds with a receptor to cause the manufacture of the same receptor is the basis of the clonal selection theory.
7. The variable region is the part that recognises specific antigens while the constant part carries out the immunologic task and is the effector.
8. C
9. A

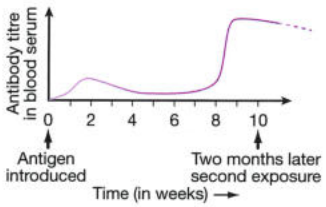
72 Immunity

1. Natural resistance refers to the barriers and the genetic resistance you are born with whereas acquired resistance is that which you build up during your lifetime.
2. Both passive and active immunity provide protection against infection. Passive immunity involves receiving antibodies and is thus short term with no memory cell production, while active immunity involves being exposed to the antigen or a form of it. The body then makes antibodies and memory cells. It is more long term.
3. Injection of gamma globulins containing antibodies provides immediate protection.
4. Immunisation is a process which stimulates the immune system to produce the immune response to fight an infection, while vaccination is a method of providing immunisation in which a form of the pathogen is introduced into the body.
5. Vaccines are either the killed pathogen; living attenuated forms of the pathogen; or a toxoid.
6. The vaccine acts as an antigen, inducing the immune response without the symptoms. The memory cells give immunological memory for a particular pathogen so a subsequent exposure to a pathogen is met by quick mass production of antibodies.
7. The smallpox scab contained the smallpox virus. If parts of the scab were introduced into a person's body it would trigger an immune response so that the body was sensitised to that particular antigen. Memory cells would form and the person would now be 'immunised' against smallpox.
8. C

73 Vaccination Programs

1. Vaccination programs are important as they help maintain herd immunity.
2. Herd immunity involves the majority of individuals in a population being vaccinated against a specific disease. This decreases the chance of transmission of the disease to the few individuals in the population who are susceptible and are not immune, e.g. herd immunity against smallpox led to its eradication and people no longer have to be immunised against smallpox.
3. Diphtheria is part of the triple antigen requiring injections at 2, 4, 6 and 18 months, school entry, 15 years and preferably every 10 years after that. Immunisation programs have greatly reduced its incidence and it is no longer a major cause of death in children.

4. The graph shows quick and large mass production of antibodies which means the person will not develop symptoms during the second exposure.



5. The traditional whooping cough vaccine causes some side effects and some parents prefer not to run the risk that their child is susceptible. Although most side effects are minor, it may cause brain damage in about 1 in 200 000 vaccines. Other parents simply forget to follow the immunisation program and lose track of the dates.
6. Those who want the destruction of the smallpox stores are concerned about its potential use by terrorists in viral biological warfare. Those against the destruction of the stores want the virus samples to be maintained for study, as their DNA may contain genes that could solve other problems such as cancer.
7. Suppression prevents tissue rejection.
8. There has been an increase in the number of children being vaccinated for poliomyelitis. In September 1997, 83% of children had received at least 3 doses of polio vaccine by the age of 12 months. Vaccination has reduced the incidence of the disease.
9. A
10. D
11. A
12. D
13. B
14. B

74 Epidemiology

1. Epidemiology is the study of disease and its prevalence in the community.
2. Pandemic means it is spread to many countries of the world; endemic refers to an epidemic always present in a particular part of the world; sporadic epidemics occur in different places at different times.
3. Epidemiology studies the cause of the disease, how it is transmitted, who is at risk, method of treatment and control, long term measures of prevention, virulence of the strain and if it changes over time, using descriptive and statistical data for analysis.
4. John Snow studied the cholera epidemic in London, interviewing families about where they obtained their water and making maps of where the outbreak reached, and he believed that cholera was spread through the water supply. His work led to the introduction of a proper water system, saving lives.
5. The work of John Snow is highly significant in our understanding of disease as he is considered to be the 'father of modern epidemiology' and showed how careful analysis and statistical mapping methods could be used to understand the spread of infectious diseases.
6. Since yellow fever was endemic to West Africa, natural selection had occurred over many generations so that the black Africans living in this area were the offspring of people who had some form of natural immunity to the disease.
7. The hepatitis A epidemic would have started with contamination through contaminated water or soil or through contact with an infected person. The disease is caused by a virus that infects the liver and can exist outside the body. The virus is resistant to heating under 85°C and freezing. Thus lack of proper hygiene and food handling procedures transmitted the pathogen from the source of the berries to the packages which were then imported into Australia.
8. To investigate the source of Legionnaire's disease authorities needed to find a place where all people affected by the disease had recently visited. Legionnaire's disease is caused by the *Legionella* bacteria (in this case *L. pneumophilia)*. Bacterial spores are water borne and the pathogen is found in water systems, e.g. cooling systems and air conditioning systems. Thus authorities needed to find a common location that had a cooling system which turned out to be the Melbourne Aquarium. The water could then be tested for the presence of the bacteria.
9. B
10. D

75 Pandemics – Influenza

1. Influenza is an infectious disease caused by a variety of strains of a RNA virus
2. The influenza pathogen is a virus that has a direct life cycle and is usually transmitted when the host coughs or sneezes without using a handkerchief and the airborne droplets carry the virus to another host. Although the influenza virus is relatively 'hardy' and can exist independently outside a host body for quite a while it can be inactivated by sunlight, disinfectant and detergents.
3. Typical 'flu' symptoms include – sore throat with inflammation of the upper respiratory tract, fever, chills, congested nose, pain in the back and limbs, headache and total exhaustion.
4. Complications can occur when the virus settles in parts of the respiratory system, e.g. adults can get pneumonia in the lungs or children can get croup in the larynx or when there are secondary bacterial infections, e.g. causing bronchitis.
5. The three main strains of influenza virus are A, B and C. Type A strains cause severe illness and have caused human pandemics. Type B cause sporadic human cases and small scale outbreaks. Type C strains rarely cause human infection and have not caused outbreaks.
6. Influenza A (H1N1) virus is a subtype of influenza A. It was the most common cause of human influenza in 2009. Some strains of H1N1 are endemic in pigs (swine influenza), in birds (avian influenza) and in humans. If transmission of the swine flu virus causes human influenza, it is called zoonotic swine flu. The outbreak of H1N1 was declared a pandemic by WHO in June 2009.
7. The host produces antibodies against each specific strain of influenza to which they have been exposed.
8. Once a particular strain has passed through a population there is a change in the structure of the virus and it becomes a new strain. The antibodies produced by humans for the previous strain are often ineffective against the new strain and people can thus get the 'flu' many times.
9. Influenza cannot be cured and treatment involves relieving the symptoms, e.g. bed rest, extra fluids, aspirin (not for children) or paracetamol to reduce fever and muscle aches.
10. Each vaccine is specific to particular strain(s) and people who rely on the vaccine for prevention need to have the injection yearly before the beginning of winter. A new version of the vaccine is developed twice a year due to the rapid changing nature of the virus.
11. To prevent the spread of H1N1 infected airborne droplets, people should cover their mouth and nose when coughing or sneezing. Some people wear face masks in winter as a preventative measure. Prevention of H1N1 involves prevention in swine, prevention of transmission to humans and prevention of its spread among humans.
12. A

76 Modelling Disease Outbreak and Spread

1. Scientists create models to show an idea or system giving a representation that makes it easier to understand or see what is happening. The models can then be used to make predictions about what could happen under specific conditions.
2. Daniel Bernoulli is most famous for Bernoulli's principle on the conservation of energy.
3. Bernoulli made a significant contribution to biology and disease as he used statistics in 1766 to analyse smallpox morbidity and mortality data in an attempt to work out the efficiency of vaccination. This was a milestone as it was the first time censored data was used for disease in statistical analysis.
4. Morbidity is the presence of the disease in patients while mortality is the number of deaths.
5. Statistical censoring means only some of the values, measurements or observations are known in the statistical analysis.
6. Variolation uses material taken from a patient with the disease to inoculate a person who does not have the disease, e.g. with smallpox rubbing a powdered smallpox scab onto the skin or a scratch mark.
7. At the time of Bernoulli's statistical analysis of the efficacy of vaccination people were using variolation procedures that had been introduced into England by Lady Mary Montagu between 1716 and 1718.

8. Edward Jenner carried out his experiment in 1796.
9. Developments in computer software with increased processing capacity and data storage have enabled the development of more complex models with computers manipulating large amounts of data to simulate events.
10. The germ theory was not fully developed until the mid 19th century and early 20th century and thus models about the spread of disease that were constructed before this time were not fully reliable.
11. Zika fever is caused by the zika virus (ZIKV) and is related to dengue and yellow fever with milder symptoms such as mild headache, macropapular rash, fever, conjunctivitis and joint pains. About one in five people who are infected with the zika virus show symptoms.
12. The zika virus is a single stranded RNA molecule. Like other flaviviruses the zika virion attaches to the host cell membrane receptors and induces virion endocytosis. Inside the cell it replicates in the host endoplasmic reticulum and new virions are transported to the Golgi apparatus to be secreted.
13. The arthropod vector is the *Aedes* mosquito which means the spread of the disease is related to the distribution of this vector.
14. Incubation time of the virus in the mosquito vector is around 10 days.
15. There was global concern about the 2015 outbreak of zika fever across the Americas as the zika virus in pregnant women is linked to the birth of children with congenital microcephaly.
16. Congenital microcephaly means a child is born with microcephaly. The child will have an abnormally small head compared to babies of the same sex and age and incomplete brain development. In severe cases of microcephaly the child can have seizures, hearing and vision problems, intellectual disability with developmental delay and decreased ability to learn and function in daily life.
17. In 2016 the World Health Organisation (WHO) designated the situation with the zika virus as a Public Health Emergency of International Concern (PHEIC).
18. Computer models were used to plot an epidemic trajectory of zika virus and the projected number of children born with zika-induced microcephaly.
19. The computer models considered human demographics, e.g. annual numbers of pregnancies, the economic index of areas, the occurrence and abundance of the *Aedes* mosquito and temperature conditions.
20. The models estimated human-mosquito ratios, the attack rate projections and the potential numbers of infections. These models quantified the potential magnitude of the zika epidemic, e.g. they predicted that Brazil would have the largest number of infections. The models also projected the numbers of children born with zika-induced microcephaly.
21. D
22. C
23. B

77 Managing Pandemics in the Asia Region

1. Epidemics occur when there is a rapid spread of a pathogen and large numbers of people are infected.
2. The epidemic becomes pandemic when the disease spreads around the world to many countries.
3. (a) Diseases of global concern include HIV/AIDS, diphtheria, malaria, measles and swine flu.
   (b) Diseases of concern in the Asia region include SARS and avian flu.
4. The Australian government provides management plans for particular diseases, e.g. *Australian Health Management Plan for Pandemic Influenza*.
5. Steps involve: 1. Recognise outbreak and prepare to investigate. 2. Verify diagnosis and confirm existence of outbreak. 3. Case definition and case. 4. Outbreak by person, place and time. 5. Work out who is at risk. 6. Hypothesis of how and why. 7. Test hypothesis. 8. Further studies. 8. Ongoing control and prevention measures.
6. UNAIDS is the Joint United Nations Program on HIV/AIDS.
7. To manage the AIDS/HIV epidemic the report advised that HIV testing, and counselling should be expanded and sustained, there should be stronger political leadership and national ownership of programs, key populations and affected communities at highest risk should be targeted and involved in all stages of the AIDS response, funding for programs needs to be maintained and there needs to be an increased push both in the media and legally to reduce discrimination and to ensure human rights and gender equality.
8. Groups with the highest risk of new infection are men who have sex with men and transgender people, people who inject drugs and female and male sex workers and their clients.
9. The World Health Day 2016 took the slogan Prevent. Treat. Beat diabetes. This disease was chosen as the incidence of diabetes is rapidly increasing and people need to become aware of the cause and prevention of diabetes.
10. D

78 Immune Deficiency Diseases

1. In an immune deficiency disease the immune system fails to protect the body from pathogens or cancer cells.
2. Immunodeficiency diseases can be classified as an inborn or primary immunodeficiency caused by a genetic or developmental defect or an acquired or secondary immunodeficiency after exposure to several chemical or biological agents.
3. Most primary immunodeficiency diseases are diagnosed before the child is one year old though in some people who have less severe cases the disorder is not picked up until they are older.
4. If a child is premature they can show signs of immunodeficiency as their organ development can be incomplete including an underdeveloped immune system.
5. A child with a primary immunodeficiency disease will have reduced ability to produce specific proteins, e.g. IgA or specific complement proteins and also cause reduced numbers or dysfunctional B cells and T cells. This means that they have a high risk of developing severe infections and often have frequent, severe respiratory infections, pneumonia, sepsis blood infections or meningitis.
6. Severe combined immunodeficiency (SCID) is a group of disorders caused by genetic mutations that reduce the ability of the immune system to fight infection.
7.

Feature	Description for SCID
Cause	By genetic mutations with different forms of SCID due to different mutations. One form is due to a mutation on the X chromosome which means that female carriers with one copy of this gene will not show symptoms while males show symptoms if their one X chromosome has the mutated gene. One mutation affects the gene for the enzyme adenosine deaminase (ADA).
Diagnosis	Diagnosis uses TREC (T cell receptor excision circle) assay to identify most forms of SCID and white blood cell counts are also used.
Symptoms	People with SCID can have non-functional helper T cells and/or defective B cells and problems with antibody production or ineffective complement proteins. The forms of SCID is due to a lack of the enzyme adenosine deaminase (ADA) which leads to an accumulation of substances toxic to both B and T cells. Children with SCID have a high risk of developing severe infections and often have frequent, severe respiratory infections, pneumonia, sepsis blood infections or meningitis.
Treatment	Treatments include using bone marrow transplants so that the child can make new white blood cells or gene therapy, e.g. with the child's own bone marrow cells genetically engineered with the needed functional gene .or enzyme replacement therapy. One of the forms of SCID is due to a lack of the enzyme adenosine deaminase (ADA) which leads to an accumulation of substances toxic to both B and T cells. Enzyme replacement therapy and gene therapy where a functional adenosine deaminase gene is inserted into the patient's own bone marrow cells have helped children with this form of SCID.

8. Early diagnosis is important so that treatments can begin and the child has less time to get a life-threatening infection.
9. Secondary immunodeficiency develops later in life and can be due to several factors, e.g. the use of immunosuppressive drugs in transplant surgery, the presence of certain cancers, e.g. Hodgkin's disease or pathogen infection, e.g. AIDS is caused by HIV infection.
10. AIDS stands for acquired immunodeficiency syndrome and is caused by infection with the human immunodeficiency virus (HIV).
11.

Feature	Description for AIDS
Cause	Is caused by infection with the human immunodeficiency virus (HIV).
Symptoms	Early symptoms of HIV infection include fever, swollen glands, sore throat, rash, fatigue, muscle and joint aches and pains and headache. Late stage infection when the immune system is weakening symptoms include rapid weight loss, recurring fever and night sweats, extreme tiredness, prolonged swelling of lymph glands, pneumonia, purplish blotches on the skin, mouth, nose, eyes, memory loss and depression.
Diagnosis	Diagnosis requires a test for HIV infection checking for specific antibodies.
Transmission	Transmission of HIV involves the transfer of body fluids containing infected cells and can occur by sexual contact with an infected partner, the use of HIV contaminated equipment, from an infected mother to child across the placenta during pregnancy or during birth, through blood transfusions or contaminated blood products. Because HIV weakens the immune system opportunistic pathogens, e.g. fungi, bacteria, viruses and parasites can take hold more easily in an AIDS patient. Viral induced cancers, e.g. Kaposi sarcoma, cervical cancer, and Burkitt's lymphoma are also likely to develop due to reduced numbers of functional helper T cells.
Treatment	Treatment involves anti-HIV drugs which interfere with the replication cycle of the virus and measures to help AIDS related illnesses to help maintain the quality of life for the patient.
Control	The main control measures are safe sex practices and not sharing needles, syringes or other injecting equipment.

12. HIV is a retrovirus and when it locates its target cell it binds to the surface molecules in this case the CD4 receptor molecule on helper T cells and injects its genes into the interior of the cell.
13. Some drugs used to treat AIDS patients do not stay effective over many years as mutations occur at each round of virus reproduction generating new strains of the virus which can be drug resistant. This means that the use of drugs keeps putting a selection pressure on the virus which is continually evolving and changing.
14. Once inside the cell HIV is uncoated and the viral RNA is copied to DNA (it is a retrovirus). The viral DNA integrates with the DNA of the helper T cell cell and becomes part of the genome. The lymphocyte uses the viral DNA as a template for reproducing the HIV RNA genome and synthesises viral proteins. The viral proteins assemble and new viral particles bud from the lymphocyte and move to infect another helper T cell cell. A depletion of helper T lymphocytes weakens the immune system and allows life-threatening diseases called opportunistic infections to occur. Helper T cells activate plasma B cells to produce antibodies and thus antibody production decreases with fewer helper T cells. Fewer T lymphocytes also means that cell-mediated immunity is reduced.
15. Using antiviral drugs that interfere with the replication cycle of the virus means that the number of HIV in the body is restricted. This means that fewer helper T cells are infected and the immune system has a chance to fight some other infections. It does not mean all HIV will be destroyed and the infection will not develop into AIDS.
16. B
17. A
18. B
19. D
20. C

79 Allergic Reactions

1. An allergy is a hypersensitivity to a non-pathogenic antigen causing an antigen-antibody reaction in some people to substances that do not cause the same response in other people.
2. An allergen is an antigen that triggers an exaggerated immune response leading to an allergy
3. Common types of allergies include hay fever, asthma, urticarial, childhood eczema and some gastrointestinal disorders.
4. Allergic inflammatory responses range from mild tissue damage to fatal reactions.
5. Basophils and mast cells release histamines.
6.

Feature	Description for pollen allergy
Cause	For each person the allergy is caused by hypersensitivity to a particular pollen from a specific plant species or to a group of plant species.
Symptoms	Common symptoms are runny or itchy nose, sore throat, watery eyes, sinus pressure, sneezing, swollen areas, e.g. under eyes, eyelids and nose, difficulty breathing and wheezing.
Diagnosis	For diagnosis of a pollen allergy it is important to take careful note of the timing of symptoms and to identify local plants and trees. Allergy tests using skin prick tests and/or blood allergen specific IgE tests will confirm the allergy.
How inflammation response is involved	When a person inhales the pollen the allergen stimulates the release of IgE from plasma cells that have been already sensitised to that antigen. The IgE attaches to the surface receptors on mast cells. This activates the mast cell to degranulate. When the granules are in the extracellular fluid they release the stored histamine and other chemicals, e.g. serotonin that cause inflammation. Dilation of blood vessels in the nose cause the area to become red and swollen and the increase in permeability of capillary membranes makes the nose run, eyes water and the irritation can cause sneezing.
Treatment	Treatment involves the use of antihistamine tablets to reduce the runny nose and watery eyes. Nasal sprays containing corticosteroids act against inflammation. Decongestant sprays clear blocked noses. Desensitisation therapy (also called allergen immunotherapy) involves giving the patient very small amounts of the allergen over a period of months or years to stimulate the production of IgG antibodies against the antigen. The IgG antibodies combine with the allergen blocking the action of IgE.
Control	Control measures for people with pollen allergies include staying inside on windy days in the pollen season, not being near lawn mowing, keeping windows closed when airborne pollen counts are high and planting suitable garden plants.

7. Common pollen allergies occur with pellitory wee (often called asthma weed), Paterson's curse *(Echium plantagineum)*, ragweed, white cypress (Murray) pine and parthenium weed. Allergic rhinitis (hay fever) and asthma can be caused by pollen.
8. Food allergy symptoms affect the mouth, stomach and intestines causing vomiting, hives, coughing, tight hoarse throat, weak pulse, swelling of the tongue and dizziness. Allergy symptoms for airborne allergens affect the eyes, nose and lungs causing sneezing, blocked, itchy or runny nose, red weeping eyes and headache.
9. The number of eosinophils increases to relieve the symptoms and remove the histamine.
10. The sequence of events would be – soluble allergens on pollen surface cross nasal mucosa → bind with sensitised plasma cell → IgE released from plasm cell → IgE combines with mast cell receptors → bound IgE combines with allergen → mast cell releases granules → histamine and other chemicals released → increased vasodilation and increased permeability capillary walls → inflammation and allergic reaction.
11. Hay fever is an allergic reaction that occurs in the nose.
12. Anaphylaxis is a dangerous allergic condition where a widespread reaction takes place very quickly with cardiac output and arterial pressure dramatically decreasing.

13. Desensitisation therapy (also called allergen immunotherapy) involves giving the patient very small amounts of the allergen over a period of months or years to stimulate the production of IgG antibodies against the antigen. The IgG antibodies combine with the allergen blocking the action of IgE.
14. (a) In anaphylaxis a person with a severe allergy is exposed to the antigen basophils in the blood and mast cells in tissues immediately outside the blood vessels release large amounts of histamine which causes extreme dilation of blood vessels and increased permeability of the walls of capillaries. This in turn causes a very large loss of fluid and proteins from the blood stream into tissues. So much plasma from the blood can cause circulatory shock and death can occur within a few minutes.
    (b) Adrenaline is used to oppose the effects of the histamine.
15. The Western Australian government has an Anaphylaxis Management Guidelines. By law each school must have a School Anaphylaxis Management Policy and inform and educate students and staff about the policy and how to deal with an anaphylactic situation.
16. D
17. D
18. A

80 Snake Antivenom Production

1. In Australia there are around 2000 incidents of snakebite per year of which around only a couple lead to death.
2. In Australia of the 172 snake species 115 are venomous – about 67%.
3. Snakes belong to the order squamata.
4. Snakes are legless and compared to lizards have an elongated body with a modified backbone.
5. Three venomous snakes found in Western Australia are dugite, tiger snake and death adder.
6. Few people currently receive fatal bites from the inland taipan as the snake is quite shy and being relatively rare and found in dry inland areas people do not often encounter the inland taipan and hence people do not receive bites. Since the development of a taipan antivenom (made for coastal taipans but works against inland taipans as well) any recent incidents have used the antivenom saving the person's life.
7. An antivenom is an antidote that counteracts the venom neutralising its effects.
8. The elapid snakes have large hollow fangs that can inject venom into they prey. The venom is produced in venom glands that are located towards the back of the upper jaw. The venom travels down venom ducts from the gland through the hollow fang to be released at the end of the fang which would have pierced the skin of the prey.
9. (a) Identify genes active when snake is refilling venom sacs → extract genes from snake DNA → put these genes into one synthetic DNA molecule that codes for the protein molecules that cause the haemorrhaging and destruction to blood vessels → inject synthetic DNA into mice → collect antibodies made by mice in response to synthetic DNA venom → purify to make serum → use serum as an antivenom testing on other mice.
    (b) Bioinformatics involves the construction, maintenance and use of databases to analyse the relationships in biological data, such as amino acid sequences or nucleotide sequences. It uses computers, software and mathematical models to process and integrate biological information from large data sets. The production of synthetic antivenom is an important development in bioinformatics as it shows how particular genes that are of interest can be identified and located to be extracted to help save lives. The vast amount of data in a genome can now be manipulated more easily and efficiently since the development of genetic engineering techniques and computer software and hardware.
10. B
11. D
12. A

81 Monoclonal Antibodies and Cancer

1. Macfarlane Burnet in his 1959 book *The Clonal Selection Theory of Acquired Immunity* proposed that immunological memory was due to one clone acting immediately to produce a specific type of antibody while the other clone remained in the immune system providing long term immunity to that antigen. These cells proliferated making many clones against a particular antigen.
2. (a) In 1975 Georges Kohler and Cesar Milstein worked out a way to produce monoclonal antibodies which is still the basic principle of fusion and selection that is used today.
    (b) Kohler and Milstein injected a protein antigen into a mouse in two doses – a primer dose and a booster dose. The immune system of the mouse identified the antigen as non-self and B cells produced antibodies specific for this antigen. These antibody producing lymphocytes were added to a culture of myeloma cancer cells and mouse hybrid cells were produced using cell fusion techniques that had previously been used in chromosome mapping investigations. The hybridoma cells were screened to find which made pure monoclonal antibodies. Since the hybridoma has the properties of a cancer cell it will not die in the tissue culture like other cells.
3. A hybridoma is a cell that has been formed by the fusion of an antibody-producing B lymphocyte and a tumour cell.
4. A line of monoclonal antibodies come from a single ancestral B cell clone and are thus highly specific as an antibody is only capable of binding to the particular epitope of the antigen to which it was sensitised.
5. The epitope (also called the antigenic determinant) is the small accessible region of an antigen to which the antibody or antigen receptor binds.
6. Developers are investigating new ways to streamline quality, reduce timelines, increase production and decrease costs.
7. One problem had been the presence of mouse antibodies which caused a reaction by the human immune system. The creation of 'chimeric' antibodies increases the amount of human protein involved in the fusion process.
8. Large amounts of particular antibodies are needed for diagnosis and treatment of particular diseases and by researchers in academic, medical and commercial fields.
9. A neoplasm (or tumour) is an abnormal new growth of cells and can be benign or malignant. A cancer is a malignant tumour.
10. Radioimmunodetection uses radiolabelled antibodies for diagnostic imaging of neoplasms and cancers whereas radioimmunotherapy combines radiation therapy and immunotherapy to treat different types of cancers.
11. Uses of monoclonal antibodies include: 1. Radioimmunodetection and radioimmunotherapy detect and treat different types of cancers. 2. Diagnostic testing, e.g. pregnancy tests checking for human chorionic gonadotropin (hCG) and ELISA testing for AIDS. 3. Treatment of viral diseases, e.g. emergency treatment for a rabies infection or hepatitis infection. 4. Classification of strains of single pathogens, e.g. to detect *Plasmodium* which causes malaria. 5. Identify and trace specific cells or molecules in an organism, e.g. which proteins are responsible for cell differentiation in the respiratory system. 6. Aid the problem of organ rejection in patients.
12. The laboratory-produced monoclonal antibodies are engineered to attach to specific parts of cancer cells. They can tag particular cells and toxin molecules can be bound to tumour-specific monoclonal antibodies to carry out a specialised search and destroy action against specific tumour cells.
13. The drug rituximab is a chimeric monoclonal antibody that attaches the CD20 proteins found on B cells from which certain lymphomas are generated. The drug causes opsonisation attracting lymphocytes and macrophages to the cancer cell. However, the drug can cause other adverse effects, e.g. cardiac arrest or cytokine release syndrome.
14. A
15. D
16. C

82 Rational Drug Design

1. Rational drug design is a process that looks for new medications based on a biological target.
2. The drug is usually a small organic molecule that is complementary in shape and charge to its target which it activates or inhibits, e.g. a target protein.
3. The drug has a beneficial therapeutic effect on the patient.
4. Drug design often involves computer modelling techniques in computer aided drug design. Research into the structure and activity of the specific target molecule allows researchers to modify the target molecule to see if the changes have therapeutic value. The 3-D shape of the target molecule gives the shape to which the potential drug needs to attach and physical models in structure based drug design are used to predict the functionality of changes in the structure of the molecule. Different parameters can be set in the computer modelling to predict properties of the changed molecule, e.g. binding ability, electronic properties before the molecule is produced. Once suitable forms have been found possible forms can be produced reducing the time, effort and cost of research.
5. There are three basic steps in drug design: 1. Target, e.g. receptor or enzyme is identified relating to a particular disease state. With knowledge about differences between normal cell proteins and diseased cell proteins, a suitable target can be chosen. The target needs to be disease modifying, e.g. enzyme activator or inhibitor. The target must not affect any other 'off-target' molecules which will cause side effects of the potential drug. 2. Target is fully characterised giving structural information about the given target. In this process the target is usually cloned and produced and purified to establish a screening assay which is used to determine the 3-D structure. 3. Molecule chosen – screening libraries are searched to find possible molecules that will suitably interact with the target. The chosen molecule is then designed so that it binds to the target to either restore normal signalling pathway or inhibit a dysfunctional biomolecule. The design should make a compound that can be taken as a drug, e.g. form an oral medicine with suitable chemical and metabolic stability and have few toxic effects.
6. (a) Relenza is an antiviral drug developed by the Australian biotech firm Biota Holdings.
   (b) Relenza is a neuraminidase inhibitor used in the treatment of influenza caused by influenza A and B viruses. Neuraminidase is viral enzyme that is part of the mechanism that causes newly formed virions in an infected host cell to be released from the host cell. Relenza acts by blocking the active site of neuraminidase using competitive inhibition to stop the host cell releasing the virions and spreading the infection.

83 Antiviral Drugs and Antibiotics

1. Both antiviral drugs and antibiotics are antimicrobials. Antibiotics are substances that kill or inhibit the growth of bacteria whereas antiviral drugs are used to cure or control virus infections.
2. Broad spectrum antibiotics are effective for killing many types of bacteria while narrow spectrum antibiotics target disease-causing bacteria.
3. Bacteriostatic antibiotics do not kill the bacteria, but inhibit their growth allowing the immune system to overcome the infection whereas bactericidal antibiotics destroy bacterial pathogens. Many bacteriostatic antibiotics interfere with protein synthesis, e.g. tetracycline, chloramphenicol, macrolides and spectinomycin. Bactericidal antibiotics can target outer cell walls, inner cell membranes or metabolic pathways of bacteria, e.g. penicillins and polymixins.
4.

Feature	Antiseptics	Disinfectants
Description of method	Antiseptics are antimicrobial chemicals that are applied to living tissue or skin to reduce the likelihood of infection, sepsis or putrefaction, e.g. alcohols are used to disinfect skin before injections. Chlorhexidine is a skin antiseptic or is used to treat inflammation of the gums.	Disinfectants are antimicrobial agents that are applied to non-living objects to destroy micro-organisms, e.g. hypochlorite is used to purify swimming pools.
Advantages	Antiseptics kill and/or prevent microbial growth and prevent infection. They are mild chemicals less effective than disinfectants but not damaging to skin and mucous membranes.	Are extremely effective in sterilising surfaces, e.g. sterilisation in hospitals of medical tools, in dental surgeries, kitchens, bathrooms and floors. Choice of disinfectant depends upon the need, e.g. some have a wide spectrum and kill nearly all organisms while others kill a small range of pathogens.
Disadvantages	Toxicity means they cannot be ingested. Since many are mild chemicals all endospores may not be killed, the resistant forms multiply leading to a population that is not affected by the antiseptic.	Cannot be used on living tissue, or ingested If all endospores are not killed, the resistant forms multiply leading to a population that is not affected by the disinfectant.
Evaluation	Each method to control microbial growth has specific applications and their advantages are related to their use, e.g. antiseptics are applied to living skin while disinfectants are applied to non-living matter. Each method also has disadvantages, e.g. disinfectants are toxic and cannot be ingested. Each method has its own value with none being more important than the other as each is required in its own application to prevent the spread of pathogens or the spoilage of food	

84 Synthetic Hormones

1. Synthetic hormones are being developed to control or treat endocrine dysfunction, e.g. for diabetes mellitus, hypothyroidism and hyperthyroidism, to improve the quality of life for individuals.
2. A synthetic hormone is a hormone that acts on target cells and is produced artificially.
3. The synthetic hormone needs to be bioidentical with the shape of the synthetic hormone and must be very similar to the 'natural' hormone as it must 'fit' with the receptor sites on the cell membrane or inside the cell.
4. Synthetic hormones can be produced using recombinant DNA technology.
5. Benefits of recombinant DNA technology include – 1. Reduction of the possibility of rejection or allergic reaction. 2. It is cheaper to produce than using animal extracted hormones. 3. Synthetic hormones can be produced in quantity. 4. There is a reduced need to extract particular hormones from farm animals for the treatment of particular human medical disorders.
6. Recombinant DNA technology to produce synthetic hormones involves: 1. Identifying the human gene for a particular hormone. 2. Isolating this gene. 3. Inserting the gene into a bacterial plasmid, e.g. insertion of the human gene into *Escherichia coli* plasmids (Ti plasmids). 4 Culturing the transformed bacteria in optimum growing conditions so that the bacteria produce large quantities of the gene product (required hormone). 5. Extracting the human hormone and packaging for medical treatments for particular disorders.
7. HRT is medical treatment that involves giving hormones to a person to replace hormones which are malfunctioning or insufficient.

8. Diabetes mellitus is an endocrine disorder where the body cannot maintain glucose homeostasis.
9. Diabetes mellitus type 1 is a type of diabetes mellitus when not enough insulin is produced leading to high blood glucose levels. Whereas diabetes mellitus type 2 is a long term progressive metabolic disorder with high blood glucose levels, insulin resistance and the body loses its ability to produce enough insulin.
10. The symptoms of diabetes include thirst due to high glucose levels, high glucose levels in urine, and problems with the eye, kidney and circulatory system.
11. The main treatment for diabetes is controlling blood glucose levels with insulin medications.
12. Insulin is a hormone produced by beta cells in the islets of Langerhans in the pancreas. It consists of two polypeptide chains that are linked by disulfide bonds.
13. Glycogenesis is the production of glycogen from glucose for storage.
14. Traditional treatment of diabetes required the injection of insulin gathered from the pancreases of pigs and calves. There were several problems with the use of animal insulin, e.g. it was difficult to extract in required quantities for the number of diabetics and was expensive, animal insulin is slightly different to human insulin which means it is detected by the immune system and can potentially cause an allergic response and using animal insulin also creates the possibility of the transmission of animal viruses into the human population. DNA recombinant technology has enabled the production of synthetic insulin using the human gene inserted into a bacterial plasmid. This has enabled a greater production of human insulin at lower cost with less chance of allergic reaction or rejection and less harm to farm animals.
15. The production of synthetic insulin also needs to be linked to an efficient and beneficial method of hormone delivery and further research is investigating new, more efficient and more accurate ways of measuring and monitoring blood glucose levels throughout the day. Patients using synthetic hormones may still have hypoglycaemic attacks and delivery needs to be timely and precise.
16. T4 and T3 are two thyroid hormones – thyroxine and triiodothyronine respectively that increase the rate of chemical reactions in most cells and thus increase metabolic rate.
17. Hypothyroidism is a condition in which the body lacks sufficient T4 and T3 which means body metabolism slows whereas hyperthyroidism is the overactivity of the thyroid gland.
18.

Feature	Hypothyroidism	Hyperthyroidism
Symptoms	Weakness, slow heart rate, weight gain or difficulty regulating weight, intolerance to cold, muscle cramps, depression, slowing of mental capabilities, constipation, dry or rough skin, deepening voice, memory loss, swollen tongue and face, goitre, cretinism in children.	Increased rate of metabolism, nervousness, anxiety, irritability, weight loss, rapid heart rate or palpitations, hand tremors, sleeping problems, excessive sweating and feeling hot and many people develop some degree of eyeball protrusion that stretches the optic nerve affecting vision.
Treatment	For limited production of T4 and T3 treatment involves increasing iodine intake with iodised salt or iodine-rich foods. Hormone replacement therapy is used if there is failure or damage to the thyroid gland, pituitary gland or hypothalamus.	Involves antithyroid drugs which interfere with the ability of the thyroid gland to use iodine, surgery removes part or all of the thyroid gland or radiotherapy using radioactive iodine which is taken up by active follicular cells and kills these cells. Treatment often leads to hypothyroidism which is then treated with hormone replacement therapy.

19. D

85 Gene Therapy

1. Gene therapy is a broad area which aims to correct genes that cause disease.
2. Gene therapy is being used in clinical trials for different types of cancer and treating children with X-SCID (severe combined immunodeficiency) or the 'bubble' children.
3.

Feature	Somatic gene therapy	Germ line gene therapy
What is changed	Body cell	Gamete, zygote or embryo
Inherited	Cannot be passed on	Can be passed on
How long it is effective	Lifetime of person	For generations

4. Two different methods of gene delivery are: (a) using virus vectors, and (b) non-viral delivery. Nasal sprays containing a viral vector carrying the therapeutic gene have been used to introduce new genes into the lungs. The target lung cells are invaded with the viral vector. Non-viral delivery includes using liposomes. Liposomes carry the therapeutic DNA inside an artificial lipid sphere and it fuses with the target cell's membrane.
5. (a) Children with SCID are called 'bubble children' as the disease makes them unable to fight infection and many have to live inside a 'bubble', e.g. a specially built room or wearing a ventilated helmet to reduce exposure to pathogens.
   (b) It is very important to provide these children with early treatment as they can easily catch infections, rapidly sicken and die.
   (c) The vector is the virus. The function of the virus is to implant the normal gene into the bone marrow.
   (d) It is important for the government to monitor clinical trials as the trials are dealing with people with a certain life expectancy and the trial should not decrease that expectancy. If results are questionable, then future actions need to be modified.
6. Using bone marrow cells that include stem cells in gene therapy has the benefit that a functional stem cell can divide over and over and produce more stem cells as well as cells that can differentiate into other types of cells. In bone marrow hematopoietic stem cells (HSC) can differentiated into red blood cells, white blood cells and platelets. These cells when used in gene therapy, e.g. for immunodeficiency diseases such as SCIDs so that the patient will be able to produce functional blood cells and will reduce the need for blood transfusions and other ongoing treatments.
7. Gene tests are a genetic procedure used to diagnose genetic disorders even if the person at present has no symptoms. This is beneficial as it can help the person by providing information about the course of the disease and the risk of future disease in their children, e.g. genetic testing for Duchenne muscular dystrophy, cystic fibrosis, sickle-cell anaemia and Huntington's disease. However, there are several potential disadvantages arising from gene testing, e.g. the emotional and psychological impact on the person; even if the disease has been diagnosed some diseases have no effective treatment or cure; privacy issues related to healthcare cover and employment. As a procedure genetic testing needs to be supported by genetic counsellors to help explain the disorders and provide information about future options. Overall gene testing is highly valuable as the diagnosis of disease can assist people and if they are supported after diagnosis they can make informed decisions about their future.

86 Gene Cloning and Cell Replacement

1. Gene cloning is a process which isolates a single gene so that multiple copies can be made, e.g. by inserting the gene into a vector such as a bacterial plasmid and then producing many copies through replication.
2. Bacterial plasmids are circular DNA molecules additional to the normal bacterial chromosome.
3. The required gene is isolated from chromosomal DNA using restriction enzymes.

4. Vectors for gene cloning include bacterial plasmids, bacterial artificial chromosomes, yeast artificial chromosomes, viruses, and mammalian cells.
5. The bacteria are often treated with calcium salts to make their membranes more permeable and allow plasmids to be taken up more easily.
6. Research into tissue and organ cloning is important as its success would mean damaged cells, e.g. in Alzheimer's or Parkinson's disease could be replaced by healthy cells and thus sufferers of these diseases could be more successfully treated than by current means.
7. Gene cloning has been used to insert the gene for therapeutic proteins into *E. coli* to produce large quantities of these proteins, e.g. manufacturing insulin, growth hormone and antihaemophilic globulin (the clotting factor missing in haemophilia).
8. A DNA library is a collection of DNA genes.
9. Selective breeding involves the choosing of particular desired characteristics in an individual and then breeding that individual for those characteristics. Over time the desired feature will be more common in the population and the frequency of that gene will increase in the gene pool while other genes may have decreased. On a large scale this can lead to a decrease in genetic diversity in that population. Gene cloning involves selecting specific strands of DNA and inserting it into a suitable vector so that it can enter the required host. This can create transgenic species, e.g. bacteria with the human gene for insulin production. Gene cloning also changes the gene pool. Cloning of the host with the desired gene produces large numbers of the new population which will also have reduced genetic diversity.
10. Gene cloning involves the isolation of segments of DNA and inserting them into a vector to make a recombinant DNA molecule. This required DNA can then be propagated in the host cell. Whole organism cloning involves producing offspring with an identical genotype to the parent, e.g. by taking nuclear DNA from the organism to be cloned, inserting it into a denucleated egg and allowing the egg to develop into an adult organism.
11. A clone library can be: 1. Used as a source of DNA that can be sequenced. 2. The sequence tagged site (STS) clones can also be used to construct a physical map of a chromosome.

87 Alzheimer's Disease and Parkinson's Disease

1. Dementia is a progressive, irreversible degeneration of the cerebral cortex that affects memory, intellect and reasoning.
2. (a) In Alzheimer's disease amyloid plaques (fibrous patches) ranging in diameter from 10 micrometres to several hundred micrometres and neurofibrillary tangles (protein filament strands) form on brain cells disrupting their functioning. The affected brain cells die leading to shrinkage (atrophy) of the brain which eventually affects the ability of the brain to function correctly.
   (b) Symptoms include memory loss, decreased ability to plan and problem solve, vagueness in conversation, becoming disoriented and taking longer to do routine tasks.
3. There are medications that can help slow down the loss of memory and thinking abilities, manage behavioural symptoms or to treat other aspects of the disease, e.g. depression and sleep disorders. Researchers are investigating ways to stop amyloid proteins forming clumps and then plaques in the brain, using monoclonal antibodies designed to lower amyloid in the brain, ways to stop the protein fibres from collapsing and twisting into tangles, vaccines to stimulate the immune system to destroy protein fibre tangles in the brain and drugs that slow down or prevent symptoms in people who have the genetic mutations for Alzheimer's disease.
4. The basal ganglia are the nuclei surrounding the thalamus that help with planning movement.
5. Parkinson's disease is a disease that affects the neurons especially the dopamine releasing neurons in the middle area of the brain that are involved in muscle movement.
6. The loss of control and coordination leads to fixed muscle tone which causes an expressionless face, stooping posture, shuffling gait and rigid muscles. There is often muscle tremor of the extremities which makes simple movements difficult and high mental effort and concentration is needed to perform simple tasks or speak clearly. These symptoms often create distress and anxiety.
7. In many cases there is no clearly identifiable cause for Parkinson's disease, though most researchers believe it is a combination of genetic and environmental factors.
8. Parkinson's disease involves the death of neurons in the midbrain which affects the circulatory system of the basal ganglia. Implantation of dopamine-secreting neurons into the basal ganglia would slow down the progressive degeneration of the brain and could even lead to the recovery of motor control.
9. The use of embryonic stem cells that have been stimulated or genetically engineered to develop into dopamine-secreting neurons would be a great aid to a person with Parkinson's disease. These cells could be transplanted into the patient with the possibility of restoring motor control and stopping the progress of the disease.
10. C

Topic Test

Section A – Multiple Choice (35 marks)

1. C	2. B	3. D	4. A	5. B
6. C	7. C	8. B	9. A	10. C
11. A	12. A	13. B	14. C	15. B
16. D	17. C	18. A	19. A	20. B
21. B	22. A	23. D	24. C	25. A
26. B	27. B	28. C	29. B	30. A
31. D	32. C	33. A	34. D	35. B

Section B – Written Response (65 marks)

36. (a) Marking Guidelines

Criteria	Marks
Outlines one correct advantage of homeostasis	1

Sample answer:
Maintaining a constant internal environment within narrow limits means that enzymes can work at optimum efficiency in cell metabolism.

(b) Marking Guidelines

Criteria	Marks
Gives four correct factors	3
Gives three correct factors	2
Gives two correct factors	1

Sample answer:
Factors kept constant in homeostasis include – carbon dioxide content, pH, water content, ion balance, temperature.

37. Marking Guidelines

Criteria	Marks
Gives action for response R and response Q	2
Gives correct response for either response R or response Q	1

Sample answer:
Response R is an action that lowers body temperature, e.g. increased rate of sweat production or dilation of muscles in walls of blood vessels.
Response Q is an action that increases body temperature, e.g. shivering and muscle contraction or vasoconstriction.

38. Marking Guidelines

Criteria	Marks
Correctly completes table filling in all sections	6
Correctly completes some parts of the table	1 to 5

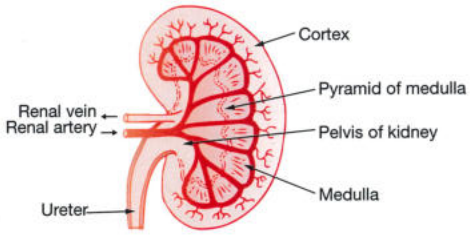
Sample answer:

Type of hormone	Structure	Example
Peptide	Consists of chains of amino acids.	Insulin, growth hormone.
Steroid	Lipid and phospholipid derived, generally synthesised from cholesterol.	Testosterone, cortisol.
Amine	Amine derived, e.g. from tyrosine, amino acids.	Thyroxine, melatonin.

39. Marking Guidelines

Criteria	Marks
Draw a cross-section of a kidney with correct labels for the cortex, the medulla, a pyramid of the medulla, the pelvis of the kidney, the renal vein, the renal artery and the ureter	4
Draws the correct shape and some labelled features	1 to 3

Sample answer:



40. (a) Marking Guidelines

Criteria	Marks
Lists all steps in the procedure, in correct sequence; identifies equipment needed; explains at least two of the actions and why they were taken	4
Lists all steps in the procedure, in correct sequence; identifies equipment needed; explains one of the actions and why they were taken OR Lists some steps in the procedure; identifies equipment needed; explains at least two of the actions and why they were taken	3
Lists some steps in the procedure; identifies equipment needed; explains one of the actions and why they were taken OR Lists all steps in the procedure, in correct sequence; identifies equipment needed	2
Lists some steps in procedure OR identifies equipment	1

Sample answer:

- A small quantity of a particular food (e.g. raw mince meat, cheese, bread) was purchased from several different shops.
- Each group investigated one particular food group and results were then pooled so that a range of foods could be covered, each group performed all the tests, and allowed an efficient time frame for the experiment.
- The food was placed in a beaker and water was added to form a dilute solution of each food.
- Samples were taken from each solution, streaked across agar plates and the plates were sealed, labelled and incubated at room temperature.
- After several days the plates were examined to see if any colonies had grown. The plates were not opened as they could contain pathogens that could infect one of the students.
- The plates were compared so that conclusions could be drawn about the food source, food type and presence of microbes.

(b) Marking Guidelines

Criteria	Marks
Identifies TWO relevant safety procedures	2
Identifies ONE relevant safety procedure	1

Sample answer:

- Wear a laboratory coat and closed-in footwear to prevent possible contamination of self or clothing.
- Properly label all cultures very carefully with name, date and what was placed on the plate so that if a plate is accidentally broken or opened, correct precautions can be taken.

41. (a) Marking Guidelines

Criteria	Marks
Identifies TWO aspect of insulin	2
Identifies one aspect of insulin	1

Sample answer:

Insulin causes body cells and liver to take up glucose with liver cells storing it as glycogen.

(b) Marking Guidelines

Criteria	Marks
Defines antagonistic hormones and shows how insulin and glucagon are antagonistic	2
Shows understanding of antagonistic hormone function	1

Sample answer:

Insulin and glucagon are antagonistic hormones as they both regulate the concentration of glucose in the blood but have opposite effects. Low blood glucose levels will cause the alpha cells of the pancreas to release glucagon into the blood which will cause liver cells to break down glycogen and release glucose into the blood. High blood glucose levels will cause the beta cells of the pancreas to release insulin into the blood which will cause the liver to take up glucose and store it as glycogen as well as body cells to take up more glucose.

42. (a) Marking Guidelines

Criteria	Marks
Correctly identifies what causes an increase in carbon dioxide levels in the blood and what detects this change	2
Correctly identifies what causes an increase in carbon dioxide levels in the blood OR what detects this change	1

Sample answer:

Increased respiration will raise carbon dioxide levels in the blood which detected by receptors in the brain stem and lungs.

(b) Marking Guidelines

Criteria	Marks
Correctly related high carbon dioxide to pH and effect of acidic conditions on homeostasis and cell metabolism	3
Identifies some aspects of high carbon dioxide in blood and cell metabolism	1 to 2

Sample answer:

High levels of carbon dioxide are toxic and damage cell metabolism as high carbon dioxide levels decrease pH. The major buffer system in human blood is the acid-base pair H_2CO_3-HCO_3^-. Carbonic acid (H_2CO_3) is a weak acid and resists changes to pH. If carbon dioxide is not removed the buffer system would fail and enzymes would not function at maximum efficiency thereby impairing cell metabolism.

43. Marking Guidelines

Criteria	Marks
Correctly relates the function of the immune system to process in gene therapy and identifies problem	3
Correctly relates the function of the immune system to process in gene therapy	2
Correctly states the function of the immune system or states process in gene therapy	1

Sample answer:
The immune system identifies any foreign object that is introduced into the body and uses a variety of means to destroy the foreign object and establish memory cells that protect against any further exposure of that type of object. The introduction of therapeutic DNA using a viral vector can stimulate the immune response so that the vector or any cell containing the vector is attacked by the immune system. The immune system is activated by any antigen which means that therapeutic DNA coated in a liposome can also activate B and T cells.

44. (a) Marking Guidelines

Criteria	Marks
Correctly identifies two chemicals removed from the blood by the kidney	2
Correctly identifies one chemical removed from the blood by the kidney	1

Sample answer:
Any two of the following – urea, salt, vitamins, water, glucose, amino acids.

(b) Marking Guidelines

Criteria	Marks
Correctly identifies two chemicals reabsorbed into the blood by the kidney	2
Correctly identifies one chemical reabsorbed into the blood by the kidney	1

Sample answer:
Any two of the following – glucose, amino acids, some vitamins and water and salts as needed

(c) Marking Guidelines

Criteria	Marks
Correctly explains the main steps in the formation of urine with delivery of blood to the kidney, filtration, reabsorption, secretion and the action of a hormone	5
Correctly identifies some aspects of the formation of urine	1 to 4

Sample answer:
Blood enters the kidney from the renal artery. Ultrafiltration, which is passive transport filters water, salts, urea, glucose, amino acids and vitamins from the glomerular capillaries into Bowman's capsule of a nephron. These substances form the filtrate which moves through the renal tubules of the nephron. Selective reabsorption occurs with useful substances from the filtrate, e.g. amino acids and glucose being actively reabsorbed and sodium ions are pumped back into to the blood in the surrounding capillaries. There is also secretion of other substances, e.g. toxins and excess ions not needed by body which are added to the filtrate. Antidiuretic hormone (ADH) controls the permeability to water of the walls of the collecting duct so that water is reabsorbed by osmosis as needed by the body. Urine is the waste-containing liquid produced by the kidneys and is collected in the urinary bladder. Urine contains water, salts, urea and other secreted substances.

45. Marking Guidelines

Criteria	Marks
Recognises presence of normal flora on healthy body; recognises that barriers exist to prevent entry of these flora to other areas of the body; identifies the condition which leads to imbalance of microflora; relates changed condition to normal flora invading the tissue and causing disease	4
Recognises presence of normal flora on healthy body; AND discusses TWO of the following – recognises that barriers exist to prevent entry of these flora to other areas of the body; identifies condition which leads to imbalance of microflora; relates changed condition to normal flora invading the tissue and causing disease	3
Recognises presence of normal flora on healthy body AND recognises that barriers exist to prevent entry of these flora to other areas of the body OR identifies the condition which leads to imbalance of microflora	2
Recognises presence of normal flora on healthy body OR recognises that barriers exist to prevent entry of these flora to other areas of the body OR identifies the condition which leads to imbalance of microflora	1

Sample answer:
The skin and membranes of the mouth, throat, intestines and vagina normally host a range of microscopic organisms, some of which are harmful and some are not. The use of antibiotics often destroys all microbes, including the useful varieties. For example, the use of penicillin can cause the environment to change and the fungus *Candida albicans* can invade normal healthy tissue. The fungus multiplies and the infection is known as candidiasis, or thrush.

46. Marking Guidelines

Criteria	Marks
Gives suitable explanations of how natural and acquired immunity develop and the duration of each type of immunity	4
Some correct infomation given	1 to 3

Sample answer:

Feature	Natural immunity	Acquired immunity
How it develops	Develops when the person is exposed to the antigen and begins to produce their own antibodies.	Develops when the person is immunised and exposed to a form of the antigen that will cause the production of antibodies without having the systems.
Short duration immunity	Occurs naturally when antibodies from mother given to baby, e.g. via milk or placenta.	Occurs artificially when antibodies given in an injection, e.g. hepatitis immunoglobulins.

47. Marking Guidelines

Criteria	Marks
Correctly names each type of neuron and outlines its function	6
Correctly identifies some neurons and their function	1 to 5

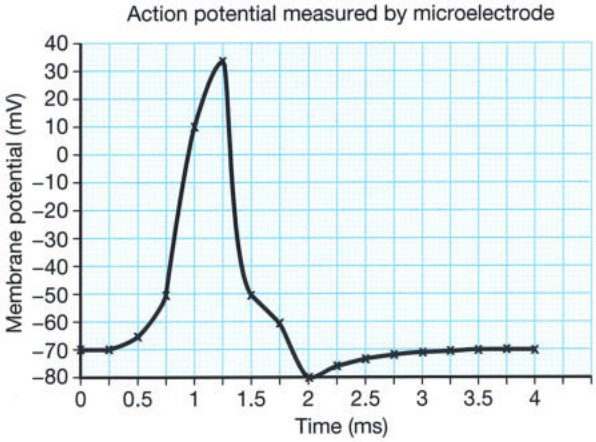
Sample answer:

Neuron	Description
	Name – sensory neuron
	Function – transmits information from the receptors, e.g. in a sense organ to connector neurons in CNS.
	Name – connector neuron (also multipolar neuron, interneuron).
	Function – are part of CNS and connect to other connector neurons, sensory neurons and to motor neurons.
	Name – motor neuron.
	Function – transmits information from connecter neuron in CNS to the effector, e.g. a muscle or a gland.

48. (a) Marking Guidelines

Criteria	Marks
Draws graph with correctly labelled axes with units, plots points and joins points	3
Draws some aspects of graph	1 to 2

Sample answer:

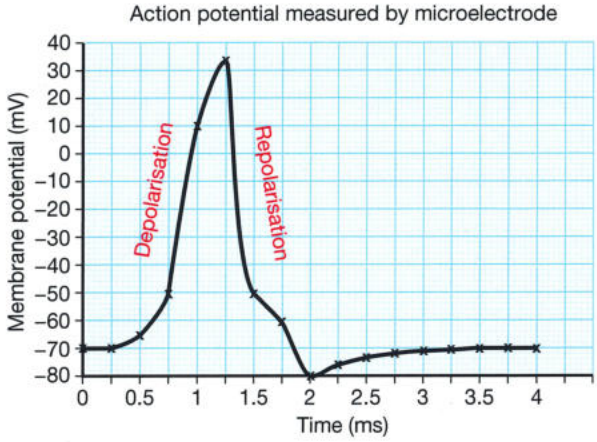


(b) Marking Guidelines

Criteria	Marks
Correctly distinguishes between depolarisation and repolarisation and marks them on the graph	3
Correctly answers some parts	1 to 2

Sample answer:

Depolarisation is the change in the cell membrane potential at the beginning of the nerve impulse when sodium ions move into the cell making the inside of the cell more positive whereas repolarisation is the change in the cell membrane potential that returns the membrane potential to a more negative value due to the movement of potassium ions out of the cell.



(c) Marking Guidelines

Criteria	Marks
Correctly explains why some stimuli do not result in an action potential by linking to all or none rule and sodium ion movement	2
Identifies stimulus is too weak to initiate impulse	1

Sample answer:

For an action potential to occur the change in membrane potential must pass the threshold value. A stimulus will cause either a full response or no response (all or none rule). If the stimulus is too small (subthreshold) insufficient numbers of sodium ions enter the neuron.

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